A CONCEPT-BASED MATHEMATICS COURSE

YEARS 3 · 4 · 5
MIDDLE YEARS PROGRAMME

STUDENT COURSEBOOK · VOLUME FOUR

MYP *Mathematics*

4

STUDENT COURSEBOOK

YEAR 4 · INQUIRY · PRACTICE · ASSESSMENT



AUTHOR

Sadikou Moukoddim

Head of Mathematics Department
Mathematics Teacher · Category 1

ABOUT THIS VOLUME

An original, inquiry-led course that builds algebraic fluency, geometric reasoning, statistical literacy and mathematical modelling. Designed as the fourth volume in a coherent five-year series.

MYP MATHEMATICS 4



A FIVE-VOLUME MATHEMATICS COURSE
FOR THE MIDDLE YEARS PROGRAMME

VOLUME FOUR

MYP *Mathematics* 4

STUDENT COURSEBOOK

Number · Algebra · Geometry and trigonometry · Statistics and probability

First edition · 2026

MYP MATHEMATICS 4

A CONCEPT-BASED COURSE FOR YEARS 3–5

MYP Mathematics 4

STUDENT COURSEBOOK

Inquiry-led teaching, rigorous practice and authentic assessment for the fourth year of a coherent five-year mathematics series.

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VOLUME 4 OF 5

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Author Sadikou Moukoddim, Head of Mathematics Department, Mathematics Teacher, Category 1.

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“Mathematics is not about numbers, equations, computations or algorithms: it is about understanding.” — William Thurston

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The course

Page numbers refer to the main body of the book. Each chapter opens with an inquiry, develops concepts through worked examples and graduated practice, and closes with mixed practice, an assessment task, a summary and reflection.

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HOW THE BOOK WORKS

Every chapter follows the same rhythm: an authentic problem, a conceptual inquiry, precise definitions, worked reasoning, graduated practice, an investigation, a modelling task and a criterion-mapped assessment.

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The chapters are sequenced so that each idea is met in a concrete or visual form before it is formalised. Later chapters deliberately reuse earlier tools: for example, factorisation from Chapter 2 becomes the engine of Chapter 5, and linear models from Chapter 4 reappear in the financial modelling of Chapter 8.

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A COHERENT FIVE-YEAR SERIES

Year 3 builds fluency with rational number, proportion, linear relationships, angle geometry and basic data.
Year 4 (this volume) extends algebra to quadratics and exponentials, formalises similarity and trigonometry, and develops statistical and probabilistic reasoning.
Year 5 adds logarithms, the sine and cosine rules, circle theorems, polynomials, series, conditional probability and vectors.

Year 4 is the bridge year: it consolidates the concrete and visual reasoning of Year 3 while demanding the abstraction and justification expected in Year 5. The course map at the front of the book makes this progression explicit.

ORIENTATION

How to use *this book*

The rhythm of a chapter

1. **Opening problem.** A real or mathematical situation that you cannot yet solve completely. It creates the need for the mathematics.
2. **Inquiry.** The statement of inquiry, key and related concepts, global context, and factual, conceptual and debatable questions frame the chapter.
3. **Prerequisite check.** A short diagnostic. If you cannot do these, use the refresher box before continuing.
4. **Concept development.** Definitions, representations, worked examples and your own explanations.
5. **Graduated practice.** Fluency, understanding, reasoning, multi-step application and extension. Do not skip the reasoning tier.
6. **Investigation.** You discover a pattern, generalise it and justify it.
7. **Modelling.** You use mathematics in a context where assumptions, units and accuracy matter.
8. **Mixed practice.** Cumulative retrieval across the whole chapter.
9. **Assessment.** A Criterion A–D task with a rubric. This is summative: complete it independently.
10. **Summary and reflection.** Check what you can do and what you still need to repair.

Components you will meet

INQUIRY
A question or problem that drives the section. It is not rhetorical: you should be able to attempt an answer by the end.

KEY IDEA
The smallest statement worth remembering. If you can explain it and use it, you have understood the concept.

WORKED EXAMPLE
A complete line of reasoning with the decisions made visible. The notes in the margin explain *why*, not just *what*.

MISCONCEPTION
A common error, stated plainly, with the reasoning that exposes it. Read these before practising.

GUIDED PRACTICE
Partially completed work. You supply the missing reasoning, then compare with a partner.

EXERCISE
Numbered practice. Tier labels indicate fluency, understanding, reasoning, application or extension.

ASSESSMENT
A summative task mapped to MYP criteria. The rubric tells you what evidence is required.

"An answer is not a solution until you can say why it must be true, what it means, and when it would fail."

USING THE ANSWER KEY
Attempt every question before checking. For a wrong answer, read the first line of the solution only, then retry. The key often shows more than one valid method; yours may differ and still be correct.

BEFORE A LESSON
Skim the inquiry and the first worked example. Write one question you want answered.

DURING A LESSON
Copy the reasoning, not just the answer. Annotate each step with *why* it is allowed.

AFTER A LESSON
Do the odd-numbered questions. Mark them. Repair one error before moving on.

THE PROGRAMME

The MYP framework *in Year 4*

The MYP is concept-based and inquiry-led. It asks students not only to calculate, but to notice structure, make generalisations, justify them and apply them responsibly. Mathematics in the MYP is organised through three key concepts—**form**, **logic** and **relationships**—and through related concepts that name the mathematical work more precisely.

KEY CONCEPT

Form

What something is like: its shape, structure, notation and pattern. In this course: algebraic form, graphical form, geometric form and distributional form.

KEY CONCEPT

Logic

What follows from what: deduction, proof, counterexample and the rules that make an argument valid.

KEY CONCEPT

Relationships

How quantities, representations and ideas connect: functions, equivalence, similarity, correlation and modelling.

Related concepts used in this volume

approximation · change · equivalence · generalisation · justification · measurement · models · patterns · quantity · representation · simplification · space · systems · validity

Global contexts

identities and relationships · orientation in space and time · personal and cultural expression · scientific and technical innovation · fairness and development · globalization and sustainability

Approaches to learning

Communication · Research · Thinking · Social · Self-management. Each chapter identifies the ATL clusters it develops and gives a concrete skill indicator.

The four assessment criteria

A **Knowing and understanding.** Select appropriate mathematics, apply it successfully, and solve problems correctly in familiar and unfamiliar contexts.

B **Investigating patterns.** Select problem-solving techniques, describe patterns as general rules, and prove or verify and justify those rules.

C **Communicating.** Use mathematical language and notation, represent information, move between representations, reason coherently and organise information logically.

D **Applying mathematics in real-life contexts.** Identify relevant elements, select and apply strategies, justify accuracy, and judge whether a solution makes sense.

The criterion summaries above are adapted, in plain language, from the MYP Mathematics framework (IB, updated 2020). The full official descriptors and achievement-level rubrics are held by the school. This book uses the criteria to design tasks and to explain what evidence a response should contain; it does not replace the official guide.

ASSESSMENT

Evidence of *understanding*

Formative and summative are not the same

Formative work is practice for learning. Exercises, guided examples, investigations and mixed practice are formative: you are expected to make errors, use the answer key and revise. **Summative** work is evidence of learning at a point in time. The assessment task at the end of each chapter is summative: it should be completed independently, without the answer key.

What strong evidence looks like

- **Criterion A:** correct selection and execution, with units and sensible accuracy.
- **Criterion B:** a table or diagram of cases, a general rule, and a verification or proof—not just three examples.
- **Criterion C:** notation used correctly; words, tables, graphs and algebra connected; a reader can follow the argument without guessing.
- **Criterion D:** assumptions stated, model applied, accuracy justified, and the answer interpreted in context.

Command terms

State / write down	Give a short answer without explanation.
Describe	Give a detailed account of the features.
Explain	Give reasons or causes; show how the result follows.
Calculate / determine	Obtain a numerical answer, showing sufficient working.
Verify / justify	Confirm with evidence; give a logical reason for a statement.
Prove	Establish truth by a general, logical argument.
Interpret / evaluate	Explain the meaning; make a judgement using evidence.
Model	Represent a situation mathematically, stating assumptions.

ACHIEVEMENT LEVELS 0–8

Each criterion is marked out of 8. The bands below are a study aid, not a substitute for the school's official rubric.

1–2	Simple, familiar situations; limited selection; support needed.
3–4	More complex familiar situations; appropriate selection; some justification.
5–6	Challenging familiar situations; successful application; general rules verified.
7–8	Challenging familiar and unfamiliar situations; precise, coherent, justified and evaluated.

Academic honesty

Collaboration is encouraged when learning, but summative work must be your own. Acknowledge data sources, tools and any help received. Copying a solution without understanding it is both dishonest and ineffective: the assessment asks you to transfer ideas to a new context.

After the task

Use the reflection page. Identify one conceptual error, one communication error and one habit to change. The answer key gives sample responses and common-error notes; compare your reasoning, not just your final answer.

SCOPE AND SEQUENCE

Year 4 *course map*

This is a school-designed Year 4 course aligned to the MYP framework. It is not a mandatory IB syllabus. The sequence assumes the Year 3 foundations listed below and deliberately reserves advanced content for Year 5.

CH.	FOCUS	STRAND	KEY CONCEPT	GLOBAL CONTEXT	CRITERIA
1	Number foundations	Number	Form	Scientific innovation	A, C
2	Algebraic expressions	Algebra	Form	Personal expression	A, B, C
3	Equations, inequalities, systems	Algebra	Relationships	Fairness and development	A, C, D
4	Functions and linear models	Algebra / geometry	Relationships	Orientation in space	A, C, D
5	Quadratic equations	Algebra	Form	Scientific innovation	A, B, C
6	Quadratic functions	Algebra / functions	Relationships	Personal expression	A, C, D
7	Patterns and sequences	Number / algebra	Form	Globalisation and sustainability	A, B, C
8	Growth and financial mathematics	Number / algebra	Relationships	Fairness and development	A, C, D
9	Congruence, similarity, transformations	Geometry	Relationships	Identities and relationships	A, B, C
10	Pythagoras and trigonometry	Geometry / trigonometry	Relationships	Orientation in space	A, B, C, D
11	Surface area and volume	Geometry / measurement	Form	Globalisation and sustainability	A, C, D
12	Statistics and distributions	Statistics	Relationships	Identities and relationships	A, C, D
13	Probability and combined events	Probability	Logic	Fairness and development	A, B, C, D
★	Capstone modelling project	Integrated	Relationships	Globalisation and sustainability	B, C, D

Assumed from Year 3

Order of operations; negative numbers; fractions, decimals and percentages; ratio and proportion; simplification of linear expressions; solving one- and two-step linear equations; plotting points and straight-line graphs; angle facts for triangles and parallel lines; area and perimeter; mean, median and mode; simple probability.

Taught in depth in Year 4

Rational exponents; surds; factorisation and algebraic fractions; inequalities and simultaneous equations; function notation and coordinate geometry; quadratics; linear and quadratic sequences; repeated percentage change, compound interest and depreciation; congruence and similarity; right-triangle trigonometry; surface area and volume; histograms, quartiles and box plots; combined-event probability.

Reserved for Year 5

Sum formulae for arithmetic and geometric series; exponential and logarithmic functions, with continuous growth and decay; solving exponential equations using logarithms; the sine and cosine rules and the area of a triangle from two sides and an included angle; trigonometry in non-right triangles in three dimensions; circle theorems; polynomial division and the factor theorem; inverse and composite functions; vectors and matrices; spread from grouped data and the normal distribution; correlation and regression; conditional probability and counting principles; quadratic inequalities in two variables and linear programming.

How the progression works

Year 4 moves from *procedure* to *structure*: students explain why a factorisation is equivalent, why a discriminant predicts the number of roots, and why an enlargement scales area by the square of the scale factor.

POWERS, EXACT FORMS AND THE LIMITS OF MEASUREMENT

Number Foundations

01

Exponents, surds and accuracy — the forms we choose to describe quantity, and the precision those forms allow us to claim.

FORM · SIMPLIFICATION · APPROXIMATION

STATEMENT OF INQUIRY

The form in which a number is expressed — exact, approximate, or as a power — determines both the precision we can claim and the operations we can perform.

KEY CONCEPT
Form

RELATED CONCEPTS
Simplification · Equivalence · Approximation · Quantity

GLOBAL CONTEXT
Scientific and technical innovation — precision, measurement and discovery

FACTUAL
What are the laws of exponents, and how do we write a number in standard form?

CONCEPTUAL
How does the form of a number reveal or conceal its structure?

DEBATABLE
Can a measurement ever be exact — and when is rounding before the final step acceptable?

IN THIS CHAPTER

1.1 Number systems and proof
1.2 Integer and rational exponents
1.3 Standard form and estimation
1.4 Surds and exact values
1.5 Accuracy, bounds and error
1.6 Investigation: how exact is exact?

OPENING PROBLEM · A CLAIM WORTH TESTING

A sheet of paper, forty-two folds

Take a sheet of paper 0.1 mm thick. Fold it in half: the stack is 0.2 mm thick. Fold again: 0.4 mm. After ten folds the stack is about 10 cm deep; after twenty folds it is about 105 m — taller than most buildings. A popular claim says that after **42 folds** the stack would reach the Moon, about $384\,400$ km away.

Is the arithmetic sound? What does the claim ignore? Work in a group of three. Record your first estimate before calculating.

1. Write an expression for the thickness after n folds, starting from 0.1 mm.
2. Complete a table for $n = 0, 1, 5, 10, 20, 30, 42$. Use $2^{10} \approx 10^3$ to estimate before you calculate.
3. Compare the thickness after 42 folds with the Earth–Moon distance. Is the claim numerically sound?
4. State at least two physical assumptions that make the claim misleading as a description of real paper.

This chapter gives you the language to answer the problem precisely: powers to describe repeated multiplication, standard form to handle very large and very small quantities, surds to keep exact values exact, and bounds to say honestly how much a measurement can be trusted.

ORIENTATION

1.0 Learning intentions

- Extend the laws of indices from positive whole-number exponents to integer and rational exponents.
- Represent very large and very small quantities in standard form, and calculate with them sensibly.
- Simplify surds and rationalise denominators to preserve exact values.
- Quantify the accuracy of a measurement using significant figures, bounds and percentage error.
- Investigate a pattern in powers, state a general rule, and justify why it must continue.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

classify numbers within the real number system and justify each classification; simplify and evaluate expressions with integer and rational exponents; convert between standard form and ordinary form and calculate to a stated accuracy; simplify surds and rationalise simple denominators; calculate upper and lower bounds and percentage error; choose between exact and approximate forms and justify the choice.

ATL SKILLS · THINKING AND SELF-MANAGEMENT

Critical thinking: before rounding, decide whether the result will be used for a comparison, a measurement or a further calculation — that choice sets the accuracy you need.
Information literacy: when you quote a large number from a calculator, record its units and its degree of accuracy; 4.4×10^5 km is a claim about rounding as much as about distance.
Organisation: apply one index law per line and keep the base visible; most errors in this chapter come from applying a law to the wrong factor, not from arithmetic.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 1.1.

1. Evaluate 2^5, 3^4, 10^3 and 5^0.
2. Evaluate $\sqrt{49}$, $\sqrt{81}$ and $\sqrt[3]{27}$.
3. Round 3.846 to 2 decimal places and to 1 significant figure.
4. Simplify $2x \cdot 3x^2$.
5. Solve $2^x = 16$.
6. Write 0.25 and 0.4 as fractions in simplest form.

REFRESHER

a^n means $a \times a \times \cdots \times a$ with n factors. The number a is the *base* and n is the *index* (or exponent). A square root asks: which positive number squared gives this value? A cube root asks: which number cubed gives this value?

STRUCTURE OF THE REAL NUMBERS

1.1 Number systems and proof

Numbers are not all of the same kind. The counting numbers, the integers, the fractions and the irrational numbers form nested sets inside the real numbers. Knowing which set a number belongs to tells you what operations are guaranteed to stay inside that set, and what kind of decimal representation to expect.

The diagram below is a map, not a definition. Read it from the outside in: every natural number is an integer; every integer is rational; every rational is real. Irrational numbers are real, but they sit outside the rational set.

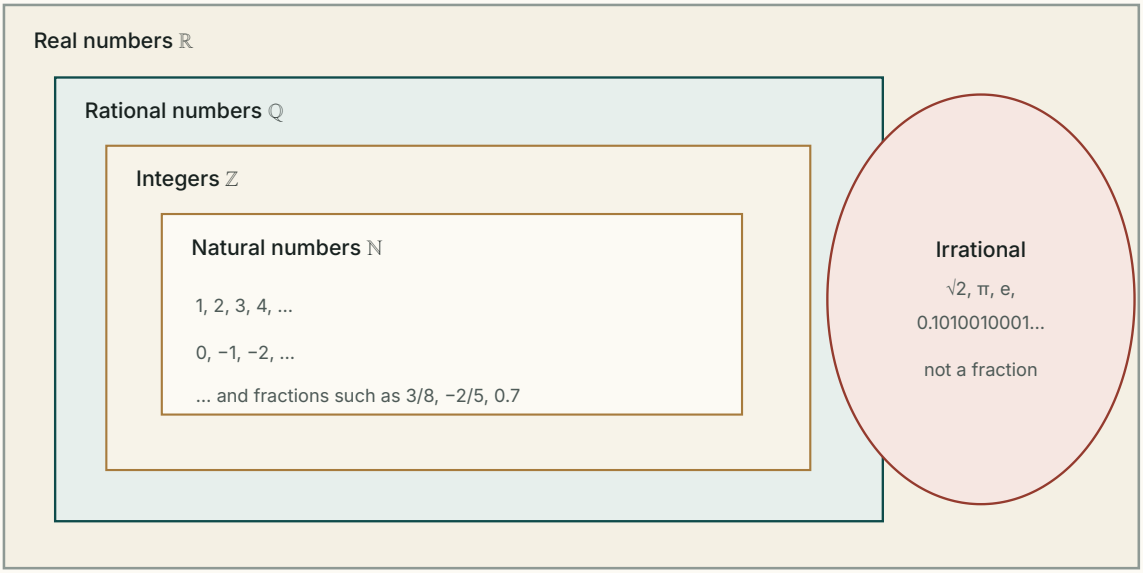


Figure 1.1 The real number system. The rationals include all terminating and recurring decimals; the irrationals have decimal expansions that never terminate and never repeat.

DEFINITION

Rational and irrational numbers

A number is **rational** if it can be written in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$. Its decimal expansion either terminates or eventually repeats a block of digits.

A number is **irrational** if it cannot be written in that form. Its decimal expansion neither terminates nor repeats. The real numbers are the union of the rational and irrational numbers.

Recurring decimals are fractions in disguise

A terminating decimal such as 0.375 is easy to convert: $0.375 = \frac{375}{1000} = \frac{3}{8}$. A recurring decimal needs a trick: multiply by a power of 10 that shifts one full repeating block, then subtract to eliminate the recurring tail.

EXAMPLE 1.1 CLASSIFYING NUMBERS

QUESTION

Classify each number as natural, integer, rational or irrational. Some belong to more than one set. Justify each answer.

$$-7, \quad 0, \quad \frac{3}{8}, \quad \sqrt{25}, \quad \sqrt{7}, \quad 0.\overline{2}, \quad \pi, \quad 0.75$$

SOLUTION

1. -7**:** an integer, and therefore rational $\left(-7 = \frac{-7}{1}\right)$. Not natural.
2. 0**:** an integer and rational $\left(0 = \frac{0}{1}\right)$. Not natural if we take $\mathbb{N} = \{1, 2, 3, \ldots\}$.
3. $\frac{3}{8}$**:** rational by definition; its decimal is 0.375, which terminates.
4. $\sqrt{25} = 5$**:** a natural number, integer and rational. The square root symbol does not make a number irrational.
5. $\sqrt{7}$**:** irrational. $2^2 = 4$ and $3^2 = 9$, so $\sqrt{7}$ lies between 2 and 3 and is not an integer; it cannot be written as a fraction of integers (proved in Example 1.3 for $\sqrt{2}$).
6. $0.\overline{2} = 0.222\ldots$**:** rational, equal to $\frac{2}{9}$.
7. π**:** irrational. $\frac{22}{7}$ is a rational approximation, not equal to π.
8. 0.75**:** rational, equal to $\frac{3}{4}$. It terminates.

> **Reasoning:** "Has a decimal" does not mean "is irrational". The test is whether the decimal terminates or repeats. $\sqrt{25}$ and 0.75 both have finite decimals, so both are rational.

EXAMPLE 1.2 FROM RECURRING DECIMAL TO FRACTION

QUESTION

Write $0.\overline{47}$ and $0.8\overline{3}$ as fractions in simplest form.

SOLUTION

1. Let $x = 0.\overline{47} = 0.474747\ldots$. The repeating block has two digits, so multiply by $10^2 = 100$:

$$100x = 47.474747\ldots$$

2. Subtract the original equation:

$$100x - x = 47.4747\ldots - 0.4747\ldots \;\Rightarrow\; 99x = 47$$

3. So $x = \dfrac{47}{99}$. It is already in simplest form because 47 is prime and does not divide 99.
4. For $0.8\overline{3} = 0.8333\ldots$, use $10x = 8.333\ldots$ and $100x = 83.333\ldots$. Subtracting gives $90x = 75$, so $x = \dfrac{75}{90} = \dfrac{5}{6}$.

Check: $47 \div 99 = 0.474747\ldots$ and $5 \div 6 = 0.8333\ldots$. The fraction and the decimal are two forms of the same number — not approximations of each other.

KEY IDEA

Every terminating or recurring decimal is rational. A decimal that continues forever without repeating is irrational. The form changes what you can see, not the value of the number.

PROOF BY CONTRADICTION

1.1a Why $\sqrt{2}$ cannot be a fraction

It is not enough to observe that a calculator gives $1.41421356\ldots$ and stops. A calculator can only show finitely many digits. To prove that $\sqrt{2}$ is irrational, we use a **proof by contradiction**: assume the opposite, and show that it forces an impossible conclusion.

EXAMPLE **1.3** PROOF THAT $\sqrt{2}$ IS IRRATIONAL

CLAIM

There is no pair of integers p and q, with $q \neq 0$, such that $\sqrt{2} = \dfrac{p}{q}$.

PROOF

1. **Assume the opposite.** Suppose $\sqrt{2} = \dfrac{p}{q}$, where p and q are integers with no common factor other than 1 (the fraction is in lowest terms), and $q \neq 0$.
2. Squaring both sides gives $2 = \dfrac{p^2}{q^2}$, so $p^2 = 2q^2$.
3. The right-hand side is even, so p^2 is even. If a square is even, its root is even; hence p is even. Write $p = 2k$ for some integer k.
4. Substituting: $(2k)^2 = 2q^2$, so $4k^2 = 2q^2$, which simplifies to $q^2 = 2k^2$. Now q^2 is even, so q is even.
5. Both p and q are even, so they have a common factor of 2. This contradicts the assumption that $\frac{p}{q}$ was in lowest terms.
6. The assumption must be false. Therefore $\sqrt{2}$ cannot be written as a fraction of integers, so $\sqrt{2}$ is irrational. ■

> **What the symbols mean:** "■" marks the end of a proof. The argument does not depend on the decimal value of $\sqrt{2}$; it works for every possible fraction, so it is a general proof, not a check of examples.

COMMON MISCONCEPTION

"$\sqrt{2} = 1.414$, so it is rational." 1.414 is a rational approximation, not $\sqrt{2}$. Squaring 1.414 gives 1.999396, not 2. The exact value is irrational; the decimal is a rounded form.

"Every number with a square root sign is irrational." $\sqrt{9} = 3$, $\sqrt{25} = 5$ and $\sqrt{0.25} = 0.5$ are all rational. A surd is irrational only when the radicand is not a perfect power of the relevant kind — and even then, the expression may simplify.

GUIDED PRACTICE · COMPLETE THE ARGUMENT

Adapt the proof of Example 1.3 to show that $\sqrt{5}$ is irrational. Fill in the missing words.

1. Assume $\sqrt{5} = \frac{p}{q}$ with p, q integers, $q \neq 0$, and no common factor.
2. Squaring gives $5q^2 =$ _____, so p^2 is a multiple of _____.
3. If $5 \mid p^2$, then $5 \mid p$ (because 5 is prime). Write $p = 5k$.
4. Substitute: $25k^2 = 5q^2$, so $q^2 =$ _____. Hence $5 \mid q$.
5. This contradicts _____. Therefore $\sqrt{5}$ is irrational.

EXERCISE 1A

NUMBER SYSTEMS · CLASSIFICATION · PROOF

FLUENCY AND UNDERSTANDING

1 List every set ($\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, irrational) to which each number belongs:
$12,\ -4,\ 0.6,\ \sqrt{16},\ \sqrt{11},\ \frac{7}{3},\ \pi$.

2 Decide whether each statement is always true, sometimes true or never true. Give a counterexample where appropriate.

a. Every integer is rational.
b. Every rational number is an integer.
c. $\sqrt{9}$ is irrational.
d. The product of two irrational numbers is irrational.

3 Write each recurring decimal as a fraction in simplest form.

a. $0.\overline{7}$
b. $0.\overline{12}$
c. $0.\overline{24}$
d. $1.5\overline{3}$

4 Use algebra to show that $0.\overline{9} = 1$. Explain why this does not mean that 0.9 and 1 are "almost" equal.

5 Find a rational number between $\frac{1}{3}$ and $\frac{1}{2}$. Then explain how to find another rational number between any two different rational numbers.

6 Classify $0.101001000100001\ldots$ (the number of zeros increases each time). Is it rational? Justify your answer.

7 Explain why 3.14 is rational even though it is often used as an approximation for π.

8 Is $2 + \sqrt{2}$ rational or irrational? Justify your answer by supposing it were rational and deriving a contradiction.

9 Give an example of two irrational numbers whose sum is rational, and two whose product is rational. Explain why your examples work.

10 The number $0.123456789101112\ldots$ is formed by writing the positive integers in order. Is it rational? Give a reason.

11 Complete the proof that $\sqrt{3}$ is irrational. You may use the fact that if $3 \mid p^2$ then $3 \mid p$.

12 **Extension.** Prove that $\sqrt{2} + \sqrt{3}$ is irrational. Start by assuming it equals a rational number r, square both sides, and use the known fact that $\sqrt{6}$ is irrational.

REASONING AND JUSTIFICATION

13 **Error analysis.** A student writes: "$\sqrt{2} = 1.4142135623730951$, so $\sqrt{2}$ is rational because it has a finite decimal." Identify the misconception, and write a one-sentence correction that a Year 4 student could understand.

REPEATED MULTIPLICATION

1.2 Integer and rational exponents

Index notation is a compression of repeated multiplication. The laws below are not arbitrary rules to memorise: each one follows from counting factors. If you forget a law, expand a small example and count.

LAWS OF INDICES · $A, B > 0$ UNLESS STATED

LAW	EXAMPLE	WHY IT WORKS
$a^m \times a^n = a^{m+n}$	$2^3 \times 2^2 = 2^5$	3 factors of 2 times 2 factors of 2 gives 5 factors.
$a^m \div a^n = a^{m-n}$	$\dfrac{5^6}{5^4} = 5^2$	Four of the six factors cancel.
$(a^m)^n = a^{mn}$	$(3^2)^4 = 3^8$	Four groups of two factors gives eight factors.
$(ab)^n = a^n b^n$	$(2x)^3 = 8x^3$	The index applies to every factor in the product.
$a^0 = 1\ (a \neq 0)$	$7^0 = 1$	$\dfrac{a^n}{a^n} = a^{n-n} = a^0$, and any non-zero number divided by itself is 1.
$a^{-n} = \dfrac{1}{a^n}$	$4^{-2} = \dfrac{1}{16}$	A negative index records a reciprocal, not a negative value.
$a^{1/n} = \sqrt[n]{a}$	$8^{1/3} = 2$	The cube root of 8 is the number whose cube is 8.
$a^{m/n} = (\sqrt[n]{a})^m$	$27^{2/3} = 3^2 = 9$	Take the root first, then the power.

EXAMPLE **1.4** SIMPLIFYING WITH INTEGER EXPONENTS

QUESTION

Simplify $\left(2x^3\right)^2 \left(3x^{-2}\right)$, leaving your answer with positive indices.

SOLUTION

1 Apply the power to each factor: $\left(2x^3\right)^2 = 2^2 \left(x^3\right)^2 = 4x^6$.

2 Multiply the two parts: $4x^6 \cdot 3x^{-2} = 12x^{6+(-2)} = 12x^4$.

3 The index is positive, so no reciprocal is needed. The simplified form is $12x^4$.

> **Reasoning:** we could write $3x^{-2} = \frac{3}{x^2}$, but simplifying to a single power is more efficient. The final index, 4, tells us there are four factors of x left after cancellation.

EXAMPLE **1.5** RATIONAL EXPONENTS AND RECIPROCALS

QUESTION

Evaluate without a calculator: (a) $16^{-3/4}$, (b) $27^{2/3}$, (c) $\left(\dfrac{9}{25}\right)^{-1/2}$.

SOLUTION

1 (a) $16^{-3/4} = \dfrac{1}{16^{3/4}}$. Since $16^{1/4} = 2$, $16^{3/4} = 2^3 = 8$. Hence $16^{-3/4} = \dfrac{1}{8}$.

2 (b) $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$. Taking the root first keeps the numbers small.

3 (c) $\left(\dfrac{9}{25}\right)^{-1/2} = \left(\dfrac{25}{9}\right)^{1/2} = \dfrac{5}{3}$. The negative sign inverts the fraction; the $1/2$ takes square roots.

Check: $(5/3)^2 = 25/9$, so the last answer is consistent. A rational exponent is a root; the denominator of the exponent is the root, and the numerator is the power.

EXAMPLE **1.6** SOLVING AN EXPONENTIAL EQUATION BY MATCHING BASES

QUESTION

Solve $4^x = 8^{x-1}$ and $9^x = \dfrac{1}{27}$.

SOLUTION

1 Write both sides with base 2: $4^x = (2^2)^x = 2^{2x}$ and $8^{x-1} = (2^3)^{x-1} = 2^{3x-3}$.

2 So $2^{2x} = 2^{3x-3}$. Powers of the same base are equal only when the indices are equal: $2x = 3x - 3$, hence $x = 3$.

3 For $9^x = \frac{1}{27}$, use base 3: $9^x = 3^{2x}$ and $\frac{1}{27} = 3^{-3}$. Thus $2x = -3$, so $x = -\frac{3}{2}$.

Check: $4^3 = 64$ and $8^{3-1} = 8^2 = 64$. Also $9^{-3/2} = 1/9^{3/2} = 1/27$.

COMMON MISCONCEPTIONS

a^{-n} is not $-a^n$. $2^{-3} = \frac{1}{8}$, not -8. The negative sign belongs to the index, and it means "reciprocal".

$(a+b)^2 \neq a^2 + b^2$. The power applies to the whole bracket, not separately. Test with $a = 3, b = 4$: $(3+4)^2 = 49$, but $3^2 + 4^2 = 25$.

$a^0 = 1$, not 0 (for $a \neq 0$). The pattern $2^3, 2^2, 2^1, 2^0$ divides by 2 each time: $8, 4, 2, 1$.

GUIDED PRACTICE · FIND THE ERROR AND REPAIR IT

A student simplifies $\dfrac{(3x^2)^3}{9x^4}$ as follows:

$$\frac{(3x^2)^3}{9x^4} = \frac{3x^6}{9x^4} = \frac{1}{3}x^{10}$$

1. Circle the first incorrect line.
2. Explain what the student did with the coefficient 3 and with the indices.
3. Write the correct simplification.
4. Check by substituting $x = 2$ into both the original expression and your answer.

EXERCISE 1B

INDEX LAWS · RATIONAL EXPONENTS · EXPONENTIAL EQUATIONS

FLUENCY

1 Simplify, leaving positive indices.

a. $x^4 \cdot x^7$

b. $\dfrac{y^9}{y^4}$

c. $(2a^3)^4$

d. $5x^2 \cdot 2x^{-5}$

2 Evaluate.

a. 3^{-2}

b. $(-2)^3$

c. $(-3)^{-2}$

d. $\left(\frac{2}{3}\right)^{-2}$

3 Evaluate without a calculator.

a. $81^{1/2}$

b. $64^{1/3}$

c. $16^{3/4}$

d. $125^{-2/3}$

4 Write each expression using a single exponent.

a. $\sqrt{x}$

b. $\dfrac{1}{x^3}$

c. $\sqrt[3]{x^2}$

d. $\dfrac{1}{\sqrt[4]{x}}$

5 Simplify.

a. $\dfrac{6x^5}{3x^{-2}}$

b. $\left(4a^2b\right)^3$

c. $\dfrac{(2x^2)^3}{4x^5}$

d. $\left(\dfrac{x^3}{2}\right)^2$

6 Solve for x.

a. $2^x = 32$

b. $3^{x+1} = 81$

c. $5^{2x} = 125$

d. $4^x = 2$

REASONING AND APPLICATION

7 Solve $4^{x+1} = 8^x$ by writing both sides with base 2.

8 Solve $9^x = \frac{1}{81}$ and $27^x = \frac{1}{9}$.

9 Explain why $a^0 = 1$ for any non-zero a, using the quotient law. Does the argument work for $a = 0$? Explain.

10 The thickness of a sheet of paper is 0.1 mm. Write an expression for the thickness after n folds. Use $n = 10$ and $n = 20$ to check your expression.

11 **Error analysis.** A student writes $(-8)^{2/3} = -4$. Decide whether the answer is correct, and justify using the definition of a rational exponent.

12 **Extension.** Solve $2^{x+2} - 2^x = 48$ by factorising 2^x.

EXERCISE 1B · CONTINUED INDEX LAWS · RATIONAL EXPONENTS · EXPONENTIAL EQUATIONS

13 **Extension.** Show that $4^n + 4^n = 2^{2n+1}$ for every integer n.

SCALE AND PRECISION

1.3 Standard form and estimation

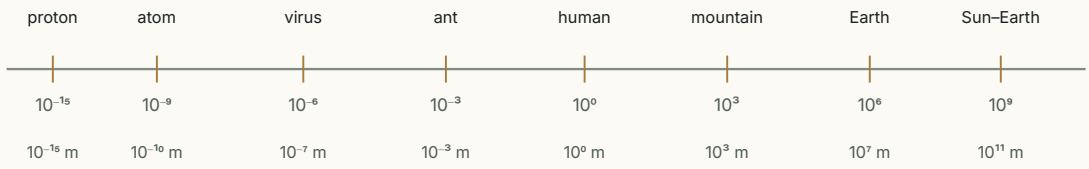
Scientific work spans distances of 10^{26} m and sizes of 10^{-15} m. Writing all the zeros is error-prone and hides the scale. **Standard form** (also called scientific notation) writes a number as $a \times 10^n$, where $1 \leq a < 10$ and n is an integer. The exponent records the order of magnitude.

DEFINITION

Standard form

A positive number is in standard form when it is written as $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$. For example, $4.7 \times 10^5 = 470\,000$ and $3.2 \times 10^{-4} = 0.00032$.

To convert a large number, move the decimal point left until one non-zero digit remains; the exponent is the number of places moved. For a small number, move the point right; the exponent is negative.



A logarithmic scale: equal distances represent equal powers of ten, not equal differences.

Figure 1.2 Powers of ten. Standard form compresses this enormous range into a coefficient and an exponent.

EXAMPLE **1.7** CONVERTING AND CALCULATING IN STANDARD FORM

QUESTION

(a) Write $0.000\,046$ and $38\,000\,000$ in standard form. (b) Calculate $\left(2.4 \times 10^{5}\right)\left(3.5 \times 10^{-2}\right)$ and $\dfrac{8.4 \times 10^{7}}{2.1 \times 10^{3}}$, giving answers in standard form.

SOLUTION

1 (a) $0.000\,046 = 4.6 \times 10^{-5}$: the decimal point moves 5 places right. $38\,000\,000 = 3.8 \times 10^{7}$: the point moves 7 places left.

2 (b) Multiply the coefficients and add the indices: $\left(2.4 \times 10^{5}\right)\left(3.5 \times 10^{-2}\right) = 8.4 \times 10^{3}$.

3 Divide the coefficients and subtract the indices: $\dfrac{8.4 \times 10^{7}}{2.1 \times 10^{3}} = 4.0 \times 10^{4}$.

Check the order of magnitude: $2.4 \times 10^{5} \approx 2.5 \times 10^{5}$, and $3.5 \times 10^{-2} \approx 3.5 \times 10^{-2}$; the product is about 8.75×10^{3}, consistent with 8.4×10^{3}.

EXAMPLE **1.8** A CALCULATION WITH UNITS AND SIGNIFICANT FIGURES

QUESTION

Light travels at 3.00×10^{8} m/s. A year is approximately 3.15×10^{7} s. Calculate the distance light travels in one year, in metres and in kilometres. Give the answer to 3 significant figures and interpret it.

SOLUTION

1 Distance = speed × time = $\left(3.00 \times 10^{8}\right)\left(3.15 \times 10^{7}\right) = 9.45 \times 10^{15}$ m.

2 In kilometres, divide by 10^{3}: 9.45×10^{12} km.

3 Both inputs are given to 3 significant figures, so the result is stated to 3 significant figures: 9.45×10^{15} m.

> **Interpretation:** this distance is called a light-year. It is a distance, not a time, despite the word "year" in its name. The calculation assumes a constant speed and a simplified year length.

Significant figures, estimation and the reasonableness of an answer

Significant figures communicate the precision of a measurement. All non-zero digits are significant; zeros between them are significant; leading zeros are not; trailing zeros after a decimal point are significant. A result cannot be more precise than the least precise measurement used to calculate it.

Before using a calculator, estimate to one significant figure. If the exact answer and the estimate differ by more than a factor of about 10, something is wrong — usually a misplaced decimal point or a bracket error. Estimation is not a substitute for calculation; it is a check on it.

EXAMPLE **1.9** ESTIMATING BEFORE CALCULATING

QUESTION

Estimate $\dfrac{4.87 \times 10^6}{2.1 \times 10^3}$ to one significant figure, then calculate it to 3 significant figures. Is the calculator answer reasonable?

SOLUTION

1. Estimate: $4.87 \times 10^6 \approx 5 \times 10^6$ and $2.1 \times 10^3 \approx 2 \times 10^3$, so the quotient is about $\frac{5}{2} \times 10^3 = 2.5 \times 10^3$.
2. Calculate: $4.87 \div 2.1 = 2.319\ldots$, so the result is 2.32×10^3 to 3 significant figures.
3. The estimate 2.5×10^3 is within about 8% of the calculated value, so the order of magnitude is reasonable.

Common error: entering 4.87×10^6 as 4.87E6 and then dividing by 2.1E3 is correct on most calculators. Writing $4.87 \times 10 \wedge 6$ without brackets may square only the 10 and then multiply, giving the wrong answer. Always bracket the exponent.

COMMON MISCONCEPTIONS

Counting zeros instead of places. $0.00042 = 4.2 \times 10^{-4}$, not 4.2×10^{-5}. The decimal point moves four places to reach 4.2.

Trailing zeros are not always significant. "2 500" is ambiguous without context; "2.5×10^3" is precise about two significant figures, and "2.500×10^3" claims four.

Rounding too early. Keep full calculator precision through a multi-step calculation and round only the final answer; otherwise small errors accumulate.

GUIDED PRACTICE · PRECISION AUDIT

A student calculates the area of a rectangle measuring 4.2 cm by 3.7 cm and writes 15.54 cm^2.

1. Is the arithmetic correct? Calculate 4.2×3.7.
2. How many significant figures does each measurement have?
3. What is the appropriate number of significant figures for the area? Give the rounded answer.
4. Explain why reporting 15.54 cm^2 overstates the precision of the measurement.

EXERCISE 1C

STANDARD FORM · SIGNIFICANT FIGURES · ESTIMATION

FLUENCY

1. Write in standard form.
   a. 0.000 003 2
   b. 7 400 000
   c. 0.056
   d. 91 000
2. Write as ordinary numbers.
   a. 5.1×10^{-4}
   b. 2.03×10^6
   c. 9.9×10^{-2}
   d. 1.2×10^0

EXERCISE 1C · CONTINUED

STANDARD FORM · SIGNIFICANT FIGURES · ESTIMATION

3 Calculate, giving answers in standard form.

a. $(3.1 \times 10^4)(2.0 \times 10^3)$

b. $(8.4 \times 10^{-5})(2.5 \times 10^{-2})$

c. $\dfrac{6.3 \times 10^8}{2.1 \times 10^2}$

d. $\dfrac{4.8 \times 10^{-3}}{1.6 \times 10^{-7}}$

4 Calculate $(2.7 \times 10^4) + (5.1 \times 10^3)$ by writing both numbers with the same power of ten.

5 Round each number to the stated accuracy.

a. 3.8472 to 2 decimal places

b. $0.004\,563$ to 2 significant figures

c. $28\,496$ to 3 significant figures

d. 9.995 to 3 significant figures

6 Estimate each answer to one significant figure, then calculate to 3 significant figures.

a. $\dfrac{38.7 \times 0.48}{0.021}$

b. $\sqrt{5.8 \times 10^5}$

c. $\dfrac{6.2 \times 10^{-4}}{3.1 \times 10^{-6}}$

REASONING AND APPLICATION

7 A grain of sand has a diameter of about 0.5 mm. The Earth's diameter is about 1.27×10^7 m. How many grains of sand would fit across the Earth? Give your answer in standard form to 2 significant figures.

8 The mass of a hydrogen atom is 1.67×10^{-27} kg. Estimate the mass of 6.02×10^{23} hydrogen atoms, giving your answer in standard form.

9 A computer reports a file size as 4.7 GB. Explain why this could mean 4.7×10^9 bytes or 4.7×2^{30} bytes. Which is larger?

10 **Error analysis.** A student writes $0.000\,72 = 7.2 \times 10^{-3}$. Identify the error, and explain how to count the places correctly.

11 **Reasoning.** Explain why a measurement given as 3.0 m has two significant figures, while 3 m has only one (or is ambiguous). Why does this matter when calculating an area?

12 **Extension.** The Sun's mass is 1.989×10^{30} kg. The mass of the Earth is 5.972×10^{24} kg. How many times heavier is the Sun? Give your answer to 3 significant figures.

13 **Extension.** A light-year is 9.46×10^{12} km. The nearest star is about 4.24 light-years away. Calculate the distance in kilometres and in metres, in standard form.

EXACTNESS AND FORM

1.4 Surds and exact values

A calculator reports $\sqrt{2} = 1.414213562\ldots$ and then stops. The display is finite; the number is not. Writing $\sqrt{2}$ keeps the exact value visible and postpones rounding until a decision about accuracy has been made. A surd is a root of a rational number that cannot be simplified to a rational number. Expressions such as $\sqrt{2}$, $3\sqrt{5}$ and $\frac{1}{\sqrt{7}}$ are exact; their decimal forms are approximations.

The rules for combining surds come from the laws of exponents. Since $\sqrt{a} = a^{1/2}$, the product law $a^m a^n = a^{m+n}$ gives $\sqrt{a}\sqrt{b} = \sqrt{ab}$. The quotient law gives $\sqrt{a/b} = \sqrt{a}/\sqrt{b}$. These rules hold for non-negative a and b, with $b \neq 0$ in the quotient.

KEY IDEA · THE ALGEBRA OF SQUARE ROOTS

For $a \geq 0$ and $b \geq 0$: $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$ and $\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$ for $b > 0$. A surd is simplified when the radicand has no square factor greater than 1 and no fraction remains inside the root. Only **like surds** — multiples of the same root — can be added or subtracted.

Why simplify a surd?

Simplification exposes structure. $\sqrt{72} = 6\sqrt{2}$ shows at once that the number is six copies of $\sqrt{2}$, so it can be combined with other multiples of $\sqrt{2}$. The decimal $8.485\ldots$ hides that relationship. In algebra, exact forms also cancel reliably: $\frac{\sqrt{72}}{\sqrt{2}} = 6$, whereas dividing rounded decimals may leave a residue such as $6.000\,001$.

EXAMPLE **1.10** SIMPLIFYING SURDS

QUESTION

Simplify, giving each answer in the form $a\sqrt{b}$ with b as small as possible.

(a) $\sqrt{72}$ (b) $\sqrt{50} + \sqrt{18}$ (c) $\sqrt{12} \times \sqrt{15}$ (d) $\sqrt{\dfrac{3}{8}}$

SOLUTION

a Find the largest square factor of 72. Since $72 = 36 \times 2$,

$$\sqrt{72} = \sqrt{36 \times 2} = \sqrt{36}\,\sqrt{2} = 6\sqrt{2}.$$

b Simplify each term first: $\sqrt{50} = 5\sqrt{2}$ and $\sqrt{18} = 3\sqrt{2}$. They are like surds, so add the coefficients:

$$\sqrt{50} + \sqrt{18} = 5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}.$$

c Multiply inside the root, then simplify:

$$\sqrt{12} \times \sqrt{15} = \sqrt{180} = \sqrt{36 \times 5} = 6\sqrt{5}.$$

d Split the fraction, then remove the root from the denominator:

$$\sqrt{\frac{3}{8}} = \frac{\sqrt{3}}{\sqrt{8}} = \frac{\sqrt{3}}{2\sqrt{2}} = \frac{\sqrt{3} \times \sqrt{2}}{2\sqrt{2} \times \sqrt{2}} = \frac{\sqrt{6}}{4}.$$

> **Decision:** in (b), do not add the radicands. $\sqrt{50} + \sqrt{18}$ is not $\sqrt{68}$. Addition requires identical roots; only the coefficients combine.

Rationalising the denominator

A fraction such as $\frac{1}{\sqrt{5}}$ is exact but awkward: division by an irrational number is hard to estimate and hard to compare. Rationalising the denominator rewrites the fraction with a rational denominator, without changing its value. Multiply by a form of 1 that removes the root. For a single root, multiply by $\frac{\sqrt{5}}{\sqrt{5}}$. For a denominator of the form $a + \sqrt{b}$, multiply by the conjugate $a - \sqrt{b}$; the difference of two squares then eliminates the root.

EXAMPLE **1.11** RATIONALISING DENOMINATORS

QUESTION

Write each expression with a rational denominator.

(a) $\dfrac{1}{\sqrt{5}}$ (b) $\dfrac{6}{2\sqrt{3}}$ (c) $\dfrac{\sqrt{3}+1}{\sqrt{3}-1}$ (d) $\dfrac{5}{\sqrt{7}-\sqrt{2}}$

SOLUTION

a Multiply numerator and denominator by $\sqrt{5}$:

$$\frac{1}{\sqrt{5}} = \frac{1 \times \sqrt{5}}{\sqrt{5} \times \sqrt{5}} = \frac{\sqrt{5}}{5}.$$

b Simplify first: $\dfrac{6}{2\sqrt{3}} = \dfrac{3}{\sqrt{3}}$. Then multiply by $\dfrac{\sqrt{3}}{\sqrt{3}}$:

$$\frac{3}{\sqrt{3}} = \frac{3\sqrt{3}}{3} = \sqrt{3}.$$

c Multiply by the conjugate $\sqrt{3}+1$:

$$\frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{(\sqrt{3}+1)^2}{3-1} = \frac{4+2\sqrt{3}}{2} = 2+\sqrt{3}.$$

d Multiply by $\sqrt{7}+\sqrt{2}$:

$$\frac{5(\sqrt{7}+\sqrt{2})}{(\sqrt{7}-\sqrt{2})(\sqrt{7}+\sqrt{2})} = \frac{5(\sqrt{7}+\sqrt{2})}{7-2} = \sqrt{7}+\sqrt{2}.$$

Check (c): $2+\sqrt{3} \approx 3.732$ and $\frac{2.732}{0.732} \approx 3.732$. The two forms agree numerically, but only the first is exact.

EXAMPLE **1.12** EXPANDING PRODUCTS WITH SURDS

QUESTION

Expand and simplify.

(a) $(2+\sqrt{3})(2-\sqrt{3})$ (b) $(\sqrt{5}+\sqrt{2})^2$ (c) $(3-\sqrt{2})(1+2\sqrt{2})$

SOLUTION

a This is a difference of two squares:

$$(2+\sqrt{3})(2-\sqrt{3}) = 2^2 - (\sqrt{3})^2 = 4 - 3 = 1.$$

b Expand the square, keeping the middle term:

$$(\sqrt{5}+\sqrt{2})^2 = 5 + 2\sqrt{10} + 2 = 7 + 2\sqrt{10}.$$

c Expand all four products, then collect like terms:

$$3 + 6\sqrt{2} - \sqrt{2} - 2(\sqrt{2})^2 = 3 + 5\sqrt{2} - 4 = -1 + 5\sqrt{2}.$$

> **Structure:** part (a) produces a rational number because the cross terms cancel. This is exactly why the conjugate removes a surd from a denominator.

EXAMPLE **1.13** COMPARING EXACT VALUES WITHOUT A CALCULATOR

QUESTION

Without using decimals, decide which is larger: $2\sqrt{3}$ or $3\sqrt{2}$. Justify your answer.

SOLUTION

1 Both numbers are positive, so their squares are in the same order. Square each one:

$$(2\sqrt{3})^2 = 4 \times 3 = 12, \qquad (3\sqrt{2})^2 = 9 \times 2 = 18.$$

2 Since $12 < 18$, it follows that $2\sqrt{3} < 3\sqrt{2}$.

3 Check with approximations: $2\sqrt{3} \approx 3.464$ and $3\sqrt{2} \approx 4.243$. The exact comparison agrees.

> **Why squaring is valid:** for positive x and y, $x < y$ if and only if $x^2 < y^2$. The implication fails for negative numbers, which is why the positivity check is part of the reasoning.

COMMON MISCONCEPTIONS

"$\sqrt{a+b} = \sqrt{a} + \sqrt{b}$." Test it: $\sqrt{9+16} = \sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$. The rule applies to products and quotients, not sums.

"$\sqrt{x^2} = x$." If $x = -3$, $\sqrt{(-3)^2} = \sqrt{9} = 3$, not -3. The correct statement is $\sqrt{x^2} = |x|$.

"$\frac{1}{\sqrt{2}} = \sqrt{2}$." Multiply by $\frac{\sqrt{2}}{\sqrt{2}}$: the correct value is $\frac{\sqrt{2}}{2}$, which is about 0.707, not 1.414.

"Every square root is a surd." $\sqrt{25} = 5$ is rational, so it is not a surd. A surd is irrational.

GUIDED PRACTICE · COMPLETE THE REASONING

Simplify $\sqrt{98}$. The largest square factor is _____, so $\sqrt{98} =$ _____$\sqrt{2}$.

Simplify $\sqrt{75} - \sqrt{12}$: first write each term as _____$\sqrt{3}$ and _____$\sqrt{3}$, then subtract to obtain _____$\sqrt{3}$.

Rationalise $\dfrac{4}{\sqrt{2}}$: multiply by $\dfrac{\sqrt{2}}{\sqrt{2}}$ to obtain $\dfrac{4\sqrt{2}}{____} =$ _____.

Expand $(\sqrt{7} - 2)(\sqrt{7} + 2)$. This is a difference of squares, so the result is _____, which is rational.

EXERCISE 1D

SURDS · EXACT VALUES · RATIONALISING

FLUENCY

1 Simplify.

a. $\sqrt{32}$

b. $\sqrt{48}$

c. $\sqrt{200}$

d. $\sqrt{180}$

2 Simplify.

a. $\sqrt{45} + \sqrt{20}$

b. $\sqrt{72} - \sqrt{50}$

c. $2\sqrt{8} + 3\sqrt{18}$

d. $\sqrt{12} + \sqrt{27} - \sqrt{48}$

3 Simplify.

a. $\sqrt{6} \times \sqrt{10}$

b. $\sqrt{18} \times \sqrt{8}$

c. $(\sqrt{5})^2$

d. $(2\sqrt{3})^2$

4 Write with a rational denominator.

a. $\dfrac{1}{\sqrt{3}}$

b. $\dfrac{5}{\sqrt{2}}$

c. $\dfrac{8}{2\sqrt{2}}$

d. $\dfrac{6}{\sqrt{12}}$

5 Write with a rational denominator.

a. $\dfrac{1}{\sqrt{5}+1}$

b. $\dfrac{2}{\sqrt{7}-\sqrt{3}}$

c. $\dfrac{\sqrt{2}+1}{\sqrt{2}-1}$

d. $\dfrac{3}{\sqrt{6}+\sqrt{3}}$

6 Expand and simplify.

a. $(1 + \sqrt{2})(3 - \sqrt{2})$

b. $(\sqrt{6} - \sqrt{2})^2$

c. $(2\sqrt{5} + 1)(2\sqrt{5} - 1)$

d. $(3\sqrt{2} - 1)^2$

EXERCISE 1D · CONTINUED

UNDERSTANDING AND REASONING

UNDERSTANDING AND REASONING

7 Explain, with a counterexample, why $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ in general. State the condition under which $\sqrt{ab} = \sqrt{a}\sqrt{b}$ is valid.

8 Is $\sqrt{x^2} = x$ always true? Explain using a positive value and a negative value of x.

EXERCISE 1D · CONTINUED

UNDERSTANDING AND REASONING

9 Decide whether each is rational or irrational. Justify each answer.

a. $\sqrt{49}$

b. $\sqrt{50}$

c. $\sqrt{\frac{1}{4}}$

d. $\sqrt{\frac{2}{3}}$

e. $(\sqrt{3})^2$

f. $\sqrt{3}+\sqrt{12}$

10 Show that $3\sqrt{2} > 2\sqrt{3}$ without using decimals. Explain why squaring both sides is a valid step here.

11 **Error analysis.** A student writes $\sqrt{45} = 5\sqrt{3}$. Identify the error, write the correct simplification, and explain how to check a surd simplification by squaring.

12 Give an example of two irrational numbers whose product is rational. Explain why your example works.

13 Rationalise and simplify $\dfrac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$.

14 A square has area $12\ \text{cm}^2$. Write its side length and perimeter in simplified surd form, then give each to 3 significant figures.

15 A rectangle has sides $\sqrt{8}\ \text{cm}$ and $\sqrt{2}\ \text{cm}$. Show that its area is exactly $4\ \text{cm}^2$ and write its perimeter in the form $a\sqrt{2}$.

EXERCISE 1D · CONTINUED

APPLICATION AND EXTENSION

APPLICATION AND EXTENSION

16 A rectangular screen has width $32\ \text{cm}$ and height $18\ \text{cm}$. Write the diagonal in simplified surd form and as a decimal to 2 decimal places.

17 The period of a pendulum is $T = 2\pi\sqrt{\dfrac{L}{g}}$. For $L = 1.2\ \text{m}$ and $g = 9.8\ \text{m/s}^2$, write T in exact form, then calculate it to 2 significant figures. Explain why the exact form is more informative before rounding.

18 **Extension.** Show that $(\sqrt{2}+1)^3 = 7+5\sqrt{2}$.

19 **Extension.** Suppose a and b are rational and $a+b\sqrt{2}=0$. Prove that $a=b=0$. Hence show that $\dfrac{1}{3-2\sqrt{2}} = 3+2\sqrt{2}$.

MEASUREMENT AND TRUST

1.5 Accuracy, bounds and error

No measurement is exact. A ruler marked in millimetres, a scale that displays $0.1\ \text{kg}$, and a timer that records hundredths of a second all report a rounded value. The reported number is the centre of an interval of possible true values. **Accuracy** describes how close a measurement is to the true value; **precision** describes how fine the reporting scale is. A measurement can be precise but inaccurate if the instrument is badly calibrated.

Two questions matter in every calculation. First, how much can the true value differ from the reported value? Second, how does that uncertainty combine when values are added, multiplied or divided? Upper and lower bounds answer the first question; interval arithmetic answers the second.

DEFINITIONS

Absolute error, percentage error and bounds

If a quantity is measured as x and its true value is X, the **absolute error** is $|x - X|$. The **percentage error** is $\dfrac{|x - X|}{X} \times 100\%$; when X is unknown, the measured value is often used as the denominator and the assumption is stated.

If a value is rounded to the nearest unit u, the true value lies in the interval $x - \dfrac{u}{2} \leq X < x + \dfrac{u}{2}$. The **lower bound** is $x - \dfrac{u}{2}$ and the **upper bound** is $x + \dfrac{u}{2}$. For a value given to a stated number of significant figures, u is the place value of the last significant figure.

EXAMPLE **1.14** ROUNDING AND SIGNIFICANT FIGURES

QUESTION

Round each number to the accuracy stated.

(a) 3.8472 to 2 d.p.

(b) 3.8472 to 3 s.f.

(c) $0.004\,563$ to 2 s.f.

(d) $28\,496$ to 3 s.f.

(e) 9.995 to 3 s.f.

SOLUTION

a The third decimal digit is 7, so round the second digit up: 3.85.

b The fourth significant digit is 7, so round the third digit up: 3.85. (The trailing zero is not needed because 5 is the last digit.)

c Leading zeros are not significant. The first two significant digits are 4 and 5; the next digit is 6, so round up: 0.0046.

d The fourth significant digit is 9, so round 8 up to 9 and replace the remaining digits with zeros: $28\,500$.

e The fourth significant digit is 5, so round the third digit up. 9.99 becomes 10.0: the trailing zero must be written to show three significant figures.

> **Significant figures:** count from the first non-zero digit. For numbers greater than 1, trailing zeros before the decimal point may be ambiguous; $28\,500$ has three significant figures here because the rounding was specified. Standard form removes the ambiguity: 2.85×10^4.

EXAMPLE **1.15** BOUNDS AND PERCENTAGE ERROR

QUESTION

A length is measured as 12.4 cm, correct to 1 decimal place. (a) Write the lower and upper bounds. (b) State the maximum absolute error. (c) Calculate the percentage error, using the measured value as the denominator.

SOLUTION

1 The unit is 0.1 cm, so half a unit is 0.05 cm.

2 Lower bound $= 12.4 - 0.05 = 12.35$ cm. Upper bound $= 12.4 + 0.05 = 12.45$ cm. The true length satisfies $12.35 \le L < 12.45$.

3 The maximum absolute error is 0.05 cm.

4 Percentage error $\approx \dfrac{0.05}{12.4} \times 100\% = 0.403\%$, or 0.40% to 2 s.f.

Interpretation: a 0.4% error is small for a length, but the same absolute error on a 1.2 cm length would be about 4%. Percentage error communicates relative precision.

EXAMPLE **1.16** BOUNDS OF A CALCULATED AREA AND PERIMETER

QUESTION

A rectangle is measured as 8.5 cm by 5.2 cm, each correct to 1 decimal place. Find the lower and upper bounds of (a) the perimeter and (b) the area.

SOLUTION

1 Each measurement has a half-unit of 0.05 cm:

$$8.45 \le w < 8.55, \qquad 5.15 \le h < 5.25.$$

2 Perimeter $= 2(w + h)$. The minimum uses both lower bounds, the maximum both upper bounds:

$$P_{\min} = 2(8.45 + 5.15) = 27.2 \text{ cm}, \qquad P_{\max} = 2(8.55 + 5.25) = 27.6 \text{ cm}.$$

3 Area $= wh$. For positive dimensions, the minimum product uses both lower bounds and the maximum uses both upper bounds:

$$A_{\min} = 8.45 \times 5.15 = 43.5175 \text{ cm}^2, \qquad A_{\max} = 8.55 \times 5.25 = 44.8875 \text{ cm}^2.$$

4 Report the interval with sensible precision: $43.5 \le A < 44.9$ cm^2, or $A = 44.2 \pm 0.7$ cm^2.

> **Different rules:** for a sum, bounds add; for a difference, the maximum is upper minus lower. For products and quotients of positive quantities, use the combination that makes the result as large or as small as possible.

EXAMPLE **1.17** BOUNDS OF A QUOTIENT

QUESTION

A runner covers 100 m, measured to the nearest metre, in 12.3 s, measured to the nearest 0.1 s. Find the lower and upper bounds of the average speed.

SOLUTION

1 Distance: $99.5 \le d < 100.5$ m. Time: $12.25 \le t < 12.35$ s.

2 Speed $= d/t$. The maximum speed uses the largest distance and the smallest time:

$$v_{\max} = \frac{100.5}{12.25} = 8.204\ldots \text{ m/s}.$$

3 The minimum speed uses the smallest distance and the largest time:

$$v_{\min} = \frac{99.5}{12.35} = 8.057\ldots \text{ m/s}.$$

4 So $8.06 \le v < 8.21$ m/s. The reported value $100/12.3 = 8.13$ m/s lies inside the interval, as it must.

Reasoning: dividing by a smaller time makes the speed larger, so the upper bound of speed pairs the upper bound of distance with the lower bound of time. Mixing the bounds the other way would not produce an extremum.

COMMON MISCONCEPTIONS

"The error is ± 1 unit." Rounding to the nearest unit gives an error of at most half a unit. A length reported as 8 cm to the nearest centimetre lies between 7.5 and 8.5 cm, not 7 and 9 cm.

"0.00450 has 2 significant figures." Leading zeros are placeholders. The digits 4, 5 and the trailing 0 are significant, so there are 3 significant figures.

"More decimal places always means more accurate." A calculator may display many digits derived from a rounded input. The output cannot be more precise than the measurements that produced it.

"Percentage error always uses the measured value." If the true value is known, use it. If not, state that the measured value is being used as the denominator.

GUIDED PRACTICE · COMPLETE THE REASONING

A mass is 6.8 kg, correct to 1 decimal place. The half-unit is _____ kg, so the lower bound is _____ kg and the upper bound is _____ kg.

The maximum percentage error is $\dfrac{\text{half-unit}}{6.8} \times 100\% \approx$ _____%.

A square has side 4.0 cm, correct to 1 decimal place. Its minimum area is _____ cm^2 and its maximum area is _____ cm^2.

Explain why the area interval is wider, in percentage terms, than the side interval:

_____.

EXERCISE 1E

SIGNIFICANT FIGURES · BOUNDS · ERROR

FLUENCY

1 Round each number to the accuracy stated.

a. 7.268 to 2 d.p.

b. $0.030\,47$ to 2 s.f.

c. $64\,820$ to 3 s.f.

d. 0.9995 to 3 s.f.

2 State the number of significant figures in each value.

a. 0.0042

b. 30.50

c. 7.00×10^3

d. $0.070\,80$

3 Write the lower and upper bounds for each measurement.

a. 14 cm to the nearest cm

b. 3.7 kg to 1 d.p.

c. 250 m to 2 s.f.

d. 0.84 s to 2 d.p.

4 A mass is recorded as 48 kg, correct to the nearest kilogram. Calculate the maximum absolute error and the maximum percentage error, using the measured value as the denominator.

5 For $a = 5.4$ and $b = 2.1$, each correct to 1 d.p., find the lower and upper bounds of $a + b$ and $a - b$.

6 A square has side 6.5 cm, correct to 1 d.p. Find the lower and upper bounds of its area.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student says that a length of 3.7 cm, correct to 1 d.p., has a maximum error of 0.1 cm. Identify the error and correct it.

8 A rope is 24 m long, correct to the nearest metre. Could it be used to mark out a square of area 36 m^2? Justify using bounds.

9 Which measurement is more precise: 3.4 cm or 34 mm? Explain carefully, distinguishing precision from accuracy.

10 A distance is reported as 100 m to 1 significant figure and also as 100 m to 3 significant figures. Write the bounds in each case and explain why the same digits can represent different precision.

11 Two students measure the same length. Student A reports 12.0 cm; Student B reports 12.00 cm. Compare the precision and the maximum percentage error of the two reports.

12 A calculator displays 3.464101615 for $2\sqrt{3}$. Explain why writing this as the final answer to a measurement question may be misleading.

APPLICATION AND EXTENSION

13 A courier measures a parcel as 42 cm $\times$ 28 cm $\times$ 15 cm, each to the nearest centimetre. Find the lower and upper bounds of its volume. A locker has usable volume $18\,000$ cm^3. Can the courier be certain the parcel fits? Justify.

14 A patient's mass is 68 kg to the nearest kilogram and height 1.72 m to the nearest centimetre. Calculate the lower and upper bounds of the body-mass index $\text{BMI} = \dfrac{m}{h^2}$, giving your answers to 1 decimal place.

15 A cube has side 4.0 cm, correct to 2 significant figures. Find the lower and upper bounds of its volume and calculate the maximum percentage error in the volume. Compare this with the maximum percentage error in the side length.

16 **Extension.** For positive quantities a and b, explain why the upper bound of ab is the product of the upper bounds, while the upper bound of $a - b$ is the upper bound of a minus the lower bound of b. Give a numerical example of each.

PATTERN AND GENERALISATION

1.6 Investigation: how exact is exact?

Exactness is not the same as having many digits. The number 2^{2026} is an integer, so it is exact, but its decimal expansion has hundreds of digits. Sometimes a single digit — the last one — can be determined exactly without computing the whole number. In this investigation you will find a pattern, explain why it must continue, and use it to make a prediction that no calculator display could check directly.

INVESTIGATION 1 · CYCLES IN THE LAST DIGIT

What is the last digit of 2^{2026}?

Complete the table by calculating successive powers of 2 and recording only the last digit. Do not use a calculator beyond 2^{10}.

n	1	2	3	4	5	6	7	8
2^n	2	4	8	16	32	64	128	256
Last digit	2	4	8	6	2	4	8	6

1. Extend the table to $n = 12$. Describe the pattern in the last digits.
2. Predict the last digit of 2^{20}, 2^{21}, 2^{22} and 2^{23}. Check your predictions using the pattern.
3. Find the last digit of 2^{2026}. Explain how you used the remainder when 2026 is divided by 4.
4. Why can the pattern never break? Suppose a power ends in 6. What is the last digit of the next power? Explain using $6 \times 2 = 12$.
5. Can 2^n ever end in 0? Use the prime factorisation of 2^n to justify your answer.
6. Repeat the investigation for powers of 3. Record the cycle of last digits and state its length. Predict the last digit of 3^{2026}.
7. Repeat for powers of 7. Compare the cycle lengths for bases 2, 3 and 7. What do you notice?
8. Explain why the last-digit cycle of a^n depends only on the last digit of a, not on the higher digits. For example, explain why 13^n and 3^n have the same last-digit pattern.

Generalise. For a base whose last digit is 2, 3, 7 or 8, the last digit of the powers repeats with period 4. Write a rule that gives the last digit of a^n in terms of the remainder when n is divided by 4. Then test your rule on 8^{101} and 7^{250}. Does the rule hold when the base ends in 0, 1, 5 or 6? Explain.

MODELLING · CRITERION D

The fold that cannot happen

Return to the opening problem. A sheet of paper is 0.1 mm thick. After n folds, the stack is 0.1×2^n mm thick. The claim is that 42 folds reach the Moon, about $384\,400$ km away.

1. **Calculate.** Use $2^{10} \approx 10^3$ to estimate, then calculate 0.1×2^{42} exactly in millimetres. Convert to kilometres and write the result in standard form to 2 significant figures.
2. **Compare.** The Moon's distance varies from about $363\,300$ km to $405\,500$ km. Does the calculated thickness reach the Moon? Does it exceed the maximum distance?
3. **Bound the claim.** The paper thickness 0.1 mm is given to 1 significant figure, so take $0.05 \leq t < 0.15$ mm. Find the lower and upper bounds of the stack after 42 folds. How does this affect the conclusion?
4. **Test the physics.** Each fold halves the area. An A4 sheet has area 0.0624 m^2. Calculate the area after 42 folds and compare it with the cross-sectional area of a single atom (about 10^{-20} m^2). What does this show?
5. **Evaluate.** Write a short judgement: is the claim numerically sound, physically possible, or both? Distinguish the arithmetic from the assumptions. State which assumption is most important and why.
6. **Reflect.** Which is more useful to a scientist: the exact arithmetic or the physical bound? Explain in two sentences.

Assumptions to examine.

- The paper is uniform and incompressible, so each fold exactly doubles the thickness.
- Each fold halves the area and leaves a single stack with no buckling, sliding or trapped air.
- The Moon's distance is treated as a fixed value, although it varies by about $42\,000$ km.
- The thickness 0.1 mm is exact, although it is reported to 1 significant figure.

Evidence to submit: a table of values, the bound calculation, the area calculation, and a paragraph of no more than 150 words that separates mathematical correctness from real-world feasibility.

MIXED PRACTICE 1

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–6 check fluency; 7–14 require reasoning; 15–20 are multi-step or unfamiliar. Allow 55 minutes.

1. Classify each number as natural, integer, rational or irrational:
$-3,\ 0,\ \frac{5}{8},\ \sqrt{36},\ \sqrt{10},\ 0.\overline{6},\ \pi$.

MIXED PRACTICE 1 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

2 Simplify.

a. $a^5 \times a^{-2}$

b. $\dfrac{12x^7}{3x^3}$

c. $\left(2x^2\right)^3$

d. $81^{-3/4}$

3 Write in standard form.

a. $0.000\,045$

b. $8\,120\,000$

c. $3.6 \times 10^{-3} \times 4.0 \times 10^{5}$

d. $\dfrac{7.2 \times 10^{-4}}{2.4 \times 10^{-7}}$

4 Simplify.

a. $\sqrt{80}$

b. $\sqrt{27} + \sqrt{12}$

c. $\sqrt{6} \times \sqrt{8}$

d. $\dfrac{10}{\sqrt{5}}$

5 Round $0.004\,678$ to 2 significant figures and $36\,249$ to 3 significant figures.

6 A length is 7.3 cm, correct to 1 decimal place. Write its lower and upper bounds and its maximum percentage error.

7 Decide whether each statement is always, sometimes or never true. Justify each answer.

a. The sum of two rational numbers is rational.

b. The sum of two irrational numbers is irrational.

c. The square of an irrational number is irrational.

d. $\sqrt{a^2} = a$ for every real number a.

8 Prove that $\sqrt{2}$ is irrational. State clearly where the assumption of lowest terms is used.

9 Solve $3^{2x-1} = 27$ and $5^{x+1} = \frac{1}{25}$.

10 Write $0.\overline{36}$ and $2.1\overline{6}$ as fractions in simplest form.

MIXED PRACTICE 1 · CONTINUED

REASONING · APPLICATION · EXTENSION

QUESTIONS 11–20

11 Rationalise $\dfrac{4}{\sqrt{7} - \sqrt{3}}$ and simplify $\dfrac{\sqrt{8} + \sqrt{18}}{\sqrt{2}}$.

12 Show that $\sqrt{5} + \sqrt{20} = 3\sqrt{5}$ and that $\left(\sqrt{5} + \sqrt{20}\right)^2 = 45$.

13 Without a calculator, arrange in ascending order: $\sqrt{10},\ 2\sqrt{3},\ 3\sqrt{2},\ \pi$. Justify each comparison.

14 A rectangle measures 12.5 cm by 7.4 cm, each to 1 decimal place. Find the lower and upper bounds of its perimeter and area.

15 The mass of a grain of rice is about 2.5×10^{-5} kg. A bag contains 4.0×10^{4} grains. Estimate the total mass in kilograms, giving your answer in standard form to 2 significant figures.

16 The distance from the Earth to the Sun is 1.496×10^{8} km. Light travels at 3.00×10^{5} km/s. Calculate the time for light to reach the Earth in minutes and seconds, to 3 significant figures.

17 **Error analysis.** A student writes $\sqrt{18} = 3\sqrt{2}$ and $\sqrt{50} = 5\sqrt{2}$, then concludes $\sqrt{18} + \sqrt{50} = 8\sqrt{2}$. The conclusion is correct. Explain why, and identify what would make the method invalid.

18 **Error analysis.** A student calculates $6.8 \times 4.2 = 28.56$ and reports 28.56 as the area of a rectangle measured to 1 decimal place. Explain why this overstates the precision and give a more appropriate report.

MIXED PRACTICE 1 · CONTINUED

REASONING · APPLICATION · EXTENSION

19 **Reasoning.** Explain why $0.\overline{9} = 1$. Does this contradict the statement that every number has a unique decimal expansion? Explain.

20 **Extension.** Prove that $\sqrt{2} + \sqrt{3}$ is irrational. Begin by assuming it equals a rational number r, square both sides, and use the fact that $\sqrt{6}$ is irrational.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING 30 minutes · 30 marks

Number foundations: short response

NO CALCULATOR SHOW WORKING

Instructions. Answer all questions. Where a question says "exactly", do not give a decimal approximation. Marks are shown in brackets.

21 Classify $\sqrt{16}$, $\sqrt{17}$, -2.5 and $0.\overline{3}$. Justify each classification. [4]

22 Simplify $\dfrac{2x^{3} \cdot 3x^{-1}}{6x^{-2}}$. [3]

23 Evaluate $27^{2/3}$ and $16^{-3/4}$ without a calculator. [4]

24 Write $0.000\,000\,72$ and 5.4×10^{6} as ordinary numbers, and $64\,000\,000$ in standard form. [3]

25 Simplify $\sqrt{98} - \sqrt{32}$ and $\sqrt{12} \times \sqrt{18}$. [4]

26 Rationalise $\dfrac{6}{\sqrt{3}}$ and $\dfrac{2}{\sqrt{5} - 1}$. [4]

27 A mass is 3.6 kg, correct to 1 decimal place. Write its bounds and calculate the maximum percentage error. [4]

28 Explain why $\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$ in general, using a numerical counterexample. [4]

CRITERION B · INVESTIGATING PATTERNS — 45 minutes · 24 marks

The last digit of powers of 7

Instructions. Work independently. Use the table to record evidence, then answer in full sentences. You may use a calculator for the arithmetic but not for the generalisation.

1. Complete a table of the last digit of 7^n for $n = 1, 2, \ldots, 8$. [4]
2. Describe the pattern and state the period. [3]
3. Predict the last digit of 7^{50}, 7^{51} and 7^{100}. Show how the remainder on division by 4 is used. [5]
4. Explain why the pattern continues forever. Your explanation should refer to what happens to the last digit when a number ending in each digit of the cycle is multiplied by 7. [6]
5. Generalise: for a base ending in 3, 7 or 9, describe how to find the last digit of a^n. Test your rule on 9^{33}. [6]

Evidence: table, description, prediction with remainders, justification, general rule and test.

CRITERION C · COMMUNICATING — 40 minutes · 20 marks

Exact or approximate? A student guide

Instructions. Write a guide of 300–400 words for a younger student who has just discovered that a calculator shows $\sqrt{2} = 1.41421356$. Your guide must explain: (i) why the decimal is not the number; (ii) why exact surd form is useful; (iii) when an approximation is appropriate; (iv) one situation where rounding too early changes a conclusion. Use at least two worked examples and correct mathematical notation.

Evidence: a titled guide with paragraphs, at least two equations, and a short summary box. Diagrams are optional but must be labelled.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Does the parcel fit?

Instructions. A courier measures a parcel as $42\text{ cm} \times 28\text{ cm} \times 15\text{ cm}$, each to the nearest centimetre. A locker has usable volume $18\,000\text{ cm}^3$. The courier must decide whether the parcel is guaranteed to fit.

1. Calculate the nominal volume. [2]
2. Write the lower and upper bounds of each dimension and calculate the lower and upper bounds of the volume. [6]
3. Compare the interval with the locker volume. Can the courier be certain? Explain. [4]
4. State two assumptions in the volume model and explain how each could affect the decision. [6]
5. Recommend a decision and a practical check the courier could use. Justify both. [6]

Evidence: bound calculations, a clear comparison, assumptions, and a justified recommendation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- The real numbers contain the rationals and the irrationals. Rationals have terminating or recurring decimals; irrationals have decimals that never terminate and never repeat.
- Proof by contradiction assumes the opposite and derives an impossibility. The proof that $\sqrt{2}$ is irrational uses lowest terms and parity.
- The index laws extend to integer and rational exponents: $a^m a^n = a^{m+n}$, $a^m / a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$, $a^{-n} = 1/a^n$, $a^{1/n} = \sqrt[n]{a}$.
- Standard form writes $a \times 10^n$ with $1 \leq a < 10$. Multiply by multiplying the coefficients and adding the exponents; divide by dividing the coefficients and subtracting the exponents.
- Surds are simplified by removing square factors. Like surds combine; $\sqrt{ab} = \sqrt{a}\sqrt{b}$. Rationalising multiplies by a form of 1, using a conjugate for a binomial denominator.
- Rounding to a unit u gives bounds $x \pm u/2$. Absolute and percentage error quantify the uncertainty. For positive quantities, bounds of products and quotients use the extreme combinations.
- Last-digit cycles show that exactness is a property of structure, not of the number of digits displayed.

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad (a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- I can classify a number as natural, integer, rational or irrational and justify it.
- I can convert a recurring decimal to a fraction.
- I can outline the proof that $\sqrt{2}$ is irrational.
- I can apply the index laws with negative and fractional exponents.
- I can solve simple exponential equations by matching bases.
- I can convert between standard form and ordinary form.
- I can calculate with numbers in standard form, including division.
- I can simplify a surd and combine like surds.
- I can rationalise a denominator, including a binomial denominator.
- I can expand and simplify products involving surds.
- I can round to a stated number of decimal places or significant figures.
- I can calculate bounds and percentage error.
- I can find bounds of a sum, difference, product or quotient.
- I can find and justify a last-digit cycle.
- I can choose between an exact form and an approximation and explain why.
- I can evaluate a model's assumptions as well as its arithmetic.

REFLECTION

Write short answers in your notebook. Use evidence from your work, not general statements.

1. Which representation — exact surd, decimal, standard form or bound — did you find most useful in this chapter? Give a question where it was the best choice.
2. Describe one error you made and the reasoning that corrected it. What will you check first next time?
3. The statement of inquiry claims that form determines the precision we can claim. Use one example from this chapter to support the claim and one example that complicates it.
4. What question about number or measurement do you still want to answer?

EQUIVALENT FORMS, FACTORS AND THE GRAMMAR OF ALGEBRA

Algebraic Expressions

02

EQUIVALENCE · STRUCTURE · FACTORISATION

Factorisation, identities and algebraic fractions — rewriting an expression until the structure you need becomes visible.

STATEMENT OF INQUIRY

An algebraic expression can be rewritten in many equivalent forms, and the form we choose reveals different structure: factors expose dimensions and zeros, while expanded forms expose coefficients and degree.

KEY CONCEPT

Form

RELATED CONCEPTS

Equivalence · Simplification · Pattern · Representation

GLOBAL CONTEXT

Scientific and technical innovation — design, optimisation and efficient form

FACTUAL

How do we expand a product of brackets, and how do we factorise an expression?

CONCEPTUAL

How can two different-looking expressions carry exactly the same information?

DEBATABLE

Is the most elegant algebraic form always the most useful one?

IN THIS CHAPTER

2.1 Expanding and simplifying
2.2 Common factors and grouping
2.3 Quadratic factorisation
2.4 Special products and identities
2.5 Algebraic fractions
2.6 Rearranging formulae

OPENING PROBLEM · DESIGN UNDER CONSTRAINT

The open box

A packaging designer starts with a square sheet of card 24 cm by 24 cm. She cuts a square of side x centimetres from each corner, folds up the four sides, and tapes the corners to make an open box.

The volume of the box is **length × width × height**. Before any calculation, predict which value of x will give the largest volume. Write your prediction down; you will test it later.

1. Write the length, width and height of the box in terms of x.
2. Write the volume V as a product of three factors.
3. Expand the product to write V as a polynomial in standard form.
4. Evaluate V when $x = 2$, $x = 4$ and $x = 6$. Which is largest?
5. What values of x make sense in this context? Justify the restriction.

Expanding turns a product into a sum; factorising turns a sum back into a product. Both are ways of changing the form of an expression without changing its value. Knowing which form to use is the central skill of this chapter.

2.0 Learning intentions

ORIENTATION

- Expand products of brackets and simplify the resulting expressions.
- Factorise expressions using common factors, grouping and quadratic methods.
- Recognise and prove algebraic identities, including the special products.
- Simplify algebraic fractions by factorising, and state the values for which they are defined.
- Rearrange formulae to change the subject, including formulae with the subject appearing twice.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

expand a product of two or more brackets without sign errors; factorise fully, choosing between a common factor, grouping, a quadratic pattern and a special product; prove an identity by expanding both sides; simplify an algebraic fraction and state its domain; rearrange a formula and explain each inverse operation; and choose the form of an expression that answers a particular question.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 2.1.

1. Expand $3(x + 4)$ and $-2(x - 5)$.
2. Simplify $4x + 3x - 2x$.
3. Find the highest common factor of 12 and 18; of $6x^2$ and $9x$.
4. Expand $(x + 3)(x + 2)$.
5. Evaluate $x^2 - 3x + 1$ when $x = -2$.
6. Solve $2x - 7 = 11$.

REFRESHER

The distributive law says $a(b + c) = ab + ac$. It is the engine behind every expansion in this chapter. The index law $x^m \cdot x^n = x^{m+n}$ tells you how powers combine: $x \cdot x^2 = x^3$, not x^2. A term is a product of a number and powers of variables; like terms have exactly the same variable part.

ATL SKILLS · THINKING AND COMMUNICATION

Critical thinking: test whether two expressions really are equivalent by expanding, substituting and comparing structure. **Transfer:** use the area model from arithmetic to organise algebraic products. **Organisation:** write one equal sign per line and keep columns aligned; most expansion errors are recording errors, not reasoning errors.

FROM PRODUCT TO SUM

2.1 Expanding and simplifying

Arithmetic gives a single answer: $3 \times 7 = 21$. Algebra keeps the structure visible: $3(x + 4)$ is a product, while $3x + 12$ is a sum. They are equal for every value of x, so they are **equivalent expressions**. Expanding is the process of turning the product form into the sum form.

The area model explains why every term in the first bracket must multiply every term in the second. A rectangle with sides $x + 3$ and $x + 4$ can be split into four smaller rectangles whose areas are x^2, $4x$, $3x$ and 12. The total area is the sum of the four parts.

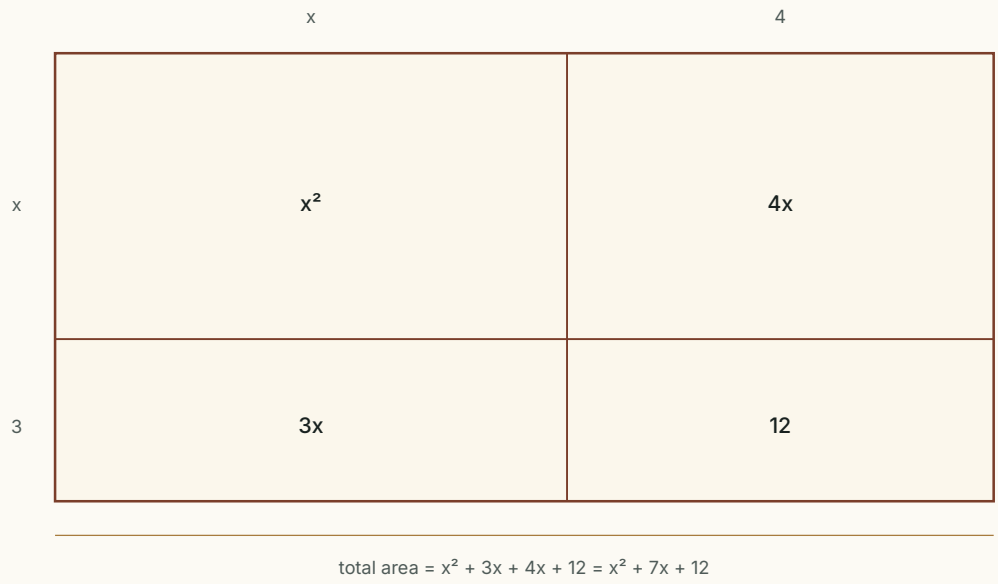


Figure 2.1 The area model for $(x+3)(x+4)$. The four regions correspond to the four products; the two x-terms combine because they are like terms.

DEFINITIONS

Term, polynomial, degree and like terms

A **term** is a product of a coefficient and one or more variables raised to whole-number powers, such as $5x^2$ or $-3ab$. A **polynomial** is a sum of terms in which every variable has a whole-number exponent. The **degree** of a polynomial is the highest power of the variable; a quadratic has degree 2. **Like terms** have identical variable parts and can be combined by adding their coefficients.

EXAMPLE 2.1 EXPANDING A SINGLE BRACKET

QUESTION

Expand and simplify.

(a) $3x(2x - 5)$

(b) $-2a(3a^2 - 4a + 1)$

(c) $4x - 2(x - 3)$

(d) $5y(2y + 1) - 3y(y - 4)$

SOLUTION

a Multiply each term inside the bracket by $3x$: $3x \cdot 2x = 6x^2$ and $3x \cdot (-5) = -15x$. So $3x(2x - 5) = 6x^2 - 15x$.

b The negative sign applies to every term: $-2a \cdot 3a^2 = -6a^3$, $-2a \cdot (-4a) = +8a^2$, $-2a \cdot 1 = -2a$. Hence $-6a^3 + 8a^2 - 2a$.

c Expand the bracket first: $4x - 2x + 6$. The minus sign multiplies both terms, so $-2 \cdot (-3) = +6$. The result is $2x + 6$.

d Expand both products: $10y^2 + 5y - (3y^2 - 12y)$. The bracket is subtracted, so change both signs: $10y^2 + 5y - 3y^2 + 12y = 7y^2 + 17y$.

> **Structure:** in part (d) the bracket must be expanded before subtracting. Writing $10y^2 + 5y - 3y^2 - 12y$ loses a sign because the whole expression $3y^2 - 12y$ is subtracted.

EXAMPLE 2.2 EXPANDING TWO BRACKETS

QUESTION

Expand and simplify $(x + 3)(x - 4)$ and $(2x - 1)(3x + 2)$.

SOLUTION

1 **First product:** multiply every term in the first bracket by every term in the second.

$$(x + 3)(x - 4) = x^2 - 4x + 3x - 12$$

2 Combine the like x-terms: $-4x + 3x = -x$. Hence $(x + 3)(x - 4) = x^2 - x - 12$.

3 **Second product:** the coefficient 2 multiplies both terms of the second bracket.

$$(2x - 1)(3x + 2) = 6x^2 + 4x - 3x - 2$$

4 Combine: $6x^2 + x - 2$.

Check: substitute $x = 1$. The original $(2 - 1)(3 + 2) = 1 \times 5 = 5$; the expansion $6 + 1 - 2 = 5$. A numerical check does not prove equivalence, but it catches most errors.

EXAMPLE **2.3** EXPANDING THREE FACTORS AND SIMPLIFYING

QUESTION

Expand $(x+1)(x^2-x+1)$, and simplify $5x-2(x-3)+(x+1)(x-4)$.

SOLUTION

1 For the first product, distribute x and then 1:

$$x(x^2-x+1)+1(x^2-x+1)=x^3-x^2+x+x^2-x+1$$

2 The $-x^2$ and $+x^2$ cancel, as do $+x$ and $-x$. The result is x^3+1.

3 For the second expression, expand each part: $5x-2x+6+x^2-3x-4$.

4 Collect like terms: $x^2+(5-2-3)x+(6-4)=x^2+2$. The x-terms cancel completely.

> **Pattern:** $(x+1)(x^2-x+1)=x^3+1$ is an identity. It is the sum of two cubes pattern, and it holds for every value of x.

KEY IDEA

Expanding is repeated use of the distributive law. Every term in one bracket multiplies every term in the other, and like terms are combined only after all products have been written. An expansion can be checked numerically but is proved by the distributive law.

COMMON MISCONCEPTIONS

"$(a+b)^2=a^2+b^2$." Test $a=3, b=4$: $(3+4)^2=49$, but $3^2+4^2=25$. The middle term $2ab$ is missing.

"A minus sign only affects the first term in the bracket." $-(2x-5)=-2x+5$, not $-2x-5$. The minus multiplies every term.

"$3x\cdot 2x=6x$." Multiply coefficients and add indices: $3x\cdot 2x=6x^2$.

GUIDED PRACTICE · COMPLETE THE EXPANSION

Fill in each blank, then check by substituting $x=2$.

$(x+5)(x-2)=x^2-_____x+_____x-10=x^2+_____x-10$

$(3x-1)^2=(3x)^2-2(3x)(_____)+(____)^2=_____x^2-_____x+1$

$4a(2a-3)-2a(a-5)=_____a^2-_____a-_____a+_____a=_____a^2-_____a$

EXERCISE 2A

EXPANDING · SIMPLIFYING · EQUIVALENCE

FLUENCY

1 Expand.

a. $5(x+3)$
b. $-3(2x-1)$
c. $4x(3x+2)$
d. $-2a(a^2-3a+1)$

2 Expand and simplify.

a. $(x+2)(x+5)$
b. $(x-4)(x+1)$
c. $(a-3)(a-6)$
d. $(2y+1)(y+4)$

EXERCISE 2A · CONTINUED EXPANDING · SIMPLIFYING · EQUIVALENCE

3 Expand and simplify.
a. $(2x-3)(3x+4)$
b. $(4a+1)(2a-5)$
c. $(3b-2)(b-3)$
d. $(5-2x)(3+x)$

4 Expand $(x+2)(x^2-3x+1)$.

5 Simplify.
a. $3(x+2)-2(x-5)$
b. $4x(2x-1)-3x(2x-4)$

6 Simplify $(x+1)(x-3)+(x+2)(x+4)$.

UNDERSTANDING AND REASONING

7 A rectangle has sides $x+7$ and $x+2$. Write its area in expanded form and in factorised form. Draw an area model and label all four regions.

8 **Error analysis.** A student writes $(x+3)(x+4)=x^2+12$. Identify the missing terms, write the correct expansion, and explain the error using an area model.

9 Show that $(x+a)(x+b)=x^2+(a+b)x+ab$ by expanding. Use the result to expand $(x+7)(x-2)$ without writing every product.

10 Is $x^2+5x+6=(x+2)(x+3)$ an identity or an equation? Explain the difference in a sentence.

APPLICATION AND EXTENSION

11 A rectangular garden is $(3x+2)$ m long and $(2x-1)$ m wide. Write its area in expanded form. Find the area when $x=4$.

12 A box has a square base of side x and height $x+5$. Its surface area is $2x^2+4x(x+5)$. Simplify this expression and explain what the two parts represent.

13 Find a numerical counterexample to the statement $(a+b)^2=a^2+b^2$. Then write the correct identity and explain why the extra term is necessary.

14 **Extension.** Expand $(x-1)(x^2+x+1)$ and $(x-1)(x^3+x^2+x+1)$. Predict the expansion of $(x-1)(x^4+x^3+x^2+x+1)$, then prove your prediction. Generalise to n terms.

FROM SUM TO PRODUCT

2.2 Common factors and grouping

Factorising is the reverse of expanding. Instead of removing brackets, we insert them. The simplest method looks for a factor common to every term. For $6x^3-9x^2$, both terms contain $3x^2$: the highest common factor of 6 and 9 is 3, and the lowest power of x present in both terms is x^2. Taking it out gives $3x^2(2x-3)$.

Expanding the result is always the check. If $3x^2(2x-3)$ does not return $6x^3-9x^2$, the factorisation is wrong. A factorisation is **complete** when no further common factor remains inside the bracket.

DEFINITION

Factorising

To **factorise** an expression is to write it as a product of factors. The **highest common factor** (HCF) of an expression is the largest factor common to every term: take the HCF of the numerical coefficients and the lowest power of each variable that appears in every term. A factorisation is complete when the factors cannot be factorised further using integer coefficients.

EXAMPLE **2.4** TAKING OUT THE HIGHEST COMMON FACTOR

QUESTION

Factorise fully.

(a) $6x^3 - 9x^2$

(b) $4a^2b - 6ab^2 + 2ab$

(c) $3x^2 - 12$

(d) $5x^2y - 10xy^2$

SOLUTION

a HCF of 6 and 9 is 3; the lowest power of x is x^2. So $6x^3 - 9x^2 = 3x^2(2x - 3)$.

b HCF of 4, 6 and 2 is 2; every term contains ab. So $4a^2b - 6ab^2 + 2ab = 2ab(2a - 3b + 1)$. The final 1 is essential: $2ab \cdot 1 = 2ab$.

c $3x^2 - 12 = 3(x^2 - 4)$. The bracket is a difference of squares, so the complete factorisation is $3(x - 2)(x + 2)$.

d HCF is $5xy$, so $5x^2y - 10xy^2 = 5xy(x - 2y)$.

Check (b): $2ab(2a - 3b + 1) = 4a^2b - 6ab^2 + 2ab$. Expanding restores the original expression term by term.

EXAMPLE **2.5** FACTORISING BY GROUPING

QUESTION

Factorise $ax + ay + bx + by$ and $x^3 + 3x^2 + 2x + 6$.

SOLUTION

1 For the first expression, group the first two terms and the last two: $a(x + y) + b(x + y)$.

2 The bracket $(x + y)$ is now a common factor: $(x + y)(a + b)$.

3 For the second, group $x^3 + 3x^2$ and $2x + 6$: $x^2(x + 3) + 2(x + 3)$.

4 Take out the common bracket: $(x + 3)(x^2 + 2)$.

> **How to choose the grouping:** group terms so that the bracket left behind is identical. If the brackets differ only by sign, factor out -1 from one group to make them match.

COMMON MISCONCEPTIONS

"Factorising means dividing the expression by something." It is a rewriting, not a division that changes the value. $6x^2 + 9x = 3x(2x + 3)$ is still the same expression.

"The common factor disappears." It moves outside the bracket. Leaving it out changes the value: $4a^2b - 6ab^2 + 2ab \neq 2a - 3b + 1$.

"$x^2 + 5x = x(x + 5)$, so x can be cancelled everywhere." Cancelling is only valid in a fraction and only for factors of the entire numerator and denominator, never for terms joined by $+$.

GUIDED PRACTICE · FACTORISE AND CHECK

Complete each factorisation, then expand your answer to check.

$8x^3 + 12x^2 = _____(_____ + _____)$

$6a^2b - 9ab^2 + 3ab = _____(_____)$

$x^2 + 4x + 3x + 12 = x(_____) + 3(_____) = (_____)(_____)$

$2x^2 - 18 = _____(x^2 - _____) = _____(_____)(_____)$

EXERCISE 2B

COMMON FACTORS · GROUPING · COMPLETENESS

FLUENCY

1 Find the highest common factor.

a. $8x^3$ and $12x^2$

b. $15a^2b$ and $25ab^2$

c. $6x^2y$ and $9xy^2$

d. $4a^3b^2$ and $10a^2b^3$

2 Factorise.

a. $5x + 15$

b. $7x^2 - 21x$

c. $8a^3 + 12a^2$

d. $9x^2y - 6xy^2$

3 Factorise fully.

a. $3x^2 - 6x + 9$

b. $4a^2b + 8ab^2 - 12ab$

c. $x^3 - x^2 + x$

d. $10x^3y^2 - 15x^2y^3$

4 Factorise by grouping.

a. $ax + ay + 3x + 3y$

b. $2x + 2y + px + py$

c. $x^3 + 4x^2 + 3x + 12$

d. $ab - 2a + 3b - 6$

5 Factorise fully.

a. $2x^2 - 18$

b. $3x^3 - 12x$

c. $5x^2 - 45$

d. $4x^3 - 16x$

UNDERSTANDING AND REASONING

6 **Error analysis.** A student writes $x^2 + 6x = x(x + 6)$, then divides both sides by x and claims $x^2 + 6x = 6$ when $x = 1$. Explain what is wrong with the division step.

7 Explain why $4x + 6 = 2(2x + 3)$ is correct, but $4x + 6 = 2 \cdot 2x + 3$ is not. Use the distributive law in your explanation.

8 A rectangle has area $6x^2 + 9x$ and width $3x$. Find its length. Explain why the length must be a polynomial.

9 Factorise $x^3 - 2x^2 + 5x - 10$ by grouping. Check by expanding.

EXERCISE 2B · CONTINUED COMMON FACTORS · GROUPING · COMPLETENESS

APPLICATION AND EXTENSION

10 The area of a rectangle is $12x^2 + 18x$. Its width is $6x$. Write its length and perimeter in terms of x.

11 Factorise $(x+1)^2 + 3(x+1)$ by treating $(x+1)$ as a single common factor. Then expand your result to check.

12 **Extension.** Factorise $x^2 - y^2 + x - y$ by grouping. Begin by writing $x^2 - y^2 = (x-y)(x+y)$.

13 **Extension.** Show that $a^2 - ab = a(a-b)$, and interpret the result as the area of a rectangle with a square removed. Draw and label a diagram.

PRODUCT AND SUM

2.3 Quadratic factorisation

A quadratic expression has the form $ax^2 + bx + c$. When $a = 1$, factorising is a search for two numbers whose product is c and whose sum is b. The identity $(x+p)(x+q) = x^2 + (p+q)x + pq$ explains why: expanding the right-hand side must reproduce the coefficients of the left.

When $a \neq 1$, the search is slightly larger, but the idea is the same: split the middle term into two terms that share a common factor with the ends, then group. This is often called the **ac method**.

DEFINITIONS

Quadratic, monic and factorable

A **quadratic expression** in x has degree 2: $ax^2 + bx + c$, with $a \neq 0$. It is **monic** when $a = 1$. A quadratic is **factorable over the integers** if it can be written as a product of two linear expressions with integer coefficients.

EXAMPLE **2.6** MONIC QUADRATICS WITH POSITIVE AND NEGATIVE COEFFICIENTS

QUESTION

Factorise $x^2 + 7x + 12$ and $x^2 - 5x - 24$.

SOLUTION

1 For $x^2 + 7x + 12$, find two numbers with product 12 and sum 7. The factor pairs of 12 are $1 \cdot 12, 2 \cdot 6, 3 \cdot 4$. Only $3 + 4 = 7$, so the numbers are 3 and 4.

2 Hence $x^2 + 7x + 12 = (x+3)(x+4)$.

3 For $x^2 - 5x - 24$, the product is -24 and the sum is -5. Because the product is negative, the numbers have opposite signs; because the sum is negative, the larger absolute value is negative. The pair is -8 and 3.

4 Hence $x^2 - 5x - 24 = (x-8)(x+3)$.

Check: $(x-8)(x+3) = x^2 + 3x - 8x - 24 = x^2 - 5x - 24$.

EXAMPLE **2.7** QUADRATICS WITH A LEADING COEFFICIENT OTHER THAN 1

QUESTION

Factorise $2x^2 + 7x + 3$ and $6x^2 - 11x - 10$.

SOLUTION

1. For $2x^2 + 7x + 3$, multiply a and c: $2 \cdot 3 = 6$. Find two numbers with product 6 and sum 7: they are 6 and 1.
2. Split the middle term: $2x^2 + 6x + x + 3$. Group: $2x(x + 3) + 1(x + 3) = (x + 3)(2x + 1)$.
3. For $6x^2 - 11x - 10$, $ac = 6 \cdot (-10) = -60$. Two numbers with product -60 and sum -11 are -15 and 4.
4. Split and group: $6x^2 - 15x + 4x - 10 = 3x(2x - 5) + 2(2x - 5) = (2x - 5)(3x + 2)$.

> **Why the ac method works:** the split terms must have the same product ac and sum b. Grouping then produces a common bracket because the ratio of the end coefficients matches the split.

COMMON MISCONCEPTIONS

"$(x + 2)(x + 3) = x^2 + 6$." The constant is $2 \cdot 3 = 6$, but there is also a middle term $(2 + 3)x = 5x$. The expansion is $x^2 + 5x + 6$.

"If the product is negative, both numbers are negative." A negative product means the signs are opposite. The sum then tells you which number has the larger absolute value.

"$2x^2 + 7x + 3 = (2x + 3)(x + 1)$ **because the ends match.**" Check by expanding: $2x^2 + 2x + 3x + 3 = 2x^2 + 5x + 3$, not $2x^2 + 7x + 3$. Always expand to verify.

INVESTIGATION · CRITERION B

Which quadratics factorise?

Consider monic quadratics of the form $x^2 + bx + 12$, where b is an integer.

1. List every pair of integers with product 12. Use each pair to write a factorised quadratic, and record the value of b.
2. Which integer values of b make $x^2 + bx + 12$ factorable over the integers? How many are there?
3. Repeat for $x^2 + bx + 7$. Explain why there are fewer values of b.
4. Test the rule "$x^2 + bx + c$ is factorable exactly when $b^2 - 4c$ is a perfect square" on at least three examples, including one that does not factorise.

Generalise and justify. If $x^2 + bx + c = (x + p)(x + q)$, explain why $p + q = b$ and $pq = c$. Hence show that $b^2 - 4c = (p - q)^2$, and explain why this proves the rule. Why must p and q have the same parity as b?

GUIDED PRACTICE · FACTORISE, THEN CHECK

Fill in the missing numbers.

$x^2 + 9x + 20 = (x + ____)(x + ____)$ because $____ \times ____ = 20$ and $____ + ____ = 9$.

$x^2 - 2x - 15 = (x - ____)(x + ____)$ because the product is negative and the sum is $____$.

$3x^2 + 10x + 8$: $ac = ____$; the split is $____x + ____x$; the factorisation is $(____)(____)$.

EXERCISE 2C

QUADRATIC FACTORISATION · PRODUCT AND SUM

FLUENCY

1 Factorise.

a. $x^2 + 8x + 15$

b. $x^2 + 9x + 20$

c. $x^2 + 6x + 8$

d. $x^2 + 10x + 21$

2 Factorise.

a. $x^2 - 7x + 12$

b. $x^2 - 3x - 10$

c. $x^2 + x - 20$

d. $x^2 - 9x + 18$

3 Factorise.

a. $2x^2 + 5x + 2$

b. $3x^2 + 10x + 8$

c. $2x^2 - 7x + 3$

d. $5x^2 - 13x + 6$

4 Factorise.

a. $6x^2 + 11x - 10$

b. $4x^2 - 4x - 15$

c. $8x^2 + 2x - 3$

d. $9x^2 - 9x - 10$

5 Factorise fully by taking out a common factor first.

a. $2x^2 - 8x + 6$

b. $3x^2 + 15x + 18$

c. $4x^2 - 4$

d. $5x^3 - 20x$

UNDERSTANDING AND REASONING

6 Explain why $x^2 + bx + c$ can factorise as $(x + p)(x + q)$ only when $p + q = b$ and $pq = c$.

7 **Error analysis.** A student factorises $x^2 - 6x + 8$ as $(x + 2)(x + 4)$. Expand the student's answer, identify the sign error, and write the correct factorisation.

8 A rectangle has area $x^2 + 11x + 30$ and width $x + 5$. Find its length and explain how factorising gives the answer directly.

9 Use the investigation rule to decide whether $x^2 + 4x + 2$ factorises over the integers. Show the test and interpret the result.

APPLICATION AND EXTENSION

10 A rectangular courtyard has area $2x^2 + 9x + 4$. Its length is $2x + 1$. Find its width and perimeter.

11 Factorise $x^4 + 5x^2 + 6$ by treating x^2 as a single variable. Check by expanding.

12 List every integer b for which $x^2 + bx + 12$ factorises over the integers. Justify your list using the investigation.

13 **Extension.** Prove that if $x^2 + bx + c = (x + p)(x + q)$ with integers p, q, then $b^2 - 4c$ is a perfect square. Explain why the converse is also true when b and c are integers.

STRUCTURE WORTH MEMORISING

2.4 Special products and identities

Some expansions occur so often that their structure is worth recognising instantly. The square of a binomial and the difference of two squares are the two most important. They are not shortcuts that replace understanding: each is proved by expanding, and each has a geometric interpretation.

An **identity** is an equation that is true for every value of the variables for which both sides are defined. The symbol $\equiv$ is sometimes used to emphasise identity, as in $(a+b)^2 \equiv a^2 + 2ab + b^2$.

IDENTITIES

The three special products

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

$$(a+b)(a-b) = a^2 - b^2$$

Each is proved by expanding the left-hand side. The first two are called **perfect squares**; the third is the **difference of two squares**.

EXAMPLE 2.8 SQUARING A BINOMIAL

QUESTION

Expand $(3x-2)^2$ and $(5a+2b)^2$.

SOLUTION

1. Use $(a-b)^2 = a^2 - 2ab + b^2$ with $a = 3x$ and $b = 2$: $(3x)^2 - 2(3x)(2) + 2^2 = 9x^2 - 12x + 4$.
2. Use $(a+b)^2$ with $a = 5a$ and $b = 2b$: $(5a)^2 + 2(5a)(2b) + (2b)^2 = 25a^2 + 20ab + 4b^2$.

> **Why the middle term is not optional:** the two cross-products ab and ba are equal, so they combine to $2ab$. Omitting it is the most common expansion error in algebra.

EXAMPLE **2.9** FACTORISING WITH THE DIFFERENCE OF TWO SQUARES

QUESTION

Factorise fully.

(a) $49x^2 - 16y^2$ (b) $x^4 - 16$

(c) $3x^3 - 27x$ (d) $2x^2 - 32$

SOLUTION

a Both terms are squares: $49x^2 = (7x)^2$ and $16y^2 = (4y)^2$. So $49x^2 - 16y^2 = (7x - 4y)(7x + 4y)$.

b $x^4 - 16 = (x^2)^2 - 4^2 = (x^2 - 4)(x^2 + 4)$. The first factor is again a difference of squares: $(x - 2)(x + 2)(x^2 + 4)$.

c Take out the common factor first: $3x(x^2 - 9) = 3x(x - 3)(x + 3)$.

d $2x^2 - 32 = 2(x^2 - 16) = 2(x - 4)(x + 4)$.

Check (b): expand $(x - 2)(x + 2)$ to get $x^2 - 4$, then multiply by $(x^2 + 4)$: $x^4 - 16$.

EXAMPLE **2.10** IDENTITIES IN CALCULATION AND PROOF

QUESTION

Use identities to calculate 101×99 and $53^2 - 47^2$. Then simplify $(a + b)^2 - (a - b)^2$.

SOLUTION

1 $101 \times 99 = (100 + 1)(100 - 1) = 100^2 - 1^2 = 10\,000 - 1 = 9999$.

2 $53^2 - 47^2 = (53 - 47)(53 + 47) = 6 \times 100 = 600$.

3 For the algebraic expression, expand each square: $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$.

Structure: the difference of squares converts a difference of two large squares into a product of two small numbers. The same identity explains why the a^2 and b^2 terms cancel in the last part.

COMMON MISCONCEPTIONS

"$(x + 3)^2 = x^2 + 9$." The square of a sum is not the sum of the squares. $(x + 3)^2 = x^2 + 6x + 9$; the area model has a missing $3x$ strip on each side.

"$x^2 - 9 = (x - 3)^2$." $(x - 3)^2 = x^2 - 6x + 9$, which is not $x^2 - 9$. The correct factorisation is $(x - 3)(x + 3)$.

"$x^2 + 9$ **factorises as** $(x + 3)(x + 3)$." A sum of squares does not factorise over the integers. Only a difference of squares does.

GUIDED PRACTICE · IDENTIFY THE PATTERN FIRST

Write the identity you will use, then complete the expansion or factorisation.

$(2x+5)^2 = _____ + _____ + _____$

$(7a-3b)(7a+3b) = _____ - _____$

$x^2 - 81 = (____ - ___)(____ + ___)$

$16x^2 - 25y^2 = (_____ - _____)(_____ + _____)$

EXERCISE 2D

SPECIAL PRODUCTS · IDENTITIES · PROOF

FLUENCY

1 Expand.

a. $(x+5)^2$

b. $(x-7)^2$

c. $(3x+2)^2$

d. $(2a-5b)^2$

2 Expand.

a. $(x+4)(x-4)$

b. $(2x+3)(2x-3)$

c. $(5a-b)(5a+b)$

d. $(x^2+1)(x^2-1)$

3 Factorise.

a. $x^2 - 36$

b. $4x^2 - 9$

c. $25a^2 - 16b^2$

d. $x^4 - 1$

4 Factorise fully.

a. $2x^2 - 32$

b. $3x^3 - 27x$

c. $5x^2 - 45$

d. $8x^2 - 50$

5 Use a difference of squares to calculate.

a. 103×97

b. $64^2 - 36^2$

c. $205^2 - 195^2$

d. 51×49

UNDERSTANDING AND REASONING

6 Simplify.

a. $(x+3)^2 - (x-3)^2$

b. $(2a+b)^2 - (2a-b)^2$

7 Show that $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$. Explain what the identity says about the two squares.

8 **Error analysis.** A student writes $(x+3)^2 = x^2 + 9$. Draw an area model to show the missing regions, and write the correct expansion.

9 Prove $(a+b)(a-b) = a^2 - b^2$ by expanding. Then explain the identity as the area of a square with a smaller square removed.

APPLICATION AND EXTENSION

10 A square frame has outer side x and inner side $x-4$. Write the area of the frame in factorised form and in expanded form. Evaluate both when $x = 10$.

11 Use the identity $(a+b)^2 = a^2 + 2ab + b^2$ to calculate 101^2 without a calculator.

12 **Extension.** Expand $(x+y)^3$ by writing it as $(x+y)(x+y)^2$. Collect terms and describe the pattern of coefficients.

13 **Extension.** Factorise $(a+b)^2 - c^2$. Hence solve $(a+b)^2 = c^2$ for a in terms of b and c, stating any restriction.

FACTORS IN THE NUMERATOR AND DENOMINATOR

2.5 Algebraic fractions

An algebraic fraction is a quotient of polynomials. The rules are the same as for numerical fractions, but factorising is the key that makes cancellation visible. You may cancel a factor that multiplies the whole numerator with the same factor in the whole denominator; you may never cancel a term joined by $+$ or $-$.

Every algebraic fraction has a **domain**: the set of values for which its denominator is not zero. Simplifying a fraction can remove a factor from the denominator, but the original restriction remains part of the expression's meaning.

DEFINITION

Algebraic fraction and domain

An **algebraic fraction** is an expression of the form $\dfrac{P(x)}{Q(x)}$, where P and Q are polynomials and $Q(x) \neq 0$. Its **domain** is all real values of x for which $Q(x) \neq 0$. Two algebraic fractions are equivalent on a domain if they agree at every point of that domain.

EXAMPLE 2.11 SIMPLIFYING BY FACTORISING

QUESTION

Simplify $\dfrac{x^2 - 9}{x^2 + x - 12}$ and state the domain.

SOLUTION

1. Factorise the numerator: $x^2 - 9 = (x - 3)(x + 3)$.
2. Factorise the denominator: $x^2 + x - 12 = (x + 4)(x - 3)$.
3. Cancel the common factor $(x - 3)$: $\dfrac{(x - 3)(x + 3)}{(x + 4)(x - 3)} = \dfrac{x + 3}{x + 4}$.
4. The original denominator is zero when $x = -4$ or $x = 3$, so the domain is $x \neq -4,\ 3$. The simplified form is used only on this domain.

> **Why the restriction stays:** at $x = 3$ the original fraction is $\frac{0}{0}$, which is undefined. The simplified expression happens to equal 1.5 there, but it is not the same function at that point.

EXAMPLE 2.12 MULTIPLYING AND DIVIDING ALGEBRAIC FRACTIONS

QUESTION

Simplify $\dfrac{x+2}{x-3} \times \dfrac{x^2-9}{x^2-4}$, and $\dfrac{x^2-1}{x^2+2x} \div \dfrac{x-1}{x}$.

SOLUTION

1. Factorise before multiplying: $\dfrac{x+2}{x-3} \times \dfrac{(x-3)(x+3)}{(x-2)(x+2)}$.
2. Cancel $(x+2)$ and $(x-3)$: the result is $\dfrac{x+3}{x-2}$, with $x \neq 3,\ -3,\ 2,\ -2$.
3. For the division, multiply by the reciprocal: $\dfrac{(x-1)(x+1)}{x(x+2)} \times \dfrac{x}{x-1}$.
4. Cancel x and $(x-1)$: $\dfrac{x+1}{x+2}$, with $x \neq 0,\ -2,\ 1$.

Check the division numerically: at $x = 2$, the original is $\frac{3}{8} \div \frac{1}{2} = \frac{3}{4}$; the simplified form is $\frac{3}{4}$.

EXAMPLE 2.13 ADDING AND SUBTRACTING ALGEBRAIC FRACTIONS

QUESTION

Write each as a single fraction: $\dfrac{3}{x+1} + \dfrac{2}{x-2}$, and $\dfrac{5}{x-3} - \dfrac{2}{x+1}$.

SOLUTION

1. The lowest common denominator is $(x+1)(x-2)$. Rewrite each fraction: $\dfrac{3(x-2)+2(x+1)}{(x+1)(x-2)}$.
2. Expand the numerator: $3x - 6 + 2x + 2 = 5x - 4$. The result is $\dfrac{5x-4}{(x+1)(x-2)}$.
3. For the subtraction, use $(x-3)(x+1)$: $\dfrac{5(x+1)-2(x-3)}{(x-3)(x+1)}$.
4. Expand carefully: $5x + 5 - 2x + 6 = 3x + 11$. The result is $\dfrac{3x+11}{(x-3)(x+1)}$.

> **Sign warning:** in the subtraction, the minus sign applies to the whole numerator $2(x-3)$, so $-2(x-3) = -2x + 6$. Writing $-2x - 6$ is a common error.

COMMON MISCONCEPTIONS

"$\dfrac{x+3}{x+5} = \dfrac{3}{5}$." The x is a term, not a factor. Cancel only factors of the whole numerator and denominator.

"$\dfrac{x^2}{x} = x^2$." $\dfrac{x^2}{x} = x$ for $x \neq 0$, because $x^2 = x \cdot x$.

"The simplified fraction has no restrictions." Restrictions come from the original denominator. They do not disappear when a factor cancels.

GUIDED PRACTICE · FACTORISE BEFORE YOU CANCEL

Simplify each fraction and state one excluded value.

$$\frac{x^2-4}{x^2+3x+2} = \frac{(____)(____)}{(____)(____)} = _____,\ x \neq ____$$

$$\frac{2}{x}+\frac{3}{x+1} = \frac{_____}{_____}$$

$$\frac{x^2-1}{x} \div \frac{x+1}{x^2} = \frac{(____)(____)}{x} \times \frac{x^2}{____} = _____$$

EXERCISE 2E

ALGEBRAIC FRACTIONS · DOMAINS · OPERATIONS

FLUENCY

1 Simplify.

a. $\frac{6x^2}{3x}$

b. $\frac{8a^3b}{12ab^2}$

c. $\frac{x^2-4}{x+2}$

d. $\frac{x^2-9}{x-3}$

2 Simplify and state the domain.

a. $\frac{x^2+5x+6}{x^2+4x+3}$

b. $\frac{2x^2-8}{x^2+x-6}$

c. $\frac{x^2-2x}{x^2-4x+4}$

d. $\frac{x^2-1}{x^2+2x+1}$

3 Multiply.

a. $\frac{x+1}{x-2} \times \frac{x-2}{x+3}$

b. $\frac{x^2-1}{x^2-4} \times \frac{x+2}{x-1}$

4 Divide.

a. $\frac{x^2-9}{x+1} \div \frac{x-3}{x+1}$

b. $\frac{x^2-4x}{x+1} \div \frac{x-4}{x+1}$

5 Write as a single fraction.

a. $\frac{1}{x}+\frac{1}{y}$

b. $\frac{2}{x+1}+\frac{3}{x-1}$

c. $\frac{4}{x-2}-\frac{1}{x+3}$

UNDERSTANDING AND REASONING

6 Simplify $\frac{1}{x-1}+\frac{1}{x+1}$ as a single fraction, then evaluate it at $x=3$. Check by evaluating the original expression.

7 **Error analysis.** A student writes $\frac{x^2-16}{x-4} = x-4$. Identify the error and write the correct simplification, including the domain.

8 Explain why $\frac{x^2-1}{x-1}$ is undefined at $x=1$, even though the simplified form $x+1$ is defined there.

9 Two resistors R_1 and R_2 in parallel satisfy $\frac{1}{R}=\frac{1}{R_1}+\frac{1}{R_2}$. Write R as a single fraction. Evaluate R when $R_1=6$ and $R_2=3$.

APPLICATION AND EXTENSION

10 Simplify $\frac{x+1}{x^2-x-6}+\frac{2}{x-3}$, stating the domain.

11 Show that $\frac{\frac{1}{a}+\frac{1}{b}}{\frac{1}{a}-\frac{1}{b}} = \frac{a+b}{b-a}$ for $a \neq 0, b \neq 0, a \neq b$.

12 A rectangle has length $\frac{x+2}{x-1}$ and width $\frac{x-1}{x+3}$. Write its area as a simplified fraction

EXERCISE 2E · CONTINUED ALGEBRAIC FRACTIONS · DOMAINS · OPERATIONS

13 **Extension.** Simplify $\dfrac{x^3-1}{x^2-1}$ by factorising both numerator and denominator completely. State the domain and explain why one factor cancels.

CHANGING THE SUBJECT

2.6 Rearranging formulae

A formula is an equation that describes a relationship. The **subject** is the variable isolated on one side. Rearranging means performing the same inverse operation on both sides until the required variable stands alone. Every step has a reason: undo addition with subtraction, undo multiplication with division, undo a square with a square root.

When the subject appears twice, first collect all terms containing it on one side, then factor it out. When the subject is inside a square root, isolate the root and square both sides, checking whether the context requires the positive or negative root.

DEFINITION

Subject of a formula

The **subject** of a formula is the variable that is expressed in terms of all the others. Changing the subject means rewriting the formula so that a different variable is isolated. The two formulae are equivalent wherever both are defined.

EXAMPLE **2.14** USING INVERSE OPERATIONS

QUESTION

Make x the subject of $y = 3x - 7$ and of $5x + 2y = 10$.

SOLUTION

1 For $y = 3x - 7$, add 7 to both sides: $y + 7 = 3x$.

2 Divide both sides by 3: $x = \dfrac{y+7}{3}$.

3 For $5x + 2y = 10$, subtract $2y$: $5x = 10 - 2y$.

4 Divide by 5: $x = \dfrac{10-2y}{5}$, or $x = 2 - \dfrac{2y}{5}$.

Check: substitute $y = 1$ into the first formula: $x = \frac{8}{3}$. The original gives $3 \cdot \frac{8}{3} - 7 = 1$.

EXAMPLE **2.15** FORMULAE FROM PHYSICS

QUESTION

Make u the subject of $s = ut + \frac{1}{2}at^2$, and make h the subject of $A = 2\pi r^2 + 2\pi rh$.

SOLUTION

1 Subtract $\frac{1}{2}at^2$: $s - \frac{1}{2}at^2 = ut$.

2 Divide by t, assuming $t \neq 0$: $u = \dfrac{s - \frac{1}{2}at^2}{t} = \dfrac{2s - at^2}{2t}$.

3 For the surface area, subtract $2\pi r^2$: $A - 2\pi r^2 = 2\pi rh$.

4 Divide by $2\pi r$, assuming $r \neq 0$: $h = \dfrac{A - 2\pi r^2}{2\pi r}$.

> **Units:** in the first formula u has units of velocity. Dividing a length by a time gives m/s, which confirms the rearrangement is dimensionally consistent.

EXAMPLE **2.16** WHEN THE SUBJECT APPEARS TWICE

QUESTION

Make x the subject of $y = \dfrac{2x + 1}{x - 3}$.

SOLUTION

1 Multiply both sides by $x - 3$: $y(x - 3) = 2x + 1$.

2 Expand the left side: $yx - 3y = 2x + 1$.

3 Collect x-terms on one side and the rest on the other: $yx - 2x = 1 + 3y$.

4 Factor out x: $x(y - 2) = 1 + 3y$.

5 Divide by $y - 2$, assuming $y \neq 2$: $x = \dfrac{1 + 3y}{y - 2}$.

Check: let $y = 3$. Then $x = \frac{10}{1} = 10$. The original gives $\frac{2(10)+1}{10-3} = \frac{21}{7} = 3$.

COMMON MISCONCEPTIONS

"Move the term to the other side and keep its sign." Rearranging is not moving; it is applying the same operation to both sides. Subtracting $2y$ is the reason the sign changes.

"Squaring both sides always preserves the solution." Squaring can introduce solutions, and taking a square root loses the sign. State whether the context requires a positive quantity.

"$y = (2x + 1)/(x - 3)$ gives $x = (2y + 1)/(y - 3)$ by swapping." Swapping letters does not change the subject correctly. Solve the equation, collecting x-terms and factorising.

GUIDED PRACTICE · ONE INVERSE OPERATION AT A TIME

Complete the rearrangement and name the operation used at each step.

$v = u + at \Rightarrow u =$ _____ (operation: _____)

$A = \frac{1}{2}bh \Rightarrow h =$ _____ (operation: _____)

$y = \frac{x+1}{x-2} \Rightarrow y(x-2) =$ _____ $\Rightarrow x =$ _____

$T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T^2 =$ _____ $\Rightarrow l =$ _____

EXERCISE 2F

REARRANGING · INVERSE OPERATIONS · SUBJECT TWICE

FLUENCY

1 Make x the subject.

a. $y = 5x + 2$

b. $y = 7 - 2x$

c. $3x + 2y = 12$

d. $4x - 3y = 15$

2 Make y the subject.

a. $2x + y = 9$

b. $5x - 2y = 8$

c. $\frac{y}{5} + 2x = 1$

d. $3(y + 2) = x$

3 Make u the subject of $s = ut + \frac{1}{2}at^2$.

4 Make a the subject of $v = u + at$.

5 Make h the subject.

a. $A = \frac{1}{2}(a + b)h$

b. $V = \pi r^2 h$

6 Make r the subject.

a. $C = 2\pi r$

b. $A = \pi r^2, r > 0$

UNDERSTANDING AND REASONING

7 Make l the subject of $T = 2\pi\sqrt{\frac{l}{g}}$.

8 Make x the subject of $y = \dfrac{x+1}{x-2}$.

9 Make x the subject of $y = \dfrac{3x-2}{x+1}$.

10 **Error analysis.** A student writes $y = 3x - 5 \Rightarrow x = \frac{y}{3} - 5$. Explain the error and write the correct rearrangement.

APPLICATION AND EXTENSION

11 Make b the subject of $a = \sqrt{b^2 - c^2}$, stating any condition on the sign of b.

12 The stopping distance of a car is modelled by $s = ut + \frac{1}{2}at^2$. Make a the subject and explain what the formula tells you when $s = 0$.

13 Make x the subject of $\dfrac{a}{x} + \dfrac{b}{y} = 1$.

14 **Extension.** Make x the subject of $y = \sqrt{\dfrac{x+1}{x-1}}$. State the domain of the rearranged formula and explain why $y \neq 1$.

MODELLING · CRITERION D

Return to the open box

The designer's square sheet is 24 cm by 24 cm, and the corner squares have side x. The volume is $V = x(24 - 2x)^2$.

1. **Expand.** Show that $V = 4x^3 - 96x^2 + 576x$.
2. **Restrict.** Explain why $0 < x < 12$. What happens physically at $x = 0$ and at $x = 12$?
3. **Tabulate.** Calculate V for $x = 1, 2, 3, \ldots, 8$. Record the results to 3 significant figures.
4. **Interpret.** Which value in your table gives the largest volume? Does the graph appear to peak between two table values? Estimate the best value of x.
5. **Evaluate.** State two assumptions in the model. How would the thickness of the card, or the need for a lid, change the problem?
6. **Communicate.** Write a recommendation to the designer in no more than 120 words, using both the expanded and factorised forms to justify your answer.

Evidence to submit: the expansion, the domain argument, a table, a sketch of the graph, and a short recommendation that distinguishes the mathematical model from the physical box.

MIXED PRACTICE 2

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–10 check fluency; 11–16 require reasoning; 17–20 are multi-step or unfamiliar. Allow 55 minutes.

1. Expand $(2x - 3)(x + 4)$.
2. Factorise $6x^2 - 15x$.
3. Factorise $x^2 - 8x + 15$.
4. Factorise $2x^2 - 7x - 4$.
5. Expand $(3x - 2)^2$.
6. Factorise $16x^2 - 25$.
7. Simplify $\dfrac{x^2 - 16}{x^2 - 3x - 4}$.
8. Write $\dfrac{3}{x + 2} + \dfrac{2}{x - 1}$ as a single fraction.
9. Make x the subject of $y = \dfrac{4x - 1}{x + 3}$.
10. Show that $(a - b)^2 = a^2 - 2ab + b^2$.
11. **Error analysis.** A student writes $3(x + 2) = 3x + 2$. Explain the error using the distributive law.
12. Prove that $(x + a)(x + b) = x^2 + (a + b)x + ab$.
13. Factorise fully $x^3 - 4x$.
14. A rectangle has area $x^2 + 7x + 10$ and width $x + 2$. Find its length and perimeter in terms of x.
15. Explain why $x^2 + 6x + 9 = (x + 3)^2$ is an identity, while $x^2 + 6x + 9 = 0$ is an equation.
16. Make t the subject of $s = \frac{1}{2}(u + v)t$.

MIXED PRACTICE 2 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

17 Simplify $\dfrac{x^2+2x+1}{x^2-1}$.

18 Factorise fully $3x^2+12x+12$.

19 **Extension.** Factorise x^4-5x^2+4 completely.

20 **Extension.** Show that $\left(x+\dfrac{1}{x}\right)^2 = x^2+2+\dfrac{1}{x^2}$ for $x \neq 0$.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING

30 minutes · 30 marks

Expressions and factorisation: short response

NO CALCULATOR | SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. Where a question says "fully", a partially factorised answer earns partial credit only.

21 Expand and simplify $4x(2x-3)-(x-2)(x+5)$. [5]

22 Factorise fully $12a^3b^2-18a^2b$. [3]

23 Factorise x^2-x-30. [3]

24 Factorise $3x^2-5x-2$. [3]

25 Expand $(2x-3)^2$ and $(2x-3)(2x+3)$. [4]

26 Simplify $\dfrac{x^2-4x}{x^2-x-12}$ and state the domain. [4]

27 Make x the subject of $y=\dfrac{3x-1}{x+2}$. [4]

28 Prove that $(a+b)^2-(a-b)^2=4ab$. [4]

CRITERION B · INVESTIGATING PATTERNS

45 minutes · 24 marks

Differences of consecutive squares

Instructions. Work independently. Record results in a table, then answer in full sentences. You may use a calculator for arithmetic but not for the generalisation.

1. Complete a table of $(n+1)^2 - n^2$ for $n = 1, 2, \ldots, 8$. [4]
2. Describe the sequence of differences. What kind of numbers appear? [3]
3. Write a formula for the difference in terms of n, and verify it for three values. [5]
4. Prove your formula by expanding $(n+1)^2 - n^2$. Explain why the result must be odd. [6]
5. Generalise: expand $(n+k)^2 - n^2$ and describe the result. Use your generalisation to calculate $101^2 - 99^2$. [6]

Evidence: a table, a description, a formula with verification, an algebraic proof, and a generalisation with one application.

CRITERION C · COMMUNICATING

40 minutes · 20 marks

Why factorising is expanding in reverse

Instructions. Write a guide of 300–400 words for a student in the year below. Your guide must explain: (i) what it means for two expressions to be equivalent; (ii) how an area model shows both expanding and factorising; (iii) two worked examples, one with a common factor and one quadratic; (iv) one common error and how to avoid it. Use correct notation and at least one labelled diagram.

Evidence: a titled guide with paragraphs, a labelled area diagram, two worked examples, and a short “check it yourself” box.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Packaging from a square sheet

Instructions. A factory cuts squares of side x centimetres from the corners of a square sheet of card with side 24 cm, then folds the sides to make an open box.

1. Write the length, width and height of the box, and hence write its volume V in factorised form. State the domain. [4]
2. Expand the volume expression. [3]
3. Calculate V for $x = 1, 2, \ldots, 8$, and identify the maximum in your table. [5]
4. Write $V = 4x(12 - x)^2$ and explain why this form shows the domain and the zeros more clearly than the expanded form. [4]
5. State two assumptions in the model and explain how each could affect the recommended value of x. [4]
6. Recommend a value of x to the factory, justify it using your evidence, and suggest one practical check before production. [4]

Evidence: a labelled diagram, the factorised and expanded forms, a table, a stated domain, assumptions, and a justified recommendation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- Expanding uses the distributive law: every term in one bracket multiplies every term in the other. Combine like terms only after all products are written.
- Factorising is expanding in reverse. Take out the highest common factor first; a factorisation is complete when no further integer factor remains.
- Grouping works when the terms can be arranged into pairs that share a bracket.
- For $x^2 + bx + c$, find two numbers with product c and sum b. For $ax^2 + bx + c$, use the ac method to split the middle term.
- The special products are $(a \pm b)^2 = a^2 \pm 2ab + b^2$ and $(a + b)(a - b) = a^2 - b^2$. An identity holds for every value of the variable.
- Cancel only factors of the whole numerator and denominator. State the domain restrictions from the original denominator.
- To change the subject, apply inverse operations to both sides. If the subject appears twice, collect and factorise it.

$$(a+b)^2 = a^2 + 2ab + b^2, \quad (a-b)^2 = a^2 - 2ab + b^2, \quad (a+b)(a-b) = a^2 - b^2$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

I can expand a product of two brackets without losing a term.

I can expand and simplify an expression with three or more terms.

I can find and take out the highest common factor.

I can factorise by grouping.

I can factorise a monic quadratic using product and sum.

I can factorise a quadratic with a leading coefficient other than 1.

I can recognise and use the three special products.

I can prove an identity by expanding both sides.

I can simplify an algebraic fraction by factorising.

I can add and subtract algebraic fractions.

I can state the domain of an algebraic fraction.

I can change the subject of a formula using inverse operations.

I can rearrange a formula in which the subject appears twice.

I can rearrange a formula involving a square root and state any sign condition.

I can choose between expanded and factorised forms to answer a question.

I can check a factorisation by expanding.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which form — expanded, factorised or a single fraction — did you find most useful in the modelling task, and why?
2. Describe one error you made in this chapter. Was it a reasoning error or a recording error? What check will you use next time?
3. The statement of inquiry says that form reveals structure. Give one example where changing the form made a problem easier and explain what became visible.
4. Look ahead to Chapter 5, where quadratic equations are solved. Why will factorising be useful there?

BALANCE, CONSTRAINT AND THE MEETING OF TWO CONDITIONS

Linear Equations

03

BALANCE · ORDER · SIMULTANEITY

Inequalities and systems of equations — using algebra to describe not one value, but a range of possible values and the point where two relationships agree.

STATEMENT OF INQUIRY

Equations and inequalities are statements of balance and constraint. Comparing two relationships reveals the exact point at which they agree, the range of values that remain possible, and the conditions under which no agreement exists.

KEY CONCEPT

Relationships

RELATED CONCEPTS

Equivalence · Change · System

GLOBAL CONTEXT

Fairness and development — collective decisions under constraints

FACTUAL

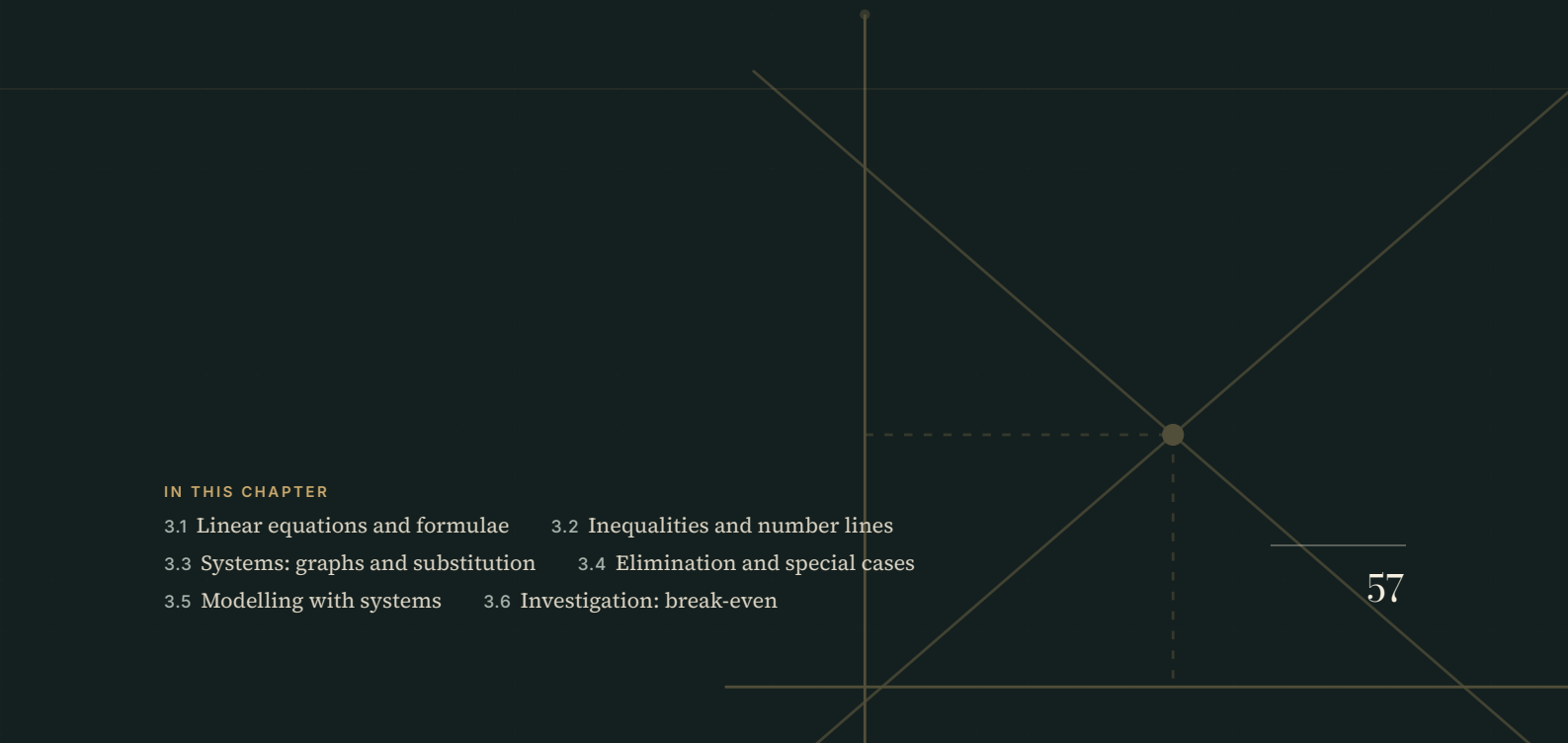
Which operations preserve the solutions of an equation or inequality?

CONCEPTUAL

What does the solution of a system represent, and why can a system have no solution?

DEBATABLE

Should a group always choose the cheapest option, or the fairest one?



IN THIS CHAPTER

3.1 Linear equations and formulae
3.2 Inequalities and number lines
3.3 Systems: graphs and substitution
3.4 Elimination and special cases
3.5 Modelling with systems
3.6 Investigation: break-even

OPENING PROBLEM · TWO QUOTES, ONE DECISION

The trip that had to be affordable

A school is hiring a coach for a one-day trip. Company A charges a booking fee of £180 plus £12 for every student. Company B charges a booking fee of £420 plus £8 for every student. The organising committee has a budget of £1000, and the coach can carry at most 72 students.

1. Write an expression for the cost from each company when n students travel.
2. For what number of students do the two companies charge the same amount? Solve an equation.
3. Which company is cheaper for 40 students? For 80 students? Explain without drawing a graph.
4. Using the £1000 budget, write an inequality for each company and find the greatest number of students each can carry.
5. Make a recommendation. What does the mathematics decide, and what must the committee decide?

An equation asserts that two quantities are equal at some unknown value. An inequality asserts an order over a range of values. A system asserts that several relationships hold at once. This chapter builds the language and the habits needed to move between these three statements without losing information.

ORIENTATION

3.0 Learning intentions

- Solve linear equations, including equations with brackets and fractions, and justify each step.
- Solve linear inequalities, represent the solution on a number line, and interpret it in context.
- Solve systems of two linear equations by graphing, substitution and elimination.
- Recognise systems with one solution, no solution or infinitely many solutions.
- Form equations, inequalities and systems from real situations, and evaluate the resulting model.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

solve a multi-step linear equation and check the solution; solve an inequality and explain why the direction reverses when multiplying or dividing by a negative number; solve a system by two different methods and verify the ordered pair; classify a system from its equations or graph; build and interpret a model involving a budget, a capacity or a break-even point; and explain the assumptions that limit a recommendation.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 3.1.

1. Solve $3x + 7 = 22$.
2. Expand $2(x - 5)$ and simplify $3(x + 2) - (x - 4)$.
3. Solve $\frac{x}{4} = 3$.
4. Plot the points $(0, -1)$ and $(3, 5)$ on the same axes.
5. Solve $5 - 2x = 1$.
6. Find the value of $4x - 3$ when $x = -2$.

REFRESHER

An equation is a balance. Whatever you do to one side, you must do to the other. Inverse operations undo each other: addition undoes subtraction, multiplication undoes division. When an expression contains brackets, expand first; when it contains fractions, multiply every term by a common denominator. A negative number multiplied by a negative number is positive.

ATL SKILLS · THINKING AND TRANSFER

Critical thinking: decide which method — graph, substitution or elimination — exposes the answer most efficiently. **Transfer:** recognise the same balance structure in a formula, an inequality and a pair of simultaneous equations. **Communication:** define variables and units before writing equations, and interpret every solution in the language of the problem, not only as a number.

PRESERVING BALANCE

3.1 Linear equations and formulae

An equation is a question disguised as a statement. The statement $4(2x - 3) = 3(x + 5) - 2$ claims that two expressions are equal for some value of x. Solving means replacing the equation, step by step, with simpler equations that have exactly the same solutions. Each step must preserve the truth of the equality for every value that made the previous line true.

Three moves are always safe: add or subtract the same expression on both sides; multiply or divide both sides by the same *non-zero* number; and replace one side by an equivalent expression. These moves are the algebra of a balance. They explain why the familiar instruction "move it to the other side and change the sign" works — it is really a subtraction performed on both sides.

DEFINITIONS

Equation, solution and equivalence

A **linear equation** in one variable can be written in the form $ax + b = 0$, with $a \neq 0$. A **solution** is a value of the variable that makes the equation true; the set of all such values is the solution set. Two equations are **equivalent** when they have exactly the same solution set.

EXAMPLE **3.1** AN EQUATION WITH BRACKETS ON BOTH SIDES

QUESTION

Solve $4(2x - 3) = 3(x + 5) - 2$.

SOLUTION

1. Expand both sides. Left: $8x - 12$. Right: $3x + 15 - 2 = 3x + 13$. The equation becomes $8x - 12 = 3x + 13$.
2. Subtract $3x$ from both sides: $5x - 12 = 13$. This collects the unknown on one side.
3. Add 12 to both sides: $5x = 25$.
4. Divide both sides by 5: $x = 5$.

Check: left side $4(10 - 3) = 28$; right side $3(10) - 2 = 28$. Both sides agree, so $x = 5$ is a solution.

EXAMPLE **3.2** AN EQUATION WITH ALGEBRAIC FRACTIONS

QUESTION

Solve $\dfrac{x + 1}{3} - \dfrac{2x - 5}{4} = 1$.

SOLUTION

1. The denominators are 3 and 4, so the lowest common multiple is 12. Multiply *every* term by 12.
2. $$12 \cdot \frac{x + 1}{3} - 12 \cdot \frac{2x - 5}{4} = 12 \cdot 1$$
3. This gives $4(x + 1) - 3(2x - 5) = 12$. The fraction line acts as a bracket, so both terms of $2x - 5$ are multiplied by 3.
4. Expand: $4x + 4 - 6x + 15 = 12$. Combine: $-2x + 19 = 12$.
5. Subtract 19: $-2x = -7$. Divide by -2: $x = \dfrac{7}{2} = 3.5$.

Check: $\frac{4.5}{3} = 1.5$ and $\frac{2}{4} = 0.5$, so the left side is $1.5 - 0.5 = 1$. Correct.

EXAMPLE **3.3** FORMING AN EQUATION FROM A DESCRIPTION

QUESTION

A rectangle is 3 cm longer than twice its width. Its perimeter is 36 cm. Find its dimensions.

SOLUTION

1. **Define:** let the width be w centimetres. Then the length is $2w + 3$ centimetres.
2. **Form:** the perimeter is $2(w + 2w + 3) = 36$. The units are consistent, so no conversion is needed.
3. **Solve:** $2(3w + 3) = 36$, so $6w + 6 = 36$, $6w = 30$, $w = 5$.
4. **Interpret:** the width is 5 cm and the length is $2(5) + 3 = 13$ cm.

Check: the perimeter is $2(5 + 13) = 36$ cm, and 13 is 3 more than twice 5. The dimensions satisfy both conditions.

EXAMPLE **3.4** REARRANGING A FORMULA

QUESTION

(a) Make F the subject of $C = \frac{5}{9}(F - 32)$. (b) Make u the subject of $s = ut + \frac{1}{2}at^2$.

SOLUTION

a. Multiply by 9: $9C = 5(F - 32)$. Divide by 5: $\frac{9C}{5} = F - 32$. Add 32: $F = \frac{9C}{5} + 32$.

b. The variable u appears in the term ut. Subtract $\frac{1}{2}at^2$ from both sides: $s - \frac{1}{2}at^2 = ut$.

c. Divide by t, provided $t \neq 0$: $u = \frac{s - \frac{1}{2}at^2}{t} = \frac{s}{t} - \frac{at}{2}$.

> **Domain note:** the rearrangement for u is valid only when $t \neq 0$. A formula and its rearrangement are equivalent only where both are defined.

KEY IDEA

Solving is a chain of equivalence-preserving moves. Expanding, collecting like terms, adding the same quantity to both sides, and multiplying or dividing by a non-zero number all preserve the solution set. A solution is not finished until it has been checked in the original equation and interpreted in the context.

COMMON MISCONCEPTIONS

"Divide every term by 2" when only one side is divisible. Dividing one side changes the equation. Both sides must be divided, and every term within a side is affected.

"The fraction line is only a division sign." It is also a bracket. In $\frac{2x-5}{4}$, the whole expression $2x-5$ is divided by 4, so multiplying by 4 gives $2x-5$, not $2x-5/4$.

"A solution is correct because it came from algebra." Every step can be valid and an arithmetic slip can still occur. Substitution into the original equation is the cheapest check available.

"x can be cancelled from both sides." Cancelling divides by x, which is invalid when $x=0$ and changes the solution set. Solve by factorising instead.

GUIDED PRACTICE · COMPLETE THE SOLUTION

Fill in each blank, then check by substitution.

$3(x-2)=5x+8$

$3x-____=5x+8$

$-2x=____$

$x=____$

$\frac{2x-1}{5}+1=\frac{x+3}{2}$

Multiply every term by ____:

$2(2x-1)+____=5(x+3)$

$4x-2+____=5x+15$

$x=____$

EXERCISE 3A

LINEAR EQUATIONS · FORMULAE · FORMING EQUATIONS

FLUENCY

1 Solve.

a. $5x-7=23$

b. $9-2x=1$

c. $4x+3=2x+11$

d. $7x-5=3x+19$

2 Solve.

a. $3(x+4)=27$

b. $5(2x-1)=35$

c. $2(x-3)=x+7$

d. $4(2x+1)=3(x+8)$

3 Solve.

a. $\frac{x}{3}+\frac{x}{4}=7$

b. $\frac{x+2}{5}=3$

c. $\frac{2x-1}{3}=x-2$

d. $\frac{x}{2}-\frac{x-1}{4}=2$

4 Solve $\frac{2x+1}{3}-\frac{x-4}{2}=1$.

5 Make x the subject.

a. $y=3x-7$

b. $y=\frac{x}{4}+2$

c. $3x+2y=12$

d. $y=\frac{2x+1}{5}$

EXERCISE 3A · CONTINUED LINEAR EQUATIONS · FORMULAE · FORMING EQUATIONS

UNDERSTANDING AND REASONING

6 Explain why adding the same expression to both sides of an equation cannot change its solution set.

7 **Error analysis.** A student solves $\frac{x+3}{2} = 5$ by writing $x + 3 = 5$ and obtaining $x = 2$. Identify the error and give the correct solution.

8 Write an equation with brackets whose solution is $x = 4$. Solve it to show that your equation works.

9 Classify each equation as having one solution, no solution, or every real number as a solution. Justify each answer.

a. $3(x + 2) = 3x + 6$

b. $2x + 5 = 2x - 1$

c. $4(x - 1) = 2(2x - 3) + 2$

APPLICATION AND EXTENSION

10 A rectangle is 4 cm longer than its width and has perimeter 44 cm. Find its dimensions.

11 The sum of three consecutive odd integers is 81. Form and solve an equation to find the integers.

12 The formula $v = u + at$ relates speed, initial speed, acceleration and time. Make a the subject, then find a when $v = 28$, $u = 4$ and $t = 6$. Include units if v and u are in m/s and t is in seconds.

13 **Extension.** Solve $\dfrac{3}{x - 2} = 4$ and state the value of x for which the equation is undefined. Explain why that value cannot be a solution.

FROM ONE VALUE TO A RANGE

3.2 Inequalities and number lines

An inequality compares two expressions instead of asserting equality. “The coach can carry at most 72 students” is not a single number; it is a condition that every value up to 72 satisfies. The solution of an inequality is therefore usually a range, drawn on a number line and written in interval notation.

Most of the balance rules carry over unchanged. Adding or subtracting the same quantity preserves the order: if $a < b$, then $a + c < b + c$. Multiplying or dividing by a positive number also preserves it. Multiplying or dividing by a negative number, however, reverses the order, because the number line is reflected about zero.

DEFINITIONS AND NOTATION

Inequality, solution set and interval notation

An **inequality** is a statement comparing two expressions with $<$, $\leq$, $>$ or $\geq$. Its solution set can be written as an inequality or as an interval: $x > -2$ is $(-2, \infty)$; $x \leq 3$ is $(-\infty, 3]$; $-1 \leq x < 3$ is $[-1, 3)$. A square bracket includes the endpoint, a round bracket excludes it. A **compound inequality** combines two conditions, as in $-1 \leq x < 3$.

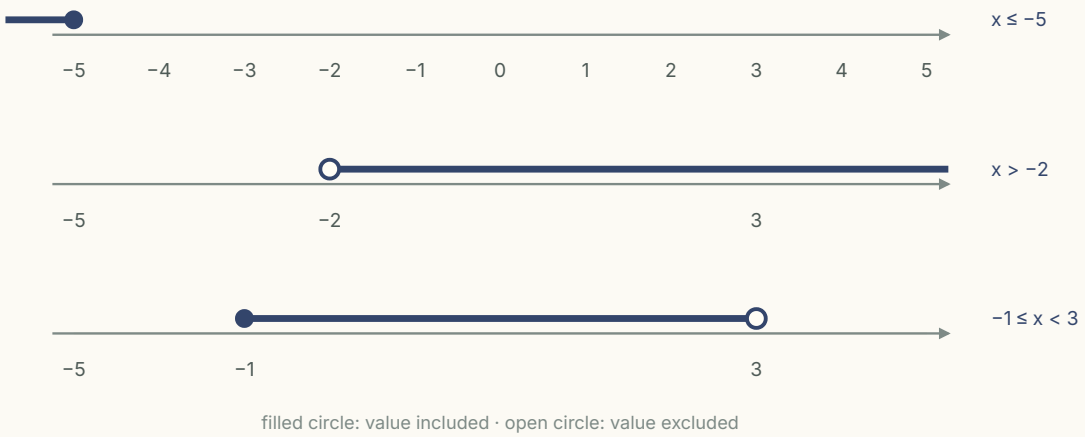


Figure 3.1 Three solution sets on the number line. A filled circle includes the endpoint; an open circle excludes it. The arrow shows that the set continues without end in that direction.

EXAMPLE **3.5** INEQUALITY WITH A NEGATIVE COEFFICIENT

QUESTION

Solve $4 - 3x \geq 19$ and represent the solution on a number line.

SOLUTION

1. Subtract 4 from both sides: $-3x \geq 15$.
2. Divide both sides by -3. Because -3 is negative, the inequality reverses: $x \leq -5$.
3. The solution is $(-\infty, -5]$. On the number line, use a filled circle at -5 and shade to the left.

Check: choose $x = -6$: $4 - 3(-6) = 22 \geq 19$. Choose $x = -4$: $4 - 3(-4) = 16$, which is not ≥ 19. The direction is correct.

EXAMPLE **3.6** INEQUALITY WITH BRACKETS ON BOTH SIDES

QUESTION

Solve $2(x - 3) < 5x + 6$.

SOLUTION

1. Expand: $2x - 6 < 5x + 6$.
2. Subtract $2x$: $-6 < 3x + 6$. Subtract 6: $-12 < 3x$.
3. Divide by 3: $-4 < x$, usually written $x > -4$. The solution is $(-4, \infty)$.

> **Reading the result:** $-4 < x$ and $x > -4$ name the same set. Write the variable on the left when you interpret the answer, because the number line is read from left to right.

EXAMPLE **3.7** A COMPOUND INEQUALITY

QUESTION

Solve $5 \leq 3x - 1 < 11$, list the integer solutions, and write the answer in interval notation.

SOLUTION

1. Apply the same operations to all three parts. Add 1: $6 \leq 3x < 12$.
2. Divide by 3: $2 \leq x < 4$. The interval is $[2, 4)$.
3. The integers satisfying this condition are 2 and 3. The value 4 is excluded.

Check: $x = 2$ gives $5 \leq 5 < 11$, true. $x = 3$ gives $5 \leq 8 < 11$, true. $x = 4$ gives $5 \leq 11 < 11$, false because the second inequality is strict.

EXAMPLE **3.8** INEQUALITY FROM A CAPACITY CONSTRAINT

QUESTION

A lift can carry at most 420 kg. It already carries equipment of mass 180 kg. Boxes of mass 24 kg are loaded. What is the greatest number of boxes that can be carried?

SOLUTION

1. Let b be the number of boxes. The total mass is $180 + 24b$, and it must not exceed 420: $180 + 24b \leq 420$.
2. Subtract 180: $24b \leq 240$. Divide by 24: $b \leq 10$.
3. The model allows any b up to 10. Since b must be a whole number, the greatest number of boxes is 10.

Interpretation: the inequality has infinitely many real solutions, but the context restricts the solution set to whole numbers. Always apply the context after solving.

KEY IDEA

Inequalities obey the same balance rules as equations except for one: multiplying or dividing by a negative number reverses the order. The solution is a set, so it must be represented — on a number line, in interval notation, or in words — and then restricted to the values the context allows.

COMMON MISCONCEPTIONS

"Multiply by a negative and keep the sign." Test $-2x < 6$. Dividing by -2 gives $x > -3$, not $x < -3$. Substitute $x = 0$: $-2(0) < 6$ is true, and 0 is greater than -3.

"A compound inequality is two separate answers." It is one set: the values satisfying both conditions simultaneously. The solution to $2 \leq x < 4$ is the overlap of $x \geq 2$ and $x < 4$.

"The inequality symbol always points to the smaller number." It points to the smaller number, but $x > -4$ is not the same as $-4 > x$. Read the symbol as a sentence: "x is greater than -4."

"Divide by x to simplify." The sign of x is unknown, so dividing by x would require two cases and often loses solutions. Collect terms instead.

GUIDED PRACTICE · SOLVE AND REPRESENT

Complete each step and draw the solution set.

$7 - 2x \geq 1$

$-2x \geq ____$

$x _____$ (reverse the symbol because _____)

$-3 < 4x + 5 \leq 13$

$____ < 4x \leq ____$

$____ < x \leq ____$

EXERCISE 3B

INEQUALITIES · NUMBER LINES · CONSTRAINTS

FLUENCY

1 Solve and write the solution in interval notation.

a. $x + 5 > 2$

b. $3x \leq 18$

c. $2x - 1 \geq 9$

d. $10 - x < 4$

2 Solve.

a. $-2x > 10$

b. $-5x \leq 20$

c. $4 - 3x < -8$

d. $7 - 2x \geq 1$

3 Solve.

a. $3(x + 1) \leq 18$

b. $2(x - 4) > x + 1$

c. $5(2x - 1) < 3(x + 6)$

d. $4(x - 2) \geq 6x - 3$

4 Solve each compound inequality and list the integer solutions.

a. $-2 \leq x + 1 < 4$

b. $3 < 2x - 1 \leq 9$

c. $-6 \leq 1 - 3x \leq 4$

UNDERSTANDING AND REASONING

5 Explain, using a numerical example, why multiplying both sides of an inequality by -1 reverses the symbol.

6 **Error analysis.** A student writes $-4x < 12 \Rightarrow x < -3$. Explain the error, write the correct answer, and check it with a value that satisfies the correct set.

7 For what values of x is $3x - 5$ at least 7? Write the answer as an inequality and on a number line.

8 Is $x = -4$ a solution of $2x + 9 \leq 1$? Justify by substitution, not by solving.

APPLICATION AND EXTENSION

9 A delivery van may carry at most 900 kg. It already carries 240 kg of equipment and crates of 55 kg. Find the greatest number of crates it can carry.

10 A phone battery has 42% charge and loses 3% per hour of video. For how many whole hours can video be watched while keeping at least 12% charge? Form and solve an inequality.

11 **Extension.** The solution of $ax + 3 \leq 7$ is $x \geq -2$. Find the value of a, showing your reasoning. Explain what happens if $a = 0$.

TWO CONDITIONS AT ONCE

3.3 Systems: graphs and substitution

A single linear equation in two variables has infinitely many solutions: every point on its line. A second equation adds a second condition. The pair of values that satisfies both is the point where the two lines meet. This is why a system of two linear equations usually has exactly one solution, and why the graph is not an illustration added after the algebra — it is the algebra made visible.

Substitution is the algebraic version of the same idea. If one equation already expresses one variable in terms of the other, replace that variable in the second equation. The result is a single equation in one unknown.

DEFINITION

System and simultaneous solution

A **system of linear equations** is a set of two or more linear equations in the same variables. A **simultaneous solution** is an ordered pair (x, y) that satisfies every equation in the system. Solving the system means finding every such pair.

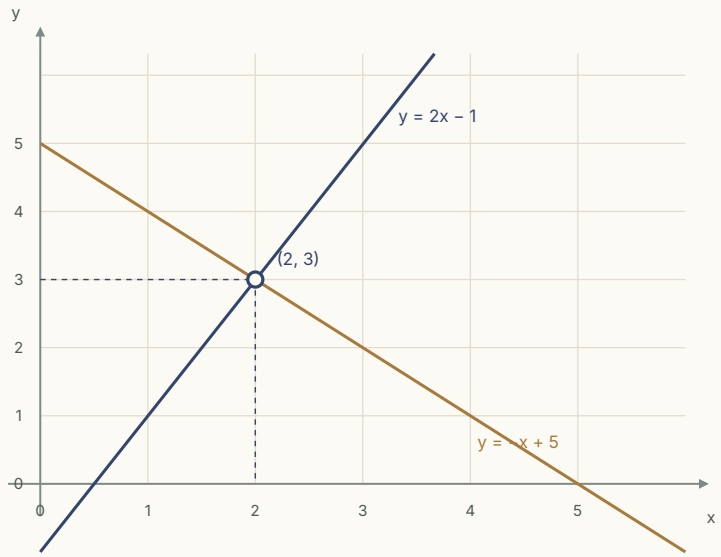


Figure 3.2 The graphs of $y = 2x - 1$ and $y = -x + 5$. The lines meet at $(2, 3)$: this pair satisfies both equations, so it is the solution of the system.

EXAMPLE **3.9** SOLVING BY SUBSTITUTION

QUESTION

Solve the system $y = x - 2$ and $3x + 2y = 16$.

SOLUTION

1. The first equation already gives y in terms of x. Substitute $x - 2$ for y in the second: $3x + 2(x - 2) = 16$.
2. Expand: $3x + 2x - 4 = 16$. Collect: $5x = 20$, so $x = 4$.
3. Substitute back into $y = x - 2$: $y = 4 - 2 = 2$.
4. The solution is $(4, 2)$.

Check both equations: $y = x - 2$ gives $2 = 4 - 2$; $3x + 2y = 3(4) + 2(2) = 16$. Both hold, so $(4, 2)$ is the simultaneous solution.

EXAMPLE **3.10** SUBSTITUTION AFTER REARRANGING

QUESTION

Solve $2x + 3y = 12$ and $x = y + 1$.

SOLUTION

1. The second equation gives x in terms of y. Substitute $y + 1$ for x: $2(y + 1) + 3y = 12$.
2. Expand and collect: $2y + 2 + 3y = 12$, so $5y = 10$ and $y = 2$.
3. Then $x = y + 1 = 3$. The solution is $(3, 2)$.

> **Choice:** substitute into the equation you did not rearrange. Substituting $x = y + 1$ into $x = y + 1$ would only return $1 = 1$ and discard the information in the other equation.

EXAMPLE **3.11** READING A SOLUTION FROM A GRAPH

QUESTION

Use Figure 3.2 to estimate the solution of $y = 2x - 1$ and $y = -x + 5$, then verify it algebraically.

SOLUTION

1. The lines meet at the point with coordinates $(2, 3)$. The graph suggests $x = 2$, $y = 3$.
2. At the intersection the two expressions for y are equal: $2x - 1 = -x + 5$.
3. Add x: $3x - 1 = 5$. Add 1: $3x = 6$, so $x = 2$. Then $y = 2(2) - 1 = 3$.
4. The exact solution is $(2, 3)$, agreeing with the graph.

> **Why graphs can mislead:** a graph read by eye gives an approximation. Algebra gives the exact value. Use the graph to understand the number of solutions and their approximate location; use algebra to establish them.

KEY IDEA

A system asks for the values that make several relationships true at once. On a graph this is an intersection point; by substitution it is the value that reduces the system to one equation in one unknown. The ordered pair must satisfy every equation, not only the one used to find it.

COMMON MISCONCEPTIONS

"Solve each equation and combine the answers." Each equation alone has infinitely many solutions. Solving them separately does not identify the pair that satisfies both.

"Substitute the expression back into the same equation." This produces a tautology such as $1 = 1$ and loses the second condition. Substitute into the other equation.

"Read the intersection as (y, x)." Coordinates are ordered pairs (x, y): the horizontal coordinate first. A point $(2, 3)$ is 2 units right and 3 units up.

"One point on one line is a solution." A solution must lie on both lines at once. Check by substituting into both equations.

GUIDED PRACTICE · SUBSTITUTION

Solve $y = 2x + 1$ and $3x + y = 16$.

Substitute: $3x + (_____) = 16$.
Collect: $____x + 1 = 16$, so $x = ____$.
Then $y = 2(___) + 1 = ____$.
The solution is $(___, ___)$. Check in *both* equations.

EXERCISE 3C

SYSTEMS · GRAPHS · SUBSTITUTION

FLUENCY

1 Solve by substitution.
   a. $y = x + 3$, $2x + y = 12$
   b. $y = 3x - 1$, $y = x + 5$
   c. $x = y - 2$, $2x + 3y = 9$
   d. $y = 2x - 3$, $4x - y = 6$

2 Solve by substitution after rearranging.
   a. $x = y + 4$, $3x + 2y = 17$
   b. $x = 2y - 1$, $x + 3y = 9$
   c. $y = 1 - 2x$, $5x - 3y = 8$

3 Use the graph of $y = x - 1$ and $y = -2x + 8$ to estimate the intersection, then verify by substitution. Show your check.

UNDERSTANDING AND REASONING

4 Explain why substituting an expression for y into the *other* equation produces a single equation in one variable.

5 **Error analysis.** A student solves $y = x + 3$ and $2x + y = 12$ by writing $2x + x + 3 = 12$, then $3x = 15$, $x = 5$, and gives the solution $(5, 8)$. Check the solution and explain the error in the final step.

6 A system has solution $(2, -1)$. Write two different linear equations that are both satisfied by this pair.

7 Two lines are parallel. How many simultaneous solutions does the system have? Explain using both the graph and the equations.

EXERCISE 3C · CONTINUED SYSTEMS · GRAPHS · SUBSTITUTION

APPLICATION AND EXTENSION

8 Two numbers have sum 27. The larger number is 3 more than twice the smaller. Write a system, solve it, and identify both numbers.

9 A rectangle has perimeter 30 cm. Its length is 3 cm more than its width. Write and solve a system to find the dimensions.

10 **Extension.** The lines $y = 2x + k$ and $y = mx + 3$ intersect at $(2, 6)$. Find k and m, and explain which equation determined each value.

REMOVING A VARIABLE

3.4 Elimination and special cases

Substitution is convenient when one variable is already isolated. When both equations are in standard form, elimination is often faster: add or subtract multiples of the equations so that one variable cancels. Adding two equations is valid because both sides are equal; adding equal quantities to equal quantities preserves equality.

If the coefficients of a variable are not opposites, multiply one or both equations by a constant first. The goal is to create a pair of coefficients such as 6 and -6, or 10 and -10, whose sum or difference removes that variable.

EXAMPLE **3.12** ELIMINATION BY ADDING

QUESTION

Solve $3x + 2y = 16$ and $5x - 2y = 0$.

SOLUTION

1 The coefficients of y are $+2$ and -2. Add the equations: $(3x + 5x) + (2y - 2y) = 16 + 0$.

2 The y-terms cancel: $8x = 16$, so $x = 2$.

3 Substitute into $5x - 2y = 0$: $10 - 2y = 0$, so $y = 5$. The solution is $(2, 5)$.

Check: $3(2) + 2(5) = 16$ and $5(2) - 2(5) = 0$. Both equations hold.

EXAMPLE **3.13** ELIMINATION AFTER SCALING BOTH EQUATIONS

QUESTION

Solve $2x + 3y = 13$ and $3x + 4y = 18$.

SOLUTION

1 Multiply the first equation by 3 and the second by 2 so that the x-coefficients both become 6: $6x + 9y = 39$ and $6x + 8y = 36$.

2 Subtract the second from the first: $(6x - 6x) + (9y - 8y) = 39 - 36$, so $y = 3$.

3 Substitute $y = 3$ into $2x + 3y = 13$: $2x + 9 = 13$, so $x = 2$. The solution is $(2, 3)$.

> **Why scaling is allowed:** multiplying an equation by a non-zero constant produces an equivalent equation. The solution set of the system is unchanged.

EXAMPLE **3.14** WHEN ELIMINATION REMOVES EVERYTHING

QUESTION

Describe the solutions of (a) $2x + 3y = 6$ and $4x + 6y = 10$; (b) $2x + 3y = 6$ and $4x + 6y = 12$.

SOLUTION

a Multiply the first equation by 2: $4x + 6y = 12$. Subtracting it from $4x + 6y = 10$ gives $0 = -2$, which is false. There is **no solution**: the lines are parallel and distinct.

b Multiplying the first equation by 2 gives exactly the second equation. Every point on $2x + 3y = 6$ satisfies both. There are **infinitely many solutions**: the two equations describe the same line.

> **Classifying a system:** one solution means the lines intersect once; no solution means they are parallel and distinct; infinitely many means they coincide. Elimination reveals which case holds through the final statement: $0 =$ non-zero, or $0 = 0$.

KEY IDEA

Elimination chooses a variable and creates opposite coefficients, then adds or subtracts. If both variables vanish, the final numerical statement decides the case: a false statement means no solution; a true statement means infinitely many. A system with no solution is called **inconsistent**; one with infinitely many is **dependent**.

COMMON MISCONCEPTIONS

"Add the equations but not the constants." The whole equation is added, including the right-hand side. In $3x + 2y = 16$ and $5x - 2y = 0$, the constants give $16 + 0 = 16$.

"Subtract only the left sides." Subtraction applies to both sides. Writing $6x + 9y = 39$ and $6x + 8y = 36$, subtract to get $y = 3$, not $y = 39 - 36$ on one side only by accident.

"$0 = 0$ means the solution is zero." It means the two equations are the same condition, so every point on the line is a solution.

"$0 = -2$ means $x = 0$." It is a contradiction. No pair of values can make it true, so the system has no solution.

GUIDED PRACTICE · CHOOSE A MULTIPLIER

Solve $3x + 4y = 18$ and $2x - 3y = -5$.

Multiply the first equation by ___ and the second by ___ to make the y-coefficients opposite.
New equations: ___x + ___y = ___ and ___x − ___y = ___.
Add: ___x = ___, so x = ___. Then find y.

EXERCISE 3D

ELIMINATION · CLASSIFICATION · CHECKING

FLUENCY

1 Solve by elimination.

a. $x + y = 9,\ x - y = 3$

b. $2x + y = 11,\ 3x - y = 9$

c. $4x + 3y = 18,\ 2x - 3y = 6$

d. $5x + 2y = 16,\ 3x - 2y = 8$

2 Solve by elimination after scaling.

a. $2x + 3y = 12,\ 3x + 2y = 13$

b. $4x + 5y = 26,\ 3x - 2y = 1$

c. $5x - 2y = 7,\ 3x + 4y = 25$

3 Classify each system as one solution, no solution, or infinitely many.

a. $3x + y = 7,\ 6x + 2y = 14$

b. $3x + y = 7,\ 6x + 2y = 10$

c. $2x - 5y = 3,\ 4x - 10y = 6$

UNDERSTANDING AND REASONING

4 Explain why multiplying an equation by 2 before eliminating does not change the solution of the system.

5 **Error analysis.** A student solves $2x + 3y = 13$ and $3x + 4y = 18$ by subtracting the equations as written and obtaining $-x - y = -5$. Explain why this does not solve the system, and complete a correct method.

6 A system reduces to $0 = 5$. What can you conclude about the graphs? What if it reduces to $0 = 0$?

APPLICATION AND EXTENSION

7 At a school fair, 100 tickets were sold. Adult tickets cost £40 and child tickets cost £25. Total revenue was £3100. Write and solve a system to find the number of each ticket sold.

8 A chemist mixes a 20%-acid solution with a 50%-acid solution to obtain 30 L of a 30%-acid solution. Write and solve a system to find the volumes used.

9 **Extension.** Find all values of k for which the system $kx + 2y = 6$ and $3x + 4y = 9$ has no solution. Explain your reasoning, then describe the case $k = 1.5$.

FROM CONDITIONS TO DECISIONS

3.5 Modelling with systems

A model begins with definitions, not equations. Name the quantities, choose units, and say what each variable represents. Then translate each condition into a sentence, and each sentence into an equation. After solving, return to the context: reject values that are impossible, round where the situation demands a whole number, and state what the solution means.

Two modelling structures recur in this chapter. The first is a **fixed-plus-rate** comparison: cost equals a fixed charge plus a rate per unit. The second is a **balance**: two quantities add to a total while a weighted combination equals another total. Both become systems of two linear equations.

EXAMPLE **3.15** A TICKET MODEL

QUESTION

100 tickets were sold for a concert. Adult tickets cost £40 and child tickets cost £25. The revenue was £3100. How many of each were sold?

SOLUTION

1. **Define:** let a be the number of adult tickets and c the number of child tickets.
2. **Form:** the numbers add to 100, so $a + c = 100$. The revenue is $40a + 25c = 3100$.
3. **Solve:** from the first equation $a = 100 - c$. Substitute: $40(100 - c) + 25c = 3100$.
4. Expand: $4000 - 40c + 25c = 3100$. Collect: $-15c = -900$, so $c = 60$. Then $a = 40$.
5. **Interpret:** 40 adult tickets and 60 child tickets were sold. Check: $40 + 60 = 100$ and $40(40) + 25(60) = 1600 + 1500 = 3100$.

EXAMPLE **3.16** A BREAK-EVEN MODEL

QUESTION

A club sells a product for £40 each. Its costs are a fixed £1800 plus £15 per unit. Write cost and revenue functions and find the break-even quantity. What is the profit at 100 units?

SOLUTION

1. For n units, the cost is $C = 1800 + 15n$ and the revenue is $R = 40n$.
2. Break-even means cost equals revenue: $1800 + 15n = 40n$.
3. Subtract $15n$: $1800 = 25n$, so $n = 72$. The break-even quantity is 72 units.
4. At 100 units, $R = 4000$ and $C = 1800 + 1500 = 3300$, so the profit is $R - C = 700$ pounds.

Interpret: below 72 units the club makes a loss; at 72 it covers costs exactly; above 72 it makes a profit. Profit is $R - C = 25n - 1800$.

EXAMPLE **3.17** A MIXTURE MODEL

QUESTION

A shop blends 20 kg of coffee worth £24 per kg from two types costing £18 per kg and £30 per kg. How many kilograms of each type are used?

SOLUTION

1. **Define:** let x kg cost £18 per kg and y kg cost £30 per kg.
2. **Form:** $x + y = 20$ for the mass, and $18x + 30y = 20 \cdot 24 = 480$ for the value.
3. Substitute $x = 20 - y$: $18(20 - y) + 30y = 480$, so $360 + 12y = 480$ and $y = 10$. Then $x = 10$.
4. The blend uses 10 kg of each type. Check: $10 + 10 = 20$ and $180 + 300 = 480$.

Weighted average: the total value is the mass times the price per kilogram. Mixing equal masses of £18 and £30 gives the midpoint price, £24, because the masses are equal.

KEY IDEA

In a modelling problem, the equations are a translation of conditions, not the starting point. Define variables with units, state each condition in words, then solve and interpret. A solution that is mathematically valid but contextually impossible — a negative number of tickets, a fractional number of people — must be rejected or rounded with justification. Break-even is the quantity at which revenue equals cost.

COMMON MISCONCEPTIONS

"Use x and y without saying what they mean." An undefined variable cannot be interpreted. Write "let a be the number of adult tickets" before writing $a + c = 100$.

"Revenue and cost are the same equation." Revenue is price times quantity; cost includes fixed and variable parts. Confusing them removes the break-even question entirely.

"A decimal answer is always acceptable." In a ticket or people problem, 60.5 tickets is not meaningful. Decide whether the context requires a whole number and round in the direction the situation demands.

"The break-even point is the maximum profit." Break-even is where profit is zero. Profit usually increases with quantity after that point, until a capacity or demand constraint intervenes.

EXERCISE 3E

MODELLING · BREAK-EVEN · INTERPRETATION

APPLICATION

1 A theatre sells 250 tickets. Adult tickets cost £30 and child tickets cost £18. Revenue is £5700. Find the number of each type sold.

2 A club's cost is £240 plus £6 per person; its revenue is £15 per person. Find the break-even number of people and the profit at 50 people.

3 A 30 L mixture is 30% acid. It is made from a 20% solution and a 50% solution. Find the volume of each.

4 A phone plan costs £18 per month plus £0.04 per minute. A second plan costs £30 per month plus £0.01 per minute. Find the number of minutes for which the plans cost the same.

5 A rectangular field has perimeter 120 m. Its length is 10 m more than twice its width. Find its dimensions.

REASONING AND EVALUATION

6 A student finds a break-even quantity of 26.7 items and reports "26.7 items". Explain why this is not a valid production quantity and state the smallest whole number that avoids a loss.

7 For the break-even model $C = 1800 + 15n$, $R = 40n$, explain how the answer changes if the selling price falls to £35. Find the new break-even quantity and compare.

8 **Extension.** A sponsor adds a fixed £600 to the revenue in Question 2. Write the new profit expression, find the new break-even number, and explain why sponsorship changes the risk but not the profit per person.

INVESTIGATION · CRITERION B

3.6 Break-even and beyond

INVESTIGATION · CRITERION B

Where does a small business stop losing money?

A student enterprise makes phone cases. The fixed costs are £240 (equipment and stall hire). Each case costs £6 in materials. The selling price is £15.

1. Copy and complete a table for $n = 0, 5, 10, \ldots, 50$, showing cost, revenue and profit.
2. Plot cost and revenue on the same axes. Estimate the break-even point from the graph.
3. Solve $15n = 240 + 6n$. Explain why the exact answer is not a whole number and what the business must do.
4. Write a formula for the break-even quantity in terms of fixed cost F, price p and variable cost v.
5. Use your formula to find the break-even quantity when the price is £12, £18 and £9. Explain the last case.
6. Find the quantity needed to make a profit of £450. Generalise to a target profit T.

Generalise and justify. Starting from $pn = F + vn$, derive $n = \dfrac{F}{p - v}$. Explain why the formula requires $p > v$, and why a real business must round the break-even quantity *up*. Then derive a formula for the quantity needed to reach a target profit T, and explain how the graph changes when F, p or v changes.

MODELLING TASK · CRITERION D PRACTICE

The festival budget

A community festival has fixed costs of £2600 and a variable cost of £4 per visitor. The entry price is £12 and a sponsor contributes £1000. The site can admit at most 350 visitors.

1. Define a variable for the number of visitors, and write expressions for total cost, total income (including the sponsor) and profit.
2. Find the break-even number of visitors, and explain what happens below that number.
3. The organisers want a profit of at least £1000. Form and solve an inequality. Is the target achievable within capacity?
4. What ticket price would achieve the £1000 target if only 300 visitors attend? Give the price to the nearest penny, then to a sensible currency unit.
5. State two assumptions in the model. How would a rainy day, or a cap on the sponsor's contribution, affect the recommendation?

Evidence to submit: definitions with units, all three expressions, the break-even calculation, the inequality solution, the price calculation, and a short recommendation that separates mathematical results from assumptions.

MIXED PRACTICE 3

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–8 check fluency; 9–14 require reasoning; 15–18 are multi-step or unfamiliar. Allow 55 minutes.

1. Solve $7x - 4 = 24$.
2. Solve $3(x - 2) = 5x + 4$.
3. Solve $\frac{x}{3} + \frac{x}{4} = 7$.
4. Solve $5 - 2x < 11$ and write the solution in interval notation.
5. Solve $-3 \leq 2x + 1 \leq 9$ and list the integer solutions.
6. Solve $y = 3x - 1$ and $y = x + 5$.
7. Solve $2x + 5y = 16$ and $x - y = 1$.
8. Solve $4x - 3y = 6$ and $2x + 3y = 12$.
9. Classify the system $2x + 3y = 7$, $4x + 6y = 20$. Justify your answer.
10. **Error analysis.** A student solves $-2x < 8$ as $x < -4$. Explain the error and give the correct solution.
11. Make y the subject of $3x + 4y = 12$.
12. The sum of two numbers is 45. One number is 5 less than three times the other. Find the numbers.
13. A lift can carry 500 kg. It carries 140 kg of equipment and boxes of 28 kg. Find the greatest number of boxes.
14. Explain why a system of two parallel distinct lines has no solution, using both algebra and graphs.
15. A club sells 180 tickets at £25 each. Costs are £900 plus £7 per ticket sold. Find the profit.

MIXED PRACTICE 3 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

16 A phone plan costs £20 plus £0.05 per minute. Another costs £32 plus £0.02 per minute. Find the break-even number of minutes.

17 **Extension.** A system has solution $(3, -2)$. One equation is $2x + y = 4$. Write a second equation whose graph is perpendicular to the first, and verify that $(3, -2)$ satisfies it.

18 **Extension.** For the cost model $C = F + vn$ and revenue $R = pn$, show that profit is $(p - v)n - F$. Explain why the break-even quantity is $F/(p - v)$ and why the business must have $p > v$.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING 30 minutes · 30 marks

Equations, inequalities and systems: short response

NO CALCULATOR SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

19 Solve $3(2x + 5) = 4(x + 8) + 3$. [3]

20 Solve $\dfrac{x + 1}{2} - \dfrac{x - 3}{3} = 2$. [3]

21 Solve $7 - 4x > 23$ and represent the solution on a number line. [3]

22 Solve $-5 < 3x + 1 \leq 10$ and state the integer solutions. [3]

23 Solve $3x + 2y = 17$ and $x - y = 4$. [4]

24 Solve $4x + 3y = 18$ and $2x - 5y = -4$. [4]

25 250 tickets were sold. Adult tickets cost £30 and child tickets cost £18. Total revenue was £5700. Find the number of each ticket sold. [5]

26 Explain why the system $2x + 3y = 7$ and $4x + 6y = 20$ has no solution. [2]

27 **Error analysis.** A student writes $-2x < 8 \Rightarrow x < -4$. Identify the error, correct it, and check the correct answer with one value. [3]

CRITERION B · INVESTIGATING PATTERNS — 45 minutes · 24 marks

Which phone plan wins?

Instructions. Work independently. Use a table and a graph, then answer in full sentences. Record all costs in pounds.

A phone company offers three plans. **Pay-as-you-go:** £0.12 per minute. **Plan A:** £18 per month plus £0.04 per minute. **Plan B:** £30 per month plus £0.01 per minute.

1. Complete a table of monthly costs for $m = 0, 100, 200, \ldots, 500$ minutes. [4]
2. Write a cost expression for each plan. [3]
3. Draw all three graphs on the same axes and label the intersections. [5]
4. Find the number of minutes at which each pair of plans costs the same. Show your equations. [6]
5. For each interval of minutes, state which plan is cheapest. Justify using your graph. [4]
6. State one assumption in the model and explain how it could change the recommendation. [2]

Evidence: a table, three equations, a labelled graph, algebraic solutions for the intersections, and interval-based recommendations.

CRITERION C · COMMUNICATING — 40 minutes · 20 marks

Why the inequality sign flips

Instructions. Write a guide of 300–400 words for a student in the year below. Your guide must explain: (i) what it means to solve an inequality; (ii) why multiplying or dividing by a negative number reverses the symbol; (iii) two worked examples, one numerical and one algebraic; (iv) one number-line diagram; (v) one common error and how to avoid it. Use correct notation throughout.

Evidence: a titled guide with paragraphs, a labelled number line, two worked examples that are checked, and a short “remember” box.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The community shuttle

Instructions. A community group is hiring a shuttle for a day. Operator A charges £240 plus £6 per passenger. Operator B charges £420 plus £3 per passenger. The shuttle can carry 120 passengers. The group charges £8 per passenger. The committee has a budget of £900 for the hire.

1. Write an expression for the cost from each operator for n passengers. [3]
2. Find the number of passengers for which the two operators charge the same amount, and the cost at that point. [4]
3. Compare the costs for 40 passengers and for 80 passengers. [3]
4. Using the £900 budget, find the greatest number of passengers each operator can carry. [4]
5. At £8 per passenger and full capacity, find the profit under each operator. Find each break-even number. [5]
6. Recommend an operator. Justify your choice using your results, state two assumptions, and explain one way fairness could change the decision. [5]

Evidence: cost expressions, the intersection calculation, comparisons with units, budget inequalities, profit and break-even results, and a justified recommendation that distinguishes cost from fairness.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- An equation is solved by equivalence-preserving moves: expand, collect, add or subtract the same quantity, multiply or divide by a non-zero number. Clear fractions by multiplying every term by the common denominator.
- Rearranging a formula is the same process with letters; state any restriction needed for the rearrangement to be valid.
- Inequalities obey the same rules, except that multiplying or dividing by a negative number reverses the symbol. Represent the solution on a number line: filled includes, open excludes.
- A system is solved by graphing, substitution or elimination, and the ordered pair must satisfy both equations. If elimination gives a false statement there is no solution; if it gives $0 = 0$ there are infinitely many.
- In modelling, define variables and units, translate conditions, solve, then interpret and evaluate. Break-even occurs when revenue equals cost.

$$n = \frac{F}{p - v} \quad (p > v) \qquad ax + b = c \Rightarrow x = \frac{c - b}{a}$$

$$a < b \Rightarrow -a > -b$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

I can solve a linear equation with brackets on both sides.

I can solve an equation containing algebraic fractions.

I can check a solution by substitution into the original equation.

I can make a variable the subject of a formula.

I can form an equation from a perimeter or age problem.

I can solve an inequality and draw its solution set.

I can explain why the symbol reverses for a negative multiplier.

I can write a solution in interval notation.

I can solve a compound inequality and list integer solutions.

I can solve a system by substitution.

I can solve a system by elimination, scaling equations if needed.

I can classify a system as one solution, none, or infinitely many.

I can interpret an intersection point in context.

I can write cost, revenue and profit expressions.

I can find and interpret a break-even quantity.

I can state assumptions and evaluate a model's limitations.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which method — graphing, substitution or elimination — did you choose most often? Was that choice based on the structure of the equations or on habit?
2. Describe one error you made in an inequality. Was it a reasoning error (the direction of the symbol) or an arithmetic error? What check will you use next time?
3. Explain in your own words why a system can have no solution even though each equation has infinitely many solutions.
4. Look ahead to Chapter 4: how does the graph of a line connect its equation, a system and a model?

COORDINATES, CHANGE AND THE GEOMETRY OF A STRAIGHT LINE

Functions and Linear Models

04

INPUT · RULE · OUTPUT

From a rule to a graph, and from a graph to a decision — measuring distance, direction and rate of change on the coordinate plane.

STATEMENT OF INQUIRY

A function is a relationship in which each input determines exactly one output. Representing that relationship as a graph turns algebraic structure into visible geometry, where gradient measures change and position supports measurement and modelling.

KEY CONCEPT
Form

RELATED CONCEPTS
Representation · Change · Pattern

GLOBAL CONTEXT
Orientation in space and time — maps, routes and rates

FACTUAL
How do we read, evaluate and interpret a function given by a rule, a table or a graph?

CONCEPTUAL
What does the gradient of a line measure, and why does it determine the line together with one point?

DEBATABLE
When is a straight-line model an acceptable description of a real situation?

IN THIS CHAPTER

4.1 Functions and notation
4.2 Distance and midpoint
4.3 Gradient and equations of lines
4.4 Parallel and perpendicular lines
4.5 Linear models and variation
4.6 Investigation: a fair fare

OPENING PROBLEM · THE FAIR FARE

Two taxis, one journey

Two taxi companies advertise different fares. Company A charges a flag-fall of £4 plus £2 per kilometre. Company B charges a flag-fall of £8 plus £1.50 per kilometre. A student has £30 for a journey across the city.

1. Write a function for the fare $A(d)$ and $B(d)$ for a journey of d kilometres.
2. Plot both functions on the same axes. What does each intercept represent?
3. For what distance do the companies charge the same fare? Solve an equation and interpret the intersection.
4. Which company is cheaper for 5 km? For 12 km? Explain using the gradients.
5. Using the £30 budget, find the greatest whole number of kilometres each company can offer.
6. Is the "same fare for the same distance" a fair pricing rule? What does a flag-fall do to short journeys?

A function is a machine with one rule and one output for each input. A graph is the machine's behaviour drawn in space. Coordinates let us measure the distance between two positions, the midpoint of a route, and the steepness of the change. This chapter connects the algebra of a rule to the geometry of a line.

4.0 Learning intentions

ORIENTATION

- Understand functions as input–output rules, and use function notation accurately.
- Calculate distances and midpoints on the coordinate plane.
- Find the gradient of a line and use it to write the equation of the line.
- Recognise parallel and perpendicular lines from their gradients.
- Build and interpret linear models, including direct variation, and evaluate their limitations.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

evaluate a function and solve a simple equation involving function notation; state a restricted domain; calculate a distance and a midpoint and use them to classify a triangle; find a gradient from two points or an equation, and write the equation of a line through a given point; identify parallel and perpendicular lines; construct a linear model from a table or description, interpret its gradient and intercept with units, and explain when extrapolation is unsafe.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 4.1.

1. Plot $(-2, 3)$ and $(4, -1)$ on a coordinate grid.
2. Evaluate $3x - 4$ when $x = 5$ and when $x = -2$.
3. Solve $2x + 5 = 17$.
4. Find the value of $6^2 + 8^2$, then its square root.
5. Find the midpoint of the numbers 2 and 10 on a number line.
6. Write the next term in the pattern 3, 7, 11, 15, … and state the rule.

REFRESHER

In the coordinate plane, a point is written (x, y): the horizontal coordinate first, then the vertical. The *gradient* of a straight line compares vertical change to horizontal change. If a quantity changes by the same amount for each unit increase of another, the relationship is linear. The square root of a positive number a^2 is a when $a \geq 0$.

ATL SKILLS · THINKING AND REPRESENTATION

Critical thinking: decide whether a table, rule or graph best answers a question. **Representation:** move fluently between the four forms — words, table, equation, graph — and check that they tell the same story. **Transfer:** use the same linear structure to model a fare, a conversion and a route.

ONE INPUT, ONE OUTPUT

4.1 Functions and function notation

A function is a rule that assigns exactly one output to each allowed input. The rule may be a formula, a table, a graph or a verbal description. What matters is the determinacy: once the input is chosen, the output is fixed. A vending machine that sometimes gives two different drinks for the same button is not a function.

Function notation names the rule. Writing $f(x) = 2x - 3$ says that the function f takes an input x, doubles it and subtracts 3. The symbol $f(4)$ is not f multiplied by 4; it is the output of the function when the input is 4. The set of allowed inputs is the domain; the set of outputs that actually occur is the range.



Figure 4.1 A function machine for $f(x) = 2x - 3$. The domain is the set of allowed inputs; the range is the set of outputs produced.

DEFINITIONS

Function, domain and range

A **function** is a relationship that assigns exactly one output to each input in its domain. The **domain** is the set of allowed inputs; the **range** is the set of outputs. The notation $f(x)$ denotes the output of f at input x. A function is **linear** when its rule can be written $f(x) = mx + c$ with $m \neq 0$.

EXAMPLE **4.1** EVALUATING AND SOLVING WITH FUNCTION NOTATION

QUESTION

For $f(x) = 2x - 3$, find $f(4)$, $f(-2)$ and $f(0)$. Solve $f(x) = 7$.

SOLUTION

1. Substitute the input for x: $f(4) = 2(4) - 3 = 5$.
2. $f(-2) = 2(-2) - 3 = -4 - 3 = -7$. The input is negative; the output is more negative.
3. $f(0) = 2(0) - 3 = -3$. This is the value of the function at the y-intercept.
4. Solve $2x - 3 = 7$: add 3, then divide by 2, giving $x = 5$. So $f(5) = 7$.

Check: $f(5) = 2(5) - 3 = 7$. The input 5 produces the required output 7.

EXAMPLE **4.2** A QUADRATIC FUNCTION AND AN EXPRESSION INPUT

QUESTION

For $g(x) = x^2 - 2x$, find $g(3)$, $g(-1)$ and $g(a+1)$.

SOLUTION

1. $g(3) = 3^2 - 2(3) = 9 - 6 = 3$.
2. $g(-1) = (-1)^2 - 2(-1) = 1 + 2 = 3$. Different inputs can produce the same output; this does not stop the rule being a function.
3. Replace every x by $(a+1)$: $g(a+1) = (a+1)^2 - 2(a+1)$.
4. Expand: $a^2 + 2a + 1 - 2a - 2 = a^2 - 1$.

> **Notation:** $g(a+1)$ means "the output when the input is $a+1$", not g times $(a+1)$. The brackets protect the whole input.

EXAMPLE **4.3** DOMAIN RESTRICTIONS

QUESTION

State the domain of each function. (a) $h(x) = \dfrac{1}{x-3}$ (b) $p(x) = \sqrt{x-2}$ (c) the cost $C(n)$ of n tickets at £12 each.

SOLUTION

a. The denominator cannot be zero, so $x - 3 \neq 0$ and the domain is $x \neq 3$.

b. A real square root requires a non-negative input: $x - 2 \geq 0$, so $x \geq 2$.

c. The number of tickets must be a whole number, so the domain is $n = 0, 1, 2, \ldots$, not all real numbers.

> **Domain from context:** mathematics may allow every real number, but the situation may allow only whole numbers, positive lengths or values within a capacity. Always state the contextual domain of a model.

KEY IDEA

A function is defined by its rule *and* its domain. Function notation names the output, not a product. The graph of a function passes the **vertical line test**: no vertical line meets it more than once, because one input cannot have two outputs.

COMMON MISCONCEPTIONS

"$f(4) = f \times 4$." The brackets denote input. $f(4)$ is a single output; $f \times 4$ would be a product of a function and a number.

"Every rule is a function." A rule such as $x^2 + y^2 = 9$ does not define y as a function of x: the input $x = 0$ gives two outputs, $y = 3$ and $y = -3$.

"The domain is always all real numbers." Denominators, square roots and context can all restrict the domain.

"Different inputs must give different outputs." A function may repeat outputs; $g(-1) = g(3) = 3$ in Example 4.2. It is the input that cannot have two outputs.

GUIDED PRACTICE · EVALUATE AND SOLVE

For $f(x) = 5 - 2x$, complete the following.

$f(3) = 5 - 2(___) = ____$

$f(-1) = 5 - 2(___) = ____$

If $f(x) = 1$, then $5 - 2x = 1$, so $x = ____$.

The domain of f is _____.

EXERCISE 4A

FUNCTIONS · NOTATION · DOMAIN

FLUENCY

1 For $f(x) = 3x + 1$, find $f(2)$, $f(0)$, $f(-4)$ and $f\left(\frac{1}{3}\right)$.

2 For $g(x) = x^2 - 4$, find $g(3)$, $g(-3)$, $g(0)$ and $g(2a)$.

3 For $h(x) = \dfrac{x+1}{2}$, find $h(5)$, $h(-3)$ and $h(0)$.

4 Solve.

a. $f(x) = 11$ for $f(x) = 4x - 1$

b. $g(x) = 0$ for $g(x) = 2x + 6$

c. $h(x) = 5$ for $h(x) = x^2 + 1$, $x \geq 0$

5 State the domain.

a. $f(x) = \dfrac{2}{x+4}$

b. $g(x) = \sqrt{x-5}$

c. $h(x) = \dfrac{1}{2x-1}$

UNDERSTANDING AND REASONING

6 Explain why $f(x) = x^2$ is a function but the relation $x = y^2$ is not a function of x.

7 **Error analysis.** A student writes $f(2) = 2x - 3 = 1$. Explain what is wrong with the notation and write the correct evaluation for $f(x) = 2x - 3$.

8 A function has domain $1 \leq x \leq 5$. What does this tell you about the graph? Sketch a possible graph.

9 Give an example of a function whose range is $y \geq 0$. Explain why no negative output can occur.

APPLICATION AND EXTENSION

10 A phone plan costs $C(m) = 18 + 0.04m$ pounds for m minutes. Find the cost of 300 minutes and solve $C(m) = 26$ to find the minutes used.

11 A ball is dropped from a height of 45 m. Its height after t seconds is $h(t) = 45 - 5t^2$. Find $h(1)$, $h(2)$ and the time at which the ball reaches the ground.

12 **Extension.** For $f(x) = ax + 3$, $f(2) = 11$. Find a, then find the value of x for which $f(x) = 0$. Explain what the second answer represents on the graph.

MEASURING ON THE PLANE

4.2 Distance and midpoint

Coordinates turn geometry into arithmetic. Two points on a horizontal line differ only in their x-coordinates; two points on a vertical line differ only in their y-coordinates. For any other pair, the horizontal and vertical differences form the legs of a right triangle, and the distance between the points is the hypotenuse.

The midpoint is the average of the coordinates. It lies exactly halfway along the segment, so its x-coordinate is the mean of the two x-coordinates and its y-coordinate is the mean of the two y-coordinates. Averaging is the same as splitting the total difference into two equal parts.

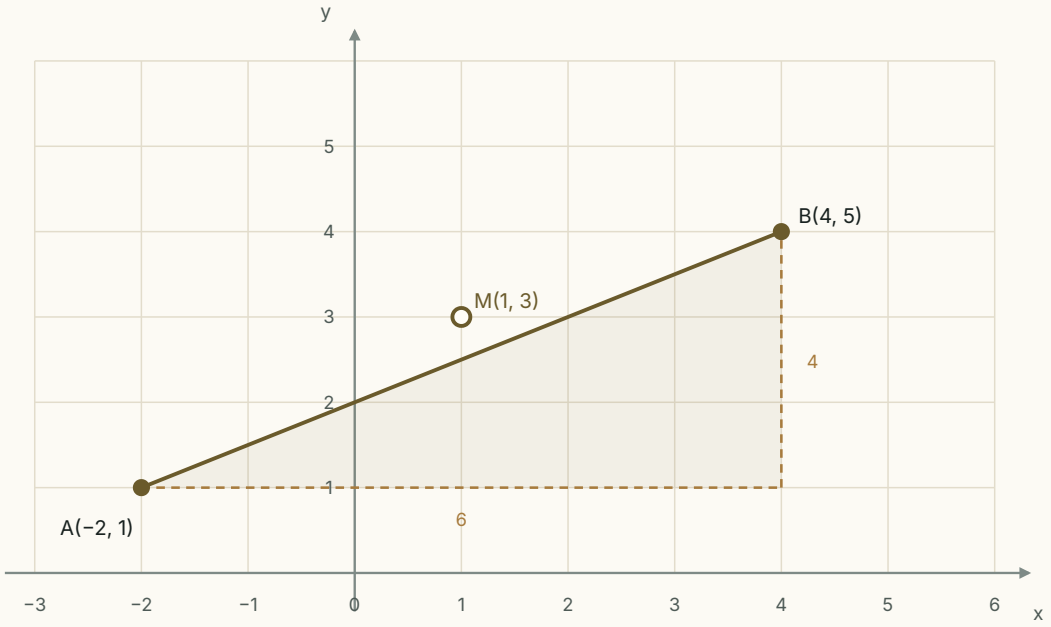


Figure 4.2 The segment AB with midpoint M. The horizontal and vertical differences form a right triangle with legs 6 and 4, so $AB = \sqrt{6^2 + 4^2} = \sqrt{52}$.

FORMULAE

Distance and midpoint

For points $A(x_1, y_1)$ and $B(x_2, y_2)$, the **distance** is

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

and the **midpoint** is

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

The distance formula follows from the Pythagorean relationship, which is proved in Chapter 10.

EXAMPLE **4.4** DISTANCE AND MIDPOINT

QUESTION

Find the distance and the midpoint of $A(-2, 1)$ and $B(4, 5)$.

SOLUTION

1. Differences: $x_2 - x_1 = 4 - (-2) = 6$ and $y_2 - y_1 = 5 - 1 = 4$.
2. Distance: $AB = \sqrt{6^2 + 4^2} = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13} \approx 7.21$.
3. Midpoint: $M\left(\frac{-2+4}{2}, \frac{1+5}{2}\right) = (1, 3)$.

Check: M lies on the segment and is the same distance from A and B. From $A(-2, 1)$ to $M(1, 3)$, the change is $(3, 2)$; from M to $B(4, 5)$, the change is also $(3, 2)$.

EXAMPLE **4.5** FINDING AN ENDPOINT FROM A MIDPOINT

QUESTION

The midpoint of AB is $M(5, 2)$ and A is $(2, -1)$. Find B.

SOLUTION

1. Let $B = (x, y)$. The midpoint formula gives $\frac{2+x}{2} = 5$ and $\frac{-1+y}{2} = 2$.
2. Multiply by 2: $2 + x = 10$, so $x = 8$; and $-1 + y = 4$, so $y = 5$.
3. Hence $B = (8, 5)$.

Check: the midpoint of $(2, -1)$ and $(8, 5)$ is $\left(\frac{10}{2}, \frac{4}{2}\right) = (5, 2)$.

EXAMPLE **4.6** CLASSIFYING A TRIANGLE FROM SIDE LENGTHS

QUESTION

Classify the triangle with vertices $P(1, 2)$, $Q(4, 6)$ and $R(7, 2)$.

SOLUTION

1. $PQ^2 = (4 - 1)^2 + (6 - 2)^2 = 9 + 16 = 25$, so $PQ = 5$.
2. $QR^2 = (7 - 4)^2 + (2 - 6)^2 = 9 + 16 = 25$, so $QR = 5$.
3. $PR^2 = (7 - 1)^2 + (2 - 2)^2 = 36$, so $PR = 6$.
4. Two sides are equal, so the triangle is isosceles. Since $5^2 + 5^2 = 50 \neq 36$, it is not right-angled.

> **Efficiency:** compare squared lengths. Only take square roots when the actual lengths are needed; squaring avoids unnecessary rounding.

KEY IDEA

Distance measures the hypotenuse of the horizontal–vertical right triangle; midpoint averages the coordinates. Both formulas are consequences of coordinate structure, not separate facts to memorise without meaning.

COMMON MISCONCEPTIONS

"Distance is $x_2 - x_1 + y_2 - y_1$." Differences are squared before addition; the distance is not the sum of the changes.

"Midpoint is the difference divided by 2." It is the *sum* divided by 2. The midpoint of 2 and 10 is 6, not 4.

"A negative coordinate can be ignored." Subtraction handles signs: $4 - (-2) = 6$, not 2. Sketching the points catches sign errors.

"The distance formula gives a negative result." Squares are non-negative, so the square root is non-negative. Distance is never negative.

GUIDED PRACTICE · DISTANCE AND MIDPOINT

For $A(-1, 2)$ and $B(5, -4)$:

Horizontal difference: $5 - (-1) =$ ____. Vertical difference: $-4 - 2 =$ ____.
Distance: $AB = \sqrt{____ + ____} = \sqrt{____} =$ ____.
Midpoint: $\left(\frac{____}{2}, \frac{____}{2}\right) = (____, ____)$.

EXERCISE 4B

DISTANCE · MIDPOINT · COORDINATE GEOMETRY

FLUENCY

1 Find the distance between the points.
   a. $(1, 2)$ and $(4, 6)$
   b. $(-2, 3)$ and $(3, -1)$
   c. $(0, 0)$ and $(5, 12)$
   d. $(-4, -2)$ and $(2, 6)$

2 Find the midpoint of the segment joining the points.
   a. $(2, 5)$ and $(8, 1)$
   b. $(-3, 4)$ and $(5, -2)$
   c. $(0, -6)$ and $(6, 0)$
   d. $(-7, -3)$ and $(-1, 9)$

3 $M(4, 1)$ is the midpoint of AB. Find B when $A = (1, -3)$.

4 Find the distance from $(3, -2)$ to the origin, in simplified surd form.

UNDERSTANDING AND REASONING

5 Explain why the midpoint formula averages the coordinates rather than subtracting them.

6 **Error analysis.** A student finds the distance between $(1, 2)$ and $(4, 6)$ as $3 + 4 = 7$. Identify the error and calculate the correct distance.

7 A segment has midpoint $(0, 0)$ and one endpoint $(3, -5)$. Find the other endpoint. What is the geometric relationship between the two endpoints?

8 Can the distance between two points be zero? What does that tell you about the points?

APPLICATION AND EXTENSION

9 The vertices of a triangle are $A(2, 1)$, $B(6, 1)$ and $C(4, 5)$. Find all three side lengths and classify the triangle.

10 A drone flies in a straight line from $P(-2, 3)$ to $Q(6, 9)$. Find the distance flown and the coordinates of the point halfway along the route.

11 **Extension.** Show that the midpoint M of $A(x_1, y_1)$ and $B(x_2, y_2)$ is equidistant from A and B. Use the distance formula and simplify.

MEASURING STEEPNESS

4.3 Gradient and equations of lines

The gradient of a straight line measures how quickly the output changes as the input increases. It is the vertical change divided by the horizontal change between any two points on the line. A gradient of 3 means that for every 1 unit to the right, the line rises 3 units. A negative gradient means the line falls as it moves right.

A line is completely determined by its gradient and one point. The **point–gradient form** $y - y_1 = m(x - x_1)$ encodes exactly that information: starting from (x_1, y_1), every other point on the line satisfies the same ratio of changes. Rearranging gives the gradient–intercept form $y = mx + c$, where c is the y-intercept.

DEFINITIONS

Gradient and equation of a line

For two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$, the **gradient** is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{vertical change}}{\text{horizontal change}}.$$

A line with gradient m through (x_1, y_1) has equation $y - y_1 = m(x - x_1)$. A vertical line has undefined gradient; a horizontal line has gradient 0.

EXAMPLE **4.7** GRADIENT FROM TWO POINTS

QUESTION

Find the gradient of the line through (a) $(1, 2)$ and $(4, 8)$; (b) $(-2, 5)$ and $(4, -1)$.

SOLUTION

a $m = \dfrac{8 - 2}{4 - 1} = \dfrac{6}{3} = 2$. The line rises 2 units for every 1 unit right.

b $m = \dfrac{-1 - 5}{4 - (-2)} = \dfrac{-6}{6} = -1$. The line falls at 45° to the horizontal, since a gradient of -1 means a drop of 1 for every 1 across.

> **Order matters consistently:** subtract the coordinates in the same order on top and bottom. Reversing both gives the same gradient; reversing only one changes the sign.

EXAMPLE **4.8** EQUATION FROM A POINT AND A GRADIENT

QUESTION

Find the equation of the line through $(3, -2)$ with gradient $\frac{2}{3}$, in the form $y = mx + c$.

SOLUTION

1. Use $y - y_1 = m(x - x_1)$ with $m = \frac{2}{3}$ and $(x_1, y_1) = (3, -2)$: $y - (-2) = \frac{2}{3}(x - 3)$.
2. Simplify: $y + 2 = \frac{2}{3}x - 2$.
3. Subtract 2: $y = \frac{2}{3}x - 4$. The y-intercept is -4.

Check: at $x = 3$, $y = \frac{2}{3}(3) - 4 = 2 - 4 = -2$, as required. The point lies on the line.

EXAMPLE **4.9** READING GRADIENT AND INTERCEPT FROM AN EQUATION

QUESTION

Find the gradient and y-intercept of $3x + 4y = 12$, and sketch the line.

SOLUTION

1. Rearrange for y: $4y = -3x + 12$, so $y = -\frac{3}{4}x + 3$.
2. The gradient is $-\frac{3}{4}$ and the y-intercept is 3.
3. The x-intercept occurs when $y = 0$: $3x = 12$, so $x = 4$. Plot $(0, 3)$ and $(4, 0)$, then draw the line through them.

> **Sketching:** two intercepts are enough for a straight line. A third point, such as $x = 4$ giving $y = 0$, checks the sketch.

KEY IDEA

Gradient is a rate: output change per unit input change. The equation $y = mx + c$ displays the rate m and the starting value c. The point–gradient form is the quickest route from a point and a gradient to an equation.

COMMON MISCONCEPTIONS

"Gradient is x over y." Gradient is vertical change over horizontal change: y difference divided by x difference.

"A negative gradient means the line goes left." The line still goes right; it falls as it goes right. Negative gradients slope downhill from left to right.

"The y-intercept is the gradient." In $y = mx + c$, m is the gradient and c is the y-intercept. In $3x + 4y = 12$, the intercept is 3, not 4.

"Vertical lines have gradient zero." A vertical line has undefined gradient because the horizontal change is zero. A horizontal line has gradient zero.

GUIDED PRACTICE · BUILD THE EQUATION

Find the equation of the line through $(2, 5)$ and $(6, 13)$.

Gradient $m = \dfrac{13 - 5}{__ - __} = \dfrac{__}{__} = ___$.

Use $y - y_1 = m(x - x_1)$: $y - ___ = ___(x - ___)$.

Simplify to $y = ___x + ___$. Check by substituting $(6, 13)$.

EXERCISE 4C

GRADIENT · EQUATIONS · INTERCEPTS

FLUENCY

1 Find the gradient of the line through the points.

a. $(0, 1)$ and $(3, 7)$

b. $(2, -1)$ and $(6, 7)$

c. $(-3, 4)$ and $(1, -4)$

d. $(5, 2)$ and $(5, 9)$

2 Find the equation of the line with the given gradient and point, in the form $y = mx + c$.

a. $m = 3$, through $(1, 4)$

b. $m = -2$, through $(2, 5)$

c. $m = \frac{1}{2}$, through $(-4, 1)$

d. $m = -\frac{2}{3}$, through $(3, -1)$

3 Find the equation of the line through each pair of points.

a. $(1, 3)$ and $(4, 9)$

b. $(-2, 3)$ and $(2, -5)$

c. $(0, 4)$ and $(5, 0)$

4 State the gradient and y-intercept.

a. $y = 4x - 7$

b. $2x + 3y = 6$

c. $5x - 2y = 10$

UNDERSTANDING AND REASONING

5 Explain what a gradient of -3 tells you about the graph, using the words *horizontal* and *vertical*.

6 **Error analysis.** A student calculates the gradient of the line through $(1, 4)$ and $(3, 10)$ as $\frac{3-1}{10-4} = \frac{1}{3}$. Explain the error and give the correct gradient.

7 A line has equation $y = 2x + c$ and passes through $(3, 1)$. Find c, then state where the line crosses the y-axis.

8 Is the point $(4, 7)$ on the line $y = 2x - 1$? Justify by substitution, then explain what the answer means geometrically.

APPLICATION AND EXTENSION

9 A ramp rises 1.2 m over a horizontal distance of 8 m. Find its gradient and interpret it as a ratio.

10 A phone plan costs £18 per month plus £0.04 per minute. Write the cost as a linear function, state the gradient with units, and interpret the y-intercept.

11 **Extension.** The line $y = mx + 3$ passes through $(2, 7)$. Find m, then find the coordinates of the point where the line crosses the x-axis. Explain how the answer relates to solving $mx + 3 = 0$.

SAME DIRECTION, RIGHT ANGLES

4.4 Parallel and perpendicular lines

Parallel lines never meet, so they must rise and fall at the same rate: their gradients are equal. **Perpendicular lines** meet at a right angle. If one line rises m units for each unit across, a line at right angles must turn the direction by $90°$; its gradient is the negative reciprocal, $-\frac{1}{m}$. The product of the two gradients is -1.

Two special cases break the reciprocal rule. A horizontal line (gradient 0) is perpendicular to a vertical line (undefined gradient). The product rule does not apply because division by zero is undefined, so these cases are identified from the geometry rather than from the formula.

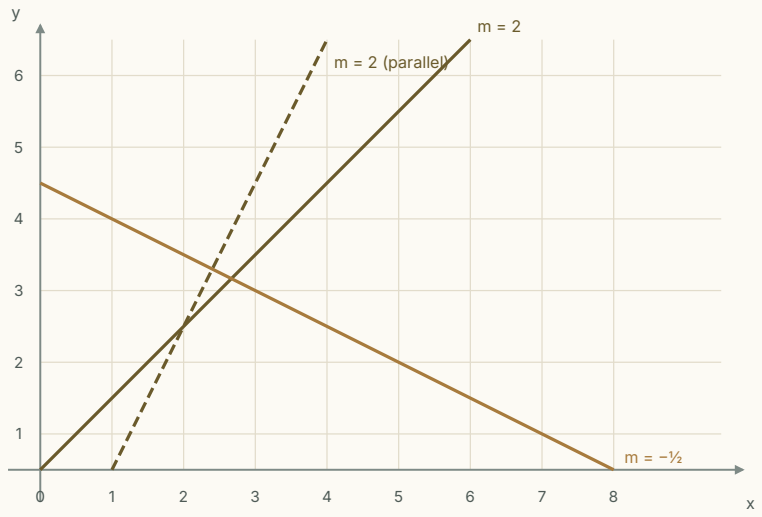


Figure 4.3 Two parallel lines have equal gradients. The line with gradient $-\frac{1}{2}$ is perpendicular to the lines with gradient 2, because $2 \times \left(-\frac{1}{2}\right) = -1$.

EXAMPLE **4.10** PARALLEL AND PERPENDICULAR LINES THROUGH A POINT

QUESTION

Find the equation of the line through $(2, -3)$ that is (a) parallel to $y = 4x - 1$; (b) perpendicular to $y = 2x + 3$.

SOLUTION

a Parallel lines have equal gradients, so $m = 4$. Using $y - y_1 = m(x - x_1)$: $y + 3 = 4(x - 2)$, so $y = 4x - 11$.

b The gradient of $y = 2x + 3$ is 2. The perpendicular gradient is $-\frac{1}{2}$. Then $y + 3 = -\frac{1}{2}(x - 2)$, so $y = -\frac{1}{2}x - 2$.

Check (b): the product of gradients is $2 \cdot \left(-\frac{1}{2}\right) = -1$, and substituting $x = 2$ gives $y = -1 - 2 = -3$, as required.

EXAMPLE **4.11** PERPENDICULAR TO A LINE IN GENERAL FORM

QUESTION

Find the equation of the line through $(1, 4)$ perpendicular to $3x - 2y = 6$.

SOLUTION

1 Rearrange $3x - 2y = 6$: $-2y = -3x + 6$, so $y = \frac{3}{2}x - 3$. The gradient is $\frac{3}{2}$.

2 The perpendicular gradient is $-\frac{2}{3}$, since $\frac{3}{2} \cdot \left(-\frac{2}{3}\right) = -1$.

3 Use the point $(1, 4)$: $y - 4 = -\frac{2}{3}(x - 1)$, so $y = -\frac{2}{3}x + \frac{14}{3}$.

> **Alternative form:** multiplying by 3 gives $3y = -2x + 14$, or $2x + 3y = 14$. Any equivalent form is acceptable unless a particular form is requested.

KEY IDEA

Parallel: $m_1 = m_2$. Perpendicular: $m_1 m_2 = -1$, so each gradient is the negative reciprocal of the other. Horizontal and vertical lines are perpendicular by geometry, even though the product rule is undefined.

COMMON MISCONCEPTIONS

"Perpendicular gradients are negatives." They are negative reciprocals: the perpendicular to gradient 3 is $-\frac{1}{3}$, not -3.

"Perpendicular gradients multiply to 1." They multiply to -1. A product of 1 would describe the same direction, not a right angle.

"Parallel lines have the same y-intercept." Equal intercepts would make them the same line. Parallel lines have equal gradients but different intercepts.

"A vertical line has gradient 0, so it is parallel to a horizontal line." A vertical line has undefined gradient and is perpendicular to a horizontal line.

GUIDED PRACTICE · PARALLEL AND PERPENDICULAR

For the line $y = -3x + 2$:

A parallel line has gradient _____.
A perpendicular line has gradient _____, because ____ × ____ = -1.
The perpendicular line through $(6, 1)$ has equation $y -$ ____ = _____$(x -$ ____$)$.

EXERCISE 4D

PARALLEL · PERPENDICULAR · COORDINATE REASONING

FLUENCY

1 State the gradient of a line parallel to each line.

a. $y = 5x - 2$
b. $y = -\frac{2}{3}x + 1$
c. $4x + y = 9$
d. $2x - 3y = 6$

2 State the gradient of a line perpendicular to each line.

a. $y = 3x + 4$
b. $y = -\frac{1}{4}x$
c. $2x + 5y = 10$
d. $y = 7$

EXERCISE 4D · CONTINUED

PARALLEL · PERPENDICULAR · COORDINATE REASONING

3 Find the equation of the line through the given point parallel to the given line.

a. $(2, 1)$, $y = 2x - 5$

b. $(-1, 4)$, $y = -x + 2$

c. $(3, -2)$, $2x + y = 5$

4 Find the equation of the line through the given point perpendicular to the given line.

a. $(4, 1)$, $y = 2x + 3$

b. $(0, 5)$, $y = \frac{1}{3}x - 1$

c. $(2, 2)$, $x + 2y = 4$

UNDERSTANDING AND REASONING

5 Explain why the product of the gradients of perpendicular lines is -1, using a rotation of 90° and a sketch.

6 **Error analysis.** A student says the line perpendicular to $y = 4x - 1$ has gradient -4. Explain the error and give the correct gradient.

7 Are the lines $y = 3x + 2$ and $6x - 2y = 7$ parallel, perpendicular or neither? Justify by comparing gradients.

8 Describe the line through $(3, -2)$ with undefined gradient and the line through the same point with gradient zero. How are they related?

APPLICATION AND EXTENSION

9 A rectangular garden has corners at $A(1, 2)$, $B(7, 2)$, $C(7, 8)$ and $D(1, 8)$. Verify that consecutive sides are perpendicular by finding gradients.

10 A wheelchair ramp is designed perpendicular to a path with gradient $\frac{1}{10}$. What is the gradient of the ramp? Is a gradient of -10 physically sensible? Explain.

11 **Extension.** The line L passes through $(1, 2)$ and is perpendicular to $3x + 4y = 12$. Find the equation of L, then find the coordinates of the point where L meets the original line.

FROM DATA TO A RULE

4.5 Linear models and variation

A **linear model** is a straight-line relationship between two quantities. Its gradient is the rate of change, with units of output per unit input; its intercept is the value of the output when the input is zero. A model is useful when the rate is approximately constant over the range of interest, and dangerous when it is extended far beyond the data that support it.

Direct variation is the special case in which the intercept is zero: $y = kx$. Doubling the input doubles the output. The constant k is the constant of variation. Many proportional relationships in measurement are of this form, but a fixed charge or a start-up cost produces a non-zero intercept and is not direct variation.

EXAMPLE **4.12** DIRECT VARIATION

QUESTION

y varies directly with x, and $y = 12$ when $x = 3$. Find the constant of variation and the value of y when $x = 10$.

SOLUTION

1. Write $y = kx$. Substitute the known pair: $12 = k(3)$, so $k = 4$.
2. The model is $y = 4x$. When $x = 10$, $y = 40$.
3. Check the ratio: $y/x = 4$ for both pairs. A constant ratio is the signature of direct variation.

> **Not direct variation:** if $y = 4x + 2$, the ratio y/x changes with x. The relationship is linear but not proportional.

EXAMPLE **4.13** BUILDING A LINEAR MODEL FROM A TABLE

QUESTION

A courier charges a fixed fee plus a rate per kilometre. The table shows the charge for several distances. Find a linear model and interpret its gradient and intercept.

DISTANCE d (km)	2	4	6	8
Charge C (£)	13	19	25	31

SOLUTION

1. Each increase of 2 km increases the charge by £6, so the gradient is $6/2 = 3$ pounds per kilometre.
2. Use $C = 3d + b$. Substituting $d = 2$, $C = 13$: $13 = 6 + b$, so $b = 7$.
3. The model is $C = 3d + 7$. The intercept £7 is the fixed fee; the gradient £3/km is the distance charge.
4. Check with $d = 8$: $C = 24 + 7 = 31$, matching the table.

EXAMPLE **4.14** INTERPRETING AND LIMITING A MODEL

QUESTION

The temperature in degrees Fahrenheit is $F = 1.8C + 32$. Interpret the gradient and intercept, then explain why this model is reliable for ordinary temperatures but extrapolation to $C = 1000$ is not physically meaningful.

SOLUTION

1. The gradient 1.8 means that each 1°C increase corresponds to a 1.8°F increase.
2. The intercept 32 means that 0°C corresponds to 32°F.
3. The relationship is exact by definition of the scales, so it is reliable wherever the temperature scale is defined. At $C = 1000$, the formula still produces a number, but no everyday substance remains in a state described by this temperature; the model is mathematically valid but contextually unsupported.

> **Extrapolation:** a linear model may be reliable inside its data range (**interpolation**) and unreliable outside it (**extrapolation**). Always ask whether the rate can reasonably stay constant.

KEY IDEA

In a linear model $y = mx + c$, m is the rate of change with units and c is the starting value. Direct variation has $c = 0$. Interpolation inside the data range is safer than extrapolation beyond it.

COMMON MISCONCEPTIONS

"Any linear relationship is proportional." Proportional means $y = kx$, with intercept zero. A fixed fee makes the relationship linear but not proportional.

"The gradient is the total cost." The gradient is the cost per additional unit, not the total. The intercept carries the fixed part.

"A model that fits three points is true everywhere." Three points show a local pattern, not a universal law. State the range over which the model is supported.

"Units can be added later." The gradient has units of output per input. Without units, the number 3 could mean £3 per km, 3 litres per minute or 3 degrees per hour.

GUIDED PRACTICE · MODEL FROM DATA

A repairer charges a call-out fee plus an hourly rate. The charge for 1 hour is £85 and for 3 hours is £145.

Gradient $= \dfrac{145 - 85}{___ - ___} = \dfrac{___}{___} = ___$ pounds per hour.

Using $C = mh + b$: $85 = ___ + ___$, so $b = ___$.

The model is $C = ___h + ___$. The call-out fee is $___$ pounds.

EXERCISE 4E

VARIATION · LINEAR MODELS · INTERPRETATION

APPLICATION

1 y varies directly with x. Find the constant and the missing value.

a. $y = 15$ when $x = 5$; find y when $x = 9$.

b. $y = 42$ when $x = 7$; find x when $y = 30$.

c. $y = 2.4$ when $x = 0.6$; find y when $x = 2$.

2 A taxi charges £4 plus £2 per kilometre. Write the fare function, state its gradient with units, and find the fare for 9 km.

3 A printer costs £120 plus £0.05 per page. Write a cost model and find the cost of printing 800 pages.

4 The table shows a linear relationship. Find the model and the missing value.

x	1	3	5	7
y	8	16	24	?

REASONING AND EVALUATION

5 Explain the difference between a linear relationship and a directly proportional one. Give one example of each.

6 A model predicts a delivery cost of £50 for 0 km. Is that reasonable? What does the intercept represent physically?

7 The population of a town is modelled by $P = 12000 + 350t$, where t is years after 2020. Interpret the gradient and intercept, and evaluate the model's reliability for $t = 100$.

8 **Extension.** A phone plan has equation $C = 18 + 0.04m$. A second plan has $C = 30 + 0.01m$. Find the number of minutes for which the costs are equal, and explain how the gradients determine which plan is cheaper for large m.

INVESTIGATION · CRITERION B

4.6 Modelling a fair fare

INVESTIGATION · CRITERION B

What makes a fare fair?

Two ride-hailing companies publish the following fares for journeys of different lengths. The distances are in kilometres; the fares are in pounds.

DISTANCE d	1	2	3	4	5	6	8
Company A	6.00	8.00	10.00	12.00	14.00	16.00	20.00
Company B	9.50	11.00	12.50	14.00	15.50	17.00	20.00

1. Plot both sets of data on the same axes. Are the points collinear? What does this tell you about the fare structure?
2. Find the gradient of each line. State its units and interpret it as a rate.
3. Find the intercept of each line. Interpret each intercept as a flag-fall or booking fee.
4. Write a function for each company's fare, and find the distance at which the fares are equal.
5. For each interval of distances, state which company is cheaper. Justify using both the graph and the functions.
6. Which fare structure is fairer for short journeys? Justify mathematically, then discuss one non-mathematical consideration.

Generalise and justify. Write the fare as $F(d) = f + rd$, where f is the flag-fall and r is the rate per kilometre. Show that the distance at which two companies charge the same fare is $d = \dfrac{f_2 - f_1}{r_1 - r_2}$. Explain why this formula requires $r_1 \neq r_2$, and why a larger flag-fall makes short journeys relatively more expensive. Test your formula on the data.

MODELLING TASK · CRITERION D PRACTICE

The courier's route

A courier depot is at $D(0, 0)$ on a coordinate grid measured in kilometres. A parcel is collected at $P(6, 2)$ and delivered to $Q(10, 6)$. Courier A charges £3 plus £1.20 per kilometre. Courier B charges £5 plus £0.90 per kilometre.

1. Plot the three points and draw the route $D \to P \to Q$. Find the total distance travelled, to 2 decimal places.
2. Find the midpoint of the direct route from D to Q. Explain what it represents.
3. Find the gradient of each leg of the route and interpret the values as directions on the grid.
4. Write a cost function for each courier and calculate the cost of the route.
5. The budget is £18. Which courier can complete the route within budget? What is the maximum distance each can cover for £18?
6. Recommend a courier, state two assumptions, and explain one factor other than cost that could change the decision.

Evidence to submit: a labelled coordinate diagram, calculations with units, both cost functions, the budget comparison, and a recommendation that distinguishes the model from the real delivery.

MIXED PRACTICE 4

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–9 check fluency; 10–15 require reasoning; 16–18 are multi-step or unfamiliar. Allow 55 minutes.

1. For $f(x) = 3x - 2$, find $f(4)$, $f(-1)$ and $f(0)$.
2. Solve $f(x) = 10$ for $f(x) = 3x - 2$.
3. State the domain of $g(x) = \dfrac{1}{x + 2}$.
4. Find the distance between $(1, 2)$ and $(5, 5)$.
5. Find the midpoint of $(-3, 4)$ and $(5, -2)$.
6. Find the gradient of the line through $(2, -1)$ and $(6, 7)$.
7. Find the equation of the line through $(0, 3)$ with gradient -2.
8. State the gradient of $4x - 2y = 8$.
9. Find the gradient of a line perpendicular to $2x + 5y = 10$.
10. **Error analysis.** A student finds the midpoint of $(2, 5)$ and $(6, 9)$ as $(4, 7)$ using differences. Explain why the answer happens to be correct here but the method is wrong.
11. y varies directly with x, and $y = 18$ when $x = 6$. Find y when $x = 10$.
12. A taxi fare is $C = 5 + 1.5d$. Interpret the gradient and intercept, and find the fare for 12 km.
13. Find the equation of the line through $(1, 2)$ parallel to $y = 3x - 4$.
14. Find the equation of the line through $(2, 1)$ perpendicular to $y = -\frac{1}{2}x + 3$.

MIXED PRACTICE 4 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

15 Classify the triangle with vertices $(0,0)$, $(4,0)$ and $(2,3)$.

16 A drone travels from $(-2,3)$ to $(6,9)$. Find the distance and the midpoint of the route.

17 **Extension.** Find k so that the line through $(k,3)$ and $(4,-1)$ has gradient 2.

18 **Extension.** Show that the midpoint of $A(x_1,y_1)$ and $B(x_2,y_2)$ is equidistant from A and B.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING

30 minutes · 30 marks

Functions and lines: short response

NO CALCULATOR | SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

19 For $f(x)=x^2-3x$, find $f(2)$ and $f(-1)$. [3]

20 State the domain of $h(x)=\dfrac{3}{x-5}$. [2]

21 Find the distance between $(-1,2)$ and $(5,10)$. [3]

22 Find the midpoint of $(3,-4)$ and $(7,2)$. [3]

23 Find the equation of the line through $(2,5)$ and $(6,-3)$. [4]

24 Find the equation of the line through $(4,-1)$ parallel to $y=2x+5$. [4]

25 Find the equation of the line through $(1,3)$ perpendicular to $y=-3x+2$. [4]

26 A model is $C=25+0.4m$. Interpret the gradient and intercept, and find C when $m=120$. [4]

27 **Error analysis.** A student finds the gradient through $(2,3)$ and $(5,9)$ as $\frac{5-2}{9-3}=\frac{1}{2}$. Identify the error and give the correct gradient. [3]

CRITERION B · INVESTIGATING PATTERNS — 45 minutes · 24 marks

A family of lines

Instructions. Work independently. Use a table, then answer in full sentences. Consider the lines $L_k : y = 2x + k$ for integer values of k.

1. Complete a table for $k = -2, -1, 0, 1, 2, 3$, showing the y-intercept and the x-intercept of each line. [5]
2. Describe the pattern of the x-intercepts in terms of k. [3]
3. Write a formula for the x-intercept of L_k, and verify it for two values of k. [4]
4. Show that the distance between the two intercepts of L_k is $\frac{\sqrt{5}}{2}|k|$. Explain each step. [6]
5. Find the area of the triangle formed by L_k and the axes, in terms of k. Describe how the area changes as k increases. [6]

Evidence: a complete table, a described pattern, a formula with verification, an algebraic derivation, and a generalised area result with interpretation.

CRITERION C · COMMUNICATING — 40 minutes · 20 marks

What gradient really measures

Instructions. Write a guide of 300–400 words for a student in the year below. Your guide must explain: (i) what gradient means as a rate of change; (ii) how to calculate it from a graph and from two points; (iii) what positive, negative, zero and undefined gradients look like; (iv) one worked example of finding the equation of a line; (v) one common error and how to avoid it. Include one labelled diagram.

Evidence: a titled guide with paragraphs, a labelled diagram, a worked example with a check, and a short “remember” box.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The clinic bus

Instructions. A rural health service maps three places on a coordinate grid measured in kilometres: a clinic at $C(2, 3)$, a market at $M(10, 9)$ and a school at $S(8, 3)$. A straight road joins the clinic and the market. A minibus will run from the clinic to a stop at the midpoint B of CM, then along a feeder road to the school. Firm A charges £5 plus £1.30 per kilometre. Firm B charges £7 plus £0.90 per kilometre. The service has £15 per trip.

1. Plot C, M and S on a coordinate grid, draw the road CM, and calculate its length. [4]
2. Find the coordinates of B, the midpoint of CM, and explain why a stop there is equally distant from the clinic and the market. [3]
3. Find the gradient of CM and interpret it in the context of the map. [3]
4. A feeder road from the school is perpendicular to CM. Find its gradient and its equation. [5]
5. Calculate the length of the route $C \to B \to S$, correct to 2 decimal places. Find the cost of the route with each firm and determine which firm stays within the £15 budget. [5]
6. Recommend a firm. Justify your recommendation using your results, state two assumptions, and explain one non-cost factor that could change the decision. [4]

Evidence: a labelled coordinate diagram, distance and midpoint calculations, the gradient with units, the perpendicular equation, both cost functions, a budget comparison, and a justified recommendation.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A function assigns exactly one output to each input. Function notation $f(x)$ names the output; the domain is the set of allowed inputs.
- Distance is $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$; midpoint is the average of the coordinates.
- Gradient is vertical change divided by horizontal change: $m=\frac{y_2-y_1}{x_2-x_1}$.
- A line with gradient m through (x_1,y_1) is $y-y_1=m(x-x_1)$; rearranged, $y=mx+c$.
- Parallel lines have equal gradients. Perpendicular gradients satisfy $m_1m_2=-1$.
- A linear model $y=mx+c$ has rate m and starting value c. Direct variation has $c=0$; extrapolation beyond the data is unsafe.

$$AB=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \qquad M\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right) \qquad m=\frac{y_2-y_1}{x_2-x_1}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- I can evaluate a function using correct notation.
- I can solve an equation written in function notation.
- I can state the domain of a function with a denominator or square root.
- I can calculate the distance between two points.
- I can find the midpoint of a segment.
- I can find a missing endpoint from a midpoint.
- I can calculate a gradient from two points.
- I can write the equation of a line from a point and a gradient.
- I can rearrange a line equation to find its gradient and intercept.
- I can identify parallel lines from their gradients.
- I can find a perpendicular gradient using the negative reciprocal.
- I can build a linear model from a table.
- I can interpret gradient and intercept with units.
- I can recognise direct variation.
- I can explain why extrapolation may be unreliable.
- I can check a point lies on a line by substitution.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which representation — rule, table, graph or words — helped you most in this chapter? Give one example where changing representation made the answer visible.
2. Describe one error you made with signs in a gradient or midpoint calculation. What check will you use next time?
3. The statement of inquiry says that a graph turns structure into visible geometry. Explain this using the fare comparison.
4. Look ahead to Chapter 5: how might a quadratic function differ from the linear functions studied here?

ROOTS, FACTORS AND THE SHAPE OF A QUESTION

Quadratics and Equations

05

FACTOR · ROOT · INTERCEPT

Expanded, factorised, completed and solved — four views of one relationship, and what each view makes visible.

STATEMENT OF INQUIRY

A quadratic can be written as an expanded expression, a product of factors or a completed square. Because these forms are equivalent, each reveals different structure — and the equation they define asks where that structure meets zero.

KEY CONCEPT

Logic

RELATED CONCEPTS

Equivalence · Generalisation · Justification

GLOBAL CONTEXT

Scientific and technical innovation — design, motion and optimisation

FACTUAL

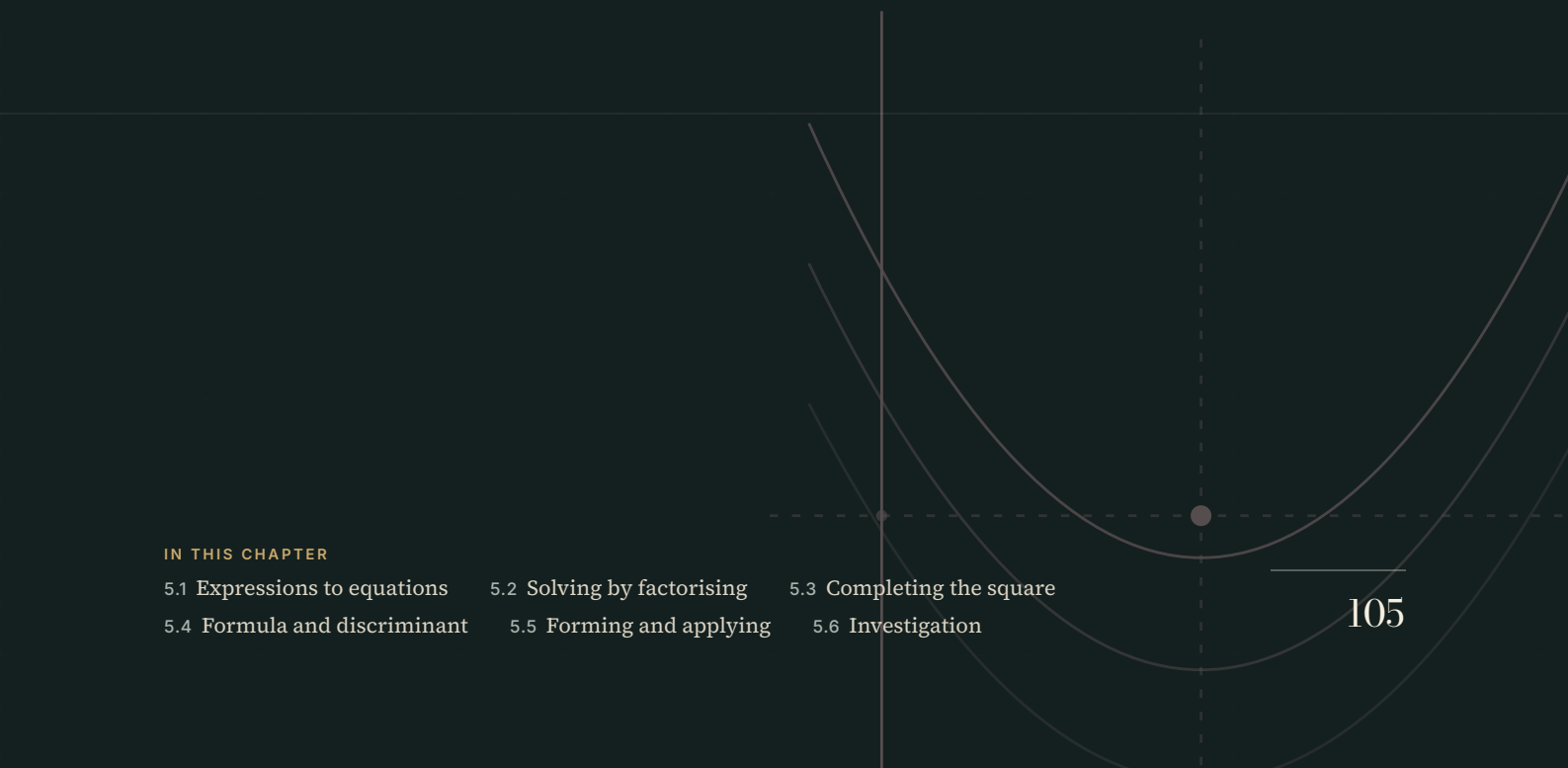
What does each form of a quadratic expression reveal about its graph and its solutions?

CONCEPTUAL

Why can a quadratic equation have two, one or no real solutions?

DEBATABLE

When is an exact surd more useful than a decimal approximation?



IN THIS CHAPTER

5.1 Expressions to equations
5.2 Solving by factorising
5.3 Completing the square
5.4 Formula and discriminant
5.5 Forming and applying
5.6 Investigation

OPENING PROBLEM · THE SCHOOL FARM PEN

Twenty-six metres of fencing

The school farm has 26 m of fencing to make a rectangular pen. The pen must use all the fencing, and its width w metres can be chosen. The area is A square metres.

1. Complete a table for $w = 1, 2, 3, 4, 5, 6$ and describe how A changes.
2. Write A as a function of w, first as a product and then in expanded form.
3. The farm committee wants an area of $40\ \text{m}^2$. Write an equation and solve it. Why do two different widths give the same pen?
4. Use a sketch or a table to estimate the greatest possible area. Then prove your estimate by completing the square.
5. A second design requires an area of at least $36\ \text{m}^2$. Solve $w(13 - w) \geq 36$ and interpret the solution on a number line.
6. Which width would you recommend, and what does the algebra not tell you about the choice?

A quadratic is the first relationship in which the rate of change is itself changing. Its graph can meet the ground twice, once or not at all, and each meeting is a solution of an equation. Factorising, completing the square and the quadratic formula are three equivalent routes to those meetings. Choosing between them is a decision about structure, not a matter of taste.

5.0 Learning intentions

ORIENTATION

- Move confidently between expanded, factorised and completed-square forms of a quadratic.
- Solve quadratic equations by factorising, by completing the square and by the formula.
- Use the discriminant to decide how many real solutions an equation has.
- Form a quadratic equation from a description and reject solutions that do not fit the context.
- Justify the choice of method and check solutions by substitution.
- Explain what the solutions mean on the graph and in the situation.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

expand and factorise a quadratic confidently; write an equation in standard form and identify its coefficients; solve a quadratic by factorising, completing the square and the formula, giving exact answers where required; calculate and interpret the discriminant; form and solve a quadratic from a real context and reject inadmissible roots; and explain why equivalent forms give the same solutions.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, read the refresher box before Section 5.1.

1. Expand $(x + 3)(x - 4)$.
2. Factorise $x^2 - 5x + 6$.
3. Solve $3x - 7 = 11$.
4. Evaluate $\sqrt{49}$ and $5^2 - 4(2)(3)$.
5. Expand $(x - 5)^2$.
6. Solve $x^2 = 16$.

REFRESHER

Expanding means multiplying every term in one bracket by every term in the other, then collecting like terms. Factorising reverses expansion: look first for a common factor, then for two numbers whose product is the constant term and whose sum is the coefficient of x. An equation is solved when the variable is isolated. The square of a negative number is positive, so $x^2 = 16$ has two solutions, 4 and -4.

ATL SKILLS · THINKING AND COMMUNICATION

Critical thinking: inspect the coefficients before choosing a method; a factorable quadratic is usually quicker by inspection. **Transfer:** recognise the same product–sum structure in arithmetic, algebra and area diagrams. **Communication:** show the step where each factor is set to zero, state whether a root is rejected, and give units with every contextual answer.

SAME OBJECT, DIFFERENT QUESTIONS

5.1 From expressions to equations

An expression such as $x^2 - 5x + 6$ is a recipe for a number: choose x, substitute, and calculate the result. An equation such as $x^2 - 5x + 6 = 0$ is a question: for which choices of x is the result zero? The expression can be evaluated at every value; the equation is true only at particular values. Confusing the two is the most common source of lost marks in this chapter.

Every quadratic equation can be rearranged into the standard form $ax^2 + bx + c = 0$. The rearranging must be done before anything is factorised or substituted into the formula, because the coefficients a, b and c are read from that form. A solution is also called a root, and a root corresponds to a point where the graph crosses the horizontal axis.

DEFINITIONS

Quadratic equation, standard form and root

A **quadratic equation** in one variable can be written as $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$. This is **standard form**. A **root** (or solution) is a value of x that makes the equation true. A quadratic equation has at most two real roots.

EXAMPLE 5.1 EXPANDING AND COLLECTING LIKE TERMS

QUESTION

Expand and simplify $(2x - 3)(x + 4)$.

SOLUTION

1. Multiply each term in the first bracket by each term in the second: $2x(x + 4) - 3(x + 4)$.
2. Distribute: $2x^2 + 8x - 3x - 12$.
3. Collect like terms: $2x^2 + 5x - 12$.

Check: substitute $x = 1$. The product is $(2 - 3)(5) = -5$; the expansion gives $2 + 5 - 12 = -5$.

EXAMPLE 5.2 REARRANGING INTO STANDARD FORM

QUESTION

Write $3x(x - 2) = 5$ in standard form and state a, b and c.

SOLUTION

1. Expand the left side: $3x^2 - 6x = 5$.
2. Subtract 5 from both sides: $3x^2 - 6x - 5 = 0$.
3. Compare with $ax^2 + bx + c = 0$: $a = 3, b = -6, c = -5$.

Sign care: the coefficient b is -6, not 6. The formula and the discriminant both depend on the sign.

EXAMPLE 5.3 VERIFYING A ROOT AND FINDING THE OTHER

QUESTION

Show that $x = -2$ is a root of $x^2 + 5x + 6 = 0$, then find the other root.

SOLUTION

1. Substitute: $(-2)^2 + 5(-2) + 6 = 4 - 10 + 6 = 0$. The equation is satisfied.
2. Factorise: two numbers with product 6 and sum 5 are 2 and 3, so $x^2 + 5x + 6 = (x + 2)(x + 3)$.
3. If $(x + 2)(x + 3) = 0$, then $x + 2 = 0$ or $x + 3 = 0$. The roots are -2 and -3.

Check: $(-3)^2 + 5(-3) + 6 = 9 - 15 + 6 = 0$. Both roots are confirmed.

EXAMPLE **5.4** SOLVING BY INVERSE OPERATIONS

QUESTION

Solve each equation: (a) $x^2 = 25$ (b) $2x^2 - 7 = 11$ (c) $3x^2 + 4 = 4$.

SOLUTION

a Take square roots of both sides: $x = \pm 5$. Both values square to 25.

b Add 7, then divide by 2: $2x^2 = 18$, so $x^2 = 9$ and $x = \pm 3$.

c Subtract 4: $3x^2 = 0$, so $x^2 = 0$ and $x = 0$. There is only one root, repeated.

> **Why $\pm$?** The equation $x^2 = 25$ asks for every number whose square is 25. Both 5 and -5 qualify, so writing only $x = 5$ loses a solution.

KEY IDEA

Solving is not the same as simplifying. An expression can be rewritten endlessly without changing its value; an equation is solved by producing the finite set of values that make it true. Every step in a solution should be reversible, so that no solution is gained or lost.

COMMON MISCONCEPTIONS

"$x^2 = 9$ means $x = 3$." It means $x = 3$ or $x = -3$. The square root symbol $\sqrt{9}$ denotes the non-negative root, but the equation has two solutions.

"I can cancel x." Dividing by x is only valid if $x \neq 0$, and $x = 0$ may be a solution. Factorise instead; see Section 5.2.

"$(x + 4)^2 = x^2 + 16$." Squaring a bracket gives a middle term: $(x + 4)^2 = x^2 + 8x + 16$. Test with $x = 1$: $25 \neq 17$.

"An equation and its expression have the same solutions." An expression has values, not solutions. Only the equation has roots.

GUIDED PRACTICE · FROM EXPRESSION TO EQUATION

Expand $(x - 7)(x + 2) =$ _____$x^2 +$ _____$x -$ _____.

Write the equation $x^2 - 5x - 14 = 0$ in factorised form: (_____)(_____) $= 0$.

The roots are $x =$ _____ and $x =$ _____. Check one root by substitution.

EXERCISE 5A

EXPRESSIONS · STANDARD FORM · ROOTS

FLUENCY

1 Expand and simplify.

a. $(x + 5)(x - 2)$

b. $(2x - 1)(x + 3)$

c. $(3x - 2)(2x + 5)$

2 Factorise.

a. $x^2 + 7x + 10$

b. $x^2 - 6x + 8$

c. $3x^2 - 15x$

EXERCISE 5A · CONTINUED

EXPRESSIONS · STANDARD FORM · ROOTS

3 Write in standard form and state a, b and c.

a. $x(2x - 3) = 4$

b. $(x - 1)(x + 4) = 2x + 3$

c. $5 - 2x^2 = x$

4 Solve.

a. $x^2 = 36$

b. $x^2 - 5 = 0$

c. $2x^2 - 8 = 0$

UNDERSTANDING AND REASONING

5 Verify that $x = 2$ is a root of $x^2 - x - 2 = 0$, then find the other root.

6 Explain the difference between "factorise $x^2 - 5x + 6$" and "solve $x^2 - 5x + 6 = 0$".

7 **Error analysis.** A student writes $(x + 4)^2 = x^2 + 16$. Expand correctly and explain the error.

8 How many solutions has $x^2 = 0$? Explain your answer using a graph or a factorisation.

APPLICATION AND EXTENSION

9 A square has area $49\ \text{cm}^2$. Find its side length. If the area is $50\ \text{cm}^2$, write the exact side length.

10 A rectangle has area $x^2 + 7x + 12$. Factorise to find expressions for its possible side lengths.

11 A ball thrown upwards has height $h = 15t - 5t^2$ metres after t seconds. Solve $h = 0$ and interpret both solutions.

12 **Extension.** The equation $(x - 1)(2x + 3) = 0$ has a root $x = 1$. Find the other root and verify it by substitution.

A PRODUCT THAT EQUALS ZERO

5.2 Solving by factorising

The zero product property is the bridge between factorising and solving. If two numbers multiply to zero, at least one of them must be zero. This is special to zero: a product can equal 12 in many ways, but a product equals zero only when a factor vanishes.

The method is therefore: rearrange the equation so that one side is zero; factorise the other side completely; set each factor equal to zero; solve the resulting linear equations. Each factor produces one root. When a factor repeats, the two roots coincide — the equation has a repeated root.

PROPERTY

Zero product property

If $pq = 0$, then $p = 0$ or $q = 0$ (or both). Consequently, if a quadratic factorises as $a(x - r)(x - s) = 0$ with $a \neq 0$, its roots are $x = r$ and $x = s$.

EXAMPLE **5.5** A MONIC QUADRATIC

QUESTION

Solve $x^2 - 7x + 12 = 0$.

SOLUTION

1. Find two numbers with product 12 and sum -7: they are -3 and -4.
2. Factorise: $(x - 3)(x - 4) = 0$.
3. Set each factor to zero: $x - 3 = 0$ or $x - 4 = 0$.
4. Hence $x = 3$ or $x = 4$.

Check: $3^2 - 7(3) + 12 = 9 - 21 + 12 = 0$ and $4^2 - 28 + 12 = 0$.

EXAMPLE **5.6** A NON-MONIC QUADRATIC

QUESTION

Solve $2x^2 + 5x - 3 = 0$.

SOLUTION

1. The product of the x^2 coefficient and the constant is $2 \times (-3) = -6$. Two numbers with product -6 and sum 5 are 6 and -1.
2. Split the middle term: $2x^2 + 6x - x - 3 = 0$.
3. Factor by grouping: $2x(x + 3) - 1(x + 3) = 0$, so $(2x - 1)(x + 3) = 0$.
4. Hence $x = \frac{1}{2}$ or $x = -3$.

Check: $x = \frac{1}{2}$: $2(\frac{1}{4}) + \frac{5}{2} - 3 = \frac{1}{2} + \frac{5}{2} - 3 = 0$.

EXAMPLE **5.7** DO NOT DIVIDE BY THE VARIABLE

QUESTION

Solve $3x^2 = 12x$.

SOLUTION

1. Rearrange: $3x^2 - 12x = 0$.
2. Take out the common factor: $3x(x - 4) = 0$.
3. Set each factor to zero: $3x = 0$ gives $x = 0$; $x - 4 = 0$ gives $x = 4$.

> **Why not divide by x?** Dividing by x assumes $x \neq 0$, which is exactly the solution that would be lost. Always factorise instead.

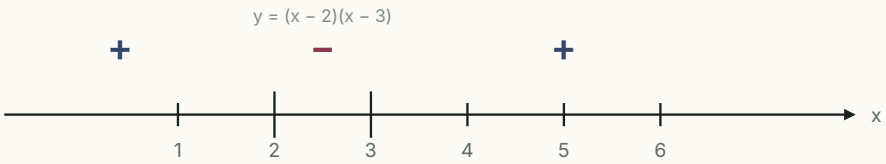


Figure 5.1 The sign of $(x-2)(x-3)$. The product is zero at the roots $x=2$ and $x=3$, negative between them and positive outside them.

KEY IDEA

A factorised quadratic displays its roots directly. The sign diagram shows that the graph is below the axis between the roots and above it outside them — a pattern that will become the parabola in Chapter 6. Factorising is the fastest method when the roots are rational.

COMMON MISCONCEPTIONS

"$(x-5)(x+5)=x=5$." Each factor can be zero, so $x=5$ or $x=-5$. Setting only one factor to zero gives half the answer.

"A repeated factor gives one root, so I made an error." $(x-3)^2=0$ has the repeated root $x=3$. One value, counted twice.

"$x^2=7x$ so $x=7$." Divide only by a non-zero number. Rearranging gives $x(x-7)=0$, so $x=0$ or $x=7$.

"I can factorise before making one side zero." The zero product property needs a product equal to zero. Move every term to one side first.

GUIDED PRACTICE · FACTORISE AND SOLVE

$x^2+2x-15=(x+___)(x-___)=0$

So $x=$ _____ or $x=$ _____.

$2x^2-8x=0$ factors as _____(_____) $=0$, giving $x=$ ___ or $x=$ ___.

EXERCISE 5B

FACTORISING · ZERO PRODUCT · ROOTS

FLUENCY

1 Solve.
a. $x^2-5x+6=0$
b. $x^2+3x-10=0$
c. $x^2-16=0$
d. $x^2+6x+9=0$

2 Solve.
a. $2x^2-7x+3=0$
b. $3x^2+10x-8=0$
c. $4x^2-9=0$

3 Solve.
a. $x^2=5x$
b. $2x^2+6x=0$
c. $3x^2=27x$

4 Solve.
a. $2x^2-8x+8=0$
b. $x^2+12x+36=0$
c. $5x^2=20$

UNDERSTANDING AND REASONING

EXERCISE 5B · CONTINUED FACTORISING · ZERO PRODUCT · ROOTS

5 **Error analysis.** A student solves $x^2 = 7x$ by dividing by x and obtains $x = 7$. Explain what is missing and give both roots.

6 Explain why $(x - 5)(x + 5) = 0$ gives the roots 5 and -5, but $(x - 5)^2 = 0$ gives only 5.

7 Write a quadratic equation whose roots are 4 and -2. Is your answer unique? Explain.

8 Write a quadratic equation with a repeated root at $x = 3$.

APPLICATION AND EXTENSION

9 A rectangle has dimensions x and $x + 3$ metres and area 54 m^2. Form and solve an equation to find the dimensions.

10 A number squared plus five times the number equals 24. Find the possible numbers.

11 Solve $(2x - 1)(x + 3) = 3x$. Give exact answers.

12 **Extension.** Solve $x^4 - 5x^2 + 4 = 0$ by treating x^2 as the unknown. How many real roots does the equation have?

BUILDING A PERFECT SQUARE

5.3 Completing the square

Factorising only works when the roots are rational. Completing the square works always, and it exposes the maximum or minimum of the expression at the same time. The idea is to rewrite $x^2 + bx$ as the square of a bracket plus or minus a constant.

Half of b becomes the number inside the bracket: $x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$. The subtraction is essential; it restores the value that squaring added. Once the expression is a square plus a constant, the equation can be solved by taking square roots.

DEFINITION

Completed-square form

A quadratic is in **completed-square form** when it is written as $a(x - h)^2 + k$. The expression is a perfect square when $k = 0$. For $a > 0$, the minimum value is k, occurring at $x = h$; for $a < 0$, the maximum value is k.

EXAMPLE **5.8** COMPLETING THE SQUARE TO SOLVE

QUESTION

Solve $x^2 + 6x - 7 = 0$ by completing the square.

SOLUTION

1 Take half of 6, which is 3: $x^2 + 6x = (x + 3)^2 - 9$.

2 Substitute: $(x + 3)^2 - 9 - 7 = 0$, so $(x + 3)^2 = 16$.

3 Take square roots: $x + 3 = \pm 4$.

4 Hence $x = -3 + 4 = 1$ or $x = -3 - 4 = -7$.

Check: $1 + 6 - 7 = 0$ and $49 - 42 - 7 = 0$. The factorised form is $(x - 1)(x + 7)$.

EXAMPLE **5.9** AN IRRATIONAL RESULT

QUESTION

Solve $x^2 - 4x + 1 = 0$, giving exact answers.

SOLUTION

1 Half of -4 is -2: $x^2 - 4x = (x - 2)^2 - 4$.

2 The equation becomes $(x - 2)^2 - 4 + 1 = 0$, so $(x - 2)^2 = 3$.

3 Take square roots: $x - 2 = \pm\sqrt{3}$.

4 Hence $x = 2 + \sqrt{3}$ or $x = 2 - \sqrt{3}$.

> **Exact or decimal?** $x \approx 3.732$ or 0.268. The exact surd retains the structure and avoids rounding error; use a decimal only when the context asks for one.

EXAMPLE **5.10** WHEN THE LEADING COEFFICIENT IS NOT 1

QUESTION

Solve $2x^2 - 8x + 3 = 0$, giving exact answers.

SOLUTION

1 Divide every term by 2: $x^2 - 4x + \frac{3}{2} = 0$.

2 Complete the square: $(x - 2)^2 - 4 + \frac{3}{2} = 0$, so $(x - 2)^2 = \frac{5}{2}$.

3 Take square roots: $x - 2 = \pm\sqrt{\frac{5}{2}} = \pm\frac{\sqrt{10}}{2}$.

4 Hence $x = 2 \pm \frac{\sqrt{10}}{2} \approx 3.58$ or 0.42.

Check the structure: $\sqrt{\frac{5}{2}} = \frac{\sqrt{5}}{\sqrt{2}} = \frac{\sqrt{10}}{2}$.

EXAMPLE **5.11** READING A MAXIMUM OR MINIMUM

QUESTION

Write $y = x^2 + 10x + 3$ in completed-square form. Hence state the minimum value of y and the value of x at which it occurs.

SOLUTION

1 Half of 10 is 5: $x^2 + 10x = (x + 5)^2 - 25$.

2 So $y = (x + 5)^2 - 25 + 3 = (x + 5)^2 - 22$.

3 A square is never negative, so $(x + 5)^2 \geq 0$ and $y \geq -22$.

4 The minimum $y = -22$ occurs when $x + 5 = 0$, that is $x = -5$.

> **Graph preview:** the point $(-5, -22)$ is the lowest point of the parabola. Functions and their graphs are developed fully in Chapter 6.

KEY IDEA

Completing the square is a rewriting, not a trick: it produces an equivalent expression. The constant k in $a(x-h)^2+k$ records the extreme value, and the bracket $x-h$ records where that extreme occurs. It also proves that a square cannot be negative — the basis of the discriminant in Section 5.4.

COMMON MISCONCEPTIONS

"$(x+3)^2 = x^2+9$." The square adds $6x$. Completing the square does not add a number inside the bracket; it subtracts the square of half the coefficient outside.

"$x^2+6x+4=0$ gives $(x+3)^2=4$." It gives $(x+3)^2=5$, because $9-4=5$. The constant moves to the other side when solving.

"I can leave the coefficient of x^2 alone." Divide by it first, every term. Otherwise the bracket does not square correctly.

GUIDED PRACTICE · COMPLETE THE SQUARE

$x^2+8x = (x+___)^2 - ____$.

Hence $x^2+8x+7=0$ becomes $(x+___)^2 = ___$, so $x = ____$ or $x = ____$.

Write $y = x^2-6x+11$ as $(x-___)^2+___$. The minimum value is $____$.

EXERCISE 5C

COMPLETING THE SQUARE · EXACT ROOTS

FLUENCY

1 Write in completed-square form.
   a. x^2+8x
   b. x^2-6x
   c. $x^2+10x+7$

2 Solve by completing the square.
   a. $x^2+8x+7=0$
   b. $x^2-6x+5=0$
   c. $x^2+2x-5=0$

3 Solve, giving exact answers.
   a. $x^2-4x-2=0$
   b. $2x^2-8x+3=0$
   c. $3x^2+6x-1=0$

4 Write $y=x^2-12x+30$ in completed-square form and state its minimum value.

UNDERSTANDING AND REASONING

5 **Error analysis.** A student solves $x^2+6x+4=0$ as $(x+3)^2=4$, giving $x=-1$ or $x=-5$. Identify the error and solve correctly.

6 Explain why completing the square can solve $x^2+6x+2=0$ even though the expression does not factorise with integers.

7 Show that $x^2+2x+5=0$ has no real solution by completing the square.

APPLICATION AND EXTENSION

8 A rectangular pen uses 20 m of fencing on all four sides. Show that its area is $A = 25-(w-5)^2$, and find the maximum possible area.

9 **Extension.** Find the values of k for which $x^2+kx+16$ is a perfect square.

A FORMULA FOR EVERY CASE

5.4 The quadratic formula and discriminant

Completing the square on the general equation $ax^2 + bx + c = 0$ produces a formula that solves every quadratic. Divide by a, complete the square, and isolate x. The result is worth memorising because it is universal: any quadratic, rational or irrational roots, one root or none.

The expression under the square root decides the nature of the roots before any root is calculated. It is called the discriminant, because it discriminates between the three possible cases: positive, zero or negative.

DEFINITIONS

The quadratic formula and the discriminant

For $ax^2 + bx + c = 0$ with $a \neq 0$, the solutions are given by the **quadratic formula**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The **discriminant** is $\Delta = b^2 - 4ac$. If $\Delta > 0$ there are two distinct real roots; if $\Delta = 0$ there is one repeated real root; if $\Delta < 0$ there are no real roots.

EXAMPLE 5.12 USING THE FORMULA WITH A SURD

QUESTION

Solve $x^2 - 4x - 1 = 0$, giving exact answers.

SOLUTION

1 Here $a = 1, b = -4, c = -1$. The discriminant is $(-4)^2 - 4(1)(-1) = 16 + 4 = 20$.

2 Substitute: $x = \frac{4 \pm \sqrt{20}}{2}$.

3 Simplify the surd: $\sqrt{20} = 2\sqrt{5}$, so $x = \frac{4 \pm 2\sqrt{5}}{2} = 2 \pm \sqrt{5}$.

Check: $2 + \sqrt{5} \approx 4.236$. Substitution gives $17.944 - 16.944 - 1 = 0$ to three decimal places.

EXAMPLE 5.13 RATIONAL ROOTS FROM THE FORMULA

QUESTION

Solve $3x^2 + 5x - 2 = 0$.

SOLUTION

1 $a = 3, b = 5, c = -2$. The discriminant is $25 - 4(3)(-2) = 25 + 24 = 49$.

2 Substitute: $x = \frac{-5 \pm \sqrt{49}}{6} = \frac{-5 \pm 7}{6}$.

3 Hence $x = \frac{2}{6} = \frac{1}{3}$ or $x = \frac{-12}{6} = -2$.

Check: the factorised form is $(3x - 1)(x + 2)$, confirming the roots.

EXAMPLE **5.14** USING THE DISCRIMINANT FIRST

QUESTION

How many real roots has $2x^2 - 4x + 5 = 0$? Do not solve the equation.

SOLUTION

1. $a = 2, b = -4, c = 5$: $\Delta = (-4)^2 - 4(2)(5) = 16 - 40 = -24$.
2. Because $\Delta < 0$, there are no real roots.
3. Completing the square shows why: $2(x - 1)^2 + 3 = 0$, and a square cannot equal a negative number.

Interpretation: the graph of $y = 2x^2 - 4x + 5$ lies entirely above the x-axis; it never meets it.

EXAMPLE **5.15** CHOOSING A PARAMETER FOR ONE ROOT

QUESTION

Find the values of k for which $x^2 + kx + 9 = 0$ has exactly one real root.

SOLUTION

1. Exactly one real root means $\Delta = 0$: $k^2 - 4(1)(9) = 0$.
2. So $k^2 = 36$, giving $k = 6$ or $k = -6$.
3. Check $k = 6$: $x^2 + 6x + 9 = (x + 3)^2$, a repeated root at $x = -3$. The case $k = -6$ similarly gives $(x - 3)^2$.

Both signs: a parameter question usually has two answers because the discriminant depends on k^2.

KEY IDEA

The formula is a consequence of completing the square, not a separate fact. The discriminant is the part of the formula that cannot be simplified without knowing the numbers: it determines how many times the parabola meets the axis, while the formula determines where.

COMMON MISCONCEPTIONS

"The formula uses b, not $-b$." The numerator begins with $-b$. With $b = -4$, $-b = 4$, which is positive — a frequent sign slip.

"Divide by $2a$ only at the end." Both terms in the numerator are divided by $2a$. Write the fraction clearly before cancelling.

"$\Delta < 0$ means there is no equation." The equation exists but has no *real* solutions. In this course we say it has no real roots.

"Use the formula before rearranging." The coefficients must be read from standard form, with zero on one side.

GUIDED PRACTICE · FORMULA AND DISCRIMINANT

For $x^2 - 6x + 4 = 0$: $a =$ ___, $b =$ ___, $c =$ ___.

$\Delta = (-6)^2 - 4(1)(4) =$ _____, so there are _____ real roots.

$x = \frac{6 \pm \sqrt{}}{2} =$ _____.

EXERCISE 5D

FORMULA · DISCRIMINANT · EXACT VALUES

FLUENCY

1 Solve using the formula.

a. $x^2 - 6x + 4 = 0$

b. $x^2 + 2x - 1 = 0$

c. $2x^2 - 3x - 1 = 0$

2 Find the discriminant and state the number of real roots.

a. $x^2 - 5x + 6 = 0$

b. $x^2 + 4x + 4 = 0$

c. $3x^2 - 2x + 1 = 0$

3 Solve.

a. $3x^2 + 5x - 2 = 0$

b. $x^2 - 4x - 1 = 0$

c. $4x^2 - 4x - 1 = 0$

4 Solve $2x^2 + 3x - 4 = 0$, giving answers correct to 2 decimal places.

UNDERSTANDING AND REASONING

5 Find k so that $x^2 + kx + 16 = 0$ has exactly one real root.

6 Find the values of k for which $2x^2 + 4x + k = 0$ has no real roots.

7 **Error analysis.** A student solves $x^2 + 3x - 1 = 0$ as $x = \frac{-3 \pm \sqrt{5}}{2}$. Identify the error and give the correct exact roots.

8 Show that using the formula on $x^2 - 7x + 12 = 0$ produces the same roots as factorising.

9 Which method would you choose for each equation? Justify in one sentence.

a. $2x^2 + 5x - 3 = 0$

b. $x^2 + 4x - 3 = 0$

c. $x^2 - 10x + 25 = 0$

APPLICATION AND EXTENSION

10 A rectangle has sides $x + 1$ and $x - 2$ and area 5 cm^2. Form an equation and find the exact value of x.

11 A ball's height is $h = 30t - 5t^2$ metres. Find the two times at which $h = 40$ m, and interpret them.

12 **Extension.** The equation $x^2 + bx + 9 = 0$ has two distinct real roots. Find the possible positive integer values of b below 10, and explain what happens for negative b.

FROM SITUATION TO EQUATION

5.5 Forming and applying equations

A quadratic model begins with a variable and a relationship, not with a formula. Define the variable in words and units, express the relationship, rearrange to standard form, solve, and then return to the situation. The last step is not optional: a negative length or a negative time may be a valid root of the equation but not a valid answer to the question.

Checking is part of modelling. Substitute the accepted root into the original description, confirm the units, and ask whether the decimal answer makes sense in context. If two roots are both admissible, the context may still prefer one — and that preference needs a reason.

EXAMPLE **5.16** CONSECUTIVE ODD INTEGERS

QUESTION

Two positive consecutive odd integers have a product of 143. Find the integers.

SOLUTION

1. Let the smaller integer be n. The next odd integer is $n + 2$.
2. Form the equation: $n(n + 2) = 143$, so $n^2 + 2n - 143 = 0$.
3. Factorise: $(n - 11)(n + 13) = 0$, so $n = 11$ or $n = -13$.
4. The integers are positive, so $n = 11$. The pair is 11 and 13.

Check: $11 \times 13 = 143$. The negative root would give -13 and -11, also a valid pair for the equation but excluded by "positive".

EXAMPLE **5.17** A RECTANGLE WITH A CONDITION

QUESTION

A rectangle is 3 m longer than it is wide and has area $40\ \text{m}^2$. Find its dimensions.

SOLUTION

1. Let the width be w metres. The length is $w + 3$ metres.
2. Area: $w(w + 3) = 40$, so $w^2 + 3w - 40 = 0$.
3. Factorise: $(w - 5)(w + 8) = 0$, so $w = 5$ or $w = -8$.
4. A width cannot be negative, so $w = 5$. The length is 8 m.

Check: $5 \times 8 = 40$ and $8 - 5 = 3$. Both conditions hold.

EXAMPLE **5.18** A PROJECTILE

QUESTION

A ball is thrown upwards. Its height after t seconds is $h = 20t - 5t^2$ metres. (a) Find when the ball is 15 m above the ground. (b) Find when it lands. (c) Find its maximum height.

SOLUTION

a Solve $20t - 5t^2 = 15$. Rearrange: $5t^2 - 20t + 15 = 0$, so $t^2 - 4t + 3 = 0$. Then $(t - 1)(t - 3) = 0$, giving $t = 1$ or $t = 3$. The ball passes 15 m on the way up and again on the way down.

b Landing means $h = 0$: $5t(4 - t) = 0$, so $t = 0$ (the throw) or $t = 4$ s.

c Complete the square: $h = -5(t^2 - 4t) = -5((t - 2)^2 - 4) = 20 - 5(t - 2)^2$. The maximum is 20 m at $t = 2$ s.

Check: at $t = 2$, $h = 40 - 20 = 20$. The maximum lies halfway between the two times at the same height, $t = 1$ and $t = 3$.

KEY IDEA

Every contextual quadratic answer needs three things: the algebraic root, the contextual test that accepts or rejects it, and the unit. A root is a number; an answer is a number with a meaning.

COMMON MISCONCEPTIONS

"Both roots are answers." Negative lengths, negative times and fractional people are usually inadmissible. State why you reject a root.

"The maximum is a root." The maximum height is not found by solving $h = 0$; it is the extreme value found by completing the square or by symmetry.

"Round early." Keep exact values until the final line. Rounding a surd too soon can change the accepted root or the check.

GUIDED PRACTICE · FORM THE EQUATION

A rectangle is 4 cm longer than its width and has area 96 cm^2. Let the width be w.

Equation: $w($_____$) = 96$, so $w^2 +$ ____$w -$ ____ $= 0$.

Factorise and solve: $w =$ ____ (reject _____). The dimensions are _____ cm by _____ cm.

EXERCISE 5E

MODELLING · CONSTRAINTS · INTERPRETATION

APPLICATION

1 Two consecutive integers have product 56. Find both possible pairs and explain why each satisfies the wording.

2 Two positive consecutive odd integers have product 99. Find the integers.

3 A rectangle is 4 m longer than it is wide and has area 96 m^2. Find its dimensions.

EXERCISE 5E · CONTINUED

MODELLING · CONSTRAINTS · INTERPRETATION

4 A right-angled triangle has legs x and $x+1$, and hypotenuse $x+2$. Form an equation and find the three side lengths.

5 A ball's height is $h = 25t - 5t^2$ metres. Find the times when it is 20 m high, and its maximum height.

6 A number plus its reciprocal equals $\frac{5}{2}$. Find the possible numbers.

7 A fence of 40 m is used for three sides of a rectangular pen; the fourth side is a wall. The area is 150 m^2. Find the possible dimensions and recommend one.

8 **Extension.** A number and three less than twice the number have product 35. Form an equation and find the possible values.

MODELLING TASK · CRITERION D PRACTICE

The market garden

A market garden uses 30 m of fencing for a rectangular plot against an existing wall, so only three sides need fencing. The width is w metres and the plot is $30 - 2w$ metres deep.

1 Write the area $A(w)$ and state the values of w that make sense.

2 Complete the square to find the width that maximises the area, and the maximum area.

3 The grower wants an area of 100 m^2. Solve $A(w) = 100$ and interpret both solutions.

4 Fencing costs £12 per metre and topsoil costs £18 per square metre. Find the cost of the maximum-area design.

5 The budget is £2,500. Is the maximum-area design affordable? What area can the budget support if fencing is fixed at 30 m?

6 State two assumptions in the model and one practical factor it ignores.

INVESTIGATION · CRITERION B

5.6 The discriminant detective

The discriminant is a single number that predicts the number of real roots. In this investigation you will gather evidence, look for a pattern, and then explain why the pattern must hold for every quadratic — not only the examples you tried.

INVESTIGATION · CRITERION B

How many times does a parabola meet the axis?

Consider the family $x^2 + bx + 9 = 0$, where b is an integer.

b	−10	−8	−6	−4	0	4	6	8	10
$\Delta = b^2 - 36$									
Number of real roots									

1. Copy and complete the table. [4]
2. Describe the pattern in the number of roots as b changes. [3]
3. Predict the number of roots for $b = 7$, $b = -6$ and $b = 0$. Test each prediction. [4]
4. Generalise: for which values of b does $x^2 + bx + 9 = 0$ have two, one or no real roots? Justify using the discriminant. [5]
5. Repeat the investigation for $x^2 + 6x + k = 0$. For which values of k are there two, one or no real roots? [4]
6. Explain why the boundary cases always produce a perfect square. [4]

Present your work: a completed table, a clear description of the pattern, a general rule with its domain of validity, and an algebraic justification that does not rely on the examples alone.

MIXED PRACTICE 5

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–8 check fluency; 9–14 require reasoning; 15–18 are multi-step or unfamiliar. Allow 55 minutes.

1. Expand $(3x - 2)(x + 5)$.
2. Factorise $2x^2 - 18$.
3. Solve $x^2 - 11x + 30 = 0$.
4. Solve $5x^2 = 45$.
5. Write $2(x - 3)^2 - 8$ in expanded form.
6. Solve $x^2 + 6x - 16 = 0$.
7. Find the discriminant of $x^2 - 3x + 5$.
8. Solve $3x^2 - 7x + 2 = 0$.
9. Explain why $x^2 = -4$ has no real solution.
10. **Error analysis.** A student writes $x^2 = 6x \Rightarrow x = 6$. Identify the lost root and explain the error.
11. A parabola crosses the x-axis at $x = -1$ and $x = 4$. Write a possible equation and explain how the discriminant confirms two roots.
12. For $2x^2 + bx + 8 = 0$, find the values of b that give exactly one real root.
13. Solve $x^2 - 2x - 4 = 0$, giving exact answers, and state which method you used and why.
14. Show that $x^2 + 8x + 20 = (x + 4)^2 + 4$, and hence explain why the equation $x^2 + 8x + 20 = 0$ has no real roots.
15. A rectangle has perimeter 26 m and area 40 m^2. Find its dimensions.
16. The revenue from selling n items at price p is $R = np$. A stall sells $300 - 15x$ items at $12 + x$ pounds each. Write R in terms of x, and find the price that maximises revenue.
17. **Extension.** The roots of $x^2 + bx + c = 0$ are p and q. Show that $p + q = -b$ and $pq = c$.

MIXED PRACTICE 5 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

18 **Extension.** Solve $2x^2 - 3x - 7 = 0$, giving exact answers, and verify that the sum of the roots is $\frac{3}{2}$.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING

35 minutes · 30 marks

Quadratics: short response

NO CALCULATOR | SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

19 Expand and simplify $(x - 5)(2x + 3)$. [3]

20 Factorise $3x^2 - 12x$. [3]

21 Solve $x^2 - 9x + 20 = 0$. [3]

22 Solve $2x^2 + 7x - 4 = 0$. [4]

23 Solve $x^2 + 4x - 1 = 0$ by completing the square. [4]

24 Solve $3x^2 - 5x - 2 = 0$ using the formula. [4]

25 Find the discriminant of $2x^2 - 3x + 4 = 0$ and state the number of real roots. [4]

26 **Error analysis.** A student solves $x^2 = 5x$ by dividing by x and writes $x = 5$. Identify the error and give both roots. [3]

27 Find k so that $x^2 + kx + 25 = 0$ has exactly one real root. [2]

CRITERION B · INVESTIGATING PATTERNS

40 minutes · 24 marks

Roots, sums and products

Instructions. Work independently. Consider the family $x^2 + bx + c = 0$ with integer b and c.

1. Complete a table for $x^2 + bx + 4 = 0$ with $b = -9, -6, -3, 0, 3, 6, 9$, showing the discriminant and the number of real roots. [5]
2. Describe the pattern and state the values of b for which the equation has two, one or no real roots. [4]
3. For the cases with two roots, calculate the sum and product of the roots. Record them in a second table. [6]
4. Generalise: express the sum and product of the roots of $x^2 + bx + c = 0$ in terms of b and c. Verify your rule with two more examples. [5]
5. Prove your rule by expanding $(x - p)(x - q)$ and comparing coefficients. Explain why the discriminant guarantees that the roots are real when they are needed. [4]

Evidence: two complete tables, a stated pattern, a general rule with verification, and an algebraic proof.

CRITERION C · COMMUNICATING

35 minutes · 20 marks

Three routes to the same roots

Instructions. Write a guide of 350–450 words for a student who has just met quadratic equations. Your guide must: (i) explain what a root is and why a quadratic can have two, one or no real roots; (ii) solve one equation by two different methods, showing that the answers agree; (iii) include a labelled sketch of three parabolas meeting the axis twice, once and not at all; (iv) give advice on choosing between factorising, completing the square and the formula; (v) identify one common error and explain how to avoid it.

Evidence: a titled guide with clear paragraphs, two complete solutions to the same equation, a labelled diagram, and a short comparison table of methods.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Pricing the school concert

Instructions. A school hall holds 400 people. At a ticket price of £12, the committee sells 300 tickets. A survey suggests that for each £1 increase in price, 15 fewer tickets are sold. Let x be the increase in pounds.

1. Write expressions for the ticket price and the number of tickets sold, and show that the revenue is $R = 3600 + 120x - 15x^2$. [4]
2. Complete the square to find the price that maximises revenue and the maximum revenue. [5]
3. The committee needs £3,600. Solve $R = 3600$ and interpret both solutions, including the number of tickets sold at each price. [4]
4. Explain why both prices produce the same revenue, and recommend one of them with a reason that is not purely financial. [4]
5. The committee hopes for £3,900. Show, using the discriminant, that this target is impossible under the model. [4]
6. State two assumptions in the model and one factor that could make the survey unreliable. [3]

Evidence: the revenue expression, a completed-square derivation, both target calculations, the discriminant argument, and a justified recommendation with assumptions.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A quadratic equation has the standard form $ax^2 + bx + c = 0$ with $a \neq 0$; a root is a value that makes the equation true.
- The zero product property: if a product is zero, at least one factor is zero — provided one side is already zero.
- Completing the square writes $x^2 + bx$ as $\left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$ and reveals the extreme value.
- The quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ solves every quadratic; the discriminant $\Delta = b^2 - 4ac$ decides how many real roots exist.
- Contextual solutions must be checked against the situation and given units; inadmissible roots are rejected with a reason.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \Delta = b^2 - 4ac \qquad (x + p)^2 = x^2 + 2px + p^2$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can expand two brackets and collect like terms.
- ☐ I can factorise a quadratic with a common factor, by inspection or by grouping.
- ☐ I can write an equation in standard form and identify a, b and c.
- ☐ I can solve a quadratic by factorising and explain the zero product property.
- ☐ I can complete the square for a monic and a non-monic quadratic.
- ☐ I can solve a quadratic by the formula and simplify a surd answer.
- ☐ I can calculate the discriminant and interpret its sign.
- ☐ I can find a parameter value that gives one repeated root.
- ☐ I can form a quadratic from a word problem and reject an inadmissible root.
- ☐ I can explain what a root means on the graph of a quadratic.
- ☐ I can check a solution by substitution and recognise a sign error.
- ☐ I can choose an efficient method and justify the choice.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which form of a quadratic — expanded, factorised or completed square — helped you most in this chapter? Give one example where changing form made the answer visible.
2. Describe one error you made with signs or with dividing by x. What check will you use next time?
3. The statement of inquiry claims that equivalent forms reveal different structure. Explain this using either the projectile example or the pen problem.
4. Look ahead to Chapter 6: how might the completed-square form help you sketch a parabola without plotting many points?

SHAPE, SYMMETRY AND THE HIGHEST POINT

Quadratic Functions

06

VERTEX · AXIS · ROOTS

Reading a curve as a set of decisions — where it turns, where it meets the ground, and what its shape makes possible.

STATEMENT OF INQUIRY

The graph of a quadratic function is a parabola whose shape and position are encoded in its equation. Equivalent algebraic forms expose different graphical features, and comparing models shows when a quadratic description is justified.

KEY CONCEPT

Form

RELATED CONCEPTS

Representation · Change · Pattern

GLOBAL CONTEXT

Orientation in space and time — arches, paths and trajectories

FACTUAL

How do the intercepts, vertex and axis of symmetry appear in each algebraic form?

CONCEPTUAL

Why does changing one parameter move the whole curve in a predictable way?

DEBATABLE

When is a quadratic model more honest than a straight line?



IN THIS CHAPTER

6.1 Parabolas and features
6.2 Vertex form
6.3 Quadratic inequalities
6.4 Optimisation
6.5 Comparing models
6.6 Investigation

OPENING PROBLEM · THE TUNNEL ARCH

Will the van fit?

A road tunnel has a parabolic cross-section. At road level it is 12 m wide, and its highest point is 6 m above the centre of the road. Measure x in metres from the left edge of the road, so the arch is modelled by $h = a(x - 6)^2 + 6$.

1. Use the point $(0, 0)$ at the left edge to find a. Write the equation of the arch.
2. State the maximum height and the axis of symmetry. Explain what the axis represents physically.
3. Find the height of the arch 2 m and 4 m from the left edge.
4. A warning sign must hang with its lower edge at least 4.8 m above the road and must span from $x = 4$ to $x = 8$. Is there room? Justify with calculations.
5. A van is 2.2 m wide and 3.2 m tall. It drives with its left side 1 m from the left edge. Show that its top-left corner would strike the arch.
6. Suggest one change to the road layout that would let the van pass, and state how the model would need to change.

A quadratic function turns twice as fast as a linear one: its rate of change is not constant, so its graph bends. That bend creates a highest or lowest point, an axis of symmetry and a pattern of intercepts. Reading those features from an equation — and reading the equation from a graph — is the core skill of this chapter.

ORIENTATION

6.0 Learning intentions

- Identify the vertex, axis of symmetry, intercepts, domain and range of a parabola.
- Use vertex form to describe and perform transformations of $y = x^2$.
- Convert between standard and vertex form by completing the square.
- Solve quadratic inequalities and represent the solution on a number line.
- Use quadratic functions to optimise quantities in real contexts.
- Compare linear and quadratic models and evaluate their limitations.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

sketch a parabola from its equation and label its key features; explain how a, h and k transform the graph of $y = x^2$; convert a quadratic to vertex form; solve a quadratic inequality and interpret the solution in context; find and interpret a maximum or minimum; and choose between linear and quadratic models with a justified reason.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, reread Sections 5.3 and 5.4 before starting.

1. Complete the square: $x^2 - 8x + 11$.
2. Solve $x^2 - 5x + 6 = 0$.
3. Find the vertex of $y = (x - 2)^2 - 7$.
4. Solve $x^2 < 9$.
5. Evaluate $(x - 3)^2 - 4$ when $x = 5$.
6. Factorise $x^2 - x - 6$.

REFRESHER

The graph of $y = x^2$ is a U-shaped curve with its lowest point at the origin. Adding a constant outside the square moves the curve up or down; adding a constant inside the bracket moves it left or right. A negative coefficient of x^2 turns the U upside down. Every parabola has a line of symmetry through its vertex.

ATL SKILLS · THINKING AND TRANSFER

Critical thinking: decide which algebraic form exposes the feature you need — roots from factors, the vertex from completed square, the y-intercept from standard form. **Transfer:** connect the same transformation rules to other function families. **Communication:** label axes with units, mark the vertex and intercepts, and interpret each feature in the language of the context.

READING THE CURVE

6.1 Parabolas and key features

A quadratic function has the form $f(x) = ax^2 + bx + c$ with $a \neq 0$. Its graph is a parabola. When $a > 0$ the parabola opens upwards and the vertex is the minimum point; when $a < 0$ it opens downwards and the vertex is the maximum. The larger the value of $|a|$, the narrower the curve; the closer a is to zero, the wider it becomes.

Every parabola is symmetric about a vertical line through its vertex. That symmetry links the roots: if the roots are r and s, the axis of symmetry is $x = \frac{r+s}{2}$, and the vertex lies on that line. This gives a reliable sketching strategy: find the roots, find the vertex, plot the y-intercept, and reflect it across the axis.

DEFINITIONS

Parabola, vertex and axis of symmetry

The graph of a quadratic function is a **parabola**. Its **vertex** is the highest or lowest point. The **axis of symmetry** is the vertical line through the vertex; for $y = ax^2 + bx + c$ it has equation $x = -\frac{b}{2a}$. The **y-intercept** is c, and the roots are the x-intercepts.

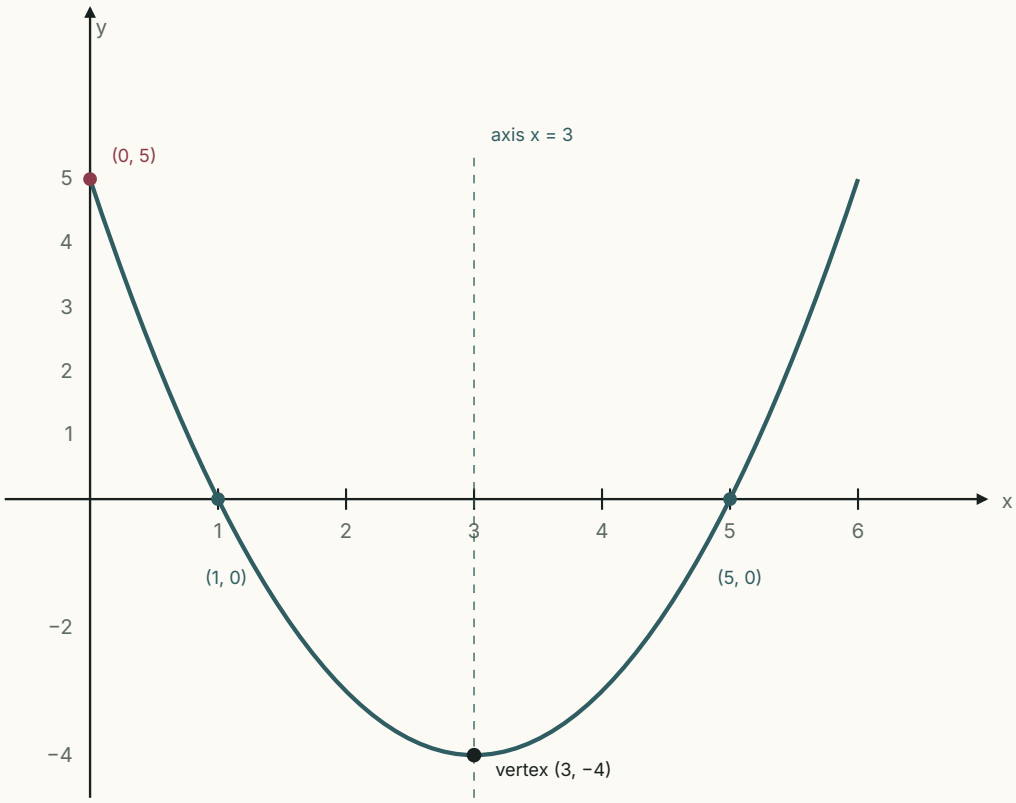


Figure 6.1 The parabola $y = x^2 - 6x + 5 = (x - 1)(x - 5)$. The roots are 1 and 5, the vertex is $(3, -4)$ on the axis $x = 3$, and the y-intercept is 5.

EXAMPLE **6.1** ANALYSING A PARABOLA FROM STANDARD FORM

QUESTION

For $y = x^2 - 6x + 5$, find the roots, vertex, axis of symmetry, y-intercept, and state the minimum value and range. Sketch the graph.

SOLUTION

1. Factorise: $x^2 - 6x + 5 = (x - 1)(x - 5)$, so the roots are $x = 1$ and $x = 5$.
2. The axis is halfway: $x = \frac{1+5}{2} = 3$.
3. At $x = 3$: $y = 9 - 18 + 5 = -4$. The vertex is $(3, -4)$.
4. The y-intercept is 5, reflected to $(6, 5)$ by symmetry. Because $a = 1 > 0$, the minimum is -4 and the range is $y \geq -4$.

Check: the vertex form is $(x - 3)^2 - 4$, confirming the same vertex.

EXAMPLE **6.2** A DOWNWARD-OPENING PARABOLA

QUESTION

For $y = -2x^2 + 8x - 3$, find the vertex and state whether it is a maximum or minimum. Find the roots correct to 2 decimal places.

SOLUTION

1. The axis is $x = -\frac{b}{2a} = -\frac{8}{2(-2)} = 2$.
2. At $x = 2$: $y = -8 + 16 - 3 = 5$. The vertex $(2, 5)$ is a maximum because $a = -2 < 0$.
3. Set $y = 0$: $2x^2 - 8x + 3 = 0$, so $x = \frac{8 \pm \sqrt{64 - 24}}{4} = 2 \pm \frac{\sqrt{10}}{2}$.
4. Numerically, $x \approx 0.42$ or $x \approx 3.58$.

Check: the roots are symmetric about $x = 2$: $2 - 1.58 = 0.42$ and $2 + 1.58 = 3.58$.

EXAMPLE **6.3** FROM FACTORISED FORM

QUESTION

The graph of $y = (x + 1)(x - 5)$ is a parabola. Find its roots, axis of symmetry, vertex and y-intercept.

SOLUTION

1. The roots are $x = -1$ and $x = 5$.
2. The axis is $x = \frac{-1 + 5}{2} = 2$.
3. At $x = 2$: $y = (3)(-3) = -9$, so the vertex is $(2, -9)$.
4. At $x = 0$: $y = (1)(-5) = -5$, so the y-intercept is -5.

Check: expanding gives $x^2 - 4x - 5$, whose axis is $x = -\frac{-4}{2} = 2$.

KEY IDEA

Each form answers a different question. Standard form gives the y-intercept and the direction of opening; factorised form gives the roots; completed-square form gives the vertex. A sketch should show all the features the equation provides, not only a plausible curve.

COMMON MISCONCEPTIONS

"The vertex is (b, c)." The y-intercept is $(0, c)$; the vertex is at $x = -\frac{b}{2a}$ and must be found by substitution.

"A parabola has two roots." It may have two, one or none. A graph that touches the axis has one repeated root; a graph above the axis has no real roots.

"The axis of symmetry is the y-axis." Only for $y = ax^2 + c$. A horizontal shift moves the axis.

"Negative a makes the graph go left." A negative a reflects the parabola vertically, so it opens downwards. Left and right movement comes from the x term, not from a.

GUIDED PRACTICE · KEY FEATURES

For $y = x^2 - 4x - 5$: factorise $(x - ____)(x + ____)$, so the roots are ____ and ____.

Axis: $x =$ _____; vertex: (____, _____); y-intercept: ____.

Because $a > 0$, the vertex is a _____ and the range is $y \geq$ ____.

EXERCISE 6A

PARABOLAS · VERTEX · INTERCEPTS

FLUENCY

1 For each parabola, state the roots, axis, vertex and y-intercept.
   a. $y = (x - 2)(x - 6)$
   b. $y = (x + 3)(x - 1)$
   c. $y = x(x - 4)$

2 Find the vertex and state whether it is a maximum or minimum.
   a. $y = x^2 - 8x + 12$
   b. $y = -x^2 + 6x - 5$
   c. $y = 2x^2 - 4x + 7$

3 State the direction of opening and the y-intercept.
   a. $y = 3x^2 - 2x + 1$
   b. $y = -0.5x^2 + 4x$
   c. $y = 7 - x^2$

4 Write the equation of the axis of symmetry for $y = ax^2 + bx + c$.

UNDERSTANDING AND REASONING

5 Explain why the axis of symmetry is the average of the two roots.

6 A parabola has roots -2 and 6. Find its axis of symmetry and the x-coordinate of its vertex.

7 **Error analysis.** A student states that the vertex of $y = x^2 - 10x + 21$ is $(10, 21)$. Identify both errors and give the correct vertex.

8 Can a parabola with a negative a have a minimum? Explain.

APPLICATION AND EXTENSION

9 A projectile's height is $h = 40t - 5t^2$. Find the maximum height and the time at which it occurs.

10 The arch of a bridge is modelled by $y = 8 - 0.5(x - 4)^2$ for $0 \leq x \leq 8$. Find the height at the centre and the width at ground level.

11 **Extension.** A parabola passes through $(0, 3)$, $(2, -1)$ and $(4, 3)$. Find its axis of symmetry, vertex and equation.

BUILDING THE CURVE FROM $Y = X^2$

6.2 Vertex form and transformations

The vertex form $y = a(x - h)^2 + k$ displays the vertex (h, k) immediately. It also records a sequence of transformations of the basic curve $y = x^2$: the factor a stretches or reflects the parabola, the subtraction h translates it horizontally, and the addition k translates it vertically.

Transformations are applied to the variable before the outside operation. In $y = (x - 4)^2$, the curve moves right by 4 because the input must be 4 larger to produce the same square. In $y = (x + 4)^2$, it moves left by 4. This inside/outside distinction is the most common source of sign errors.

DEFINITION

Vertex form

The **vertex form** of a quadratic function is $y = a(x - h)^2 + k$, with $a \neq 0$. The vertex is (h, k) and the axis of symmetry is $x = h$. If $a > 0$ the vertex is a minimum; if $a < 0$ it is a maximum.

EXAMPLE **6.4** DESCRIBING TRANSFORMATIONS

QUESTION

Describe how the graph of $y = -2(x + 1)^2 + 3$ is obtained from $y = x^2$, and state its vertex, axis and maximum value.

SOLUTION

1. The factor -2 reflects the curve in the x-axis and stretches it vertically by a factor of 2.
2. The $(x + 1)$ shifts the curve 1 unit left; the $+3$ shifts it 3 units up.
3. The vertex is $(-1, 3)$ and the axis is $x = -1$.
4. Because $a < 0$, the vertex is a maximum; the maximum value is 3.

Check: at $x = -1$, $y = 0 + 3 = 3$; at $x = 0$, $y = -2 + 3 = 1$, so the y-intercept is 1.

EXAMPLE **6.5** CONVERTING TO VERTEX FORM

QUESTION

Write $y = x^2 - 8x + 11$ in vertex form. Hence state the vertex, axis and minimum value.

SOLUTION

1. Complete the square: $x^2 - 8x = (x - 4)^2 - 16$.
2. So $y = (x - 4)^2 - 16 + 11 = (x - 4)^2 - 5$.
3. The vertex is $(4, -5)$, the axis is $x = 4$, and the minimum value is -5.

Check: expand $(x - 4)^2 - 5 = x^2 - 8x + 16 - 5 = x^2 - 8x + 11$.

EXAMPLE **6.6** VERTEX FORM WITH A LEADING COEFFICIENT

QUESTION

Write $y = 2x^2 + 4x + 1$ in vertex form and state its minimum value.

SOLUTION

1. Factor out 2 from the x-terms: $y = 2(x^2 + 2x) + 1$.
2. Complete the square inside: $x^2 + 2x = (x + 1)^2 - 1$.
3. Substitute: $y = 2((x + 1)^2 - 1) + 1 = 2(x + 1)^2 - 2 + 1 = 2(x + 1)^2 - 1$.
4. The vertex is $(-1, -1)$, so the minimum value is -1.

Check: $2(x + 1)^2 - 1 = 2x^2 + 4x + 2 - 1 = 2x^2 + 4x + 1$.

EXAMPLE **6.7** SKETCHING FROM VERTEX FORM

QUESTION

Sketch $y = (x - 3)^2 - 4$, showing the vertex, axis and intercepts.

SOLUTION

1 Vertex $(3, -4)$, axis $x = 3$, opening upwards.

2 Set $y = 0$: $(x - 3)^2 = 4$, so $x - 3 = \pm 2$ and the roots are 1 and 5.

3 Set $x = 0$: $y = 9 - 4 = 5$, so the y-intercept is 5; by symmetry $(6, 5)$ is also on the curve.

4 Plot the vertex and intercepts and draw a smooth symmetric curve through them.

> **Sketching standard:** mark the vertex, the axis, the intercepts and one reflected point. The curve should be smooth and symmetric, not a series of straight segments.

KEY IDEA

The sign inside the bracket is the opposite of the direction of the shift: $(x - h)^2$ moves right by h. The value of a controls width and direction, never horizontal position. Completing the square is the bridge from standard to vertex form.

COMMON MISCONCEPTIONS

"$y = (x + 3)^2$ moves right 3." It moves left 3, because the input must be -3 to make the bracket zero.

"$a = 2$ moves the graph up 2." It stretches the graph vertically. Only the constant outside the square translates it vertically.

"The vertex of $y = 2(x - 1)^2 + 5$ is $(2, 5)$." The vertex is $(1, 5)$; the 2 affects width, not position.

"Completing the square changes the value." It is an identity; expanding returns the original expression.

GUIDED PRACTICE · VERTEX FORM

Write $y = x^2 + 6x + 2$ in vertex form: $y = (x + ____)^2 - ____$.

The vertex is $(____, ____)$, the axis is $x = ____$, and the minimum value is $____$.

Describe the transformation of $y = x^2$ that produces $y = (x - 2)^2 + 7$:

_____.

EXERCISE 6B

VERTEX FORM · TRANSFORMATIONS · SKETCHING

FLUENCY

1 State the vertex and axis.

a. $y = (x - 4)^2 + 2$

b. $y = (x + 1)^2 - 6$

c. $y = -3(x - 2)^2 + 1$

2 Write in vertex form.

a. $y = x^2 - 10x + 24$

b. $y = x^2 + 4x - 3$

c. $y = 2x^2 - 12x + 13$

EXERCISE 6B · CONTINUED

VERTEX FORM · TRANSFORMATIONS · SKETCHING

3 Describe the transformations from $y = x^2$.

a. $y = (x - 5)^2$

b. $y = x^2 + 4$

c. $y = -x^2$

d. $y = 3(x + 2)^2 - 1$

4 State whether each vertex is a maximum or minimum, and give its value.

a. $y = 4(x - 1)^2 - 9$

b. $y = -(x + 2)^2 + 8$

c. $y = 0.5(x - 3)^2$

UNDERSTANDING AND REASONING

5 Explain why the graph of $y = (x - 7)^2$ is 7 units to the right of $y = x^2$, not 7 units to the left.

6 **Error analysis.** A student writes $y = x^2 - 6x + 1 = (x - 3)^2 + 1$. Identify the error and give the correct vertex form.

7 Two parabolas have the same vertex $(2, -3)$ but different values of a. Describe how their graphs differ.

8 Write a quadratic function whose maximum value is 6 at $x = -4$. Is your answer unique?

APPLICATION AND EXTENSION

9 A ball's height is $h = 12t - 5t^2$. Write h in vertex form and find the maximum height.

10 A parabolic arch has equation $y = -0.25(x - 6)^2 + 9$. Find its vertex and its width at ground level.

11 **Extension.** The graph of $y = a(x - h)^2 + k$ passes through $(0, 5)$, has vertex $(2, 1)$. Find a, h and k, and sketch the graph.

WHERE THE CURVE IS ABOVE OR BELOW

6.3 Quadratic inequalities

A quadratic inequality asks where a parabola lies above or below the x-axis. The sign of a product such as $(x - 3)(x + 2)$ changes only at its zeros, so the roots divide the number line into intervals. Testing one value in each interval determines the sign throughout that interval.

This is the same sign diagram introduced in Chapter 5, now used as a solution method. For a strict inequality the boundary roots are excluded; for a non-strict inequality they are included. Writing the answer as an inequality or in interval notation is equally valid, provided the notation is clear.

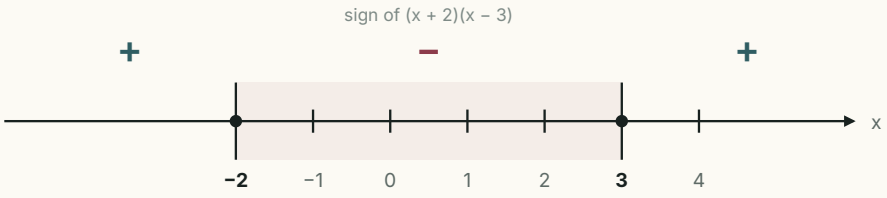


Figure 6.2 The sign of $x^2 - x - 6 = (x - 3)(x + 2)$. The product is negative between the roots and positive outside them.

EXAMPLE **6.8** A NON-STRICT INEQUALITY

QUESTION

Solve $x^2 - x - 6 \leq 0$.

SOLUTION

1. Factorise: $(x - 3)(x + 2) \leq 0$. The roots are -2 and 3.
2. Test intervals: at $x = -3$, $(-6)(-1) > 0$; at $x = 0$, $(-3)(2) < 0$; at $x = 4$, $(1)(6) > 0$.
3. The product is negative or zero between the roots, including the roots themselves.
4. Hence $-2 \leq x \leq 3$.

Graph check: the parabola $y = x^2 - x - 6$ lies on or below the axis between its roots.

EXAMPLE **6.9** A "GREATER THAN" INEQUALITY

QUESTION

Solve $x^2 \geq 9$.

SOLUTION

1. Rearrange: $x^2 - 9 \geq 0$, so $(x - 3)(x + 3) \geq 0$.
2. The roots are -3 and 3; the product is positive outside the roots.
3. Hence $x \leq -3$ or $x \geq 3$.

> **Do not "square root" an inequality.** Writing $x \geq 3$ loses the negative interval; a square root removes the sign information.

EXAMPLE **6.10** REVERSING A NEGATIVE LEADING COEFFICIENT

QUESTION

Solve $-x^2 + 4x - 3 > 0$.

SOLUTION

1. Multiply by -1 and reverse the inequality: $x^2 - 4x + 3 < 0$.
2. Factorise: $(x - 1)(x - 3) < 0$.
3. The product is negative between the roots: $1 < x < 3$.

Check: at $x = 2$, $-4 + 8 - 3 = 1 > 0$; at $x = 0$, $-3 > 0$ is false.

EXAMPLE **6.11** AN INEQUALITY IN CONTEXT

QUESTION

A stall's profit, in pounds, is $P = -x^2 + 10x - 16$, where x is the price increase in pounds. For which price increases is the profit positive?

SOLUTION

1 Solve $P > 0$: $-x^2 + 10x - 16 > 0$, so $x^2 - 10x + 16 < 0$.

2 Factorise: $(x - 2)(x - 8) < 0$.

3 Hence $2 < x < 8$: the stall makes a positive profit for price increases between £2 and £8.

Check: at $x = 5$, $P = -25 + 50 - 16 = 9 > 0$; at $x = 1$, $P = -1 + 10 - 16 = -7 < 0$.

KEY IDEA

The roots are the only places where the sign can change. Testing one point per interval is enough, provided the test points are chosen correctly. A quick sketch of the parabola confirms whether the solution is "between" or "outside" the roots.

COMMON MISCONCEPTIONS

"$x^2 < 9$ **means** $x < 3$." It means $-3 < x < 3$. Both signs matter.

"$x^2 > 9$ **means** $x > 3$." It also includes $x < -3$.

"Multiply by -1 without changing the inequality." Multiplying or dividing by a negative number reverses the inequality sign.

"Include the roots for a strict inequality." For $<$ or $>$, the roots are excluded; for $\leq$ or $\geq$, they are included.

GUIDED PRACTICE · INEQUALITIES

Solve $x^2 - 5x + 4 < 0$: factorise $(___)(___) < 0$, so the roots are $___$ and $___$.

The solution is $___ < x < ___$.

Solve $x^2 - 16 \geq 0$: the solution is $x \leq ___$ or $x \geq ___$.

EXERCISE 6C

QUADRATIC INEQUALITIES · SIGN DIAGRAMS

FLUENCY

1 Solve.
   a. $(x - 2)(x - 6) < 0$
   b. $(x + 1)(x - 4) \leq 0$
   c. $(x + 3)(x + 1) > 0$

2 Solve.
   a. $x^2 - 9 \leq 0$
   b. $x^2 - 25 > 0$
   c. $x^2 < 16$

3 Solve.
   a. $x^2 - 7x + 10 \geq 0$
   b. $x^2 + 2x - 8 < 0$
   c. $x^2 - 6x + 9 > 0$

4 Solve $2x^2 - 5x - 3 \leq 0$.

EXERCISE 6C · CONTINUED QUADRATIC INEQUALITIES · SIGN DIAGRAMS

UNDERSTANDING AND REASONING

5 Explain why $x^2 < 9$ cannot be solved by writing $x < 3$ alone.

6 **Error analysis.** A student solves $-x^2 + 6x - 5 > 0$ as $(x-1)(x-5) > 0$, giving $x < 1$ or $x > 5$. Identify the error and give the correct solution.

7 For which values of x is the parabola $y = x^2 - 4x$ below the x-axis?

8 Explain the difference between the solutions of $x^2 - 4 \leq 0$ and $x^2 - 4 < 0$.

APPLICATION AND EXTENSION

9 A projectile's height is $h = 24t - 5t^2$. For what times is the projectile above 20 m?

10 The profit of a stall is $P = -x^2 + 12x - 20$. For which price increases is $P \geq 0$?

11 **Extension.** Find the values of k for which $x^2 - 6x + k > 0$ for all real x.

FINDING THE BEST VALUE

6.4 Optimisation and modelling

Many design questions ask for the largest area, the greatest revenue or the highest point of a path. If the quantity is quadratic, its maximum or minimum is the vertex. Completing the square turns the question into a statement about a square: a square is smallest when it is zero, so the vertex is found without drawing.

The domain matters. A mathematical maximum may occur outside the range of values that make sense — a negative width, or a time before the object was launched. The best answer is the best *admissible* value, so check the endpoints of the contextual domain as well as the vertex.

EXAMPLE **6.12** MAXIMISING AN AREA

QUESTION

A rectangular pen is made with 40 m of fencing on all four sides. Find the dimensions that maximise the area, and the maximum area.

SOLUTION

1 Let the width be w. The length is $20 - w$ because the perimeter is 40.

2 Area $A = w(20 - w) = 20w - w^2 = 100 - (w - 10)^2$.

3 A square is non-negative, so $A \leq 100$, with equality when $w = 10$.

4 The pen is 10 m by 10 m, with maximum area $100\ \text{m}^2$.

Check: $w = 10$ gives length 10 and perimeter 40.

EXAMPLE **6.13** MAXIMISING REVENUE

QUESTION

A hall sells 200 tickets at £20 each. A survey predicts that each £1 reduction in price sells 20 more tickets. Find the price that maximises revenue and the maximum revenue.

SOLUTION

1. Let the reduction be x pounds. Price $= 20 - x$; tickets $= 200 + 20x$.
2. Revenue $R = (20 - x)(200 + 20x) = 4000 + 200x - 20x^2$.
3. Complete the square: $R = -20(x^2 - 10x) + 4000 = -20(x - 5)^2 + 4500$.
4. The maximum revenue is £4,500 when $x = 5$, so the price is £15.

Check: at £15, 300 tickets are sold and $15 \times 300 = 4500$.

EXAMPLE **6.14** A PROJECTILE WITH AN INITIAL HEIGHT

QUESTION

A ball is thrown from 1 m above the ground with height $h = 1 + 8t - 5t^2$ metres after t seconds. Find its maximum height and the time it lands.

SOLUTION

1. Complete the square: $h = -5(t^2 - 1.6t) + 1 = -5((t - 0.8)^2 - 0.64) + 1$.
2. So $h = 4.2 - 5(t - 0.8)^2$. The maximum is 4.2 m at $t = 0.8$ s.
3. Landing: $1 + 8t - 5t^2 = 0$, so $5t^2 - 8t - 1 = 0$.
4. $t = \frac{8 \pm \sqrt{64 + 20}}{10} = \frac{4 \pm \sqrt{21}}{5}$. The positive value is $t \approx 1.72$ s.

Check: at $t = 1.72$, $h \approx 1 + 13.76 - 14.79 \approx 0$.

KEY IDEA

Optimisation is not only “find the vertex”. It is: form a function of one variable, identify the admissible domain, find the extreme, and interpret it. A maximum outside the domain is not an answer; the endpoint may be.

COMMON MISCONCEPTIONS

“The vertex is always the answer.” If the domain is restricted, the best admissible value may be an endpoint.

“Maximum area means maximum perimeter.” The perimeter is fixed in these problems; the variables are the dimensions.

“Revenue is price times number sold at the original price.” If the price changes, the number sold changes too; both factors must use the same variable.

“The maximum height is the root.” The maximum is the vertex value; the roots are where the height is zero.

GUIDED PRACTICE · OPTIMISATION

A pen uses 24 m of fencing on all four sides. Let the width be w; the length is _____.

Area $A = w($_____$) =$ _____ $- (w -$ ____$)^2$.

The maximum area is _____ m^2, when $w =$ _____.

EXERCISE 6D

OPTIMISATION · DOMAINS · INTERPRETATION

APPLICATION

1 A rectangle has perimeter 36 m. Find the dimensions that maximise its area.

2 A ball's height is $h = 30t - 5t^2$. Find its maximum height and when it lands.

3 A farmer uses 60 m of fencing for three sides of a rectangular field, with a wall as the fourth side. Find the dimensions that maximise the area.

4 A theatre sells 400 tickets at £10. Each £1 increase in price loses 20 sales. Find the price that maximises revenue.

5 A rectangular garden is 2 m longer than it is wide and has area 80 m^2. Find its dimensions.

6 The height of a tunnel arch is $h = 6 - 0.24x^2$ metres, for $-5 \le x \le 5$. Find the maximum height and the width of the arch at ground level.

7 A company's profit is $P = -2x^2 + 24x - 54$ thousand pounds, where x is the number of units produced in hundreds. Find the maximum profit.

8 **Extension.** A shop's daily profit is $P = 120x - 10x^2$ pounds for $7 \le x \le 12$, where x is the price in pounds. Find the price that maximises profit and explain why the vertex is not the answer.

MODELLING TASK · CRITERION D PRACTICE

The school mural

A mural is to be painted inside a parabolic arch. The arch has height 5 m at the centre and width 8 m at the ground, modelled by $y = 5 - kx^2$ for $-4 \le x \le 4$.

1. Use the ground edge $(4, 0)$ to find k.
2. Find the height of the arch 1 m and 3 m from the centre.
3. The mural must be at least 2.5 m tall over its whole width. Solve $5 - kx^2 \ge 2.5$ and state the greatest possible width.
4. A rectangular mural 3 m tall has its base on the ground and is centred. Find its greatest width and the cost of painting it at £28 per square metre.
5. The artist wants a 6 m wide mural. Does it meet the height requirement? Justify.
6. State two assumptions and one reason the real arch might differ from the model.

LINEAR OR QUADRATIC?

6.5 Comparing models

A linear model changes by a constant amount per step; a quadratic model changes by an amount that itself changes. Over a short interval the two can look similar, which is why a straight line may be an acceptable local approximation of a curve. Over a longer interval they diverge, often dramatically.

Comparing models means more than finding which graph is higher. It means asking which model represents the mechanism, which fits the data over the required range, and what each predicts outside that range. The intersection of two models is the point where their predictions agree.

EXAMPLE **6.15** COMPARING TWO COST MODELS

QUESTION

Printer A costs £15 plus £3 per copy. Printer B costs £0.20 per copy squared plus £5. For how many copies do the costs agree, and which printer is cheaper for 10 copies? For 20 copies?

SOLUTION

1. Models: $A(n) = 15 + 3n$ and $B(n) = 0.2n^2 + 5$.
2. Set equal: $0.2n^2 - 3n - 10 = 0$, so $n^2 - 15n - 50 = 0$.
3. $n = \frac{15 \pm \sqrt{225 + 200}}{2} = \frac{15 \pm \sqrt{425}}{2}$, giving $n \approx 17.81$ or a negative value.
4. At $n = 10$: $A = 45$, $B = 25$, so B is cheaper. At $n = 20$: $A = 75$, $B = 85$, so A is cheaper.

> **Interpretation:** B has a lower fixed cost but grows faster. The crossover near 18 copies is where the two pricing structures agree.

EXAMPLE **6.16** INTERSECTING A LINE AND A PARABOLA

QUESTION

Find the points where the line $y = 2x + 1$ meets the parabola $y = x^2 - 2$. For which x is the line above the parabola?

SOLUTION

1. Set $x^2 - 2 = 2x + 1$, so $x^2 - 2x - 3 = 0$.
2. Factorise: $(x - 3)(x + 1) = 0$, so $x = 3$ or $x = -1$.
3. The points are $(3, 7)$ and $(-1, -1)$.
4. Test $x = 0$: line 1, parabola -2, so the line is above between the intersections: $-1 < x < 3$.

Check: at $x = 3$, both give 7; at $x = -1$, both give -1.

KEY IDEA

Setting two models equal finds where they agree; testing one point between the intersections decides which is larger there. The number of intersections is the number of real solutions of the resulting equation — and the discriminant tells you that before you solve.

COMMON MISCONCEPTIONS

"A quadratic is always a better model because it fits more points." More parameters can fit noise. The mechanism and the domain must justify the model.

"The graphs cross once." A line and a parabola can meet twice, once (tangent) or not at all.

"The model with the lower starting value is cheaper." The slope determines the long-run comparison.

"Extrapolating a quadratic is safe because the equation continues." The equation continues; the situation may not.

GUIDED PRACTICE · COMPARE AND SOLVE

Solve $x^2 - 3x = 2x - 4$: rearrange $x^2 - ____x + ____ = 0$.

The solutions are $x = ____$ and $x = ____$, so the graphs meet at $____$ points.

At $x = 0$, the quadratic value is $____$ and the line value is $____$, so the $_____$ is above near $x = 0$.

EXERCISE 6E

COMPARING · INTERSECTING · EVALUATING

FLUENCY

1 Find the points where the line and parabola meet.
   a. $y = x + 2, y = x^2$
   b. $y = 3x - 2, y = x^2$
   c. $y = 2x + 3, y = x^2 - 1$

2 For each pair, state which graph is above at $x = 0$.
   a. $y = x^2 + 3, y = 2x + 1$
   b. $y = x^2 - 2, y = x$

3 Model A is $y = 10 + 2x$; model B is $y = 0.5x^2$. Find where they agree.

4 Explain how the discriminant tells you the number of intersections of a line and a parabola.

UNDERSTANDING AND REASONING

5 Two models of a population are $P = 500 + 40t$ and $P = 0.5t^2 + 100$. Find when they predict the same population.

6 **Error analysis.** A student finds the intersection of $y = x^2$ and $y = 4$ as $x = 2$. What is missing?

7 Give one reason a linear model might be preferred to a quadratic model even when the quadratic fits the data slightly better.

8 A quadratic model predicts a company's profit becomes negative after 12 years. Why should this prediction be treated cautiously?

APPLICATION AND EXTENSION

9 A taxi charges £8 plus £2 per km; a ride-share charges £0.50 per km squared. For what distance do they charge the same? Which is cheaper for 6 km?

10 A ball's height is $h = 20t - 5t^2$; a drone's height is $h = 10 + 2t$. Find when they are at the same height.

11 **Extension.** Show that the line $y = 2x + c$ meets the parabola $y = x^2 + 1$ twice when $c > 0$, once when $c = 0$, and not at all when $c < 0$.

INVESTIGATION · CRITERION B

6.6 Arches and trajectories

A water jet in a park follows a parabolic path. The table records its height at different horizontal distances from the nozzle. In this investigation you will use the symmetry of the data to locate the vertex, build a model, and test it at points that were not measured.

INVESTIGATION · CRITERION B

The fountain arc

The height h metres of a water jet is measured at horizontal distances x metres from the nozzle.

DISTANCE x (m)	0	1	2	4	6	8	9	10
Height h (m)	0	1.8	3.2	4.8	4.8	3.2	1.8	0

1. Plot the points. Describe the symmetry and estimate the coordinates of the vertex.
2. Use the symmetry to write the model in the form $h = a(x - 5)^2 + 5$. Find a using the point $(0, 0)$.
3. Test your model at $x = 1$, $x = 4$ and $x = 9$. Comment on the fit.
4. Use the model to predict the height at $x = 3$ and at $x = 7$. Are these predictions interpolations or extrapolations?
5. Solve $h = 4$ and interpret both solutions.
6. Explain how the model would change if the nozzle pressure increased so that the jet reached a height of 6 m at the same horizontal distance.

Generalise and justify. For a symmetric set of data with vertex (p, q), explain why the model can be written $h = a(x - p)^2 + q$. Show how to find a from any other data point, and state the condition under which the model predicts a real landing point.

MIXED PRACTICE 6

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–9 check fluency; 10–15 require reasoning; 16–18 are multi-step or unfamiliar. Allow 55 minutes.

1 Find the vertex of $y = x^2 - 10x + 21$.

2 Write $y = x^2 + 8x + 10$ in vertex form.

3 State the axis of symmetry of $y = -2x^2 + 12x - 7$.

4 Solve $x^2 - 5x - 24 = 0$.

5 Solve $x^2 < 25$.

6 Solve $x^2 - 3x - 10 \geq 0$.

7 Find the maximum value of $y = 8 - 2(x - 3)^2$.

8 Find where $y = x^2 - 4$ meets $y = 3x$.

9 State the range of $y = (x + 2)^2 - 6$.

10 **Error analysis.** A student says the vertex of $y = 3(x - 4)^2 + 1$ is $(12, 1)$. Explain the error.

11 A parabola has roots -3 and 7. Find its axis of symmetry and the x-coordinate of its vertex.

12 Explain why a quadratic inequality can have a solution made of two separate intervals.

13 A ball's height is $h = 18t - 5t^2$. Find its maximum height and when it lands.

14 Find the values of k for which $y = x^2 + kx + 9$ has a minimum value of 0.

15 A rectangle has perimeter 28 m. Show that its area is at most 49 m^2.

16 A company's revenue is $R = (30 - x)(500 + 25x)$. Find the price that maximises revenue.

17 **Extension.** The line $y = x + c$ is tangent to $y = x^2 - 3x + 5$. Find c and the point of contact.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 30 marks

Quadratic graphs: short response

NO CALCULATOR SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

1. For $y = x^2 - 4x - 12$, state the roots, axis of symmetry and vertex. [4]
2. Write $y = x^2 - 6x + 11$ in vertex form. [3]
3. Describe the transformations of $y = x^2$ that produce $y = -2(x + 3)^2 + 4$. [4]
4. State the maximum value of $y = 5 - 3(x + 2)^2$. [2]
5. Solve $x^2 - 2x - 8 \leq 0$. [4]
6. Solve $x^2 > 16$. [3]
7. Find the point where $y = x^2 - 2x$ meets $y = x + 4$. [4]
8. Explain why the graph of $y = (x - 5)^2 + 2$ has no x-intercepts. [3]
9. **Error analysis.** A student writes $x^2 < 9 \Rightarrow x < 3$. Identify the error and give the complete solution. [3]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

The family $y = x^2 + bx$

Instructions. Work independently. Consider the family $y = x^2 + bx$ for integer values of b.

1. Complete a table for $b = -4, -2, 0, 2, 4$, showing the roots, the axis of symmetry and the vertex. [6]
2. Describe how the axis of symmetry and the vertex change as b increases. [4]
3. Generalise: write the roots, axis and vertex in terms of b. [5]
4. Verify your general rule for $b = 6$ by completing the square. [4]
5. Describe the locus of the vertex as b varies, and explain why this locus is a parabola. [5]

Evidence: a complete table, a clear description of the pattern, general formulas, and an algebraic justification of the vertex locus.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Reading a parabola

Instructions. Write a 350–450 word guide for a student who is about to sketch quadratic graphs. Your guide must: (i) explain how the values of a, h and k in $y = a(x - h)^2 + k$ affect the graph; (ii) show how to find the vertex, axis and intercepts; (iii) include a labelled sketch of one upward- and one downward-opening parabola; (iv) explain how the discriminant predicts the number of x-intercepts; (v) give one common sketching error and how to avoid it.

Evidence: a titled guide with clear paragraphs, two labelled sketches, a step-by-step method and a short summary table.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The greenhouse arch

Instructions. The cross-section of a greenhouse is a parabolic arch. Its height at the centre is 6 m and its width at ground level is 10 m. The arch is modelled by $y = 6 - 0.24x^2$ for $-5 \le x \le 5$, with x and y in metres.

1. Find the height of the arch at $x = 2$ and at $x = 4$. [4]
2. Solve $y = 0$ and explain what the solutions represent. [3]
3. A shelf must be at least 2.5 m above the ground. Solve $6 - 0.24x^2 \ge 2.5$ and state the usable width of the greenhouse at that height. [5]
4. A worker 1.9 m tall must walk where the clearance is at least 2.1 m. Find the greatest possible distance from the centre line. [4]
5. The designer wants the arch to be at least 4 m high at $x = \pm 3$. Show that the current model fails, and find the value of k in $y = 6 - kx^2$ that would satisfy the requirement. [5]
6. State two assumptions in the model and one reason a real greenhouse might differ. [3]

Evidence: substitution calculations, a solution of the quadratic equation, an inequality with its interval, a corrected parameter, and an evaluation of the model.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- Every quadratic function has a parabolic graph with a vertex and a vertical axis of symmetry.
- Standard form gives the y-intercept; factorised form gives the roots; vertex form $y = a(x - h)^2 + k$ gives the vertex.
- The axis of a parabola with roots r and s is $x = \frac{r+s}{2}$.
- Quadratic inequalities are solved by finding the roots and testing the sign in each interval.
- The vertex gives the maximum or minimum; the admissible domain decides whether that value is a valid answer.
- Comparing models requires attention to intersections, long-run behaviour and the limits of extrapolation.

$$y = a(x - h)^2 + k \qquad x = -\frac{b}{2a}$$

$$y = x^2 - 6x + 5 = (x - 1)(x - 5) = (x - 3)^2 - 4$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can identify the vertex, axis and intercepts of a parabola from its equation.
- ☐ I can sketch a parabola and label its key features.
- ☐ I can explain the effect of a, h and k in vertex form.
- ☐ I can convert standard form to vertex form by completing the square.
- ☐ I can solve a quadratic inequality using a sign diagram.
- ☐ I can include or exclude roots correctly for strict and non-strict inequalities.
- ☐ I can find and interpret a maximum or minimum in context.
- ☐ I can check whether an optimal value lies inside the admissible domain.
- ☐ I can find the intersection of a line and a parabola.
- ☐ I can compare linear and quadratic models and justify a choice.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which representation — equation, table, graph or context — made the idea of a vertex clearest to you? Explain with an example.
2. Describe one sign error you made with a horizontal shift or an inequality. What check will you use next time?
3. The statement of inquiry claims that equivalent forms expose different features. Illustrate this with the three forms of one quadratic.
4. Look ahead to Chapter 7: how could a quadratic model describe the growth of a pattern that has a constant second difference?

STRUCTURE, CHANGE AND THE NEXT TERM

Patterns & Sequences

07

TERM · DIFFERENCE · RULE

From counting a row of matches to predicting a hundredth term — how differences expose the rule behind a pattern.

STATEMENT OF INQUIRY

A sequence is a relationship between position and value. Its differences reveal the structure of that relationship, and a justified general rule turns a finite pattern into a prediction.

KEY CONCEPT
Relationships

RELATED CONCEPTS
Pattern · Generalisation · Representation

GLOBAL CONTEXT
Orientation in space and time — cycles, growth and prediction

FACTUAL
How do first and second differences reveal the type of a sequence?

CONCEPTUAL
Why does a constant second difference force a quadratic rule?

DEBATABLE
How far beyond the data can a sequence model be trusted?

IN THIS CHAPTER

7.1 Describing patterns
7.2 Linear sequences
7.3 Quadratic sequences
7.4 Recursive rules
7.5 Modelling
7.6 Investigation

OPENING PROBLEM · THE AMPHITHEATRE

How many seats?

A new amphitheatre has 24 seats in the first row, and every row behind it has 3 more seats than the row in front. Let r_n be the number of seats in row n.

1. Write the first six terms of the sequence r_n.
2. Find the number of seats in row 20 without drawing all the rows.
3. Write a rule for r_n in terms of n, and use it to check your answer to step 2.
4. Which row has exactly 60 seats? Is there a row with exactly 100 seats? Justify both answers.
5. Let S_n be the total number of seats in the first n rows. Complete a table of S_n for $n = 1, 2, 3, 4, 5$, then find the first and second differences of the totals.
6. The architect claims that the first 20 rows will seat about 1,000 people. Explain why the total must be less than $20 \times 81 = 1{,}620$, then estimate whether 1,000 is reasonable.

A sequence is more than a list of numbers. When the step from one term to the next follows a rule, that rule can be written, tested and used. This chapter moves from spotting a pattern to proving that it must continue — and from a formula for one term to a model for a whole situation.

7.0 Learning intentions

ORIENTATION

- Describe a sequence by its terms, its differences and a rule.
- Find and use the nth term of an arithmetic sequence.
- Recognise a quadratic sequence from its second differences and find its rule.
- Write and interpret recursive rules, including spreadsheet formulas.
- Model real situations with sequences and respect the domain of the model.
- Evaluate predictions, including the risks of extrapolation.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

extend a pattern and explain the rule in words; find the nth term of a linear sequence and decide whether a number belongs to it; use second differences to find a quadratic rule; write a recursive rule and translate between recursive and explicit forms; model a context with a sequence and justify a prediction beyond the given data.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit Chapter 4 before starting.

1. Evaluate $3n + 5$ when $n = 7$.
2. Continue the pattern $4, 9, 14, 19, \ldots$ for two more terms.
3. Find the difference $38 - 17$.
4. Solve $2n + 1 = 31$.
5. Expand $(n + 1)(n + 2)$.
6. Solve $x^2 + 3x - 10 = 0$.

REFRESHER

A sequence is an ordered list, so the position of each term matters. The first differences are found by subtracting each term from the next. If those differences are constant, the sequence grows by equal steps; if they change in a regular way, the sequence has a richer structure.

ATL SKILLS · THINKING AND COMMUNICATION

Critical thinking: test a proposed rule against every term you know, not only the first two. **Transfer:** connect a difference table to the graphs of linear and quadratic functions. **Communication:** state whether a rule is recursive or explicit, define n, and give the domain of a model.

READING A LIST

7.1 Describing and extending patterns

A **sequence** is an ordered list of numbers. The numbers are called **terms**, and their positions are labelled $n = 1, 2, 3, \ldots$. The term in position n is written u_n, so u_1 is the first term, u_2 the second, and so on. A finite sequence stops; an infinite sequence continues without end.

A pattern can be described in two ways. A **recursive** rule tells you how to get from one term to the next, such as "add 4". An **explicit** rule tells you how to calculate any term directly from its position n, such as $u_n = 4n + 2$. Both describe the same sequence, but they answer different questions.

Differences are the quickest diagnostic. Write the terms in a row, subtract neighbours to get the first differences, then subtract again to get the second differences. Constant first differences signal a linear pattern; constant second differences signal a quadratic one; a constant ratio signals geometric growth.

DEFINITIONS

Sequence, term and differences

A **sequence** is an ordered list of numbers. Each number is a **term**, and u_n denotes the term in position n. The **first differences** are $u_2 - u_1, u_3 - u_2, \ldots$; the **second differences** are the differences of the first differences. An **arithmetic sequence** has a constant first difference. A **geometric sequence** has a constant ratio between consecutive terms.

EXAMPLE **7.1** EXTENDING A SEQUENCE

QUESTION

For the sequence $6, 10, 14, 18, \ldots$, find the next two terms and describe the rule in two ways.

SOLUTION

1. First differences: $10 - 6 = 4$, $14 - 10 = 4$, $18 - 14 = 4$. The difference is constant.
2. Recursive rule: start at 6 and add 4 to each term. The next terms are $18 + 4 = 22$ and $22 + 4 = 26$.
3. Explicit rule: the terms are 4 more than the multiples of 4, so $u_n = 4n + 2$. Check: $u_1 = 6$, $u_2 = 10$.



Figure 1

4 matches

Figure 2

7 matches

Figure 3

10 matches

Figure 7.1 Each new square shares one side with the previous square, so each step adds three matches, not four.

EXAMPLE **7.2** A DIAGRAM PATTERN

QUESTION

Figure n is made from n squares in a row, as shown. Find the number of matches in Figure 8, and explain why the pattern is linear.

SOLUTION

1. Count: $M_1 = 4$, $M_2 = 7$, $M_3 = 10$. The first difference is 3 each time.
2. Each new square adds three matches: two new sides plus one side that replaces a shared side. So $M_{n+1} = M_n + 3$.
3. Continue: $M_4 = 13$, $M_5 = 16$, $M_6 = 19$, $M_7 = 22$, $M_8 = 25$.
4. The constant difference means the growth is linear: each figure adds the same number of matches.

Check: the explicit rule is $M_n = 3n + 1$. For $n = 8$, $3(8) + 1 = 25$.

EXAMPLE **7.3** CLASSIFYING BY DIFFERENCES AND RATIOS

QUESTION

Classify each sequence as arithmetic, geometric or quadratic. Give the next term.
(a) $3, 7, 11, 15, \ldots$ (b) $3, 6, 12, 24, \ldots$ (c) $2, 5, 10, 17, \ldots$

SOLUTION

1 (a) First differences are $4, 4, 4$: constant, so the sequence is arithmetic. The next term is $15 + 4 = 19$.

2 (b) Ratios are $2, 2, 2$: constant, so the sequence is geometric. The next term is $24 \times 2 = 48$.

3 (c) First differences are $3, 5, 7$; second differences are $2, 2$. Constant second difference means quadratic. The next first difference is 9, so the next term is $17 + 9 = 26$.

Check: (c) fits $u_n = n^2 + 1$, giving $2, 5, 10, 17, 26$.

EXERCISE 7A

PATTERNS · DIFFERENCES · CLASSIFICATION

FLUENCY

1 Write the next three terms.
   a. $7, 11, 15, \ldots$
   b. $30, 26, 22, \ldots$
   c. $2, 6, 18, \ldots$
   d. $1, 4, 9, 16, \ldots$
   e. $1, 3, 6, 10, \ldots$

2 Find the first differences and state whether each sequence is linear.
   a. $4, 11, 18, 25$
   b. $20, 17, 14, 11$
   c. $2, 5, 9, 14$
   d. $3, 7, 12, 18$

3 Match each rule to a sequence: $u_n = 5n$, $u_n = n^2$, $u_n = 2^n$, $u_n = \frac{n(n+1)}{2}$.
   a. $1, 4, 9, 16$
   b. $5, 10, 15, 20$
   c. $2, 4, 8, 16$
   d. $1, 3, 6, 10$

UNDERSTANDING AND REASONING

4 Explain why a sequence with constant first difference is linear.

5 A pattern has terms $5, 9, 13, \ldots$. Explain the rule and find the 12th term.

6 **Error analysis.** A student says $2, 4, 8, 16$ has common difference 2. Identify the error and describe the sequence correctly.

7 A cross pattern has $1, 5, 9, 13, \ldots$ dots. Find the 10th term and explain the rule.

APPLICATION AND EXTENSION

8 A stack has 1 block in the top row, 3 in the next, 5 in the next, and so on. How many blocks are in row 8? Explain why every term is odd.

9 A bacterial culture starts with 500 cells and doubles every hour. List the first five terms. Is the sequence arithmetic? Explain.

10 **Extension.** The sequence $1, 2, 4, 7, 11, \ldots$ has first differences $1, 2, 3, 4$. Find the next two terms and the second differences. What type of rule do you expect?

COMMON MISCONCEPTIONS

"The prettiest continuation is the right one." A pattern rule must be testable. Several different rules can produce the same first four terms, so state the rule you are using before claiming the next term.

"A sequence is just a set of numbers." A set forgets order and repetition; a sequence keeps both. The list $2, 2, 2$ is a perfectly good sequence of three terms, and $1, 2, 3$ differs from $3, 2, 1$.

"The first term is the zeroth term." In this book n starts at 1 unless a rule says otherwise. u_n is a label for the term in position n, not a power of u.

"Two terms are enough to find the rule." Two terms fit infinitely many rules. They decide a unique answer only once you know the *type* of rule — linear, quadratic or geometric — that the situation calls for.

EQUAL STEPS

7.2 Linear sequences and the nth term

An arithmetic sequence adds the same number d, called the **common difference**, at each step. Its explicit rule is $u_n = dn + c$. The coefficient of n is the common difference, and the constant c adjusts the starting value. Since $u_1 = d + c$, we have $c = u_1 - d$.

This rule answers two different questions. Given n, it produces the term. Given a value, it produces an equation for the position: if $u_n = V$, solve $dn + c = V$. A valid position must be a positive integer. If the solution is not an integer, the value is not a term of the sequence.

DEFINITION

The nth term of an arithmetic sequence

For an arithmetic sequence with first term u_1 and common difference d, the **nth term** is $u_n = u_1 + (n - 1)d$, equivalently $u_n = dn + c$ with $c = u_1 - d$.

EXAMPLE **7.4** FINDING A POSITIVE NTH TERM

QUESTION

Find the nth term of $5, 9, 13, 17, \ldots$.

SOLUTION

1. The common difference is $9 - 5 = 4$, so the rule has the form $u_n = 4n + c$.
2. Use the first term: $5 = 4(1) + c$, so $c = 1$.
3. Hence $u_n = 4n + 1$.

Check: $u_3 = 4(3) + 1 = 13$, which matches the third term.

EXAMPLE **7.5** A NEGATIVE COMMON DIFFERENCE

QUESTION

Find the nth term of $20, 17, 14, 11, \ldots$.

SOLUTION

1. The common difference is $17 - 20 = -3$, so $u_n = -3n + c$.
2. Use $u_1 = 20$: $20 = -3 + c$, so $c = 23$.
3. Hence $u_n = 23 - 3n$.

Check: $u_4 = 23 - 12 = 11$, as required.

EXAMPLE **7.6** IS A NUMBER A TERM?

QUESTION

Is 83 a term of the sequence $5, 9, 13, 17, \ldots$? If not, find the largest term below 83.

SOLUTION

1. The rule is $u_n = 4n + 1$. Set $4n + 1 = 83$.
2. Then $4n = 82$, so $n = 20.5$. This is not a positive integer, so 83 is not a term.
3. The neighbouring positions are $n = 20$ and $n = 21$: $u_{20} = 81$ and $u_{21} = 85$.
4. The largest term below 83 is 81.

Check: the terms near 83 are $77, 81, 85$, so 83 is skipped.

EXAMPLE **7.7** A SAVINGS MODEL

QUESTION

An account starts with £120 and £35 is added at the end of each week. Let T_n be the balance after n weeks. Write a rule for T_n, find the balance after 12 weeks, and find the first week in which the balance exceeds £1,000.

SOLUTION

1. The balance grows by £35 each week, so $T_n = 35n + 120$.
2. After 12 weeks: $T_{12} = 35(12) + 120 = 420 + 120 = £540$.
3. Solve $35n + 120 > 1000$: $35n > 880$, so $n > 25.14$.
4. The first whole week is $n = 26$, with balance $35(26) + 120 = £1,030$.

Check: week 25 gives £995, just below £1,000.

KEY IDEA

In $u_n = dn + c$, the value d is the step size and c is what the formula would give at $n = 0$. A negative d makes the sequence decrease; a zero d makes it constant. Always check the rule against the first two or three terms before using it.

EXERCISE 7B

LINEAR RULES · POSITION · CONTEXT

FLUENCY

1 Find the nth term.

a. $4, 7, 10, 13, \ldots$

b. $11, 15, 19, 23, \ldots$

c. $2, 9, 16, 23, \ldots$

d. $50, 45, 40, 35, \ldots$

2 Find the 15th term.

a. $u_n = 6n - 1$

b. $u_n = 3 - 2n$

c. $u_n = 0.5n + 4$

3 Which term of $7, 11, 15, 19, \ldots$ is 71?

4 Is 100 a term of $4n + 3$? Justify your answer.

UNDERSTANDING AND REASONING

5 A sequence has 3rd term 13 and 8th term 33. Find its nth term.

6 A phone plan costs £15 plus £0.08 per minute. Write a rule for the cost of m minutes, find the cost of 200 minutes, and find the minutes for a £45 bill.

7 **Error analysis.** A student writes the nth term of $8, 11, 14, \ldots$ as $3n + 8$. Explain the error and give the correct rule.

8 Explain why the common difference is the coefficient of n in $u_n = dn + c$.

APPLICATION AND EXTENSION

9 A sequence has $u_n = 6n - 4$. Find the first term greater than 100.

10 A tree is 1.8 m tall and grows 0.25 m each year. Write a rule for its height after n years. When does it first exceed 4 m?

11 **Extension.** A sequence has $u_n = an + b$. If $u_5 = 18$ and $u_{12} = 39$, find a and b.

12 **Extension.** For the sequences $2n + 5$ and $3n - 1$, find the term number at which they are equal.

COMMON MISCONCEPTIONS

"The nth term is dn." For a linear sequence $u_n = dn + c$, where $c = u_1 - d$ is the term in position 0. Using dn alone shifts every term by d.

"u_{n+1} means $u \times (n + 1)$." The subscript names a position, it is not a multiplier. u_{n+1} is the next term after u_n.

"The rule is checked as soon as it gives the term you used to build it." That is circular. Build the rule from two terms, then test it on a third term you have not used.

"A common difference must be positive." Sequences can decrease: $20, 17, 14, \ldots$ has $d = -3$ and $u_n = 23 - 3n$. A negative d is a smaller amount, not an impossible one.

CHANGING DIFFERENCES

7.3 Quadratic sequences and differences

When the first differences of a sequence are not constant but change by a constant amount, the sequence is **quadratic**. Its first differences form an arithmetic sequence, and its second differences are constant. The rule has the form $u_n = an^2 + bn + c$, which is a quadratic function evaluated at the counting numbers.

The second difference determines a. Expanding $a(n+1)^2 + b(n+1) + c - (an^2 + bn + c)$ gives $2an + (a+b)$; the next first difference is $2a$ larger, so the constant second difference is $2a$. Hence a is half of the constant second difference. Subtract an^2 from each term, and the remainders follow a linear rule.

DEFINITION

Quadratic sequence

A **quadratic sequence** has a constant non-zero second difference. Its nth term has the form $u_n = an^2 + bn + c$ with $a \neq 0$, and the constant second difference is $2a$.

EXAMPLE **7.8** FINDING A QUADRATIC RULE

QUESTION

Find the nth term of $2, 7, 16, 29, 46, \ldots$.

SOLUTION

1. First differences: $5, 9, 13, 17$. Second differences: $4, 4, 4$.
2. Second difference $= 2a = 4$, so $a = 2$. Subtract $2n^2$ from each term.
3. Remainders: $0, -1, -2, -3, -4$, which follow $1 - n$.
4. Hence $u_n = 2n^2 - n + 1$.

Check: $u_4 = 2(16) - 4 + 1 = 29$, matching the fourth term.

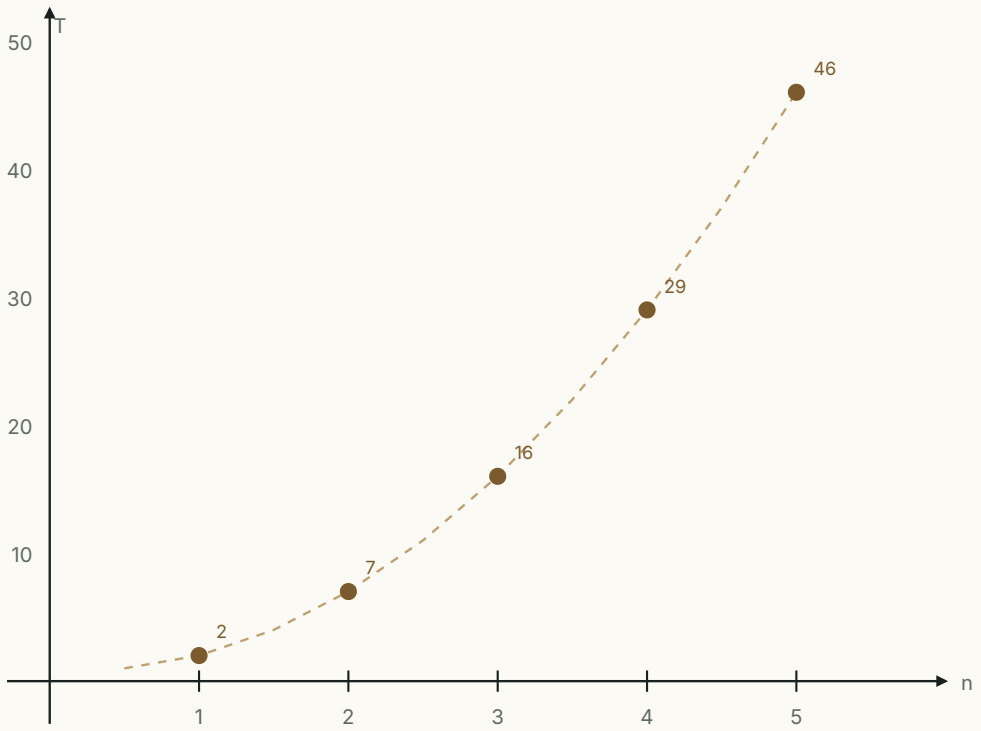


Figure 7.2 The terms of $u_n = 2n^2 - n + 1$ lie on the parabola $y = 2x^2 - x + 1$, but a sequence is defined only at whole-number positions.

EXAMPLE **7.9** A NEGATIVE SECOND DIFFERENCE

QUESTION

Find the nth term of $20, 17, 12, 5, -4, \ldots$.

SOLUTION

1. First differences: $-3, -5, -7, -9$. Second differences: $-2, -2, -2$.
2. Second difference $= 2a = -2$, so $a = -1$. Subtract $-n^2$, which is the same as adding n^2.
3. Remainders: $21, 21, 21, 21, 21$, so the remainder is constant 21.
4. Hence $u_n = 21 - n^2$.

Check: $u_5 = 21 - 25 = -4$, as required.

EXAMPLE **7.10** MODELLING A TOTAL WITH A QUADRATIC SEQUENCE

QUESTION

The amphitheatre has $r_n = 3n + 21$ seats in row n. The totals S_n of the first n rows are $24, 51, 81, 114, 150, \ldots$. Find a rule for S_n, then calculate the seating in the first 20 rows and compare it with the architect's estimate of 1,000.

SOLUTION

1 First differences of S_n: $27, 30, 33, 36$. Second differences: $3, 3, 3$.

2 So $2a = 3$, giving $a = \frac{3}{2}$. Subtract $\frac{3}{2}n^2$: $22.5, 45, 67.5, 90, 112.5$.

3 The remainders are $22.5n = \frac{45}{2}n$. Hence $S_n = \frac{3}{2}n^2 + \frac{45}{2}n$.

4 For $n = 20$: $S_{20} = \frac{3}{2}(400) + \frac{45}{2}(20) = 600 + 450 = 1{,}050$.

5 The estimate of 1,000 is close but about 5% low; 1,050 seats is the model's prediction.

Check: $S_5 = 150$, matching the table. The total is a quadratic sequence because the row sizes themselves grow linearly.

EXERCISE 7C

SECOND DIFFERENCES · QUADRATIC RULES

FLUENCY

1 Find the second difference and state whether the sequence is quadratic.
   a. $3, 6, 11, 18, \ldots$
   b. $1, 6, 15, 28, \ldots$
   c. $10, 7, 6, 7, \ldots$
   d. $2, 9, 20, 35, \ldots$

2 Find the nth term.
   a. $1, 4, 9, 16, \ldots$
   b. $2, 8, 18, 32, \ldots$
   c. $5, 12, 23, 38, \ldots$
   d. $3, 9, 19, 33, \ldots$

3 A sequence begins $4, 10, 18, 28, \ldots$. Find its n th term and its 20th term.

UNDERSTANDING AND REASONING

4 Explain why a constant second difference means the sequence is quadratic.

5 **Error analysis.** A student says $2, 5, 10, 17$ is linear because the first differences are $3, 5, 7$. Identify the error and find the nth term.

6 The totals of the first n rows of a stand are $18, 38, 60, 84, \ldots$. Find a rule for the total and hence the total for 15 rows.

7 A sequence has $u_n = n^2 + kn + 3$ and third term 21. Find k.

APPLICATION AND EXTENSION

8 Which term of $2, 7, 16, 29, \ldots$ is 142? If none, find the two neighbouring terms.

9 A tiling design with $n^2 + 4n$ tiles is used for design n. Find the number of tiles in design 10, and the design that uses 96 tiles.

10 **Extension.** A quadratic sequence has second difference 6, first term 4 and second term 13. Find its nth term.

11 **Extension.** Prove that the second difference of $an^2 + bn + c$ is always $2a$.

COMMON MISCONCEPTIONS

"The second difference is the coefficient of n^2." For $an^2 + bn + c$ the second difference is $2a$, so a is half the second difference — for $2, 7, 16, 29, \ldots$ the second difference is 4 and $a = 2$.

"Take one set of differences and use the linear rule." Once, and the sequence still has changing differences. Take differences again; only a constant *second* difference makes the rule quadratic.

"Growing terms mean a quadratic." Growth alone says nothing. $1, 2, 4, 8, \ldots$ grows faster than any quadratic but is geometric, and its second differences are not constant.

"Two terms fix a quadratic rule." A quadratic needs three conditions: three terms, or two terms together with the second difference.

ONE STEP AT A TIME

7.4 Recursive rules and spreadsheets

A **recursive rule** defines each term from the previous term or terms. It needs at least one starting value. Subscript notation is compact: $u_{n+1} = u_n + 4$ means "add 4 to get the next term", while $u_1 = 7$ supplies the start. Some processes are naturally recursive: a balance grows by a fixed deposit, a population doubles, or a pattern adds a fixed number of tiles.

Recursive rules are easy to enter into a spreadsheet. Put the starting value in one cell, write the next cell as a formula referring to the cell above, then fill down. The spreadsheet performs the recursion for you. An explicit rule is better when you need a distant term directly, because it avoids calculating every intermediate term.

DEFINITION

Recursive and explicit rules

A **recursive rule** expresses u_{n+1} (or a later term) in terms of earlier terms, together with one or more initial terms. An **explicit rule** expresses u_n directly in terms of n.

EXAMPLE **7.11** FROM RECURSIVE TO EXPLICIT

QUESTION

A sequence is defined by $u_1 = 7$ and $u_{n+1} = u_n + 4$. Write its first five terms and find the 20th term.

SOLUTION

1. Apply the rule repeatedly: $7, 11, 15, 19, 23$.
2. The sequence is arithmetic with $d = 4$, so $u_n = 4n + c$.
3. Using $u_1 = 7$: $7 = 4 + c$, so $c = 3$. Hence $u_n = 4n + 3$.
4. Therefore $u_{20} = 4(20) + 3 = 83$.

Check: the 20th step is 19 additions of 4 after the first term: $7 + 19(4) = 83$.

EXAMPLE **7.12** A MULTIPLICATIVE RECURSION

QUESTION

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n$. Write its first five terms and find the 10th term.

SOLUTION

1. Each term is double the previous one: $3, 6, 12, 24, 48, \ldots$.
2. The ratio between consecutive terms is constant, so the sequence is geometric with ratio 2.
3. From term 1 to term 10 there are 9 multiplications by 2: $u_{10} = 3 \times 2^9$.
4. $2^9 = 512$, so $u_{10} = 1{,}536$.

Check: the sequence doubles rapidly; $u_6 = 96$, $u_7 = 192$, $u_8 = 384$, $u_9 = 768$, $u_{10} = 1{,}536$.

EXAMPLE **7.13** A SPREADSHEET RECURSION

QUESTION

An account starts with £120 and receives £35 at the end of every week. Write a recursive rule and a spreadsheet formula for the balance after n weeks, then find the balance after 20 weeks.

SOLUTION

1. Let T_n be the balance after n weeks. Then $T_0 = 120$ and $T_{n+1} = T_n + 35$.
2. In a spreadsheet, put 120 in cell B2 for week 0, then enter `=B2+35` in B3 and fill down. Column A holds the week numbers.
3. The explicit rule is $T_n = 120 + 35n$, since 35 is added n times.
4. After 20 weeks: $T_{20} = 120 + 35(20) = £820$.

Check: the recursive and explicit rules agree at $n = 1$: $155 = 120 + 35$.

KEY IDEA

A recursive rule describes a process; an explicit rule describes the result at any position. Use recursion when the next value depends naturally on the current one, and an explicit rule when you need a far-away term or a quick comparison.

EXERCISE 7D

RECURSION · SPREADSHEETS · EXPLICIT FORM

FLUENCY

1. Write the first five terms.
   a. $u_1 = 3,\ u_{n+1} = u_n + 6$
   b. $u_1 = 20,\ u_{n+1} = u_n - 4$
   c. $u_1 = 2,\ u_{n+1} = 3u_n$
   d. $u_1 = 1,\ u_2 = 2,\ u_{n+2} = u_{n+1} + u_n$
2. For $u_1 = 3,\ u_{n+1} = u_n + 6$, find u_{20} and an explicit rule.
3. Write a recursive rule for $5, 9, 13, 17, \ldots$.

UNDERSTANDING AND REASONING

EXERCISE 7D · CONTINUED

RECURSION · SPREADSHEETS · EXPLICIT FORM

4 An account starts at £500 and receives £45 each week. Write a recursive rule and an explicit rule for the balance after n weeks. Find the balance after 10 weeks and the first week it exceeds £1,100.

5 A spreadsheet has 7 in cell B2 and the formula =B2+3 in B3, filled down to B10. Find the value in B10 and explain what the fill-down does.

6 The Fibonacci sequence begins $1, 1, 2, 3, 5, \ldots$ Write its recursive rule, find the next five terms, and explain why an explicit rule is harder to guess.

7 **Error analysis.** A student writes $u_{n+1} = u_n + 3$ for $2, 6, 18, 54, \ldots$. Explain the error and write a correct recursive rule.

8 A bacterial population starts at 200 and doubles every hour. Write a recursive rule. Find the population after 5 hours.

APPLICATION AND EXTENSION

9 Explain one advantage of an explicit rule and one advantage of a recursive rule.

10 **Extension.** A sequence has $u_1 = 2$ and $u_{n+1} = u_n^2$. Write the first four terms. Is the sequence arithmetic, geometric or neither? Explain.

11 **Extension.** A sequence has $u_1 = 5$ and $u_{n+1} = 2u_n - 1$. Find u_5 and show that $u_n = 2^{n+1} + 1$.

COMMON MISCONCEPTIONS

"$u_{n+1} = 2u_n$ gives the powers of 2." Doubling needs a starting value. With $u_1 = 3$ the sequence is $3, 6, 12, 24, \ldots$, which is twice a power of 2, not a power of 2.

"A recursive rule finds the 100th term." It finds term 100 only after 99 previous steps. That is exactly why an explicit rule is worth deriving — and why a spreadsheet, which repeats the step for you, is useful.

"Every cell in a recursive spreadsheet column can hold the same formula referring to row 1." Each cell must refer to the cell immediately above it, and only the seed row is typed by hand. Copying a fixed reference gives the same number in every row.

"$u_{n+1} = u_n + n$ is arithmetic." The amount added changes with n, so the differences are not constant; the sequence is quadratic, growing like $\frac{n(n-1)}{2}$ above its seed.

FROM RULE TO PREDICTION

7.5 Modelling with sequences

A sequence model turns a context into a rule for the term number. The domain matters: term numbers are positive integers, and most contexts impose a smaller range, so a mathematically valid term such as a fractional row may be impossible. Linear models grow by equal steps; quadratic models turn and then accelerate. Interpolation uses a rule inside the data range; extrapolation uses it outside, and is a claim about the future that needs a reason to believe the pattern continues.

EXAMPLE **7.14** STACKED CANS

QUESTION

Cans are stacked in rows: 1 can on top, 2 in the next row, 3 in the next, and so on. The total after n rows is $T_n = \frac{n(n+1)}{2}$. How many rows are needed for 210 cans?

SOLUTION

1. Set $\frac{n(n+1)}{2} = 210$, so $n(n+1) = 420$.
2. Expand and factorise: $n^2 + n - 420 = 0$, so $(n+21)(n-20) = 0$.
3. The solutions are $n = -21$ and $n = 20$. Only the positive integer is admissible, so 20 rows are needed.

Check: $T_{20} = \frac{20 \times 21}{2} = 210$, and $T_{19} = 190$, so 20 rows are required.

EXAMPLE **7.15** SOLVING FOR A ROW NUMBER

QUESTION

The amphitheatre total is $S_n = \frac{3}{2}n^2 + \frac{45}{2}n$. How many rows are needed to seat at least 800 people?

SOLUTION

1. Solve $\frac{3}{2}n^2 + \frac{45}{2}n = 800$. Multiply by 2: $3n^2 + 45n = 1600$.
2. So $3n^2 + 45n - 1600 = 0$. The discriminant is $45^2 + 4(3)(1600) = 2025 + 19{,}200 = 21{,}225$.
3. $\sqrt{21{,}225} \approx 145.69$, so $n = \frac{-45 + 145.69}{6} \approx 16.78$.
4. n must be a whole number, so 16 rows are not enough. Check: $S_{16} = 744$ and $S_{17} = 816$.
5. Hence 17 rows are needed.

Check: $S_{17} = \frac{3}{2}(289) + \frac{45}{2}(17) = 433.5 + 382.5 = 816 \geq 800$.

EXAMPLE **7.16** EVALUATING A MODEL

QUESTION

A shop's weekly sales, in hundreds of pounds, are $120, 150, 190, 240$ in weeks 1 to 4. A quadratic model is $P_n = 5n^2 + 15n + 100$. Predict week 10 and evaluate the prediction.

SOLUTION

1. First differences: $30, 40, 50$. Second difference: 10, so $2a = 10$ and $a = 5$.
2. Subtract $5n^2$: $115, 130, 145, 160$, which is $15n + 100$.
3. So $P_n = 5n^2 + 15n + 100$. For $n = 10$: $500 + 150 + 100 = 750$, or £75,000.
4. This is an extrapolation six weeks beyond the data. A quadratic cannot keep rising forever in a real market: competition, capacity and demand eventually change. The prediction is a hypothesis, not a fact.

Check: the model reproduces the four data values at $n = 1, 2, 3, 4$.

KEY IDEA

A sequence model is a claim about a relationship, not just a curve through points. State the domain, keep units, round only at the end, and distinguish a prediction inside the data from one beyond it.

EXERCISE 7E

MODELLING · PREDICTION · EVALUATION

FLUENCY

1 Find the 10th and 20th triangular numbers $T_n = \frac{n(n+1)}{2}$.

2 A display has 15 rows of cans, with 15 cans in the bottom row, 14 in the next, and so on up to 1. How many cans are there?

3 The heights of a plant in weeks 1 to 4 are $12, 18, 26, 36$ cm. Find a quadratic model, predict the height in week 8, and explain why the prediction may fail.

4 A phone plan costs £25 plus £0.10 per GB of data. Write a linear rule for the monthly cost of g GB, find the cost for 30 GB, and find the data allowance for a £40 bill.

UNDERSTANDING AND REASONING

5 The number of diagonals of a polygon with n sides is $D_n = \frac{n(n-3)}{2}$. Find D_{12}, and find the polygon with 35 diagonals.

6 Explain why the number of rows in a stack must be a whole number, and why a quadratic model can produce a non-whole answer.

7 A model predicts a company's profit will grow quadratically forever. Give two reasons this prediction should be treated cautiously.

8 Explain the difference between interpolation and extrapolation, using an example from this chapter.

APPLICATION AND EXTENSION

9 A sequence model for a bike rack is $d_n = 4n + 6$ docks in row n. How many docks are in row 9, and which row has 70 docks?

10 A tiling pattern uses $n^2 + 3n$ tiles in design n. Find the design with 180 tiles and the number of tiles in design 15.

11 **Extension.** The total number of handshakes among n people is $\frac{n(n-1)}{2}$. Find the number of people needed for 66 handshakes, and explain why the answer must be an integer.

COMMON MISCONCEPTIONS

"S_n and u_n are the same thing." u_n is one row, S_n is the running total of the first n. They are linked by $u_n = S_n - S_{n-1}$.

"A row of 4.7 is a valid answer." Positions are whole numbers. When a rule returns $n = 4.7$, report row 4 and row 5, test both against the question, and say which one answers it.

"The model keeps working however far ahead I look." A rule fitted to five rows claims nothing about row 200. State the range over which the situation still matches the rule, and revisit it when new rows appear.

"The formula is the answer." A formula with no units and no statement of what n counts cannot be checked, so it cannot be defended in Criterion C.

INVESTIGATION · CRITERION B

7.6 Figures that grow

Figure n is a rectangular array of dots with $n + 1$ rows and $n + 2$ columns. Count the dots, record the results, and look for structure before you write a formula.

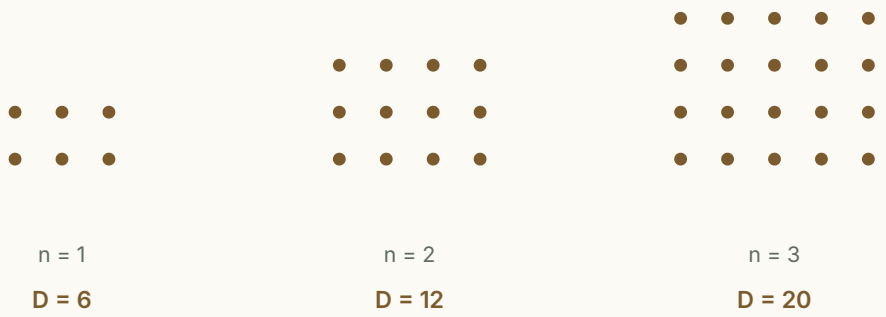


Figure 7.3 Rectangular dot arrays with $n + 1$ rows and $n + 2$ columns.

INVESTIGATION · CRITERION B

Counting a growing rectangle

Let D_n be the number of dots in Figure n.

1. Count the dots for $n = 1, 2, 3, 4$. Record the results in a table.
2. Find the first and second differences. What does the constant second difference tell you?
3. Write D_n in the form $(n + 1)(n + 2)$, expand it, and explain why the expansion matches your difference table.
4. Predict the number of dots in Figure 10. Is this interpolation or extrapolation?
5. Which figure has exactly 210 dots? Solve an equation and explain why only one solution is admissible.
6. Describe how the pattern would change if Figure n had $n + 3$ rows and $n + 2$ columns. Write the new rule and its second difference.

Generalise and justify. Explain why a rectangular array with rows and columns that both grow linearly produces a quadratic sequence. Use the structure of the array, not just the numbers, to justify the coefficient of n^2.

MIXED PRACTICE 7

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–9 check fluency; 10–15 require reasoning; 16–18 are multi-step or unfamiliar. Allow 55 minutes.

1. Write the next three terms of $4, 11, 18, \ldots$.
2. Find the nth term of $9, 14, 19, 24, \ldots$.
3. Find the 20th term of $u_n = 5n - 2$.
4. Is 92 a term of $7, 12, 17, 22, \ldots$? Justify.

MIXED PRACTICE 7 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

5 Find the nth term of $3, 8, 15, 24, \ldots$.

6 Find the nth term of $12, 9, 6, 3, \ldots$.

7 Find the first and second differences of $2, 6, 12, 20, 30, \ldots$.

8 Find the nth term of $2, 6, 12, 20, \ldots$.

9 For $u_1 = 5,\ u_{n+1} = 3u_n - 2$, find u_4.

10 For $u_1 = 2,\ u_2 = 3,\ u_{n+2} = u_{n+1} + u_n$, find u_6.

11 A sequence has $u_n = 4n + 7$. Which term is 63?

12 Explain why $1, 4, 9, 16, \ldots$ is not arithmetic.

13 **Error analysis.** A student writes the nth term of $6, 10, 14, \ldots$ as $4n + 6$. Identify the error and give the correct rule.

14 A pattern uses $n^2 + 2n$ tiles in design n. Find the tiles in design 12, and the design that uses 120 tiles.

15 A stack has rows of $1, 3, 6, 10, \ldots$ cans. Find the number in row 15.

16 An account starts at £250 and receives £40 each week. When does it first exceed £1,000?

17 **Extension.** A quadratic sequence has second difference 8, first term 3 and second term 14. Find its nth term.

18 **Extension.** Show that $n^2 + 3n + 2$ has constant second difference 2, and find the first term greater than 500.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and are assessed against the criteria shown. Your teacher will tell you which tasks are required and how long you have.

CRITERION A · KNOWING AND UNDERSTANDING — 35 minutes · 30 marks

Sequences: short response

NO CALCULATOR | SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

19 Find the nth term of $7, 12, 17, 22, \ldots$. [3]

20 Find the 30th term of $u_n = 4n - 3$. [2]

21 Find the nth term of $20, 17, 14, 11, \ldots$. [3]

22 Find the nth term of $1, 5, 13, 25, \ldots$. [4]

23 Find the first and second differences of $3, 9, 19, 33, \ldots$, and state the type of sequence. [4]

24 A sequence has 4th term 18 and 9th term 38. Find its nth term. [4]

25 For $u_1 = 4,\ u_{n+1} = 2u_n + 1$, find u_5. [3]

26 Explain why a sequence with constant second difference cannot be arithmetic. [3]

27 A tiling pattern uses $n^2 + 3n$ tiles in design n. Find the tiles in design 15 and the design that uses 180 tiles. [4]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

Matches in a square grid

Instructions. Work independently. A square grid of side n is made from unit squares. Let M_n be the number of matches needed to draw the grid.

1. Draw the grids for $n = 1, 2, 3, 4$ and count the matches. Record your results in a table. [6]
2. Find the first and second differences. Describe the pattern. [4]
3. Find a formula for M_n. Verify it for $n = 4$. [5]
4. Predict M_{12} and explain why your formula should continue to hold. [4]
5. The perimeter of the grid is $4n$. Explain why the perimeter is linear while the number of matches is quadratic. [5]

Evidence: a complete table, difference calculations, a general formula, a verification, and a structural explanation of the linear versus quadratic growth.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining a sequence

Instructions. Write a 300–400 word explanation for a student who has missed the lessons on sequences. Your explanation must: (i) distinguish arithmetic, geometric and quadratic sequences using differences or ratios; (ii) find the nth term of $4, 9, 14, 19, \ldots$ and of $3, 8, 15, 24, \ldots$; (iii) include a table of differences and a labelled sketch of the terms; (iv) explain why checking $n = 1$ is essential; (v) give one common error and how to avoid it.

Evidence: clear paragraphs, correct notation, a difference table, a labelled sketch and a worked check.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The bike-share station

Instructions. A bike-share station is built in rows of docks. Row 1 has 10 docks, and each row has 4 more docks than the row in front. Let d_n be the number of docks in row n, and S_n the total number of docks in the first n rows.

1. Find d_8. [4]
2. Write a rule for d_n. [4]
3. Which row has 70 docks? [4]
4. Find a quadratic rule for S_n, and use it to find the total number of docks in 12 rows. [5]
5. A grant funds 500 docks. How many complete rows can be built? Justify your answer. [4]
6. The designer claims that 20 rows will provide 1,000 docks. Check the claim and state two assumptions in the model. [3]

Evidence: a linear rule, a quadratic total, an inequality or equation solved with integer constraints, and an evaluation of the claim.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A sequence is an ordered list; u_n denotes the term in position n.
- First differences reveal arithmetic growth; second differences reveal quadratic growth.
- The nth term of an arithmetic sequence is $u_n = dn + c$, with common difference d.
- For a quadratic sequence, the second difference is $2a$ when $u_n = an^2 + bn + c$.
- A triangular number $T_n = \frac{n(n+1)}{2}$ is the sum $1 + 2 + \cdots + n$.
- A recursive rule gives the next term from previous terms; an explicit rule gives any term directly.
- Sequence models have whole-number domains, and extrapolation needs justification beyond a good fit.

$$u_n = u_1 + (n-1)d \qquad \text{second difference} = 2a$$

$$T_n = \frac{n(n+1)}{2} \qquad u_n = an^2 + bn + c$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can extend a pattern and explain its rule in words.
- ☐ I can find and use the nth term of an arithmetic sequence.
- ☐ I can decide whether a number belongs to a sequence and justify the decision.
- ☐ I can use second differences to recognise a quadratic sequence.
- ☐ I can find the rule of a quadratic sequence from its terms.
- ☐ I can write a recursive rule and translate it into an explicit rule.
- ☐ I can use a spreadsheet formula to generate a sequence.
- ☐ I can model a context with a sequence and state its domain.
- ☐ I can solve for a term number and reject inadmissible solutions.
- ☐ I can evaluate a prediction made beyond the data.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which representation — a list, a difference table, a diagram or a formula — made a pattern clearest to you? Explain with an example.
2. Describe a time you predicted a term and then found the pattern was different. What check would have caught the error earlier?
3. The statement of inquiry links difference structure to justified prediction. Illustrate this link with one linear and one quadratic sequence.
4. Look ahead to Chapter 8: how would you describe a sequence that multiplies by the same factor each year, and why is that different from adding the same amount?

MULTIPLIERS, INTEREST AND LONG-TERM DECISIONS

Growth & Financial Mathematics

08

RATE · MULTIPLIER · MODEL

Why a small percentage, repeated year after year, can change a decision — and how a single multiplier makes saving, borrowing and depreciation comparable.

STATEMENT OF INQUIRY

Repeated multiplication by a constant factor models growth and decay. Multipliers turn percentage changes into a form that can be compared, projected and questioned — but every financial model rests on assumptions that must be stated.

KEY CONCEPT
Relationships

RELATED CONCEPTS
Change · Quantity · Model

GLOBAL CONTEXT
Fairness and development — access, saving and long-term planning

FACTUAL
How does a percentage change become a multiplier?

CONCEPTUAL
Why does compound growth eventually overtake simple growth?

DEBATABLE
Is the lowest headline rate always the best financial choice?

IN THIS CHAPTER

8.1 Repeated change
8.2 Compound interest
8.3 Depreciation
8.4 Comparing offers
8.5 Model limits
8.6 Investigation

OPENING PROBLEM · THE LONG GAME

A fund and a machine

A school community fund invests £2,500 at 6% per year, with interest added to the fund at the end of each year. In the same month, the school buys equipment for £2,500; the equipment loses 12% of its value each year.

1. Complete a table showing the fund after 0, 1, 2, 3, 4, 5 and 6 years.
2. Explain why each year's amount is 1.06 times the previous amount, not £150 more.
3. In which year does the fund first exceed £3,500?
4. A simpler account pays 6% of the original £2,500 each year. Compare the two accounts after 6 years and explain the difference.
5. Find the equipment's value after 3 years, and the first year in which it is worth less than £1,500.
6. The treasurer says: "A 6% gain and a 12% loss roughly cancel." Use the multipliers to test this claim for the first year, and explain why percentages of different amounts cannot simply be added or subtracted.

Percentages describe one change. Finance asks what happens when a change repeats. A single multiplier captures that repetition, so a table of yearly values becomes a formula, and a formula becomes a decision. The same mathematics describes savings, prices, populations and declining resources — which is why its assumptions matter as much as its arithmetic.

8.0 Learning intentions

ORIENTATION

- Convert a percentage change into a multiplier and combine successive changes.
- Calculate compound interest and depreciation using powers.
- Reverse a percentage change to recover an original amount.
- Compare financial offers using total cost and effective rates.
- State the assumptions and limitations of a financial model.
- Estimate and evaluate doubling times for growth and decay.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

express a percentage increase or decrease as a multiplier; calculate a value after repeated change; distinguish simple from compound interest; reverse a percentage change without subtracting the percentage; compare offers by total cost over a stated period; identify assumptions that could make a prediction wrong.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit percentage work in Chapter 2 and powers in Chapter 1 before starting.

1. Find 15% of 80.
2. Increase 240 by 5%.
3. Decrease 60 by 12%.
4. Evaluate 1.08^2.
5. Solve $2^n = 32$.
6. Write 0.85 as a percentage decrease.

REFRESHER

A percentage is a fraction of 100, so 7% means 0.07. Increasing by 7% gives the original amount plus 0.07 of it, which is 1.07 times the original. Decreasing by 7% leaves 0.93 of it. The number 1.07 or 0.93 is called a multiplier, and multiplying is faster and safer than finding the change and adding it.

ATL SKILLS · THINKING, RESEARCH AND COMMUNICATION

Critical thinking: test whether a quoted rate is simple or compound, annual or monthly, before comparing. **Research:** read the total cost and the assumptions, not only the headline rate. **Communication:** state the period, the rate basis and the units; distinguish a percentage change from a change in percentage points.

ONE MULTIPLIER, MANY STEPS

8.1 Repeated percentage change

A percentage change acts on the amount that exists at that moment. When several changes happen in succession, each one uses the result of the previous change as its base. The multipliers therefore multiply: a 10% rise followed by a 20% fall has combined multiplier $1.10 \times 0.80 = 0.88$, a net fall of 12%.

The order of two changes does not affect the final multiplier, because multiplication is commutative. What does matter is whether a percentage is taken of the original amount or of a changed amount. This is why successive percentages cannot be added, and why "percentage" and "percentage point" are different units.

DEFINITIONS

Multiplier and percentage change

For a change of $r\%$, write r as a decimal. An increase has **multiplier** $1 + r$ and a decrease has multiplier $1 - r$. For successive changes with multipliers $m_1, m_2, \ldots, m_k$, the overall multiplier is $m_1 m_2 \cdots m_k$. The percentage change from an original amount P to a new amount A is $\frac{A-P}{P} \times 100\%$.

EXAMPLE **8.1** UP, THEN DOWN

QUESTION

A jacket costs £80. Its price is increased by 15%, and later the new price is reduced by 15%. Find the final price and the overall percentage change.

SOLUTION

1. A 15% increase has multiplier 1.15: $80 \times 1.15 = 92$.
2. A 15% decrease has multiplier 0.85: $92 \times 0.85 = 78.20$.
3. The combined multiplier is $1.15 \times 0.85 = 0.9775$.
4. 0.9775 is 97.75%, so the overall change is a fall of 2.25%. The final price is £78.20.

Check: 2.25% of 80 is 1.80, and $80 - 1.80 = 78.20$. The fall is not 0% because the 15% reduction acts on the larger price of £92.

EXAMPLE **8.2** REPEATED GROWTH

QUESTION

A city has 240,000 inhabitants. Its population grows by 2.5% each year. Find the population after 3 years, to the nearest person.

SOLUTION

1. The yearly multiplier is $1 + 0.025 = 1.025$.
2. After 3 years: $240{,}000 \times 1.025^3$.
3. $1.025^3 = 1.076891$, so the population is $258{,}453.75$.
4. To the nearest person: 258,454. The increase is 18,454.

Check: three years of simple 2.5% growth would add $3 \times 6{,}000 = 18{,}000$; compounding adds slightly more because each year's growth also grows.

EXAMPLE **8.3** MIXED CHANGES

QUESTION

The price of a ticket is £45. It rises by 12%, then falls by 8%. Find the final price and the overall percentage change.

SOLUTION

1. First change: $45 \times 1.12 = 50.40$.
2. Second change: $50.40 \times 0.92 = 46.368$.
3. Combined multiplier: $1.12 \times 0.92 = 1.0304$, an increase of 3.04%.
4. The final price is £46.37 to the nearest penny.

Check: 3.04% of 45 is 1.368, and $45 + 1.368 = 46.368$.

COMMON MISCONCEPTIONS

"Add the percentages." A 20% rise followed by a 10% rise is not 30%. The multipliers give $1.20 \times 1.10 = 1.32$, so the increase is 32%. The second 10% is taken of a larger amount, so it contributes more than 10% of the original.

"A rise and an equal fall cancel." They do not: $1.10 \times 0.90 = 0.99$. The fall is calculated on the increased amount, so the net result is 1% below the start.

EXERCISE 8A

MULTIPLIERS · SUCCESSIVE CHANGE

FLUENCY

1 Write the multiplier for each change: (a) $+7\%$ (b) -18% (c) $+2.5\%$ (d) -0.5%.

2 Increase £240 by 15%, then decrease £240 by 15%. Compare the two results.

3 A jacket costs £85. Its price rises by 8% and then falls by 10%. Find the final price and the overall percentage change.

4 A population of 18,000 grows by 1.8% per year for 2 years. Find the new population.

5 A price falls by 20% and then rises by 25%. Show that the final price equals the original, and explain why.

6 Which is larger: a 30% rise followed by a 30% fall, or a 25% fall followed by a 25% rise? Show the multipliers.

UNDERSTANDING AND REASONING

7 **Error analysis.** A student claims that a 10% rise followed by a 10% fall returns to the starting value. Show the correct result and explain the error.

8 A phone battery loses 4% of its charge each hour. Write the hourly multiplier and find the fraction of charge left after 6 hours.

9 A sale offers 30% off, then a further 20% off the reduced price. Explain why this is not 50% off, and find the true discount.

10 **Extension.** Two successive increases of 10% give 21%, not 20%. Show that two successive increases of $r\%$ give a single increase of $2r + r^2/100$ percent, and explain where the extra part comes from.

INTEREST ON INTEREST

8.2 Compound interest and multipliers

Simple interest is paid only on the original amount. Compound interest is paid on the original amount and on interest already credited, so the balance grows by the same multiplier each period. After n years the factor is $(1 + r)^n$, not $1 + nr$. The difference is small at first and large over time.

The same formula works for any repeated percentage change: population, prices, savings or a growing colony. The rate and the period must match — a monthly rate needs a number of months, not years.

DEFINITIONS

Principal, simple and compound interest

The **principal** is the original amount invested or borrowed. **Simple interest** is calculated only on the principal: $A = P(1 + nr)$. **Compound interest** is calculated on the principal and on interest already added: $A = P(1 + r)^n$, where r is the rate per period as a decimal and n is the number of periods.

EXAMPLE **8.4** COMPOUND INTEREST

QUESTION

£6,000 is invested at 4.5% per year, compounded annually. Find the value after 8 years and the interest earned.

SOLUTION

1. The annual multiplier is $1 + 0.045 = 1.045$.
2. After 8 years: $A = 6000 \times 1.045^8$.
3. $1.045^8 = 1.422101$, so $A = 8532.60$.
4. Interest earned: $8532.60 - 6000 = 2532.60$.

Check: simple interest would give $6000 \times 0.045 \times 8 = 2160$, so compounding adds £372.60.

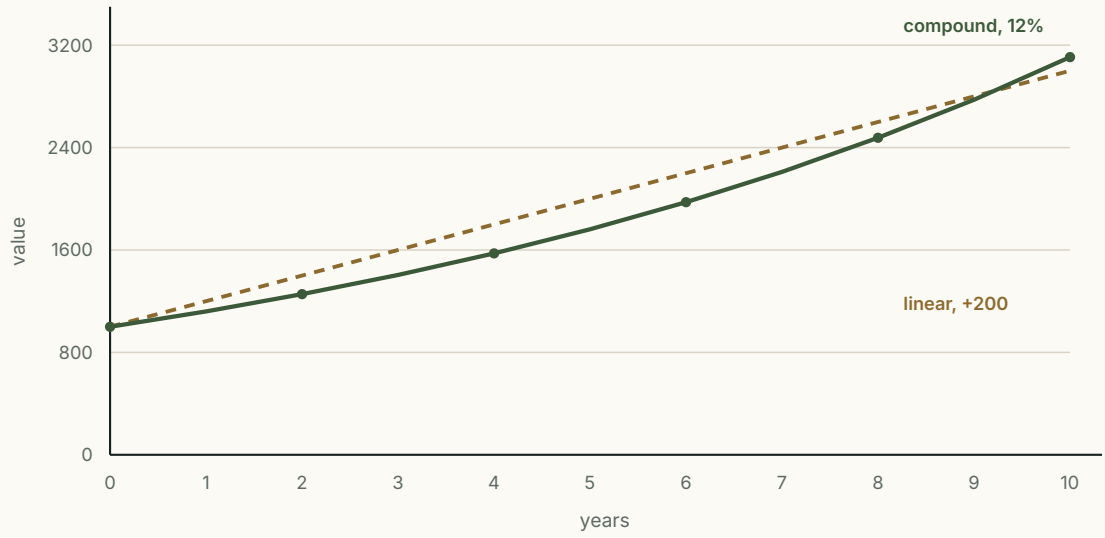


Figure 8.1 Linear and compound growth from the same start of 1,000. The compound curve is below the straight line at first, then crosses it between years 9 and 10 and pulls away.

EXAMPLE **8.5** SIMPLE VERSUS COMPOUND

QUESTION

£4,000 is invested at 3.5% per year for 5 years. Compare the value under simple interest and under compound interest.

SOLUTION

1 Simple interest: $A = 4000(1 + 5 \times 0.035) = 4000 \times 1.175 = 4700$.

2 Compound interest: $A = 4000 \times 1.035^5$.

3 $1.035^5 = 1.187686$, so $A = 4750.75$.

4 Compound interest is greater by $4750.75 - 4700 = 50.75$.

Check: the gap is small after 5 years because the interest on interest is still a small part of the total; at higher rates or longer times it grows quickly.

EXAMPLE **8.6** A MONTHLY RATE

QUESTION

An account pays 1.5% each month. Find the effective annual rate, and the value of £1,000 after one year.

SOLUTION

1 The monthly multiplier is $1 + 0.015 = 1.015$.

2 Twelve months give $1.015^{12} = 1.195618$.

3 The effective annual rate is $19.5618\% \approx 19.56\%$.

4 £1,000 grows to £1,195.62.

Check: twelve simple payments of 1.5% would give 18%, so monthly compounding adds about 1.56 percentage points on top.

EXERCISE 8B

COMPOUND INTEREST · POWERS

FLUENCY

1 Find the value of £2,000 invested at 5% per year, compounded annually, after 3 years.

2 Find the value of £5,000 invested at 3.2% per year for 6 years.

3 £1,200 is invested at 2.8% per year for 4 years. Find the interest earned.

4 £850 is invested at 3.5% per year for 3 years. Find the value under simple interest and under compound interest, and the difference.

5 £3,200 is invested at 4.2% per year for 4 years. Find its value.

UNDERSTANDING AND REASONING

6 How many complete years does £2,000 at 6% per year need to exceed £2,500? Use a table of yearly values.

7 £1,500 is invested at 1.2% per month for 12 months. Find its value.

EXERCISE 8B · CONTINUED

COMPOUND INTEREST · POWERS

8 The rule of 72 estimates a doubling time as $72 \div r$ years. At 6%, it gives 12 years. Verify this by calculating 1.06^{12}.

9 **Error analysis.** A student calculates the value of £3,000 at 5% for 8 years as $3000 \times 1.05 \times 8$. Explain the error and give the correct expression.

10 **Extension.** An account pays 1.5% per month. Show that its effective annual rate is about 19.56%, and explain why this is greater than $12 \times 1.5\%$.

COMMON MISCONCEPTIONS

"A 6% increase means multiplying by 0.06." It means multiplying by 1.06. The factor 0.06 is the interest added, not the new balance.

"Three years of compound interest is three times one year's interest." Each year's interest is calculated on a bigger balance: on £2,500 at 6% the yearly amounts are £150, then £159, then £168.54.

"2500×1.06^3 equals $2500 \times 1.06 \times 3$." An exponent repeats a multiplier; it is not a factor you can pull out. The two results are £2,977.54 and £7,950 — the second is not a balance in any account.

"6% a year for 3 years is an 18% increase." $1.06^3 = 1.191$, so the total increase is 19.1%. The extra 1.1% is the interest earned on the interest.

CH 08

WORKING BACKWARDS

8.3 Depreciation and reverse percentages

Assets such as vehicles, machinery and computers usually lose value over time. A constant percentage decrease is called depreciation, and it uses the same multiplier structure as compound interest, with $1 - r$ in place of $1 + r$. Appreciation is the corresponding increase.

Reverse percentage problems give the value after a change and ask for the value before it. The original amount is found by dividing by the multiplier, never by subtracting the percentage. A discount of 15% leaves 85%, so £68 after the discount means the original was $68 \div 0.85 = 80$.

DEFINITIONS

Depreciation and reverse percentage

Depreciation is a repeated percentage decrease: $A = P(1 - r)^n$. **Appreciation** is a repeated increase. In a **reverse percentage** problem, the original amount is $P = A \div m$, where m is the multiplier that produced A.

EXAMPLE **8.7** DEPRECIATION OF A VEHICLE

QUESTION

A car costs £22,000 and loses 13% of its value each year. Find its value after 3 years, to the nearest pound.

SOLUTION

1 The yearly multiplier is $1 - 0.13 = 0.87$.

2 $A = 22{,}000 \times 0.87^3$.

3 $0.87^3 = 0.658503$, so $A = 14{,}487.066$.

4 The value is £14,487, a fall of £7,513.

Check: 13% of 22,000 is 2,860; three simple losses would be 8,580, so the actual fall is smaller because each loss is taken on a smaller value.

EXAMPLE **8.8** REACHING A TARGET VALUE

QUESTION

A machine worth £16,000 loses 10% of its value each year. Find its value after 4 years, and the first year in which it is worth less than £8,000.

SOLUTION

1 After 4 years: $16{,}000 \times 0.9^4 = 16{,}000 \times 0.6561 = 10{,}497.60$.

2 We need $16{,}000 \times 0.9^n < 8{,}000$, so $0.9^n < 0.5$.

3 $0.9^6 = 0.5314$ and $0.9^7 = 0.4783$.

4 The first whole year is $n = 7$: $16{,}000 \times 0.9^7 = 7{,}652.75$, using the unrounded power, which is below £8,000.

Check: at $n = 6$ the value is £8,503.06, still above £8,000, so 7 is the first year.

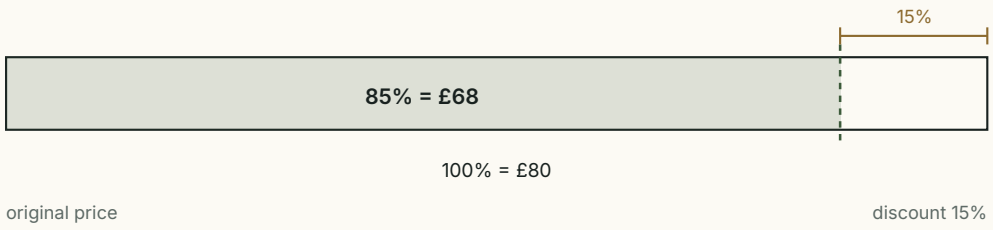


Figure 8.2 A reverse percentage as a bar model. The discounted price is 85% of the original; dividing by 0.85 recovers 100%.

EXAMPLE **8.9** FINDING THE ORIGINAL PRICE

QUESTION

After a 15% discount, a coat costs £68. Find the original price and the amount of the discount.

SOLUTION

1. A 15% discount leaves 85%, so the multiplier is 0.85.
2. Original $\times 0.85 = 68$, so original $= 68 \div 0.85$.
3. $68 \div 0.85 = 80$. The original price was £80.
4. The discount is $80 - 68 = 12$, which is 15% of 80.

Check: $80 \times 0.85 = 68$. Subtracting 15% of 68 would give 57.80, which is wrong because the 15% applies to the original price.

EXAMPLE **8.10** TAX AND SUCCESSIVE DISCOUNTS

QUESTION

(a) A price including 20% tax is £84. Find the price before tax. (b) After successive discounts of 20% and 15%, an item costs £51. Find its original price.

SOLUTION

1. (a) The multiplier is 1.20, so the price before tax is $84 \div 1.20 = 70$.
2. (b) The combined multiplier is $0.80 \times 0.85 = 0.68$.
3. Original $= 51 \div 0.68 = 75$.
4. Check the discounts: $75 \times 0.80 = 60$ and $60 \times 0.85 = 51$.

Check: 20% tax on 70 is 14, and $70 + 14 = 84$. The two discounts total 32%, not 35%, because the second applies to the reduced price.

EXERCISE 8C

DEPRECIATION · REVERSE PERCENTAGES

FLUENCY

1. A car worth £16,000 loses 15% of its value each year. Find its value after 3 years.
2. A laptop worth £1,200 loses 25% of its value each year. Find its value after 2 years.
3. A machine is worth £8,100 after 2 years of 10% annual depreciation. Find its original value.
4. After a 12% discount, a bag costs £66. Find its original price.
5. A price including 20% tax is £756. Find the price before tax.
6. A town's population falls by 2% per year and is 24,000 after 5 years. Find the population 5 years earlier.

UNDERSTANDING AND REASONING

7. A salary rises by 4% to £31,200. Find the salary before the rise.
8. After successive discounts of 25% and 12%, an item costs £66. Find its original price.

EXERCISE 8C · CONTINUED DEPRECIATION · REVERSE PERCENTAGES

9 **Error analysis.** To reverse a 20% increase, a student subtracts 20% from the final value. Explain why this is wrong and give the correct method.

10 **Extension.** An investment grows from £5,000 to £5,408 in 2 years at a constant annual rate. Find the annual rate.

COMMON MISCONCEPTIONS

"A 12% fall followed by a 12% rise restores the original." The multipliers are 0.88 and 1.12, whose product is 0.9856: the value ends 1.44% below where it began, because the rise is taken on the smaller base.

"To undo a 20% increase, subtract 20%." Increasing by 20% multiplies by 1.2; the reverse operation is dividing by 1.2, which is a fall of $\frac{1}{6} \approx 16.7\%$.

"12% a year for 3 years is a 36% fall." The base shrinks every year, so the value is $0.88^3 = 0.6815$, a fall of 31.9%. Adding the percentages overstates the loss.

"A depreciating asset loses a fixed amount each year." That is straight-line depreciation, a different model. A percentage model never reaches zero; a fixed-amount model does.

BEYOND THE HEADLINE RATE

8.4 Comparing financial offers

A headline rate is only one part of an offer. A fair comparison uses the same time period, includes fees and bonuses, and asks what the money is worth at the end. A lower rate with a large bonus can win in the short term and lose later; a lower monthly payment can hide a longer term and a larger total cost.

Effective rates convert different compounding patterns to one yearly scale. Total cost of credit is the sum of all payments minus the amount borrowed. Both are needed: the effective rate compares the price of money, while the total cost shows what will actually leave the account.

EXAMPLE **8.11** TWO SAVINGS OFFERS

QUESTION

£5,000 is available to save. Offer A pays 4.0% per year, compounded annually. Offer B pays 3.5% per year and adds a £80 bonus at the start. Complete a table for 6 years and state when A overtakes B.

SOLUTION

1. A: 5000×1.04^n. B: 5080×1.035^n.
2. Year 1: A £5,200.00; B £5,257.80. Year 2: A £5,408.00; B £5,441.82. Year 3: A £5,624.32; B £5,632.29.
3. Year 4: A £5,849.29; B £5,829.42. B leads for three years, then A leads.
4. At year 10, A is £7,401.22 and B is £7,165.84, so A is ahead by £235.38.

Check: B's £80 bonus is 1.6% of £5,000, while A's rate advantage is 0.5% per year, so the bonus is overtaken after a little more than three years.

EXAMPLE **8.12** CONTRACT OR BUY OUTRIGHT?

QUESTION

A phone can be bought for £620 plus £12 per month for data, or obtained on a 24-month contract at £35 per month including data. Compare the two options over 24 months and over 48 months.

SOLUTION

1. Contract over 24 months: $35 \times 24 = 840$.
2. Buy over 24 months: $620 + 12 \times 24 = 620 + 288 = 908$.
3. The contract is cheaper by £68 over 24 months.
4. Over 48 months, keeping the bought phone: $620 + 12 \times 48 = 1{,}196$. Two contracts: $2 \times 840 = 1{,}680$. Buying is cheaper by £484.

Check: the contract has the lower monthly payment (£35 against £37.83), but the phone is not owned at the end and a second contract is needed to keep the service.

COMPARISON CHECKLIST

Use the same period; include fees, bonuses and taxes; check whether the rate is simple or compound; check whether payments are monthly or annual; compare both the monthly payment and the total cost; state what happens at the end of the term.

EXAMPLE **8.13** THE COST OF BORROWING

QUESTION

A loan of £12,000 is offered in two ways over 24 months. Offer A: flat interest at 4.5% per year. Offer B: interest-free, with a 3% arrangement fee. Compare the monthly payment and the total cost.

SOLUTION

1. Offer A interest: $12{,}000 \times 0.045 \times 2 = 1{,}080$. Total $= 13{,}080$.
2. Offer A monthly payment: $13{,}080 \div 24 = 545$.
3. Offer B fee: $12{,}000 \times 0.03 = 360$. Total $= 12{,}360$.
4. Offer B monthly payment: $12{,}360 \div 24 = 515$. B is cheaper by £720 in total and £30 per month.

Check: $545 \times 24 = 13{,}080$ and $515 \times 24 = 12{,}360$. The fee is paid at the start, so B needs more cash up front even though its total is lower.

EXERCISE 8D

COMPARING OFFERS · TOTAL COST

FLUENCY

1. £1,200 is offered at 0.4% per month or at 5% per year. Which is better after one year? Show both multipliers.
2. A phone costs £560 plus £9 per month for data, or £32 per month on a 24-month contract. Find the total cost of each over 24 months.
3. A £9,000 loan is offered at flat 6% per year over 3 years, or interest-free with a 4% fee over 36 months. Find the monthly payment and total cost of each.
4. £3,000 is saved for 5 years. Offer A pays 4.2% compound; Offer B pays 3.9% plus a £150 bonus at the start. Find the value of each.

UNDERSTANDING AND REASONING

5. An account pays 0.3% per month; another pays 3.7% per year. Which effective annual rate is higher? Justify with a calculation.
6. A purchase of £1,000 is paid in 24 monthly instalments of £45. Find the total cost of credit and express it as a percentage of the purchase price.
7. **Error analysis.** A shopper compares two loans only by their monthly payments. Explain why this can lead to a worse choice, and state two other quantities to compare.
8. An advertisement says “3.9% APR”. List three pieces of information that are still needed before the total cost of a loan can be calculated.
9. **Extension.** Offer A pays 4.0% compound on £4,000. Offer B pays 3.6% plus a £60 bonus. Complete a table for 8 years and state when A overtakes B.

COMMON MISCONCEPTIONS

"The offer with the smaller percentage is the cheaper offer." Percentages are taken on different bases. Compare the actual amounts paid, over the same period, for the same goods and the same conditions.

"30% off and then a further 20% off is 50% off." Successive discounts multiply: $0.7 \times 0.8 = 0.56$, so the price is 44% below the original, not 50%.

"A price of £24 including 20% tax means the pre-tax price was £19.20." £19.20 is 20% *off* £24. The tax was added to the original, so the original is $24 \div 1.2 = £20$.

"A deposit is part of the discount." A deposit is money paid early, usually deducted from the balance still owed. Only a comparison that includes deposit, fees, interest and total paid can decide which offer is better.

WHEN THE CURVE MEETS THE WORLD

8.5 Modelling decisions and limitations

A financial model is a simplification. It assumes a constant rate, regular compounding, no unexpected fees and a population or market that can keep growing. Each assumption is a place where the prediction can fail. A responsible answer states the assumption, tests the sensitivity of the result, and says what would change the conclusion.

Exponential growth cannot continue forever in a limited system. Savings accounts are limited by contribution, populations by resources, and products by the number of possible customers. Recognising a ceiling is part of interpreting the model, not an optional extra.

EXAMPLE **8.14** A MODEL THAT BREAKS

QUESTION

A video has 500 views and the number of views doubles every 3 days. Model the views after t days. Predict day 30, then evaluate the prediction if the platform has 200,000 active users.

SOLUTION

1. Doubling every 3 days gives $V = 500 \times 2^{t/3}$.
2. At $t = 30$: $V = 500 \times 2^{10} = 500 \times 1{,}024 = 512{,}000$.
3. The platform has only 200,000 active users, so 512,000 views would require repeated viewing far beyond the audience.
4. At day 24, $V = 500 \times 2^{8} = 128{,}000$; at day 27, $V = 500 \times 2^{9} = 256{,}000$. The model exceeds the audience between days 24 and 27.

Check: doubling is a good local description while the audience is much larger than the number of viewers; it fails when the ceiling is approached. A model with a ceiling would grow quickly at first and then level off.

EXAMPLE 8.15 REAL VERSUS NOMINAL CHANGE

QUESTION

A salary of £30,000 rises by 3% per year while prices rise by 2.5% per year. Find the yearly real multiplier and the change in real value after 10 years.

SOLUTION

1. Salary multiplier 1.03; price multiplier 1.025.
2. Real multiplier per year: $1.03 \div 1.025 = 1.004878$, about $+0.49\%$ per year.
3. After 10 years: $1.004878^{10} = 1.0499$, a real increase of about 5.0%.
4. The salary becomes £40,317.49, but its purchasing power is about £31,496 at today's prices.

Check: the nominal rise is 34.4%, the price rise is 28.0%, and the difference in purchasing power is about 5% — not 6.4%, because the percentages apply to different bases.

EXAMPLE 8.16 AUDITING A PREDICTION

QUESTION

A student predicts a pension fund by calculating $150 \times 12 \times 30 \times 1.06^{30} \approx 310{,}000$. The plan is to save £150 per month for 30 years at 6% per year. Explain why the prediction overstates the fund.

SOLUTION

1. The calculation treats all £54,000 of contributions as if invested at the start for 30 full years.
2. In reality the contributions arrive month by month. The first payment earns almost 30 years of growth; the last earns almost none.
3. So the true fund is smaller than £310,000. The exact value requires summing a geometric series of monthly payments, which is developed in Year 5.
4. The model also assumes a constant 6% return, no fees and no inflation. Each assumption should be stated with the prediction.

Check: the prediction is an upper bound, not an estimate. An upper bound can be useful, but it must not be presented as the expected outcome.

EXERCISE 8E

MODELS · ASSUMPTIONS · EVALUATION

FLUENCY

1. A video has 800 views and doubles every 4 days. Write a model for the views after t days, and predict the views at day 24.
2. A salary rises by 4% while prices rise by 3%. Find the real multiplier and the real change over 12 years.
3. A population grows by 2.5% per year. Use the rule of 72 to estimate its doubling time, then find the first year in which it is at least double.
4. A website claims house prices rise 8% per year. By what factor would prices grow in 25 years? Give one reason the claim may not hold.

UNDERSTANDING AND REASONING

5. List three assumptions made when a savings balance is modelled by $A = P(1 + r)^n$.

EXERCISE 8E · CONTINUED

MODELS · ASSUMPTIONS · EVALUATION

6 Explain why a constant percentage growth model must fail once a product has almost all possible customers.

7 A claim says "prices double every 10 years". Find the implied annual percentage increase, and explain whether the claim is plausible.

8 A credit card charges 22% per year on a balance of £1,500. If no payment is made, find the balance after 2 years and the interest added.

9 **Extension.** A linear model adds £200 per year to an initial 1,000; an exponential model multiplies by 1.12 each year. Find the first year in which the exponential model is larger, using a table.

COMMON MISCONCEPTIONS

"A model that fits the last five years will fit the next five." It fits only if the rate stays fixed and the change still applies at the same intervals. State the range over which you expect the model to hold.

"Rounding each year's balance does no harm." A rounding error of £0.40 in year 3 is a different percentage in year 12. Carry the unrounded value and round only the answer the question asks for.

"6% per year compounded monthly is still exactly 6% per year." Monthly compounding applies $\frac{6}{12} = 0.5\%$ twelve times: $1.005^{12} = 1.0617$, so the balance grows by 6.17% over the year.

"An estimate is not worth calculating." A model is judged by whether its error changes the decision. If two offers differ by £40 and the assumptions move the answer by £10, the calculation still decides.

INVESTIGATION · CRITERION B

8.6 Doubling and halving

Doubling times turn a percentage rate into a length of time that people can interpret. The rule of 72 is a quick estimate: doubling time $\approx 72 \div r$ years for a rate of $r\%$. It is accurate enough for mental arithmetic at everyday rates, and its error is itself worth investigating.

RATE (% PER YEAR)	FIRST WHOLE YEAR WITH MULTIPLIER ≥ 2	$72 \div r$	DIFFERENCE
3	24	24.0	0.0
4	18	18.0	0.0
5	15	14.4	+0.6
6	12	12.0	0.0
8	10	9.0	+1.0
10	8	7.2	+0.8
12	7	6.0	+1.0

INVESTIGATION · CRITERION B

How good is the rule of 72?

1. Verify the table values for $r = 6$, $r = 8$ and $r = 12$ by calculating powers.
2. For each rate, compare the exact first whole year with $72 \div r$. Describe the pattern of the differences.
3. Compare the rules $70 \div r$ and $72 \div r$ with exact doubling times found by trial, to one decimal place. Which rule is closer for rates of 3–4%? Which is closer for rates of 8–12%?
4. At 10% depreciation per year, find the first whole year in which an asset is worth less than half its value. Compare with $72 \div 10$.
5. Explain why the doubling time is roughly inversely proportional to the rate. Use the idea that a small rate r gives a yearly multiplier close to $1 + r$.
6. Evaluate the rule at 50% per year. Explain why it fails badly there, and state the range of rates for which it is a useful estimate.

Generalise and justify. Show that a doubling time of n years at rate r satisfies $(1 + r)^n = 2$. Use logarithms or a numerical table to justify your estimate, and explain why the constant is near 70 rather than 100.

MIXED PRACTICE 8

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–8 check fluency; 9–15 require reasoning; 16–18 are multi-step or unfamiliar. Allow 55 minutes.

1 Write the multiplier for an increase of 6.5%.

2 Increase £350 by 12%.

3 Decrease £480 by 35%.

4 Find the value of £2,500 at 4% compound interest after 3 years.

5 Find the value of £7,500 at 2.5% compound interest after 4 years.

6 A machine worth £12,000 depreciates by 18% per year. Find its value after 2 years.

7 After a 25% discount, an item costs £96. Find its original price.

8 A price including 20% tax is £66. Find the price before tax.

9 £5,000 is invested at 5% for 5 years. Compare simple and compound interest.

10 How many complete years does £3,000 at 4.5% need to exceed £3,600?

11 A price rises by 15% and then falls by 15%. Find the overall percentage change.

12 **Error analysis.** A student says 5% per year for 10 years gives a 50% increase. Explain why this is wrong for compound interest and give the correct multiplier.

13 £2,000 is invested for 6 years at 4% compound, or at 4.4% simple. Which gives more, and by how much?

14 A salary rises by 3.5% while prices rise by 2.8%. Find the real percentage change.

15 A car loses 20% of its value in the first year and 15% in the second. Find the overall percentage loss.

16 A loan of £8,000 is repaid in 36 monthly payments of £245. Find the total cost of credit.

17 **Extension.** An investment doubles in 12 years. Find its annual percentage rate, to 2 decimal places.

MIXED PRACTICE 8 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

18 **Extension.** At 4.5% per year the rule of 72 gives 16 years. Find the first whole year in which an investment doubles, then explain why the exact doubling time is slightly less than 16 years.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and assessed against the criteria shown. Your teacher will confirm which tasks are required and how long you have.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Growth and decay: short response	30	35 min
B · Investigating patterns	The lily pond	24	40 min
C · Communicating	Explaining compound growth	20	35 min
D · Applying mathematics in real-life contexts	Buy or subscribe?	24	45 min
Total	All four tasks	98	155 min

The tasks begin overleaf. Read the instructions and evidence requirements before you start, and show the reasoning that supports each answer.

CRITERION A · KNOWING AND UNDERSTANDING

35 minutes · 30 marks

Growth and decay: short response

CALCULATOR | SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

19 Write the multiplier for each change: (a) $+4.5\%$ (b) -12% (c) $+0.8\%$ (d) -30%. [4]

20 Find the value of £4,500 invested at 3.5% per year, compounded annually, after 6 years. [4]

21 A vehicle worth £18,000 depreciates by 15% per year. Find its value after 3 years. [4]

22 After an 18% increase, a price is £29,500. Find the original price. [4]

23 £2,000 is invested at 6% for 4 years. Find the value under simple interest and under compound interest, and the difference. [5]

24 Using a table of yearly values, find the first complete year in which £3,000 at 8% per year exceeds £6,000. [4]

25 **Error analysis.** A student claims that two successive 20% increases give a 40% increase. Identify the error and give the correct percentage. [3]

26 Explain why 0.9×1.1 is less than 1, using the idea of the base of each percentage. [2]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

The lily pond

Instructions. Work independently. A lily patch covers 1 m² on day 0 and doubles its area every day. The pond has an area of 1,024 m².

1. Record the covered area for days 0 to 6 in a table, and describe the pattern. [6]
2. Write a rule for the area A_d on day d, and explain why the rule has this structure. [4]
3. On which day is the pond completely covered? On which day is it half covered? Justify both answers. [5]
4. If the patch starts at 4 m² instead of 1 m², on which day is the pond covered? Explain using the rule. [4]
5. Explain why the model cannot apply after the pond is full, and describe how the graph would change if the lily were limited by space. [5]

Evidence: a complete table, a rule with a structural explanation, correct day numbers with justification, and an evaluation of the model's limit.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

Explaining compound growth

Instructions. Write a 300–400 word explanation for a student who has missed the lessons on financial mathematics. Your explanation must: (i) distinguish simple and compound interest with a worked example on £2,000 at 5% for 3 years; (ii) explain the multiplier method and why successive percentages cannot be added; (iii) show how to reverse a 20% increase; (iv) state two assumptions of a compound-interest model; (v) give one common error and how to avoid it.

Evidence: clear paragraphs, correct notation, a side-by-side calculation, and an explicit statement of assumptions.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Buy or subscribe?

Instructions. A bicycle can be bought for £850, with maintenance of £60 per year and a resale value of £300 after 4 years. A subscription costs £25 per month and includes maintenance. The bicycle has a battery that loses 8% of its capacity each year.

1. Find the total cost of buying and running the bicycle for 4 years, allowing for the resale value. [5]
2. Find the total cost of the subscription for 4 years. [4]
3. Compare the two options and state one non-financial factor that could change the decision. [5]
4. Find the battery capacity after 4 years as a percentage of its original capacity. [4]
5. Recommend an option for a student who will keep the bicycle for 4 years. Justify your recommendation using your calculations, and state two assumptions. [6]

Evidence: totals with units, a like-for-like comparison over the same period, a capacity calculation, and a justified recommendation with assumptions.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A percentage change of $r\%$ has multiplier $1 + r/100$; successive changes multiply.
- Compound interest uses $A = P(1 + r)^n$; simple interest uses $A = P(1 + nr)$.
- Depreciation uses $A = P(1 - r)^n$, the same structure with a multiplier below 1.
- To reverse a percentage change, divide by the multiplier.
- Compare offers over the same period, using total cost and the effective annual rate.
- A model is only as good as its assumptions; exponential growth cannot continue past a ceiling.
- Doubling time is approximately $72 \div r$ years for a rate of $r\%$ per year.

$$A = P(1 + r)^n \qquad A = P(1 - r)^n$$

$$\text{multiplier} = 1 + \frac{r}{100} \qquad P = \frac{A}{m}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can convert a percentage change into a multiplier.
- ☐ I can combine two or more successive changes.
- ☐ I can calculate compound interest over a number of years.
- ☐ I can calculate depreciation to a given year.
- ☐ I can find the interest earned, not just the final value.
- ☐ I can reverse a percentage change by dividing by the multiplier.
- ☐ I can compare offers using total cost over the same period.
- ☐ I can distinguish simple and effective annual rates.
- ☐ I can state two assumptions of a financial model.
- ☐ I can estimate a doubling time and evaluate the estimate.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which representation — a multiplier, a table or a graph — made compound growth clearest to you? Explain with an example.
2. Describe a time you compared two offers and the headline rate gave the wrong impression. What calculation corrected it?
3. The statement of inquiry links repeated multiplication to justified decisions. Illustrate this link with one saving example and one borrowing example.
4. Look ahead to Chapter 9: how could a scale factor be treated as a multiplier, and what changes when the multiplier applies to length, area and volume?

WHAT SURVIVES A MOVE, A TURN AND A CHANGE OF SIZE

Congruence, Similarity & Transformations

09

FORM · SPACE · SCALE

A figure can be moved, turned, flipped or enlarged. Some properties survive every one of these changes; others survive only when the size is left alone. Knowing which is which turns shape into a measuring instrument.

STATEMENT OF INQUIRY

Form can be preserved while size changes. Congruence keeps every length and angle; similarity keeps angles and multiplies every length by one factor. That single factor controls area and volume in a predictable way, so ratios can measure objects that cannot be reached.

KEY CONCEPT

Form

RELATED CONCEPTS

Space · Equivalence · Representation

GLOBAL CONTEXT

Scientific and technical innovation — scale models, design and measurement at a distance

FACTUAL

Which properties survive a translation, a rotation, a reflection and an enlargement?

CONCEPTUAL

Why does one scale factor affect length, area and volume differently?

DEBATABLE

Can a scale model ever be a reliable substitute for the object it represents?

IN THIS CHAPTER

9.1 Transformations 9.2 Similarity 9.3 Enlargement 9.4 Coordinates 9.5 Area and volume 9.6 Investigation

OPENING PROBLEM · THE SAME SHAPE, A DIFFERENT SIZE

A mural, a flagpole and a tiled floor

A photograph measuring 12 cm by 18 cm is to be reproduced as a wall mural 3.6 m by 5.4 m. In the courtyard, the floor is covered with congruent triangular tiles. The caretaker needs the height of a flagpole that is too tall to climb, so she measures its shadow instead: a vertical stick 2.0 m long casts a shadow 2.4 m long, and at the same moment the flagpole casts a shadow 11.4 m long.

1. Show that the mural is an enlargement of the photograph with scale factor 30, and find the factor by which the area increases.
2. Paint covers about 4 m^2 per litre. Find the area of the mural and the quantity of paint needed, to the nearest 0.1 litre.
3. Why are the photograph and the mural similar but not congruent? Name one measurement that would be enough to show they are similar.
4. Use the stick and its shadow to find the flagpole's height. Explain why the two triangles formed by the sun's rays are similar, and which angle they share.
5. A tile on the floor is lifted, turned through 90° and laid in a new position. Which of its measurements change? Which transformations would produce a congruent copy of a tile?
6. The printer charges by the length of the longest side, but the signwriter charges by area. Explain why doubling the longest side doubles the printer's charge but multiplies the signwriter's charge by four.

A photograph and a mural share every angle and every ratio of lengths; they differ only in size. A shadow and the object casting it form a pair of triangles with the same angles, which is why a stick can measure a flagpole. Transformations give this everyday reasoning a precise language: they separate what an object *is* from where it happens to sit, how it is turned, or how large it is drawn.

ORIENTATION

9.0 Learning intentions

- Identify and describe translations, rotations, reflections and enlargements.
- Use the properties of congruent figures to justify that lengths and angles are preserved.
- Test triangles for similarity using angle and side criteria, and use the scale factor to find unknown lengths.
- Construct and describe enlargements, including fractional and negative scale factors.
- Apply coordinate rules for transformations and combinations of transformations.
- Use the relationships between length, area and volume scale factors in modelling.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

describe a transformation precisely; decide whether two figures are congruent or similar and justify the decision; find a missing length using a scale factor; enlarge a figure from a given centre; apply coordinate rules and combine two transformations; calculate an area or volume after scaling and work backwards from an area or volume ratio.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit ratio in Chapter 2, powers and roots in Chapter 1, and angle facts in Chapter 6 before starting.

1. Solve $\frac{x}{6} = \frac{4}{3}$.
2. Simplify the ratio $12 : 18$.
3. Find the area of a rectangle measuring 4 cm by 7 cm.
4. Evaluate 3^2, 2^3 and $\sqrt{49}$.
5. Find $\sqrt[3]{64}$.
6. State the equation of the x-axis.

REFRESHER

A ratio compares two quantities of the same kind. If two lengths are in the ratio $3 : 5$, then one is $3/5$ of the other, and every corresponding length in the two figures is multiplied by the same factor. Angles are not lengths, so they are never multiplied by that factor: equal angles stay equal. In similar figures the angles match and the sides are in a constant ratio; in congruent figures the sides are equal as well.

ATL SKILLS · THINKING, RESEARCH AND COMMUNICATION

Critical thinking: distinguish what a transformation preserves from what it changes, and test a claim of similarity by checking angles and ratios separately. **Research:** choose measurement methods for inaccessible objects and evaluate their sources of error. **Communication:** name vertices in corresponding order and state the scale factor with its meaning.

MOVING WITHOUT CHANGING

9.1 Transformations and congruence

A **transformation** changes the position, orientation or size of a figure. The original figure is the object or pre-image; the result is the image. Vertices of the image are written with a dash, so A maps to A'. A transformation that keeps every distance the same is an **isometry**: translations, rotations and reflections are isometries, while an enlargement is not, unless its scale factor is 1 or -1.

Two figures are **congruent** when one can be moved onto the other by an isometry. Congruent figures have equal corresponding sides and equal corresponding angles. A reflection produces a congruent image but reverses orientation: a clockwise labelling becomes anticlockwise. Congruence does not require orientation to be preserved, only lengths and angles.

DEFINITIONS

Transformation, isometry and congruence

A **transformation** maps every point of a figure to an **image** point. An **isometry** preserves distance, so every length and angle is unchanged. Two figures are **congruent** if an isometry maps one exactly onto the other; we write $\triangle ABC \cong \triangle DEF$ with vertices in corresponding order.

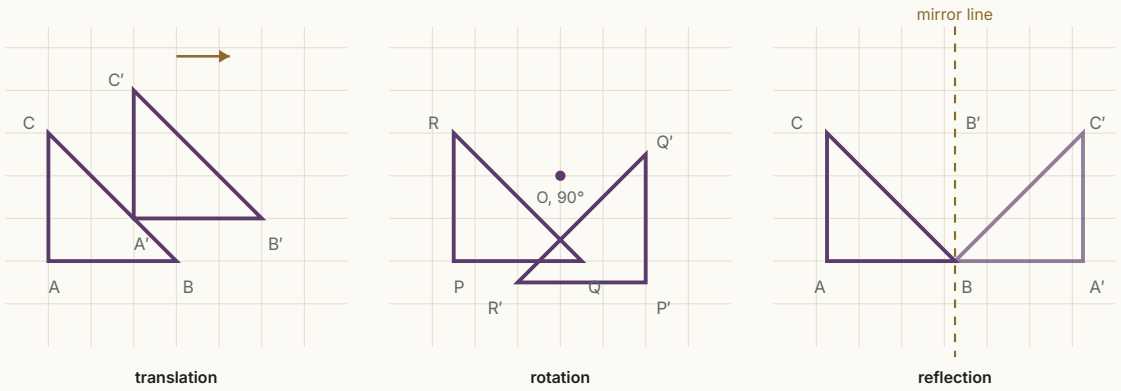


Figure 9.1 Three isometries. Each image is congruent to its original: all side lengths and angles are unchanged. Translation and rotation preserve orientation; reflection reverses it.

EXAMPLE 9.1 TRANSLATION ON A GRID

QUESTION

Triangle ABC has vertices $A(2, 1)$, $B(5, 1)$ and $C(2, 4)$. It is translated by the vector $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$. Find the image coordinates and verify that the image is congruent to ABC.

SOLUTION

1. Add the vector to each vertex: $A' = (2 + 4,\ 1 - 3) = (6, -2)$.
2. Similarly $B' = (9, -2)$ and $C' = (6, 1)$.
3. $AB = 5 - 2 = 3$ and $A'B' = 9 - 6 = 3$; $AC = 4 - 1 = 3$ and $A'C' = 1 - (-2) = 3$.
4. $BC = \sqrt{3^2 + 3^2} = \sqrt{18}$ and $B'C' = \sqrt{3^2 + 3^2} = \sqrt{18}$. Angles are unchanged, so $\triangle A'B'C' \cong \triangle ABC$.

Check: a translation slides every point by the same vector, so the figure cannot be stretched or turned. Comparing the three side lengths is enough because three sides fix a triangle.

EXAMPLE 9.2 ROTATION ABOUT THE ORIGIN

QUESTION

Rotate the triangle with vertices $A(2, 1)$, $B(5, 1)$ and $C(2, 4)$ through 90° clockwise about the origin. Find the image and explain why it is congruent to the original.

SOLUTION

1. A quarter-turn clockwise maps (x, y) to $(y, -x)$.
2. $A' = (1, -2)$, $B' = (1, -5)$, $C' = (4, -2)$.
3. $AB = 3$ and $A'B' = 5 - 2 = 3$; $AC = 3$ and $A'C' = 4 - 1 = 3$.
4. A rotation is an isometry: it preserves every distance and angle, so the image is congruent. Orientation is also preserved.

Check: the distance of A from the origin is $\sqrt{2^2 + 1^2} = \sqrt{5}$, and A' is also $\sqrt{1^2 + 2^2} = \sqrt{5}$ from the origin, as a rotation requires.

EXAMPLE 9.3 REFLECTION IN THE LINE $y = x$

QUESTION

Reflect the triangle with vertices $A(2, 1)$, $B(5, 1)$ and $C(2, 4)$ in the line $y = x$. Find the image. The image is congruent to the original, yet the labelling runs the other way round. Explain why this is not a contradiction.

SOLUTION

1. Reflection in $y = x$ swaps the coordinates: $(x, y) \mapsto (y, x)$.
2. $A' = (1, 2)$, $B' = (1, 5)$, $C' = (4, 2)$.
3. All three side lengths are unchanged: $AB = A'B' = 3$, $AC = A'C' = 3$, $BC = B'C' = \sqrt{18}$.
4. Congruence requires equal lengths and angles, not the same sense of turn. A reflection reverses orientation, so the image is a mirror copy.

Check: the midpoint of AA' is $(1.5, 1.5)$, which lies on $y = x$, and AA' has gradient -1, perpendicular to the mirror line. Both conditions define a reflection.

COMMON MISCONCEPTIONS

"A reflection keeps everything the same." It keeps lengths and angles, but it reverses orientation. The letters A, B, C taken clockwise become anticlockwise in the image, so a reflected object cannot be slid and turned back onto the original unless it is symmetrical.

"Congruent means identical." Congruent figures have the same size and shape, not the same position or orientation. Two identical tiles remain congruent wherever they are laid, provided the tile is not stretched.

EXERCISE 9A

ISOMETRIES · CONGRUENCE

FLUENCY

1. Describe the translation that maps $A(2, 3)$ to $A'(5, -1)$.
2. Triangle $P(1, 1)$, $Q(4, 1)$, $R(1, 3)$ is translated by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$. Find the image coordinates.
3. Reflect the point $(3, 4)$ in the x-axis, then in the y-axis.
4. Rotate $(5, 2)$ through $90°$ clockwise about the origin.
5. Which of these are isometries: translation, rotation, reflection, enlargement with scale factor 3?

UNDERSTANDING AND REASONING

6. Two triangles have sides $3, 4, 5$ and $6, 8, 10$. Are they congruent, similar, or both? Explain.
7. **Error analysis.** A student writes: "A reflection makes the figure a different size." Explain what the student has confused, and correct the statement.
8. Two reflections are made in a pair of parallel lines 4 cm apart. Describe the single transformation that has the same effect, and state its distance.
9. Which single transformation is equivalent to a reflection in the x-axis followed by a reflection in the y-axis? Justify your answer with coordinates.

EXERCISE 9A · CONTINUED ISOMETRIES · CONGRUENCE

10 **Extension.** Two reflection lines meet at 35°. Show that the combined reflection is a rotation, state its angle, and explain why the angle is twice the angle between the lines.

SAME SHAPE, NEW SIZE

9.2 Similarity and similar triangles

Similar figures have the same shape but not necessarily the same size. Every pair of corresponding angles is equal, and every pair of corresponding sides is multiplied by the same number. That number is the **scale factor**, written k. Congruence is the special case $k = 1$.

For triangles, checking all three angles and all three sides is unnecessary. If two angles match, the third must match too, because angles in a triangle total 180°. This gives the AA test, the most useful of the similarity tests, especially when lines are parallel or angles are shared.

DEFINITIONS

Similar figures and the tests for similar triangles

Two figures are **similar** if corresponding angles are equal and corresponding lengths are in the same ratio: $\frac{A'B'}{AB} = \frac{B'C'}{BC} = \frac{C'A'}{CA} = k$. Triangles are similar if **AA** (two angles equal), **SSS** (all three pairs of sides in the same ratio) or **SAS** (two pairs of sides in the same ratio with the included angles equal). Write $\triangle ABC \sim \triangle DEF$ with corresponding vertices in order.

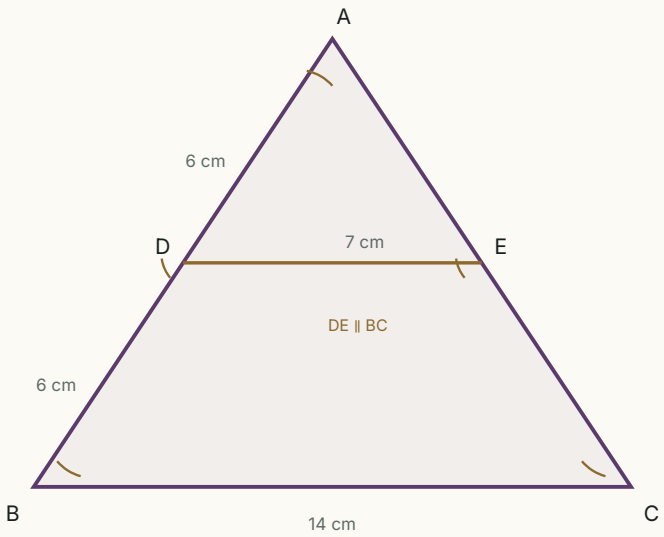


Figure 9.2 Because $DE \parallel BC$, the corresponding angles at B and D, and at C and E, are equal. Angle A is shared, so $\triangle ADE \sim \triangle ABC$ by AA. Here the scale factor is $1/2$.

EXAMPLE **9.4** FINDING MISSING SIDES

QUESTION

Triangles ABC and DEF are similar, with $AB = 6$ cm, $BC = 9$ cm, $AC = 7.5$ cm and $DE = 4$ cm. Find EF and DF.

SOLUTION

1. Match corresponding sides using the vertex order: $AB \leftrightarrow DE$, so the scale factor from ABC to DEF is $k = \frac{DE}{AB} = \frac{4}{6} = \frac{2}{3}$.
2. $EF = k \times BC = \frac{2}{3} \times 9 = 6$ cm.
3. $DF = k \times AC = \frac{2}{3} \times 7.5 = 5$ cm.
4. Check: $\frac{6}{4} = \frac{9}{6} = \frac{7.5}{5} = 1.5$, so each ratio is the same.

Check: the image is smaller, so $k < 1$. Every side of DEF should be two-thirds of the matching side of ABC, not two-thirds of an arbitrary side.

EXAMPLE **9.5** PROVING SIMILARITY WITH PARALLEL LINES

QUESTION

In triangle ABC, D lies on AB and E lies on AC, with $DE \parallel BC$. Given $AD = 4$ cm, $DB = 6$ cm and $DE = 5$ cm, prove that $\triangle ADE \sim \triangle ABC$ and find BC.

SOLUTION

1. Angle A is shared by both triangles.
2. Since $DE \parallel BC$, the corresponding angles satisfy $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$.
3. Two equal angles are enough (AA), so $\triangle ADE \sim \triangle ABC$.
4. $AB = AD + DB = 10$ cm, so $k = AD/AB = 4/10 = 0.4$. Then $BC = DE/k = 5/0.4 = 12.5$ cm.

Check: $DE/BC = 5/12.5 = 0.4$, the same as AD/AB, as similarity requires. Note that DB is not a side of the small triangle, so it must not be used in the ratio.

EXAMPLE **9.6** A RAMP WITH TWO SUPPORTS

QUESTION

A straight ramp rises at a constant angle from level ground. Two vertical posts support it. The first post is 1.2 m high and the second is 1.8 m high. The posts are 2.4 m apart. Find the distance from the foot of the ramp to the first post.

SOLUTION

1. Let x metres be the distance from the foot to the first post. Each post, the ground and the ramp form a right-angled triangle.
2. The two triangles share the angle at the foot of the ramp, and both have a right angle, so they are similar by AA.
3. Corresponding heights are in the same ratio as corresponding horizontal distances: $1.2 : x = 1.8 : (x + 2.4)$.
4. Cross-multiply: $1.2(x + 2.4) = 1.8x$, so $1.2x + 2.88 = 1.8x$ and $x = 4.8$.

Check: the first post is 4.8 m from the foot and the second is 7.2 m away. The ratios are $1.2/4.8 = 0.25$ and $1.8/7.2 = 0.25$, as required.

EXERCISE 9B

SIMILAR TRIANGLES · SCALE FACTOR

FLUENCY

1. Are two triangles with angles $50°, 60°, 70°$ and $50°, 70°, 60°$ similar? Give a reason.
2. Triangles are similar with corresponding sides 6 cm and 4 cm. A second side of the larger triangle is 9 cm. Find the matching side of the smaller triangle.
3. In similar triangles ABC and DEF, $AB = 8$, $BC = 12$ and $DE = 6$. Find EF.
4. A photograph 10 cm by 15 cm is enlarged to 14 cm by 21 cm. Are the two rectangles similar? Give the scale factor.

UNDERSTANDING AND REASONING

5. Explain why knowing two angles of a triangle is enough to test for similarity.
6. In triangle ABC, $DE \parallel BC$ with $AD = 3$ cm, $DB = 5$ cm and $DE = 6$ cm. Find BC.
7. **Error analysis.** A student matches the longest side of one triangle with the shortest side of the other. Explain why this can give a wrong scale factor, and state how the vertex order prevents it.
8. A vertical pole 1.8 m high casts a shadow 2.4 m long. At the same time a tree casts a shadow 9.6 m long. Find the height of the tree, and state one assumption.
9. Two triangles have a pair of vertically opposite angles. A second pair of angles is equal because two lines are parallel. Prove the triangles are similar.
10. **Extension.** In triangle ABC, $DE \parallel BC$ with $AD : DB = 2 : 3$. Find the ratio $DE : BC$ and the ratio of the areas of $\triangle ADE$ and $\triangle ABC$.

COMMON MISCONCEPTIONS

"Similar shapes have equal sides." Similar shapes have equal *angles* and sides in the same ratio. Equal sides, as well as equal angles, is the stricter condition of congruence.

"Any two rectangles are similar." Only if their side ratios match. $12\text{ cm} \times 18\text{ cm}$ and $10\text{ cm} \times 15\text{ cm}$ are both $2:3$ and are similar; 12×18 and 10×16 are $2:3$ against $5:4$ and are not.

"One equal angle is enough to prove two triangles similar." Infinitely many triangles share a single angle. AA similarity needs two pairs of equal angles — and the third pair then follows.

"Match the shortest side of one with the longest of the other." Corresponding sides lie opposite equal angles, so the longest pairs with the longest and the shortest with the shortest.

SCALING FROM A POINT

9.3 Enlargement and scale factors

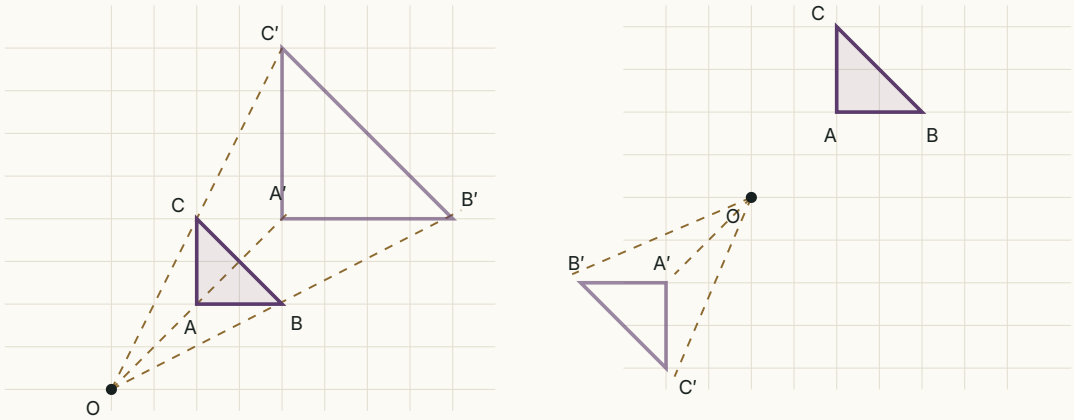
An enlargement needs two pieces of information: a centre and a scale factor. Every vertex of the image lies on the line through the centre and the original vertex, at a distance multiplied by k. Angles are unchanged, every side is multiplied by $|k|$, and corresponding sides remain parallel.

When $k > 1$ the image is larger and lies beyond the original from the centre. When k is between 0 and 1 the image is smaller and lies between the centre and the original. When $k < 0$ the image lies on the opposite side of the centre, as though the figure had been enlarged and then rotated through $180°$.

DEFINITIONS

Enlargement, centre and scale factor

An **enlargement** with **centre** O and scale factor $k \neq 0$ maps each point P to P' on the line OP, with $OP' = |k| \times OP$. If $k > 0$, P' lies on the same side of O as P; if $k < 0$, it lies on the opposite side. The image is similar to the original with scale factor $|k|$, and it is congruent only when $|k| = 1$.



k = 2: beyond the centre, same orientation

k = −1: opposite the centre, turned 180°

Figure 9.3 Enlargements from O. With $k = 2$ the image is twice as far from O on the same side. With $k = -1$ it is the same size but opposite O, which is equivalent to a half-turn. All lengths are multiplied by $|k|$, and all angles are unchanged.

EXAMPLE **9.7** ENLARGING FROM A CENTRE

QUESTION

Triangle ABC has vertices $A(2, 1)$, $B(3, 3)$ and $C(2, 4)$. Enlarge it from $O(1, 1)$ with scale factor 2. Find the image coordinates and show that $A'B' = 2AB$.

SOLUTION

1 Work with vectors from the centre. $\vec{OA} = (1, 0)$, so $\vec{OA'} = 2(1, 0) = (2, 0)$ and $A' = (1 + 2,\ 1 + 0) = (3, 1)$.

2 $\vec{OB} = (2, 2)$, so $\vec{OB'} = (4, 4)$ and $B' = (5, 5)$.

3 $\vec{OC} = (1, 3)$, so $\vec{OC'} = (2, 6)$ and $C' = (3, 7)$.

4 $AB = \sqrt{1^2 + 2^2} = \sqrt{5}$ and $A'B' = \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5}$, so $A'B' = 2AB$.

Check: each image point is twice as far from $O(1, 1)$ as its original. For example $OA = \sqrt{1^2 + 0^2} = 1$ and $OA' = 2$. The area of the image should be $2^2 = 4$ times the original area.

EXAMPLE **9.8** FRACTIONAL AND NEGATIVE SCALE FACTORS

QUESTION

Use the triangle with vertices $A(2,1)$, $B(3,3)$ and $C(2,4)$, and the origin O. (a) Enlarge it with scale factor $1/2$. (b) Enlarge it with scale factor -2. Describe where each image lies and whether its orientation is preserved.

SOLUTION

1. (a) Multiply each coordinate by $1/2$: $A'(1, 0.5)$, $B'(1.5, 1.5)$, $C'(1, 2)$. The image is smaller and lies between O and the original.
2. (b) Multiply each coordinate by -2: $A'(-4, -2)$, $B'(-6, -6)$, $C'(-4, -8)$. The image is larger and lies on the opposite side of O.
3. In (a) the sides are halved, so $A'B' = \sqrt{5}/2$; in (b) they are doubled, so $A'B' = \sqrt{20} = 2\sqrt{5}$.
4. Angles are unchanged in both cases. A negative factor turns the figure through $180°$, which preserves orientation, so the labelling still runs the same way round.

Check: the origin is not the midpoint of AA' here, because $k = -2$, not -1. Instead A' lies on the opposite ray from A and $OA' = 2\,OA$: $OA = \sqrt{5}$ and $OA' = \sqrt{20} = 2\sqrt{5}$.

EXAMPLE **9.9** FINDING THE CENTRE AND FACTOR

QUESTION

Triangle ABC with $A(2,2)$, $B(4,2)$, $C(2,5)$ is enlarged to $A'(3,3)$, $B'(6,3)$, $C'(3,7.5)$. Find the scale factor and the centre of enlargement.

SOLUTION

1. $AB = 2$ and $A'B' = 3$, so the scale factor is $k = \frac{3}{2} = 1.5$.
2. The centre lies on the line joining each vertex to its image. Line AA' passes through $(2,2)$ and $(3,3)$, so its equation is $y = x$.
3. Line CC' passes through $(2,5)$ and $(3,7.5)$, with gradient 2.5, so $y = 2.5x$.
4. Solving $y = x$ and $y = 2.5x$ gives $x = 0$, $y = 0$. The centre is the origin and $k = 1.5$.

Check: $OA = \sqrt{2^2 + 2^2} = \sqrt{8}$ and $OA' = \sqrt{3^2 + 3^2} = \sqrt{18} = 1.5\sqrt{8}$, so $OA' = 1.5\,OA$. The image coordinates are 1.5 times the original coordinates, confirming the origin as centre.

EXERCISE 9C

ENLARGEMENTS · CENTRES · SCALE FACTORS

FLUENCY

1. Enlarge the triangle with vertices $(2,1)$, $(4,1)$ and $(2,3)$ from the origin with scale factor 3.
2. Enlarge the same triangle from the origin with scale factor $\frac{1}{2}$.
3. Enlarge the point $(2,3)$ from the origin with scale factor -1.
4. A length of 5 cm becomes 12.5 cm under an enlargement. Find the scale factor.
5. Enlarge the triangle $(2,1)$, $(4,1)$ and $(2,3)$ from the centre $(1,1)$ with scale factor 2.

EXERCISE 9C · CONTINUED ENLARGEMENTS · CENTRES · SCALE FACTORS

UNDERSTANDING AND REASONING

6 The area of a triangle is 12 cm^2. Its image under an enlargement has area 108 cm^2. Find the scale factor.

7 Where does the centre of an enlargement lie in relation to corresponding vertices? Explain how to locate it with a ruler.

8 **Error analysis.** A student says that a scale factor of $\frac{1}{2}$ makes the image twice as large. Explain the error and describe the correct image.

9 Describe the position and orientation of the image when the scale factor is -2, compared with the original.

10 **Extension.** Triangle $A(1, 1)$, $B(3, 1)$, $C(1, 2)$ is enlarged from $O(2, 3)$ with scale factor -2. Find the image coordinates and verify that the side lengths are doubled.

COMMON MISCONCEPTIONS

"A scale factor less than 1 is a reflection." A factor with $0 < k < 1$ shrinks the image towards the centre. A *negative* factor puts the image on the opposite side of the centre, which is what flips it.

"The centre of enlargement does not matter." It is the point whose distances to every vertex are multiplied. The same factor with a different centre produces a different image in a different place.

"To enlarge by 2, double the coordinates." That works only when the centre is the origin. Otherwise double the *difference* from the centre: a vertex 3 right and 1 above the centre moves to 6 right and 2 above it.

"A scale factor of 2 doubles the area." It doubles every length, so area is multiplied by $2^2 = 4$ and volume by $2^3 = 8$ — the subject of Section 9.5.

RULES ON A GRID

9.4 Transformations on the coordinate plane

On a coordinate grid, each transformation can be written as a rule that changes x and y. The rules make combinations easy to compute, and they reveal which transformations are equivalent to one another. A translation adds to the coordinates; a reflection or rotation changes their signs or swaps them; an enlargement centred at the origin multiplies them.

The order of two transformations matters. Reflecting and then rotating generally gives a different image from rotating and then reflecting. Where the lines or centres coincide, however, two transformations can collapse into one: two reflections in parallel lines give a translation, and two reflections in intersecting lines give a rotation.

COORDINATE RULES · ANGLES MEASURED ANTICLOCKWISE

TRANSFORMATION	RULE	IMAGE OF $(3, -2)$
Translation by $\begin{pmatrix} a \\ b \end{pmatrix}$	$(x, y) \mapsto (x + a,\ y + b)$	$\begin{pmatrix} 2 \\ 5 \end{pmatrix}$: $(5, 3)$
Reflection in the x-axis	$(x, y) \mapsto (x, -y)$	$(3, 2)$
Reflection in the y-axis	$(x, y) \mapsto (-x, y)$	$(-3, -2)$
Reflection in $y = x$	$(x, y) \mapsto (y, x)$	$(-2, 3)$
Reflection in $y = -x$	$(x, y) \mapsto (-y, -x)$	$(2, -3)$
Rotation 90° anticlockwise about O	$(x, y) \mapsto (-y, x)$	$(2, 3)$
Rotation 90° clockwise about O	$(x, y) \mapsto (y, -x)$	$(-2, -3)$
Rotation 180° about O	$(x, y) \mapsto (-x, -y)$	$(-3, 2)$
Enlargement centre O, factor k	$(x, y) \mapsto (kx, ky)$	$k = 2$: $(6, -4)$

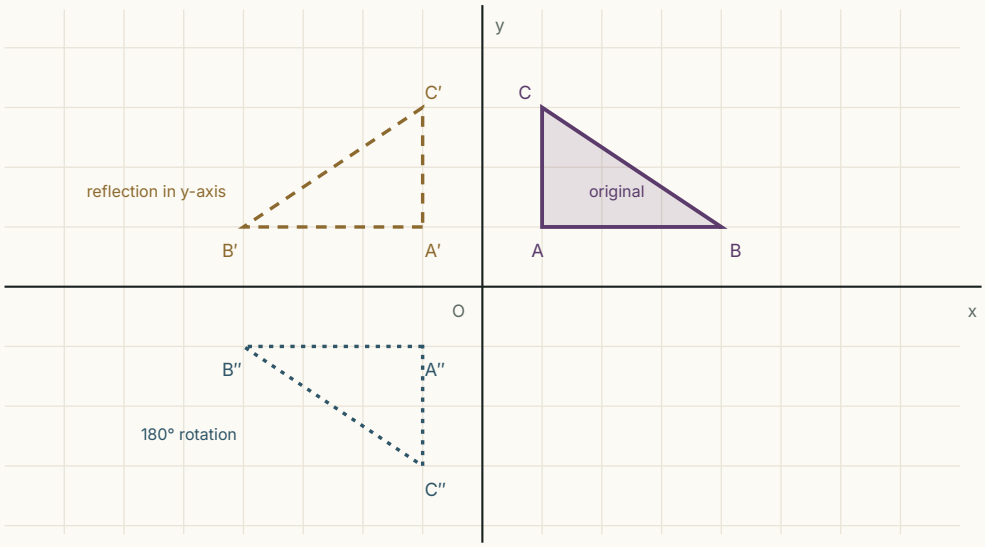


Figure 9.4 The same triangle under two transformations. The reflection keeps every length but reverses orientation; the half-turn keeps orientation but moves every point to the opposite side of O. Both images are congruent to the original.

EXAMPLE **9.10** APPLYING THREE RULES

QUESTION

Triangle PQR has vertices $P(-2, 3)$, $Q(1, 3)$ and $R(-2, 7)$. Find the image under (a) a translation by $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$, (b) a reflection in the y-axis, (c) a rotation of 90° anticlockwise about the origin.

SOLUTION

1. (a) Add the vector to each point: $P'(1, -2)$, $Q'(4, -2)$, $R'(1, 2)$.
2. (b) Change the sign of each x-coordinate: $P'(2, 3)$, $Q'(-1, 3)$, $R'(2, 7)$.
3. (c) Apply $(x, y) \mapsto (-y, x)$: $P'(-3, -2)$, $Q'(-3, 1)$, $R'(-7, -2)$.
4. Both (a) and (c) are isometries and preserve orientation; the reflection in (b) reverses it.

Check: in (c), $PQ = 3$ and $P'Q' = 1 - (-2) = 3$. In (b), the image of a point on the y-axis would not move, and no vertex lies there, so all three images have changed side.

EXAMPLE **9.11** IDENTIFYING A TRANSFORMATION

QUESTION

The triangle with vertices $A(1, 2)$, $B(3, 2)$, $C(1, 5)$ maps to $A'(2, -1)$, $B'(2, -3)$, $C'(5, -1)$. Describe the transformation.

SOLUTION

1. The side lengths are unchanged: $AB = 2$ and $A'B' = 2$, $AC = 3$ and $A'C' = 3$. So the transformation is an isometry.
2. Test a quarter-turn clockwise, $(x, y) \mapsto (y, -x)$: $A(1, 2) \mapsto (2, -1) = A'$, and $B(3, 2) \mapsto (2, -3) = B'$, $C(1, 5) \mapsto (5, -1) = C'$.
3. All three vertices agree, so the transformation is a rotation of 90° clockwise about the origin.
4. Orientation is preserved, which rules out a reflection.

Check: $OA = \sqrt{1^2 + 2^2} = \sqrt{5}$ and $OA' = \sqrt{2^2 + 1^2} = \sqrt{5}$. Distances from the origin are unchanged, consistent with a rotation about O.

EXAMPLE **9.12** COMBINING TRANSFORMATIONS

QUESTION

Describe the single transformation equivalent to a rotation of 90° anticlockwise about O followed by a reflection in the x-axis.

SOLUTION

1. Apply the rotation: $(x, y) \mapsto (-y, x)$.
2. Reflect the result in the x-axis: $(-y, x) \mapsto (-y, -x)$.
3. The rule $(x, y) \mapsto (-y, -x)$ is a reflection in the line $y = -x$.
4. Test with $(2, 1)$: rotation gives $(-1, 2)$, then reflection gives $(-1, -2)$; a direct reflection in $y = -x$ gives $(-1, -2)$, as required.

Check: two transformations, one reversing orientation and one preserving it, produce a single reversing transformation. A reflection is consistent with that count; a rotation would not be.

EXERCISE 9D

COORDINATE RULES · COMPOSITION

FLUENCY

1. Translate $(3, -1)$ by $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$.
2. Reflect $(4, -3)$ in the line $y = x$.
3. Rotate $(2, 5)$ through 180° about the origin.
4. Reflect $(6, 2)$ in the line $y = -x$.
5. Enlarge $(3, -2)$ from the origin with scale factor 1.5.

UNDERSTANDING AND REASONING

6. Which single transformation maps $(2, 3)$ to $(-3, -2)$? Justify your answer by testing the rules.
7. Describe the single transformation equivalent to a reflection in the x-axis followed by a reflection in the y-axis.
8. A rotation of 90° clockwise is followed by a reflection in the x-axis. Compare this with the reverse order using the point $(1, 2)$.
9. A point is unchanged by a reflection. Where must it lie? Give one point unchanged by a reflection in $y = x$.
10. **Extension.** Find the image of $(3, 1)$ under a rotation of 90° clockwise about the point $(1, 1)$. Describe a method for rotating about a centre that is not the origin.

COMMON MISCONCEPTIONS

"$(x, y) \mapsto (-x, y)$ is a reflection in the line $y = x$." It is a reflection in the y-axis. Reflection in $y = x$ swaps the coordinates: $(x, y) \mapsto (y, x)$.

"A $90°$ clockwise turn maps (x, y) to $(-y, x)$." That is the anticlockwise rule. Clockwise gives $(y, -x)$: test it on $(1, 0)$, which must land at $(0, -1)$.

"The order of transformations does not matter." Take $(2, 1)$, translate by $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, then rotate $90°$ anticlockwise about the origin: the result is $(-1, 5)$. Rotating first gives $(2, 2)$. Two different answers, so the order is part of the description.

"A rule that changes both coordinates is a rotation." Read the rule: adding a fixed pair to every point is a translation, changing signs is a reflection, and only a rotation has a centre and an angle.

LENGTH, AREA, VOLUME

9.5 Similarity in area and volume

When every length is multiplied by k, area is multiplied by k^2 and volume by k^3. A rectangle 4 cm by 6 cm has area 24 cm²; doubling the sides gives 8 cm by 12 cm and area 96 cm², four times as much. The extra factor appears because area is measured in squares: each square of the original grid becomes k^2 squares in the image.

The same reasoning applies to any shape, not only rectangles. For similar solids, every face area is multiplied by k^2 and the volume by k^3. This is why a model at $1 : 50$ needs $1/2500$ of the paint but only $1/125\,000$ of the material, and why a small increase in a container's dimensions adds much more capacity than its surface.

DEFINITIONS

Area and volume scale factors

If two figures are similar with length scale factor k, then the **area scale factor** is k^2 and the **volume scale factor** is k^3. Conversely, if two similar figures have area ratio $A_2 : A_1$, then $k = \sqrt{A_2/A_1}$; if they have volume ratio $V_2 : V_1$, then $k = \sqrt[3]{V_2/V_1}$.

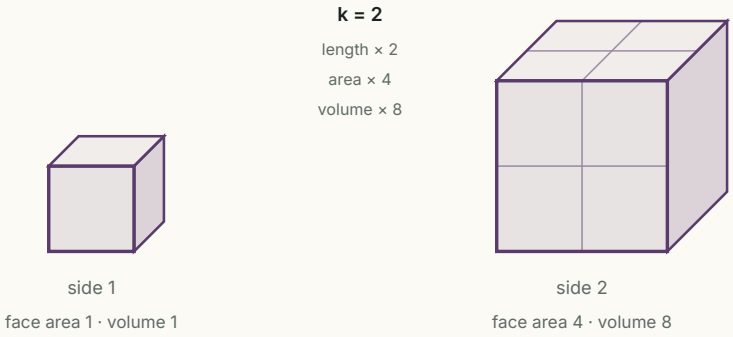


Figure 9.5 Doubling every length squares the area factor and cubes the volume factor. A face of the large cube holds four copies of a face of the small cube, and the large cube holds eight copies of the small one.

EXAMPLE **9.13** AREAS OF SIMILAR TRIANGLES

QUESTION

Two similar triangles have corresponding sides 3 cm and 5 cm. The area of the smaller triangle is 18 cm². Find the area of the larger triangle.

SOLUTION

1 The length scale factor is $k = \frac{5}{3}$.

2 The area scale factor is $k^2 = \frac{25}{9}$.

3 Area $= 18 \times \frac{25}{9} = 50$ cm².

Check: $50 \div 18 = 2.777\ldots = \frac{25}{9}$, and $\sqrt{\frac{25}{9}} = \frac{5}{3}$, returning the side ratio.

EXAMPLE **9.14** SIMILAR CANS

QUESTION

Two cylindrical cans are similar, with heights 8 cm and 20 cm. The smaller can has surface area 90 cm² and volume 120 cm³. Find the surface area and volume of the larger can.

SOLUTION

1 Length scale factor: $k = \frac{20}{8} = 2.5$.

2 Area scale factor: $k^2 = 6.25$. Surface area $= 90 \times 6.25 = 562.5$ cm².

3 Volume scale factor: $k^3 = 15.625$. Volume $= 120 \times 15.625 = 1875$ cm³.

Check: $1875 \div 120 = 15.625 = 2.5^3$, and $562.5 \div 90 = 6.25 = 2.5^2$. The volume has grown much faster than the surface area.

EXAMPLE **9.15** A SCALE MODEL

QUESTION

A model car is built at a scale of $1:24$. The real car needs 4.8 L of paint for its body, and its body panels have total area 6.0 m². The model is 18 cm long. Find the real length of the car, the area of the model's body panels, and the quantity of paint needed for the model.

SOLUTION

1. Real length $= 18 \times 24 = 432$ cm $= 4.32$ m.
2. Area scale factor $= \left(\frac{1}{24}\right)^2 = \frac{1}{576}$, so model area $= \frac{6.0}{576} = 0.010416\ldots$ m², about 104 cm².
3. Paint scales with area: $4.8 \times \frac{1}{576} = 0.00833\ldots$ L, about 8.3 mL.

Check: 4.8 L divided by 576 is about 8 mL, a few teaspoons. Painting a model with a brush would waste far more than the theoretical amount, which is itself a limitation of the model.

EXAMPLE **9.16** WORKING BACKWARDS FROM VOLUME

QUESTION

Two similar solid containers have volumes 64 cm³ and 216 cm³. Find the ratio of their heights and the ratio of their surface areas.

SOLUTION

1. The volume ratio is $216:64 = 27:8$, so $k^3 = \frac{27}{8}$.
2. Taking cube roots, $k = \frac{3}{2}$. The heights are in the ratio $3:2$.
3. Surface areas are in the ratio $k^2 = \frac{9}{4}$, that is $9:4$.

Check: $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$ and $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$, matching both ratios.

EXAMPLE **9.17** WHICH PIZZA IS BETTER VALUE?

QUESTION

A 30 cm pizza costs £9 and a 40 cm pizza costs £14. Assuming both have the same thickness, which is better value?

SOLUTION

1. The pizzas are similar circles. The length scale factor is $k = \frac{40}{30} = \frac{4}{3}$.
2. The area factor is $k^2 = \frac{16}{9} \approx 1.78$, so the large pizza has about 1.78 times the food.
3. At the same value per unit area, it should cost $9 \times \frac{16}{9} =$ £16.
4. It costs £14, so the 40 cm pizza is better value.

Check: cost per cm² is $\frac{9}{706.9} \approx$ £0.0127 for the small pizza and $\frac{14}{1256.6} \approx$ £0.0111 for the large one. The large pizza is cheaper per unit area.

EXERCISE 9E

AREA AND VOLUME SCALE FACTORS

FLUENCY

1 Two similar squares have sides 4 cm and 7 cm. Find the ratio of their areas.

2 Two similar cubes have sides 2 cm and 6 cm. Find the ratio of their volumes.

3 A model is built at 1 : 20. Find the area scale factor and the volume scale factor.

4 Two similar triangles have length scale factor 4. Find the area scale factor.

5 Two similar solids have volumes 8 cm^3 and 125 cm^3. Find the length scale factor.

UNDERSTANDING AND REASONING

6 All the lengths of a cuboid are doubled. Explain why the volume becomes eight times as large, using the formula for the volume of a cuboid.

7 A model bridge is built at 1 : 50. Its deck has area 0.072 m^2. The real deck has area 180 m^2. Verify that the area scale factor is correct.

8 **Error analysis.** A student claims that if the sides of a cube are tripled, the volume is tripled. Explain the error and give the correct factor.

9 Two similar solids have surface areas 25 cm^2 and 64 cm^2. Find the ratio of their volumes.

10 **Extension.** A cone is enlarged so that its volume becomes 27 times larger. What happens to its height and its surface area? Justify both answers.

COMMON MISCONCEPTIONS

"Length, area and volume all scale by k." They scale by k, k^2 and k^3. Doubling the edges of a box multiplies its surface by 4 and its capacity by 8.

"The area factor is the square root of the length factor." It is the other way round: a length factor of k gives an area factor of k^2, and you take the square root when you are given areas and need the length factor.

"Two solids with the same volume are similar." Equal volume says nothing about the ratios of their dimensions. Similarity is a statement about every corresponding length.

"Scaling is just a way of making pictures bigger." It changes what a quantity measures: paint, wrapping paper and heat loss follow the area factor, while fill, mass and capacity follow the volume factor.

INVESTIGATION · CRITERION B

9.6 Shadows and scale

On a sunny day, every vertical object and its shadow form a right-angled triangle. The sun's rays are parallel, so all these triangles share the same angle with the ground. They are therefore similar, and the ratio of height to shadow length is the same for every object — a ruler and a shadow can measure a building.

A class measured four vertical sticks at the same moment. The results are shown in the table. The measurements are rounded to the nearest 0.1 m.

SHADOW MEASUREMENTS TAKEN AT THE SAME TIME

OBJECT	HEIGHT h (m)	SHADOW L (m)	$\frac{h}{L}$
Stick 1	1.0	1.2	0.833
Stick 2	1.5	1.8	0.833
Stick 3	2.0	2.4	0.833
Stick 4	2.5	3.0	0.833
Tree	?	9.6	—

INVESTIGATION · CRITERION B

How far can a shadow measure?

1. Copy and complete the table, adding a row for a flagpole with shadow 11.4 m.
2. Describe the pattern in the ratio $\frac{h}{L}$. Explain why the ratio is constant, referring to the angles in the triangles.
3. Write a rule for the height H of any vertical object in terms of its shadow L and the constant ratio.
4. Use your rule to find the height of the tree and of the flagpole. Show the similar triangles explicitly in a diagram.
5. The class repeats the measurement one hour later and obtains a different ratio. Explain why. State two other assumptions of the method and how each could be tested.
6. Estimate the uncertainty in the tree's height if each shadow is measured to the nearest 0.1 m. Does the method become more or less reliable for a very tall object? Justify your answer.

Generalise and justify. Show that for two vertical objects measured at the same time, $\frac{H}{h} = \frac{L}{l}$, where h and l are the height and shadow of a reference stick. Explain why the result depends on the rays being parallel, and describe how the graph of H against L would look.

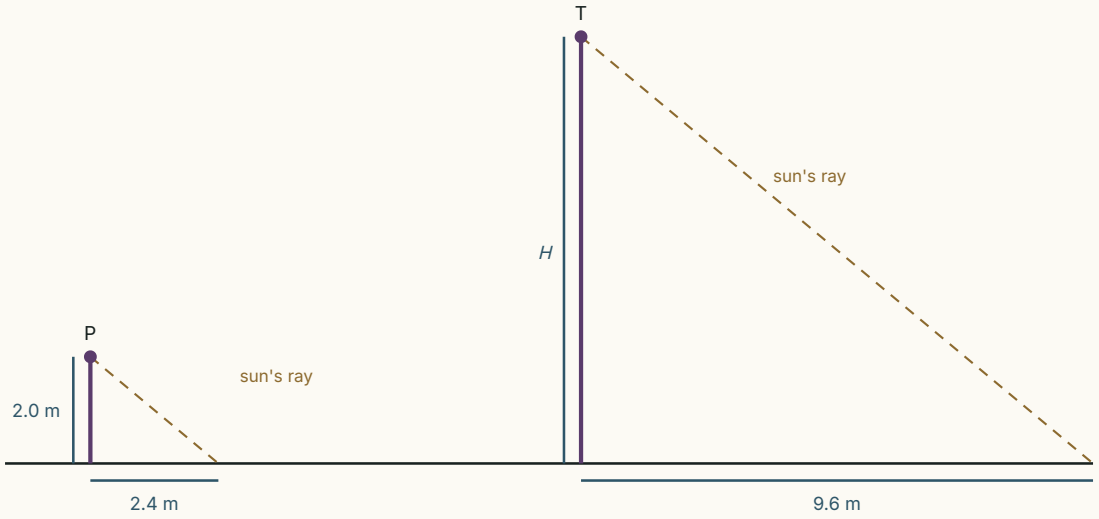


Figure 9.6 The stick and the tree form similar right-angled triangles because the sun's rays are parallel. The ratio $\text{height}/\text{shadow}$ is the same for both, so the tree's height H can be calculated from its shadow. The diagram is drawn to scale.

MIXED PRACTICE 9

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–6 check fluency; 7–16 require reasoning; 17 and 18 are extension tasks asking for proof. Allow 55 minutes.

FLUENCY

1 Translate the point $(1, -2)$ by $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$.

2 Reflect $(-3, 4)$ in the y-axis, then in the line $y = x$.

3 Rotate $(5, 2)$ through $90°$ anticlockwise about the origin.

4 Triangles have sides $5, 6, 7$ and $5, 6, 8$. Are they congruent? Explain.

5 Triangles have angles $50°, 60°, 70°$ and $50°, 70°, 60°$. Are they similar? Explain.

6 Similar triangles have corresponding sides 8 cm and 12 cm. A second side of the smaller triangle is 6 cm. Find the matching side of the larger triangle.

UNDERSTANDING AND REASONING

7 An enlargement has scale factor 3. Find the area and volume scale factors.

8 Two similar solids have volumes 125 cm^3 and 343 cm^3. Find the ratio of their surface areas.

9 A 1.6 m person casts a shadow 2.0 m long. A building casts a shadow 25 m long. Find the height of the building.

10 A map has scale $1 : 25\,000$. A rectangular park measures 4 cm by 6 cm on the map. Find its actual area in km^2.

11 **Error analysis.** A student says that doubling the sides of a rectangle doubles its area. Explain the error and give the correct factor.

12 Describe the single transformation equivalent to a reflection in the x-axis followed by a rotation of $90°$ clockwise about the origin. Support your answer with the point $(2, 3)$.

13 An enlargement with scale factor -1.5 maps $(2, -4)$ to P'. Find P' and describe where it lies relative to the centre.

14 In similar triangles, sides 4 cm and 6 cm correspond to 6 cm and x cm, with a third pair 8 cm and 12 cm. Find x.

MIXED PRACTICE 9 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

15 A price of £80 is increased by 25% and then reduced by 20%. Find the final price and the overall percentage change.

16 Solve $3(x - 2) = 2x + 5$.

17 **Extension.** Two straight lines intersect at O. A figure is reflected in one line and then in the other. Prove that the result is a rotation about O through twice the angle between the lines.

18 **Extension.** In trapezium $ABCD$, $AB \parallel DC$ and the diagonals meet at E. Show that $\triangle ABE \sim \triangle CDE$. If $AB = 12$ cm and $DC = 8$ cm, find the ratio of the areas of the two triangles.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and assessed against the criteria shown. Your teacher will confirm which tasks are required and how long you have.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Transformations and scale: short response	30	35 min
B · Investigating patterns	Growing triangles	24	40 min
C · Communicating	What survives a transformation?	20	35 min
D · Applying mathematics in real-life contexts	The footbridge model	24	45 min
Total	All four tasks	98	155 min

The tasks begin overleaf. Read the instructions and evidence requirements before you start, and show the reasoning that supports each answer.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 30 marks

Transformations and scale: short response

CALCULATOR SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

1 Find the translation vector that maps $A(2, 3)$ to $A'(5, -1)$. [2]

2 Reflect the point $(3, -2)$ in the x-axis and in the y-axis. [3]

3 Rotate $(4, 1)$ through 90° clockwise about the origin. [2]

4 Decide whether each pair is congruent, similar but not congruent, or neither. Give a reason for each. (a) Rectangles 3×4 and 6×8. (b) Triangles with sides $5, 12, 13$ and $5, 12, 13$. (c) Squares of side 4 cm and 4 cm. [6]

5 Triangles ABC and DEF are similar, with $AB = 9$ cm, $BC = 12$ cm and $DE = 6$ cm. Find EF. [4]

6 A figure is enlarged with scale factor 2.5. A side of length 8 cm becomes 20 cm, and the original area is 48 cm^2. Find the new area. [4]

7 Two similar solids have volumes 27 cm^3 and 64 cm^3. Find the ratio of their heights and the ratio of their surface areas. [5]

8 **Error analysis.** A student claims that doubling every side of a rectangle doubles its area. Use a 2×3 rectangle to show the error, and state the correct area factor. [4]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

Growing triangles

Instructions. Work independently. Stage 0 is a right-angled triangle with sides 3 cm, 4 cm and 5 cm. Each stage is an enlargement of the previous stage with scale factor 2.

1 Record the side lengths, perimeter and area for stages 0, 1, 2 and 3 in a table. [6]

2 Write a rule for the perimeter P_n of stage n, and explain why the rule has this form. [4]

3 Write a rule for the area A_n of stage n, and explain how the area scale factor produces it. [5]

4 Explain why the area grows by a factor of 4 at each stage while the sides grow by a factor of 2. [5]

5 Find the first stage whose area exceeds 1000 cm^2. Justify your answer with calculations. [4]

Evidence: a complete table, correct general rules, an explanation linking length and area factors, and a justified stage number.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

What survives a transformation?

Instructions. Write a 300–400 word explanation for a student who has missed the lessons on shape. Your explanation must: (i) define congruent and similar; (ii) give one example of each using coordinates; (iii) explain why area is multiplied by k^2 when lengths are multiplied by k, using a rectangle; (iv) state one common error and how to avoid it; (v) explain one real situation where similarity is more useful than congruence.

Evidence: clear definitions, a worked coordinate example, a labelled diagram or rectangle calculation, and a conclusion suited to the audience.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The footbridge model

Instructions. A footbridge is 60 m long, 3 m wide and has a deck 0.3 m thick. A scale model is built at $1:50$. The real bridge needs 54 m^3 of concrete, and the model material costs £0.02 per cm^3.

1. Find the length, width and thickness of the model in metres, and in centimetres. [6]
2. Find the area of the real deck and the area of the model deck. Verify that the area ratio is $1:2500$. [5]
3. Find the volume of concrete in the model, in cm^3, using the volume scale factor. [5]
4. Find the cost of the model material. [4]
5. Evaluate the model as a way of testing the bridge's strength. State two assumptions or limitations and explain how each affects the usefulness of the model. [4]

Evidence: dimensions with units, area and volume calculations linked to the scale factors, a cost with units, and an evaluation that distinguishes geometric accuracy from structural behaviour.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- Translations, rotations and reflections are isometries: they preserve all lengths and angles, so images are congruent.
- Similar figures have equal corresponding angles and corresponding lengths in a constant ratio k.
- AA, SSS and SAS are the standard tests for similar triangles; AA is usually the fastest.
- An enlargement is fixed by its centre and scale factor; a negative factor places the image on the opposite side of the centre.
- Coordinate rules give each transformation as a mapping of (x, y).
- If lengths scale by k, areas scale by k^2 and volumes by k^3; use square and cube roots to work backwards.

$$\frac{A'B'}{AB} = \frac{B'C'}{BC} = \frac{C'A'}{CA} = k$$

$$\frac{\text{area}'}{\text{area}} = k^2 \qquad \frac{\text{volume}'}{\text{volume}} = k^3$$

$$k = \frac{\text{image length}}{\text{original length}}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can describe a translation, rotation and reflection precisely.
- ☐ I can decide whether two figures are congruent and justify it.
- ☐ I can use AA, SSS or SAS to show that triangles are similar.
- ☐ I can find a missing length using a scale factor.
- ☐ I can enlarge a figure from a given centre.
- ☐ I can work with fractional and negative scale factors.
- ☐ I can apply coordinate rules and combine two transformations.
- ☐ I can calculate an area or volume after scaling.
- ☐ I can work backwards from an area or volume ratio.
- ☐ I can use similar triangles to measure an inaccessible height.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which representation — a diagram, a coordinate rule or a ratio — made similarity clearest to you? Give an example.
2. Describe a time when you used a ratio and had to check that two lengths really corresponded. What went wrong or right?
3. The statement of inquiry claims that one factor controls shape, area and volume. Give one example where this helped you predict a result before calculating it.
4. Look ahead to Chapter 10: why will similar triangles and Pythagoras' theorem together allow you to solve right-angled triangles that cannot be drawn accurately?

HOW FAR, HOW HIGH, HOW STEEP

10 Pythagoras, Trigonometry & Measurement

SPACE · MEASUREMENT · REASONING

A right-angled triangle connects three lengths and two acute angles. One relationship links the lengths; three ratios link lengths to angles. Together they turn a tape measure and a protractor into instruments for measuring what cannot be reached.

STATEMENT OF INQUIRY

In every right-angled triangle the sides obey one fixed square relationship, and the ratios of pairs of sides depend only on the acute angle. One measured length and one measured angle therefore determine every other length in the triangle, so inaccessible distances can be calculated from accessible ones.

KEY CONCEPT

Relationships

RELATED CONCEPTS

Space · Measurement · Equivalence

GLOBAL CONTEXT

Scientific and technical innovation — surveying, navigation and safe design

FACTUAL

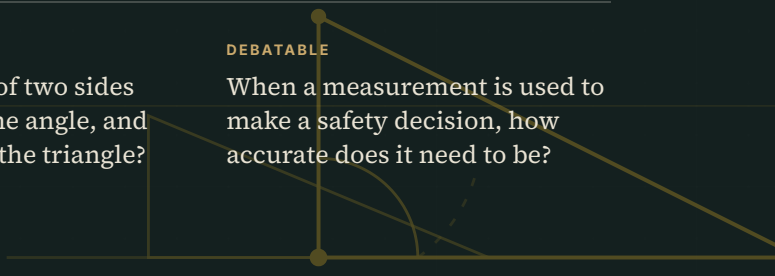
What relationship connects the three sides of a right-angled triangle? What does each trigonometric ratio compare?

CONCEPTUAL

Why does a ratio of two sides depend only on the angle, and not on the size of the triangle?

DEBATABLE

When a measurement is used to make a safety decision, how accurate does it need to be?



IN THIS CHAPTER

10.1 Pythagoras
10.2 Three dimensions
10.3 Trig ratios
10.4 Solving triangles
10.5 Elevation and bearings
10.6 Investigation

OPENING PROBLEM · THREE MEASUREMENTS, ONE TRIANGLE

A ramp, a roof and a tower

A community centre is renovating its entrance. The architect proposes a ramp that rises 0.75 m over a horizontal run of 9.0 m. The new roof has a half-span of 3.6 m and a rise of 2.4 m at the ridge. Across the road, a surveyor needs the height of a radio tower: standing 85 m from its base on level ground, she measures an angle of elevation of 28° to the top, with her instrument 1.5 m above the ground.

1. Find the length of the ramp surface along the slope, to the nearest centimetre.
2. Write the ramp's steepness as a ratio of rise to run in the form $1 : n$, and as a percentage.
3. Find the length of a roof rafter running from the eaves to the ridge. Find the pitch of the roof, the angle it makes with the horizontal.
4. Find the height of the tower above the instrument, then the height above the ground.
5. Which of questions 1–4 can be answered using only the two given lengths? Which needed an angle as well? Explain what changes when an angle is available.
6. Building rules recommend a ramp gradient no steeper than $1 : 12$. Is the proposed ramp acceptable? What would happen to the gradient if the rise stayed at 0.75 m but the run were shortened to 6.0 m?

Two lengths and a right angle are enough to fix a triangle completely: the third length follows from the square relationship we call Pythagoras' theorem. But a length and an angle also fix the triangle, through ratios that depend only on the angle. The first idea measures across a diagonal; the second measures up a height. Surveyors, navigators, roofers and engineers use both, and the same two ideas explain why a ladder at too steep an angle is dangerous and why a map bearing is written with three figures.

ORIENTATION

10.0 Learning intentions

- State, apply and prove Pythagoras' theorem, including its converse.
- Give exact answers using surds and give sensible rounded answers with units.
- Use Pythagoras' theorem in three dimensions and to find distances between points.
- Define sine, cosine and tangent for an acute angle and explain why they depend only on the angle.
- Solve right-angled triangles, choosing the ratio that connects the known and unknown quantities.
- Interpret angles of elevation and depression and use three-figure bearings.
- Model a measurement problem and evaluate the sources and size of error in the result.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

apply $a^2 + b^2 = c^2$ and its converse with confidence; explain a proof of the theorem; find a space diagonal of a cuboid; write the three ratios for a labelled triangle; choose sine, cosine or tangent from the information given; find a missing side or angle and check the result; read and write three-figure bearings; calculate a height from a distance and an angle; and state how measurement error affects a conclusion.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit squares and roots in Chapter 1, solving equations in Chapter 3, and similar triangles in Chapter 9 before starting.

1. Evaluate $9^2 + 12^2$.
2. Find $\sqrt{169}$.
3. Solve $0.6 = \frac{x}{9}$.
4. Two angles of a triangle are 90° and 37°. Find the third.
5. Round 0.04726 to 3 significant figures.
6. Which side is the longest in a right-angled triangle, and where is it?

REFRESHER

Squaring and taking a square root are inverse operations: $\sqrt{49} = 7$ because $7^2 = 49$. A square root of a number that is not a perfect square is a **surd**; for example $\sqrt{13}$ is exact, while 3.61 is an approximation. When a calculation has several steps, keep the exact value or an extra decimal place until the final step, then round. Do not distribute a square root over a sum: $\sqrt{a^2 + b^2}$ is not $a + b$. Most calculator work in this chapter uses degrees, so check that the display shows **DEG**, not RAD.

ATL SKILLS · THINKING, RESEARCH AND COMMUNICATION

Critical thinking: decide whether a problem is determined by lengths alone or needs an angle, and test whether an answer is physically reasonable. **Research:** plan a field measurement, record distances and angles consistently, and identify the largest source of error. **Communication:** draw a labelled diagram before calculating, state which angle you are using, and give bearings and angles with their units.

THE SQUARE RELATIONSHIP

10.1 Pythagoras' theorem and proof

A right-angled triangle has one angle of 90°. The side opposite that angle is the longest side and is called the **hypotenuse**; the other two sides are the **legs**. The theorem connects their lengths in a way that holds for every right-angled triangle, whatever its size or orientation.

Draw a $3, 4, 5$ triangle and squares on each side. The square on the hypotenuse has area 25, and the two smaller squares have areas 9 and 16. Their areas add to 25. The same pattern appears for $6, 8, 10$, for $5, 12, 13$ and for triangles whose sides are not whole numbers. The relationship was used by Babylonian, Indian and Chinese surveyors long before the Greek mathematician Pythagoras, but it is his name that the theorem now carries in European mathematics.

DEFINITIONS

Pythagoras' theorem and its converse

In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides. With legs a and b and hypotenuse c,

$$a^2 + b^2 = c^2$$

Converse: if the sides of a triangle satisfy $a^2 + b^2 = c^2$, where c is the longest side, then the triangle is right-angled, with the right angle opposite c. A set of three whole-number sides that satisfies the theorem is a **Pythagorean triple**, such as $(3, 4, 5)$ or $(5, 12, 13)$.

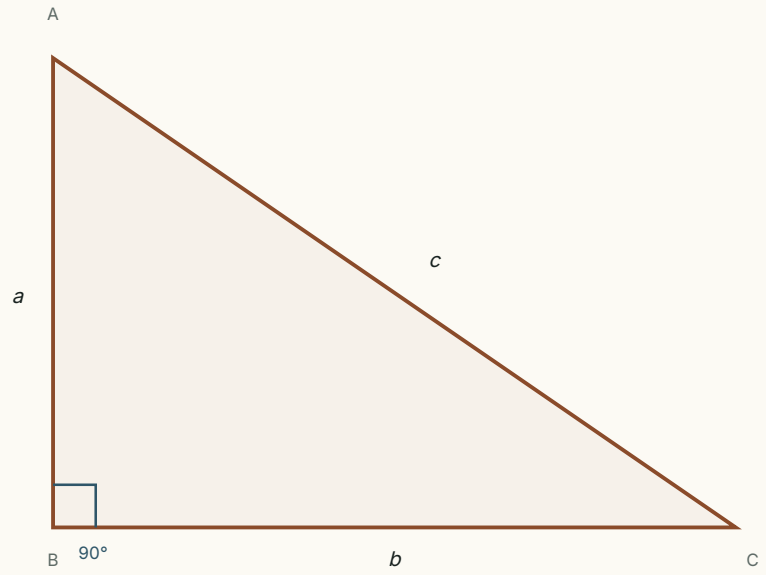


Figure 10.1 The right angle is at B. The hypotenuse c is opposite the right angle and is always the longest side. The legs a and b are adjacent to the right angle.

The theorem is a statement about areas, not only about lengths. The square built on the hypotenuse can be cut up and rearranged into the two smaller squares. The proof below uses four copies of the same triangle placed inside a large square.

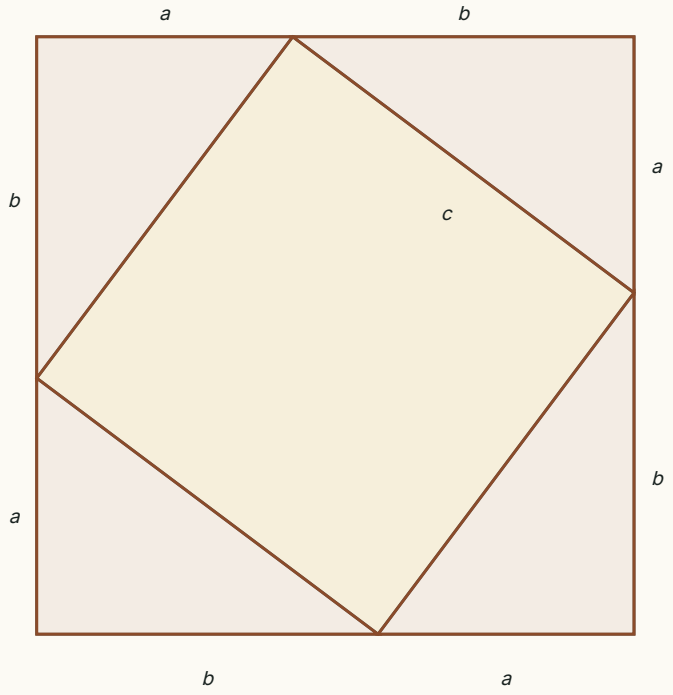


Figure 10.2 Four congruent copies of a right-angled triangle with legs a, b and hypotenuse c fit inside a square of side $a + b$. The uncovered inner square has side c. Comparing two ways of calculating the area proves the theorem.

EXAMPLE **10.1** PROOF BY DISSECTION

QUESTION

Use Figure 10.2 to prove that $a^2 + b^2 = c^2$.

SOLUTION

1. The large square has side $a + b$, so its area is $(a + b)^2 = a^2 + 2ab + b^2$.
2. The same square contains four triangles, each of area $\frac{1}{2}ab$, plus the inner square of area c^2. Its area is also $4 \times \frac{1}{2}ab + c^2 = 2ab + c^2$.
3. Equating the two expressions: $a^2 + 2ab + b^2 = 2ab + c^2$.
4. Subtract $2ab$ from both sides: $a^2 + b^2 = c^2$.

Check: the proof does not depend on the particular values of a and b; it uses only that the four triangles are congruent and that the inner figure is a square. Try $a = 3$, $b = 4$: the large square has area 49, the four triangles total 24, and the inner square has area 25.

There is a second proof, which uses the similarity from Chapter 9. Drop a perpendicular from the right angle to the hypotenuse, splitting it into parts p and q. Each smaller triangle is similar to the original, so $\frac{a}{c} = \frac{p}{a}$ and $\frac{b}{c} = \frac{q}{b}$. Hence $a^2 = pc$ and $b^2 = qc$. Adding gives $a^2 + b^2 = c(p + q) = c^2$, because $p + q = c$.

EXAMPLE **10.2** FINDING THE HYPOTENUSE

QUESTION

The legs of a right-angled triangle are 9 cm and 12 cm. Find the hypotenuse.

SOLUTION

1. The hypotenuse is opposite the right angle, so $c^2 = 9^2 + 12^2$.
2. $c^2 = 81 + 144 = 225$.
3. $c = \sqrt{225} = 15$.

Check: 15 cm is longer than each leg, as the hypotenuse must be, and $9^2 + 12^2 = 15^2$.

EXAMPLE **10.3** FINDING A LEG

QUESTION

A right-angled triangle has hypotenuse 13 m and one leg 5 m. Find the other leg.

SOLUTION

1. The unknown is a leg, so subtract: $b^2 = 13^2 - 5^2$.
2. $b^2 = 169 - 25 = 144$.
3. $b = 12$ m.

Check: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$. The result is shorter than the hypotenuse, as required.

EXAMPLE **10.4** EXACT SURD ANSWERS

QUESTION

Find the hypotenuse of a right-angled triangle with legs 5 cm and 7 cm. Give an exact answer and an approximation to 2 decimal places.

SOLUTION

1. $c^2 = 5^2 + 7^2 = 25 + 49 = 74$.
2. $c = \sqrt{74}$, which cannot be simplified because 74 has no square factor.
3. $\sqrt{74} \approx 8.6023$, so $c \approx 8.60$ cm.

Check: $8.60^2 = 73.96$, close to 74; the exact value is slightly larger. Do not write $c = 5 + 7 = 12$.

EXAMPLE **10.5** USING THE CONVERSE

QUESTION

Decide whether a triangle with sides 8 cm, 15 cm and 17 cm is right-angled. Then test sides 6 cm, 7 cm and 9 cm.

SOLUTION

1. The longest side is 17, so test whether $8^2 + 15^2 = 17^2$.
2. $64 + 225 = 289$ and $17^2 = 289$. The equality holds, so the triangle is right-angled.
3. For $6, 7, 9$: $6^2 + 7^2 = 36 + 49 = 85$, but $9^2 = 81$. Since $85 \neq 81$, the triangle is not right-angled.

Check: the converse must use the longest side as c. Testing $7^2 + 9^2$ against 6^2 would compare the wrong side and prove nothing.

EXAMPLE **10.6** AN ISOSCELES TRIANGLE

QUESTION

An isosceles triangle has equal sides 13 cm and base 10 cm. Find its height and area.

SOLUTION

1. The perpendicular from the apex to the base bisects the base, creating two right-angled triangles with hypotenuse 13 cm and one leg 5 cm.
2. $h^2 = 13^2 - 5^2 = 169 - 25 = 144$, so $h = 12$ cm.
3. Area $= \frac{1}{2} \times 10 \times 12 = 60$ cm².

Check: each half-triangle is a $5, 12, 13$ triangle, and $5^2 + 12^2 = 13^2$. The height is less than the equal side, as it must be.

COMMON MISCONCEPTION

"Add the two sides to get the hypotenuse." The theorem adds the *squares* of the legs, then takes the square root: $a^2 + b^2 = c^2$, not $a + b = c$. For legs 9 and 12, the hypotenuse is 15, not 21.

"The hypotenuse is the side at the bottom." The hypotenuse is opposite the right angle, wherever that angle sits. Rotate the page and the hypotenuse rotates with it.

"If $a^2 + b^2 = c^2$, any side can be c." In the theorem, c must be the hypotenuse. When testing a triangle with the converse, always substitute the longest side for c.

EXERCISE 10A

PYTHAGORAS' THEOREM

FLUENCY

1 The legs of a right-angled triangle are 6 cm and 8 cm. Find the hypotenuse.

2 A right-angled triangle has legs 5 m and 12 m. Find the hypotenuse.

3 A right-angled triangle has hypotenuse 17 cm and one leg 8 cm. Find the other leg.

4 A right-angled triangle has hypotenuse 26 m and one leg 10 m. Find the other leg.

5 Find the hypotenuse of a right-angled triangle with legs 7 cm and 24 cm.

6 Find the exact hypotenuse of a right-angled triangle with legs 1 m and 1 m, and give the answer to 2 decimal places.

7 Find the exact hypotenuse of a right-angled triangle with legs 2 cm and 3 cm, and give the answer to 2 decimal places.

8 Decide whether each triangle is right-angled: (a) $9, 12, 15$; (b) $7, 8, 11$. Show your working.

UNDERSTANDING AND REASONING

9 A rectangular field measures 18 m by 24 m. Find the length of its diagonal.

10 A 5 m ladder leans against a wall with its foot 3 m from the wall. How far up the wall does it reach?

11 A television screen is 48 cm high and 64 cm wide. Find the diagonal, and explain why the diagonal is longer than either side.

12 An isosceles triangle has base 16 cm and equal sides 10 cm. Find its height and area.

13 Find the distance between the points $(1, 2)$ and $(7, 10)$. Draw a sketch showing the right-angled triangle you used.

14 Give the exact hypotenuse of a right-angled triangle with legs 4 cm and 6 cm, simplified as far as possible.

15 **Error analysis.** A student writes $\sqrt{6^2 + 8^2} = 6 + 8 = 14$. Explain the error and give the correct value.

16 **Extension.** The sides of a right-angled triangle are three consecutive whole numbers. Show that the only possibility is $3, 4, 5$.

17 **Extension.** A rectangle has sides x and $x + 7$, and diagonal $x + 8$. Find x and the side lengths.

DIAGONALS IN SPACE

10.2 Pythagoras in three dimensions

A box contains many right-angled triangles. The diagonal of the base is the hypotenuse of a right-angled triangle drawn on the floor. That diagonal then becomes a leg of a second right-angled triangle rising to the opposite top corner. Applying the theorem twice gives the **space diagonal**, the longest straight line that fits inside a cuboid.

For a cuboid with length l, width w and height h, the base diagonal is $\sqrt{l^2 + w^2}$. The space diagonal d satisfies $d^2 = (l^2 + w^2) + h^2$, so $d^2 = l^2 + w^2 + h^2$. The order of the three measurements does not matter.

KEY RELATIONSHIP

Space diagonal of a cuboid

For a cuboid with length l, width w and height h,

$$d^2 = l^2 + w^2 + h^2$$

so $d = \sqrt{l^2 + w^2 + h^2}$. The space diagonal joins two opposite vertices and passes through the interior. It is longer than any edge and longer than any face diagonal.

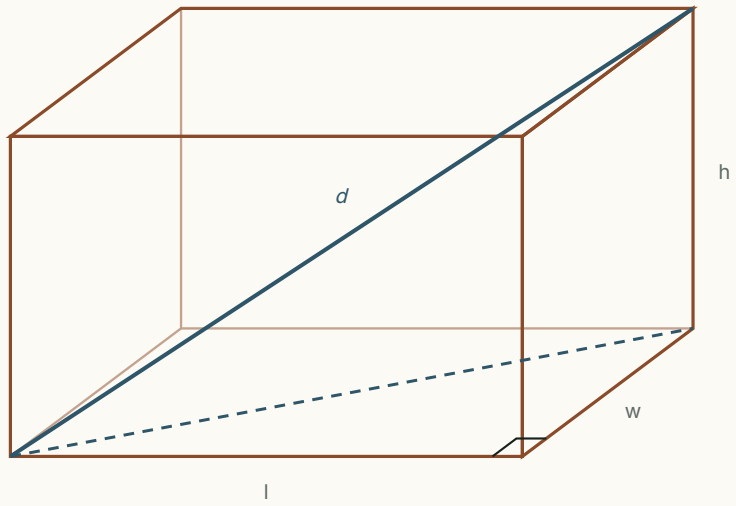


Figure 10.3 The base diagonal is found from l and w. That diagonal and the height h form a second right-angled triangle, whose hypotenuse is the space diagonal d.

EXAMPLE 10.7 SPACE DIAGONAL OF A CUBOID

QUESTION

A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of its space diagonal.

SOLUTION

1. The base diagonal is $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$ cm.
2. The space diagonal satisfies $d^2 = 5^2 + 12^2 = 25 + 144 = 169$.
3. $d = 13$ cm.

Check: the direct formula gives $d^2 = 3^2 + 4^2 + 12^2 = 9 + 16 + 144 = 169$, the same result. The space diagonal is longer than the base diagonal, as expected.

EXAMPLE 10.8 THE LONGEST POLE IN A ROOM

QUESTION

A room measures 5.0 m by 4.2 m by 2.6 m. Find the length of the longest straight pole that can fit inside the room, to 2 decimal places.

SOLUTION

1. The pole must lie along the space diagonal: $d^2 = 5.0^2 + 4.2^2 + 2.6^2$.
2. $d^2 = 25 + 17.64 + 6.76 = 49.40$.
3. $d = \sqrt{49.40} \approx 7.03$ m.

Check: the floor diagonal alone is $\sqrt{25 + 17.64} \approx 6.53$ m, and the pole is longer because it also rises 2.6 m. A pole of 7.1 m would not fit.

EXAMPLE 10.9 DISTANCE BETWEEN POINTS IN SPACE

QUESTION

Find the distance between $A(1, 2, 3)$ and $B(4, 6, 15)$.

SOLUTION

1. The differences in the three coordinate directions are $4 - 1 = 3$, $6 - 2 = 4$ and $15 - 3 = 12$.
2. These are the three perpendicular edges of a cuboid with A and B at opposite corners, so $AB^2 = 3^2 + 4^2 + 12^2$.
3. $AB^2 = 9 + 16 + 144 = 169$, so $AB = 13$.

Check: the same coordinate differences can be used in any order. The distance is positive and greater than each coordinate difference, as it must be.

EXAMPLE **10.10** A SHIPPING CRATE

QUESTION

A crate measures 30 cm by 40 cm by 120 cm. A rigid metal rod is packed inside. Find the greatest possible length of the rod, and explain why it cannot be longer.

SOLUTION

1 The floor diagonal is $\sqrt{30^2 + 40^2} = \sqrt{2500} = 50$ cm.

2 The space diagonal is $\sqrt{50^2 + 120^2} = \sqrt{2500 + 14\,400} = \sqrt{16\,900} = 130$ cm.

3 A straight rod longer than 130 cm would have endpoints farther apart than any two points in the crate. A longer rod could only fit if it were bent, which a rigid rod cannot be.

Check: the direct formula gives $30^2 + 40^2 + 120^2 = 900 + 1600 + 14\,400 = 16\,900$, and $\sqrt{16\,900} = 130$.

COMMON MISCONCEPTION

"Add the three edges to get the space diagonal." The formula adds the three *squared* edges and then takes the square root: $d = \sqrt{l^2 + w^2 + h^2}$, not $l + w + h$.

"Use the height with one edge." The height is perpendicular to the base, not to a single edge. It must be combined with the whole base diagonal, which already uses both base edges.

"The space diagonal is just the base diagonal." The base diagonal lies flat on the floor. The space diagonal rises from it to the opposite top corner and is always longer.

EXERCISE 10B

THREE DIMENSIONS

FLUENCY

1 Find the space diagonal of a cuboid measuring 1 cm by 2 cm by 2 cm.

2 Find the space diagonal of a cuboid measuring 2 cm by 3 cm by 6 cm.

3 Find the space diagonal of a cuboid measuring 4 m by 4 m by 7 m.

4 Find the space diagonal of a cuboid measuring 5 cm by 5 cm by 10 cm, to 2 decimal places.

5 Find the space diagonal of a cuboid measuring 8 cm by 9 cm by 12 cm.

6 Find the distance between $(0, 0, 0)$ and $(2, 3, 6)$.

7 Find the distance between $(1, 1, 1)$ and $(4, 5, 13)$.

UNDERSTANDING AND REASONING

8 A room measures 6.0 m by 3.5 m by 2.4 m. Find the longest straight pole that fits inside, to 2 decimal places.

9 A box measures 30 cm by 40 cm by 120 cm. Find its space diagonal.

10 A suitcase measures 55 cm by 40 cm by 20 cm. Will a 75 cm umbrella fit inside? Justify your answer.

11 A cube has edge 5 cm. Find the exact length of its space diagonal and an approximation to 2 decimal places.

12 **Error analysis.** A student calculates $d = \sqrt{4^2 + 3^2} + 2$ for a cuboid 4 cm by 3 cm by 2 cm. Explain what the student has done and give the correct answer.

13 A cuboid has a square base 6 cm by 6 cm and space diagonal 11 cm. Find its height.

EXERCISE 10B · CONTINUED THREE DIMENSIONS

14 **Extension.** A room is 4 m long, 3 m wide and 2.5 m high. A cable runs from one floor corner to the opposite ceiling corner, but it must stay on the walls. Find the shortest wall route and compare it with the space diagonal.

ANGLES INTO RATIOS

10.3 Trigonometric ratios

Pythagoras' theorem works when two sides are known and no angle is needed. But a surveyor often knows a distance and an angle, not two distances. The bridge is provided by similarity: in every right-angled triangle with the same acute angle, the ratio of any two sides is the same, whatever the size of the triangle. That constant ratio depends only on the angle.

Fix an acute angle θ. In a right-angled triangle containing θ, the side opposite θ is called the **opposite**, the side next to θ that is not the hypotenuse is the **adjacent**, and the longest side is the **hypotenuse**. The three trigonometric ratios are given special names.

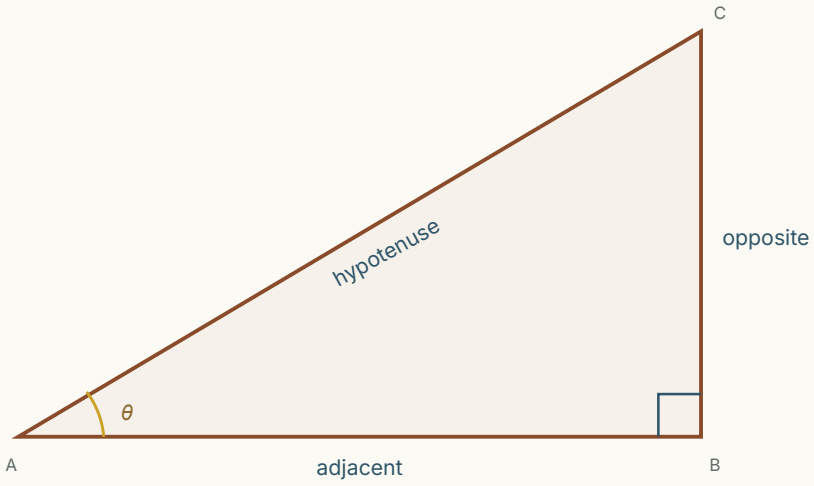


Figure 10.4 The labels opposite, adjacent and hypotenuse are always read from the angle θ, not from the position of the triangle on the page. Here the right angle is at B and θ is at A.

DEFINITIONS

Sine, cosine and tangent

For an acute angle θ in a right-angled triangle:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

The ratios are numbers without units. The mnemonic **SOH CAH TOA** records which sides belong to each ratio. Because the hypotenuse is the longest side, $\sin\theta$ and $\cos\theta$ are always less than 1 for an acute angle; $\tan\theta$ can be any positive number. The inverse functions $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ return the angle when the ratio is known.

THE RATIOS DEPEND ONLY ON THE ANGLE

TRIANGLE	OPPOSITE	ADJACENT	HYPOTENUSE	$\sin\theta$	$\cos\theta$	$\tan\theta$
Small	3	4	5	0.6	0.8	0.75
Large	6	8	10	0.6	0.8	0.75

The two triangles in the table are similar, with scale factor 2. Every side of the large triangle is twice the corresponding side of the small one, so each ratio is unchanged. This is why a table of trigonometric ratios can be printed once: the angle alone determines the ratio, and the size of the triangle only determines the scale.

EXAMPLE **10.11** WRITING THE THREE RATIOS

QUESTION

In a right-angled triangle the sides are 5 cm, 12 cm and 13 cm. The angle θ is opposite the 5 cm side. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$ as fractions and as decimals.

SOLUTION

1. The hypotenuse is the longest side, 13 cm. The side opposite θ is 5 cm, so the adjacent side is 12 cm.
2. $\sin\theta = \frac{5}{13} \approx 0.385$.
3. $\cos\theta = \frac{12}{13} \approx 0.923$.
4. $\tan\theta = \frac{5}{12} \approx 0.417$.

Check: $\sin\theta$ and $\cos\theta$ are both less than 1, and $\tan\theta = \frac{0.385}{0.923} \approx 0.417$. Also $5^2 + 12^2 = 13^2$, so the triangle is right-angled.

EXAMPLE 10.12 FINDING BOTH LEGS FROM THE HYPOTENUSE

QUESTION

A right-angled triangle has hypotenuse 14 cm and an angle of 38°. Find the lengths of the opposite and adjacent sides, to 2 decimal places.

SOLUTION

1. The opposite side and the hypotenuse are connected by sine: $\sin 38^\circ = \frac{\text{opp}}{14}$.
2. $\text{opp} = 14 \sin 38^\circ \approx 14 \times 0.6157 = 8.62$ cm.
3. The adjacent side and the hypotenuse are connected by cosine: $\text{adj} = 14 \cos 38^\circ \approx 14 \times 0.7880 = 11.03$ cm.

Check: $8.62^2 + 11.03^2 \approx 74.3 + 121.7 = 196.0 \approx 14^2$. The two legs are shorter than the hypotenuse, as required.

EXAMPLE 10.13 FINDING A SIDE FROM THE ADJACENT SIDE

QUESTION

In a right-angled triangle, an angle of 57° has an adjacent side of 9.5 m. Find the opposite side and the hypotenuse, to 2 decimal places.

SOLUTION

1. Opposite and adjacent are connected by tangent: $\tan 57^\circ = \frac{\text{opp}}{9.5}$.
2. $\text{opp} = 9.5 \tan 57^\circ \approx 9.5 \times 1.5399 = 14.63$ m.
3. For the hypotenuse, use cosine: $\cos 57^\circ = \frac{9.5}{\text{hyp}}$, so $\text{hyp} = \frac{9.5}{\cos 57^\circ} \approx \frac{9.5}{0.5446} = 17.44$ m.

Check: the opposite side (14.63 m) is shorter than the hypotenuse (17.44 m), and $9.5^2 + 14.63^2 \approx 90.25 + 214.0 = 304.3$, while $17.44^2 \approx 304.2$. The small difference is rounding.

EXAMPLE 10.14 FINDING AN ANGLE FROM TWO SIDES

QUESTION

A right-angled triangle has opposite side 7 cm and hypotenuse 15 cm. Find the angle opposite the 7 cm side, to 1 decimal place.

SOLUTION

1. Opposite and hypotenuse give sine: $\sin\theta = \frac{7}{15} \approx 0.4667$.
2. Use the inverse sine: $\theta = \sin^{-1}(0.4667) \approx 27.8^\circ$.

Check: $\sin 28^\circ \approx 0.4695$, close to 0.4667, so 27.8° is reasonable. The angle is acute, as the triangle requires.

EXAMPLE **10.15** USING THE INVERSE TANGENT

QUESTION

In a right-angled triangle, the side opposite angle θ is 11 m and the adjacent side is 7 m. Find θ, to 1 decimal place.

SOLUTION

1 Opposite and adjacent give tangent: $\tan\theta = \frac{11}{7} \approx 1.5714$.

2 $\theta = \tan^{-1}(1.5714) \approx 57.5°$.

Check: the opposite side is longer than the adjacent side, so the angle should be greater than $45°$. $57.5°$ satisfies this.

EXACT VALUES YOU WILL MEET AGAIN

Two special triangles give exact ratios. A $45°$-$45°$-$90°$ triangle is half a square, so its sides are in the ratio $1 : 1 : \sqrt{2}$. A $30°$-$60°$-$90°$ triangle is half an equilateral triangle, so its sides are in the ratio $1 : \sqrt{3} : 2$. Hence $\sin 45° = \cos 45° = \frac{\sqrt{2}}{2}$, $\tan 45° = 1$; $\sin 30° = \frac{1}{2}$, $\cos 30° = \frac{\sqrt{3}}{2}$, $\tan 30° = \frac{1}{\sqrt{3}}$; and $\sin 60° = \frac{\sqrt{3}}{2}$, $\cos 60° = \frac{1}{2}$, $\tan 60° = \sqrt{3}$. A calculator gives decimals for these, but the exact forms show the structure.

COMMON MISCONCEPTION

"Sine is opposite over adjacent." That ratio is tangent. Use SOH CAH TOA: sine uses the hypotenuse, tangent does not.

"The calculator answer is the angle in degrees, so it must be right." A calculator in radian mode returns a different number for the same ratio. Check for **DEG** before using sin, cos or tan.

"sin θ can be 1.4." For an acute angle, the opposite side is shorter than the hypotenuse, so $\sin\theta$ is between 0 and 1. A result above 1 signals an error in labelling or in the calculation.

"Ratios have units." A ratio divides a length by a length, so the units cancel. The sine of an angle is a pure number.

EXERCISE 10C

TRIGONOMETRIC RATIOS

FLUENCY

1 A right-angled triangle has sides 3 cm, 4 cm and 5 cm. For the angle opposite the 3 cm side, write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

2 A right-angled triangle has sides 8 cm, 15 cm and 17 cm. For the angle opposite the 15 cm side, write the three ratios.

3 Use a calculator to find $\sin 23°$, $\cos 23°$ and $\tan 23°$, to 4 decimal places.

4 Find the opposite side when the hypotenuse is 12 cm and the angle is $41°$.

5 Find the adjacent side when the hypotenuse is 20 m and the angle is $36°$.

6 Find the opposite side when the adjacent side is 7 cm and the angle is $52°$.

7 Find θ when the opposite side is 4 cm and the hypotenuse is 9 cm.

8 Find θ when the adjacent side is 12 m and the hypotenuse is 15 m.

9 Find θ when the opposite side is 9 cm and the adjacent side is 5 cm.

10 Without a calculator, state $\sin 45°$, $\cos 45°$ and $\tan 45°$.

EXERCISE 10C · CONTINUED TRIGONOMETRIC RATIOS

UNDERSTANDING AND REASONING

11 Explain why $\sin\theta$ cannot be greater than 1 for an acute angle θ.

12 **Error analysis.** A student calculates $\tan 45°$ and obtains 1.6198. Explain what has probably happened and how to fix it.

13 Without a calculator, evaluate $\sin 30° \times \cos 60°$. Explain your reasoning.

14 A ramp makes an angle of $2°$ with the horizontal. Find $\sin 2°$ and $\tan 2°$, and explain why the two values are almost equal for a small angle.

15 Show that $\tan\theta = \frac{\sin\theta}{\cos\theta}$ for any acute angle in a right-angled triangle.

16 **Extension.** If $\sin\theta = 0.35$ for an acute angle, find $\cos\theta$ and $\tan\theta$ without finding θ first. (Use a right-angled triangle with hypotenuse 1.)

CHOOSING THE RIGHT RATIO

10.4 Solving right-angled triangles

To **solve** a right-angled triangle means to find every unknown side and angle. A right angle is already known, so at most two more pieces of information are needed. The method depends on what is given:

- two sides and no angle: use Pythagoras' theorem;
- one side and one acute angle: use sine, cosine or tangent;
- two sides and you need an angle: use the inverse trigonometric function.

Always draw and label the triangle first. Mark the angle you are using and name the three sides relative to it. Then choose the ratio that contains the two sides involved in the problem — one known, one unknown. Solving the equation is the last step, not the first.

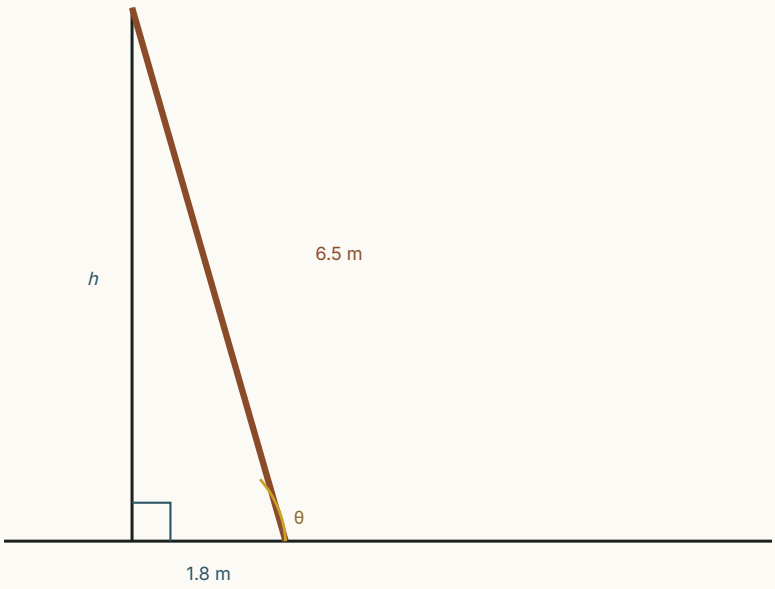


Figure 10.5 The ladder, the wall and the ground form a right-angled triangle. The angle θ is between the ladder and the ground; the height h is opposite θ, and the 1.8 m distance is adjacent to it.

EXAMPLE **10.16** A LADDER AGAINST A WALL

QUESTION

A 6.5 m ladder leans against a vertical wall with its foot 1.8 m from the wall. Find the height it reaches and the angle it makes with the ground.

SOLUTION

1. The ladder is the hypotenuse. The height h and the 1.8 m distance are the legs, so use Pythagoras: $h^2 = 6.5^2 - 1.8^2$.
2. $h^2 = 42.25 - 3.24 = 39.01$, so $h \approx 6.25$ m.
3. For the angle at the ground, the adjacent side is 1.8 m and the hypotenuse is 6.5 m: $\cos\theta = \frac{1.8}{6.5}$.
4. $\theta = \cos^{-1}(0.2769) \approx 73.9°$.

Check: the ladder is steep, so the angle should be close to $90°$. Also $1.8^2 + 6.25^2 = 3.24 + 39.06 = 42.30 \approx 6.5^2$.

EXAMPLE 10.17 ANGLE OF ELEVATION OF A FLAGPOLE

QUESTION

A flagpole 24 m tall casts a shadow 31 m long on level ground. Find the angle of elevation of the Sun, to 1 decimal place.

SOLUTION

1. The pole, its shadow and the ray of sunlight form a right-angled triangle. The pole is opposite the Sun's angle and the shadow is adjacent.
2. $\tan\theta = \frac{24}{31} \approx 0.7742$.
3. $\theta \approx 37.7^\circ$.

Check: the pole is shorter than its shadow, so the angle should be less than 45°. 37.7° is consistent.

EXAMPLE 10.18 DESIGNING A RAMP

QUESTION

A ramp is 5.2 m long along its slope and makes an angle of 8° with the horizontal. Find its rise and its horizontal run, to the nearest centimetre.

SOLUTION

1. The ramp length is the hypotenuse. The rise is opposite the 8° angle: $\text{rise} = 5.2\sin 8^\circ \approx 5.2 \times 0.1392 = 0.72$ m.
2. The run is adjacent: $\text{run} = 5.2\cos 8^\circ \approx 5.2 \times 0.9903 = 5.15$ m.
3. Check with Pythagoras: $0.72^2 + 5.15^2 \approx 0.52 + 26.52 = 27.04$, and $5.2^2 = 27.04$.

Check: for a small angle, the run is almost as long as the ramp and the rise is small. The rise is about 14% of the run, consistent with an 8° angle.

EXAMPLE 10.19 AN ISOSCELES ROOF TRUSS

QUESTION

An isosceles triangle has equal sides 13 m and base 10 m. Find its height, its apex angle and its base angles.

SOLUTION

1. The altitude bisects the base, giving a right-angled triangle with hypotenuse 13 m and adjacent side 5 m. Height: $h = \sqrt{13^2 - 5^2} = 12$ m.
2. Half the apex angle has $\sin\alpha = \frac{5}{13}$, so $\alpha \approx 22.6^\circ$ and the apex angle is about 45.2°.
3. Each base angle is $90^\circ - \alpha \approx 67.4^\circ$.

Check: $45.2^\circ + 2 \times 67.4^\circ = 180.0^\circ$. The base angles are equal, as an isosceles triangle requires.

EXAMPLE **10.20** AN ANGLE INSIDE A CUBOID

QUESTION

A cuboid measures 3 cm by 4 cm by 12 cm. Find the angle between its space diagonal and its base.

SOLUTION

1 The base diagonal is $\sqrt{3^2 + 4^2} = 5$ cm.

2 The space diagonal, the base diagonal and the vertical edge form a right-angled triangle. The height 12 cm is opposite the required angle and the base diagonal 5 cm is adjacent.

3 $\tan\theta = \frac{12}{5} = 2.4$, so $\theta \approx 67.4°$.

Check: the cuboid is tall relative to its base, so the diagonal should rise steeply. The angle is between $0°$ and $90°$, as required.

COMMON MISCONCEPTION

"Use sine because it is the first ratio." Choose the ratio from the sides you have and need, not from the order of the formulae. Label the sides first, then ask which ratio contains both.

"If two sides are known, always use trigonometry." If no angle is given or needed, Pythagoras is usually shorter and avoids rounding.

"$\sin^{-1}$ means divide by sine." $\sin^{-1}(x)$ is the angle whose sine is x, not $\frac{1}{\sin x}$. Use the inverse function key, not a reciprocal.

"Round at every step." Rounding a side to 1 decimal place and then using it to find an angle can change the answer. Keep at least 4 significant figures until the final line.

EXERCISE 10D

SOLVING RIGHT-ANGLED TRIANGLES

FLUENCY

1 A right-angled triangle has hypotenuse 13 cm and an angle of $32°$. Find the opposite and adjacent sides.

2 An angle of $41°$ has an adjacent side of 18 m. Find the opposite side.

3 An angle of $27°$ has an opposite side of 9 cm. Find the hypotenuse.

4 Find θ when the opposite side is 5 cm and the hypotenuse is 9 cm.

5 Find θ when the adjacent side is 14 m and the hypotenuse is 18 m.

6 Find θ when the opposite side is 11 cm and the adjacent side is 16 cm.

7 An 8 m ladder makes an angle of $70°$ with the ground. How high does it reach?

8 A kite string 40 m long makes an angle of $55°$ with the horizontal. Find the height of the kite.

UNDERSTANDING AND REASONING

9 An isosceles triangle has base 12 cm and equal sides 10 cm. Find its height, apex angle and base angles.

10 **Error analysis.** A student writes $\sin\theta = \frac{12}{20}$, so $\theta = 0.6°$. Explain the error and give the correct angle.

11 A cuboid measures 3 cm by 4 cm by 12 cm. Find the angle between its space diagonal and its base.

12 Solve $5\sin\theta = 2.5$ for the acute angle θ.

EXERCISE 10D · CONTINUED SOLVING RIGHT-ANGLED TRIANGLES

13 **Extension.** A right-angled triangle has hypotenuse 1 and one angle of $30°$. Find the exact lengths of both legs.

14 **Extension.** From a point on level ground, the angle of elevation of a tower is $30°$. Show that moving to one-third of the original distance from the tower makes the angle of elevation $60°$. Explain why the angle does not triple.

ANGLES IN THE FIELD

10.5 Angles of elevation, depression and bearings

Surveyors and navigators describe direction with angles measured from a horizontal or from north. An **angle of elevation** is measured upwards from the horizontal, from the observer's eye to an object above. An **angle of depression** is measured downwards from the horizontal to an object below. Both are measured between the line of sight and the horizontal, never between the line of sight and the vertical.

Because the horizontal lines at the top of a cliff and at the bottom are parallel, the angle of depression from the top equals the angle of elevation from the bottom: they are alternate angles. This is often the quickest way to place an angle inside the right-angled triangle you want to solve.

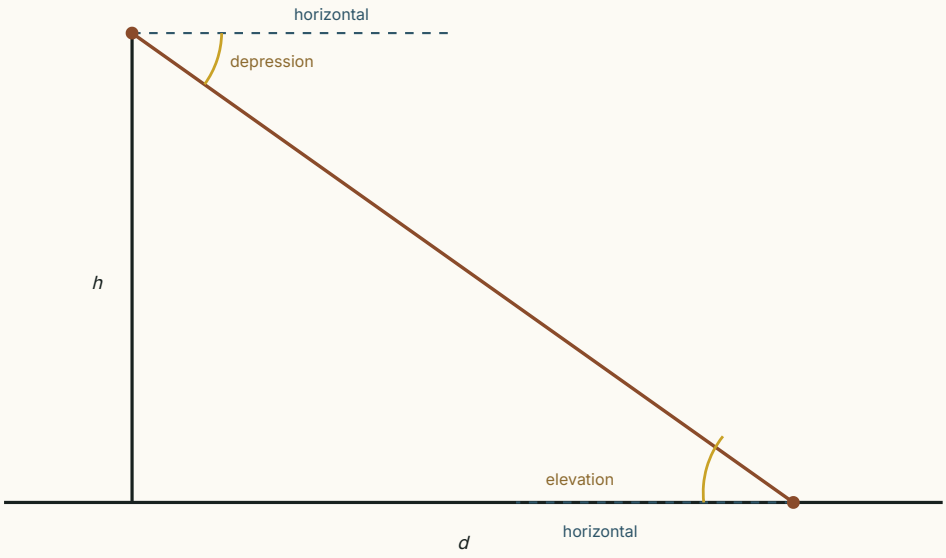


Figure 10.6 The angle of depression at the top and the angle of elevation at the boat are equal because the two horizontal lines are parallel. The height h, the horizontal distance d and the line of sight form a right-angled triangle.

DEFINITIONS

Three-figure bearings

A **bearing** gives direction as an angle measured clockwise from north. It is written with three figures, so north is $000°$, east is $090°$, south is $180°$ and west is $270°$. A bearing such as $047°$ is between north and east; a bearing of $312°$ is between north and west. The **back bearing**, the direction from the destination back to the start, differs by $180°$: add $180°$ if the bearing is less than $180°$, and subtract $180°$ if it is greater.

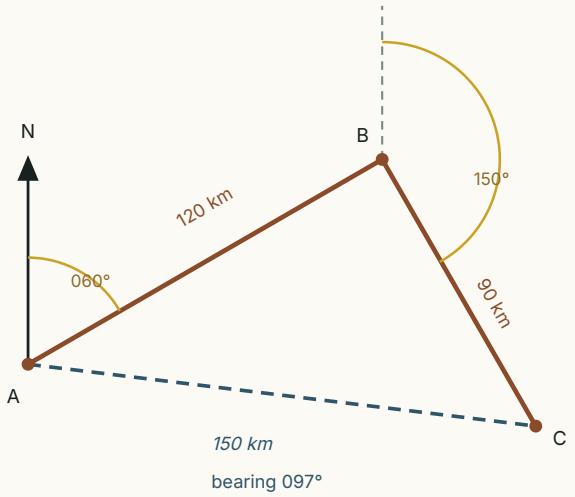


Figure 10.7 The two legs of the journey are perpendicular, because a turn from bearing $060°$ to bearing $150°$ is a turn of $90°$. The resultant displacement from A to C can therefore be found with Pythagoras and one trigonometric ratio.

EXAMPLE **10.21** ANGLE OF DEPRESSION FROM A CLIFF

QUESTION

A cliff is 45 m high. From the top, the angle of depression of a boat is $21°$. Find the horizontal distance from the boat to the foot of the cliff, to the nearest metre.

SOLUTION

1 The angle of depression equals the angle of elevation from the boat, so the angle inside the triangle at the boat is also $21°$.

2 The height 45 m is opposite this angle and the horizontal distance d is adjacent: $\tan 21° = \frac{45}{d}$.

3 $d = \frac{45}{\tan 21°} \approx \frac{45}{0.3839} = 117.2$ m, so the boat is about 117 m from the foot of the cliff.

Check: the angle is small, so the distance should be much larger than the height. $117 \times \tan 21° \approx 44.9$ m, close to 45 m.

EXAMPLE **10.22** DRONE ABOVE A LANDING SITE

QUESTION

A drone hovers 120 m above the ground. Its camera looks down at a landing pad with an angle of depression of 34°. Find the horizontal distance to the pad and the straight-line distance from the drone to the pad.

SOLUTION

1. The horizontal distance is adjacent to 34°, and the height is opposite: $d = \frac{120}{\tan 34^\circ} \approx 177.9$ m.
2. The straight-line distance is the hypotenuse: $s = \frac{120}{\sin 34^\circ} \approx 214.6$ m.

Check: the line of sight is longer than both the height and the horizontal distance. $177.9^2 + 120^2 \approx 31\,650 + 14\,400 = 46\,050$, and $214.6^2 \approx 46\,053$.

EXAMPLE **10.23** A TWO-LEG JOURNEY

QUESTION

A ship sails 120 km on a bearing of 060°, then 90 km on a bearing of 150°. Find its distance and bearing from the starting point.

SOLUTION

1. The turn from 060° to 150° is 90°, so the two legs and the resultant form a right-angled triangle with legs 120 km and 90 km.
2. Resultant distance $= \sqrt{120^2 + 90^2} = \sqrt{14\,400 + 8100} = \sqrt{22\,500} = 150$ km.
3. The angle between the first leg and the resultant has $\tan\alpha = \frac{90}{120} = 0.75$, so $\alpha \approx 36.9^\circ$.
4. The first leg is on bearing 060°, so the resultant bearing is $060^\circ + 36.9^\circ = 096.9^\circ$, or 097° to three figures.

Check: the ship has travelled east and slightly south overall, so the bearing should be a little east of 090°. 097° is consistent. The resultant 150 km is shorter than the total distance travelled, 210 km.

EXAMPLE **10.24** BEARINGS AND A RIGHT ANGLE

QUESTION

A lighthouse L is 8 km due north of a port P. A boat is due east of the lighthouse and on a bearing of 060° from the port. Find the boat's distance from the lighthouse and from the port.

SOLUTION

1. Draw PL north and LB east, so $\angle PLB = 90^\circ$. The bearing 060° at P makes $\angle LPB = 60^\circ$.
2. LB is opposite 60° and $PL = 8$ km is adjacent: $LB = 8\tan 60^\circ \approx 13.86$ km.
3. PB is the hypotenuse: $PB = \frac{8}{\cos 60^\circ} = 16$ km.

Check: $8^2 + 13.86^2 \approx 64 + 192.1 = 256.1$, and $16^2 = 256$. The boat is east of L and south of the line PL, consistent with the bearing.

COMMON MISCONCEPTION

"Measure the angle of depression from the vertical." Angles of elevation and depression are measured from the horizontal. An angle from the vertical would give a different value, and the alternate-angle relationship would no longer apply.

"A bearing can have two figures." Bearings are written with three figures: east is $090°$, not $90°$. The leading zero preserves the convention that all bearings have the same form.

"The back bearing is always $180°$ more." Add $180°$ only when the bearing is less than $180°$. For $248°$, subtract: $248° - 180° = 068°$.

"Turn left to go from $060°$ to $150°$." Bearings increase clockwise. The turn from $060°$ to $150°$ is $90°$ clockwise, which is why the triangle is right-angled.

EXERCISE 10E

ELEVATION, DEPRESSION AND BEARINGS

FLUENCY

1 From a point 25 m from the base of a building, the angle of elevation of the top is $36°$. Find the height of the building.

2 From the top of an 80 m cliff, the angle of depression of a boat is $28°$. Find the horizontal distance to the boat.

3 A plane flies 150 km on a bearing of $030°$, then 120 km on a bearing of $120°$. Find its distance and bearing from the start.

4 Find the back bearing of $215°$.

5 A point B is 14 km from A on a bearing of $072°$. How far is B east and north of A?

6 A boat is due east of a lighthouse and on a bearing of $045°$ from a point 500 m due south of the lighthouse. Find the boat's distance from that point.

UNDERSTANDING AND REASONING

7 From a drone 55 m above the ground, the angle of depression of a person is $32°$. Find the horizontal distance from the drone to the person.

8 A ramp is 12 m long and makes an angle of $5°$ with the horizontal. Find its rise and run.

9 A tree casts a shadow 18 m long when the angle of elevation of the Sun is $43°$. Find the height of the tree.

10 **Error analysis.** A student draws an angle of depression from the top of a cliff to a boat, but measures it from the vertical. Explain how this changes the triangle and why the calculated distance would be wrong.

11 Two points A and B lie on the same horizontal line from the foot of a tower, with B farther away. From A the angle of elevation is $45°$, and from B it is $30°$. If $AB = 20$ m, find the height of the tower.

12 **Extension.** Explain why a bearing of $225°$ points south-west, and find the bearing of the direction exactly opposite. Sketch both directions on a compass rose.

INVESTIGATION · CRITERION B

10.6 Measuring the inaccessible

A **clinometer** measures an angle of elevation. Hold it at eye level, sight the top of the object, and read the angle from the horizontal. If you also measure the horizontal distance to the object and your eye height, the height follows from one tangent ratio. A class used this method to estimate the height of a flagpole from four positions on level ground, all on the same side.

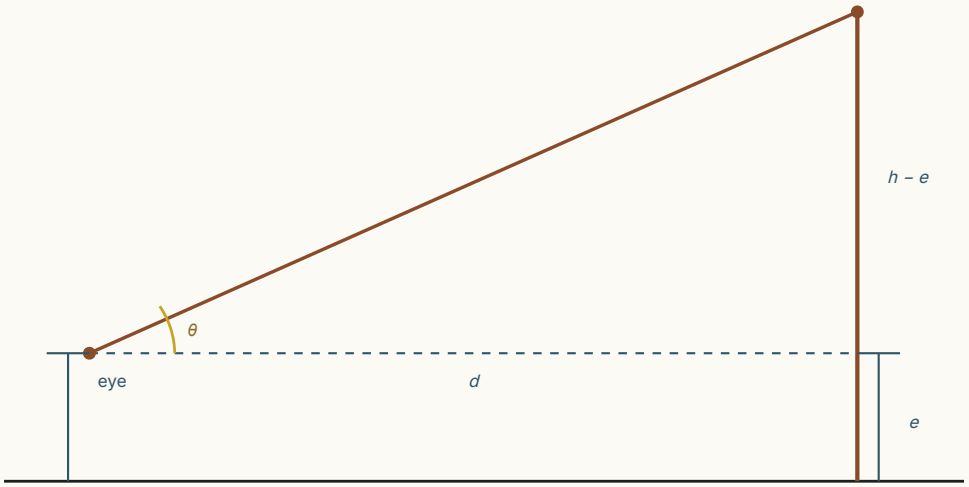


Figure 10.8 The observer's eye is at height e. The line of sight makes an angle θ with the horizontal at that height, so the part of the flagpole above eye level is $d\tan\theta$. The total height is $h = e + d\tan\theta$.

FLAGPOLE MEASUREMENTS FROM FOUR POSITIONS (EYE HEIGHT 1.5 M)

POSITION	DISTANCE d (m)	ANGLE OF ELEVATION θ	RISE $d\tan\theta$ (m)	HEIGHT h (m)
1	10	46°	—	—
2	15	35°	—	—
3	20	28°	—	—
4	25	23°	—	—

INVESTIGATION · CRITERION B

How tall is the flagpole?

1. Copy and complete the table, calculating $d\tan\theta$ and then $h = 1.5 + d\tan\theta$ to 2 decimal places.
2. Find the mean of the four heights and the range. Do the four estimates agree closely enough to trust the method?
3. Explain why the four values differ even though the flagpole did not change. Refer to the measurement of the angle, the measurement of the distance and the assumption that the ground is level.
4. Show that the method gives $h = e + d\tan\theta$, where e is eye height. Draw and label the triangle you used.
5. Suppose each angle could be wrong by 1°. By how much would the height change at $d = 10$ m and at $d = 25$ m? Which distance gives the more reliable estimate?
6. State two assumptions of the method and describe a test that would reveal whether each assumption is reasonable. Then suggest one improvement to the procedure.

Generalise and justify. For a fixed angle θ, explain why the graph of h against d is a straight line with gradient $\tan\theta$ and intercept e. What does the gradient mean in this context, and why must the intercept equal the observer's eye height?

MIXED PRACTICE 10

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–6 check fluency; 7–13 require reasoning; 14–18 are multi-step or unfamiliar. Allow 55 minutes.

FLUENCY

1 The legs of a right-angled triangle are 6 cm and 8 cm. Find the hypotenuse.

2 A right-angled triangle has hypotenuse 41 cm and one leg 9 cm. Find the other leg.

3 Find the space diagonal of a cuboid measuring 2 cm by 3 cm by 6 cm.

4 For the angle opposite the 15 cm side in an $8, 15, 17$ triangle, write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

5 Find x when $x = 9\tan 34^\circ$.

6 Find the acute angle θ when $\sin\theta = 0.42$.

UNDERSTANDING AND REASONING

7 A 4.8 m ladder makes an angle of 72° with the ground. How high does it reach?

8 Find the back bearing of 248°.

9 From a 70 m cliff, the angle of depression of a boat is 18°. Find the horizontal distance to the boat.

10 **Error analysis.** A student writes $5^2 + 12^2 = 169$, so the hypotenuse is 169. Explain the error and give the correct answer.

11 Find the space diagonal of a cuboid measuring 5 cm by 5 cm by 10 cm, to 2 decimal places.

12 A right-angled triangle has legs 6 cm and 8 cm. A similar triangle has hypotenuse 30 cm. Find the area ratio of the two triangles.

13 A ramp has a gradient of $1 : 8$. Find its angle with the horizontal, and the length of the ramp needed to rise 1.1 m.

MIXED PRACTICE 10 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

MULTI-STEP AND UNFAMILIAR

14 A tree casts a shadow 14 m long when the angle of elevation of the Sun is $38°$. Find the height of the tree.

15 A plane flies 80 km due north, then 60 km due east. Find its distance and bearing from the start.

16 Solve $2\sin\theta = 0.9$ for the acute angle θ.

17 Without a calculator, evaluate $\sin 45° \times \cos 45°$ and $\tan 60° \times \tan 30°$.

18 **Extension.** A right-angled triangle has perimeter 60 cm and hypotenuse 25 cm. Find the lengths of the two legs.

CHAPTER ASSESSMENT

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and assessed against the criteria shown. Your teacher will confirm which tasks are required and how long you have.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Right triangles and measurement: short response	30	35 min
B · Investigating patterns	Triples from odd numbers	24	40 min
C · Communicating	A field guide to measuring height	20	35 min
D · Applying mathematics in real-life contexts	The ramp and the solar panel	24	45 min
Total	All four tasks	98	155 min

The tasks begin overleaf. Read the instructions and evidence requirements before you start, and show the reasoning that supports each answer.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 30 marks

Right triangles and measurement: short response

CALCULATOR SHOW WORKING

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

1 Find the hypotenuse of a right-angled triangle with legs 7 cm and 24 cm. [2]

2 Find the other leg of a right-angled triangle with hypotenuse 41 cm and one leg 9 cm. [2]

3 Decide whether a triangle with sides 10 cm, 24 cm and 26 cm is right-angled. Show your working. [2]

4 Find the space diagonal of a cuboid measuring 2 cm by 3 cm by 6 cm. [3]

5 In a $5, 12, 13$ triangle, write $\sin\theta$, $\cos\theta$ and $\tan\theta$ for the angle θ opposite the 5 cm side. [3]

6 Find x when $x = 14\sin 38^\circ$, to 2 decimal places. [3]

7 Find the acute angle θ when $\tan\theta = \frac{7}{11}$, to 1 decimal place. [3]

8 A 7 m ladder makes an angle of 64° with the ground. Find the height it reaches, to 2 decimal places. [3]

9 Find the back bearing of 137°. [2]

10 **Error analysis.** A student writes $8^2 + 6^2 = 100$, so the hypotenuse is 100. Explain the error and give the correct value. [3]

11 From the top of a 60 m cliff, the angle of depression of a boat is 25°. Find the horizontal distance from the boat to the foot of the cliff, to the nearest metre. [4]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

Triples from odd numbers

Instructions. Work independently. Whole-number solutions of $a^2 + b^2 = c^2$ are called Pythagorean triples. The table shows a pattern that starts with each odd number n.

FIRST LEG n	SECOND LEG	HYPOTENUSE
3	4	5
5	12	13
7	24	25

1. Complete the table for $n = 9$, 11 and 13. [5]
2. Describe the pattern in words. Use it to predict the triple for $n = 15$. [4]
3. Write a rule that gives the triple for any odd number n. [5]
4. Prove algebraically that your rule always produces a right-angled triangle. [6]
5. Explain why the rule requires n to be odd if both legs are to be whole numbers. Use your rule to find the triple whose hypotenuse is 145. [4]

Evidence: a complete table, a clearly stated rule, an algebraic proof, and a justified explanation of the odd-number condition.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

A field guide to measuring height

Instructions. Write a 300–400 word field guide for a student who has never used a clinometer. Your guide must: (i) explain what to measure and how to stand; (ii) include a labelled diagram; (iii) derive the formula $h = e + d\tan\theta$; (iv) include one worked example with numbers; (v) explain two sources of error and one way to reduce each; (vi) state one situation in which the method would fail.

Evidence: a clear procedure, a correctly labelled diagram, a formula derived rather than merely quoted, a worked example with units, and an evaluation that distinguishes measurement error from a broken assumption.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

The ramp and the solar panel

Instructions. A community centre is installing a wheelchair ramp and a solar panel. The ramp is to rise 1.2 m over a horizontal run of 12 m. Building guidance recommends a gradient no steeper than $1 : 12$. The solar panel is 1.7 m long and 1.0 m wide, mounted flat on a roof with pitch 28°. Its efficiency is 19%, the average annual irradiation is $1050\ \text{kWh/m}^2$, and system losses are estimated at 8%.

1. Find the length of the ramp along its slope, its gradient in the form $1 : n$, and its angle with the horizontal. [6]
2. Find the minimum horizontal run that meets the $1 : 12$ guidance for a rise of 1.2 m, and the corresponding ramp length. Compare it with the proposed design. [6]
3. Find the vertical rise and the horizontal projection of the 1.7 m panel on the 28° roof. [4]
4. Estimate the panel's annual energy output in kWh, allowing for efficiency and system losses. [5]
5. Evaluate two assumptions in your estimate and explain how each could change the result. [3]

Evidence: a labelled diagram, lengths and angles with units, a gradient compared correctly with the guidance, an energy calculation that shows each factor, and an evaluation that distinguishes geometric accuracy from real-world performance.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few facts; applies procedures with major errors; needs support to select a method.	Recalls some facts; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies facts and procedures accurately in familiar situations; explains some steps.	Selects and applies appropriate mathematics accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- In a right-angled triangle, $a^2 + b^2 = c^2$, where c is the hypotenuse; the converse identifies right angles.
- The space diagonal of a cuboid satisfies $d^2 = l^2 + w^2 + h^2$.
- Sine, cosine and tangent compare pairs of sides and depend only on the acute angle.
- Choose Pythagoras when two sides are involved; choose a trigonometric ratio when an angle and a side are involved.
- Angles of elevation and depression are measured from the horizontal and are equal when taken from opposite ends of a line of sight between parallel horizontals.
- Bearings are measured clockwise from north, written with three figures; a back bearing differs by $180°$.
- Field measurements always carry error; the effect on a calculated height grows with distance.

$$a^2 + b^2 = c^2 \qquad d^2 = l^2 + w^2 + h^2$$

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}}$$

$$h = e + d\tan\theta$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can state and apply Pythagoras' theorem, including its converse.
- ☐ I can explain a proof of Pythagoras' theorem.
- ☐ I can find a space diagonal of a cuboid and a distance between points.
- ☐ I can write sine, cosine and tangent for a labelled triangle.
- ☐ I can explain why the ratios depend only on the angle.
- ☐ I can choose the correct ratio and solve a right-angled triangle.
- ☐ I can use inverse trigonometric functions to find an angle.
- ☐ I can interpret angles of elevation and depression.
- ☐ I can read, write and reverse three-figure bearings.
- ☐ I can calculate a height from field measurements and assess the error.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which idea was more useful to you: the square relationship or the trigonometric ratios? Give a problem where the other idea would have been useless.
2. Describe a mistake you made with a ratio — choosing the wrong sides, using the wrong calculator mode, or rounding too early — and how you found it.
3. The statement of inquiry says that one length and one angle determine the triangle. Test this claim on a triangle you have not seen before: what exactly must be known, and what must be assumed?
4. Look ahead to Chapter 11: a solid has surface area and volume as well as lengths. Which of the relationships in this chapter do you expect to reappear when a solid is enlarged?

HOW MUCH IT HOLDS, HOW MUCH IT TAKES

Surface Area & Volume

11

FORM · SPACE · QUANTITY

Every solid has two measures that pull in opposite directions: the surface that must be covered and the space that can be filled. A tank, a carton and a dome are designed by balancing the two against cost, strength and the material available.

STATEMENT OF INQUIRY

The form of a solid fixes both the surface that must be covered and the space it encloses. Because area and volume scale differently, the shape that holds the most for the least material is a mathematical optimum — one that real designs must balance against strength, stacking and manufacture.

KEY CONCEPT

Form

RELATED CONCEPTS

Space · Measurement · Quantity

GLOBAL CONTEXT

Globalisation and sustainability — packaging, water storage and material efficiency

FACTUAL

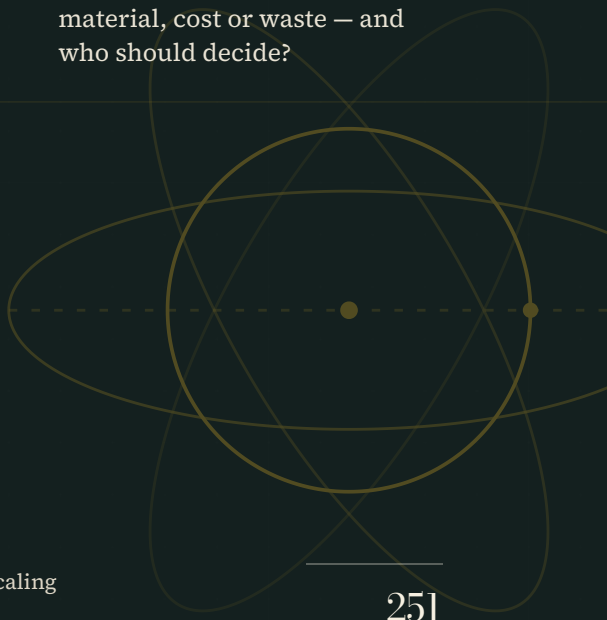
Which quantities describe a solid, and how are they calculated for a prism, a cone and a sphere?

CONCEPTUAL

Why does doubling every length multiply surface area by four but volume by eight?

DEBATABLE

Should a package minimise material, cost or waste — and who should decide?



IN THIS CHAPTER

11.1 Prisms and cylinders
11.2 Pyramids and cones
11.3 Spheres
11.4 Scaling
11.5 Packaging and cost
11.6 Investigation

OPENING PROBLEM · THREE VESSELS, ONE BUDGET

Three vessels, one material budget

A school is improving its outdoor science area. It needs a cylindrical water tank with diameter 2.2 m and height 3.0 m, a conical hopper for collecting compost with radius 1.2 m and slant height 2.0 m, and a spherical gas holder with radius 1.5 m.

1. Find the capacity of the cylindrical tank in cubic metres, then in litres, correct to 3 significant figures.
2. Find the area of sheet metal needed for the tank, including a lid. Compare it with the area of the curved surface alone.
3. Find the perpendicular height of the conical hopper, its capacity, and the area of sheet needed for its curved surface.
4. Find the surface area and the volume of the spherical gas holder.
5. For each vessel, calculate volume divided by surface area. Which vessel stores the most per square metre of material?
6. The tank height is doubled while its radius is unchanged. Predict what happens to its capacity and to the material needed. Then calculate both, and explain why they do not change by the same factor.

Area is measured in square units and volume in cubic units, so a shape cannot be compared with its own covering by a single number. The useful comparison is a ratio: how much space a solid encloses for each unit of surface. That ratio is large for a sphere and small for a flat sheet, and it explains why a small raindrop dries faster than a large one and why a drinks can is taller than it is wide.

ORIENTATION

11.0 Learning intentions

- Identify the faces, edges and vertices of prisms, pyramids, cylinders, cones and spheres.
- Calculate the surface area and volume of prisms and cylinders, including composite solids.
- Use the slant height, perpendicular height and radius of a pyramid or cone correctly.
- Calculate the surface area and volume of spheres, hemispheres and solids built from them.
- Apply length, area and volume scale factors to similar solids and explain the differences.
- Compare packaging designs using surface area to volume ratio and cost per unit volume.
- Model a storage or packaging problem, state its assumptions and evaluate the design.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

use a net to justify a surface-area formula; calculate the surface area and volume of a prism or cylinder; find the slant height of a cone or pyramid and use it in πrl or a triangular face; calculate the surface area and volume of a sphere and a hemisphere; solve a composite-solid problem without double-counting a hidden face; apply k^2 and k^3 to similar solids; compare packages using surface area to volume ratio and cost per litre; and justify a design recommendation with calculations and stated assumptions.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit area in Chapter 7, Pythagoras' theorem in Chapter 10, and similarity in Chapter 9 before starting.

1. Find the area of a circle with radius 5 cm, in terms of π.
2. Find the area of a triangle with base 8 cm and height 6 cm.
3. Find the hypotenuse of a right-angled triangle with legs 5 cm and 12 cm.
4. Convert $2.5\ \text{m}^3$ to litres.
5. Simplify $3^2 + 4^2$.
6. A square has side 7 cm. Find its area and its perimeter.

REFRESHER

Keep π exact until the final step. A calculator's π key is an approximation, so writing 36π is more accurate than writing 113.1. Area is measured in square units and volume in cubic units: $1\ \text{m}^2 = 10\,000\ \text{cm}^2$, $1\ \text{m}^3 = 1\,000\,000\ \text{cm}^3$, and $1\ \text{litre} = 1000\ \text{cm}^3 = 0.001\ \text{m}^3$. When a formula contains r, check whether the problem gave a radius or a diameter. Never mix units in one calculation.

ATL SKILLS · THINKING, RESEARCH AND COMMUNICATION

Critical thinking: decide whether a solid is a prism, a pyramid or a composite, and choose the formula that matches its structure. **Research:** collect real dimensions and prices, and record the units and date of the data. **Communication:** draw and label the solid or its net, state the formula before substituting, and justify a design recommendation with evidence rather than opinion.

A CONSTANT CROSS-SECTION

11.1 Prisms and cylinders

A **prism** is a solid with two identical parallel faces, called bases, and rectangular sides joining corresponding edges. A **cylinder** is the same idea with a circular base. Every slice taken parallel to the base is congruent to the base, which is why the volume depends only on the area of that slice and the distance between the bases.

If the base has area A and perimeter P, and the prism has length L measured perpendicular to the base, then the volume is AL. The surface is two bases plus a rectangle of width P and length L, so the total surface area is $2A + PL$. For a cylinder of radius r and height h, the base area is πr^2, the circumference is $2\pi r$, and the curved surface unrolls into a rectangle of dimensions $2\pi r$ by h. The capacity of a container is the volume it can hold, usually measured in litres.

KEY RELATIONSHIPS

Prisms and cylinders

For a prism with base area A, base perimeter P and perpendicular length L:

$$V = AL \qquad S = 2A + PL$$

For a cylinder with radius r and height h:

$$V = \pi r^2 h \qquad S = 2\pi r^2 + 2\pi r h$$

The first term in S is the two circular ends; the second is the curved surface. For an open cylinder, omit one or both ends as the problem requires.

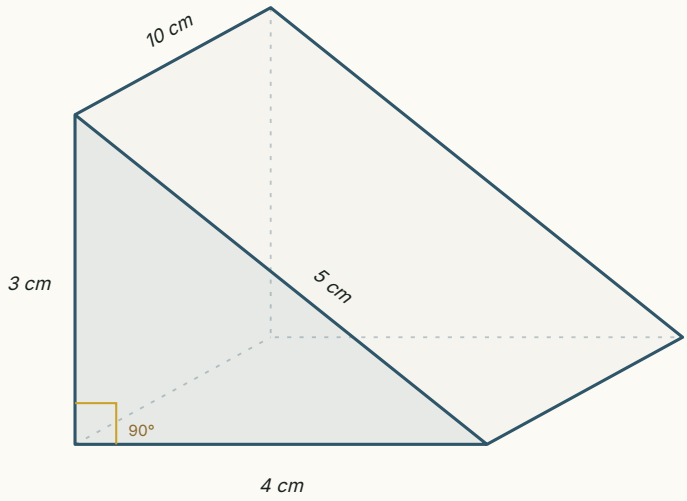


Figure 11.1 The triangular prism has a right-angled base with legs 3 cm and 4 cm and hypotenuse 5 cm. Its length is 10 cm. Its volume is the base area multiplied by the length; its surface is two triangular bases plus three rectangles.

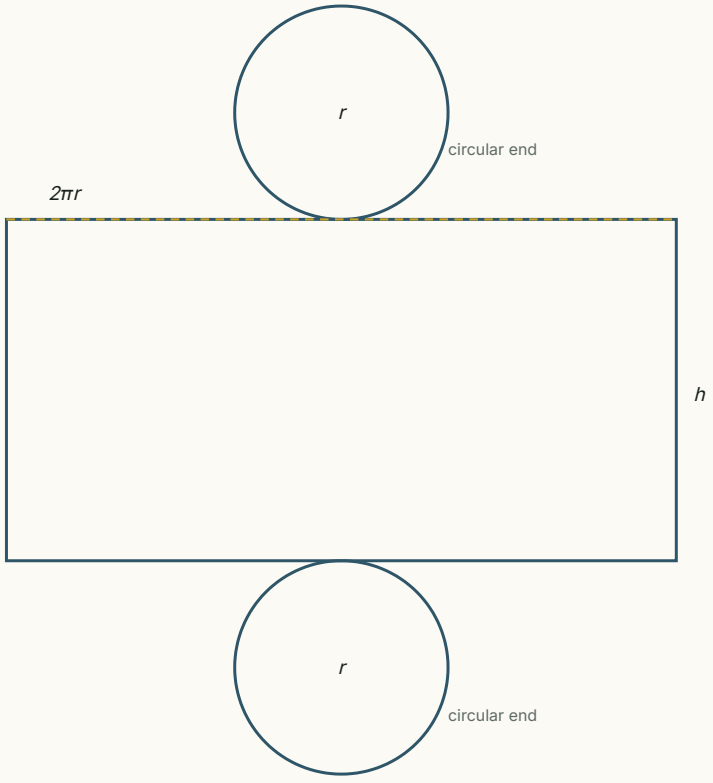


Figure 11.2 The net of a cylinder: the curved surface unrolls into a rectangle whose width is the circumference $2\pi r$. The two circular ends are separate faces. The net shows why the total surface area is $2\pi r^2 + 2\pi rh$, not simply the area of the rectangle.

EXAMPLE 11.1 A TRIANGULAR PRISM

QUESTION

The prism in Figure 11.1 has a right-angled triangular base with legs 3 cm and 4 cm, hypotenuse 5 cm, and length 10 cm. Find its volume and total surface area.

SOLUTION

1. Base area: $A = \frac{1}{2} \times 3 \times 4 = 6 \text{ cm}^2$.
2. Volume: $V = AL = 6 \times 10 = 60 \text{ cm}^3$.
3. Base perimeter: $P = 3 + 4 + 5 = 12 \text{ cm}$.
4. Surface area: $S = 2A + PL = 2(6) + 12(10) = 12 + 120 = 132 \text{ cm}^2$.

Check: the three rectangular faces have areas 30, 40 and 50 cm^2, totalling 120 cm^2. Adding the two triangular ends, $2 \times 6 = 12$, gives 132 cm^2.

EXAMPLE 11.2 A CLOSED CYLINDER

QUESTION

A closed cylinder has radius 7 cm and height 20 cm. Find its volume and surface area. Give exact answers in terms of π, then approximate to 3 significant figures.

SOLUTION

1. Volume: $V = \pi r^2 h = \pi(7)^2(20) = 980\pi \text{ cm}^3 \approx 3080 \text{ cm}^3$.
2. Two ends: $2\pi r^2 = 2\pi(49) = 98\pi \text{ cm}^2$.
3. Curved surface: $2\pi rh = 2\pi(7)(20) = 280\pi \text{ cm}^2$.
4. Total: $S = 98\pi + 280\pi = 378\pi \text{ cm}^2 \approx 1190 \text{ cm}^2$.

Check: the curved surface is much larger than the ends because the cylinder is tall. The exact values 980π and 378π are preferable; 3080 and 1190 are rounded to 3 significant figures.

EXAMPLE 11.3 WORKING BACKWARDS FROM CAPACITY

QUESTION

A cylindrical container must hold 2.0 litres and be 25 cm tall. Find its radius and its total surface area, to 3 significant figures.

SOLUTION

1. Convert: $2.0 \text{ litres} = 2000 \text{ cm}^3$.
2. $2000 = \pi r^2(25)$, so $r^2 = \frac{2000}{25\pi} = \frac{80}{\pi} \approx 25.46$.
3. $r \approx 5.05 \text{ cm}$, so the diameter is about 10.1 cm.
4. $S = 2\pi r^2 + 2\pi rh = 2\pi\left(\frac{80}{\pi}\right) + 2\pi(5.046)(25) \approx 160 + 793 = 953 \text{ cm}^2$.

Check: the two ends contribute exactly 160 cm^2 because $2\pi r^2 = 2\pi(80/\pi)$. The curved surface is about five times larger, as expected for a tall narrow container.

EXAMPLE 11.4 A TRAPEZOIDAL CHANNEL

QUESTION

An open drainage channel has a trapezium cross-section with parallel sides 0.8 m and 1.2 m, depth 0.5 m, and length 6.0 m. Find its volume and the area of its lining, which covers the bottom, the two sloping sides and the two ends but not the top.

SOLUTION

1. Cross-section area: $A = \frac{0.8+1.2}{2} \times 0.5 = 0.5\ \text{m}^2$.
2. Volume: $V = 0.5 \times 6.0 = 3.0\ \text{m}^3$.
3. Each sloping side has horizontal run $\frac{1.2-0.8}{2} = 0.2\ \text{m}$, so its length is $\sqrt{0.5^2 + 0.2^2} = \sqrt{0.29} \approx 0.539\ \text{m}$.
4. Lining area: bottom $0.8 \times 6.0 = 4.8$, two sides $2 \times 0.539 \times 6.0 = 6.47$, two ends $2 \times 0.5 = 1.0$. Total $\approx 12.3\ \text{m}^2$.

Check: the volume is small because the channel is shallow, but the lining is large because it is long. The top is excluded, so the lining is less than the full prism surface $2A + PL$, which would include the water surface.

EXAMPLE 11.5 PAINTING A TANK

QUESTION

A cylindrical tank has diameter 2.2 m and height 3.0 m. It is painted on the outside, including the top. One litre covers $6.0\ \text{m}^2$ per coat, and two coats are required. Paint costs £14.50 per litre and is sold only in whole litres. Find the cost.

SOLUTION

1. Radius $r = 1.1$ m. Surface area: $S = 2\pi(1.1)^2 + 2\pi(1.1)(3.0) = 2.42\pi + 6.60\pi = 9.02\pi \approx 28.3\ \text{m}^2$.
2. Two coats cover $2 \times 28.3 = 56.7\ \text{m}^2$.
3. Paint needed: $56.7 \div 6.0 = 9.45$ litres, so 10 litres must be bought.
4. Cost: $10 \times £14.50 = £145$.

Check: rounding up to whole litres adds about half a litre of unused paint. In a real quotation, the painter would also allow for spillage and for the area occupied by pipes or a ladder.

COMMON MISCONCEPTION

"The volume of a prism is the perimeter times the length." The perimeter gives the length of the rectangle in the net, not the area of the base. Volume uses the base *area*: $V = AL$.

"A cylinder's surface area is $2\pi rh$." That is only the curved surface. A closed cylinder also has two circular ends, each of area πr^2. Read whether the cylinder is open or closed.

"A taller cylinder always holds more." Doubling the height doubles the volume, but doubling the radius multiplies it by four. Compare the actual dimensions before deciding.

"$1\ \text{m}^3 = 100\ \text{cm}^3$." A cubic metre is $100 \times 100 \times 100 = 1\,000\,000\ \text{cm}^3$, and $1\ \text{cm}^3 = 1\ \text{mL}$.

EXERCISE 11A

PRISMS AND CYLINDERS

FLUENCY

1 A cuboid measures 8 cm by 5 cm by 4 cm. Find its surface area and volume.

2 A cube has edge 6 cm. Find its surface area and volume.

3 A triangular prism has a 5–12–13 right-angled base and length 20 cm. Find its volume and surface area.

4 A cylinder has radius 4 cm and height 9 cm. Find its volume and total surface area, in terms of π.

5 A cylinder has diameter 10 cm and height 12 cm. Find its volume and total surface area, to 3 significant figures.

6 A cylinder has radius 2.5 cm and height 8 cm. Find its curved surface area and volume.

7 A cuboid measures 10 cm by 10 cm by 20 cm. Find its surface area and volume.

8 A prism has cross-section area 24 cm^2, cross-section perimeter 22 cm and length 15 cm. Find its volume and surface area.

UNDERSTANDING AND REASONING

9 A room measures 2.5 m by 1.8 m by 1.2 m high. Its four walls and ceiling are painted, but not the floor. Find the painted area and the paint needed for two coats at 8.0 m^2 per litre.

10 A cylindrical water tank has diameter 1.8 m and height 2.4 m. Find its capacity in litres and its total external surface area, including the top.

11 A label wraps completely around a can of radius 3.5 cm and height 11 cm. Find the area of the label.

12 An open-top box measures 30 cm by 20 cm by 15 cm. Find the area of material needed and the capacity in litres.

13 **Error analysis.** A student calculates the surface area of a cylinder with $r = 5$ cm and $h = 10$ cm as $2\pi(5)(10) = 100\pi$. Explain the error and give the correct value.

14 A cylinder holds 1.0 litre and has radius 4.0 cm. Find its height and total surface area, to 3 significant figures.

15 A water trough is a half-cylinder of radius 1.0 m and length 5.0 m. Find its volume and the area of its curved surface plus its two semicircular ends.

16 **Extension.** A cylinder has height equal to its diameter. Show that $S = 6\pi r^2$ and $V = 2\pi r^3$. Find r and S when $V = 1.0$ litre.

17 **Extension.** An open-top box has a square base of side x cm and height h cm. Its surface area is 1200 cm^2. Show that $V = 300x - \frac{x^3}{4}$, and evaluate V for $x = 20$, 25 and 30. Which gives the greatest volume?

APEX AND SLANT

11.2 Pyramids and cones

A **pyramid** has a polygonal base and triangular faces that meet at a single point, the apex. A **cone** is the circular version. Unlike a prism, a pyramid does not have a constant cross-section: slices parallel to the base shrink towards the apex, and the volume is one third of the prism that would surround it.

Two different heights matter. The **perpendicular height** h is the vertical distance from the apex to the base and is used in the volume formula. The **slant height** l runs from the apex to the midpoint of a base edge (for a pyramid) or to the rim (for a cone), and is used in surface area. For a cone, l, r and h form a right-angled triangle, so $l^2 = r^2 + h^2$. Cutting a pyramid or cone with a plane parallel to its base leaves a **frustum**: the part between the base and the cut. Its volume is the difference between the original solid and the smaller similar solid removed from the top.

KEY RELATIONSHIPS

Pyramids and cones

For a pyramid with base area A, base perimeter P, perpendicular height h and slant height l:

$$V = \tfrac{1}{3}Ah \qquad S = A + \tfrac{1}{2}Pl$$

For a cone with radius r, perpendicular height h and slant height l:

$$l^2 = r^2 + h^2 \qquad V = \tfrac{1}{3}\pi r^2 h \qquad S = \pi r^2 + \pi r l$$

The term $\pi r l$ is the curved surface; it unrolls into a sector of a circle of radius l. The formula $S = A + \frac{1}{2}Pl$ applies to a right pyramid whose triangular faces all have the same slant height.

For a cone frustum with base radii R and r and height h:

$$V = \tfrac{\pi h}{3}\left(R^2 + Rr + r^2\right)$$

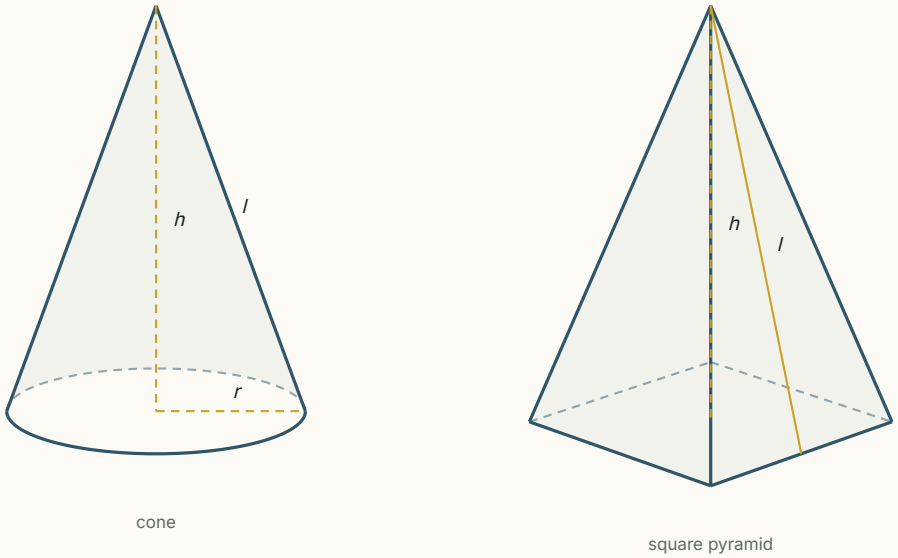


Figure 11.3 In both solids the perpendicular height h is vertical, while the slant height l follows a face. For the cone, $l^2 = r^2 + h^2$; for the pyramid, $l^2 = h^2 + \left(\frac{s}{2}\right)^2$ when the base has side s.

EXAMPLE 11.6 A SQUARE PYRAMID

QUESTION

A right pyramid has a square base of side 6 cm and perpendicular height 4 cm. Find its slant height, total surface area and volume.

SOLUTION

1. The slant height reaches the midpoint of a base edge, a horizontal distance $6 \div 2 = 3$ cm from the centre: $l = \sqrt{4^2 + 3^2} = 5$ cm.
2. Each triangular face has area $\frac{1}{2} \times 6 \times 5 = 15$ cm^2, so four faces total 60 cm^2.
3. Base area $= 36$ cm^2, so $S = 36 + 60 = 96$ cm^2.
4. Volume $= \frac{1}{3} \times 36 \times 4 = 48$ cm^3.

Check: the surrounding prism would have volume $36 \times 4 = 144$ cm^3, and the pyramid is exactly one third of it, as the cone formula predicts.

EXAMPLE 11.7 A CONE

QUESTION

A cone has radius 5 cm and perpendicular height 12 cm. Find its slant height, total surface area and volume. Give exact answers in terms of π, then approximate to 3 significant figures.

SOLUTION

1. Slant height: $l = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$ cm.
2. Curved surface: $\pi r l = \pi(5)(13) = 65\pi$ cm^2. Base: 25π cm^2.
3. Total surface area: 90π cm$^2 \approx 283$ cm^2.
4. Volume: $\frac{1}{3}\pi(25)(12) = 100\pi$ cm$^3 \approx 314$ cm^3.

Check: the cone is one third of the cylinder with the same base and height, whose volume would be 300π. The curved area is nearly three times the base, as expected because the cone is tall.

EXAMPLE 11.8 DESIGNING A CONE FROM ITS VOLUME

QUESTION

A conical hopper must hold 500 cm^3 and has base radius 6 cm. Find its perpendicular height and total surface area, to 3 significant figures.

SOLUTION

1. $500 = \frac{1}{3}\pi(36)h = 12\pi h$, so $h = \frac{500}{12\pi} \approx 13.3$ cm.
2. $l = \sqrt{6^2 + 13.26^2} = \sqrt{36 + 175.8} \approx 14.6$ cm.
3. $S = \pi(36) + \pi(6)(14.56) \approx 113 + 274 = 388$ cm^2.

Check: $\frac{1}{3}\pi(36)(13.26) = 499.8$, close to 500. The slant height is only slightly longer than the perpendicular height because the radius is small compared with the height.

EXAMPLE 11.9 A FRUSTUM

QUESTION

A bucket is a frustum of a cone with base radii 8 cm and 12 cm and perpendicular height 20 cm. Find its capacity in litres, and the area of sheet needed for its curved surface and base.

SOLUTION

1. For a frustum, $V = \frac{\pi h}{3}\left(R^2 + Rr + r^2\right)$.
2. Substitute: $V = \frac{20\pi}{3}(144 + 96 + 64) = \frac{20\pi}{3}(304) \approx 6367\ \text{cm}^3$.
3. Capacity $= 6.37$ litres.
4. Slant height: $l = \sqrt{h^2 + (R - r)^2} = \sqrt{400 + 16} \approx 20.4$ cm.
5. Curved area $= \pi(R + r)l = \pi(20)(20.40) \approx 1282\ \text{cm}^2$. Base $= \pi(8)^2 \approx 201\ \text{cm}^2$. Total $\approx 1483\ \text{cm}^2$.

Check: the capacity lies between the volumes of cylinders of radius 8 and 12 with height 20, which are about 4021 and 9048 cm³. The result 6367 cm³ is reasonable and closer to the larger cylinder because the top is wider.

EXAMPLE 11.10 A SILO

QUESTION

A silo is a cylinder of radius 2.0 m and height 5.0 m with a cone of height 1.5 m on top. Find its volume and total external surface area, including the base but not the hidden circle between the cylinder and cone.

SOLUTION

1. Cylinder volume: $\pi(2)^2(5) = 20\pi$. Cone volume: $\frac{1}{3}\pi(4)(1.5) = 2\pi$. Total $22\pi \approx 69.1\ \text{m}^3$.
2. Cylinder curved area: $2\pi(2)(5) = 20\pi$.
3. Cone slant height: $\sqrt{2^2 + 1.5^2} = 2.5$ m, so curved area $= \pi(2)(2.5) = 5\pi$.
4. Base: 4π. Total $= 29\pi \approx 91.1\ \text{m}^2$.

Check: the circle where the cone meets the cylinder is inside the solid and is not painted, so it is excluded. Adding 4π for it would overcount the surface.

COMMON MISCONCEPTION

"Use the height in πrl." The curved surface uses the slant height. For a cone with $r = 4$ and $h = 3$, the slant height is 5, so the curved area is 20π, not 12π.

"The volume of a pyramid is $\frac{1}{2}Ah$." The one-third factor comes from comparing the pyramid with the surrounding prism; a triangle's area formula does not transfer to a solid.

"A cone's surface area is πrl." That is only the curved part. A closed cone also has a circular base of area πr^2.

"The slant height is always the height." They are equal only in the limiting case of a vertical side. In general l is longer than h, and confusing them overstates the volume but understates the surface.

EXERCISE 11B

PYRAMIDS AND CONES

FLUENCY

1 A square pyramid has base side 8 cm and perpendicular height 3 cm. Find its slant height, surface area and volume.

2 A square pyramid has base side 10 cm and perpendicular height 12 cm. Find its slant height, surface area and volume.

3 A cone has radius 3 cm and height 4 cm. Find its slant height, surface area and volume.

4 A cone has radius 6 cm and height 8 cm. Find its slant height, surface area and volume.

5 A cone has diameter 14 cm and height 24 cm. Find its surface area and volume.

6 A square pyramid has base side 6 cm and slant height 5 cm. Find its height and volume.

7 A cone has slant height 13 cm and radius 5 cm. Find its height and volume.

8 A frustum has base radii 4 cm and 6 cm and height 9 cm. Find its volume.

UNDERSTANDING AND REASONING

9 A square-based tent has base side 4.0 m and height 2.5 m. Find the area of canvas needed for the four sloping sides, allowing 10% extra for seams.

10 A conical pile of sand has height 3.0 m and base diameter 8.0 m. Find its volume.

11 An ice-cream cone has radius 3 cm and height 12 cm. Find its capacity and the area of the curved wafer surface.

12 **Error analysis.** A student calculates the curved area of a cone with $r = 4$ cm and $h = 3$ cm as $\pi(4)(3) = 12\pi$. Explain the error and give the correct value.

13 A square pyramid has base side 9 cm and volume 324 cm^3. Find its height and slant height.

14 A conical container holds 1.0 litre and has radius 5.0 cm. Find its height and total surface area, to 3 significant figures.

15 **Extension.** A cone is cut by a plane parallel to its base, halfway up its height. Show that the small cone has $\frac{1}{8}$ of the volume and that the frustum has $\frac{7}{8}$.

16 **Extension.** A cone and a cylinder have the same radius 6 cm and the same height 8 cm. Find the ratio of their volumes and the ratio of their total surface areas.

17 **Extension.** A conical roof has diameter 6.0 m and height 2.0 m. Find the area of its curved surface, allowing 12% extra for overlap.

CURVED IN EVERY DIRECTION

11.3 Spheres and composite solids

A **sphere** is the set of points at a fixed distance r from a centre. Its surface has no edges and no flat faces, so it is described by two formulas rather than a net. A **hemisphere** is half a sphere cut through the centre; its surface consists of a curved half-sphere plus a flat circular base.

A **composite solid** is made by joining simpler solids. Many manufactured objects are composites: a capsule is a cylinder with two hemispherical ends, a bowl is a hemisphere, a silo may be a cylinder with a hemispherical cap. To find the surface area of a composite, add only the faces that remain exposed; a face where two parts are joined is hidden and must not be counted.

KEY RELATIONSHIPS

Spheres and hemispheres

For a sphere of radius r:

$$S = 4\pi r^2 \qquad V = \tfrac{4}{3}\pi r^3$$

For a hemisphere of radius r:

$$\textbf{curved surface} = 2\pi r^2 \qquad \textbf{total surface} = 3\pi r^2 \qquad V = \tfrac{2}{3}\pi r^3$$

The total surface of a hemisphere includes the flat circular base. For an open bowl, use only the curved surface.

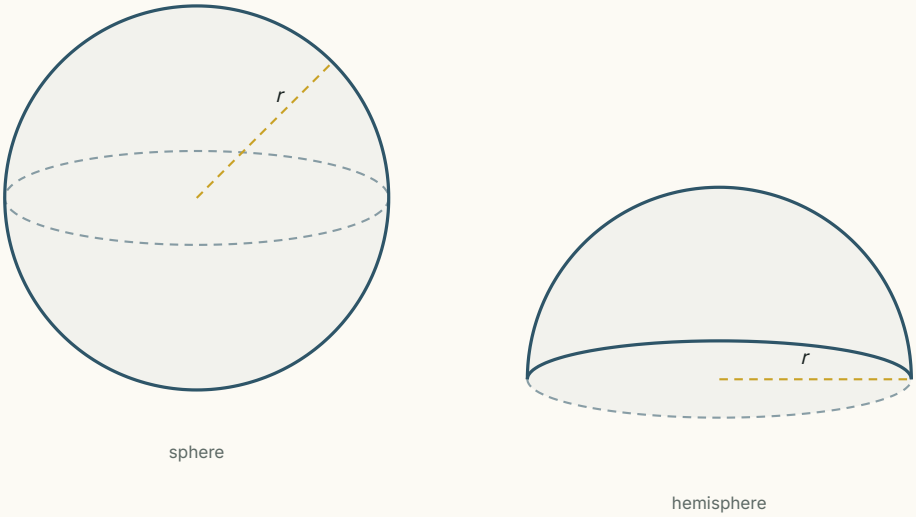


Figure 11.4 A sphere is determined by its radius. A hemisphere has the same curved surface as half a sphere, $2\pi r^2$, plus the flat circular base of area πr^2.

EXAMPLE **11.11** A SPHERE

QUESTION

A sphere has radius 6 cm. Find its surface area and volume, exactly and to 3 significant figures.

SOLUTION

1 $S = 4\pi(6)^2 = 144\pi \text{ cm}^2 \approx 452 \text{ cm}^2$.

2 $V = \tfrac{4}{3}\pi(6)^3 = \tfrac{4}{3}\pi(216) = 288\pi \text{ cm}^3 \approx 905 \text{ cm}^3$.

Check: the sphere fits inside a cube of edge 12 cm with volume 1728 cm³. The sphere occupies $288\pi \div 1728 = \tfrac{\pi}{6} \approx 52\%$ of the cube, which is the expected fraction.

EXAMPLE **11.12** FINDING THE RADIUS FROM THE VOLUME

QUESTION

A spherical tank has volume $1.00\ \text{m}^3$. Find its radius and surface area, to 3 significant figures.

SOLUTION

1 $1 = \frac{4}{3}\pi r^3$, so $r^3 = \frac{3}{4\pi} \approx 0.2387$.

2 $r = \sqrt[3]{0.2387} \approx 0.620\ \text{m}$.

3 $S = 4\pi(0.620)^2 \approx 4.84\ \text{m}^2$.

Check: the tank is about 1.24 m across and holds 1000 litres. Its surface area is less than 5 m², which is why spherical tanks are efficient for storage.

EXAMPLE **11.13** A CAPSULE

QUESTION

A capsule is made from a cylinder of radius 1.5 cm and height 4.0 cm with a hemisphere at each end. Find its volume and total surface area, to 3 significant figures.

SOLUTION

1 Cylinder volume: $\pi(1.5)^2(4) = 9\pi\ \text{cm}^3$.

2 Two hemispheres make one sphere: $\frac{4}{3}\pi(1.5)^3 = 4.5\pi\ \text{cm}^3$.

3 Total volume $13.5\pi \approx 42.4\ \text{cm}^3$.

4 Cylinder curved area: $2\pi(1.5)(4) = 12\pi$. Ends: $4\pi(1.5)^2 = 9\pi$.

5 Total surface area $21\pi \approx 66.0\ \text{cm}^2$.

Check: there are no flat circular faces on the outside: the circles where the cylinder meets each hemisphere are internal and are not part of the surface.

EXAMPLE **11.14** A HEMISPHERICAL BOWL

QUESTION

A hemispherical bowl has internal radius 12 cm and wall thickness 0.5 cm. Find its capacity in litres, and the total area that must be glazed on the inside and outside.

SOLUTION

1 Capacity: $V = \frac{2}{3}\pi(12)^3 = 1152\pi \approx 3619\ \text{cm}^3 = 3.62$ litres.

2 Inner curved area: $2\pi(12)^2 = 288\pi$.

3 Outer radius $= 12.5$ cm, outer curved area $= 2\pi(12.5)^2 = 312.5\pi$.

4 Total glazed area $= 600.5\pi \approx 1887\ \text{cm}^2$.

Check: the outer area is only slightly larger because the wall is thin: $(12.5/12)^2 \approx 1.085$. The rim, a ring of area $\pi(12.5^2 - 12^2) \approx 12.3\ \text{cm}^2$, is not included in this total.

EXAMPLE **11.15** A DOME

QUESTION

A hemispherical dome has radius 4.5 m. It is covered with panels, allowing 8% extra for waste, and painted with two coats at $5.0\ \text{m}^2$ per litre. Paint costs £12 per litre, sold in whole litres. Find the number of litres and the cost.

SOLUTION

1 Curved surface: $2\pi(4.5)^2 = 40.5\pi \approx 127.2\ \text{m}^2$.

2 With waste: $127.2 \times 1.08 \approx 137.4\ \text{m}^2$.

3 Two coats cover $274.8\ \text{m}^2$, so paint needed $= 274.8 \div 5.0 \approx 55$ litres.

4 Cost: $55 \times £12 = £660$.

Check: the flat base of the dome is not painted, so the curved area is used. Including the base would add $20.25\pi \approx 63.6\ \text{m}^2$, which would overstate the paint by about half.

COMMON MISCONCEPTION

"A sphere's surface area is $2\pi r^2$." That is the curved surface of a hemisphere. A full sphere has twice as much: $4\pi r^2$.

"The volume of a hemisphere is half of $\frac{4}{3}\pi r^3$." It is half, which is $\frac{2}{3}\pi r^3$ — but the surface area is not half of $4\pi r^2$ once the flat base is included.

"A composite solid's surface is the sum of the surfaces of its parts." Faces that are glued together are hidden. Add only exposed faces.

"The radius is the diameter." A sphere drawn across its widest point gives the diameter; halve it before using any formula.

EXERCISE 11C

SPHERES AND COMPOSITE SOLIDS

FLUENCY

1 A sphere has radius 3 cm. Find its surface area and volume, in terms of π.

2 A sphere has diameter 10 cm. Find its surface area and volume, to 3 significant figures.

3 A sphere has radius 2.5 cm. Find its surface area and volume, to 3 significant figures.

4 A hemisphere has radius 6 cm. Find its total surface area and volume.

5 A hemisphere has radius 4 cm. Find its total surface area and volume.

6 A sphere has volume $36\pi\ \text{cm}^3$. Find its radius and surface area.

7 A sphere has surface area $100\pi\ \text{cm}^2$. Find its radius and volume.

8 A solid is a cylinder of radius 3 cm and height 10 cm with a hemisphere on top. Find its volume and total surface area.

UNDERSTANDING AND REASONING

9 A ball has diameter 22 cm. Find its surface area and volume, to 3 significant figures.

10 A hemispherical dome has diameter 8.0 m. Find its curved surface area and the paint needed for two coats at $4.0\ \text{m}^2$ per litre.

11 A sphere has volume 1.0 litre. Find its radius and surface area, to 3 significant figures.

EXERCISE 11C · CONTINUED SPHERES AND COMPOSITE SOLIDS

12 **Error analysis.** A student writes $S = 2\pi r^2$ for a sphere of radius 10 cm. Explain the error and give the correct value.

13 The Earth is approximately a sphere of radius 6371 km. Find its surface area, and the area covered by water if 71% of the surface is ocean.

14 A hemispherical bowl has radius 9 cm. Find the total area to be painted on both the inside and the outside.

15 A solid is a hemisphere of radius 6 cm with a cone of height 8 cm on top. Find its volume and total surface area.

16 **Extension.** A sphere of radius r fits exactly inside a cube of edge $2r$. Show that both the surface-area ratio and the volume ratio are $\frac{\pi}{6}$, and interpret the result.

17 **Extension.** Two spheres have radii r and $2r$. Find the ratios of their surface areas and volumes. If the smaller sphere has volume $10\ \text{cm}^3$, find the volume of the larger.

WHEN EVERY LENGTH CHANGES

11.4 Scaling surface area and volume

If every length of a solid is multiplied by the same factor k, the solid is enlarged but its shape is unchanged. Each area is a product of two lengths, so it is multiplied by k^2. Each volume is a product of three lengths, so it is multiplied by k^3. The number k is the **linear scale factor**; k^2 is the area scale factor and k^3 is the volume scale factor.

This is the same relationship you met for similar plane figures in Chapter 9, now extended into three dimensions. It explains why a model is much lighter than the object it represents, and why a larger tank holds disproportionately more than the material needed to build it.

KEY RELATIONSHIP

Scaling similar solids

If two solids are similar with linear scale factor k, then

$$\text{surface area ratio} = k^2 \qquad \text{volume ratio} = k^3$$

Conversely, if the volumes are in the ratio $V_2 : V_1$, then $k = \sqrt[3]{V_2/V_1}$. Once k is known, the surface-area ratio is k^2 and any missing length can be found by multiplying by k.

EXAMPLE 11.16 A CUBE ENLARGED

QUESTION

A cube has edge 4 cm. It is enlarged so that its edge becomes 6 cm. Find the factors by which its surface area and volume increase, and verify with the actual values.

SOLUTION

1. Linear scale factor: $k = 6 \div 4 = 1.5$.
2. Surface area factor: $k^2 = 2.25$. Volume factor: $k^3 = 3.375$.
3. Original: $S = 6(4)^2 = 96\ \text{cm}^2$, $V = 4^3 = 64\ \text{cm}^3$.
4. New: $S = 6(6)^2 = 216\ \text{cm}^2$, $V = 6^3 = 216\ \text{cm}^3$. Indeed $96 \times 2.25 = 216$ and $64 \times 3.375 = 216$.

Check: the surface area grew by 125%, but the volume grew by 237.5%. A larger solid has proportionally less surface for its volume.

EXAMPLE 11.17 TWO SIMILAR BOTTLES

QUESTION

Two bottles are similar. Their heights are 20 cm and 25 cm. The smaller holds 500 mL and needs a label of area $120\ \text{cm}^2$. Find the capacity and label area of the larger bottle.

SOLUTION

1. $k = 25 \div 20 = 1.25$.
2. Capacity factor: $k^3 = 1.953125$, so capacity $= 500 \times 1.953125 \approx 977\ \text{mL}$.
3. Label factor: $k^2 = 1.5625$, so area $= 120 \times 1.5625 = 187.5\ \text{cm}^2$.

Check: a 25% increase in height almost doubles the capacity but increases the label area by only about 56%. This is why larger bottles usually cost less per litre to package.

EXAMPLE 11.18 A SCALE MODEL

QUESTION

A model car is built at a scale of $1 : 24$. The real car is 4.32 m long and has a bonnet area of $0.90\ \text{m}^2$. Its engine capacity is 2.4 litres. Find the model's length, bonnet area and engine capacity.

SOLUTION

1. Model length $= 432\ \text{cm} \div 24 = 18\ \text{cm}$.
2. Area factor $= \left(\frac{1}{24}\right)^2 = \frac{1}{576}$, so bonnet area $= 0.90\ \text{m}^2 \div 576 \approx 0.00156\ \text{m}^2 = 15.6\ \text{cm}^2$.
3. Volume factor $= \left(\frac{1}{24}\right)^3 = \frac{1}{13\,824}$, so engine capacity $= 2.4\ \text{L} \div 13\,824 \approx 0.000174\ \text{L} = 0.174\ \text{cm}^3$.

Check: the model engine is smaller than a thimble, which is why scale models cannot reproduce the physics of a real engine. The ratios, not the absolute quantities, are what the model preserves.

EXAMPLE **11.19** WORKING BACKWARDS FROM VOLUMES

QUESTION

Two similar solids have volumes $250\ \text{cm}^3$ and $2000\ \text{cm}^3$. The smaller has surface area $150\ \text{cm}^2$. Find the linear scale factor and the surface area of the larger solid.

SOLUTION

1 Volume ratio $= 2000 \div 250 = 8$, so $k^3 = 8$ and $k = 2$.

2 Surface-area ratio $= k^2 = 4$.

3 Larger surface area $= 150 \times 4 = 600\ \text{cm}^2$.

Check: doubling every length multiplies volume by 8 but surface area by only 4. The larger solid is therefore relatively less exposed per unit of volume.

EXAMPLE **11.20** A 10% INCREASE

QUESTION

Every edge of a cube is increased by 10%. Find the percentage increase in its surface area and in its volume.

SOLUTION

1 $k = 1.10$.

2 Surface area factor $= 1.10^2 = 1.21$, an increase of 21%.

3 Volume factor $= 1.10^3 = 1.331$, an increase of 33.1%.

Check: a modest change in length produces a much larger change in volume. In design, a 10% taller container can hold a third more, which is rarely obvious from the drawing.

COMMON MISCONCEPTION

"If the lengths double, the volume doubles." Every dimension contributes to volume, so doubling all three multiplies volume by $2^3 = 8$, not by 2.

"Surface area and volume scale by the same factor." They do not: area is two-dimensional and volume three-dimensional, so the factors are k^2 and k^3.

"A scale of $1 : 24$ means the model is 24 times smaller in every measurement." It means every length is $\frac{1}{24}$ of the real length; areas are $\frac{1}{576}$ and volumes $\frac{1}{13\,824}$.

"Use the surface-area ratio to find a volume ratio." Reverse the operation: take the square root of an area ratio to find k, then cube it for volume.

EXERCISE 11D SCALING SOLIDS

FLUENCY

1 A cube has edge 3 cm. Its edge is doubled. Find the factors for surface area and volume.

2 A cube has edge 5 cm. Its edge is tripled. Find the factors for surface area and volume.

EXERCISE 11D · CONTINUED

SCALING SOLIDS

3 Two similar prisms have heights 8 cm and 12 cm. Find the ratios of their surface areas and volumes.

4 Two similar cylinders have radii 4 cm and 10 cm. Find the ratios of their surface areas and volumes.

5 Two similar solids have volumes 27 cm^3 and 64 cm^3. Find the ratio of their surface areas.

6 Two similar solids have surface areas 25 cm^2 and 100 cm^2. Find the ratio of their volumes.

7 A model is built at a scale of 1 : 50. Find the area and volume factors.

8 A model swimming pool is built at 1 : 20. The model holds 250 cm^3. Find the real capacity in cubic metres.

UNDERSTANDING AND REASONING

9 A cube has volume 64 cm^3. Find its surface area. The edge is then doubled; find the new surface area.

10 A sphere has radius 3 cm. Its radius is doubled. Find the factors for surface area and volume.

11 **Error analysis.** A student says that doubling every dimension of a cuboid doubles its volume. Explain the error and state the correct factor.

12 Two similar solids have volumes 500 cm^3 and 4000 cm^3. The smaller has surface area 300 cm^2. Find the larger surface area.

13 A cylinder has radius 3 cm and height 5 cm. It is enlarged by a factor of 2. Find its new surface area and volume.

14 The surface area of a cube increases by 44%. Find the percentage increase in its volume.

15 Two similar cones have volumes in the ratio 8 : 27. Find the ratios of their heights and surface areas.

16 A raindrop is approximately spherical. Compare a drop of radius 1 mm with one of radius 2 mm: find their surface areas, volumes and surface-area-to-volume ratios, and explain which evaporates faster.

17 **Extension.** Two similar solids have surface areas 36 cm^2 and 81 cm^2. The smaller has volume 24 cm^3. Find the larger volume.

18 **Extension.** A cube has volume 1 litre. Its volume is doubled. Find the new edge length and surface area, to 3 significant figures, and explain why the surface area does not double.

GEOMETRY MEETS A BUDGET

11.5 Packaging, cost and design

For a fixed volume, the amount of material needed depends on the shape. A sphere has the smallest surface area for a given volume, but it rolls, wastes space when stacked and is difficult to fill. A cube is easy to stack but uses more material than a cylinder of the same volume. Real packaging is a compromise between mathematical efficiency and practical constraints.

Two ratios make the comparison precise. The **surface-area-to-volume ratio** S/V measures how much material is needed per unit of contents; a smaller ratio is more efficient. The **cost per unit volume** divides the total cost of material by the capacity. Both ratios must be used with the same units, and neither captures strength, printing area or transport.

EXAMPLE 11.21 TWO CARTONS OF ONE LITRE

QUESTION

Carton A is a cube of edge 10 cm. Carton B measures 20 cm by 10 cm by 5 cm. Both hold 1 litre. Compare their surface areas and their material cost at £0.012 per $100\ \text{cm}^2$.

SOLUTION

1. Carton A: $S = 6(10)^2 = 600\ \text{cm}^2$.
2. Carton B: $S = 2(20 \times 10 + 20 \times 5 + 10 \times 5) = 2(200 + 100 + 50) = 700\ \text{cm}^2$.
3. B uses $700 - 600 = 100\ \text{cm}^2$ more, an increase of $100 \div 600 \approx 16.7\%$.
4. Cost: A $= 6 \times £0.012 = £0.072$; B $= 7 \times £0.012 = £0.084$.

Check: both hold the same volume, so the difference is entirely due to shape. The cube is more efficient, but the flat carton may stack better or fit a shelf — a practical advantage that the surface area does not measure.

EXAMPLE 11.22 THE OPTIMAL CYLINDER FOR ONE LITRE

QUESTION

A cylindrical can must hold exactly 1 litre. Among all such cylinders, the one with the least surface area has height equal to twice its radius. Show that its radius is about 5.42 cm. Find its surface area and compare it with the cube of Example 11.21.

SOLUTION

1. With $h = 2r$, $V = \pi r^2(2r) = 2\pi r^3 = 1000$.
2. $r^3 = \frac{1000}{2\pi} \approx 159.15$, so $r \approx 5.419$ cm and $h \approx 10.84$ cm.
3. $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 4\pi r^2 = 6\pi r^2 \approx 553.6\ \text{cm}^2$.
4. The cube needs $600\ \text{cm}^2$, so the cylinder uses about 7.7% less material.

Check: $6\pi(5.419)^2 = 553.6$, and $\pi(5.419)^2(10.84) = 1000$. The optimal can is as tall as it is wide, because $h = 2r$. Real cans are usually taller and narrower because manufacturing, stacking and drinking comfort matter too.

THREE CONTAINERS WITH CAPACITY 500 ML

CONTAINER	DIMENSIONS	SURFACE AREA (cm²)	MATERIAL COST AT £0.009 PER 100 CM²
Cube	7.94 × 7.94 × 7.94 cm	378	£0.034
Cylinder, $h = 2r$	r = 4.30 cm, h = 8.60 cm	349	£0.031
Flat box	15 × 10 × 3.33 cm	467	£0.042

EXAMPLE 11.23 LABEL COST PER LITRE

QUESTION

A can has radius 3.3 cm and height 12 cm. Its label covers 80% of the curved surface and costs £0.00035 per square centimetre. Find the label cost and the label cost per litre of capacity.

SOLUTION

1. Curved area: $2\pi(3.3)(12) = 79.2\pi \approx 248.8 \text{ cm}^2$.
2. Label area: $0.80 \times 248.8 \approx 199.0 \text{ cm}^2$.
3. Cost: $199.0 \times £0.00035 \approx £0.070$.
4. Capacity: $\pi(3.3)^2(12) \approx 410.5 \text{ cm}^3 = 0.4105 \text{ L}$, so cost per litre $= £0.070 \div 0.4105 \approx £0.17$.

Check: the label is a small part of the total package cost, but at production volumes of millions of cans even a fraction of a penny matters. Designers reduce label area or use thinner paper to save material.

EXAMPLE 11.24 THE SAME MATERIAL, DIFFERENT SHAPES

QUESTION

A sphere, a cube and a cylinder with height equal to twice its radius all have surface area 600 cm^2. Find their volumes and identify the shape that holds the most.

SOLUTION

1. Cube: $6s^2 = 600$, so $s = 10 \text{ cm}$ and $V = 1000 \text{ cm}^3$.
2. Sphere: $4\pi r^2 = 600$, so $r^2 = 150/\pi \approx 47.75$, $r \approx 6.91 \text{ cm}$, and $V = \frac{4}{3}\pi r^3 \approx 1382 \text{ cm}^3$.
3. Cylinder with $h = 2r$: $6\pi r^2 = 600$, so $r^2 = 100/\pi \approx 31.83$, $r \approx 5.64 \text{ cm}$, and $V = 2\pi r^3 \approx 1128 \text{ cm}^3$.

Check: the sphere holds about 38% more than the cube for the same material, and the cylinder about 13% more. This is the mathematical case for spherical tanks, balanced against the practical difficulties of building and transporting them.

COMMON MISCONCEPTION

"The cheapest package is the one with the smallest surface area." Material cost is only one part; printing, assembly, transport, shelf space and waste also matter.

"A larger package always costs less per litre." It often does, because the surface-to-volume ratio falls, but only if the material and production costs stay the same. A shaped or double-walled container can reverse the advantage.

"Surface area and volume can be compared directly." They have different units. Compare the ratio S/V, and keep the units consistent.

"The optimal shape is always a sphere." A sphere is optimal for a free-standing solid, but not for a can that must stand, stack, print and ship.

EXERCISE 11E

PACKAGING AND COST

FLUENCY

1 A cuboid measures 12 cm by 8 cm by 6 cm. Find its surface area and volume. A cube with the same volume has what surface area? Which is more efficient?

2 Two cylinders each hold 1 litre. One has $h = 2r$; the other has $h = 4r$. Find both surface areas and compare.

3 A carton measures 5 cm by 5 cm by 10 cm. Find its surface area. A cube with the same volume has what surface area?

4 Using the table above, find the material cost per litre for each of the three 500 mL containers.

5 A room measures 6.0 m by 4.0 m by 2.5 m. Its four walls and ceiling are painted. Find the area, the paint needed for two coats at 7.0 m^2 per litre, and the cost at £9.50 per litre.

6 A label covers 85% of the curved surface of a can of radius 3.5 cm and height 11 cm. Find the label area.

UNDERSTANDING AND REASONING

7 **Error analysis.** A shopper compares a 1.5 L bottle at £1.20 with a 2 L bottle at £1.50 and says the 1.5 L bottle is cheaper because its price is lower. Explain the error and calculate the correct comparison.

8 A drinks can holds 330 mL and has radius 3.3 cm. Find its height, surface area and material area per litre.

9 An open box measures 20 cm by 20 cm by 15 cm. Find its surface area and volume. How much material does a lid add?

10 Explain why a sphere is the most efficient shape for a fixed surface area, and why it is rarely used for a drinks container.

11 **Extension.** A cylinder holds 1 litre. Compare the surface areas when $h = 2r$ and when $h = 3r$. Which is smaller?

12 **Extension.** A cube of edge 10 cm is cut into eight cubes of edge 5 cm. Find the total surface area before and after, and explain the change.

INVESTIGATION · CRITERION B

11.6 The most efficient package

A square sheet of card 20 cm by 20 cm is turned into an open box by cutting a square of side x from each corner and folding up the sides. The base then has side $20 - 2x$, and the height is x. The volume is $V = x(20 - 2x)^2$. Changing x changes both the base and the height, so the volume does not grow steadily.

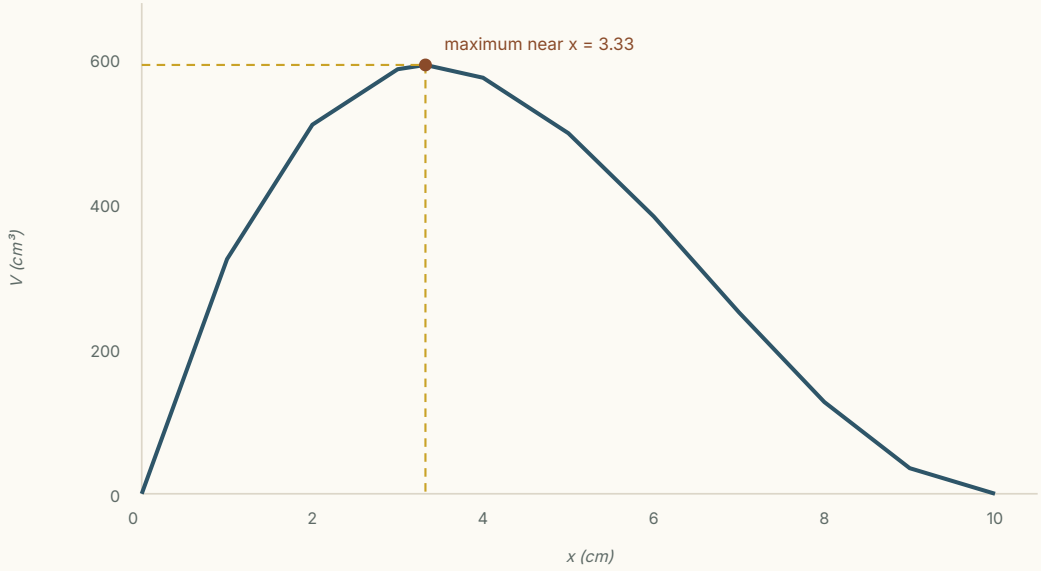


Figure 11.5 The volume rises as the corner squares grow, reaches a maximum when x is about one sixth of the sheet, and then falls. Beyond $x = 10$ cm the base would have zero or negative side, so the model stops there.

INVESTIGATION · CRITERION B

How much can one sheet hold?

1. Explain why the base side is $20 - 2x$ and why $0 < x < 10$.
2. Copy and complete the table for $x = 1, 2, 3, 3.5, 4, 5, 6$. Calculate V to the nearest whole number.
3. Plot the points on the axes of Figure 11.5 and join them with a smooth curve. Estimate the value of x that gives the greatest volume.
4. Show that the maximum volume is about $593\ \text{cm}^3$ and occurs near $x = 3.33$ cm. Compare the base side and the height at the maximum.
5. Repeat the reasoning for a square sheet of side S. Show that the maximum occurs at $x = S/6$, and that the maximum volume is $\frac{2S^3}{27}$. Test your formula with $S = 20$.
6. Compare this open box with a closed cylinder of the same volume. Find the cylinder's surface area and explain why the comparison is not completely fair.

Generalise and justify. The table suggests a single maximum rather than several. Explain why the volume must be zero at both ends of the interval and positive in between, and why a smooth curve with one peak is consistent with the data. Then state two practical reasons why a manufacturer might not choose the mathematical maximum.

OPEN BOX FROM A 20 CM SQUARE SHEET

x (cm)	BASE SIDE (cm)	VOLUME (cm³)
1	18	324
2	16	512
3	14	588
3.5	13	591.5
4	12	576
5	10	500
6	8	384

MIXED PRACTICE 11

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–6 check fluency; 7–14 require reasoning; 15–18 are multi-step or unfamiliar. Allow 55 minutes.

FLUENCY

1 A cuboid measures 6 cm by 4 cm by 3 cm. Find its surface area and volume.

2 A cylinder has radius 5 cm and height 10 cm. Find its volume and total surface area.

3 A cone has radius 4 cm and height 3 cm. Find its slant height, surface area and volume.

4 A sphere has radius 5 cm. Find its surface area and volume, exactly and to 3 significant figures.

5 A square pyramid has base side 6 cm and height 4 cm. Find its slant height, surface area and volume.

6 A hemisphere has radius 3 cm. Find its total surface area and volume.

UNDERSTANDING AND REASONING

7 A room measures 5 m by 4 m by 3 m. Its walls and ceiling are painted, but not the floor. Find the painted area and the paint needed for two coats at 8 m^2 per litre.

8 A cylindrical tank has diameter 2.0 m and height 3.0 m. Find its capacity in litres and its total surface area including the top.

9 A cone has radius 5 cm and height 12 cm. Find its total surface area and volume, in terms of π.

10 A frustum has base radii 6 cm and 9 cm and height 8 cm. Find its volume.

11 A solid is a cylinder of radius 4 cm and height 6 cm with a hemisphere on top. Find its volume and total surface area.

12 **Error analysis.** A student calculates the curved area of a cone with $r = 5$ cm and $h = 12$ cm as 60π. Explain the error and give the correct value.

13 A model building is made at $1 : 20$. Find the area and volume factors. The real building has volume 3.2 m^3; find the model volume in cm^3.

14 Two similar solids have surface areas 49 cm^2 and 196 cm^2. Find the ratio of their volumes.

MULTI-STEP AND UNFAMILIAR

15 A cube has edge 10 cm. Its edge is increased to 12 cm. Find the percentage increase in surface area and in volume.

16 A cylinder holds 1 litre and has height equal to twice its radius. Find its radius and surface area, to 3 significant figures.

MIXED PRACTICE 11 · CONTINUED

CUMULATIVE RETRIEVAL · NO NOTES

17 **Extension.** A sphere of radius 6 cm fits exactly inside a cylinder of radius 6 cm and height 12 cm. Find the volume of the empty space and the surface area of each solid.

18 **Extension.** A cone is cut by a plane parallel to its base so that the small cone removed has half the volume of the original. Find the fraction of the height at which the cut is made, to 3 significant figures.

Chapter assessment

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and assessed against the criteria shown. Your teacher will confirm which tasks are required and how long you have.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Solids and their measures: short response	30	35 min
B · Investigating patterns	The shape of a one-litre can	24	40 min
C · Communicating	A design brief for a water tank	20	35 min
D · Applying mathematics in real-life contexts	Harvesting rainwater	24	45 min
Total	All four tasks	98	155 min

The tasks begin overleaf. Read the instructions and evidence requirements before you start, and show the reasoning that supports each answer.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 30 marks

Solids and their measures: short response

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

1 A cuboid measures 9 cm by 6 cm by 4 cm. Find its surface area and volume. [3]

2 A cylinder has radius 3 cm and height 8 cm. Find its total surface area and volume, in terms of π. [3]

3 A cone has radius 6 cm and height 8 cm. Find its slant height, surface area and volume. [4]

4 A sphere has radius 7 cm. Find its surface area and volume, to 3 significant figures. [3]

5 A square pyramid has base side 8 cm and height 3 cm. Find its slant height, surface area and volume. [4]

6 A hemisphere has radius 5 cm. Find its total surface area and volume. [3]

7 A frustum has base radii 6 cm and 10 cm and height 12 cm. Find its volume. [3]

8 A capsule is a cylinder of radius 2 cm and height 6 cm with a hemisphere at each end. Find its volume and surface area. [4]

9 A cube has edge 5 cm. Its edge is doubled. State the factors by which its surface area and volume increase. [2]

10 A model is built at $1:25$. Find the area and volume factors. The real object has volume $1.25\ \text{m}^3$; find the model volume in cm^3. [3]

11 **Error analysis.** A student writes $S = 2\pi r^2$ for the surface area of a sphere of radius 6 cm. Explain the error and give the correct value. [3]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 24 marks

The shape of a one-litre can

Instructions. Work independently. A cylindrical can must hold 1 litre. Its height h is chosen as a multiple of its radius: $h = kr$. Different values of k give different surface areas. The table shows k and the corresponding radius and total surface area.

k	RADIUS (cm)	SURFACE AREA (cm²)
1	6.83	586
2	5.42	554
3	4.73	563
4	4.30	581
5	3.99	601

1. Verify the radius and surface area for $k = 2$. [4]
2. Describe how the surface area changes as k increases from 1 to 5. [3]
3. Estimate the value of k that gives the smallest surface area, and explain your reasoning. [3]
4. Show that, for a fixed volume V, the surface area is $S = 2\pi r^2 + \frac{2V}{r}$, and that with $V = 1000$ this becomes $S = 2\pi r^2 + \frac{2000}{r}$. [5]
5. Use the table to predict the surface area for $k = 6$. Calculate the radius and check your prediction. [4]
6. Explain why the optimum is close to $h = 2r$, and why a real can might not use exactly this ratio. [5]

Evidence: a verified row, a clear description of the pattern, a justified estimate, an algebraic derivation, a checked prediction, and an evaluation of the model's practical limits.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

A design brief for a water tank

Instructions. Write a 300–400 word design brief for a school that needs to store 10 m^3 of water. Compare a cylindrical tank with a cuboid tank of the same capacity. Your brief must: (i) state the dimensions you propose; (ii) calculate the surface area of each option; (iii) compare material use and footprint; (iv) explain at least two assumptions; (v) recommend one option with a justified reason. Use correct units and include a labelled sketch.

Evidence: a labelled sketch, clear calculations with units, a comparison that uses surface area rather than volume alone, stated assumptions, and a recommendation that follows from the evidence.

CRITERION D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Harvesting rainwater

Instructions. A school roof is a rectangle 12 m by 8 m. The average annual rainfall is 620 mm, and 85% of the rain that falls on the roof can be collected. A cylindrical tank has diameter 3.0 m and height 2.5 m. Water costs £1.80 per cubic metre, and the installed tank costs £2400.

1. Find the roof area and the volume of water collected in an average year, in cubic metres. [5]
2. Find the capacity of the tank and the number of times it could be filled by the annual harvest. [5]
3. Find the annual saving if all collected water replaces bought water, and the payback time for the tank. [5]
4. Evaluate three assumptions in your calculation, including one about rainfall and one about how the water is used. [5]
5. Write a recommendation to the school. Your recommendation must distinguish financial payback from other benefits. [4]

Evidence: a clear conversion of rainfall to volume, a capacity comparison, a payback calculation with units, an evaluation that distinguishes measured data from assumptions, and a recommendation that weighs more than one criterion.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few formulas; applies them with major errors; needs support to identify a solid.	Recalls some formulas; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies formulas accurately in familiar situations; explains some steps.	Selects and applies formulas accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Records isolated results; describes a pattern only with help.	Records results systematically; describes a pattern and makes a simple prediction.	Describes and verifies a pattern; generalises with some justification; tests the rule.	Describes, verifies and justifies a general rule; explains why it must continue; tests boundary cases.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A prism or cylinder has a constant cross-section: $V = AL$ and $S = 2A + PL$.
- For a cylinder, $V = \pi r^2 h$ and $S = 2\pi r^2 + 2\pi rh$; decide whether the ends are included.
- For a pyramid or cone, $V = \frac{1}{3}Ah$; the surface uses the slant height, not the perpendicular height.
- For a cone, $l^2 = r^2 + h^2$, $S = \pi r^2 + \pi rl$ and $V = \frac{1}{3}\pi r^2 h$.
- For a sphere, $S = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$; a hemisphere adds a flat base to its curved surface.
- In a composite solid, add only the faces that remain exposed.
- Similar solids with linear scale factor k have area ratio k^2 and volume ratio k^3.
- For a fixed volume, the surface-area-to-volume ratio measures material efficiency; the sphere is the mathematical optimum, but practical constraints decide real designs.

$$V = AL \qquad S = 2A + PL$$

$$V = \pi r^2 h \qquad S = 2\pi r^2 + 2\pi rh$$

$$V = \tfrac{1}{3}Ah \qquad S = A + \tfrac{1}{2}Pl$$

$$V = \tfrac{1}{3}\pi r^2 h \qquad S = \pi r^2 + \pi rl \qquad l^2 = r^2 + h^2$$

$$S = 4\pi r^2 \qquad V = \tfrac{4}{3}\pi r^3$$

$$\text{similar solids: } S_2 : S_1 = k^2 \qquad V_2 : V_1 = k^3$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can identify a prism, pyramid, cylinder, cone and sphere from its properties.
- ☐ I can calculate the surface area and volume of a prism and a cylinder.
- ☐ I can decide whether a cylinder is open or closed before calculating its surface area.
- ☐ I can find and use the slant height of a cone or pyramid.
- ☐ I can calculate the surface area and volume of a sphere and a hemisphere.
- ☐ I can find the surface area of a composite solid without double-counting hidden faces.
- ☐ I can apply k^2 and k^3 to similar solids and work backwards from a volume ratio.
- ☐ I can compare packages using surface area, volume and cost per litre.
- ☐ I can convert between cm^3, m^3 and litres correctly.
- ☐ I can model a storage problem, state its assumptions and evaluate the design.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which formula did you find hardest to use correctly, and what did you do to check it?
2. Describe a mistake you made with units — mixing cm and m, or confusing cm^2 with cm^3 — and how you found it.
3. The statement of inquiry claims that form fixes both surface and volume. Use two shapes with the same volume to test this claim.
4. Look ahead to Chapter 12: data can be described by its centre and its spread. How might the idea of a ratio, used here to compare shapes, help you compare two data sets fairly?

WHAT THE AVERAGE HIDES

Statistics: Distributions & Data

12

RELATIONSHIPS · REPRESENTATION · QUANTITY

A mean is a single number, but a class is a spread of individuals. To describe a data set honestly you need its centre, its spread and its shape — and an account of how the data were collected.

STATEMENT OF INQUIRY

The way data are collected, grouped and displayed determines what can be concluded from them. A distribution is described by its centre, its spread and its shape together; a single average conceals the variation on which real decisions depend.

KEY CONCEPT
Relationships

RELATED CONCEPTS
Statistics · Representation · Quantity

GLOBAL CONTEXT
Identities and relationships — how data about groups shape decisions about individuals

FACTUAL
How are data collected, classified, grouped and displayed?

CONCEPTUAL
How do measures of centre and spread combine to describe a distribution?

DEBATABLE
When does a statistical summary inform a decision, and when does it mislead?



IN THIS CHAPTER

12.1 Collecting data
12.2 Histograms
12.3 Cumulative frequency
12.4 Spread and outliers
12.5 Comparing
12.6 Investigation

OPENING PROBLEM · TWO CLASSES, ONE AVERAGE

Two classes, one average

Two Year 4 mathematics classes sit the same 100-mark test. The nine students in Class A score:

55, 60, 65, 70, 70, 70, 75, 80, 85

The nine students in Class B score:

40, 50, 60, 65, 70, 75, 80, 90, 100

1. Calculate the mean, median and mode of each class.
2. Calculate the range of each class.
3. Draw a dot plot for each class on the same scale from 40 to 100.
4. Which class performed better? Justify your answer using the numbers you calculated.
5. A pass mark is 60. What proportion of each class passed?
6. What else would you want to know before writing a report comparing the two classes?

Both classes have a mean of 70 and a median of 70. If the report contained only the average, the two classes would look identical. Yet Class B contains both the strongest and the weakest students, while Class A is tightly clustered around the middle. The average is not wrong; it is simply incomplete. This chapter is about the rest of the story: where the data come from, how they are spread out, and how to display them without distorting what they say.

ORIENTATION

12.0 Learning intentions

- Distinguish populations from samples, and primary from secondary data.
- Classify variables as qualitative or quantitative, and as discrete or continuous.
- Recognise bias and select an appropriate sampling method, including a stratified sample.
- Organise data into frequency tables and display them in histograms, including unequal class widths.
- Use cumulative frequency curves and box plots to find and display quartiles.
- Calculate and interpret range, interquartile range and standard deviation, and identify outliers.
- Compare distributions and evaluate statistical claims and graphs critically.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

design or critique a simple data-collection plan; classify a variable and choose a suitable display; construct a grouped frequency table and a histogram; find quartiles from a cumulative frequency table or curve and draw a box plot; calculate range, interquartile range and standard deviation; identify an outlier using the $1.5 \times \text{IQR}$ rule; compare two distributions using centre, spread and shape; and explain when a graph or a summary is misleading.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit averages in Chapter 3 and percentages in Chapter 5 before starting.

1. Find the mean of 4, 7, 9 and 10.
2. Find the median of 3, 8, 5, 9 and 4.
3. Find the mode of 2, 3, 3, 5, 7, 7, 7.
4. Find the range of 12, 18, 9, 24 and 15.
5. Write $\frac{3}{8}$ as a percentage.
6. Write $-2.5,\ 1.8,\ -0.4,\ 2.1$ in ascending order.

REFRESHER

The symbol $\sum$ means "add up": $\sum x$ is the sum of all the data values. A data set with n values has mean $\bar{x} = \frac{\sum x}{n}$. The median is the middle value when the data are in order; if n is even, average the two middle values. The mode is the most frequent value and may be absent or repeated. Discrete data can be counted (number of siblings); continuous data are measured and can take any value in an interval (height, time). Grouped data lose the individual values, so calculations from a frequency table are estimates.

ATL SKILLS · THINKING, RESEARCH AND COMMUNICATION

Critical thinking: ask who collected the data, from whom, when and why before accepting a summary. **Research:** record the source, date and definition of every secondary data set you use. **Communication:** choose a display that matches the data type, label axes with units, and state the centre and spread together rather than quoting an average alone.

WHERE THE NUMBERS COME FROM

12.1 Collecting and classifying data

Statistics begins before any calculation. The **population** is the whole group about which a question is asked; a **sample** is a part of it that is actually measured. A **census** collects data from every member of the population, while a **survey** usually collects data from a sample. Samples are cheaper and faster, but a conclusion about the population is only as trustworthy as the way the sample was chosen.

Data collected by the person who will use them are **primary data**; data collected by someone else are **secondary data**. Primary data can be designed to answer the exact question, but collecting it takes time. Secondary data may be large and free, but its definitions, coverage and accuracy are outside your control.

Each characteristic that is recorded is a **variable**. A **qualitative** variable records a category; a **quantitative** variable records a number, which may be **discrete** (a count) or **continuous** (a measurement).

KEY TERMS

Populations, samples and bias

Population: the complete set of individuals or items of interest. **Sample:** a subset of the population selected for measurement.

Random sample: every member of the population has a known, non-zero chance of selection. **Bias** is a systematic distortion that makes the sample unrepresentative of the population.

Stratified sample: the population is divided into groups (strata) and a random sample is taken from each, in proportion to the group size.

Reliability concerns whether repeating the data collection would give similar results; **validity** concerns whether the data actually measure what they claim to measure.

TYPES OF VARIABLE

TYPE	DESCRIPTION	EXAMPLES
Qualitative (categorical)	Describes a category or quality; not a number that can be calculated with.	Eye colour; favourite subject; mode of transport to school.
Quantitative discrete	A count; takes separate, usually whole-number values.	Number of siblings; books read in a month; goals scored.
Quantitative continuous	A measurement; can take any value in an interval, limited only by the measuring instrument.	Height; mass; time to run 100 m; temperature.

EXAMPLE **12.1** CLASSIFYING VARIABLES

QUESTION

Classify each variable as qualitative, discrete or continuous: (a) shoe size, (b) favourite colour, (c) time to run 100 m, (d) number of emails received in a day, (e) month of birth, (f) mass of a parcel.

SOLUTION

1. (a) Shoe size is a measurement reported in fixed steps, so it is quantitative discrete in practice.
2. (b) Favourite colour names a category, so it is qualitative.
3. (c) Time is measured and can take any value in an interval, so it is continuous.
4. (d) A count of emails is a whole number, so it is discrete.
5. (e) A month is a category, even though it can be written as a number, so it is qualitative.
6. (f) Mass is measured and can take any value, so it is continuous.

Check: the test is not whether the value is written as a number but what the number means. A month written as 7 is still a label; a shoe size of 7.5 is a measurement.

EXAMPLE 12.2 FINDING THE BIAS

QUESTION

A student wants to know how much homework Year 9 students receive. She surveys 30 students who are in the library at lunchtime and asks: "Don't you agree that homework should be reduced because it takes up too much family time?" Identify three sources of bias and suggest an improvement.

SOLUTION

1. The sample is biased: students in the library at lunchtime may be more studious or have different homework habits from the whole year group.
2. The question is leading: "Don't you agree…" invites agreement and gives a reason that supports one answer.
3. The question asks about opinions, but the stated aim is to measure the amount of homework, so it lacks validity for that aim.
4. Improvement: take a random sample from the full Year 9 list, ask a neutral question such as "How many minutes of homework did you do last night?", and record the answers anonymously.

Check: a larger sample from the same biased group would not fix the problem. Bias is about who is included and how the question is phrased, not only about size.

EXAMPLE 12.3 A STRATIFIED SAMPLE

QUESTION

A school has 720 students: Year 7 has 180, Year 8 has 200, Year 9 has 170 and Year 10 has 170. A stratified sample of 72 students is needed. How many should be chosen from each year group?

SOLUTION

1. The sampling fraction is $72 \div 720 = 0.1$, so one student in ten is chosen from each stratum.
2. Year 7: $180 \times 0.1 = 18$. Year 8: $200 \times 0.1 = 20$.
3. Year 9: $170 \times 0.1 = 17$. Year 10: $170 \times 0.1 = 17$.
4. Check: $18 + 20 + 17 + 17 = 72$.

Check: each year group contributes in proportion to its size, so the sample has the same year-group mix as the school. Within each year group the students should still be chosen at random.

COMMON MISCONCEPTION

"A random sample means choosing anyone who is convenient." Convenience is the opposite of randomness. A random sample gives every member a known chance of selection; a convenience sample gives some members no chance at all.

"A bigger sample is always more reliable." A large but biased sample can be worse than a small random one. Size reduces random variation, not bias.

"Primary data is always better than secondary data." Secondary data may be larger, cheaper and more comparable than anything you could collect, provided its source and definitions are sound.

"Discrete data can only be whole numbers." Discrete means countable in separate steps. Shoe sizes and film ratings can take half-values but are still discrete.

EXERCISE 12A

COLLECTING AND CLASSIFYING DATA

FLUENCY

1 Classify each variable as qualitative, discrete or continuous: (a) number of pets, (b) height, (c) blood group, (d) temperature, (e) number of goals, (f) postcode.

2 State whether each is primary or secondary data: (a) a class tally of favourite sports, (b) rainfall figures from a government website, (c) a survey you conduct in the canteen, (d) a table of results from a textbook.

3 A school wants to know how students travel to school. Identify the population and a possible sample.

4 Explain the difference between a census and a sample, and give one advantage of each.

5 Write one example of (a) discrete data, (b) continuous data, (c) qualitative data, from your own daily life.

6 Explain why "the number of goals in a match" is discrete but "the time until the first goal" is continuous.

UNDERSTANDING AND REASONING

7 A survey asks: "Should the school day start later so that students can get enough sleep?" Explain why this question may produce biased answers, and rewrite it neutrally.

8 A researcher surveys passengers at an airport about the best way to travel to the city. Explain why the sample may be biased.

9 A school has 900 students in Years 7–9: 300 in Year 7, 320 in Year 8 and 280 in Year 9. A stratified sample of 90 is taken. Find the number from each year.

10 A student measures the heights of 20 classmates and finds a mean of 1.62 m. Explain why this mean may not estimate the mean height of all students in the school.

11 **Error analysis.** A student says: "I asked 100 people, so my sample must be representative." Explain the error.

12 A website reports that 80% of teenagers prefer online lessons. It gives no information about who was asked. List three questions you would ask before trusting the claim.

13 Explain the difference between reliability and validity, using a faulty balance that always reads 0.5 kg too high as an example.

14 A survey of 40 students finds that 24 walk to school. Can you conclude that 60% of the school walks? Explain.

EXERCISE 12A · CONTINUED COLLECTING AND CLASSIFYING DATA

MULTI-STEP AND UNFAMILIAR

15 **Extension.** Design a short survey to estimate the average time students spend on homework each night. State the population, the sampling method, two questions you would ask, and two sources of bias you would try to avoid.

16 **Extension.** A school has 840 students. A stratified sample of 84 is planned, with strata by year group and by gender. Explain why two-stage stratification may be better than year group alone, and state one practical difficulty.

17 **Extension.** A company emails a satisfaction survey only to customers who have contacted its support line in the past month. Explain why the results are likely to underestimate overall satisfaction, and suggest a better sampling frame.

SHAPE OF THE WHOLE

12.2 Frequency distributions and histograms

Raw data are hard to read. A list of 60 reaction times says little until it is organised. A **frequency table** groups the values into **class intervals** and records how many values fall in each interval. For continuous data the intervals must cover the whole range without gaps or overlaps: 200–250, 250–300 and so on. Each class runs from its lower boundary up to, but not including, its upper boundary, so 200–250 means $200 \leq x < 250$. Each interval has a **midpoint**, used to estimate the mean when the original values are no longer available.

A **histogram** displays **grouped** continuous data with bars that touch, because the classes are adjacent. If the class widths are equal, the height of a bar can represent frequency. If the widths are unequal, height must represent **frequency density**, so that the area of each bar — not its height — is proportional to frequency.

KEY RELATIONSHIPS

Grouped data and histograms

Class midpoint: the average of the lower and upper boundaries of a class.

Class boundaries: the values where one class ends and the next begins. With classes 0–10 and 10–20, the shared boundary is 10; a value of exactly 10 belongs to the second class.

Estimated mean from a frequency table: $\bar{x} \approx \frac{\sum fx}{\sum f}$, where x is the midpoint and f the frequency.

Frequency density: $\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$.

In a histogram, the area of each bar is proportional to the frequency. With equal widths, this is equivalent to height representing frequency.

TABLE 12.1 REACTION TIMES OF 60 STUDENTS

CLASS (ms)	TALLY	FREQUENCY f	MIDPOINT x
200–250	||	2	225
250–300	|||| |	6	275
300–350	|||| |||| ||||	14	325
350–400	|||| |||| |||| |||	18	375
400–450	|||| |||| ||	12	425
450–500	|||| |||	8	475

EXAMPLE **12.4** ESTIMATING THE MEAN FROM A TABLE

QUESTION

Use Table 12.1 to estimate the mean reaction time.

SOLUTION

1 Multiply each midpoint by its frequency: $2(225) = 450$, $6(275) = 1650$, $14(325) = 4550$, $18(375) = 6750$, $12(425) = 5100$, $8(475) = 3800$.

2 Total frequency $\sum f = 60$. Total $\sum fx = 450 + 1650 + 4550 + 6750 + 5100 + 3800 = 22\,300$.

3 Estimated mean $= \frac{22\,300}{60} \approx 372$ ms, to 3 significant figures.

Check: the estimate lies inside the modal class, 350–400 ms, and close to its midpoint. It is an estimate because the midpoint stands in for every value in its class.

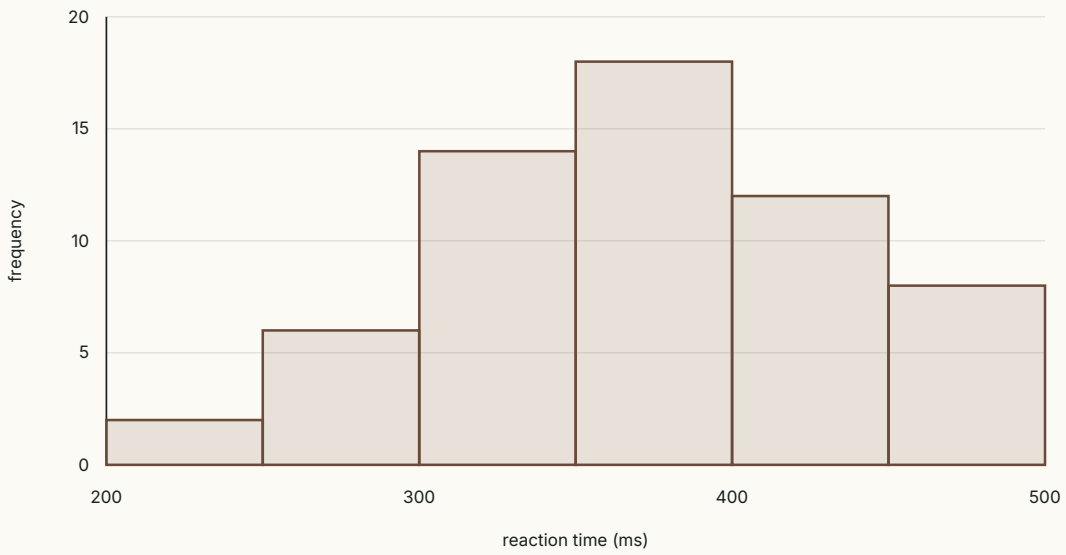


Figure 12.1 The histogram of the 60 reaction times. Bars touch because the classes are adjacent intervals of time. The tallest bar is the modal class, 350–400 ms, but height alone is not the whole story: the distribution also has a longer tail of slower times.

EXAMPLE 12.5 READING A HISTOGRAM

QUESTION

Use Figure 12.1. (a) How many students had a reaction time below 300 ms? (b) What fraction had a time of at least 400 ms? (c) Describe the shape of the distribution.

SOLUTION

1. (a) The first two bars represent $2 + 6 = 8$ students.
2. (b) The last two bars represent $12 + 8 = 20$ students, so the fraction is $\frac{20}{60} = \frac{1}{3}$.
3. (c) The distribution has one clear peak at 350–400 ms. It rises quickly from the lower times and falls more gradually towards the higher times, with a small tail above 450 ms.

Check: the frequencies add to $2 + 6 + 14 + 18 + 12 + 8 = 60$, the stated sample size, so no bar has been missed.

TABLE 12.2 AGES OF 80 MUSEUM VISITORS

AGE (years)	FREQUENCY f	CLASS WIDTH	FREQUENCY DENSITY
0–10	12	10	1.2
10–20	16	10	1.6
20–30	20	10	2.0
30–50	20	20	1.0
50–70	8	20	0.4
70–90	4	20	0.2

EXAMPLE 12.6 UNEQUAL CLASS WIDTHS

QUESTION

Use Table 12.2. (a) Explain why the 30–50 bar should not be drawn twice as tall as the 20–30 bar. (b) How many visitors were under 30? (c) Why can the table not tell you exactly how many were between 25 and 35?

SOLUTION

1. (a) The 30–50 class is twice as wide as 20–30, so its 20 visitors are spread over twice the interval. Its frequency density is $20 \div 20 = 1.0$, compared with $20 \div 10 = 2.0$. Equal heights would misrepresent the concentration.
2. (b) $12 + 16 + 20 = 48$ visitors were under 30.
3. (c) The table records only class totals. The 20 visitors aged 20–30 could lie anywhere in that interval, and the same is true of the 30–50 group, so the number between 25 and 35 is unknown.

Check: the areas of the bars are proportional to frequency: $10(1.2) = 12$, $10(1.6) = 16$, $10(2.0) = 20$, $20(1.0) = 20$, $20(0.4) = 8$, $20(0.2) = 4$.

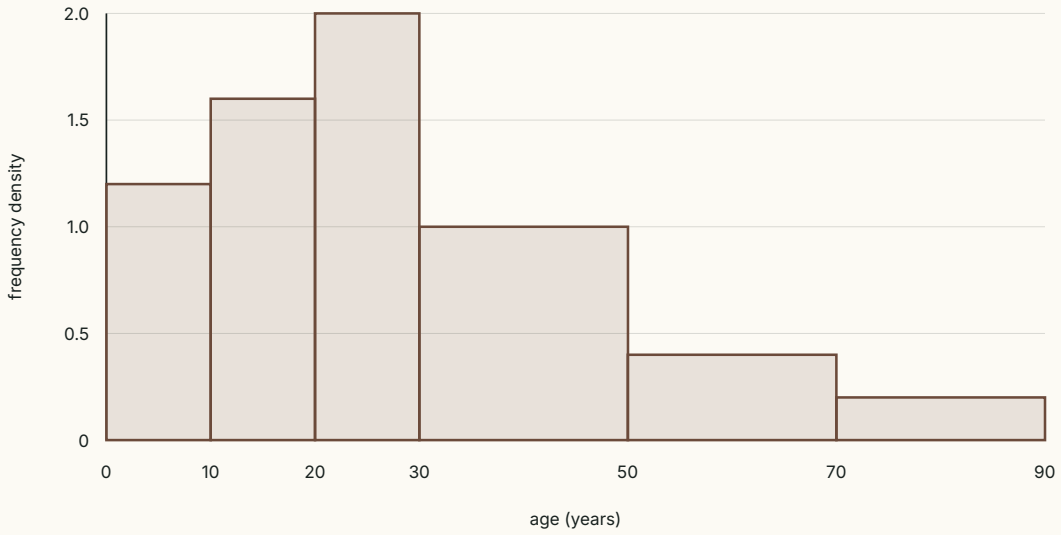


Figure 12.2 The 20 visitors aged 20–30 are concentrated in a narrow interval, so that bar has the greatest frequency density. The 30–50 class has the same frequency but twice the width, so its bar is half as tall.

COMMON MISCONCEPTION

"In a histogram, the tallest bar is the most common class." This is true only when all classes have the same width. With unequal widths, compare frequency density or area, not height.

"Histogram bars should have gaps." Gaps belong to bar charts for categories. A histogram represents continuous classes, so the bars touch at the class boundaries.

"The midpoint is an actual data value." The midpoint is a representative used for calculation. No student necessarily had a reaction time of exactly 325 ms.

"You can read exact values from a histogram." Grouping loses the individual values. The histogram shows the overall shape, not the exact data.

EXERCISE 12B

FREQUENCY DISTRIBUTIONS AND HISTOGRAMS

FLUENCY

1 The masses of 30 parcels, in kg, are grouped as 0–1 (4), 1–2 (9), 2–3 (11), 3–4 (6). Find the midpoint of each class and estimate the mean mass.

2 A grouped table has classes 10–20, 20–30, 30–40 with frequencies 5, 12, 8. Find the estimated mean.

3 Explain why the class 20–30 has boundaries 20 and 30, and why the interval is written $20 \leq x < 30$.

4 A histogram has equal class widths of 5. One bar has height 8. What area represents a frequency of 8? What does height represent?

5 A class of width 10 has frequency 15. Find its frequency density.

6 A class of width 20 has frequency density 0.8. Find its frequency.

UNDERSTANDING AND REASONING

7 Heights of 50 students are grouped in 10 cm classes from 140 to 190. Draw a possible histogram and explain why the bars touch.

8 A histogram shows a distribution with a long tail to the right. What does this suggest about the mean compared with the median?

EXERCISE 12B · CONTINUED

FREQUENCY DISTRIBUTIONS AND HISTOGRAMS

9 **Error analysis.** A student draws a histogram with unequal class widths and makes the bar heights equal to the frequencies. Explain the error and state what should be used instead.

10 Use Table 12.2. Estimate the number of visitors aged 20–50 and the number aged 50 or over.

11 Explain why a frequency table cannot tell you the exact median unless the original data are available.

12 A frequency table shows 0–10 (12), 10–20 (16), 20–30 (20), 30–50 (20), 50–70 (8), 70–90 (4). Why is it misleading to say "most visitors are in their twenties" from the frequencies alone?

13 Draw two histograms with the same frequencies but different class widths. Explain how the shape changes.

14 A data set is grouped into 0–4, 5–9, 10–14. Explain why using these labels creates gaps and how to fix them.

MULTI-STEP AND UNFAMILIAR

15 **Extension.** A histogram has classes 0–10, 10–30, 30–50 with frequencies 10, 25, 15. Calculate the frequency densities, sketch the histogram, and estimate the mean using midpoints.

16 **Extension.** The table shows waiting times at a clinic: 0–10 (8), 10–20 (14), 20–30 (18), 30–40 (10). Estimate the mean and explain why the estimate may be too high or too low.

17 **Extension.** Design a grouped frequency table for 40 reaction times with classes of width 20 ms. Explain how you chose the starting value and why equal widths make the histogram easier to read.

WHERE THE DATA ACCUMULATE

12.3 Cumulative frequency and box plots

Some questions are about totals rather than individual values. A headteacher may not need only the average journey to school; she may ask how many students travel for more than half an hour. A **cumulative frequency** answers questions of this kind. It is a running total: it counts how many values are at or below a given boundary. Adding one column to a grouped frequency table turns it into a cumulative frequency table.

The cumulative frequency is plotted against the **upper boundary** of each class, because every value in that class is at or below that boundary. The points are joined by a smooth curve that never falls: as the boundary moves to the right, the count can only stay the same or increase. Where the curve is steep, the data are concentrated; where it is flat, few values lie in that interval.

Quartiles divide an ordered data set into four parts of roughly equal size. The **lower quartile** Q_1 has about one quarter of the data below it, the **median** Q_2 has half below it, and the **upper quartile** Q_3 has three quarters below it. For grouped data these values are read from the curve at cumulative frequencies $\frac{n}{4}$, $\frac{n}{2}$ and $\frac{3n}{4}$. The **interquartile range** is the width of the middle half of the data:

$$\text{IQR} = Q_3 - Q_1$$

A **box plot**, or box-and-whisker plot, displays the **five-number summary**: minimum, Q_1, median, Q_3 and maximum. The box spans Q_1 to Q_3, a line inside marks the median, and the whiskers reach to the smallest and largest values that are not outliers. An **outlier** is a value that lies far from the rest of the data. A common rule flags any value more than $1.5 \times \text{IQR}$ below Q_1 or above Q_3; such values are plotted separately and investigated, not automatically deleted.

Quartiles can be defined in slightly different ways. This book excludes the median when the data are split into halves; if a half contains an even number of values, its quartile is the mean of its two middle values. Other conventions can shift a quartile by a small amount, so state the method when the difference matters.

KEY RELATIONSHIPS

Cumulative frequency and the five-number summary

For n ordered values, plot the cumulative frequency at each class's upper boundary. Read the quartiles from the curve at:

$$Q_1 \text{ at } \tfrac{n}{4} \quad \text{median at } \tfrac{n}{2} \quad Q_3 \text{ at } \tfrac{3n}{4}$$

Then $\text{IQR} = Q_3 - Q_1$. Outlier fences are $Q_1 - 1.5\,\text{IQR}$ and $Q_3 + 1.5\,\text{IQR}$.

The box plot shows the five-number summary: the box contains the middle 50% of the values, and each whisker reaches to the most extreme value that is not an outlier. A box plot does not show the sample size or how many values lie in each part of the box.

TABLE 12.3 TRAVEL TIMES OF 40 STUDENTS

TIME (minutes)	FREQUENCY f	CUMULATIVE FREQUENCY
0–10	4	4
10–20	9	13
20–30	13	26
30–40	8	34
40–50	4	38
50–60	2	40

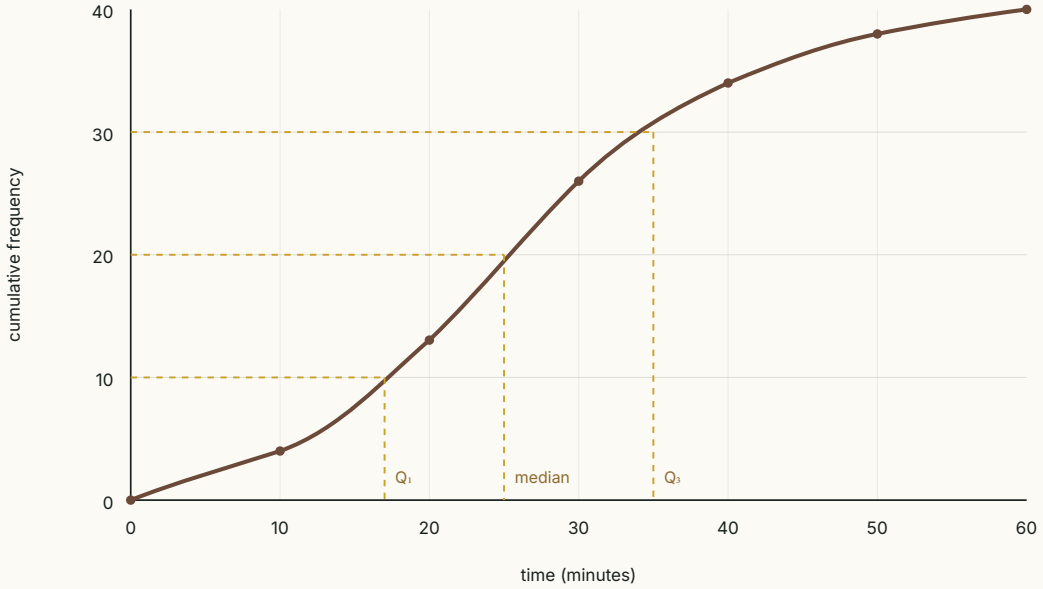


Figure 12.3 The cumulative frequency curve for Table 12.3. Reading across at 10, 20 and 30 on the vertical axis, then down, gives $Q_1 \approx 17$ min, the median ≈ 25 min and $Q_3 \approx 35$ min. The steepest part of the curve is the 20–30 class, where most journeys lie.

EXAMPLE **12.7** READING QUARTILES FROM A CUMULATIVE FREQUENCY CURVE

QUESTION

Table 12.3 shows the travel times of 40 students. (a) Complete the cumulative frequency column. (b) State the positions of the quartiles. (c) Use Figure 12.3 to estimate Q_1, the median and Q_3. (d) Find the interquartile range and interpret it.

SOLUTION

1 (a) Add the frequencies in order: $4,\ 4+9=13,\ 13+13=26,\ 26+8=34,\ 34+4=38,\ 38+2=40$.

2 (b) With $n=40$, the lower quartile is at position $\frac{40}{4}=10$, the median at $\frac{40}{2}=20$, and the upper quartile at $\frac{3(40)}{4}=30$.

3 (c) Read across from 10, 20 and 30 on the cumulative frequency axis to the curve, then down to the time axis. The readings are $Q_1 \approx 17$ min, median ≈ 25 min and $Q_3 \approx 35$ min.

4 (d) $\text{IQR} \approx 35-17=18$ min. The middle half of the students travel between about 17 and 35 minutes.

Check: the median should lie in the class 20–30, because the cumulative frequency reaches 13 at 20 minutes and 26 at 30 minutes. The estimate 25 is inside that class, and $Q_1 < \text{median} < Q_3$, as required.

EXAMPLE **12.8** A FIVE-NUMBER SUMMARY AND A BOX PLOT

QUESTION

The test scores of 20 students, in order, are
42, 55, 58, 61, 63, 65, 68, 70, 72, 74, 76, 78, 80, 82, 85, 88, 90, 92, 95, 98.
(a) Find the five-number summary. (b) Check for outliers using the $1.5 \times \text{IQR}$ rule. (c) Draw a box plot and describe the distribution.

SOLUTION

1 (a) Minimum $= 42$, maximum $= 98$. The median is the mean of the 10th and 11th values: $\frac{74+76}{2} = 75$.

2 The lower half has 10 values, so Q_1 is the mean of its 5th and 6th values: $\frac{63+65}{2} = 64$. The upper half gives $Q_3 = \frac{85+88}{2} = 86.5$.

3 $\text{IQR} = 86.5 - 64 = 22.5$. The fences are $64 - 1.5(22.5) = 30.25$ and $86.5 + 1.5(22.5) = 120.25$.

4 (b) Both 42 and 98 lie inside the fences, so there are no outliers. The whiskers reach the minimum and maximum.

5 (c) The box plot is shown in Figure 12.4. The median lies nearer Q_1 than Q_3, and the upper whisker is longer than the lower whisker, so the scores are slightly skewed to the right: a few students scored well above the middle group.

Check: $Q_1 < \text{median} < Q_3$, and the median is roughly in the middle of the box, not at an edge. The five values 42, 64, 75, 86.5 and 98 divide the 20 scores into four groups of five.

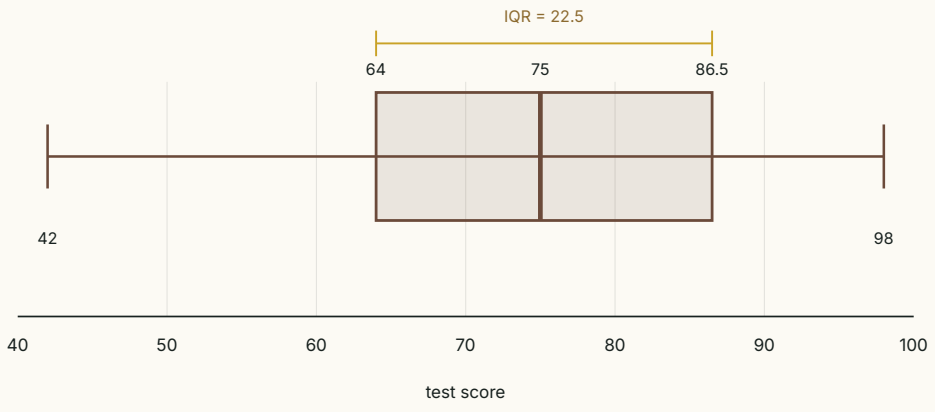


Figure 12.4 The box plot of the 20 test scores in Example 12.8. The box contains the middle 50% of the scores, from 64 to 86.5. The longer right whisker shows that the top quarter of the class is more spread out than the bottom quarter.

TABLE 12.4 HOMEWORK TIME OF TWO CLASSES

SUMMARY	CLASS 9A (min)	CLASS 9B (min)
Minimum	10	15
Lower quartile Q_1	25	30
Median	35	45
Upper quartile Q_3	50	55
Maximum	70	60
Interquartile range	25	25
Range	60	45

EXAMPLE **12.9** COMPARING TWO FIVE-NUMBER SUMMARIES

QUESTION

Table 12.4 summarises the homework time reported by two classes. Compare the two distributions. Which class typically spends longer on homework, and which has the more consistent middle group?

SOLUTION

1. The medians are 35 minutes for 9A and 45 minutes for 9B, so a typical student in 9B reports 10 minutes more homework.
2. The interquartile ranges are both 25 minutes, so the middle half of each class is equally spread out. The medians differ, but the consistency of the middle group is similar.
3. The ranges are 60 minutes for 9A and 45 minutes for 9B. Class 9A has a longer upper whisker: its maximum is 20 minutes above Q_3, compared with 5 minutes for 9B. A few students in 9A report very long homework times.
4. The shape also differs. In 9A, $Q_3 - \text{median} = 15$ and $\text{median} - Q_1 = 10$, so the upper half is more spread out. In 9B, the lower half is more spread out, because $\text{median} - Q_1 = 15$ and $Q_3 - \text{median} = 10$.
5. Conclusion: 9B typically spends longer on homework, but 9A contains the most extreme values. The medians and IQRs give a clearer comparison than the means alone would, because the long homework times in 9A would pull its mean upwards.

Check: the IQR is the same for both classes even though their ranges differ, which shows why range alone is a poor description of spread. The comparison is based on reported times, so it also depends on how honestly and consistently students estimated them.

COMMON MISCONCEPTION

"Plot the cumulative frequency at the midpoint of each class." The cumulative total counts everything up to the end of the class, so it belongs at the upper boundary. Plotting at the midpoint shifts the whole curve to the left.

"The interquartile range is the range of the middle half, so it uses all the data." It uses only Q_1 and Q_3. That is why it is resistant to extreme values, but it also ignores the values inside the middle half.

"A box plot shows the sample size and the shape exactly." It shows five numbers. Two data sets with the same box plot can have different sample sizes and different values between the quartiles.

"An outlier should always be removed." An outlier may be a recording error, or it may be a genuine and important value. Check the source and the measurement before deciding.

EXERCISE 12C

CUMULATIVE FREQUENCY AND BOX PLOTS

FLUENCY

1 A grouped table has classes 0–10 (5), 10–20 (8), 20–30 (12), 30–40 (5). Write the cumulative frequencies.

2 An ordered data set has 20 values. State the positions of the median, the lower quartile and the upper quartile.

3 Find the five-number summary of 3, 5, 6, 8, 9, 11, 12, 14, 15, 18, 20.

4 A box plot has minimum 12, $Q_1 = 18$, median 24, $Q_3 = 31$ and maximum 45. Find the IQR, the range and the outlier fences.

5 A grouped table has classes 0–10 (6), 10–20 (10), 20–30 (14), 30–40 (7), 40–50 (3). State which class contains the median, and explain your reasoning.

6 A cumulative frequency curve gives $Q_1 = 22$, median $= 31$ and $Q_3 = 44$. Find the interquartile range.

UNDERSTANDING AND REASONING

7 Explain why a cumulative frequency curve can never go down as the variable increases.

8 Explain why the cumulative frequency is plotted at the upper class boundary rather than at the midpoint.

9 A student says: "The median of grouped data is the midpoint of the modal class." Explain why this is only an estimate, and when it may be inaccurate.

10 A box plot has a much longer upper whisker than lower whisker. Describe two possible explanations.

11 Two box plots have the same median and the same IQR but different ranges. What is different about the two data sets?

12 Can two data sets have the same five-number summary but different individual values? Explain with an example or a reason.

13 A point is marked as an outlier beyond the whisker. List three checks you would make before removing it from the data set.

14 Explain why the box contains the middle 50% of the data, but each whisker does not necessarily contain 25%.

EXERCISE 12C · CONTINUED CUMULATIVE FREQUENCY AND BOX PLOTS

MULTI-STEP AND UNFAMILIAR

15 The stem-and-leaf plot shows 16 waiting times in minutes.

3 | 2 5 7
4 | 1 4 6 8
5 | 0 2 2 5 7 9
6 | 1 3 4

Write the ordered data, find the five-number summary, check for outliers and draw a box plot.

16 A clinic records the waiting times of 50 patients: 0–5 min (6), 5–10 (12), 10–15 (14), 15–20 (15), 20–25 (3). Complete a cumulative frequency table, draw the curve, and estimate the quartiles and IQR.

17 **Extension.** Two data sets are
A: 1, 2, 7, 8, 9, 10, 11, 16, 17
B: 1, 8, 8, 8, 9, 10, 10, 10, 17.
Show that they have the same median and range but different interquartile ranges. Explain what this example shows about using range alone to describe spread.

18 **Extension.** A data set has 20 values. Explain why replacing the largest value by a much larger value cannot change the median, Q_1 or Q_3, but does change the mean and the range.

HOW FAR VALUES SIT FROM THE CENTRE

12.4 Spread, outliers and robust measures

Two data sets can have the same mean and still describe very different situations. One class may have marks clustered around 70, while another has many marks near 50 and 90. A measure of centre is therefore incomplete without a measure of spread. The **range** is the difference between the largest and smallest values. It is quick to calculate but uses only two observations, so a single extreme value can change it dramatically.

The interquartile range is more stable because it ignores the lowest and highest quarters. A third measure, the **standard deviation**, uses every value. It measures the typical distance of the data from the mean. For values $x_1, x_2, \ldots, x_n$ with mean $\bar{x}$, the standard deviation is

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

The quantity inside the square root, before taking the root, is the **variance** σ^2. Squaring prevents positive and negative deviations from cancelling, and it gives extra weight to values far from the mean. The standard deviation has the same units as the data, which makes it easier to interpret than the variance.

A measure is **robust** if a few extreme values do not change it much. The median and interquartile range are robust; the mean, range and standard deviation are not. This matters when data are skewed. In a right-skewed distribution, a few unusually large values pull the mean towards the tail, while the median stays near the bulk of the data. Report the median and IQR when the distribution is skewed or contains genuine outliers, and report the mean and standard deviation when the distribution is roughly symmetric with no outliers. In every case, state the sample size and the context.

KEY RELATIONSHIPS

Measures of spread and resistance

For data with mean $\bar{x}$ and n values:

$$\text{range} = x_{\max} - x_{\min} \qquad \text{IQR} = Q_3 - Q_1 \qquad \sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

A value is flagged as an outlier if it lies below $Q_1 - 1.5\,\text{IQR}$ or above $Q_3 + 1.5\,\text{IQR}$.

Robust: median, Q_1, Q_3, IQR. **Not robust:** mean, range, standard deviation. Adding the same constant to every value leaves the spread unchanged: the range, IQR and standard deviation stay the same, while the mean increases by that constant.

Some software divides the sum of squares by $n - 1$ instead of n when the data are a sample used to estimate a wider population. The difference is small for large n, but it explains why two calculators can give slightly different standard deviations.

TABLE 12.5 CHOOSING A SUMMARY

SHAPE OF THE DATA	REPORT	WHY
Roughly symmetric, no outliers	Mean and standard deviation	Both use all the data and describe a typical value and spread efficiently.
Skewed, or contains outliers	Median and interquartile range	These are not pulled far by a few extreme values.
Any numerical data	Five-number summary, with range as extra detail	Shows the centre, the middle half and the extremes together.
Categorical data	Mode and frequencies	A mean has no meaning for categories such as colours or blood groups.

EXAMPLE **12.10** CALCULATING A STANDARD DEVIATION

QUESTION

Five students scored 4, 6, 7, 9 and 14 on a quiz out of 15. Calculate the mean and the standard deviation.

SOLUTION

1. Mean: $\bar{x} = \frac{4+6+7+9+14}{5} = \frac{40}{5} = 8$.
2. Deviations from the mean: $4 - 8 = -4$, $6 - 8 = -2$, $7 - 8 = -1$, $9 - 8 = 1$ and $14 - 8 = 6$, giving a total of zero, as it must be.
3. Squares of the deviations: $16,\ 4,\ 1,\ 1,\ 36$, with total 58.
4. Variance: $\sigma^2 = \frac{58}{5} = 11.6$.
5. Standard deviation: $\sigma = \sqrt{11.6} \approx 3.41$, to 3 significant figures.

Check: the standard deviation is in marks, the same unit as the data. A value of 3.41 marks means that typical scores lie roughly 3.4 marks from the mean of 8. Three of the five scores, 6, 7 and 9, are within one standard deviation of the mean.

EXAMPLE **12.11** SAME MEAN, DIFFERENT SPREAD

QUESTION

Student A scored 4, 6, 7, 9 and 14. Student B scored 6, 7, 8, 9 and 10. Both have mean 8. Compare their ranges, interquartile ranges and standard deviations, and say which student is more consistent.

SOLUTION

1. Student A: range $14 - 4 = 10$; $Q_1 = 6$, $Q_3 = 9$, so $\mathrm{IQR} = 3$; $\sigma \approx 3.41$ from Example 12.10.
2. Student B: range $10 - 6 = 4$; $Q_1 = 7$, $Q_3 = 9$, so $\mathrm{IQR} = 2$.
3. For B, the deviations are $-2, -1, 0, 1, 2$, whose squares total $4 + 1 + 0 + 1 + 4 = 10$. So $\sigma^2 = \frac{10}{5} = 2$ and $\sigma \approx 1.41$.
4. Every measure of spread is larger for A: range 10 versus 4, IQR 3 versus 2, standard deviation 3.41 versus 1.41.
5. Student B is more consistent. The equal means hide the fact that A has one very low score and one very high score, while B's scores lie within 2 marks of the mean.

Check: both sets sum to 40, so the equal means are not a coincidence. The standard deviation of A is more than twice that of B even though the means are identical, which shows why a mean should never be reported alone.

EXAMPLE **12.12** WHAT ONE OUTLIER DOES

QUESTION

Ten students scored
60, 65, 70, 72, 74, 76, 78, 80, 85, 90.
A late student then scores 20. Compare the mean, median, range and interquartile range before and after the new score.

SOLUTION

1. Before: the sum is 750, so the mean is 75. The median is $\frac{74+76}{2} = 75$. The lower half is 60, 65, 70, 72, 74, so $Q_1 = 70$; the upper half gives $Q_3 = 80$. Hence $\mathrm{IQR} = 10$, range $= 90 - 60 = 30$.
2. After adding 20, there are 11 values: 20, 60, 65, 70, 72, 74, 76, 78, 80, 85, 90. The sum is 770, so the mean is $\frac{770}{11} = 70$.
3. The median is now the 6th value, 74. The lower half is 20, 60, 65, 70, 72, so $Q_1 = 65$; the upper half gives $Q_3 = 80$. Hence $\mathrm{IQR} = 15$ and the range is $90 - 20 = 70$.
4. Changes: mean $75 \to 70$ (down 5); median $75 \to 74$ (down 1); range $30 \to 70$ (up 40); IQR $10 \to 15$ (up 5).
5. The new score is an outlier: $Q_3 + 1.5\,\mathrm{IQR} = 80 + 22.5 = 102.5$ and $Q_1 - 1.5\,\mathrm{IQR} = 65 - 22.5 = 42.5$, and 20 is below 42.5. The mean and range react strongly; the median and IQR react much less, although adding a value also changes the positions of the quartiles.

Check: the median moved by only 1 mark because it depends on position, not on the size of the extreme value. Robust does not mean unaffected: the IQR changed by 5 because the extra value changed the quartile positions.

EXAMPLE **12.13** CHOOSING A SUMMARY FOR SKEWED DATA

QUESTION

Nine houses on one street sold for these prices, in thousands of pounds:
180, 195, 210, 225, 240, 260, 290, 320, 1450.
Calculate the mean and median, identify any outlier, and recommend which average a newspaper should quote.

SOLUTION

1. The sum is 3370, so the mean is $\frac{3370}{9} \approx 374.4$, or about £374,000.
2. The median is the 5th value, £240,000. The lower half is 180, 195, 210, 225, so $Q_1 = 202.5$; the upper half is 260, 290, 320, 1450, so $Q_3 = 305$. Thus $\text{IQR} = 102.5$, or £102,500.
3. The upper fence is $305 + 1.5(102.5) = 458.75$. The price of 1450 is far above it, so it is an outlier. It is a genuine sale, not an error: one large house on the street.
4. The mean of £374,000 is above Q_3 and £134,000 above the median. It describes no typical house; it has been pulled towards the expensive one.
5. A newspaper should quote the median, £240,000, with the IQR of £102,500. This describes a typical price and the spread of the middle half without letting one exceptional sale dominate the summary.

Check: the mean is greater than the median, which signals right skew. Removing the outlier would give a mean of $\frac{1920}{8} = 240$, while the median of the remaining eight values is £232,500. Even so, removing a genuine sale would misrepresent the street. The better response is to choose a robust summary, not to delete data.

COMMON MISCONCEPTION

"The standard deviation is the average of the deviations." That average is always zero, because the positive and negative deviations cancel. The standard deviation is the square root of the average of the squared deviations.

"A standard deviation of zero means there are no data." It means every value is identical and equal to the mean; the data have no spread.

"The range tells you the standard deviation." Two sets with the same range can have very different standard deviations, depending on how the values are distributed between the extremes.

"Robust measures are not affected by outliers at all." They are much less affected, but adding a value still changes the sample size and can shift a quartile. Robustness is a matter of degree.

EXERCISE 12D

SPREAD, STANDARD DEVIATION AND OUTLIERS

FLUENCY

1. Find the range and interquartile range of 4, 7, 9, 12, 15, 18.
2. Find the mean and standard deviation of 2, 4, 6, 8, 10.
3. Find the standard deviation of 5, 5, 5, 5, 5. Explain your answer without calculating every square.
4. Without calculating, state which set has the larger standard deviation: A = 9, 10, 10, 10, 11 or B = 1, 5, 10, 15, 19.

EXERCISE 12D · CONTINUED SPREAD, STANDARD DEVIATION AND OUTLIERS

5 A data set has $Q_1 = 20$ and $Q_3 = 40$. Find the IQR and the outlier fences.

6 A data set has mean 50 and standard deviation 4. Describe where most values are likely to lie, and state one reason this is not a guarantee.

UNDERSTANDING AND REASONING

7 Explain why the deviations $x - \bar{x}$ always add to zero.

8 Explain why the deviations are squared before they are averaged.

9 Every value in a data set is increased by 10. Describe the effect on the mean, standard deviation, range and IQR.

10 Every value is doubled. Describe the effect on the mean, standard deviation, range and IQR.

11 **Error analysis.** A student calculates $\frac{\sum(x-\bar{x})^2}{n} = 36$ and writes "the standard deviation is 36". Explain the error and give the correct value.

12 Incomes in a town are strongly right-skewed. Which summary would you report: mean and standard deviation, or median and IQR? Justify your choice.

13 An outlier has been identified in a data set. List three checks you would make before deciding whether to keep it.

14 Two classes have the same mean test score. One has standard deviation 2 and the other 8. What can you say, and what else would you want to know before judging which class performed better?

MULTI-STEP AND UNFAMILIAR

15 The values 1, 2, 3 and 4 occur with frequencies 2, 5, 2 and 1. Find the mean and standard deviation of the 10 values.

16 Find five values with mean 10 and standard deviation 2. Explain why your answer is not the only possible set.

17 Two machines fill bags labelled 500 g. Machine A produces a mean of 498 g with standard deviation 2 g; machine B produces a mean of 500 g with standard deviation 6 g. A bag outside 500 ± 4 g is rejected. Compare the machines and recommend one, stating any assumption you make.

18 **Extension.** Show algebraically that adding a constant c to every value leaves the standard deviation unchanged, and that multiplying every value by k multiplies the standard deviation by $|k|$.

READING BEYOND THE HEADLINE

12.5 Comparing distributions and interpreting critically

A comparison is only fair when the groups, units and conditions are comparable. "Class 9B did better" may mean that its mean was higher, that its median was higher, that more students passed, or that its weakest students did better. These are different claims. A sound comparison reports the **centre**, the **spread**, the **shape** and the **sample size**, and it says clearly what was measured.

Parallel box plots place two or more distributions on the same scale. Compare the medians to see which group is typically higher, the boxes to compare the middle halves, and the whiskers to compare the extremes. Overlapping boxes are important: they show that many individuals in the lower-median group are above many individuals in the higher-median group. A difference in medians must be judged against the spread. A one-hour difference is substantial when the IQR is half an hour, but modest when the IQR is three hours. Box plots are poor at showing shape in detail, because they hide gaps, clusters and a second peak; use a histogram or dot plot when shape matters.

Graphs can mislead without containing a false number. A **truncated axis** that starts at 60 rather than 0 makes a small change look dramatic. Inconsistent scales, 3D effects, and pictures whose area or volume grows faster than the quantity they represent all exaggerate differences. A selective time range can hide a long-term trend, and an unrepresentative sample can make a confident claim about a population that was never measured. The question is not only "What does the graph show?" but "What has been left out?"

Two variables can be associated without one causing the other. **Correlation** is a statistical relationship; **causation** means that changing one variable produces a change in the other. An observational study can reveal an association, but a third variable — a **confounder** — may explain it. To support a causal claim, we need a well-designed experiment or a study that measures and accounts for the plausible confounders.

ATL SKILLS · CRITICAL THINKING AND COMMUNICATION

Ask about the source: who collected the data, from whom, when, and for what purpose? **Ask about the measure:** what exactly was recorded, in what units, and how consistently? **Ask about the sample:** how large was it, and how was it selected? **Ask about the display:** does the axis start at zero, are the scales consistent, and is the comparison like-for-like? **Ask about the summary:** is a mean being used where a median is more appropriate, and is any measure of spread reported? **Communicate carefully:** use "is associated with" for observational data and reserve "causes" for evidence that can support it.

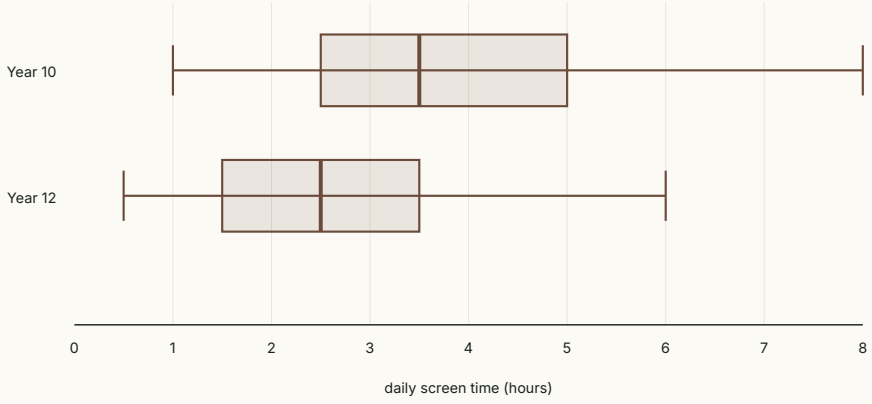


Figure 12.5 Daily screen time reported by two year groups. The Year 10 median is one hour higher and its box is slightly wider, but the boxes overlap between 2.5 and 3.5 hours. The two distributions are not cleanly separated: many Year 12 students report more screen time than many Year 10 students.

EXAMPLE **12.14** COMPARING TWO BOX PLOTS

QUESTION

Use Figure 12.5 to compare the screen time of the two year groups. Refer to centre, spread, shape and overlap, and state one limitation of the comparison.

SOLUTION

1. Centre: the medians are about 3.5 hours for Year 10 and 2.5 hours for Year 12, so the typical Year 10 student reports one hour more.
2. Spread: the interquartile ranges are about $5.0 - 2.5 = 2.5$ hours for Year 10 and $3.5 - 1.5 = 2.0$ hours for Year 12. The middle half of Year 10 is slightly more spread out, but the difference is small.
3. Extremes: the ranges are about $8.0 - 1.0 = 7.0$ hours and $6.0 - 0.5 = 5.5$ hours. Year 10 has the highest reported value, 8 hours.
4. Shape and overlap: both box plots have a longer upper whisker, suggesting right skew. The boxes overlap from 2.5 to 3.5 hours, so the groups are not separated; a substantial number of Year 12 students report more screen time than a substantial number of Year 10 students.
5. Conclusion: Year 10 reports a higher typical screen time, but the difference is modest compared with the spread within each group. The data are self-reported, and the figure does not state how the samples were chosen, so the comparison describes these samples rather than proving a difference between all Year 10 and Year 12 students.

Check: every claim is tied to a feature of the graph. "Year 10 has a higher median" is supported; "all Year 10 students use more screens" is not, because the distributions overlap.

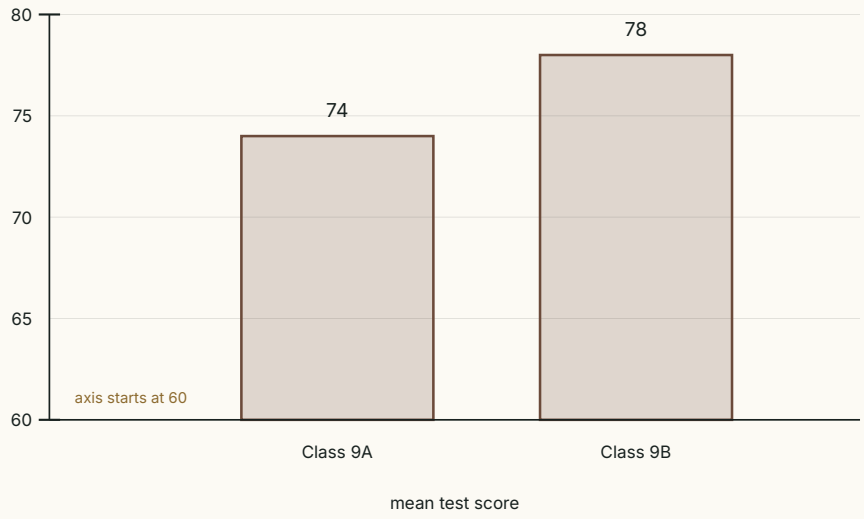


Figure 12.6 A bar chart whose vertical axis starts at 60 rather than 0. The means differ by 4 marks, but the visible scale is only 20 marks tall, so the right-hand bar looks much larger than it is. Redrawn from zero, the bars would have heights in the ratio $78 : 74$, a difference of about 5%.

EXAMPLE **12.15** CRITIQUING A GRAPH AND A CLAIM

QUESTION

A newsletter prints Figure 12.6 with the caption “Class 9B is clearly better at mathematics.” Identify three problems with the graph or the claim, and rewrite the claim fairly.

SOLUTION

1. The vertical axis starts at 60, so the bars are not drawn from zero. For a bar chart, bar length encodes the quantity, so truncating the axis exaggerates the 4-mark difference.
2. Only the means are shown. There is no measure of spread, no sample size and no indication of which test was used, so the reliability and practical size of the difference cannot be judged.
3. “Clearly better at mathematics” goes beyond the evidence. One test measures one set of tasks on one occasion; it does not measure all of mathematics, and a difference between two classes may reflect sampling, teaching, prior attainment or test conditions.
4. Fair rewrite: “In this test, the mean score of Class 9B was 4 marks higher than that of Class 9A. The graph does not show the spread of the scores or the sizes of the two samples, so the size and consistency of the difference are not yet clear.”
5. A better display would use dot plots or box plots on an axis starting at 0, with the sample size and a measure of spread shown for each class.

Check: the numbers in the graph are not false; the problem is the scale and the missing context. A critical reader asks what the graph emphasises and what it omits.

EXAMPLE **12.16** ASSOCIATION OR CAUSATION?

QUESTION

A study of 500 students finds that those who eat breakfast average 6 marks higher in a test. A news website reports: "Eating breakfast raises test scores." Evaluate the claim.

SOLUTION

1. The study is observational: it compares students who already do or do not eat breakfast. It shows an association, not a causal effect.
2. Plausible confounders include sleep, family income, morning routine, parental support and prior attainment. Students who eat breakfast may differ in many ways that also affect test scores.
3. The size of the difference also matters. Six marks may be large or small compared with the spread of scores, and the report gives no standard deviation, no sample details and no information about how breakfast was measured.
4. "Raises" is a causal verb. A fair report would say that breakfast eaters scored higher on average, and that the study does not show why.
5. A stronger design would randomly assign students to breakfast conditions, or follow students over time while measuring the confounders. Even then, ethical and practical limits would need to be considered.

Check: the association is real in the sample, but the causal claim needs more than a correlation. The evaluation separates what the data show from what the headline asserts.

COMMON MISCONCEPTION

"If the medians differ, the groups are completely different." The boxes and whiskers often overlap. A higher median means the typical value is higher, not that every value is higher.

"A truncated axis is always dishonest." Sometimes a narrow range is necessary to show small changes, but the axis must be clearly labelled, and bar charts are especially misleading when truncated because their lengths are read as amounts.

"Correlation proves causation." An association can be produced by a confounder, by reverse causation or by coincidence. Causation requires evidence from a suitable design.

"The average tells you what most people do." An average can sit between two clusters and describe nobody. Always ask about the shape of the distribution.

EXERCISE 12E

COMPARING AND INTERPRETING

FLUENCY

1. Use Figure 12.5. State the median and interquartile range for each year group.
2. Which year group has the higher median? Which has the greater interquartile range?
3. A bar chart's vertical axis starts at 90 instead of 0. Describe the effect on the apparent size of a change from 94 to 96.
4. Class A has mean 60 and standard deviation 8; Class B has mean 65 and standard deviation 3. Which class is more consistent? Explain.
5. State two questions you would ask before trusting a graph in a news report.

EXERCISE 12E · CONTINUED COMPARING AND INTERPRETING

6 Match each phrase to "association" or "causal claim": (a) "Students who sleep more tend to score higher." (b) "More sleep improves scores." (c) "Screen time is linked to lower wellbeing." (d) "Reducing screen time causes better sleep."

UNDERSTANDING AND REASONING

7 Explain why two overlapping box plots do not show a clear difference between the groups, even when their medians differ.

8 Explain why a difference in means is not meaningful without a measure of spread.

9 Describe two ways a 3D pie chart can mislead a reader.

10 A survey at a sports club finds that 85% of members exercise weekly. Explain why this result should not be used to estimate the proportion for all students in the town.

11 Ice-cream sales and drowning incidents both rise in summer. Does buying ice cream cause drowning? Explain, naming a confounding variable.

12 Two newspapers report different "average" house prices for the same town. Explain how both could be correct.

13 A graph shows a 2% change but the vertical axis covers only 48% to 52%. Explain why the visual change looks large and what should be added to the graph.

14 A student says: "The mean is 70, so most students scored 70." Explain why this may be false.

MULTI-STEP AND UNFAMILIAR

15 School A reports a median score of 68 with IQR 18 from 200 students. School B reports a median of 74 with IQR 9 from 25 students. Compare the schools and recommend what a researcher should do next. State two limitations of the comparison.

16 Write a 120-word comparison of the two distributions in Figure 12.5. Use centre, spread, shape and overlap, and state one limitation of the data.

17 Redesign Figure 12.6 so that it is fair. Describe the axis, the scale, the labels and any additional information you would include.

18 **Extension.** Design a study to test whether a later school start time improves students' sleep. State the variables, the comparison group, how you would reduce confounding, and two ethical issues you would need to address.

INVESTIGATION · CRITERION B AND C

12.6 Is the school day too long?

Arguments about school start times often use sleep data. The claim is simple: teenagers do not get enough sleep, and an early start is part of the reason. The evidence is harder. The table records the hours of sleep reported by 28 students on a school night. Fourteen students said they used a screen in bed; fourteen said they did not. The groups were not assigned: students chose their own habits. The comparison is therefore observational, and any difference is an association rather than proof of cause.

TABLE 12.6 HOURS OF SLEEP ON A SCHOOL NIGHT

STUDENT	SCREEN IN BED (hours)	NO SCREEN (hours)
1	5.5	6.5
2	6.0	7.0
3	6.0	7.0
4	6.5	7.5
5	6.5	7.5
6	6.5	7.5
7	7.0	8.0
8	7.0	9.0
9	7.0	9.0
10	7.5	9.5
11	8.0	9.5
12	8.0	9.5
13	8.5	10.0
14	9.0	10.5

INVESTIGATION · CRITERION B AND C

Does a screen in bed shorten sleep?

1. State the population and the sample. Is the study observational or experimental? Explain how you know.
2. For each group, find the five-number summary, check for outliers using the $1.5 \times \text{IQR}$ rule, and draw parallel box plots on the same axis.
3. Compare the two groups using centre, spread, shape and overlap. Write two sentences that a reader who has not seen the table would understand.
4. A report claims: "Using a screen in bed reduces sleep by 1.5 hours." Which part of this claim is supported by the data, and which part is not? Refer to specific statistics.
5. Identify two confounding variables that could affect both screen use and sleep. For each, explain how it could produce the observed difference.
6. Design a better study. State how you would assign participants, how you would measure sleep, how large a sample you would aim for, and one ethical issue you would need to address.
7. Generalise: write three conditions that evidence would need to satisfy before you would accept a causal claim that screen use shortens sleep.

Generalise and justify. The two groups differ by 1.5 hours in median sleep, but the boxes overlap slightly and the data are self-reported. Explain why a difference between groups is not the same as an effect of the screen. Then describe what a randomized study would add, and why random assignment is more convincing than comparing students who already differ in many ways.

MIXED PRACTICE 12

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–6 check fluency; 7–14 require reasoning; 15–18 are multi-step. Allow 55 minutes.

FLUENCY

1 Classify each variable as qualitative, discrete or continuous: (a) number of siblings, (b) height, (c) favourite subject, (d) number of goals scored, (e) time to run 100 m, (f) number of books read in a month.

2 Explain the difference between a census and a sample, and give one advantage of each.

3 Find the mean, median, mode and range of 4, 7, 7, 9, 12, 15.

4 Estimate the mean of the grouped data: 0–20 (5), 20–40 (9), 40–60 (6).

5 A class of width 5 has frequency 12. A class of width 10 has frequency 15. Find both frequency densities and state which bar is taller in a histogram.

6 A box plot has minimum 10, $Q_1 = 16$, median 22, $Q_3 = 30$ and maximum 42. Find the IQR, the range and the outlier fences.

UNDERSTANDING AND REASONING

7 A grouped table has classes 0–10 (4), 10–20 (7), 20–30 (9), 30–40 (5). Write the cumulative frequencies and state the class containing the median.

8 Find the standard deviation of 3, 5, 7, 9, 11.

9 Every value in a data set is increased by 10. Describe the effect on the mean, standard deviation, range and interquartile range.

10 A data set has $Q_1 = 20$ and $Q_3 = 38$. Find the outlier fences and decide whether 70 is an outlier. What would you do before removing it?

11 Two classes have mean score 65. Class A has standard deviation 3 and Class B has standard deviation 11. Which class is more consistent? What else would you want to know?

12 A graph of test scores has a vertical axis from 70 to 75. Explain why a change from 71 to 74 may look dramatic, and how the graph should be improved.

13 A study finds that students who attend tutoring score higher on average. A newspaper says tutoring "boosts" scores. Explain why this is an association rather than proof of cause, and name one confounder.

14 An online survey about school lunches is completed only by students who visit the school website. Explain why the results may be biased.

MULTI-STEP AND UNFAMILIAR

15 Design a short survey to estimate the average number of hours Year 10 students spend on homework each week. State the population, a sampling method, two neutral questions, and two sources of bias you would try to avoid.

16 A histogram has classes 0–10 (10), 10–20 (20) and 20–40 (20). Calculate the frequency densities, sketch the histogram, and explain which class has the greatest concentration of data.

17 A town reports a mean house price of £320,000 and a median of £280,000. Explain what this suggests about the shape of the distribution, and recommend which average a buyer should use. Give one reason.

18 **Extension.** Show that the two data sets
A: 0, 5, 5, 5, 10
B: 0, 0, 5, 10, 10
have the same mean and range but different standard deviations. Explain why range alone is an incomplete description of spread.

Chapter assessment

The tasks below are summative. Unlike the exercises, they are completed independently, without correction from a partner, and assessed against the criteria shown. Your teacher will confirm which tasks are required and how long you have.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Describing and displaying data: short response	30	35 min
C · Communicating	A report on screens and sleep	20	35 min
D · Applying mathematics in real-life contexts	Sleep, screens and the school day	24	45 min
Total	All three tasks	74	115 min

The tasks begin overleaf. Read the instructions and evidence requirements before you start, and show the reasoning that supports each answer.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 30 marks

Describing and displaying data: short response

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

1 Classify each variable as qualitative, discrete or continuous: (a) number of pets; (b) mass of a parcel; (c) blood group; (d) number of goals scored; (e) temperature; (f) postcode. [2]

2 Explain the difference between a census and a sample, and give one advantage of each. [3]

3 A student surveys 30 students in the library at lunchtime and asks: “Don’t you agree that homework should be reduced because it takes too much family time?” Identify two sources of bias and suggest one improvement. [3]

4 Find the mean, median, mode and range of 12, 15, 15, 18, 20, 22, 25, 28. [3]

5 Estimate the mean of the grouped data: 0–10 (4), 10–20 (9), 20–30 (7). [3]

6 A histogram has classes 0–5 (frequency 12), 5–15 (frequency 15) and 15–25 (frequency 10). Calculate the three frequency densities and state which bar is tallest. Explain why. [3]

7 A grouped table has classes 0–10 (8), 10–20 (16), 20–30 (12), 30–40 (4). Write the cumulative frequencies and estimate Q_1, the median and Q_3. Hence find the IQR. [3]

8 Find the five-number summary of 3, 4, 6, 7, 8, 10, 11, 13, 15, 16, 18. [3]

9 Calculate the standard deviation of 4, 6, 8, 10, 12. [3]

10 A data set has $Q_1 = 25$ and $Q_3 = 45$. Find the outlier fences and decide whether 80 is an outlier. [2]

11 Class A has mean 70 and standard deviation 4. Class B has mean 70 and standard deviation 10. Compare the two classes in one or two sentences. [2]

CRITERION C · COMMUNICATING 35 minutes · 20 marks

A report on screens and sleep

Instructions. A parent group claims that "students who use a screen in bed sleep far less than students who do not". Write a 350–450 word report for the school council using Table 12.7. Calculate the median, interquartile range and range of each group, then set out your findings. Your report must:

1. state the question and identify the data source;
2. describe both distributions using the median, interquartile range and range;
3. comment on the overlap between the two distributions;
4. explain one limitation of the data;
5. recommend whether the phrase "far less" is supported, with a justified reason.

Evidence: a clear question, accurate statistics with units, a comparison that uses spread as well as centre, a limitation that is specific to these data, and a recommendation that follows from the evidence rather than from opinion. Use headings and a short concluding paragraph; a labelled sketch or table is optional.

TABLE 12.7 SLEEP HOURS USED IN THE ASSESSMENT

STUDENT	SCREEN IN BED (hours)	NO SCREEN (hours)
1	5.5	6.5
2	6.0	7.0
3	6.0	7.0
4	6.5	7.5
5	6.5	7.5
6	6.5	7.5
7	7.0	8.0
8	7.0	9.0
9	7.0	9.0
10	7.5	9.5
11	8.0	9.5
12	8.0	9.5
13	8.5	10.0
14	9.0	10.5

D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Sleep, screens and the school day

Instructions. A school is considering moving its start time from 08:15 to 08:45. Table 12.7 records the sleep of 28 students, 14 of whom used a screen in bed. The groups were not randomly assigned. Answer all questions and show your working.

1. For each group, find the five-number summary and the interquartile range. Compare the two groups using centre and spread. [6]
2. A report claims: "A later start will add 30 minutes of sleep for every student." Evaluate this claim using the data and the context. Distinguish what the data show from what the claim asserts. [5]
3. Design a study to test whether a later start time increases students' sleep. State the variables, the comparison group, how sleep would be measured, a suitable sample size, and one ethical issue. [6]
4. Evaluate two assumptions or limitations of your recommended approach, including one about measurement and one about the students studied. [4]
5. Write a recommendation to the school leadership team. Distinguish the statistical evidence from other considerations such as transport, family schedules and cost. [3]

Evidence: correct summary statistics, a comparison that uses spread, a design with a clear comparison group, limitations tied to the method, and a recommendation that separates evidence from values.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few terms or formulas; applies them with major errors; needs support to choose a display.	Recalls some terms and formulas; applies familiar procedures with some accuracy; selects a method when prompted.	Recalls and applies formulas accurately in familiar situations; explains some steps.	Selects and applies formulas accurately and efficiently; explains reasoning and checks the reasonableness of results.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- Data are collected from a population or a sample; a biased sample cannot be rescued by making it larger.
- Variables are qualitative or quantitative, and quantitative variables are discrete or continuous.
- A frequency table groups continuous data; classes must not overlap, and unequal widths require frequency density rather than frequency for the bar height.
- Cumulative frequency counts values at or below each upper boundary; read Q_1, the median and Q_3 at $\frac{n}{4}$, $\frac{n}{2}$ and $\frac{3n}{4}$.
- The five-number summary gives a box plot; the box holds the middle 50% and the whiskers reach the most extreme non-outliers.
- An outlier lies beyond $Q_1 - 1.5\,\text{IQR}$ or $Q_3 + 1.5\,\text{IQR}$; investigate it before deciding what to do.
- The mean and standard deviation use every value but are not robust; the median and IQR are robust to extreme values.
- Compare distributions using centre, spread, shape, overlap and sample size, always in context.
- Truncated axes, inconsistent scales and omitted context can mislead; an association does not prove causation.

$$\bar{x} = \frac{\sum x}{n} \qquad \text{estimated mean} = \frac{\sum fx}{\sum f} \qquad \text{frequency density} = \frac{f}{\text{class width}}$$

$$Q_1 \text{ at } \frac{n}{4} \qquad \text{median at } \frac{n}{2} \qquad Q_3 \text{ at } \frac{3n}{4}$$

$$\text{IQR} = Q_3 - Q_1 \qquad \text{fences: } Q_1 - 1.5\,\text{IQR},\ Q_3 + 1.5\,\text{IQR}$$

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \qquad \text{range} = x_{\max} - x_{\min}$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can distinguish populations from samples and identify bias in a data-collection plan.
- ☐ I can classify a variable and choose an appropriate display.
- ☐ I can construct a grouped frequency table and a histogram, including unequal class widths.
- ☐ I can build a cumulative frequency table and read quartiles from the curve.
- ☐ I can draw and interpret a box plot from a five-number summary.
- ☐ I can find and interpret the range, interquartile range and standard deviation.
- ☐ I can identify an outlier and explain why it should be investigated, not deleted automatically.
- ☐ I can compare two distributions using centre, spread, shape and overlap.
- ☐ I can explain how a graph or a summary can mislead.
- ☐ I can distinguish an association from a causal claim and name a possible confounder.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which summary statistic did you find hardest to interpret, and what did you do to check your understanding?
2. Describe a time you nearly accepted a graph or a claim too quickly. Which question revealed the problem?
3. The statement of inquiry claims that how data are collected, grouped and displayed determines what conclusions are possible. Use the investigation to test this claim.
4. Look ahead to Chapter 13: probability also begins with a list of possible outcomes and uses long-run relative frequency. How might a large number of trials make an experimental probability more trustworthy?

THE MATHEMATICS OF CHANCE

13 Probability: Combined Events

LOGIC · PROBABILITY · EQUIVALENCE

A single coin is easy. Two dice, a sequence of draws or a game with several stages is harder — not because the idea changes, but because the list of possibilities grows. Probability is the discipline of listing, counting and combining outcomes without losing any.

STATEMENT OF INQUIRY

Uncertainty becomes calculable when every possible outcome is represented and the rules for combining events are applied consistently. Fairness is a mathematical property of a game, not a matter of luck.

KEY CONCEPT
Logic

RELATED CONCEPTS
Probability · Equivalence · Systems

GLOBAL CONTEXT
Fairness and development — how chance, risk and rules shape opportunities

FACTUAL
How are outcomes listed, counted and assigned probabilities?

CONCEPTUAL
How do the rules for combined events follow from the structure of a sample space?

DEBATABLE
When is a game, a test or a decision genuinely fair?

IN THIS CHAPTER

13.1 Sample spaces
13.2 Tables and Venn
13.3 Tree diagrams
13.4 Independence
13.5 Expected frequency
13.6 Investigation

OPENING PROBLEM · A GAME WITH TWO DICE

A game with two dice

Two fair dice are rolled and the two scores are added. You win if the total is 7. Your friend wins if the total is 8. No money changes hands — only the right to choose the next game.

1. List every possible total and how many ways it can occur.
2. Find the probability that the total is 7 and the probability that it is 8.
3. If the game is played 72 times, how many wins would you expect each player to have?
4. Is the game fair? Explain using the probabilities, not the results of one game.
5. Change one rule so that both players have the same probability of winning.
6. Your friend says: "There are 11 possible totals, so the chance of a 7 must be $\frac{1}{11}$." Explain why this reasoning is wrong.

A total of 7 can happen in six ways; a total of 8 in only five. The game is not fair, and no amount of playing changes that. Probability begins with a complete list of equally likely outcomes — not with the totals, which are not equally likely, and not with intuition about "lucky" numbers. This chapter builds that list, then develops rules for combining events across two or more stages.

13.0 Learning intentions

ORIENTATION

- List a sample space systematically and calculate theoretical probabilities from equally likely outcomes.
- Use the complement rule and the bounds $0 \leq P(A) \leq 1$.
- Read and construct two-way tables and Venn diagrams, and use the addition rule for combined events.
- Use tree diagrams to combine probabilities across two or more stages, with and without replacement.
- Distinguish independent from dependent events and test for independence.
- Calculate expected frequencies and expected values, and use them to judge whether a game is fair.
- Simulate a chance experiment and compare experimental relative frequency with theoretical probability.
- Evaluate claims about risk and fairness using probability, assumptions and evidence.

SUCCESS CRITERIA · BY THE END OF THE CHAPTER I CAN

list a sample space without missing or repeating outcomes; calculate and interpret a probability; use complements and the addition rule; read probabilities from a two-way table or Venn diagram; draw and use a tree diagram; decide whether two events are independent; calculate an expected frequency and the expected value of a simple game; and explain, with calculations, whether a game is fair.

PREREQUISITE CHECK · 10 MINUTES

Do these without a calculator. If you miss more than two, revisit fractions in Chapter 2 and percentages in Chapter 5 before starting.

1. Write $\frac{3}{8}$ as a decimal and as a percentage.
2. Simplify $\frac{18}{24}$.
3. Calculate $1 - \frac{2}{7}$.
4. Calculate 0.6×0.6.
5. Write 35% as a fraction in simplest form.
6. List all two-letter arrangements of the letters A and B, allowing repeats.
7. How many outcomes are there when two coins are tossed?
8. One card is drawn from a standard pack of 52. How many hearts are there?

REFRESHER

A probability is a number from 0 to 1. It can be written as a fraction, a decimal or a percentage: $\frac{1}{4} = 0.25 = 25\%$. An impossible event has probability 0, a certain event has probability 1, and all the probabilities of the separate outcomes add up to 1. "At least one" is often easiest to handle through the complement: $P(\text{at least one}) = 1 - P(\text{none})$.

ATL SKILLS · THINKING, COMMUNICATION AND RESEARCH

Critical thinking: check whether outcomes are equally likely before dividing by the number of outcomes. **Communication:** state the sample space and the event in words before calculating, and give probabilities with their meaning in context. **Research:** when a claim about risk quotes a probability, ask for the denominator, the time period and the population.

LISTING WHAT CAN HAPPEN

13.1 Sample spaces and probability rules

A **probability experiment** is a process with an uncertain result: rolling a die, spinning a spinner, drawing a card. One possible result is an **outcome**. The set of all possible outcomes is the **sample space**, usually written S. An **event** is a subset of the sample space — one or more outcomes that share the property we are interested in.

For a fair die, $S = \{1, 2, 3, 4, 5, 6\}$. The event "roll an even number" is $A = \{2, 4, 6\}$. If the outcomes are **equally likely**, the probability of an event is the number of outcomes in the event divided by the number of outcomes in the sample space:

$$P(A) = \frac{n(A)}{n(S)}.$$

The word “fair” is doing important work here. A coin or die is fair when its outcomes are equally likely; a loaded die is not, and dividing by 6 would then be wrong. When outcomes are not equally likely, probabilities must come from data or from a model of the physical process.

KEY TERMS

Events and complements

Sample space S: the set of all possible outcomes. **Event A:** a subset of S. **Complement A':** the event that A does not happen.

Impossible event: $P = 0$. **Certain event:** $P = 1$. For every event, $0 \le P(A) \le 1$ and $P(A') = 1 - P(A)$.

Mutually exclusive events: events that cannot happen at the same time, so $P(A \text{ and } B) = 0$.

Theoretical probability comes from counting equally likely outcomes. **Experimental probability**, or relative frequency, comes from data: successes ÷ trials.

TABLE 13.1 SUMS OF TWO FAIR DICE

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

EXAMPLE **13.1** A SINGLE FAIR DIE

QUESTION

A fair six-sided die is rolled once. Find: (a) $P(\text{even})$; (b) $P(\text{prime})$; (c) the probability of a score greater than 4; (d) the probability of a score that is not prime.

SOLUTION

1 The sample space is $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$ and the outcomes are equally likely.

2 (a) Even numbers are 2, 4 and 6, so $P(\text{even}) = \frac{3}{6} = \frac{1}{2}$.

3 (b) The primes are 2, 3 and 5. Note that 1 is not prime. So $P(\text{prime}) = \frac{3}{6} = \frac{1}{2}$.

4 (c) Values greater than 4 are 5 and 6, so $P(\text{greater than } 4) = \frac{2}{6} = \frac{1}{3}$.

5 (d) By the complement rule, $P(\text{not prime}) = 1 - P(\text{prime}) = 1 - \frac{1}{2} = \frac{1}{2}$. The non-primes are 1, 4 and 6.

Check: every probability lies between 0 and 1, and each event has been listed explicitly. Listing prevents the common mistake of counting 1 as prime.

EXAMPLE **13.2** TWO COINS

QUESTION

Two fair coins are tossed. (a) Write down the sample space. (b) Find the probability of exactly one head. (c) Find the probability of at least one head. (d) Find the probability that the coins show the same face.

SOLUTION

1 Take the coins in order: $S = \{HH, HT, TH, TT\}$, so $n(S) = 4$. HT and TH are different outcomes because the first coin is different.

2 (b) Exactly one head occurs for HT and TH, so $P(\text{exactly one head}) = \frac{2}{4} = \frac{1}{2}$.

3 (c) At least one head occurs for HH, HT and TH, so the probability is $\frac{3}{4}$. Equivalently, $1 - P(TT) = 1 - \frac{1}{4} = \frac{3}{4}$.

4 (d) The same face occurs for HH and TT, so $P(\text{same}) = \frac{2}{4} = \frac{1}{2}$.

Check: the four outcomes are equally likely. Writing $S = \{HH, HT, TT\}$ would be wrong because HT and TH each have probability $\frac{1}{4}$.

EXAMPLE **13.3** COUNTERS IN A BAG

QUESTION

A bag contains 5 red, 3 blue and 2 green counters. One counter is taken at random. Find: (a) $P(\text{red})$, (b) $P(\text{not green})$, (c) $P(\text{red or blue})$, (d) the probability that the counter is not red.

SOLUTION

1 There are $5 + 3 + 2 = 10$ counters, all equally likely to be taken.

2 (a) $P(\text{red}) = \frac{5}{10} = \frac{1}{2}$.

3 (b) Two counters are green, so $P(\text{not green}) = 1 - \frac{2}{10} = \frac{8}{10} = \frac{4}{5}$.

4 (c) Red or blue covers $5 + 3 = 8$ counters, so $P(\text{red or blue}) = \frac{8}{10} = \frac{4}{5}$.

5 (d) $P(\text{not red}) = 1 - \frac{5}{10} = \frac{1}{2}$. The non-red counters are the 3 blue and 2 green counters.

Check: the complement of "red" is "blue or green", not "blue". Complements are about everything outside the event.

EXAMPLE **13.4** A STANDARD PACK OF CARDS

QUESTION

One card is drawn at random from a standard pack of 52. Find: (a) $P(\text{heart})$, (b) $P(\text{king})$, (c) $P(\text{red queen})$, (d) $P(\text{not a face card})$. Face cards are jacks, queens and kings.

SOLUTION

1 (a) There are 13 hearts, so $P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$.

2 (b) There are 4 kings, so $P(\text{king}) = \frac{4}{52} = \frac{1}{13}$.

3 (c) Only the queen of hearts and the queen of diamonds are red queens, so $P(\text{red queen}) = \frac{2}{52} = \frac{1}{26}$.

4 (d) There are $4 \times 3 = 12$ face cards, so 40 cards are not face cards. Hence $P(\text{not a face card}) = \frac{40}{52} = \frac{10}{13}$.

Check: "red queen" is not "red or queen". The words and, or and not change the event, so translate the phrase into a list before counting.

COMMON MISCONCEPTION

"Two possible outcomes means each has probability $\frac{1}{2}$." Only if the outcomes are equally likely. A biased coin can have $P(H) = 0.7$ and $P(T) = 0.3$.

"A probability of 0.3 means exactly 3 successes in 10 trials." It means that in a very large number of trials the relative frequency should be close to 0.3. Short runs can deviate substantially.

"The complement of 'at least one' is 'none' — but 'at least one' means one or more, not exactly one." Read the wording carefully; "exactly" and "at least" are different events.

KEY RELATIONSHIPS · PROBABILITY FROM EQUALLY LIKELY OUTCOMES

For a finite sample space with equally likely outcomes, $P(A) = \frac{n(A)}{n(S)}$. Also $P(A') = 1 - P(A)$, $P(\varnothing) = 0$ and $P(S) = 1$. For mutually exclusive events A and B, $P(A \text{ or } B) = P(A) + P(B)$.

EXERCISE 13A

SAMPLE SPACES AND SIMPLE PROBABILITY

FLUENCY

1 A fair die is rolled. Write the sample space and find the probability of rolling: (a) a 3, (b) an odd number, (c) a number less than 3, (d) a number greater than 6.

2 A fair spinner has eight equal sectors numbered 1 to 8. Find: (a) $P(5)$, (b) $P(\text{even})$, (c) $P(\text{a multiple of 3})$, (d) $P(\text{not 8})$.

3 A bag contains 6 red, 4 blue and 5 yellow counters. One is taken at random. Find: (a) $P(\text{red})$, (b) $P(\text{blue})$, (c) $P(\text{not yellow})$, (d) $P(\text{red or blue})$.

4 Two fair coins are tossed. List the sample space and find: (a) $P(\text{two tails})$, (b) $P(\text{at least one tail})$, (c) $P(\text{exactly one tail})$.

5 A card is drawn from a standard pack. Find: (a) $P(\text{spade})$, (b) $P(\text{ace})$, (c) $P(\text{black card})$, (d) $P(\text{not a heart})$.

6 The probability that a bus is late is 0.18. Find the probability that it is not late.

EXERCISE 13A · CONTINUED SAMPLE SPACES AND SIMPLE PROBABILITY

7 Explain why the sample space for two coins contains four outcomes, not three.

8 A student says: "The probability of rain is 0.4, so the probability of no rain is 0.4 because there are two possibilities." Identify the error and give the correct probability.

9 Two dice are rolled. Using Table 13.1, find: (a) $P(\text{sum} = 7)$, (b) $P(\text{sum} = 8)$, (c) $P(\text{sum at least } 10)$, (d) $P(\text{sum is even})$.

10 A fair die is rolled 60 times. How many times would you expect to roll a number greater than 4? Explain your reasoning.

11 Is it possible for an event to have probability 1.2? Explain using the definition of probability.

12 In a class of 25 students, 10 walk to school, 9 take the bus and 6 come by car. One student is chosen at random. Find the probability that the student does not walk.

MULTI-STEP AND UNFAMILIAR

13 A bag contains n counters, of which 7 are red. A counter is taken at random and $P(\text{red}) = 0.35$. Find n.

14 In a school, 240 students are asked their favourite sport. 96 choose football, 72 choose basketball and the rest choose another sport. One student is chosen at random. Find: (a) $P(\text{football})$, (b) $P(\text{not basketball})$, (c) $P(\text{another sport})$.

15 **Extension.** A three-sided spinner has faces labelled A, B and C. The spinner is spun twice. (a) List the sample space. (b) Find $P(\text{same letter twice})$. (c) Find the probability of at least one A. (d) Explain why the outcomes of the sample space are equally likely only if the spinner is fair.

16 **Extension.** A student claims that because the probability of rolling a 6 is $\frac{1}{6}$, rolling a die six times guarantees exactly one 6. Evaluate this claim using a complete sentence and a counterexample.

TWO PROPERTIES AT ONCE

13.2 Two-way tables and Venn diagrams

Many questions involve two characteristics: whether a student studies French and whether they study Spanish; whether a person owns a cat and whether they own a dog. A **two-way table** arranges counts by both characteristics. Its row and column totals are the **marginal totals**; the count in each interior cell is a **joint frequency**.

A **Venn diagram** shows the same information as overlapping regions. The overlap is the **intersection** $A \cap B$, read "A and B". The combined region is the **union** $A \cup B$, read "A or B", where "or" includes both. The region outside both circles is the complement of the union.

If the overlap is counted twice when the two events are added, the double count must be removed:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

When A and B cannot both happen, $P(A \cap B) = 0$ and the events are **mutually exclusive**; then the probabilities simply add.

KEY TERMS

Union, intersection and complement

$A \cap B$: both A and B occur. $A \cup B$: A or B or both occur. $(A \cup B)'$: neither A nor B occurs.

Mutually exclusive: $A \cap B = \varnothing$.

Addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

In a two-way table, the total of all four interior cells is the size of the whole sample. A probability is found by dividing a cell or a row/column total by that grand total.

TABLE 13.2 LANGUAGES STUDIED BY 120 STUDENTS

	FRENCH	NO FRENCH	TOTAL
Spanish	25	45	70
No Spanish	20	30	50
Total	45	75	120

EXAMPLE **13.5** PROBABILITIES FROM A TWO-WAY TABLE

QUESTION

Table 13.2 shows the languages studied by 120 students. One student is chosen at random. Find: (a) $P(\text{Spanish})$, (b) $P(\text{French})$, (c) $P(\text{both})$, (d) $P(\text{Spanish or French})$, (e) $P(\text{neither})$.

SOLUTION

1 (a) 70 students study Spanish, so $P(\text{Spanish}) = \frac{70}{120} = \frac{7}{12}$.

2 (b) 45 students study French, so $P(\text{French}) = \frac{45}{120} = \frac{3}{8}$.

3 (c) 25 study both, so $P(\text{both}) = \frac{25}{120} = \frac{5}{24}$.

4 (d) Using the addition rule, $\frac{70}{120} + \frac{45}{120} - \frac{25}{120} = \frac{90}{120} = \frac{3}{4}$. Equivalently, $25 + 45 + 20 = 90$ students study at least one language.

5 (e) 30 students study neither, so $P(\text{neither}) = \frac{30}{120} = \frac{1}{4}$. This is also $1 - \frac{3}{4}$.

Check: $70 + 45 - 25 = 90$, and $90 + 30 = 120$. The arithmetic of the table and the addition rule agree.

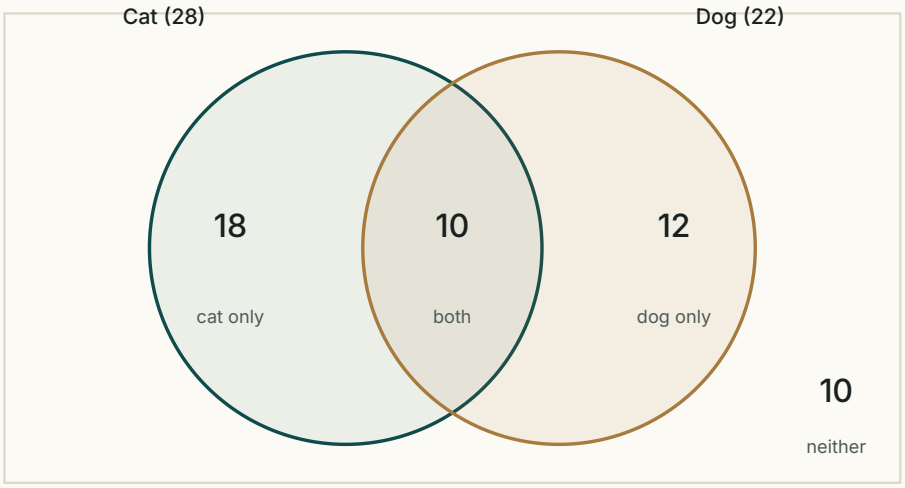


Figure 13.1 A Venn diagram for 50 people. The overlap contains the 10 people who own both animals. The regions outside the circles contain the 10 people who own neither.

EXAMPLE **13.6** READING A VENN DIAGRAM

QUESTION

In a group of 50 people, 28 own a cat, 22 own a dog and 10 own both. (a) Complete the Venn diagram in Figure 13.1. (b) Find the probability that a person chosen at random owns exactly one of the two animals. (c) Find the probability that the person owns neither.

SOLUTION

1 The overlap is 10. Cat only is $28 - 10 = 18$; dog only is $22 - 10 = 12$.

2 The number owning at least one animal is $18 + 10 + 12 = 40$, so $50 - 40 = 10$ own neither.

3 (b) Exactly one animal means cat only or dog only: $18 + 12 = 30$, so $P(\text{exactly one}) = \frac{30}{50} = \frac{3}{5}$.

4 (c) $P(\text{neither}) = \frac{10}{50} = \frac{1}{5}$.

Check: $\frac{3}{5} + \frac{1}{5} + \frac{10}{50} = 1$. The three regions “exactly one”, “both” and “neither” partition the group.

EXAMPLE **13.7** USING THE ADDITION RULE WITH CARDS

QUESTION

One card is drawn from a standard pack of 52. Find the probability that it is a heart or a king. Explain why the probabilities cannot simply be added.

SOLUTION

1. There are 13 hearts and 4 kings, but the king of hearts belongs to both events.
2. Using the addition rule: $P(H \cup K) = P(H) + P(K) - P(H \cap K) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$.
3. Adding $\frac{13}{52} + \frac{4}{52}$ would count the king of hearts twice and give $\frac{17}{52}$, which is too large.

Check: list the favourable cards: 12 hearts that are not kings, plus 4 kings, gives 16. The count agrees with the addition rule.

EXAMPLE **13.8** TESTING A CLAIM FROM A TABLE

QUESTION

In a survey of 200 students, 110 play a sport, 80 play a musical instrument and 40 do both. A student says: "Most students who play a sport also play an instrument." Evaluate the claim.

SOLUTION

1. Of the 110 students who play a sport, 40 also play an instrument. The relevant fraction is $\frac{40}{110} = \frac{4}{11} \approx 0.36$.
2. Only about 36% of sport players also play an instrument, so "most" is false.
3. The claim confuses $P(\text{instrument} \mid \text{sport})$ with $P(\text{both})$ or with $P(\text{sport} \mid \text{instrument})$. The denominator matters: among the 80 musicians, 40 play a sport, which is 50%, still not "most" of all students.
4. A correct statement is: "About 36% of students who play a sport also play an instrument."

Check: the conditional probability uses the group named after "of" or "given" as the denominator. Here the group is sport players, so the denominator is 110.

COMMON MISCONCEPTION

"$P(A \text{ or } B) = P(A) + P(B)$ always." This is true only when the events are mutually exclusive. If they overlap, subtract the intersection.

"Mutually exclusive means independent." They are different ideas. Mutually exclusive events cannot occur together; independent events do not influence each other. In fact, if two events with non-zero probability are mutually exclusive, they are dependent: knowing one occurred tells you the other did not.

"The number outside the circles is zero." Always check whether some members belong to neither set; that region is often the key to the total.

KEY RELATIONSHIPS · COMBINING TWO EVENTS

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$. Also $P(\text{neither}) = 1 - P(A \cup B)$, and $P(A \text{ only}) = P(A) - P(A \cap B)$.

EXERCISE 13B

TWO-WAY TABLES AND VENN DIAGRAMS

FLUENCY

1 A survey of 80 students records whether they own a phone and whether they own a tablet. 55 own a phone, 30 own a tablet and 20 own both. Draw a two-way table and find $P(\text{phone})$, $P(\text{tablet})$ and $P(\text{both})$.

2 In a group of 40 people, 18 speak French, 15 speak Spanish and 7 speak both. Find: (a) $P(\text{French or Spanish})$, (b) $P(\text{neither})$, (c) $P(\text{French only})$.

3 A card is drawn from a standard pack. Find $P(\text{red or ace})$. Explain why the answer is not $\frac{26}{52} + \frac{4}{52}$.

4 In a class of 30, 17 play football, 12 play tennis and 6 play both. Draw a Venn diagram and find the number who play neither.

5 A two-way table shows 120 people: 48 read a newspaper daily, 36 read a magazine daily and 12 read both. Find $P(\text{newspaper only})$ and $P(\text{neither})$.

6 Two events A and B are mutually exclusive with $P(A) = 0.3$ and $P(B) = 0.45$. Find $P(A \cup B)$ and $P(\text{neither})$.

UNDERSTANDING AND REASONING

7 Explain why a Venn diagram and a two-way table can show exactly the same information. Give one advantage of each.

8 A student calculates $P(A \cup B) = 0.6 + 0.5 = 1.1$ and concludes that the events are impossible. Identify the error and state the smallest possible value of $P(A \cap B)$.

9 In a group of 60 students, 25 study history, 30 study geography and 12 study neither. Find the number who study both subjects.

10 Give an example of two events that are mutually exclusive and two events that are not, using a single roll of a fair die.

11 A survey finds that 70% of students own a phone, 40% own a laptop and 25% own both. Find the percentage who own at least one and the percentage who own neither.

12 A Venn diagram has $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Find $P(A \cup B)$, $P(A \text{ only})$ and $P(\text{neither})$.

MULTI-STEP AND UNFAMILIAR

13 In a year group of 180 students, 108 study a language, 72 study an art subject and 36 study both. A student is chosen at random. Find $P(\text{language or art})$, $P(\text{neither})$ and $P(\text{language only})$.

14 A questionnaire is completed by 250 people. 140 have a gym membership, 90 have a swimming membership and 50 have both. (a) Draw a two-way table. (b) Find the probability that a person has exactly one membership. (c) A claim says “more than half of gym members also swim”. Evaluate the claim.

15 **Extension.** In a group of 100 people, 45 own a bicycle, 38 own a scooter and 20 own neither. Find the number who own both. Then find $P(\text{bicycle only})$.

16 **Extension.** Explain why $P(A \cup B) \leq P(A) + P(B)$ for any two events. Under what condition does equality hold?

ONE STAGE AFTER ANOTHER

13.3 Tree diagrams

When an experiment has two or more stages, a **tree diagram** shows the stages in order. Each **branch** represents one possible outcome at that stage, and its probability is written on the branch. A complete path from the start to an end is one outcome of the whole experiment. To find the probability of a path, **multiply** the probabilities along its branches. To find the probability of an event made of several paths, **add** the probabilities of those paths.

Second-stage probabilities may be the same at every branch, or they may depend on what happened first. Drawing **with replacement** restores the object, so the second-stage probabilities repeat. Drawing **without replacement** removes an object, so both the numerator and the denominator can change.

A useful check: the probabilities at every set of branches leaving one point must add to 1, and the probabilities of all complete paths must also add to 1.

KEY TERMS

Tree diagrams and replacement

Tree diagram: a diagram in which each stage of an experiment branches into its possible outcomes. **Path:** one sequence of outcomes from start to finish.

Multiplication rule for paths: $P(\text{path}) =$ product of the branch probabilities along it. **Addition rule for paths:** if several paths make up an event, add their probabilities.

With replacement: the object is returned, so the composition is unchanged. **Without replacement:** the object is not returned, so later probabilities use the reduced totals.

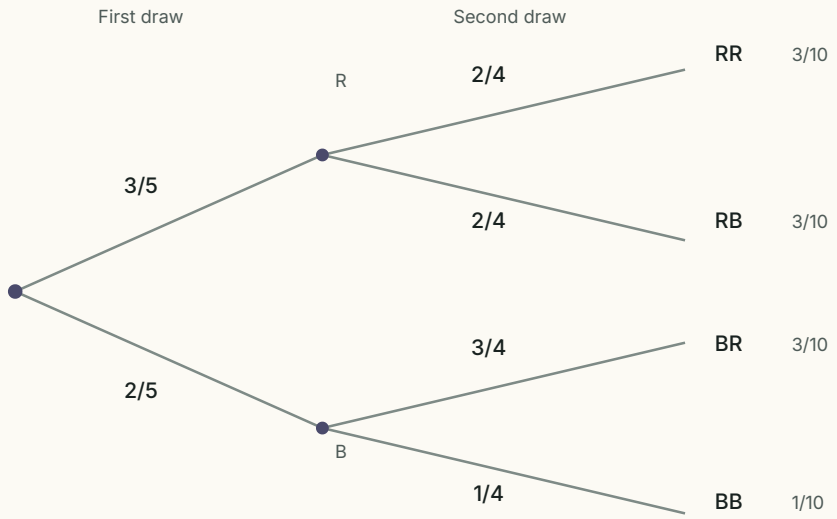


Figure 13.2 Two counters drawn without replacement from a bag containing 3 red and 2 blue counters. After a red counter is removed, 2 red and 2 blue remain; after a blue counter is removed, 3 red and 1 blue remain.

EXAMPLE 13.9 TWO DRAWS WITHOUT REPLACEMENT

QUESTION

A bag contains 3 red and 2 blue counters. Two counters are taken at random without replacement. (a) Draw a tree diagram. (b) Find $P(\text{both red})$. (c) Find $P(\text{one of each colour})$. (d) Find $P(\text{at least one red})$.

SOLUTION

1. (a) The tree is shown in Figure 13.2. First draw: $P(R) = \frac{3}{5}$, $P(B) = \frac{2}{5}$. If red is taken, 2 red and 2 blue remain; if blue is taken, 3 red and 1 blue remain.
2. (b) $P(RR) = \frac{3}{5} \times \frac{2}{4} = \frac{6}{20} = \frac{3}{10}$.
3. (c) One of each is RB or BR: $\frac{3}{5} \times \frac{2}{4} + \frac{2}{5} \times \frac{3}{4} = \frac{6}{20} + \frac{6}{20} = \frac{12}{20} = \frac{3}{5}$.
4. (d) At least one red is the complement of BB: $1 - \frac{2}{5} \times \frac{1}{4} = 1 - \frac{2}{20} = \frac{18}{20} = \frac{9}{10}$.

Check: $\frac{3}{10} + \frac{3}{10} + \frac{3}{10} + \frac{1}{10} = 1$. The four paths are mutually exclusive and exhaust the sample space.

EXAMPLE 13.10 TWO INDEPENDENT SPINS

QUESTION

A spinner lands on A with probability 0.6 and on B with probability 0.4. It is spun twice. (a) Draw a tree diagram. (b) Find $P(\text{two A})$. (c) Find $P(\text{exactly one A})$. (d) Find $P(\text{at least one A})$.

SOLUTION

1. (a) Each first branch splits into A (0.6) and B (0.4); each of those splits again into A (0.6) and B (0.4), because the spinner does not remember the first spin.
2. (b) $P(AA) = 0.6 \times 0.6 = 0.36$.
3. (c) Exactly one A is AB or BA: $0.6 \times 0.4 + 0.4 \times 0.6 = 0.24 + 0.24 = 0.48$.
4. (d) $P(\text{at least one A}) = 1 - P(BB) = 1 - 0.4 \times 0.4 = 1 - 0.16 = 0.84$.

Check: $0.36 + 0.24 + 0.24 + 0.16 = 1$. Also $P(\text{exactly one A}) + P(\text{two A}) = 0.48 + 0.36 = 0.84$, matching part (d).

EXAMPLE **13.11** A TREE WHOSE LATER BRANCHES DEPEND ON THE FIRST STAGE

QUESTION

The probability of rain on Saturday is 0.3. If it rains on Saturday, the probability of rain on Sunday is 0.6; if it does not rain on Saturday, the probability of rain on Sunday is 0.2. Find: (a) $P(\text{rain on both days})$, (b) $P(\text{rain on Sunday})$, (c) $P(\text{rain on at least one day})$.

SOLUTION

1. First branches: rain 0.3, no rain 0.7. Second branches: after rain, 0.6 rain and 0.4 no rain; after no rain, 0.2 rain and 0.8 no rain.
2. (a) $P(\text{rain both}) = 0.3 \times 0.6 = 0.18$.
3. (b) Rain on Sunday happens along two paths: $0.3 \times 0.6 + 0.7 \times 0.2 = 0.18 + 0.14 = 0.32$.
4. (c) Rain on at least one day is $0.18 + 0.3 \times 0.4 + 0.7 \times 0.2 = 0.18 + 0.12 + 0.14 = 0.44$. Equivalently, $1 - P(\text{no rain both}) = 1 - 0.7 \times 0.8 = 0.44$.

Check: the path probabilities sum to $0.18 + 0.12 + 0.14 + 0.56 = 1$. The four paths describe all combinations of the two days.

COMMON MISCONCEPTION

"Add along the branches." Multiply along a path; add across paths that make up the event. Adding along a path can produce a probability greater than 1.

"The second draw always uses the original total." Only with replacement. Without replacement, update both the number of the chosen colour and the total.

"Every path has the same probability." Paths are equally likely only when every branch probability is equal. In Figure 13.2, RR and BB have different probabilities.

EXERCISE 13C

TREE DIAGRAMS

FLUENCY

1. A bag contains 4 green and 3 yellow counters. Two are drawn without replacement. Draw a tree diagram and find $P(\text{both green})$.
2. Using the same bag as Question 1, find $P(\text{one of each colour})$ and $P(\text{at least one yellow})$.
3. A fair coin is tossed three times. Draw a tree diagram and find $P(\text{three heads})$, $P(\text{exactly two heads})$ and $P(\text{at least one tail})$.
4. A biased coin has $P(H) = 0.7$. It is tossed twice. Find $P(HH)$, $P(\text{exactly one head})$ and $P(\text{at least one tail})$.
5. A bag contains 5 red and 3 blue counters. Two are drawn with replacement. Find $P(RR)$, $P(RB)$ and $P(\text{same colour})$
6. A spinner lands on red with probability 0.25, blue with probability 0.45 and green with probability 0.3. It is spun twice. Find $P(\text{red then blue})$ and $P(\text{green both times})$.

UNDERSTANDING AND REASONING

7. Explain the difference between drawing with replacement and without replacement, using a bag of counters as an example.
8. A bag contains 5 red and 3 blue counters, and two are drawn without replacement. A student writes $\frac{5}{5}$ on the second red branch of the tree diagram. Explain what is wrong and give the correct fraction.

EXERCISE 13C · CONTINUED TREE DIAGRAMS

9 In a two-stage experiment, the path probabilities are 0.12, 0.18, 0.28 and 0.42. Do these form a complete tree? Explain.

10 A bag has 2 red and 6 blue counters. Two are drawn without replacement. Find the probability that both are blue and the probability that at least one is red.

11 A test is taken by two students independently. The probability that each passes is 0.8. Find the probability that both pass and the probability that exactly one passes.

12 Why is it useful to check that the probabilities of all complete paths add to 1?

MULTI-STEP AND UNFAMILIAR

13 A box contains 6 working and 2 faulty batteries. Two are taken without replacement. (a) Draw a tree diagram. (b) Find $P(\text{both working})$. (c) Find $P(\text{at least one faulty})$.

14 On any day, the probability that a student cycles to school is 0.4. If they cycle one day, the probability they cycle the next day is 0.7; if they do not cycle, the probability they cycle the next day is 0.2. Find the probability that they cycle on both days and the probability that they cycle on exactly one of the two days.

15 **Extension.** A bag contains r red and b blue counters. Two are drawn without replacement. Write expressions for $P(RR)$, $P(RB)$ and $P(\text{one of each colour})$. Check that your expressions give the values in Example 13.9 when $r = 3$ and $b = 2$.

16 **Extension.** Three counters are drawn without replacement from a bag containing 3 red and 3 blue counters. Find the probability that all three are the same colour and the probability that exactly two are red.

DOES ONE EVENT CHANGE THE OTHER?

13.4 Independent and dependent events

Two events are **independent** when the occurrence of one does not change the probability of the other. Tossing two coins, spinning a spinner twice, or rolling a die and flipping a coin are independent: the first result leaves the second unchanged. Drawing counters without replacement creates **dependent** events, because the first draw changes what remains.

For independent events, the multiplication rule is exact:

$$P(A \cap B) = P(A) \times P(B).$$

If this equation fails, the events are dependent. A **conditional probability** $P(B \mid A)$ is the probability of B given that A has happened. It is the natural language for dependent events: the probability of a second heart, given that the first card was a heart, is $\frac{12}{51}$, not $\frac{13}{52}$.

KEY TERMS

Independence and conditional probability

Independent events: $P(A \cap B) = P(A)P(B)$, equivalently $P(B \mid A) = P(B)$.
Dependent events: the occurrence of one changes the probability of the other.

Conditional probability: $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)}$, provided (P(A)gt0). It restricts the sample space to the outcomes in A.

Warning: mutually exclusive and independent are not the same. Mutually exclusive events cannot occur together; independent events do not affect each other.

EXAMPLE **13.12** TESTING INDEPENDENCE WITH DICE

QUESTION

Two fair dice are rolled. Let A be "the first die is even" and B be "the second die is 6". Are A and B independent?

SOLUTION

1 $P(A) = \frac{3}{6} = \frac{1}{2}$ and $P(B) = \frac{1}{6}$.

2 Both occur when the first die is 2, 4 or 6 and the second is 6: 3 outcomes out of 36, so $P(A \cap B) = \frac{3}{36} = \frac{1}{12}$.

3 $P(A) \times P(B) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$, which equals $P(A \cap B)$. The events are independent.

Check: knowing that the first die is even tells you nothing about the second die. The dice do not communicate.

EXAMPLE **13.13** TESTING INDEPENDENCE WITH A TWO-WAY TABLE

QUESTION

In a group of 200 students, 110 play a sport, 80 play an instrument and 40 do both. Let S be "plays a sport" and M be "plays an instrument". Are S and M independent?

SOLUTION

1 $P(S) = \frac{110}{200} = 0.55$, $P(M) = \frac{80}{200} = 0.40$ and $P(S \cap M) = \frac{40}{200} = 0.20$.

2 If independent, $P(S \cap M)$ would be $0.55 \times 0.40 = 0.22$.

3 The actual value 0.20 is less than 0.22, so the events are dependent. Students who play a sport are slightly less likely than others to play an instrument.

4 The conditional probability is $P(M \mid S) = \frac{40}{110} \approx 0.364$, compared with $P(M) = 0.40$.

Check: independence is a numerical test, not a judgement about whether the activities "seem related". The gap here is small but the counts are exact for this group.

EXAMPLE **13.14** TWO CARDS WITHOUT REPLACEMENT

QUESTION

Two cards are drawn from a standard pack without replacement. Find: (a) $P(\text{both hearts})$, (b) $P(\text{second is a heart} \mid \text{first is a heart})$, (c) the probability with replacement instead.

SOLUTION

1 (a) $P(\text{both hearts}) = \frac{13}{52} \times \frac{12}{51} = \frac{156}{2652} = \frac{1}{17} \approx 0.0588$.

2 (b) Given that the first card is a heart, 12 hearts remain among 51 cards, so $P(\text{second heart} \mid \text{first heart}) = \frac{12}{51} = \frac{4}{17}$.

3 (c) With replacement, the pack is restored: $\frac{13}{52} \times \frac{13}{52} = \frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = 0.0625$.

Check: (0.0588t0.0625). Removing a heart makes a second heart slightly less likely, which is exactly what dependence means.

EXAMPLE **13.15** INDEPENDENT FAULTS IN A PRODUCTION LINE

QUESTION

A machine produces items of which 2% are defective, independently of one another. Two items are chosen at random. Find: (a) $P(\text{both defective})$, (b) $P(\text{neither defective})$, (c) $P(\text{at least one defective})$.

SOLUTION

1 (a) Independence allows multiplication: $0.02 \times 0.02 = 0.0004$, or 4 in 10,000.

2 (b) Each item is non-defective with probability 0.98, so $0.98 \times 0.98 = 0.9604$.

3 (c) $P(\text{at least one defective}) = 1 - 0.9604 = 0.0396$, about 4%.

Check: "at least one" is not $0.02 + 0.02 = 0.04$, because that double-counts the case where both are defective. The complement method avoids the double count.

COMMON MISCONCEPTION

"Independent means mutually exclusive." Independent events can happen together; in fact, if $P(A)$ and $P(B)$ are both positive and the events are independent, they must be able to happen together.

"Without replacement, the probability is the same because the bag still has the same colours." The colours remain, but the counts and the total change. Use the new totals.

"$P(A \cap B) = P(A)P(B)$ **always."** Only for independent events. For dependent events, use a tree or the conditional probability.

EXERCISE 13D

INDEPENDENT AND DEPENDENT EVENTS

FLUENCY

1 $P(A) = 0.5$, $P(B) = 0.4$ and A, B are independent. Find $P(A \cap B)$ and $P(A \cup B)$.

2 $P(C) = 0.6$, $P(D) = 0.3$ and $P(C \cap D) = 0.18$. Are C and D independent? Explain.

3 A fair coin is tossed and a fair die is rolled. Find $P(\text{head and 6})$ and $P(\text{tail and even})$.

4 Two cards are drawn without replacement. Find $P(\text{both aces})$.

EXERCISE 13D · CONTINUED INDEPENDENT AND DEPENDENT EVENTS

5 The probability that a student is late on any day is 0.1, independently of other days. Find the probability that the student is late on two consecutive days.

6 $P(E) = 0.7$ and $P(F) = 0.2$. If E and F are mutually exclusive, find $P(E \cup F)$. Can they also be independent? Explain.

UNDERSTANDING AND REASONING

7 Explain the difference between independent and mutually exclusive events using a single roll of a die.

8 A student says: "The first coin landed heads, so the second coin is more likely to land tails." Identify the misconception and give the correct probability.

9 A bag contains 4 red and 6 blue counters. Two are drawn without replacement. Find $P(\text{second is red} \mid \text{first is blue})$.

10 In a survey, $P(\text{owns a bike}) = 0.6$, $P(\text{owns a helmet}) = 0.5$ and $P(\text{owns both}) = 0.3$. Are the events independent? Justify with a calculation.

11 Why does drawing without replacement make the second draw dependent on the first, while drawing with replacement does not?

12 A test for a rare disease is 95% accurate. Does that mean 95% of people who test positive have the disease? Explain why not, in one or two sentences.

MULTI-STEP AND UNFAMILIAR

13 A bag contains 3 red, 2 green and 5 blue counters. Two are drawn without replacement. Find $P(\text{both blue})$ and $P(\text{different colours})$.

14 In a school, 60% of students walk, 45% bring a packed lunch and 30% do both. (a) Find the probability a student does neither. (b) Are walking and bringing a packed lunch independent? Justify.

15 **Extension.** Two events A and B satisfy $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$. Find $P(A \cap B)$ and decide whether A and B are independent.

16 **Extension.** A fair die is rolled twice. Are "the sum is 7" and "the first die is 3" independent? Show your reasoning with probabilities.

WHAT SHOULD HAPPEN IN THE LONG RUN

13.5 Expected frequency, simulation and fairness

Probability predicts proportions, not individual results. If an event has probability p and the experiment is repeated n times, the **expected frequency** is $n \times p$. A fair die is expected to show a 6 about 100 times in 600 rolls, but no one expects exactly 100 every time.

When a game has several possible payoffs, multiply each payoff by its probability and add. The result is the **expected value**. A game is **fair** when the expected net gain is zero: over many plays, neither player has an advantage. In a raffle, the expected net gain of a ticket is the expected prize minus the cost of the ticket.

A **simulation** imitates a chance experiment many times, often with random numbers or a spreadsheet. The **relative frequency** of an event in the simulation should approach its theoretical probability as the number of trials grows. Simulation is useful when the sample space is too large or too complicated to list completely.

KEY TERMS

Expectation and fairness

Expected frequency: $n \times P(A)$, where n is the number of trials. **Expected value:** $\sum x\,P(X = x)$, where x runs over the possible payoffs.

Fair game: expected net gain $= 0$. **Biased game:** expected net gain is non-zero; one player profits in the long run.

Simulation: repeated imitation of an experiment. **Relative frequency:** successes divided by trials; it estimates probability and becomes more stable as n increases.

EXAMPLE **13.16** EXPECTED FREQUENCIES FOR A SPINNER

QUESTION

A spinner has $P(\text{red}) = 0.5$, $P(\text{blue}) = 0.3$ and $P(\text{green}) = 0.2$. It is spun 250 times. (a) Find the expected frequency of each colour. (b) The actual frequencies are 118 red, 82 blue and 50 green. Comment.

SOLUTION

1 (a) Red: $250 \times 0.5 = 125$. Blue: $250 \times 0.3 = 75$. Green: $250 \times 0.2 = 50$.

2 (b) The observed counts 118, 82 and 50 are close to 125, 75 and 50, but not identical. Random variation is expected; the relative frequencies 0.472, 0.328 and 0.2 are close to the model probabilities.

3 A single run of 250 spins is not proof that the spinner is biased. A larger number of spins would give a more stable comparison.

Check: $118 + 82 + 50 = 250$, and the expected frequencies also add to 250. The probabilities of all outcomes must sum to 1.

EXAMPLE **13.17** THE EXPECTED VALUE OF A DIE GAME

QUESTION

A game costs £3 to play. A fair die is rolled; if it shows 6, the player wins £12; otherwise the player wins nothing. (a) Find the expected winnings and the expected net gain. (b) Find the prize that would make the game fair.

SOLUTION

1 (a) Expected winnings $= 12 \times \frac{1}{6} + 0 \times \frac{5}{6} = 2$ pounds.

2 Expected net gain $= 2 - 3 = -1$ pound. On average the player loses £1 per game.

3 (b) For a fair game the expected winnings must equal the £3 cost: $p \times \frac{1}{6} = 3$, so $p = 18$. A prize of £18 makes the expected net gain zero.

Check: a fair game does not mean that every player breaks even; it means the average net gain over many plays is zero.

EXAMPLE **13.18** A SCHOOL RAFFLE

QUESTION

A school sells 200 raffle tickets at £1 each. There is one prize worth £150. (a) Find the expected net gain of one ticket. (b) Find the expected profit for the school. (c) Comment on whether the raffle is fair.

SOLUTION

1. (a) Expected prize $= 150 \times \frac{1}{200} = 0.75$. Expected net gain $= 0.75 - 1 = -0.25$, so a ticket loses 25 pence on average.
2. (b) The school receives £200 and pays £150, so the expected profit is £50. Equivalently, $200 \times 0.25 = 50$.
3. (c) The raffle is not fair to ticket buyers: their expected net gain is negative. It is a fundraiser, so this is intentional, but the school should state the odds and the purpose clearly.

Check: the expected net gains of all players and the organizer add to zero. The buyers lose £50 in expectation; the school gains £50.

EXAMPLE **13.19** READING A SIMULATION

QUESTION

A student simulates rolling a fair die. The relative frequency of a 6 after 10, 50, 100 and 500 rolls is 0.2, 0.16, 0.15 and 0.168. Compare these with the theoretical probability and explain the pattern.

SOLUTION

1. The theoretical probability is $\frac{1}{6} \approx 0.167$.
2. After only 10 rolls, 0.2 is not far from 0.167 in absolute terms, but a small sample can produce a noticeable difference.
3. As the number of rolls increases, the relative frequency moves closer to 0.167: 0.15 at 100 rolls and 0.168 at 500 rolls.
4. The pattern is consistent with the law of large numbers: relative frequency tends to settle near the theoretical probability as trials increase, but it is not guaranteed to hit it exactly.

Check: a simulation is evidence about a model, not proof that the die is fair. The model itself — equally likely faces — must also be justified.

COMMON MISCONCEPTION

"Expected means guaranteed." An expected frequency is an average over many repetitions, not a prediction for one run.

"A fair game means nobody can win." A fair game can have winners and losers; the expected net gain is zero over many plays.

"If a coin has landed heads five times, tails is due." Independent trials have no memory. The probability of tails remains $\frac{1}{2}$.

"The simulation proves the probability." A simulation estimates it. The theoretical model supplies the probability; the simulation checks whether the model is plausible.

KEY RELATIONSHIPS · EXPECTATION

Expected frequency $= nP(A)$. Expected value $= \sum x\,P(X = x)$. Expected net gain = expected winnings − cost. A game is fair when the expected net gain is 0. The sum of the probabilities of all outcomes is 1.

EXERCISE 13E

EXPECTED FREQUENCY AND FAIRNESS

FLUENCY

1 A fair die is rolled 300 times. Find the expected number of 4s.

2 A spinner has $P(\text{win}) = 0.35$. It is spun 400 times. Find the expected number of wins.

3 A bag has 7 red and 3 blue counters. One counter is drawn 500 times with replacement. Find the expected number of red counters.

4 A game costs £2. You win £10 with probability 0.1 and nothing otherwise. Find the expected net gain.

5 A raffle sells 500 tickets at £2 each with one prize of £300. Find the expected net gain of one ticket.

6 A coin is tossed 80 times. Find the expected number of heads and the expected number of tails.

UNDERSTANDING AND REASONING

7 Explain why the expected number of 6s in 60 rolls of a fair die is 10, but you would not be surprised to see 13.

8 A game has expected net gain −£0.50. What does this mean for a player who plays 100 times? For the organizer?

9 A student says: "The expected value is 2.5, so I will win 2.5 pounds." Explain the error.

10 A simulation of 20 trials gives a relative frequency of 0.4 for an event with theoretical probability 0.25. Is this necessarily evidence that the model is wrong? Explain.

11 Why must the expected frequencies of all outcomes add to the number of trials?

12 A game is fair. What can you say about the expected winnings and the cost of playing?

MULTI-STEP AND UNFAMILIAR

13 A game uses a fair die: you win £6 if you roll a 1 or 2, win £2 if you roll a 3, and win nothing otherwise. It costs £3 to play. Find the expected net gain and state whether the game is fair.

EXERCISE 13E · CONTINUED EXPECTED FREQUENCY AND FAIRNESS

14 A school fair game costs £2. A spinner lands on "win" with probability 0.2 and pays £7; otherwise it pays nothing. (a) Find the expected net gain. (b) Find the payout that would make the game fair. (c) The school wants to raise £200 from 500 plays. Is the original game suitable? Explain.

15 **Extension.** A bag contains 4 red and 6 blue counters. You win £5 for a red counter and pay £2 for a blue counter. Find the expected net gain per draw. What should the red payout be for a fair game?

16 **Extension.** Design a two-dice game that costs £1 to play and has an expected net gain of zero. Show the sample space counts and the expected value.

INVESTIGATION · CRITERION B AND C

13.6 Is the game fair?

A game at a school fair uses two fair dice. The player wins if the total is less than 7. The stallholder wins if the total is 7 or more. The stallholder says the game is fair "because both players have a good chance". Before accepting or rejecting that claim, count the outcomes. A game is fair only when the expected net gain is the same for both players — or zero for each.

INVESTIGATION · CRITERION B AND C

Less than seven, or seven and above

17 Use Table 13.1 to count the outcomes with total less than 7 and the outcomes with total 7 or more. Check that the counts add to 36.

18 Find the probability that the player wins and the probability that the stallholder wins.

19 The game is played 120 times. Find the expected number of wins for the player and for the stallholder.

20 Simulate 40 games using two dice, a calculator or a spreadsheet. Record the player's wins in a frequency table and calculate the relative frequency.

21 Compare your simulated relative frequency with the theoretical probability. Explain why they are unlikely to be identical.

22 Propose a rule that makes the game fair. Justify your rule with a sample-space count, not with the result of a few games.

23 A student plays 10 games, wins 3 and says: "That proves the game is fair." Explain why 10 games are weak evidence.

Generalise and justify. Write three conditions that a game must satisfy before you would call it fair. Then explain how the expected value of a game connects the probabilities of its outcomes to the long-run profit or loss of each player.

MIXED PRACTICE 13

CUMULATIVE RETRIEVAL · NO NOTES

Attempt all questions without notes. Questions 1–6 check fluency; 7–14 require reasoning; 15–19 are multi-step. Allow 55 minutes.

FLUENCY

1 A bag contains 5 red, 3 blue and 2 green counters. One is drawn. Find $P(\text{red})$, $P(\text{not green})$ and $P(\text{red or blue})$.

2 The probability that a train is delayed is 0.12. Find the probability that it is not delayed.

3 In a group of 100 people, 60 drink tea, 45 drink coffee and 25 drink both. Find the probability that a person drinks tea or coffee and the probability that they drink neither.

4 A bag has 3 red and 2 blue counters. Two are drawn with replacement. Find $P(RR)$, $P(RB)$ and $P(\text{same colour})$.

5 $P(A) = 0.6$, $P(B) = 0.5$ and the events are independent. Find $P(A \cap B)$ and $P(A \cup B)$.

6 A spinner has $P(\text{win}) = 0.4$. It is spun 350 times. Find the expected number of wins.

UNDERSTANDING AND REASONING

7 Explain why the sample space for two coins has four outcomes, not three.

8 A student calculates $P(A \cup B) = 0.7 + 0.6 = 1.3$. Explain what has gone wrong and state the smallest possible value of $P(A \cap B)$.

9 Two cards are drawn without replacement. Find the probability that both are kings.

10 Explain why $P(\text{at least one head in three tosses}) = 1 - P(\text{three tails})$.

11 A game costs £3 and pays £10 with probability 0.25. Find the expected net gain and state whether the game is fair.

12 A simulation of 100 trials gives a relative frequency of 0.28 for an event with theoretical probability 0.25. Is this good evidence that the model is wrong? Explain.

13 In a two-way table, $P(A) = 0.45$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$. Find $P(A \cup B)$, $P(\text{neither})$ and $P(A \text{ only})$.

14 A tree has first branches 0.3 and 0.7. The second branches after the first outcome are 0.4/0.6 and 0.2/0.8. Find the probability of the path with first outcome 0.3 and second outcome 0.6.

MULTI-STEP AND UNFAMILIAR

15 A bag contains 4 red and 6 blue counters. Three are drawn without replacement. Find the probability that exactly two are red.

16 A game uses a fair die. You win £6 if you roll a 1 or 2, £3 if you roll a 3, and nothing otherwise. It costs £3 to play. Find the expected net gain and the fair cost of playing.

17 A school raffle sells 400 tickets at £2 each with one prize of £250. Find the expected net gain of one ticket and the expected profit for the school.

18 Solve $3(2x - 5) = 4x + 7$. Check your answer.

19 **Extension.** Two events satisfy $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.8$. Find $P(A \cap B)$ and decide whether A and B are independent. Justify both answers.

Chapter assessment

The tasks below are summative. They are completed independently and assessed against the criteria shown. Your teacher will confirm which tasks are required and how long you have.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Combined events: short response	30	40 min
B · Investigating patterns	At least one six	20	30 min
C · Communicating	A report on a dice game	20	35 min
D · Applying mathematics in real-life contexts	Designing a fair fundraiser	24	45 min
Total	All four tasks	94	150 min

The tasks begin overleaf. Read the instructions and evidence requirements before you start, and show the reasoning that supports each answer.

CRITERION A · KNOWING AND UNDERSTANDING — 40 minutes · 30 marks

Combined events: short response

Instructions. Answer all questions. Marks are shown in brackets. A correct answer without working may not receive full credit.

1 Two fair coins are tossed. List the sample space and find the probability of two heads. [2]

2 A bag contains 6 red and 4 blue counters. One is drawn at random. Find $P(\text{red})$ and $P(\text{not blue})$. [3]

3 In a group of 150 students, 90 play a sport, 60 play a musical instrument and 30 do both. Find $P(\text{sport or instrument})$ and $P(\text{neither})$. [3]

4 In a group of 40 people, 18 own a cat, 14 own a dog and 6 own both. Find $P(\text{cat only})$ and $P(\text{neither})$. [3]

5 A bag contains 4 green and 3 yellow counters. Two are drawn without replacement. Find $P(\text{both green})$ and $P(\text{one of each colour})$. [3]

6 $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Decide whether A and B are independent. Show your reasoning. [3]

7 Two cards are drawn from a standard pack without replacement. Find the probability that both are hearts. [3]

8 An event has probability 0.28. Find its expected frequency in 250 trials. [3]

9 A game costs £2. A fair die is rolled; the player wins £8 if the result is 5 or 6 and nothing otherwise. Find the expected net gain and state whether the game is fair. [3]

10 $P(E) = 0.37$. Find $P(\text{not } E)$. Explain why an event cannot have probability 1.2. [2]

11 A student says: "A game has two outcomes, win or lose, so the probability of winning is $\frac{1}{2}$." Evaluate the statement. [2]

CRITERION B · INVESTIGATING PATTERNS 30 minutes · 20 marks

At least one six

Instructions. A fair die is rolled n times. Investigate the probability of getting at least one six. Show your method and generalise.

1. Complete a table for $n = 1, 2, 3, 4, 5, 6$, giving the probability of no six and the probability of at least one six. Give values to three decimal places. [5]
2. Write a formula for $P(\text{at least one six})$ in terms of n. Explain how the complement produces the formula. [4]
3. Find the smallest n for which the probability of at least one six is greater than 0.9. Show your trials. [4]
4. Verify your answer to Question 3 by calculating the probability directly for that n and for $n - 1$. [3]
5. State two assumptions of the model and explain how the probability changes as n becomes very large. [4]

Evidence: a complete table, a general formula, a systematic search, a verification, and a statement of assumptions that refers to independence and fairness.

CRITERION C · COMMUNICATING 35 minutes · 20 marks

A report on a dice game

Instructions. A stallholder offers this game: a player pays £2 and rolls two fair dice. If the total is 7, the player receives £5; otherwise the player receives nothing. The stallholder claims: “The prize is more than double the cost, so the game is generous.” Write a 350–450 word report for the school council that evaluates the claim. Your report must:

1. state the sample space and the probability of a total of 7;
2. calculate the expected winnings and the expected net gain of one play;
3. explain what “fair” means in this context;
4. evaluate the stallholder’s claim using your calculations;
5. recommend a prize that would make the game fair, with a justification.

Evidence: an explicit sample-space count, correct probabilities with units, a clear distinction between the prize and the expected gain, and a recommendation that follows from the calculation. Use headings and a short concluding paragraph.

D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 45 minutes · 24 marks

Designing a fair fundraiser

Instructions. Your class is running a stall at the school fair. The stall costs £40 for the day. You may charge £2 per play and award a prize worth £4. The class wants to raise at least £300 and wants the game to feel fair to players. Design a game using two fair dice or a spinner, then analyse it.

1. Define the sample space and the winning event, and calculate the probability of winning. [6]
2. Calculate the expected net gain per play for a player and the expected profit per play for the stall. [5]
3. Choose a number of plays and show that the expected profit, after the stall cost, reaches at least £300. [5]
4. Simulate 100 plays using your model and compare the simulated relative frequency with the theoretical probability. [4]
5. Recommend whether the game should be used. Discuss fairness, the prize budget and one ethical issue in how the game is presented. [4]

Evidence: a complete sample space, correct expected values, a profit calculation that includes the stall cost, a simulation with a comparison, and a recommendation that distinguishes evidence from opinion.

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few terms or rules; applies them with major errors; needs support to list outcomes.	Recalls some terms and rules; applies familiar procedures with some accuracy; lists some outcomes.	Recalls and applies rules accurately in familiar situations; explains some steps.	Selects and applies rules accurately and efficiently; explains reasoning and checks the reasonableness of results.
B · Investigating patterns	Attempts a table or pattern with major gaps; generalisation is absent.	Completes part of the pattern; offers a limited generalisation.	Describes the pattern correctly and states a general rule with some justification.	Derives and justifies a general rule, verifies it and evaluates the assumptions.
C · Communicating	Uses limited mathematical language; representation is unclear or incomplete.	Uses some correct language and representations; explanation is partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is coherent, complete and suited to the audience.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation is absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics accurately; justifies assumptions; evaluates the model and its limitations critically.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Chapter summary

- A sample space lists every possible outcome; an event is a subset of it. Theoretical probability uses equally likely outcomes: $P(A) = n(A)/n(S)$.
- For any event, $0 \le P(A) \le 1$ and $P(A') = 1 - P(A)$. "At least one" is usually found through its complement, "none".
- Two-way tables and Venn diagrams show the same information. The overlap is $A \cap B$; the union $A \cup B$ includes either or both.
- The addition rule is $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. It becomes a simple sum only for mutually exclusive events.
- In a tree diagram, multiply along a path and add across paths. Without replacement, update the totals at each stage.
- Events are independent exactly when $P(A \cap B) = P(A)P(B)$. Conditional probability uses the condition as the denominator.
- Expected frequency is $nP(A)$; expected value is $\sum xP(x)$. A game is fair when the expected net gain is zero.
- A simulation estimates probability; relative frequency becomes more stable as the number of trials increases.
- Mutually exclusive and independent are different ideas. Fairness is a mathematical property of the rules and payoffs, not a feeling about one game.

$$P(A) = \frac{n(A)}{n(S)} \qquad P(A') = 1 - P(A) \qquad 0 \le P(A) \le 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A)P(B) \quad \text{(independent)} \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

$$\text{expected frequency} = nP(A) \qquad \text{expected value} = \sum x\,P(x)$$

Revision checklist

Tick each statement only when you can do it without notes and explain why the method works. If you cannot, name the worked example you will revisit.

- ☐ I can list a sample space without missing or repeating outcomes.
- ☐ I can calculate a theoretical probability and use the complement rule.
- ☐ I can read probabilities from a two-way table and a Venn diagram.
- ☐ I can use the addition rule and explain why the overlap must be subtracted.
- ☐ I can draw and use a tree diagram for two or more stages.
- ☐ I can decide whether to use replacement and adjust the second-stage probabilities.
- ☐ I can test whether two events are independent.
- ☐ I can calculate an expected frequency and the expected value of a simple game.
- ☐ I can use a simulation and interpret relative frequency.
- ☐ I can explain what makes a game fair and evaluate a claim about fairness.

REFLECTION

Answer in your notebook. Use examples from this chapter, not general statements.

1. Which method — listing, a table, a Venn diagram or a tree — helped you most, and for which type of problem?
2. Describe a probability mistake you made or nearly made. What check would have caught it?
3. The statement of inquiry says fairness is a mathematical property, not a matter of luck. Use the investigation to explain what this means.
4. Look ahead to the capstone: how could probability help you evaluate a design decision when the data are uncertain?

CAPSTONE · INTERDISCIPLINARY MODELLING

Designing a Sustainable Space

C

FORM · RELATIONSHIPS · SYSTEMS

An unused courtyard becomes a shaded outdoor classroom. The brief contains numbers, but the decision does not: dimensions, cost, rainfall, sunlight and student needs pull in different directions. This capstone asks you to build a model, test it and defend a design with evidence.

STATEMENT OF INQUIRY

A sustainable design is a mathematical argument. Claims about space, cost, shade and water become defensible only when they are modelled, tested against data and evaluated for their assumptions.

KEY CONCEPT
Form

RELATED CONCEPTS
Measurement · Space · Change

GLOBAL CONTEXT
Globalization and sustainability — designing places that meet present needs without exhausting future resources

FACTUAL
Which measurements, costs and environmental data determine whether a courtyard design is feasible?

CONCEPTUAL
How do geometric, statistical and probabilistic models constrain one another in a design decision?

DEBATABLE
Should a design optimise for cost, comfort or environmental impact when these goals conflict?

IN THIS CAPSTONE

C.1 The brief C.2 Data and constraints C.3 Analysis C.4 Assessment

OPENING PROBLEM · THE COURTYARD BRIEF

A courtyard with four competing demands

The school has an unused courtyard 18 m long and 12 m wide. The student council has £18,000 to turn it into an outdoor learning space. The brief requires at least 24 seats, at least 20 m^2 of shade, at least 40% permeable surface, and a rainwater tank for the planters. Your design must also keep a 1.5 m accessible route around the space.

1. Draw the courtyard at a scale of 1:50. State the dimensions of your drawing.
2. A first proposal uses an 8 m by 6 m deck, a 1.5 m wide path of length 38.5 m, planters 16.5 m long and 1.2 m wide, and a tank with radius 0.8 m. Calculate the permeable area and check the 40% requirement.
3. Two triangular shade sails each have base 5 m and height 4.5 m. Check the 20 m^2 shade requirement. If the posts are 3.2 m high and the Sun is at 62°, how long is a shadow?
4. The roof feeding the tank has area 120 m^2. Annual rainfall is 620 mm and 85% is collected. Find the annual harvest. The planters need 87 L per day; how long can a 2,000 L tank supply them?
5. Use the cost data in Table C.3 to estimate the cost, including 10% contingency. Compare it with the £18,000 budget.
6. Propose at least two changes that bring the design within budget without breaking a constraint. Justify your choices with calculations.

No single chapter answers this brief. Geometry fixes the available space; trigonometry and measurement describe shade; statistics summarise what students actually want; probability estimates rain and demand; algebra turns the brief into constraints; financial mathematics tests the budget. The capstone is not a new topic. It is the point at which the topics become one argument.

ORIENTATION

C.0 What you will produce

The capstone has two products: a **design portfolio** built through Sections C.1–C.3, and a **summative proposal** completed in Section C.4. The portfolio is where you investigate, calculate and revise. The proposal is where you assemble the strongest evidence and communicate a recommendation.

LEARNING INTENTIONS

By the end of the capstone you will be able to:

- translate a design brief into variables, constraints and an objective;
- build and use geometric, statistical, probabilistic and financial models;
- select and justify an appropriate level of accuracy for a real context;
- test a model against data and examine the effect of changed assumptions;
- compare designs using explicit criteria and weighted scores;
- communicate a recommendation with evidence, uncertainty and limitations.

Success criteria

You can meet the capstone standard when you can:

1. state the units, variables and assumptions of each model you use;
2. carry out multi-step calculations with correct order of operations and rounding;
3. interpret a result in the context of the courtyard, not only in symbols;
4. identify which constraint is active and which has slack;
5. evaluate a claim by testing it against data and a second model;
6. write a recommendation that a non-mathematician can follow and check.

PREREQUISITE CHECK

SKILLS FROM CHAPTERS 1–13

No calculator for Questions 1–4. If you cannot do a question, revise the chapter named beside it.

1 Expand and simplify $3(2x - 5) - 2(x - 4)$. *(Chapter 2)*

2 Solve $2x + 7 = 3(x - 1)$. *(Chapter 3)*

3 Factorise $x^2 - 7x + 12$. *(Chapter 5)*

4 Find the gradient of the line through $(1, 4)$ and $(5, 12)$. *(Chapter 4)*

5 A rectangle is 8 m by 6 m. Find its area and the length of its diagonal. *(Chapters 10–11)*

6 A population of 240 increases by 15%. Find the new population. *(Chapter 8)*

7 Find the mean and range of 4, 7, 9, 11, 14. *(Chapter 12)*

8 Two fair coins are tossed. Find the probability of at least one head. *(Chapter 13)*

Approaches to learning

Critical thinking. Distinguish a measured fact, a modelled estimate and an assumption. Every number in this capstone belongs to one of those categories; label them.

Information literacy. Record the source and date of every data set. A rainfall average without a location or period is not evidence.

Collaboration. Share calculations, but write your own interpretation. A group can collect data; a proposal is individually defended.

Transfer. The modelling cycle is the same in physics, geography and economics. Name the stage you are in when you get stuck.

FROM BRIEF TO MODEL

C.1 The brief and the modelling cycle

A design brief is written in the language of need: “more shade”, “within budget”, “accessible”, “sustainable”. A mathematical model replaces those phrases with quantities that can be measured, calculated and checked. The replacement is never neutral. Choosing what to measure, and what to leave out, is the first design decision.

The modelling cycle in Figure C.1 describes the work. You **formulate** a model by naming variables and assumptions; you **solve** it with the mathematics you know; you **interpret** the result in the real situation; you **evaluate** whether the model is fit for purpose; and you **communicate** the result to an audience that will act on it. Evaluation usually sends you back to formulation. A first model is rarely the final one.

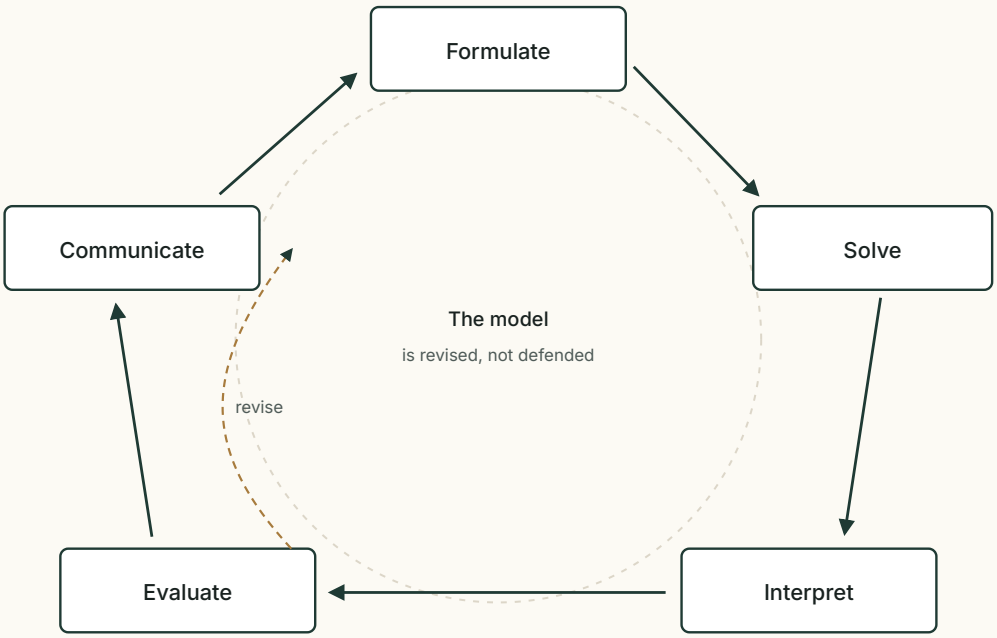


Figure C.1 The modelling cycle. Evaluation is not the end of the process: it is the point at which the assumptions are tested and the model is revised.

KEY TERMS

The vocabulary of a design model

Variable: a quantity that can change, such as the deck width w. **Parameter:** a fixed value in a particular model, such as the cost of decking per square metre.

Constraint: a condition the design must satisfy, such as $C \leq 18000$ or $n \geq 4$. **Objective:** the quantity you want to maximise or minimise, such as shade area or cost.

Assumption: something you accept as true for the model, such as “rainfall is evenly distributed across the year”. **Feasible:** a design that satisfies every constraint. **Sensitivity:** how much the conclusion changes when an assumption or parameter changes.

EXAMPLE C.1 READING A SCALE DRAWING

QUESTION

A scale drawing uses 1:50. A rectangular deck measures 7.2 cm by 4.8 cm on the plan. Find its real dimensions and area. Find the area of the deck on the plan, and verify the area scale factor.

SOLUTION

1. Each centimetre on the plan represents 50 cm in reality. So $7.2 \times 50 = 360$ cm $= 3.6$ m and $4.8 \times 50 = 240$ cm $= 2.4$ m.
2. The real area is $3.6 \times 2.4 = 8.64$ m².
3. The plan area is $7.2 \times 4.8 = 34.56$ cm².
4. Lengths scale by 50, so areas scale by $50^2 = 2500$. Check: $34.56 \times 2500 = 86{,}400$ cm² $= 8.64$ m².

Check: a plan area in cm² is multiplied by 0.25 to give m². Here $34.56 \times 2500 = 86{,}400$ cm², and since $10{,}000$ cm² $= 1$ m², this is 8.64 m².

EXAMPLE C.2 CHECKING THE PERMEABLE-SURFACE CONSTRAINT

QUESTION

The courtyard is 18 m by 12 m. The proposal has an 8 m by 6 m deck, a paved path of area 57.75 m², a tank footprint of 2.01 m² and planted beds of area 19.8 m². The brief requires at least 40% permeable surface. Does the proposal satisfy it?

SOLUTION

1. Courtyard area: $18 \times 12 = 216$ m². The 40% requirement is $0.40 \times 216 = 86.4$ m².
2. Impermeable area: deck $8 \times 6 = 48$ m², path 57.75 m², tank 2.01 m². Total $48 + 57.75 + 2.01 = 107.76$ m².
3. Permeable area: $216 - 107.76 = 108.24$ m². This includes the planted beds and the remaining grass.
4. Percentage: $\frac{108.24}{216} \times 100 = 50.1\%$, which is at least 40%. The constraint is satisfied with 21.84 m² to spare.

Check: $107.76 + 108.24 = 216$. The two categories partition the courtyard, so no area is counted twice.

EXAMPLE **C.3** FROM PROSE TO CONSTRAINTS

QUESTION

Write inequalities for these requirements. (a) At least 24 seats, with 6 seats at each table, using n tables. (b) A total cost C of at most £18,000. (c) At least 20 m² of shade from s sails, each covering 11.25 m². (d) A deck width w between 2 m and 6 m inclusive.

SOLUTION

1 (a) $6n \geq 24$, so $n \geq 4$. Since n is a whole number, $n \in \{4, 5, 6, \ldots\}$.

2 (b) $C \leq 18{,}000$. The inequality includes the contingency: the estimate must be the full amount the school will spend.

3 (c) $11.25s \geq 20$, so $s \geq 1.78$. With whole sails, $s \geq 2$.

4 (d) $2 \leq w \leq 6$. The lower bound keeps a usable deck; the upper bound preserves the path and planting.

Check: each inequality has the same units on both sides: seats, pounds, square metres, metres. A constraint without units cannot be checked.

EXERCISE C.1

BRIEF, SCALE AND CONSTRAINTS

FLUENCY

1 A plan uses a scale of 1:50. A rectangle measures 9.4 cm by 6.2 cm. Find its real dimensions and area.

2 A plan is drawn at 1:200. A path on the plan is 3.5 cm long. Find its real length in metres.

3 The courtyard is 18 m by 12 m. Find the dimensions of a 1:50 drawing and the smallest standard sheet that will hold it with a 2 cm margin on every side.

4 A proposal has a 7.5 m by 6 m deck, a path of area 54 m², a tank footprint of 2.0 m² and planted beds of 17.4 m². Find the permeable area and percentage, and check the 40% requirement.

UNDERSTANDING AND REASONING

5 Write an inequality for each requirement: (a) at least 30 seats at 6 per table; (b) a cost of no more than £18,000; (c) at least 20 m² of shade; (d) a path width of at least 1.5 m.

6 Explain the difference between an assumption and a constraint. Give one example of each from the courtyard brief.

7 A student says: “The plan is 1:50, so the real area is 50 times the plan area.” Explain the error and give the correct factor.

8 A scale model is built at 1:25. The real deck is 8 m by 6 m. Find the model dimensions and the ratio of model area to real area.

MULTI-STEP AND UNFAMILIAR

9 A plan is 1:50. A circular tank has diameter 1.6 m. Find its diameter on the plan, its plan area in cm², and its real area in m².

10 **Extension.** The brief says “at least 40% permeable”. A designer proposes a deck of area A, a path of 57.75 m² and a tank of 2.01 m². Write an inequality for A, solve it, and interpret the result in terms of the deck dimensions.

EVIDENCE BEFORE DESIGN

C.2 Data, constraints and assumptions

A design model is only as good as the data it uses. This section assembles three kinds of evidence: a **primary survey** of how students would use the space, **secondary weather data** for rain and temperature, and **supplier quotations** for materials. Each has a different status. Survey counts are exact for the students asked but may not represent the school; weather averages describe a past period, not next year; quotations are prices at a stated date.

Before calculating, ask four questions of every data set. **Who or what was measured? How was the sample chosen? Over what period? To what precision?** A number without those answers cannot be evaluated, only used.

KEY TERMS

Data quality

Primary data: collected for this project, such as the student survey. **Secondary data:** collected by someone else, such as rainfall records.

Sample: the subset measured. **Bias:** a systematic tendency for the sample to differ from the population. **Reliability:** the extent to which repeated measurements or samples give similar results.

Precision: the detail with which a value is recorded. **Accuracy:** how close a value is to the true value. A precise number can still be inaccurate.

TABLE C.1 PREFERRED USE OF THE COURTYARD, BY YEAR GROUP (180 STUDENTS SURVEYED)

	QUIET STUDY	GROUP WORK	EATING	OTHER	TOTAL
Year 9	22	18	14	6	60
Year 10	16	24	14	6	60
Year 11	16	21	17	6	60
Total	54	63	45	18	180

EXAMPLE **C.4** READING A TWO-WAY TABLE

QUESTION

Using Table C.1, find: (a) the percentage who prefer group work; (b) $P(\text{Year 10} \mid \text{group work})$; (c) $P(\text{group work} \mid \text{Year 10})$; (d) whether year group and preference are independent.

SOLUTION

1. (a) Group work: 63 of 180, so $\frac{63}{180} = 0.35 = 35\%$.
2. (b) Of the 63 who prefer group work, 24 are in Year 10: $\frac{24}{63} = \frac{8}{21} \approx 0.381$.
3. (c) Of the 60 Year 10 students, 24 prefer group work: $\frac{24}{60} = 0.4$.
4. (d) If independent, $P(\text{Year 10} \cap \text{group work})$ would equal $P(\text{Year 10})P(\text{group work}) = \frac{1}{3} \times 0.35 \approx 0.117$. The actual value is $\frac{24}{180} \approx 0.133$, so the variables are not independent. Year 10 students are somewhat more likely than the whole sample to prefer group work.

Check: $P(\text{group work} \mid \text{Year 10}) = 0.4$ is greater than the overall 0.35, which agrees with the dependence found in part (d).

EXAMPLE **C.5** SUMMARISING CATEGORICAL DATA

QUESTION

The survey also asked whether shade is essential, desirable or not important. Of 180 students, 108 said essential, 54 desirable and 18 not important. Display the results in a bar chart and a pie chart. Which display communicates the message more clearly to the student council?

SOLUTION

1. Percentages: essential $108/180 = 60\%$, desirable $54/180 = 30\%$, not important $18/180 = 10\%$.
2. Pie-chart angles: $0.60 \times 360^\circ = 216^\circ$, $0.30 \times 360^\circ = 108^\circ$, $0.10 \times 360^\circ = 36^\circ$.
3. A bar chart makes the ordering and the 60% share easy to compare; a pie chart emphasises that the three categories form one whole. For a short council report, the bar chart is easier to read at a glance.
4. Either display must state the sample size (180), the year groups surveyed and the date. Without those, the percentages cannot be judged.

Check: $216^\circ + 108^\circ + 36^\circ = 360^\circ$, and $60\% + 30\% + 10\% = 100\%$.

TABLE C.2 RAIN DAYS BY MONTH, TEN-YEAR AVERAGE FOR THE SCHOOL'S REGION

MONTH	J	F	M	A	M	J	J	A	S	O	N	D
Rain days	11	9	9	8	8	7	6	6	7	9	11	11

A “rain day” is a day with at least 1 mm of rain. The total is 102 rain days per year. Annual rainfall is 620 mm. The longest observed dry spell in the record is 17 days.

EXAMPLE **C.6** FROM WEATHER DATA TO PROBABILITY

QUESTION

Using Table C.2: (a) find the probability that a randomly chosen day is a rain day; (b) estimate the number of rain days in a 190-day school year; (c) find the probability that a five-day school week has no rain day, and hence at least one.

SOLUTION

1 (a) $P(\text{rain}) = \frac{102}{365} \approx 0.279$, or about 0.28. Then $P(\text{no rain}) = 1 - 0.279 = 0.721$.

2 (b) Expected rain days $= 190 \times \frac{102}{365} \approx 53.1$. This is a long-run average, not a prediction for a particular year.

3 (c) Assuming days are independent, $P(\text{no rain in 5 days}) = 0.721^5 \approx 0.194$. So $P(\text{at least one rain day}) = 1 - 0.194 = 0.806$, about 81%.

Check: the probability of a dry week (0.194) is close to one in five. A single dry week is not evidence that the climate is changing, and a single wet week is not evidence that the design is unnecessary.

The independence assumption in part (c) is doubtful: rain days cluster in wet spells. The calculation therefore gives a **baseline**, not a precise forecast. A more careful model would use the observed frequency of dry spells of different lengths. For design purposes, the baseline is enough to justify planning for rain; it is not enough to predict a particular week.

TABLE C.3 SUPPLIER QUOTATIONS FOR THE FIRST PROPOSAL

ITEM	UNIT	UNIT COST	QUANTITY	AMOUNT
Composite decking	m²	£38.00	48	£1,824.00
Two shade sails, installed	lot	£3,200.00	1	£3,200.00
Raised planter beds	m	£85.00	16.5	£1,402.50
Soil	50 L bag	£6.00	179	£1,074.00
Picnic tables, 6 seats	each	£380.00	6	£2,280.00
2,000 L tank and diverter	lot	£850.00	1	£850.00
Paved accessible path	m²	£48.00	57.75	£2,772.00
Labour and installation	lot	£3,600.00	1	£3,600.00

EXAMPLE **C.7** HARVEST AND DEMAND

QUESTION

The roof draining to the tank has area 120 m². Annual rainfall is 620 mm and the runoff coefficient is 0.85. (a) Find the annual volume collected. (b) The planters need 87 L per day. How many days can a full 2,000 L tank supply them? (c) Compare annual supply with annual demand over 180 watering days.

SOLUTION

1 (a) $620\text{ mm} = 0.62\text{ m}$, so the volume is $120 \times 0.62 = 74.4\text{ m}^3$ of rain. At 85% runoff: $74.4 \times 0.85 = 63.24\text{ m}^3 = 63{,}240\text{ L}$.

2 (b) $2000 \div 87 \approx 23.0$ days. The tank can bridge a dry spell of up to about 23 days, which exceeds the observed maximum of 17 days.

3 (c) Annual demand $= 87 \times 180 = 15{,}660$ L. Supply is $\frac{63{,}240}{15{,}660} \approx 4.0$ times demand, but the tank cannot store a year's supply at once; surplus overflows. Storage, not annual rainfall, is the limiting factor.

Check: $2{,}000 \div 87 = 22.99$, so 23 full days. If demand rises to 120 L per day, the tank lasts only $2000 \div 120 = 16.7$ days, less than the 17-day dry spell.

EXAMPLE **C.8** ESTIMATING THE COST

QUESTION

Use Table C.3. (a) Find the subtotal. (b) Add a 10% contingency. (c) Compare the total with the £18,000 budget and state the overrun.

SOLUTION

1 (a) Subtotal: $1824 + 3200 + 1402.50 + 1074 + 2280 + 850 + 2772 + 3600$. Pairing: $1824 + 3200 = 5024$; $1402.50 + 1074 = 2476.50$; $2280 + 850 = 3130$; $2772 + 3600 = 6372$. Total $= 5024 + 2476.50 + 3130 + 6372 = 17{,}002.50$.

2 (b) Contingency $= 0.10 \times 17{,}002.50 = 1{,}700.25$. Total $= 17{,}002.50 + 1{,}700.25 = 18{,}702.75$.

3 (c) Overrun $= 18{,}702.75 - 18{,}000 = 702.75$. The first proposal is £702.75 over budget, before any price rise.

Check: the contingency is 10% of the subtotal, not 10% of the budget. It is part of the £18,702.75, so it cannot also absorb an overrun.

EXERCISE C.2

DATA, PROBABILITY AND COST

FLUENCY

1 In the survey, 63 of 180 students prefer group work. Find the percentage and the angle this category would occupy in a pie chart.

2 108 of 180 students say shade is essential. Find the percentage. If the school has 240 students in the same proportions, how many would say essential?

EXERCISE C.2 · CONTINUED DATA, PROBABILITY AND COST

3 Using Table C.1, find $P(\text{Year 10})$, $P(\text{group work})$, $P(\text{Year 10} \cap \text{group work})$ and $P(\text{Year 10} \mid \text{group work})$.

4 Using Table C.2, find the probability that a day is not a rain day, the expected number of rain days in 190 school days, and the probability of a dry five-day week.

UNDERSTANDING AND REASONING

5 Find the modal number of rain days, the range and the mean from Table C.2. Explain why the mean is not necessarily the number of rain days in a typical month.

6 A roof of area 150 m^2 receives 580 mm of rain and collects 80% of it. Find the annual harvest. A 2,500 L tank supplies 95 L per day; how long does it last?

7 A simplified estimate has a subtotal of £9,776. Add 10% contingency and compare with the £18,000 budget.

8 A supplier announces an 8% price rise. Recalculate the subtotal and total in Question 7. Does the contingency cover the rise? Explain.

MULTI-STEP AND UNFAMILIAR

9 A survey is carried out only among students in the library at lunchtime. Give one reason the results may be biased for a courtyard design, and describe a better sampling method.

10 **Extension.** A 2,000 L tank supplies 120 L per day. The longest dry spell is 17 days. Does the tank cope? How much extra storage would give a 20% safety margin?

COMMON ERRORS IN A DESIGN PORTFOLIO

"Every number in my portfolio is a measurement." Only some are measured. Label each value as measured, calculated from a model, or assumed, so a reader can see which claims would change if an assumption did.

"Percentages taken on different totals can be added." The permeable fraction is a share of $216\ \text{m}^2$; the contingency is a share of the subtotal; the rainfall figure is a share of the roof catchment. Different bases, different quantities — never combine them directly.

"A change that saves money cannot break another constraint." Every revision must be re-tested against all four: at least 24 seats, at least $20\ \text{m}^2$ of shade, at least 40% permeable surface and a total within £18,000. Cutting a table saves £380 and still leaves 30 seats; cutting three more does not.

"Rounding the areas early is harmless." The permeable surface is $108.24\ \text{m}^2$, which is 50.1% of the courtyard; rounded to $108\ \text{m}^2$ it reads as exactly 50.0%. The difference is small, but the 40% test is settled by the unrounded comparison, so round only the figure you report.

FROM NUMBERS TO A DECISION

C.3 Analysis, evaluation and communication

Analysis is not "doing the calculations". It is choosing which calculation answers the question. A design has several objectives — cost, shade, capacity, water, accessibility — and they do not all point the same way. The analyst's job is to make the trade-offs visible, then test whether the conclusion survives a reasonable change in the assumptions.

This section works through four analyses: the shape that maximises deck area for a fixed edging length; the geometry of shade; a budget revision; and a sensitivity test. Each one ends with an interpretation, because a number that is not interpreted cannot support a decision.

EXAMPLE **C.9** MAXIMISING DECK AREA WITH A FIXED PERIMETER

QUESTION

A rectangular deck is edged with exactly 30 m of trim. Let the width be x metres. (a) Write the length and area in terms of x. (b) Find the dimensions that maximise the area. (c) The courtyard is 12 m wide and a 1.5 m path must run along one side. Does the maximum-area deck fit?

SOLUTION

1 (a) The perimeter is 30, so $2(x + \ell) = 30$, giving $\ell = 15 - x$.
The area is $A = x(15 - x) = 15x - x^2$.

2 (b) Complete the square:
$A = -(x^2 - 15x) = -\left((x - 7.5)^2 - 56.25\right) = 56.25 - (x - 7.5)^2$. The maximum is 56.25 m² at $x = 7.5$, so the deck is a 7.5 m square.

3 (c) A 7.5 m deck plus a 1.5 m path needs 9.0 m across the 12 m width, so it fits, leaving 3.0 m for planting. The square is feasible, but it may not be the best layout: it reduces the frontage available for shade sails and seating.

Check: $7.5 + 7.5 + 7.5 + 7.5 = 30$ m, and $7.5^2 = 56.25$. At $x = 6$, the area is $6 \times 9 = 54$ m², only 2.25 m² less, so the optimum is shallow; layout may matter more than the last 2 m².

EXAMPLE **C.10** SHADE, SHADOW AND CABLE LENGTH

QUESTION

A shade post is 3.2 m high. (a) Find the length of its shadow when the Sun's elevation is 62°, 35° and 20°. (b) Two posts are 6 m apart and their tops differ in height by 1.2 m. Find the cable length and its angle to the horizontal. (c) Each sail is a right triangle with base 5 m and height 4.5 m. Two sails are installed. Check the 20 m² requirement.

SOLUTION

1 (a) The shadow s satisfies $\tan\theta = \frac{3.2}{s}$, so $s = \frac{3.2}{\tan\theta}$. At 62°: $s = \frac{3.2}{1.8807} \approx 1.70$ m. At 35°: $s = \frac{3.2}{0.7002} \approx 4.57$ m. At 20°: $s = \frac{3.2}{0.3640} \approx 8.79$ m.

2 (b) Cable length $= \sqrt{6^2 + 1.2^2} = \sqrt{37.44} \approx 6.12$ m. Angle $= \tan^{-1}\left(\frac{1.2}{6}\right) \approx 11.3°$.

3 (c) One sail: $\frac{1}{2} \times 5 \times 4.5 = 11.25$ m². Two sails: 22.5 m², which exceeds the 20 m² requirement by 2.5 m².

Check: at low Sun the shadow is much longer than the sail. A horizontal sail shades the area beneath it at midday, but cannot guarantee shade at 20° elevation without vertical screens or planting. The 20 m² target measures sail area, not guaranteed shade at every hour.

EXAMPLE **C.11** REVISING THE BUDGET WITHOUT BREAKING A CONSTRAINT

QUESTION

The first proposal costs £18,702.75 including 10% contingency, which is £702.75 over budget. Propose a revision that saves at least £703 while keeping at least 24 seats, 20 m² of shade and the permeable-surface requirement.

SOLUTION

1. Remove one 6-seat table: saves £380. Seats fall from 36 to 30, still above 24.
2. Reduce the deck from 48 m² to 45 m², for example from 8 m by 6 m to 7.5 m by 6 m: saves $3 \times 38 = £114$.
3. Reduce the planter length from 16.5 m to 14.5 m: saves $2 \times 85 = £170$. The soil volume falls from 8.91 m³ to 7.83 m³, so 157 bags replace 179: saves $22 \times 6 = £132$.
4. Total saving $= 380 + 114 + 170 + 132 = £796$. New subtotal $= 17{,}002.50 - 796 = 16{,}206.50$. Contingency $= £1{,}620.65$. New total $= £17{,}827.15$, which is £172.85 under budget.

Check: the permeable area gains 3 m² where the deck shrinks and loses 2.4 m² where the shortened planters are paved over, a net gain of 0.6 m², so permeable surface is $108.24 + 0.6 = 108.84$ m² or 50.4% — still above the 40% requirement. Seats remain 30 and shade remains 22.5 m², so no constraint is broken.

EXAMPLE **C.12** SENSITIVITY AND A WEIGHTED DECISION

QUESTION

(a) The revised total is £17,827.15. If material costs rise by 8%, what is the new total? (b) The tank supplies 87 L per day. If demand rises by 40%, does a 2,000 L tank still cover a 17-day dry spell? (c) Score the two designs using the weights cost 30%, shade 25%, capacity 20%, sustainability 15%, accessibility 10%.

SOLUTION

1. (a) New subtotal $= 16{,}206.50 \times 1.08 = 17{,}503.02$. With 10% contingency, total $= 17{,}503.02 \times 1.10 = 19{,}253.32$. The project is now £1,253.32 over budget.
2. (b) New demand $= 87 \times 1.40 = 121.8$ L per day. Tank life $= 2000 \div 121.8 \approx 16.4$ days, which is less than 17. The tank no longer covers the observed dry spell; storage must increase or demand must fall.
3. (c) Design A (first proposal): cost 2, shade 4, capacity 5, sustainability 4, accessibility 5. Weighted score $= 0.30(2) + 0.25(4) + 0.20(5) + 0.15(4) + 0.10(5) = 3.70$.
4. Design B (revised): cost 4, shade 4, capacity 3, sustainability 4, accessibility 5. Weighted score $= 0.30(4) + 0.25(4) + 0.20(3) + 0.15(4) + 0.10(5) = 3.90$. Design B wins on the stated weights, mainly because it is within budget.

Check: the scores are judgments, not measurements. Changing the weights can change the winner: if capacity were weighted 40%, Design A would win. A decision matrix makes the values explicit; it does not remove them.

Evaluating a design argument

A strong proposal separates four things that are easily confused. A **measurement** is read from an instrument or a record. An **estimate** is calculated from a model. An **assumption** is accepted without proof. A **judgment** is a value choice, such as weighting cost above capacity. State which is which. A reader who disagrees with your judgment can still accept your measurements; a reader who cannot tell them apart cannot evaluate either.

Test the conclusion, not only the arithmetic. Ask: Which constraint is active? How much slack is there? What would have to change for the recommendation to reverse? A design that is £172.85 under budget is fragile; a design that is £2,000 under budget is robust. Both may be correct, but they support different levels of confidence.

Communicating the recommendation

Write for the person who will approve the money, not for your teacher. Use a short structure: **the brief** in one sentence; **the recommendation** in one sentence; **the evidence** in three or four calculations; **the trade-offs** in a short paragraph; **the limitations** and the next check. Put the full calculations in an appendix. Label every table and figure with units, source and date. Round to the precision the data support: costs to the nearest pound, areas to 0.1 m^2, probabilities to two decimal places.

Uncertainty is not a weakness in a proposal. “The tank covers a 17-day dry spell at current demand, but not if demand rises by 40%” is more useful than a single number presented as certain. The council can act on the first statement; it can only guess at the second.

EXERCISE C.3

OPTIMISATION, SHADE AND EVALUATION

FLUENCY

1 A rectangular deck has perimeter 24 m. Write its area in terms of the width x, and find the dimensions that maximise the area.

2 A rectangular deck has perimeter 36 m. Find the maximum area and the dimensions that give it.

3 A post is 2.8 m high. Find the length of its shadow when the Sun's elevation is 58°.

4 Two posts are 7 m apart and their tops differ in height by 1.5 m. Find the cable length and its angle to the horizontal.

UNDERSTANDING AND REASONING

5 A simplified estimate has subtotal £14,800 and a 12% contingency. Find the total and compare it with a budget of £16,500.

6 A revision removes 2 tables at £380 each, reduces the deck by 4 m^2 at £38 per m^2, and reduces the planters by 3 m at £85 per metre. The soil reduction is 1.62 m^3, sold in 50 L bags at £6 each. Find the total saving.

7 The revised total is £17,827.15. Material costs rise by 6%. Find the new total and the overrun against £18,000.

8 A decision matrix uses weights 30% cost, 25% shade, 20% capacity, 15% sustainability and 10% accessibility. Design A scores 4, 3, 4, 5, 4; Design B scores 3, 4, 4, 4, 5. Calculate both weighted scores.

EXERCISE C.3 · CONTINUED

OPTIMISATION, SHADE AND EVALUATION

MULTI-STEP AND UNFAMILIAR

9 A student writes: "The design is sustainable because it has a rainwater tank." Evaluate this claim in three or four sentences, referring to at least two other factors.

10 **Extension.** A 2,000 L tank supplies 100 L per day. A 19-day dry spell is expected. Does the tank cope? How much extra storage gives a 15% safety margin?

11 **Extension.** The shade target is 20 m^2. Two triangular sails have base 5 m and height 4.5 m. Find the percentage by which the shade exceeds the target. If one sail is removed, by what percentage does the shade fall short?

CAPSTONE PRACTICE

CUMULATIVE RETRIEVAL · CHAPTERS 1–13

No notes. Allow 50 minutes. Questions 1–5 check fluency; 6–10 require reasoning and application.

FLUENCY

1 Solve $3(x - 4) = 2x + 5$.

2 Factorise $2x^2 - 8x$ and solve $2x^2 - 8x = 0$.

3 Find the vertex and the roots of $y = x^2 - 6x + 5$.

4 A subtotal is £18,000 and a 7.5% contingency is added. Find the contingency and the total.

5 £4,000 is invested at 3.5% compound interest per year for 3 years. Find the final amount.

UNDERSTANDING AND REASONING

6 A ladder 5 m long makes an angle of 68° with the ground. Find the height it reaches and the distance of its foot from the wall.

7 A cylindrical tank has radius 0.8 m and height 2.2 m. Find its volume in litres.

8 Find the mean, median, range and interquartile range of 12, 15, 15, 18, 21, 27.

9 The probability of rain on a day is 0.28. Find the probability of two dry days and the probability of at least one wet day.

10 A path is 1.5 m wide and has area 57.75 m^2. Find its length. If the width becomes 1.8 m with the same length, find the new area and the extra cost at £48 per m^2.

Capstone assessment

The tasks below are summative. They are completed independently and assessed against the criteria shown. Sections C.1–C.3 are your portfolio: you may use your own calculations, plans and notes as evidence, but the submitted proposal must be your own work.

ASSESSMENT AT A GLANCE

CRITERION	TASK	MARKS	TIME
A · Knowing and understanding	Courtyard calculations	20	35 min
B · Investigating patterns	The deck and the path	20	40 min
C · Communicating	Design proposal	20	45 min
D · Applying mathematics in real-life contexts	Sustainable space in use	20	50 min
Total	All four tasks	80	170 min

The tasks begin overleaf. Read the evidence requirements before you start. Marks are for reasoning and communication as well as for correct values.

CRITERION A · KNOWING AND UNDERSTANDING 35 minutes · 20 marks

Courtyard calculations

Instructions. Answer all questions. Show the substitution and the units. A correct answer without working may not receive full credit.

1 A plan uses a scale of 1:50. A rectangle measures 8.6 cm by 5.4 cm on the plan. Find its real dimensions and area. [2]

2 The courtyard is 18 m by 12 m. A design has a deck of 45 m², a path of 57.75 m² and a tank footprint of 2.01 m². Find the permeable area and percentage, and check the 40% requirement. [3]

3 Two triangular sails each have base 5 m and height 4.5 m. Find the total shade area and the percentage by which it exceeds 20 m². [2]

4 A roof of area 100 m² receives 600 mm of rain and collects 80%. Find the annual harvest in litres. [2]

5 A 2,000 L tank supplies 95 L per day. For how many whole days can it supply the planters? [2]

6 A subtotal is £15,400. Add 10% contingency and compare the total with a budget of £18,000. [2]

7 Solve $2(x - 3) = x + 7$. [1]

8 Find the vertex of $y = x^2 - 8x + 12$. [2]

9 Find the mean and median of 8, 10, 10, 12, 15, 17. [2]

10 The probability of rain on a day is 0.3. Find the probability of five dry days and the probability of at least one wet day. [2]

CRITERION B · INVESTIGATING PATTERNS 40 minutes · 20 marks

The deck and the path

Instructions. A rectangular deck is edged with exactly 30 m of trim. Let the width be x metres. Work systematically and show the generalisation.

1. Complete a table for $x = 4, 5, 6, 7, 7.5, 8, 9, 10$, giving the length, the area and the perimeter. [5]
2. Write the length and the area as functions of x. State the domain that makes sense for a deck. [4]
3. Complete the square to find the maximum area and the value of x that gives it. [4]
4. Generalise: for a rectangle of fixed perimeter P, show that the maximum area is $\frac{P^2}{16}$ and occurs when the rectangle is a square. [4]
5. Apply your rule to perimeters of 24 m and 36 m. [2]
6. The courtyard is 12 m wide and a 1.5 m path must run along one side. Explain why the square deck may not be the best layout even though it maximises area. [1]

Evidence: a complete table, an algebraic model, a completed-square derivation, a general rule with a proof, two applications, and a layout constraint.

CRITERION C · COMMUNICATING 45 minutes · 20 marks

Design proposal

Instructions. Write a 600–750 word proposal to the student council recommending one design for the courtyard. Use your portfolio calculations. Your proposal must include:

1. the brief and the four constraints, stated in one short paragraph;
2. a labelled plan at 1:50, with the main dimensions and areas;
3. calculations for permeable area, shade, rainwater and total cost;
4. a comparison of two options using at least three criteria;
5. a recommendation that names the trade-offs and states one limitation;
6. a source note for the survey, weather and cost data.

Evidence: a structured proposal with headings, a labelled plan, a cost table, a decision matrix, correct units and a recommendation that follows from the evidence. Calculations may be placed in an appendix.

D · APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS 50 minutes · 20 marks

Sustainable space in use

Instructions. The revised design has 30 seats, 22.5 m^2 of shade, a 2,000 L tank and a total cost of £17,827.15. The survey found that 63 of 180 students prefer group work and would use the space twice a week. Answer using the data from this capstone.

1. Estimate the average number of group-work visits per school day. Compare it with the 30 seats and comment on capacity. [4]
2. Using 102 rain days in 365, estimate the rain days in a 190-day school year and the probability of a dry five-day week. Explain what this means for the shade design. [4]
3. The tank supplies 87 L per day. Test whether it covers a 17-day dry spell if demand rises by 30%. [3]
4. Material costs rise by 6%. Find the new total and the overrun. Propose one change that closes the gap without reducing the seats below 24. [4]
5. Explain one accessibility or fairness issue in the design, and how it could be addressed. [3]
6. State whether you would approve the design. Justify your decision with two pieces of evidence and one limitation. [2]

Evidence: calculations with units, an interpretation of each result in the courtyard context, a cost revision, an ethical consideration, and a recommendation that distinguishes evidence from judgment.

CAPSTONE RUBRIC

CRITERION	1–2	3–4	5–6	7–8
A · Knowing and understanding	Recalls few formulae or units; applies them with major errors; limited justification.	Recalls some formulae and units; applies them with some accuracy; explains some steps.	Applies formulae and units accurately in familiar situations; justifies most steps and checks reasonableness.	Selects and applies formulae accurately and efficiently; explains precision, units and reasonableness critically.
B · Investigating patterns	Attempts a table or pattern with major gaps; generalisation absent.	Completes part of the pattern; offers a limited generalisation.	Describes the pattern correctly and states a general rule with some justification.	Derives and proves the general rule; tests it and evaluates the assumptions.
C · Communicating	Uses limited mathematical language; representation unclear or incomplete.	Uses some correct language and representations; explanation partly coherent.	Uses appropriate language and representations; explanation is coherent and mostly accurate.	Uses precise language and well-chosen representations; explanation is complete, audience-aware and critical.
D · Applying mathematics	Identifies few relevant elements; applies a method with major errors; evaluation absent.	Identifies some relevant elements; applies a method with some accuracy; offers a limited evaluation.	Selects relevant elements and applies mathematics accurately; evaluates the result in context.	Selects and applies mathematics critically; justifies assumptions and evaluates the model, its limitations and its ethical dimension.

The rubric summarises the school's Year 4 achievement descriptors. The official IB criteria and boundaries are held by the school; this table is a student-facing guide, not an official IB rubric.

Capstone summary

- A design brief becomes a model when needs are translated into variables, constraints and an objective.
- The modelling cycle is iterative: formulate, solve, interpret, evaluate, communicate, then revise.
- Scale drawings multiply lengths by the scale factor and areas by its square. At 1:50, areas scale by 2500.
- Measurement and trigonometry turn geometry into quantities: area, volume, shadow length, cable length and gradient.
- Statistics summarise what people want; probability estimates how often weather or demand will test the design.
- A budget is a constraint with a contingency; the contingency is part of the cost, not spare money.
- Sensitivity analysis asks what changes the conclusion. A design with little slack is fragile.
- Communication separates measurement, estimate, assumption and judgment, and states uncertainty honestly.

$$A_{\text{plan}} \times k^2 = A_{\text{real}} \qquad \text{shadow} = \frac{h}{\tan\theta} \qquad V = \pi r^2 h$$

$$\text{cost} = \text{subtotal} \times (1 + \text{contingency rate})$$

$$A = x(P/2 - x), \quad A_{\max} = \frac{P^2}{16} \text{ at } x = \frac{P}{4}$$

$$P(\text{at least one}) = 1 - (1 - p)^n \qquad \text{expected frequency} = np$$

Revision checklist

Tick each statement only when you can do it without notes and explain the reasoning. If you cannot, name the section you will revisit.

- ☐ I can translate a brief into variables, constraints and an objective.
- ☐ I can read and produce a scale drawing, including area scale factors.
- ☐ I can calculate areas, volumes and percentages with correct units.
- ☐ I can use trigonometry to find a shadow, a cable length or an angle.
- ☐ I can summarise a data set and interpret a two-way table.
- ☐ I can estimate a probability from data and use the complement rule.
- ☐ I can build a cost model with a contingency and revise it.
- ☐ I can complete the square and interpret the maximum of a quadratic model.
- ☐ I can test a conclusion under changed assumptions.
- ☐ I can write a recommendation that separates evidence, assumption and judgment.

REFLECTION

Answer in your notebook. Use specific calculations and decisions from your portfolio.

1. Which part of the modelling cycle did you find hardest, and what did you do about it?
2. Describe a decision where two constraints conflicted. How did you decide which to prioritise?
3. The statement of inquiry calls a design “a mathematical argument”. Has your view of argument changed? Explain with an example.
4. Look ahead to Year 5: which additional mathematics — logarithms, the sine and cosine rules, series and summation formulae, vectors — would strengthen this design analysis?

REFERENCE · ANSWERS, REASONING AND ASSESSMENT GUIDANCE

Answers *and guidance*

This section contains the answers to every numbered question in the coursebook, together with the reasoning that makes each answer checkable. Worked steps are given wherever a final value alone would hide the method: algebraic manipulation, graph reading, proof structure, units and rounding.

Answers are grouped by chapter and follow the order of the exercises. Question numbers match the printed numbers exactly, including questions that continue across a page. Lettered parts appear in the same order as in the question.

What the entries contain

Most entries give the result and one or two decisive steps. A note in italics explains a common error, an alternative valid method, or a point of interpretation. For investigations, the entry states the result the pattern should reach. For open assessment tasks, the entry gives marking guidance and a sample response rather than a single "correct" answer.

Using the key responsibly

An answer is not a substitute for a solution. If your value differs, compare the method before changing your work: a sign error, an unrounded intermediate value, or a misread scale can produce a plausible but wrong result. Where a question asks you to explain, justify or evaluate, a correct number without reasoning does not meet the criterion.

> **Assessment tasks.** The Criterion A–D tasks are marked with the published MYP criteria, not against a single model answer. The guidance shows the evidence a response should contain and how the marks are usually distributed. A sample response illustrates one satisfactory route; other valid approaches can score full marks.

Conventions

Unless a question states otherwise, give non-exact answers to three significant figures, use π from your calculator, and include units in context. Angles are in degrees. Probability answers are given as fractions in simplest form where the outcomes are equally likely, and as decimals or percentages where the context uses them.

CHAPTER 1 · NUMBER FOUNDATIONS

Opening problem · a claim worth testing

1 Every fold doubles the thickness, so $T = 0.1 \times 2^n$ mm after n folds: a geometric sequence with first term 0.1 and common ratio 2 ($n = 0$ is the unfolded sheet).

2 Estimate with $2^{10} \approx 10^3$, then calculate. $n = 0$: 0.1 mm. $n = 1$: 0.2 mm. $n = 5$: $0.1 \times 32 = 3.2$ mm. $n = 10$: estimate $0.1 \times 10^3 = 100$ mm, exact $0.1 \times 1024 = 102.4$ mm ≈ 10 cm. $n = 20$: estimate $0.1 \times 10^6 = 100$ m, exact $0.1 \times 1\,048\,576 = 104\,857.6$ mm ≈ 105 m. $n = 30$: estimate $0.1 \times 10^9 = 10^8$ mm $= 100$ km, exact $0.1 \times 1\,073\,741\,824 = 1.07 \times 10^8$ mm ≈ 107 km. $n = 42$: estimate $0.1 \times 4 \times 10^{12} = 4 \times 10^{11}$ mm $= 4 \times 10^5$ km, exact $0.1 \times 4\,398\,046\,511\,104 = 4.398 \times 10^{11}$ mm $= 4.40 \times 10^5$ km.

The estimates fall short because $2^{10} = 1024$ is 2.4% above 10^3, and the error compounds: the exact values are 4.9% above the estimates at $n = 20$, 7.4% at $n = 30$ and 10.0% at $n = 42$.

3 4.40×10^5 km against the mean Earth–Moon distance $384\,400$ km $= 3.844 \times 10^5$ km: the stack exceeds it by 5.5×10^4 km, about 14%, and it also exceeds the greatest distance 4.055×10^5 km. The arithmetic of the claim is sound. The estimate 4×10^5 km clears the mean distance but not the greatest, so only the unrounded value settles it.

4 (i) That the paper is uniform and incompressible, so each fold exactly doubles the thickness: the model ignores the crease radius, the sliding of layers and the air trapped in the stack, and beyond a few folds a real stack cannot be bent at all. (ii) That each fold halves the area and still leaves a usable sheet: an A4 sheet of 0.0624 m^2 would end with a footprint of $0.0624 \div 2^{42} = 1.4 \times 10^{-14}$ m^2, about a million times the cross-section of an atom, so the stack would have to be microscopic in area. (iii) That 0.1 mm is exact: it is 1 s.f., so the thickness lies between 0.05 mm and 0.15 mm, and the lower bound gives $0.05 \times 2^{42} = 2.2 \times 10^5$ km, which does not reach the Moon. (i) and (ii) are the physical assumptions that make the claim misleading; (iii) shows that even the arithmetic is not certain within the precision given.

Prerequisite check

1 $2^5 = 32$, $3^4 = 81$, $10^3 = 1000$, and $5^0 = 1$ because any non-zero number divided by itself is 1.

2 $\sqrt{49} = 7$, $\sqrt{81} = 9$, $\sqrt[3]{27} = 3$ since $3^3 = 27$.

3 2 d.p.: the third decimal digit is 6, so 3.85. 1 s.f.: the first digit is 3 and the next is 8, so 4.

4 $2x \cdot 3x^2 = 6x^{1+2} = 6x^3$.

5 $16 = 2^4$, so $x = 4$.

6 $0.25 = \dfrac{25}{100} = \dfrac{1}{4}$ and $0.4 = \dfrac{4}{10} = \dfrac{2}{5}$.

Exercise 1A

1 $12 \in \mathbb{N}, \mathbb{Z}, \mathbb{Q}$; $-4 \in \mathbb{Z}, \mathbb{Q}$; $0 \in \mathbb{Z}, \mathbb{Q}$ (zero is a whole number, but not natural in this course); $0.6 = \dfrac{3}{5} \in \mathbb{Q}$; $\sqrt{16} = 4 \in \mathbb{N}, \mathbb{Z}, \mathbb{Q}$; $\sqrt{11}$ is irrational only, since $3^2 < 11 < 4^2$ and 11 is not a perfect square; $\dfrac{7}{3} \in \mathbb{Q}$; π is irrational only.

2 (a) Always: $a = \dfrac{a}{1}$ with integers $a, 1$. (b) Sometimes: $\dfrac{1}{2}$ is rational but not an integer. (c) Never: $\sqrt{9} = 3$ is rational. (d) Sometimes: $\sqrt{2} \cdot \sqrt{2} = 2$ is rational, but $\sqrt{2} \cdot \sqrt{3} = \sqrt{6}$ is irrational.

3 (a) $x = 0.\overline{7}$: $10x = 7.\overline{7}$, so $9x = 7$, $x = \dfrac{7}{9}$. (b) $99x = 12$, $x = \dfrac{12}{99} = \dfrac{4}{33}$. (c) $99x = 24$, $x = \dfrac{24}{99} = \dfrac{8}{33}$. (d) $100x - 10x = 153.\overline{3} - 15.\overline{3} = 138$, so $90x = 138$, $x = \dfrac{138}{90} = \dfrac{23}{15}$.

Common error: treating $1.5\overline{3}$ as $1.5 \times 0.\overline{3}$; the non-recurring part 1.5 is a whole part that must be carried through the subtraction, not multiplied.

4 Let $x = 0.\overline{9}$. Then $10x = 9.\overline{9}$, so $9x = 9.\overline{9} - 0.\overline{9} = 9$ and $x = 1$. The difference $1 - 0.\overline{9}$ is smaller than every positive number, so it is 0: the two numerals name one number, not two numbers almost equal.

5 $\dfrac{1}{3} = \dfrac{8}{24}$ and $\dfrac{1}{2} = \dfrac{12}{24}$, so $\dfrac{9}{24} = \dfrac{3}{8}$ lies between (also $\dfrac{5}{12}$). In general, for rationals $a < b$, the average $\dfrac{a+b}{2}$ is rational and satisfies $a < \dfrac{a+b}{2} < b$; repeating gives infinitely many.

6 Irrational. The digits never settle into a repeating block: between successive long runs of zeros the run length increases, so no finite block can repeat forever. A rational number would have a terminating or eventually periodic decimal.

7 $3.14 = \dfrac{314}{100} = \dfrac{157}{50}$, a fraction of integers with a terminating decimal, so it is rational. It is an approximation to π, not equal to π, and being close to an irrational number does not make it irrational.

8 Suppose $2 + \sqrt{2} = r$ with $r \in \mathbb{Q}$. Then $\sqrt{2} = r - 2$, a difference of rationals, hence rational. This contradicts the known irrationality of $\sqrt{2}$. Therefore $2 + \sqrt{2}$ is irrational.

9 Sum: $\sqrt{2} + (-\sqrt{2}) = 0$, because $-\sqrt{2}$ is the additive inverse of $\sqrt{2}$. Product: $\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$, because the radicands multiply to a perfect square; similarly $\sqrt{2} \cdot \sqrt{2} = 2$.

10 Irrational. The decimal is not eventually periodic: the block of zeros between consecutive integers grows without limit, so no fixed block can repeat for ever. Hence it cannot equal any fraction $\dfrac{a}{b}$.

11 Assume $\sqrt{3} = \frac{p}{q}$ in lowest terms, $q \neq 0$. Then $p^2 = 3q^2$, so $3 \mid p^2$ and hence $3 \mid p$; write $p = 3k$. Then $9k^2 = 3q^2$, so $q^2 = 3k^2$, giving $3 \mid q$. Both p and q are divisible by 3, contradicting lowest terms. Hence $\sqrt{3}$ is irrational.

12 Suppose $\sqrt{2} + \sqrt{3} = r \in \mathbb{Q}$. Squaring: $5 + 2\sqrt{6} = r^2$, so $\sqrt{6} = \frac{r^2 - 5}{2}$, which is rational. But $\sqrt{6}$ is irrational: from $\sqrt{6} = \frac{p}{q}$ in lowest terms, $6q^2 = p^2$ forces $2 \mid p$, then $3q^2 = 2k^2$ forces $2 \mid q$, a contradiction. Hence no rational r exists.

The contradiction is with the irrationality of $\sqrt{6}$, not with the value of r; this is why the known fact must be quoted explicitly.

13 The display is finite because the screen is finite, not because the decimal ends. 1.4142135623730951 is a rounded rational approximation; squaring that rounded value gives 2.0000000000000001, not exactly 2. Correction: "The calculator shows only the first few digits of a decimal that never ends and never repeats, so $\sqrt{2}$ is irrational."

Exercise 1B

1 (a) x^{11}. (b) y^5. (c) $2^4 a^{12} = 16a^{12}$. (d) $10x^{2+(-5)} = 10x^{-3} = \frac{10}{x^3}$.

2 (a) $\frac{1}{3^2} = \frac{1}{9}$. (b) -8. (c) $\frac{1}{(-3)^2} = \frac{1}{9}$. (d) $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$.

3 (a) 9. (b) 4. (c) $16^{3/4} = (16^{1/4})^3 = 2^3 = 8$. (d) $125^{-2/3} = \frac{1}{(\sqrt[3]{125})^2} = \frac{1}{25}$.

4 (a) $x^{1/2}$. (b) x^{-3}. (c) $x^{2/3}$. (d) $x^{-1/4}$.

5 (a) $2x^{5-(-2)} = 2x^7$. (b) $4^3 a^6 b^3 = 64a^6 b^3$. (c) $\frac{8x^6}{4x^5} = 2x$. (d) $\frac{x^6}{4}$.

6 (a) $2^x = 2^5$, $x = 5$. (b) $3^{x+1} = 3^4$, $x = 3$. (c) $5^{2x} = 5^3$, $2x = 3$, $x = \frac{3}{2}$. (d) $2^{2x} = 2^1$, $x = \frac{1}{2}$.

7 $2^{2x+2} = 2^{3x}$, so $2x + 2 = 3x$ and $x = 2$. Check: $4^3 = 64 = 8^2$.

8 $3^{2x} = 3^{-4}$, so $x = -2$. $3^{3x} = 3^{-2}$, so $x = -\frac{2}{3}$.

9 For $a \neq 0$, $\frac{a^n}{a^n} = a^{n-n} = a^0$ by the quotient law, and any non-zero number divided by itself is 1, so $a^0 = 1$. For $a = 0$ the step becomes $\frac{0^n}{0^n} = \frac{0}{0}$, which is undefined, so the argument fails and 0^0 is not defined by this law.

10 Thickness $= 0.1 \times 2^n$ mm. $n = 10$: $0.1 \times 1024 = 102.4$ mm ≈ 10 cm. $n = 20$: $0.1 \times 1\,048\,576 = 104\,857.6$ mm ≈ 105 m. Both agree with the opening problem.

11 The answer is wrong. $(-8)^{2/3} = \left(\sqrt[3]{-8}\right)^2 = (-2)^2 = 4$, because the denominator 3 takes the cube root first and the numerator 2 then squares it. A real square cannot be negative, so -4 is impossible; it is $-\left(8^{2/3}\right)$, not $(-8)^{2/3}$.

Common error: applying the negative sign of the base after squaring. Take the root first, then square.

12 $2^x(2^2 - 1) = 48$, so $3 \cdot 2^x = 48$, $2^x = 16$, $x = 4$. Check: $64 - 16 = 48$.

13 $4^n + 4^n = 2 \cdot 4^n = 2 \cdot 2^{2n} = 2^{2n+1}$, since adding a quantity to itself doubles it and $4^n = (2^2)^n = 2^{2n}$.

Exercise 1C

1 (a) 3.2×10^{-6}. (b) 7.4×10^6. (c) 5.6×10^{-2}. (d) 9.1×10^4.

2 (a) 0.00051. (b) 2 030 000. (c) 0.099. (d) 1.2.

3 (a) $3.1 \times 2.0 \times 10^7 = 6.2 \times 10^7$. (b) $21.0 \times 10^{-7} = 2.1 \times 10^{-6}$. (c) 3.0×10^6. (d) 3.0×10^4.

4 $5.1 \times 10^3 = 0.51 \times 10^4$, so the sum is $(2.7 + 0.51) \times 10^4 = 3.21 \times 10^4$. Adding the coefficients without matching powers would give the wrong order of magnitude.

5 (a) 3.85. (b) 0.0046. (c) 28 500. (d) 10.0: the fourth digit 5 rounds 9.99 up, and the trailing zero must be kept to show 3 s.f.

6 (a) Estimate $\frac{40 \times 0.5}{0.02} = 1000$; calculating gives $884.57\ldots$, so $885 = 8.85 \times 10^2$. (b) Estimate $\sqrt{6} \times 10^5 \approx 8 \times 10^2$; calculating gives $761.577\ldots$, so 762. (c) Estimate $\frac{6 \times 10^{-4}}{3 \times 10^{-6}} = 2 \times 10^2$; exact $200 = 2.00 \times 10^2$.

7 0.5 mm $= 5 \times 10^{-4}$ m. $\frac{1.27 \times 10^7}{5 \times 10^{-4}} = 2.54 \times 10^{10}$, so 2.5×10^{10} grains to 2 s.f.

8 $\left(1.67 \times 10^{-27}\right)\left(6.02 \times 10^{23}\right) = 10.05 \times 10^{-4} = 1.01 \times 10^{-3}$ kg, about 1 g.

9 Decimal definition: 4.7×10^9 bytes. Binary definition: $4.7 \times 2^{30} = 4.7 \times 1\,073\,741\,824 = 5.05 \times 10^9$ bytes. The binary value is larger, by about 3.5×10^8 bytes (roughly 7%).

10 $0.00072 = 7.2 \times 10^{-4}$, not 10^{-3}. The point must move until exactly one non-zero digit stands before it: $0.00072 \to 00007.2$, four places, so the exponent is -4.

Count the places moved, not the zeros written; the leading zeros in 0.00072 are three, but four moves are needed.

11 In 3.0 m the trailing zero after the point is significant, so the value has 2 s.f. and half-unit 0.05 m. In 3 m the single digit is significant, giving 1 s.f. and half-unit 0.5 m, ten times coarser. So $3.0^2 = 9.0$ m^2 (2 s.f.), with bounds $2.95^2 \le A < 3.05^2$, i.e. $8.70 \le A < 9.30$ m^2; while $3^2 = 9$ m^2 to 1 s.f. (the digit 9, not 10), with the far wider bounds $2.5^2 \le A < 3.5^2$, i.e. $6.25 \le A < 12.25$ m^2. One extra significant figure divides the possible area range by about ten, so the extra digit must be kept before squaring, not invented afterwards.

12 $\dfrac{1.989 \times 10^{30}}{5.972 \times 10^{24}} = 0.33305\ldots \times 10^6 = 3.33 \times 10^5$, so the Sun is 3.33×10^5 times heavier (about 333 000 times).

13 $4.24 \times 9.46 \times 10^{12} = 40.11 \times 10^{12} = 4.01 \times 10^{13}$ km. In metres multiply by 10^3: 4.01×10^{16} m.

Exercise 1D

1 (a) $\sqrt{16 \cdot 2} = 4\sqrt{2}$. (b) $\sqrt{16 \cdot 3} = 4\sqrt{3}$. (c) $\sqrt{100 \cdot 2} = 10\sqrt{2}$. (d) $\sqrt{36 \cdot 5} = 6\sqrt{5}$.

2 (a) $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$. (b) $6\sqrt{2} - 5\sqrt{2} = \sqrt{2}$. (c) $4\sqrt{2} + 9\sqrt{2} = 13\sqrt{2}$. (d) $2\sqrt{3} + 3\sqrt{3} - 4\sqrt{3} = \sqrt{3}$.

3 (a) $\sqrt{60} = 2\sqrt{15}$. (b) $\sqrt{144} = 12$. (c) 5. (d) $4 \cdot 3 = 12$.

4 (a) $\dfrac{\sqrt{3}}{3}$. (b) $\dfrac{5\sqrt{2}}{2}$. (c) $\dfrac{4}{\sqrt{2}} = 2\sqrt{2}$. (d) $\dfrac{6}{2\sqrt{3}} = \dfrac{3}{\sqrt{3}} = \sqrt{3}$.

5 (a) $\dfrac{\sqrt{5}-1}{5-1} = \dfrac{\sqrt{5}-1}{4}$. (b) $\dfrac{2(\sqrt{7}+\sqrt{3})}{7-3} = \dfrac{\sqrt{7}+\sqrt{3}}{2}$. (c) $\dfrac{(\sqrt{2}+1)^2}{2-1} = 3 + 2\sqrt{2}$. (d) $\dfrac{3(\sqrt{6}-\sqrt{3})}{6-3} = \sqrt{6} - \sqrt{3}$.

6 (a) $3 - \sqrt{2} + 3\sqrt{2} - 2 = 1 + 2\sqrt{2}$. (b) $6 - 2\sqrt{12} + 2 = 8 - 4\sqrt{3}$. (c) $20 - 1 = 19$. (d) $18 - 6\sqrt{2} + 1 = 19 - 6\sqrt{2}$.

7 Take $a = 9$, $b = 16$: $\sqrt{9+16} = 5$, but $\sqrt{9} + \sqrt{16} = 7$. The product rule needs $a \ge 0$, $b \ge 0$: $\sqrt{ab} = \sqrt{a}\sqrt{b}$ holds there because both sides are non-negative and their squares are equal to ab.

8 No. $\sqrt{x^2} = |x|$. For $x = 3$: $\sqrt{9} = 3 = x$. For $x = -3$: $\sqrt{(-3)^2} = \sqrt{9} = 3 \ne -3$. The square root symbol always returns the non-negative root.

9 (a) Rational: 7. (b) Irrational: $5\sqrt{2}$. (c) Rational: $\dfrac{1}{2}$. (d) Irrational: $\dfrac{\sqrt{6}}{3}$. (e) Rational: 3. (f) Irrational: $3\sqrt{3}$.

10 Squaring gives $(3\sqrt{2})^2 = 18$ and $(2\sqrt{3})^2 = 12$. Both numbers are positive, and for positive x, y, $x > y \iff x^2 > y^2$, so $18 > 12$ proves $3\sqrt{2} > 2\sqrt{3}$.

11 $\sqrt{45} = \sqrt{9 \cdot 5} = 3\sqrt{5}$, not $5\sqrt{3}$: the square factor 9 leaves 5 under the root. Check by squaring: $(3\sqrt{5})^2 = 9 \cdot 5 = 45$, whereas $(5\sqrt{3})^2 = 25 \cdot 3 = 75$.

To check any simplification, square it and compare with the radicand.

12 $\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$. In general $\sqrt{a}\sqrt{b} = \sqrt{ab}$, so the product is rational whenever ab is a perfect square; here $2 \times 8 = 16$.

13 Multiply by $\dfrac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$: $\dfrac{(\sqrt{5}+\sqrt{3})^2}{5-3} = \dfrac{8 + 2\sqrt{15}}{2} = 4 + \sqrt{15}$.

14 Side $= \sqrt{12} = 2\sqrt{3} \approx 3.46$ cm. Perimeter $= 4 \times 2\sqrt{3} = 8\sqrt{3} \approx 13.9$ cm.

15 Area $= \sqrt{8} \cdot \sqrt{2} = \sqrt{16} = 4$ cm^2. Perimeter $= 2(\sqrt{8} + \sqrt{2}) = 2(2\sqrt{2} + \sqrt{2}) = 6\sqrt{2}$ cm.

16 Diagonal $= \sqrt{32^2 + 18^2} = \sqrt{1024 + 324} = \sqrt{1348} = \sqrt{4 \cdot 337} = 2\sqrt{337} \approx 36.72$ cm.

17 $T = 2\pi\sqrt{\dfrac{1.2}{9.8}} = 2\pi\sqrt{\dfrac{6}{49}} = \dfrac{2\pi\sqrt{6}}{7} \approx 2.2$ s to 2 s.f. The exact form keeps π and $\sqrt{6}$ visible, so the rounding decision can be made once, at the end; an early decimal would be squared and divided inside the root and would inflate the error.

18 $(\sqrt{2}+1)^3 = (\sqrt{2}+1)(3+2\sqrt{2}) = 3\sqrt{2} + 4 + 3 + 2\sqrt{2} = 7 + 5\sqrt{2}$.

19 If $b \ne 0$ then $\sqrt{2} = -\dfrac{a}{b} \in \mathbb{Q}$, contradicting the irrationality of $\sqrt{2}$; hence $b = 0$, and then $a = 0$. For the reciprocal, $(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 9 - 8 = 1$, so $3 + 2\sqrt{2}$ is the reciprocal of $3 - 2\sqrt{2}$. Equivalently, if $\dfrac{1}{3 - 2\sqrt{2}} = a + b\sqrt{2}$, multiplying out gives $(3a - 4b - 1) + (3b - 2a)\sqrt{2} = 0$; the lemma forces $3b - 2a = 0$ and $3a - 4b = 1$, so $b = 2$, $a = 3$.

Exercise 1E

1 (a) 7.27. (b) 0.030. (c) 64 800. (d) 1.00: the fourth digit 5 rounds 0.999 up, and two zeros must be written.

2 (a) 2. (b) 4 (the final zero after the decimal point counts). (c) 3. (d) 4.

3 (a) $13.5 \le x < 14.5$ cm. (b) $3.65 \le m < 3.75$ kg. (c) The unit is 10 m, so $245 \le d < 255$ m. (d) $0.835 \le t < 0.845$ s.

4 Maximum absolute error $= 0.5$ kg. Maximum percentage error $= \dfrac{0.5}{48} \times 100\% = 1.04\%$.

5 $5.35 \le a < 5.45$ and $2.05 \le b < 2.15$. Sum: $7.40 \le a + b < 7.60$. Difference: minimum $5.35 - 2.15 = 3.20$, maximum $5.45 - 2.05 = 3.40$, so $3.20 \le a - b < 3.40$.

6 $6.45 \le s < 6.55$, so $6.45^2 = 41.6025$ and $6.55^2 = 42.9025$. Hence $41.6 \le A < 42.9$ cm^2.

7 The maximum error is half the unit, 0.05 cm, not 0.1 cm. The length satisfies $3.65 \le L < 3.75$ cm.

Common error: quoting the unit as the error. Rounding to the nearest 0.1 cm moves a value by at most 0.05 cm.

8 $23.5 \le L < 24.5$ m. A square of area 36 m^2 needs perimeter exactly 24 m, i.e. $L \ge 24$. The lower bound 23.5 m gives a maximum area of $\left(\dfrac{23.5}{4}\right)^2 = 34.5$ m^2 < 36 m^2, so the rope cannot be guaranteed to work.

9 They are equally precise: 3.4 cm to 1 d.p. and 34 mm to the nearest mm both have half-unit $0.05\text{ cm} = 0.5\text{ mm}$, so both give $3.35 \le L < 3.45$ cm. Precision is set by the half-unit, not by the name of the unit. Neither report tells us the accuracy, which depends on whether the ruler is correctly calibrated.

10 1 s.f.: unit 100 m, so $50 \le d < 150$ m. 3 s.f.: unit 1 m, so $99.5 \le d < 100.5$ m. The digits are the same but the claims differ by a factor of 100; only the stated accuracy fixes the meaning.

11 A: half-unit 0.05 cm, maximum percentage error $\dfrac{0.05}{12.0} \times 100\% = 0.42\%$. B: half-unit 0.005 cm, error 0.042%. B is ten times more precise and its maximum percentage error is ten times smaller.

12 The display is a ten-digit rounding of an irrational number. Any length used with it was measured to far fewer digits, so the result cannot carry ten; the answer should be rounded, here to 3.46. Quoting all ten digits claims a precision the measurement never had.

13 Bounds 41.5–42.5, 27.5–28.5, 14.5–15.5 cm. $V_{\min} = 41.5 \times 27.5 \times 14.5 = 16\,548.125\text{ cm}^3$; $V_{\max} = 42.5 \times 28.5 \times 15.5 = 18\,774.375\text{ cm}^3$. The nominal volume is $17\,640\text{ cm}^3$. Since $V_{\max} > 18\,000$, the courier cannot be certain it fits.

14 $67.5 \le m < 68.5$ kg, $1.715 \le h < 1.725$ m. $\text{BMI}_{\min} = \dfrac{67.5}{1.725^2} = 22.684\ldots$, $\text{BMI}_{\max} = \dfrac{68.5}{1.715^2} = 23.290\ldots$. To 1 d.p. and rounded outward, $22.6 \le \text{BMI} < 23.3$.

Round a lower bound down and an upper bound up, otherwise the interval no longer contains every possible value.

15 $3.95 \le s < 4.05$ cm. $V_{\min} = 3.95^3 = 61.63\text{ cm}^3$, $V_{\max} = 4.05^3 = 66.43\text{ cm}^3$ (to 2 d.p.), against a nominal 64 cm^3. Maximum percentage error $\approx \dfrac{66.43 - 64}{64} \times 100\% = 3.8\%$. The side has maximum error $\dfrac{0.05}{4.0} = 1.25\%$, about one third as large, because volume depends on the cube of the side.

16 For positive a, b, increasing either factor increases the product, so the largest product pairs the largest values: with $a = 5.4$, $b = 2.1$, $ab < 5.45 \times 2.15 = 11.7175$. For a difference, a larger b makes the result smaller, so the largest difference pairs the largest a with the smallest b: $a - b < 5.45 - 2.05 = 3.40$.

Investigation · Cycles in the last digit

→ **1.** Continuing the table: $2^9 = 512$ (2), $2^{10} = 1024$ (4), $2^{11} = 2048$ (8), $2^{12} = 4096$ (6). The last digits run $2, 4, 8, 6$ and then repeat.

→ **2.** 2^{20} ends in 6, 2^{21} in 2, 2^{22} in 4, 2^{23} in 8, because $20, 24$ are multiples of 4 and the cycle restarts there.

→ **3.** $2026 = 4 \times 506 + 2$, remainder 2. The second entry of the cycle $2, 4, 8, 6$ is 4, so 2^{2026} ends in 4.

→ **4.** The last digit of 2^{n+1} depends only on the last digit of 2^n. From 6: $6 \times 2 = 12$, so the next last digit is 2; from 2 gives 4, from 4 gives 8, from 8 gives 6. The four states map in a closed loop, so the pattern cannot break.

→ **5.** No. 2^n has only 2 as a prime factor, but a number ending in 0 is divisible by $10 = 2 \times 5$ and so by 5. Since 5 never divides 2^n, no power of 2 can end in 0.

→ **6.** Powers of 3: $3, 9, 7, 1$ repeating, cycle length 4. $2026 = 4 \times 506 + 2$, remainder 2, so 3^{2026} ends in 9.

→ **7.** Powers of 7: $7, 9, 3, 1$ repeating, also length 4. All three bases give period 4 and each cycle uses four different digits, but the cycles are not rotations of one another: bases 3 and 7 share the digit set $\{1, 3, 7, 9\}$ and one cycle is the other read backwards (since $3 \times 7 = 21$ ends in 1, 7^n and 3^{4-n} have the same last digit), while base 2 runs through the disjoint set $\{2, 4, 8, 6\}$.

→ **8.** Write $a = 10k + d$. Every term of $(10k + d)^n$ except d^n contains a factor 10, so it contributes 0 to the last digit. Hence a^n and d^n have the same last digit; for example 13^n ends as 3^n does.

→ **Generalise.** Write $n = 4q + r$, $0 \le r \le 3$. If $r = 1, 2, 3$, take the r-th digit of the cycle; if $r = 0$, take the fourth. For 8 the cycle is $8, 4, 2, 6$: $101 = 4 \times 25 + 1$, so 8^{101} ends in 8. For 7, $250 = 4 \times 62 + 2$, so 7^{250} ends in 9. Bases ending 0, 1, 5, 6 have a one-term cycle (0, 1, 5, 6 respectively), so every power ends in that digit; the remainder rule still applies with a cycle of length 1.

Modelling task · The fold that cannot happen

→ **1.** 0.1×2^{42}. Since $2^{10} \approx 10^3$, $2^{40} \approx 10^{12}$ and $2^{42} \approx 4 \times 10^{12}$, giving about 4×10^{11} mm $= 4 \times 10^5$ km. Exactly, $2^{42} = 4\,398\,046\,511\,104$, so the stack is 4.398×10^{11} mm $= 4.4 \times 10^5$ km to 2 s.f.

→ **2.** 4.4×10^5 km exceeds both the mean distance 3.844×10^5 km and the maximum 4.055×10^5 km, by about 3.4×10^4 km. The arithmetic of the claim is sound.

→ **3.** $0.05 \times 2^{42} = 2.2 \times 10^5$ km and $0.15 \times 2^{42} = 6.6 \times 10^5$ km. The lower bound is below the Moon's minimum distance 3.633×10^5 km, so within the stated precision the stack might not reach the Moon; the claim is numerically plausible but not certain.

→ **4.** Area $= \dfrac{0.0624}{2^{42}} = \dfrac{0.0624}{4.398 \times 10^{12}} = 1.42 \times 10^{-14}$ m^2, a square about 1.2×10^{-7} m ($0.12\ \mu$m) on a side. That is roughly a million times the cross-section of an atom, about the size of a virus: the stack would have to be microscopic in area, far thinner than any real sheet, so the folding is physically impossible even though the arithmetic works.

→ **5.** The claim is numerically sound and physically impossible. The arithmetic correctly doubles the thickness; it ignores that paper is a fixed number of atoms thick, cannot compress, and cannot bend through 42 folds. The most important assumption is exact doubling with no compression or buckling, because a real fold consumes length and traps air, so the true thickness grows more slowly.

→ **6.** The physical bound is more useful: it tells a scientist whether the model's conclusion applies to the world. Exact arithmetic only reports what the model says, and here a model with impossible assumptions predicts a true but unusable result.

Mixed practice 1

1 $-3 \in \mathbb{Z}, \mathbb{Q}$. $0 \in \mathbb{Z}, \mathbb{Q}$ (natural only if $0 \in \mathbb{N}$). $\frac{5}{8} \in \mathbb{Q}$. $\sqrt{36} = 6 \in \mathbb{N}, \mathbb{Z}, \mathbb{Q}$. $\sqrt{10}$ irrational. $0.\overline{6} = \frac{2}{3} \in \mathbb{Q}$. π irrational.

2 (a) a^3. (b) $4x^4$. (c) $8x^6$. (d) $81^{-3/4} = \frac{1}{(\sqrt[4]{81})^3} = \frac{1}{27}$.

3 (a) 4.5×10^{-5}. (b) 8.12×10^6. (c) $14.4 \times 10^2 = 1.44 \times 10^3$. (d) 3.0×10^3.

4 (a) $4\sqrt{5}$. (b) $3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3}$. (c) $\sqrt{48} = 4\sqrt{3}$. (d) $\frac{10\sqrt{5}}{5} = 2\sqrt{5}$.

5 $0.004678 \to 0.0047$ (2 s.f.); $36\,249 \to 36\,200$ (3 s.f.).

6 $7.25 \le L < 7.35$ cm; maximum percentage error $= \frac{0.05}{7.3} \times 100\% = 0.68\%$.

7 (a) Always: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$. (b) Sometimes: $\sqrt{2} + (-\sqrt{2}) = 0$, but $\sqrt{2} + \sqrt{3}$ is irrational. (c) Sometimes: $(\sqrt{2})^2 = 2$, but $(\sqrt[3]{2})^2$ is irrational. (d) Sometimes: $\sqrt{a^2} = |a|$, so the statement holds whenever $a \ge 0$, but $a = -3$ gives 3, not -3; as a claim about *every* real a it is false.

8 Assume $\sqrt{2} = \frac{p}{q}$ in lowest terms with $q \neq 0$. Then $p^2 = 2q^2$, so p^2 is even and hence $p = 2k$. Substituting gives $4k^2 = 2q^2$, so $q^2 = 2k^2$ and q is even. Both are even, contradicting "lowest terms" (no common factor). The assumption is false, so $\sqrt{2}$ is irrational. The lowest-terms condition is used only at the final contradiction.

9 $3^{2x-1} = 3^3$, so $2x - 1 = 3$, $x = 2$. $5^{x+1} = 5^{-2}$, so $x = -3$.

10 $0.\overline{36} = \frac{36}{99} = \frac{4}{11}$. For $2.1\overline{6}$: $100x - 10x = 216.\overline{6} - 21.\overline{6} = 195$, so $x = \frac{195}{90} = \frac{13}{6}$.

11 $\frac{4(\sqrt{7}+\sqrt{3})}{7-3} = \sqrt{7} + \sqrt{3}$. $\frac{\sqrt{8}+\sqrt{18}}{\sqrt{2}} = \frac{2\sqrt{2}+3\sqrt{2}}{\sqrt{2}} = 5$.

12 $\sqrt{20} = 2\sqrt{5}$, so $\sqrt{5} + \sqrt{20} = 3\sqrt{5}$. Then $(3\sqrt{5})^2 = 9 \cdot 5 = 45$; equivalently $5 + 2\sqrt{100} + 20 = 45$.

13 $\pi \approx 3.142$, $\sqrt{10} \approx 3.162$, $2\sqrt{3} \approx 3.464$, $3\sqrt{2} \approx 4.243$, so the order is $\pi < \sqrt{10} < 2\sqrt{3} < 3\sqrt{2}$. Exactly: $\pi^2 \approx 9.87 < 10$; $10 < 12 = (2\sqrt{3})^2$; $12 < 18 = (3\sqrt{2})^2$.

14 $12.45 \le l < 12.55$, $7.35 \le w < 7.45$. $P_{\min} = 2(12.45 + 7.35) = 39.6$ cm, $P_{\max} = 2(12.55 + 7.45) = 40.0$ cm. $A_{\min} = 12.45 \times 7.35 = 91.5075$, $A_{\max} = 12.55 \times 7.45 = 93.4975$, so $91.5 \le A < 93.5$ cm^2.

15 $(2.5 \times 10^{-5})(4.0 \times 10^4) = 10.0 \times 10^{-1} = 1.0 \times 10^0$ kg, i.e. 1.0 kg to 2 s.f.

16 $t = \frac{1.496 \times 10^8}{3.00 \times 10^5} = 498.67$ s $= 8$ min 19 s (3 s.f.: 8.31 min).

17 Both surds reduce to multiples of $\sqrt{2}$: $3\sqrt{2}$ and $5\sqrt{2}$, so the coefficients add. The method would be invalid if the simplified surds were not like — for example $\sqrt{8} + \sqrt{12} = 2\sqrt{2} + 2\sqrt{3}$ cannot collapse to one term — or if $\sqrt{18} + \sqrt{50}$ were replaced by $\sqrt{68}$.

18 Both measurements have 2 s.f., so the product can carry only 2 s.f.: $6.8 \times 4.2 = 28.56 \to 29$ cm^2. The bounds confirm this: $6.75 \times 4.15 = 28.0$ and $6.85 \times 4.25 = 29.1$, an interval about 1 cm^2 wide. Reporting 28.56 claims hundredth-of-a-unit precision that the inputs cannot support.

19 $x = 0.\overline{9}$ gives $10x = 9.\overline{9}$, so $9x = 9$ and $x = 1$. Decimal expansions are unique except that every terminating decimal has a second representation ending in recurring 9s; $1.000\ldots$ and $0.999\ldots$ are two numerals for the same real number, so uniqueness is not contradicted once that convention is excluded.

20 Suppose $\sqrt{2} + \sqrt{3} = r \in \mathbb{Q}$. Squaring gives $5 + 2\sqrt{6} = r^2$, so $\sqrt{6} = \frac{r^2 - 5}{2} \in \mathbb{Q}$. But $\sqrt{6}$ is irrational: from $6q^2 = p^2$ in lowest terms, $2 \mid p$, say $p = 2k$, so $3q^2 = 2k^2$, forcing $2 \mid q$, a contradiction. Hence no rational r exists.

Criterion A · Knowing and understanding

1 $\sqrt{16} = 4$: natural, integer and rational [1]. $\sqrt{17}$: irrational, because the square root of an integer is rational only when the integer is a perfect square, and $16 < 17 < 25$ [1]. $-2.5 = -\frac{5}{2}$: rational, not an integer [1]. $0.\overline{3} = \frac{1}{3}$: rational, since the decimal is recurring [1].

2 $\frac{2x^3 \cdot 3x^{-1}}{6x^{-2}} = \frac{6x^2}{6x^{-2}} = x^{2-(-2)} = x^4$. [3: numerator 1, quotient law 1, answer 1]

3 $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$ [2]. $16^{-3/4} = \frac{1}{\left(16^{1/4}\right)^3} = \frac{1}{2^3} = \frac{1}{8}$ [2].

4 $5.4 \times 10^6 = 5\,400\,000$ [1]. $64\,000\,000 = 6.4 \times 10^7$ [1]. $0.000\,000\,72$ is already an ordinary number $= 7.2 \times 10^{-7}$ in standard form [1].

The question asks for the first value "as an ordinary number", but it is already ordinary; both forms are given so the intended conversion is covered.

5 $\sqrt{98} - \sqrt{32} = 7\sqrt{2} - 4\sqrt{2} = 3\sqrt{2}$ [2]. $\sqrt{12} \times \sqrt{18} = \sqrt{216} = \sqrt{36 \cdot 6} = 6\sqrt{6}$ [2].

6 $\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} = 2\sqrt{3}$ [2]. $\frac{2}{\sqrt{5}-1} = \frac{2(\sqrt{5}+1)}{5-1} = \frac{\sqrt{5}+1}{2}$ [2].

7 $3.55 \le m < 3.65$ kg [2]. Maximum absolute error 0.05 kg, so percentage error $= \dfrac{0.05}{3.6} \times 100\% = 1.4\%$ [2].

8 With $a = 9$, $b = 16$: $\sqrt{a+b} = \sqrt{25} = 5$, but $\sqrt{a} + \sqrt{b} = 3 + 4 = 7$ [2]. Squaring $\sqrt{a} + \sqrt{b}$ gives $a + b + 2\sqrt{ab}$, which exceeds $a + b$ whenever $a, b > 0$, so the two expressions differ by $2\sqrt{ab}$ [2].

Marking guidance. Total 30. Award method marks for a correct law or factorisation even where arithmetic slips; do not award the answer mark if the form requested ("exactly", standard form, rational denominator) is not used. Q1 needs one mark per number, combining classification and justification. Q5 and Q6 award 1 per simplified term or rationalised denominator and 1 for the final form.

Criterion B · Investigating patterns

1 $n = 1$ to 8: 7, 9, 3, 1, 7, 9, 3, 1. [4: 1 mark for 4 correct, 2 for 6, 3 for 7, 4 for all 8]

2 The last digits repeat the block 7, 9, 3, 1. The period is 4. [3: pattern 1, block stated 1, period 1]

3 7^{50}: $50 = 4 \times 12 + 2$, remainder 2, so last digit 9 [2]. 7^{51}: remainder 3, last digit 3 [2]. 7^{100}: remainder 0, so the fourth entry, 1 [1].

4 Multiplying by 7 sends the last digit $7 \to 9$ ($7 \times 7 = 49$), $9 \to 3$ ($9 \times 7 = 63$), $3 \to 1$ ($3 \times 7 = 21$) and $1 \to 7$ ($1 \times 7 = 7$). Each step depends only on the previous last digit, and the four states map in a closed loop, so after every four powers the same digit returns and the cycle repeats for ever. [6: mapping 2, closure/determinism 2, induction to all n 1, clarity 1]

5 Rule: for a base ending in 3, 7 or 9, find $r = n \bmod 4$, replacing $r = 0$ by 4, and take the r-th entry of that base's repeating cycle, written to length 4: 3 gives 3, 9, 7, 1; 7 gives 7, 9, 3, 1; 9 gives 9, 1, 9, 1 (true period 2). Test: 9^{33}, $33 = 4 \times 8 + 1$, so the first entry, 9; the period-2 check (33 odd) gives 9 as well. [6: rule 3, cycle for 9 noted 1, test with remainder 1, answer 1]

→ **Sample response (Q4).** "The last digit of 7^n is decided entirely by the last digit of 7^{n-1}, because the higher digits contribute multiples of 10. Starting from 7, the four possible last digits are visited in the order $7 \to 9 \to 3 \to 1 \to 7$. No step can leave this loop, so the sequence of last digits must repeat every four powers, for ever."

Marking guidance. Total 24: Q1 4, Q2 3, Q3 5, Q4 6, Q5 6. For 7–8 the response must justify why the cycle cannot break (a dependence argument), not merely observe repetition; a table alone caps Q4 at 3. For 5–6 a correct rule with a correct test but no statement of the period earns 4–5. Accept 7^{100} justified as remainder 0 mapping to the fourth entry.

Criterion C · Communicating

→ **Requirements.** A 300–400 word guide with a title, four labelled parts (i)–(iv), at least two worked equations and a summary box. (i) the decimal is a finite display of an infinite, non-repeating expansion; (ii) exact form preserves value and structure, e.g. $\sqrt{72} = 6\sqrt{2}$; (iii) measurement contexts need decimals to a stated accuracy; (iv) early rounding can reverse a conclusion.

→ **Sample (opening and one example).** "Your calculator shows $\sqrt{2} = 1.41421356$, then stops. It stops because the screen is finite, not because the decimal is. The true expansion never ends and never repeats, which is exactly why $\sqrt{2}$ cannot be written as a fraction. The displayed number is a rational approximation: square it and you get $1.99999999\ldots$, not 2. Exact form keeps the meaning visible. For instance $\sqrt{72} = 6\sqrt{2}$ and $\sqrt{50} = 5\sqrt{2}$, so $\sqrt{72} - \sqrt{50} = \sqrt{2}$ at once; the decimals 8.485 and 7.071 leave a residue of $0.000\ldots$ that hides the exact answer."

→ **Sample (part iv).** "Rounding too early changes conclusions. Forty-two folds of 0.1 mm give $0.1 \times 2^{42} = 4.40 \times 10^5$ km, beyond the Moon's maximum distance 4.06×10^5 km. Replacing 2^{42} by 4×10^{12} gives 4.0×10^5 km, which would wrongly suggest the stack falls short. The estimate was useful as a check, but only the unrounded value settles the claim."

Marking guidance. Total 20: (i) 4, (ii) 4, (iii) 3, (iv) 5, notation and worked examples 4. Award full marks for (i) only if the response distinguishes "the calculator stops" from "the decimal stops" and links non-repeating decimals to irrationality. In (iv) the example must show a conclusion, not just a numerical difference. Deduct one mark for a missing summary box, and one for consistent misuse of notation such as $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.

Criterion D · Applying mathematics in real-life contexts

1 $V = 42 \times 28 \times 15 = 17\,640$ cm^3. [2]

2 $41.5 \le l < 42.5$, $27.5 \le w < 28.5$, $14.5 \le h < 15.5$ cm [3]. $V_{\min} = 41.5 \times 27.5 \times 14.5 = 16\,548$ cm^3; $V_{\max} = 42.5 \times 28.5 \times 15.5 = 18\,774$ cm^3 [3].

3 The interval $16\,548 \le V < 18\,774\ \text{cm}^3$ straddles the locker volume $18\,000\ \text{cm}^3$. The nominal value fits, but the upper bound exceeds the locker by $774\ \text{cm}^3$, so the courier cannot be certain. [4: comparison 1, straddle stated 1, interpretation 1, use of both bounds 1]

4 Assumptions: (1) the parcel is an exact rectangular prism, whereas real parcels bulge, so the true volume could exceed $V_{\max}$ [3]; (2) the three measurements are independent and the locker's usable volume is exactly $18\,000\ \text{cm}^3$, whereas packing orientation, flaps and shelf clearance may waste space, reducing the usable volume [3].

5 Recommendation: do not promise that it fits; re-measure to the nearest millimetre, which shrinks the interval by a factor of 10 and would give $V_{\max} \approx 17\,752\ \text{cm}^3 < 18\,000$ [3]. Practical check: place the parcel in a sample locker, or use an internal gauge, because a test fit captures bulging and clearance that a volume model cannot [3].

→ **Sample judgement.** "The nominal volume $17\,640\ \text{cm}^3$ is below $18\,000\ \text{cm}^3$, but the data do not justify a guarantee: the upper bound $18\,774\ \text{cm}^3$ exceeds the locker. I would re-measure to the millimetre and test-fit a prototype before promising delivery."

Marking guidance. Total 24: Q1 2, Q2 6, Q3 4, Q4 6, Q5 6. In Q2 award 1 per correct dimension bound and 1 per correct extreme product. In Q3 a bare "yes" or "no" earns at most 1; the mark is for comparing the interval, not the nominal value. In Q4 each assumption needs a stated effect on the decision, not just a list. In Q5 full marks require both a decision that follows from the bounds and a check that addresses the model's weakness.

CHAPTER 2 · ALGEBRAIC EXPRESSIONS AND FACTORISATION

Opening problem · design under constraint

1 Each corner cut removes x cm from both ends of a side, so the base is square with length $= 24 - 2x$ cm and width $= 24 - 2x$ cm, and the folded-up side gives height $= x$ cm.

2 Volume $=$ length $\times$ width $\times$ height, so $V = (24 - 2x)(24 - 2x)(x) = x(24 - 2x)^2$ cm^3. Taking a factor 2 out of each bracket gives the equivalent $V = 4x(12 - x)^2$.

3 $(24 - 2x)^2 = 24^2 - 2(24)(2x) + (2x)^2 = 576 - 96x + 4x^2$, so $V = x(576 - 96x + 4x^2) = 4x^3 - 96x^2 + 576x$, in standard form with the powers in descending order. Check with $4x(12 - x)^2 = 4x(144 - 24x + x^2) = 576x - 96x^2 + 4x^3$, the same polynomial.

4 $x = 2$: $V = 2(20)^2 = 800$ cm^3. $x = 4$: $V = 4(16)^2 = 1024$ cm^3. $x = 6$: $V = 6(12)^2 = 864$ cm^3. The largest of the three is $x = 4$ cm, with 1024 cm^3. Using the expanded form as a check, $4(4)^3 - 96(4)^2 + 576(4) = 256 - 1536 + 2304 = 1024$. Going from $x = 2$ to $x = 4$ gains 224 cm^3, but going from $x = 4$ to $x = 6$ loses 160 cm^3, so the volume turns around near $x = 4$: a taller box is not automatically a bigger one, because the base shrinks as x grows.

5 $0 < x < 12$. A cut of $x = 0$ makes no box at all, so $x > 0$; and the base side $24 - 2x$ must stay positive, so $x < 12$. At $x = 12$ the two corner cuts meet in the middle of each side, the base has zero width and $V = 0$; for $x > 12$ the cuts overlap and no box exists. In the unrestricted polynomial those values are legal, which is why the context, not the algebra, sets the domain.

Prerequisite check

1 $3(x + 4) = 3x + 12$. $-2(x - 5) = -2x + 10$, because the negative multiplies both terms.

2 $4x + 3x - 2x = (4 + 3 - 2)x = 5x$.

3 $\text{HCF}(12, 18) = 6$. $\text{HCF}(6x^2, 9x) = 3x$: the HCF of 6 and 9 is 3, and the lowest power of x present is x^1.

4 $(x + 3)(x + 2) = x^2 + 2x + 3x + 6 = x^2 + 5x + 6$.

5 $(-2)^2 - 3(-2) + 1 = 4 + 6 + 1 = 11$.

6 $2x = 18$, so $x = 9$.

Exercise 2A

1 (a) $5x + 15$. (b) $-6x + 3$. (c) $4x \cdot 3x + 4x \cdot 2 = 12x^2 + 8x$. (d) $-2a^3 + 6a^2 - 2a$.

2 (a) $x^2 + 7x + 10$. (b) $x^2 + x - 4x - 4 = x^2 - 3x - 4$. (c) $a^2 - 6a - 3a + 18 = a^2 - 9a + 18$. (d) $2y^2 + 8y + y + 4 = 2y^2 + 9y + 4$.

3 (a) $6x^2 + 8x - 9x - 12 = 6x^2 - x - 12$. (b) $8a^2 - 20a + 2a - 5 = 8a^2 - 18a - 5$. (c) $3b^2 - 9b - 2b + 6 = 3b^2 - 11b + 6$. (d) $15 + 5x - 6x - 2x^2 = 15 - x - 2x^2$.

4 $(x + 2)(x^2 - 3x + 1) = x^3 - 3x^2 + x + 2x^2 - 6x + 2 = x^3 - x^2 - 5x + 2$.

5 (a) $3x + 6 - 2x + 10 = x + 16$. (b) $8x^2 - 4x - 6x^2 + 12x = 2x^2 + 8x$.

6 $(x^2 - 2x - 3) + (x^2 + 6x + 8) = 2x^2 + 4x + 5$.

7 Factorised: $(x + 7)(x + 2)$. Expanded: $x^2 + 9x + 14$. The area model splits the rectangle into four regions with areas x^2, $2x$, $7x$ and 14; the two x-terms combine to $9x$.

8 The cross terms $4x$ and $3x$ are missing. Correct expansion: $x^2 + 4x + 3x + 12 = x^2 + 7x + 12$. The area model has four regions; the student counted only the x^2 square and the 12 square and omitted the two rectangular strips.

Common error: writing only the "first" and "last" products of two brackets.

9 $(x + a)(x + b) = x^2 + bx + ax + ab = x^2 + (a + b)x + ab$. With $a = 7$, $b = -2$: sum 5, product -14, so $(x + 7)(x - 2) = x^2 + 5x - 14$.

10 An identity: the two sides are equal for every value of x, since expanding the right side reproduces the left. An equation is a condition: $x^2 + 5x + 6 = 0$ is true only for $x = -2$ or $x = -3$.

11 Area $= (3x + 2)(2x - 1) = 6x^2 - 3x + 4x - 2 = 6x^2 + x - 2$ m^2. When $x = 4$: $6(16) + 4 - 2 = 98$ m^2. Check: $14 \times 7 = 98$.

12 $2x^2 + 4x(x + 5) = 2x^2 + 4x^2 + 20x = 6x^2 + 20x$. The $2x^2$ is the pair of square faces (top and base); $4x(x + 5)$ is the four rectangular side faces, each of area $x(x + 5)$.

13 Take $a = 3$, $b = 4$: $(3 + 4)^2 = 49$, but $3^2 + 4^2 = 25$. The correct identity is $(a + b)^2 = a^2 + 2ab + b^2$. The extra $2ab$ is the sum of the two cross products ab and ba, which the false statement omits.

14 $(x - 1)(x^2 + x + 1) = x^3 + x^2 + x - x^2 - x - 1 = x^3 - 1$. $(x - 1)(x^3 + x^2 + x + 1) = x^4 - 1$. Prediction: $(x - 1)(x^4 + x^3 + x^2 + x + 1) = x^5 - 1$. Proof: multiplying by x gives $x^5 + x^4 + x^3 + x^2 + x$, subtracting the bracket leaves $x^5 - 1$, since every intermediate power cancels. In general $(x - 1)(x^{n-1} + x^{n-2} + \cdots + x + 1) = x^n - 1$.

Exercise 2B

1 (a) $4x^2$. (b) $5ab$. (c) $3xy$. (d) $2a^2b^2$.

2 (a) $5(x + 3)$. (b) $7x(x - 3)$. (c) $4a^2(2a + 3)$. (d) $3xy(3x - 2y)$.

3 (a) $3(x^2 - 2x + 3)$. (b) $4ab(a + 2b - 3)$. (c) $x(x^2 - x + 1)$. (d) $5x^2y^2(2x - 3y)$.

4 (a) $a(x + y) + 3(x + y) = (x + y)(a + 3)$. (b) $2(x + y) + p(x + y) = (x + y)(2 + p)$. (c) $x^2(x + 4) + 3(x + 4) = (x + 4)(x^2 + 3)$. (d) $a(b - 2) + 3(b - 2) = (b - 2)(a + 3)$.

5 (a) $2(x^2 - 9) = 2(x - 3)(x + 3)$. (b) $3x(x^2 - 4) = 3x(x - 2)(x + 2)$. (c) $5(x^2 - 9) = 5(x - 3)(x + 3)$. (d) $4x(x^2 - 4) = 4x(x - 2)(x + 2)$.

6 Dividing by x is valid only when $x \neq 0$, and it does not turn the expression into 6: $\frac{x^2+6x}{x} = x+6$, because $\frac{x^2}{x} = x$ and $\frac{6x}{x} = 6$. At $x = 1$ the original expression equals $1+6 = 7$, not 6. The "6" comes from replacing $\frac{x^2}{x}$ by 1 instead of x.

Common error: cancelling a term rather than a factor; only factors of the whole expression may be cancelled.

7 The distributive law multiplies every term inside the bracket by 2: $2(2x+3) = 2\cdot 2x + 2\cdot 3 = 4x+6$. Writing $2\cdot 2x+3$ multiplies only the first term, leaving the 3 unchanged.

8 Length $= \frac{6x^2+9x}{3x} = \frac{3x(2x+3)}{3x} = 2x+3$. The division is exact because $3x$ is a factor of the whole area, so the quotient is a polynomial; no remainder term survives.

9 $x^3-2x^2+5x-10 = x^2(x-2)+5(x-2) = (x-2)(x^2+5)$. Check: $x^3+5x-2x^2-10$, which is the original.

10 Length $= \frac{12x^2+18x}{6x} = 2x+3$. Perimeter $= 2\left((2x+3)+6x\right) = 2(8x+3) = 16x+6$.

11 Treat $(x+1)$ as one factor: $(x+1)^2+3(x+1) = (x+1)\left[(x+1)+3\right] = (x+1)(x+4)$. Expanding: x^2+5x+4, which matches $x^2+2x+1+3x+3$.

12 $x^2-y^2+x-y = (x-y)(x+y)+(x-y) = (x-y)(x+y+1)$. Expanding gives back x^2-y^2+x-y.

13 $a^2-ab = a(a-b)$. A square of side a has area a^2; removing a rectangle a by b leaves a rectangle of dimensions a and $a-b$, whose area is $a(a-b)$. The label on the remaining rectangle is $a-b$ along one side and a along the other.

Exercise 2C

1 (a) $(x+3)(x+5)$, since $3+5=8$ and $3\cdot 5=15$. (b) $(x+4)(x+5)$. (c) $(x+2)(x+4)$. (d) $(x+3)(x+7)$.

2 (a) $(x-3)(x-4)$. (b) $(x-5)(x+2)$. (c) $(x+5)(x-4)$. (d) $(x-3)(x-6)$.

3 (a) $ac=2$, split $4x+x$: $(2x+1)(x+2)$. (b) $ac=24$, split $4x+6x$: $(3x+4)(x+2)$. (c) $ac=6$, split $-6x-x$: $(2x-1)(x-3)$. (d) $ac=30$, split $-10x-3x$: $(5x-3)(x-2)$.

4 (a) $ac=-60$, split $15x-4x$: $(2x+5)(3x-2)$. (b) $ac=-60$, split $-10x+6x$: $(2x-5)(2x+3)$. (c) $ac=-24$, split $6x-4x$: $(4x+3)(2x-1)$. (d) $ac=-90$, split $-15x+6x$: $(3x-5)(3x+2)$.

5 (a) $2(x^2-4x+3) = 2(x-1)(x-3)$. (b) $3(x^2+5x+6) = 3(x+2)(x+3)$. (c) $4(x^2-1) = 4(x-1)(x+1)$. (d) $5x(x^2-4) = 5x(x-2)(x+2)$.

6 Expanding gives $(x+p)(x+q) = x^2+(p+q)x+pq$. For this to equal x^2+bx+c for every x, the coefficients of like powers must match, so $p+q=b$ and $pq=c$. Conversely, any integers with that sum and product produce the factorisation.

7 The student's answer expands to x^2+6x+8, which has $+6x$ instead of $-6x$. Both constants must be negative: product $+8$ with sum -6 gives -2 and -4, so $x^2-6x+8 = (x-2)(x-4)$.

8 Area $= x^2+11x+30 = (x+5)(x+6)$, so the length is $x+6$. Factorising exposes the two linear factors directly: since area = width × length, the factor that is not the width must be the length.

9 $b^2-4c = 16-8 = 8$, not a perfect square, so x^2+4x+2 does not factorise over the integers. Checking factor pairs of 2: $1+2=3\neq 4$, so no integer pair works; the roots are $-2\pm\sqrt{2}$.

10 $2x^2+9x+4 = (2x+1)(x+4)$, so the width is $x+4$. Perimeter $= 2\left((2x+1)+(x+4)\right) = 2(3x+5) = 6x+10$.

11 Let $X = x^2$: $X^2+5X+6 = (X+2)(X+3)$, so $x^4+5x^2+6 = (x^2+2)(x^2+3)$. Check: $x^4+3x^2+2x^2+6 = x^4+5x^2+6$.

12 $b \in \{7, 8, 13, -7, -8, -13\}$. The pairs with product 12 are $(1,12)$, $(2,6)$, $(3,4)$ and their negatives, giving sums $13, 8, 7$ and $-13, -8, -7$.

13 If $x^2+bx+c = (x+p)(x+q)$ then $b = p+q$ and $c = pq$, so $b^2-4c = (p+q)^2-4pq = p^2-2pq+q^2 = (p-q)^2$, a perfect square. Conversely, if $b^2-4c = k^2$ with k an integer, then k has the same parity as b (their squares differ by $4c$), so $p = \frac{b+k}{2}$ and $q = \frac{b-k}{2}$ are integers with sum b and product $\frac{b^2-k^2}{4} = c$.

Exercise 2D

1 (a) $x^2+10x+25$. (b) $x^2-14x+49$. (c) $9x^2+12x+4$. (d) $4a^2-20ab+25b^2$.

2 (a) x^2-16. (b) $4x^2-9$. (c) $25a^2-b^2$. (d) x^4-1.

3 (a) $(x-6)(x+6)$. (b) $(2x-3)(2x+3)$. (c) $(5a-4b)(5a+4b)$. (d) $(x^2-1)(x^2+1) = (x-1)(x+1)(x^2+1)$.

4 (a) $2(x-4)(x+4)$. (b) $3x(x-3)(x+3)$. (c) $5(x-3)(x+3)$. (d) $2(4x^2-25) = 2(2x-5)(2x+5)$.

5 (a) $(100+3)(100-3) = 10\,000-9 = 9991$. (b) $(64-36)(64+36) = 28\times 100 = 2800$. (c) $(205-195)(205+195) = 10\times 400 = 4000$. (d) $(50+1)(50-1) = 2500-1 = 2499$.

6 (a) $\left[(x+3)-(x-3)\right]\left[(x+3)+(x-3)\right] = 6\cdot 2x = 12x$. (b) $(2b)(4a) = 8ab$.

7 $(a^2+2ab+b^2)+(a^2-2ab+b^2) = 2a^2+2b^2 = 2(a^2+b^2)$. The cross terms cancel: the identity says the squares of the sum and of the difference add to twice the sum of the separate squares.

8 $(x+3)^2$ is the area of a square of side $x+3$, made of x^2, two rectangles $3x$ and the square 9: x^2+6x+9. The student kept only the large square and the corner square, omitting the two $3x$ strips.

Common error: squaring each term separately; the two cross terms are the missing strips.

9 $(a+b)(a-b) = a^2 - ab + ab - b^2 = a^2 - b^2$. Geometrically, a square of side a with a square of side b removed from one corner leaves an L-shape; cutting along the inner edge and moving one piece gives a rectangle of dimensions $(a+b)$ and $(a-b)$, so the area is $(a+b)(a-b)$.

10 $x^2 - (x-4)^2 = (x-(x-4))\,(x+(x-4)) = 4(2x-4) = 8x - 16 = 8(x-2)$. At $x = 10$: $80 - 16 = 64$, and $100 - 36 = 64$.

11 $101^2 = (100+1)^2 = 100^2 + 2(100)(1) + 1^2 = 10\,000 + 200 + 1 = 10\,201$.

12 $(x+y)^3 = (x+y)(x^2+2xy+y^2) = x^3 + 2x^2y + xy^2 + x^2y + 2xy^2 + y^3 = x^3 + 3x^2y + 3xy^2 + y^3$. The coefficients $1, 3, 3, 1$ are symmetric, matching the two cross terms that combine at each level.

13 $(a+b)^2 - c^2 = (a+b-c)(a+b+c)$. So $(a+b)^2 = c^2$ gives $a+b = \pm c$, hence $a = c-b$ or $a = -c-b$. If a, b, c are lengths, $a > 0$ requires $c > b$ for the first solution, and the second is inadmissible. The $\pm$ comes from the difference of squares itself, not from an irreversible squaring step, so no root has been introduced: the negative branch is rejected only because a length cannot be negative.

Exercise 2E

1 (a) $2x$. (b) $\dfrac{2a^2}{3b}$. (c) $\dfrac{(x-2)(x+2)}{x+2} = x-2,\ x \neq -2$. (d) $x+3,\ x \neq 3$.

2 (a) $\dfrac{(x+2)(x+3)}{(x+1)(x+3)} = \dfrac{x+2}{x+1},\ x \neq -1, -3$. (b) $\dfrac{2(x-2)(x+2)}{(x+3)(x-2)} = \dfrac{2(x+2)}{x+3},\ x \neq 2, -3$. (c) $\dfrac{x(x-2)}{(x-2)^2} = \dfrac{x}{x-2},\ x \neq 2$. (d) $\dfrac{(x-1)(x+1)}{(x+1)^2} = \dfrac{x-1}{x+1},\ x \neq -1$.

3 (a) $\dfrac{x+1}{x+3},\ x \neq 2, -3$. (b) $\dfrac{(x-1)(x+1)(x+2)}{(x-2)(x+2)(x-1)} = \dfrac{x+1}{x-2},\ x \neq 2, -2, 1$.

4 (a) $\dfrac{x^2-9}{x-3} = x+3,\ x \neq -1, 3$. (b) $\dfrac{x(x-4)}{x-4} = x,\ x \neq -1, 4$.

5 (a) $\dfrac{y+x}{xy}$. (b) $\dfrac{2(x-1)+3(x+1)}{(x+1)(x-1)} = \dfrac{5x+1}{(x+1)(x-1)}$. (c) $\dfrac{4(x+3)-(x-2)}{(x-2)(x+3)} = \dfrac{3x+14}{(x-2)(x+3)}$.

6 $\dfrac{(x+1)+(x-1)}{(x-1)(x+1)} = \dfrac{2x}{x^2-1}$. At $x = 3$: $\dfrac{6}{8} = \dfrac{3}{4}$. Check: $\dfrac{1}{2} + \dfrac{1}{4} = \dfrac{3}{4}$.

7 $(x^2 - 16) = (x-4)(x+4)$, so the fraction is $x+4$ for $x \neq 4$. The student cancelled x^2 with x and wrote $x-4$, treating terms as factors; only the factor $(x-4)$ may be cancelled.

State the restriction $x \neq 4$ even after cancelling: it comes from the original denominator.

8 At $x = 1$ the original denominator is $1 - 1 = 0$, so the expression is $\dfrac{0}{0}$, which is undefined. The simplified form $x+1$ is defined there and equals 2, but simplification is an equivalence only on the original domain; it cannot restore a value the original expression never had.

9 $\dfrac{1}{R} = \dfrac{R_2 + R_1}{R_1 R_2}$, so $R = \dfrac{R_1 R_2}{R_1 + R_2}$. With $R_1 = 6$, $R_2 = 3$: $R = \dfrac{18}{9} = 2\,\Omega$.

10 $\dfrac{x+1}{(x-3)(x+2)} + \dfrac{2(x+2)}{(x-3)(x+2)} = \dfrac{3x+5}{(x-3)(x+2)}$, $x \neq 3, -2$.

11 Multiply numerator and denominator by ab: $\dfrac{b+a}{b-a}$. This is valid because $a, b \neq 0$, and the denominator $b - a \neq 0$ because $a \neq b$.

12 Area $= \dfrac{x+2}{x-1} \cdot \dfrac{x-1}{x+3} = \dfrac{x+2}{x+3}$, with $x \neq 1$ and $x \neq -3$.

13 $\dfrac{x^3-1}{x^2-1} = \dfrac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \dfrac{x^2+x+1}{x+1},\ x \neq 1, -1$. The factor $(x-1)$ cancels because it divides the whole numerator (difference of cubes) and the whole denominator (difference of squares); the value $x = 1$ remains excluded.

Exercise 2F

1 (a) $x = \dfrac{y-2}{5}$. (b) $2x = 7 - y$, $x = \dfrac{7-y}{2}$. (c) $3x = 12 - 2y$, $x = 4 - \dfrac{2y}{3}$. (d) $4x = 15 + 3y$, $x = \dfrac{15+3y}{4}$.

2 (a) $y = 9 - 2x$. (b) $2y = 5x - 8$, $y = \dfrac{5x-8}{2}$. (c) $\dfrac{y}{5} = 1 - 2x$, $y = 5 - 10x$. (d) $y + 2 = \dfrac{x}{3}$, $y = \dfrac{x}{3} - 2$.

3 $s - \dfrac{1}{2}at^2 = ut$, so $u = \dfrac{s - \frac{1}{2}at^2}{t} = \dfrac{2s - at^2}{2t},\ t \neq 0$.

4 $at = v - u$, so $a = \dfrac{v-u}{t},\ t \neq 0$.

5 (a) $2A = (a+b)h$, so $h = \dfrac{2A}{a+b}$. (b) $h = \dfrac{V}{\pi r^2},\ r \neq 0$.

6 (a) $r = \dfrac{C}{2\pi}$. (b) $r^2 = \dfrac{A}{\pi}$, so $r = \sqrt{\dfrac{A}{\pi}}$ because $r > 0$.

7 $\dfrac{T}{2\pi} = \sqrt{\dfrac{l}{g}}$, so $\dfrac{T^2}{4\pi^2} = \dfrac{l}{g}$ and $l = \dfrac{gT^2}{4\pi^2}$.

8 $y(x-2) = x+1$, so $yx - 2y = x + 1$. Then $x(y-1) = 1 + 2y$, giving $x = \dfrac{1+2y}{y-1},\ y \neq 1$.

9 $y(x+1) = 3x - 2$, so $yx + y = 3x - 2$. Then $x(y-3) = -2 - y$, giving $x = \dfrac{y+2}{3-y},\ y \neq 3$.

10 The student divided only the term $3x$ by 3 and left the 5 untouched. The correct order is to undo the subtraction first: $3x = y + 5$, then $x = \dfrac{y+5}{3}$. Their answer gives $x = \dfrac{y}{3} - 5$, which is not equivalent.

Undo operations in reverse order: addition and subtraction before multiplication and division.

11 Squaring: $a^2 = b^2 - c^2$, so $b^2 = a^2 + c^2$ and $b = \pm\sqrt{a^2 + c^2}$. Since a is a principal square root, $a \geq 0$ and $|b| \geq |c|$; if b must be positive (a length), take $b = \sqrt{a^2 + c^2}$, and if $b < 0$, take the negative root.

12 $2s = 2ut + at^2$, so $a = \dfrac{2s - 2ut}{t^2} = \dfrac{2(s-ut)}{t^2},\ t \neq 0$. When $s = 0$: $t\left(u + \dfrac{at}{2}\right) = 0$, so either $t = 0$ (the start) or $t = -\dfrac{2u}{a}$: the object has returned to its starting point, which requires a and u to have opposite signs.

13 Multiply by xy: $ay + bx = xy$. Then $ay = x(y - b)$, so $x = \dfrac{ay}{y - b}$, with $y \neq b$ and $y \neq 0$.

14 $y^2 = \dfrac{x + 1}{x - 1}$, so $y^2(x - 1) = x + 1$ and $x(y^2 - 1) = y^2 + 1$. Hence $x = \dfrac{y^2 + 1}{y^2 - 1}$. Since y is a square root, $y \geq 0$, and $y \neq 1$; the domain is $y \geq 0$, $y \neq 1$. If $y = 1$, then $y^2 = 1$ and the equation becomes $x - 1 = x + 1$, i.e. $-1 = 1$, which has no solution: the line $y = 1$ is an asymptote the graph never reaches.

Investigation · Which quadratics factorise?

→ **1.** Pairs with product 12: $(1, 12)$, $(2, 6)$, $(3, 4)$, $(-1, -12)$, $(-2, -6)$, $(-3, -4)$. Factorisations: $(x + 1)(x + 12)$, $(x + 2)(x + 6)$, $(x + 3)(x + 4)$, $(x - 1)(x - 12)$, $(x - 2)(x - 6)$, $(x - 3)(x - 4)$, with $b = 13, 8, 7, -13, -8, -7$.

→ **2.** The factorable values are $b = \pm 7, \pm 8, \pm 13$: six values.

→ **3.** For $x^2 + bx + 7$: pairs $(1, 7)$ and $(-1, -7)$, so $b = 8$ or $b = -8$, only two values. Since 7 is prime it has just one positive factor pair (and its negative), whereas 12 has three, so there are fewer ways to choose p and q.

→ **4.** $x^2 + 7x + 12$: $b^2 - 4c = 49 - 48 = 1 = 1^2$, factorises as $(x + 3)(x + 4)$. $x^2 + 6x + 8$: $36 - 32 = 4 = 2^2$, factorises as $(x + 2)(x + 4)$. $x^2 + 2x - 15$: $4 + 60 = 64 = 8^2$, factorises as $(x + 5)(x - 3)$. $x^2 + 4x + 2$: $16 - 8 = 8$, not a square, and it does not factorise. The rule holds in all four cases.

→ **Generalise.** Expanding $(x + p)(x + q) = x^2 + (p + q)x + pq$ and comparing coefficients gives $p + q = b$, $pq = c$. Then $b^2 - 4c = (p + q)^2 - 4pq = p^2 - 2pq + q^2 = (p - q)^2 \geq 0$, a perfect square, so the condition is necessary. It is also sufficient: if $b^2 - 4c = k^2$, then $k^2 = b^2 - 4c$ shows k and b have the same parity, so $p = \dfrac{b + k}{2}$ and $q = \dfrac{b - k}{2}$ are integers with $p + q = b$ and $pq = c$. Parity: $p + q = b$ fixes the parity of the pair, and b and $p - q$ always share parity because $b + (p - q) = 2p$ is even; that is what makes the halves integral.

Modelling task · Return to the open box

→ **1.** $V = x(24 - 2x)^2 = x(576 - 96x + 4x^2) = 4x^3 - 96x^2 + 576x$.

→ **2.** $x > 0$ gives a positive height, and $24 - 2x > 0$ gives $x < 12$, so $0 < x < 12$. At $x = 0$ the sheet is flat and the volume is 0; at $x = 12$ the base has zero width, the sheet folds double and the volume is again 0.

→ **3.** $x = 1$: 484; 2: 800; 3: 972; 4: 1024; 5: 980; 6: 864; 7: 700; 8: 512 cm^3. To 3 significant figures, as the task asks, these are $484, 800, 972, 1.02 \times 10^3, 980, 864, 700, 512$: only $x = 4$ needs rounding, and standard form shows the three figures unambiguously. Check: $4(24 - 8)^2 = 4 \times 256 = 1024$.

→ **4.** The largest table value is 1024 at $x = 4$. The values rise to 1024 and fall symmetrically (972, 1024, 980), so the peak is at or very near $x = 4$; the graph has its vertex at $(4, 1024)$ and zeros at $x = 0$ and $x = 12$ (double). Best estimate: $x = 4$ cm, giving about 1024 cm^3.

→ **5.** Assumptions: the corner squares are exact and the sides fold vertically; the card has negligible thickness and no lid is needed. Card thickness reduces the internal volume and a lid would consume extra card, lowering the height for a given x, so the optimum would shift slightly below 4 cm.

→ **6.** Recommendation: "Cut 4 cm squares. The expanded form $V = 4x^3 - 96x^2 + 576x$ lets me evaluate candidates, and the factorised form $V = x(24 - 2x)^2$ shows at once that the volume is zero at $x = 0$ and $x = 12$, so the maximum must lie between. At $x = 4$ the volume is 1024 cm^3, about 1.02 litres. This is a mathematical maximum for an ideal box; before production, test one box made from the actual card."

Mixed practice 2

1 $2x^2 + 8x - 3x - 12 = 2x^2 + 5x - 12$.

2 $3x(2x - 5)$.

3 $(x - 3)(x - 5)$, since $-3 - 5 = -8$ and $(-3)(-5) = 15$.

4 $ac = -8$, split $-8x + x$: $2x^2 - 8x + x - 4 = (2x + 1)(x - 4)$.

5 $9x^2 - 12x + 4$.

6 $(4x - 5)(4x + 5)$.

7 $\dfrac{(x - 4)(x + 4)}{(x - 4)(x + 1)} = \dfrac{x + 4}{x + 1}$, $x \neq 4, -1$.

8 $\dfrac{3(x - 1) + 2(x + 2)}{(x + 2)(x - 1)} = \dfrac{5x + 1}{(x + 2)(x - 1)}$.

9 $y(x + 3) = 4x - 1$, so $yx - 4x = -1 - 3y$ and $x(y - 4) = -(1 + 3y)$. Hence $x = \dfrac{1 + 3y}{4 - y}$, $y \neq 4$.

10 $(a - b)^2 = (a - b)(a - b) = a^2 - ab - ba + b^2 = a^2 - 2ab + b^2$.

11 The distributive law multiplies both terms by 3, so $3(x + 2) = 3x + 6$. Writing $3x + 2$ multiplies only the x; the 2 is left unscaled. Test $x = 10$: $3 \times 12 = 36$, but $30 + 2 = 32$.

12 $(x + a)(x + b) = x^2 + bx + ax + ab = x^2 + (a + b)x + ab$, because bx and ax are like terms and the constants multiply.

13 $x^3 - 4x = x(x^2 - 4) = x(x - 2)(x + 2)$.

14 $x^2 + 7x + 10 = (x + 2)(x + 5)$, so the length is $x + 5$. Perimeter $= 2\,((x + 2) + (x + 5)) = 4x + 14$.

15 The identity is true for every value of x, since both sides expand to $x^2 + 6x + 9$. The equation is a condition on x, satisfied only by $x = -3$; substituting other values makes it false.

16 $2s = (u + v)t$, so $t = \dfrac{2s}{u + v}$, $u + v \neq 0$.

17 $\dfrac{(x+1)^2}{(x-1)(x+1)} = \dfrac{x+1}{x-1}$, $x \neq 1, -1$.

18 $3(x^2+4x+4) = 3(x+2)^2$.

19 $x^4 - 5x^2 + 4 = (x^2-1)(x^2-4) = (x-1)(x+1)(x-2)(x+2)$.

20 $\left(x + \dfrac{1}{x}\right)^2 = x^2 + 2 \cdot x \cdot \dfrac{1}{x} + \dfrac{1}{x^2} = x^2 + 2 + \dfrac{1}{x^2}$, since $x \cdot \dfrac{1}{x} = 1$ for $x \neq 0$.

Criterion A · Knowing and understanding

1 $4x(2x-3) = 8x^2 - 12x$ [2]. $(x-2)(x+5) = x^2 + 3x - 10$ [1]. Subtracting the whole bracket: $8x^2 - 12x - x^2 - 3x + 10 = 7x^2 - 15x + 10$ [2].

2 HCF of 12 and 18 is 6; the lowest powers are a^2 and b. $12a^3b^2 - 18a^2b = 6a^2b(2ab - 3)$. [3: factor 1, inside terms 1, complete 1]

3 Product -30, sum -1: the pair is -6 and 5. $x^2 - x - 30 = (x-6)(x+5)$. [3]

4 $ac = 3(-2) = -6$; the pair with sum -5 is -6 and 1. $3x^2 - 6x + x - 2 = 3x(x-2) + (x-2) = (3x+1)(x-2)$. [3]

5 $(2x-3)^2 = 4x^2 - 12x + 9$ [2]. $(2x-3)(2x+3) = 4x^2 - 9$ [2].

6 $\dfrac{x(x-4)}{(x-4)(x+3)} = \dfrac{x}{x+3}$, with $x \neq 4, -3$. [4: factorising 2, cancellation 1, domain 1]

7 $y(x+2) = 3x - 1$, so $yx + 2y = 3x - 1$ and $x(y-3) = -1 - 2y$. Hence $x = \dfrac{1+2y}{3-y}$, $y \neq 3$. [4: clearing 1, collecting 1, factorising 1, answer 1]

8 $(a+b)^2 - (a-b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$. [4: each expansion 1, subtraction 1, answer 1]

> **Marking guidance.** Total 30. Award method marks for a correct first step (a correct expansion, a correct factor pair, a correct common denominator) even if later arithmetic slips, but no answer mark for a partly factorised result where "fully" is required. In Q1 the subtraction of the second bracket must change both signs. In Q6 and Q7 the domain restriction carries a mark; omitting it caps the question at 3.

Criterion B · Investigating patterns

1 $n = 1$ to 8: $3, 5, 7, 9, 11, 13, 15, 17$. [4]

2 The differences are the odd numbers, in order, increasing by 2 each time. The n-th difference is the n-th odd number, $2n+1$. [3: odd identified 1, order/constant difference 1, link to n 1]

3 Difference $= 2n + 1$. Verify: $n = 1$ gives 3, and $4 - 1 = 3$; $n = 4$ gives 9, and $25 - 16 = 9$; $n = 8$ gives 17, and $81 - 64 = 17$. [5: formula 2, verification 3]

4 $(n+1)^2 - n^2 = (n^2 + 2n + 1) - n^2 = 2n + 1$. Since $2n$ is even, $2n + 1$ is odd for every integer n. The n^2 terms cancel, leaving a term that grows linearly, which is why the second differences are constant. [6: expansion 2, cancellation 1, odd argument 2, structure 1]

5 $(n+k)^2 - n^2 = n^2 + 2nk + k^2 - n^2 = k(2n+k)$, a multiple of k. For $k = 2$ this is $4n + 4 = 4(n+1)$, always a multiple of 4. $101^2 - 99^2$ with $n = 99$, $k = 2$: $2(99)(2) + 4 = 400$. [6: expansion 2, factorised form 2, substitution 1, answer 1]

→ **Sample response (Q4).** "Expanding, $(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$. The square terms cancel exactly, so no approximation is involved. As $2n$ is even for every integer n, adding 1 always gives an odd number; the result cannot be even, and it increases by 2 at each step."

> **Marking guidance.** Total 24: Q1 4, Q2 3, Q3 5, Q4 6, Q5 6. In Q3 the formula alone earns at most 2; verification at three values is required. In Q4 a numerical check is not a proof: full marks require the algebraic cancellation and the parity argument. In Q5 accept $2nk + k^2$ if the factorisation and the application to $101^2 - 99^2$ are both correct; a response that only computes 400 without the general form earns 2.

Criterion C · Communicating

→ **Requirements.** A 300–400 word guide with a title, (i) a definition of equivalence, (ii) an area model used in both directions, (iii) one common-factor and one quadratic worked example, (iv) one common error with a check, plus a labelled diagram and a "check it yourself" box.

→ **Sample (opening and area model).** "Two expressions are equivalent when they take the same value for every value of the variable. $3(x+4)$ and $3x + 12$ are equivalent; $3x + 4$ is not. The area model shows why. A rectangle with sides $x + 3$ and $x + 4$ splits into four regions of area x^2, $4x$, $3x$ and 12. Adding gives $x^2 + 7x + 12$, which is expanding. Reading the same diagram backwards — recognising the two side lengths $x + 3$ and $x + 4$ — is factorising. Nothing is created or destroyed; the area is unchanged."

→ **Sample (worked examples and error).** "Common factor: $6x^2 + 9x$ has HCF $3x$, so $6x^2 + 9x = 3x(2x+3)$; expanding returns the original. Quadratic: for $x^2 - 5x - 24$ find two numbers with product -24 and sum -5, namely -8 and 3, so $x^2 - 5x - 24 = (x-8)(x+3)$. Common error: writing $(x+3)^2 = x^2 + 9$. The two $3x$ strips are missing; the correct expansion is $x^2 + 6x + 9$. Check it yourself: substitute $x = 1$ into both forms — $16 = 16$ — and expand every factorisation before trusting it."

Marking guidance. Total 20: (i) 4, (ii) 5, (iii) 6 (3 per example), (iv) 3, notation and diagram 2. The area model must be used for both directions, not just described. Award full marks for (iv) only if the correction is given alongside the error. Deduct one mark for an unlabelled diagram and one for missing domain or expansion checks where the guide claims to verify a result.

Criterion D · Applying mathematics in real-life contexts

1 Length = width = $24 - 2x$, height = x, so $V = x(24 - 2x)^2$. The domain is $0 < x < 12$, since both the height and the base side must be positive. [4: dimensions 2, volume 1, domain 1]

2 $V = x(576 - 96x + 4x^2) = 4x^3 - 96x^2 + 576x$. [3: expansion of square 2, distribution 1]

3 $x = 1$: 484; 2: 800; 3: 972; 4: 1024; 5: 980; 6: 864; 7: 700; 8: 512 cm^3. Maximum 1024 cm^3 at $x = 4$. [5: table 3, maximum value 1, value of x 1]

4 $x(24 - 2x)^2 = 4x(12 - x)^2$. The factors show the zeros immediately: $x = 0$ and $x = 12$ (the second repeated), so the volume is zero at both ends of the physical range and positive only for $0 < x < 12$. The expanded form hides both the zeros and the restriction, which must be recovered by solving a cubic. [4]

5 (1) The card has zero thickness, so the inner dimensions equal the outer ones; real card reduces the internal volume and a thicker card lowers the optimum slightly [2]. (2) The folds are perfectly vertical and the corners close exactly; in practice creasing consumes a little card and the box bulges, so the achievable volume for $x = 4$ is less than 1024 cm^3 [2].

6 Recommend $x = 4$ cm: it maximises the table and the vertex of the graph, giving 1024 $\text{cm}^3 \approx 1.02$ litres, about 5.4% more than at $x = 3$ and 4.5% more than at $x = 5$ [3]. Practical check: cut and fold one prototype from the production card and measure its capacity, since thickness and creasing shift the true optimum [3].

→ **Sample recommendation.** "I recommend 4 cm squares. The table peaks at 1024 cm^3, and the factorised form shows the volume is zero at both ends of the range, so the maximum lies inside. The gain over 3 cm is 52 cm^3, small enough that a prototype test is worthwhile before tooling."

Marking guidance. Total 24: Q1 4, Q2 3, Q3 5, Q4 4, Q5 4, Q6 4. In Q1 the domain must be justified, not merely stated. In Q3 award the maximum marks only for a complete table with units. In Q4 the comparison must name what the expanded form conceals. In Q5 and Q6 each assumption or check needs its effect on the recommendation; a generic "human error" earns nothing.

CHAPTER 3 · LINEAR EQUATIONS, INEQUALITIES AND SYSTEMS

Opening problem · two quotes, one decision

1 For n students, Company A costs $C_A = 180 + 12n$ and Company B costs $C_B = 420 + 8n$, both in pounds. The variable n is a whole number with $0 \le n \le 72$, the coach's capacity.

2 Solve $180 + 12n = 420 + 8n$: subtract $8n$ from both sides to get $180 + 4n = 420$, then subtract 180: $4n = 240$, so $n = 60$. Check in both: $180 + 12(60) = £900$ and $420 + 8(60) = £900$. The two quotes are equal at 60 students.

3 40 students: $C_A = 180 + 480 = £660$ and $C_B = 420 + 320 = £740$, so A is cheaper by £80. 80 students: $C_A = 180 + 960 = £1140$ and $C_B = 420 + 640 = £1060$, so B is cheaper by £80. Without a graph, use the difference $C_A - C_B = (180 + 12n) - (420 + 8n) = 4n - 240$. It is negative for $n < 60$, zero at $n = 60$, and positive for $n > 60$: A's £240 smaller booking fee is repaid by B's £4 dearer per student once $4n = 240$, that is after 60 students.

80 students is more than one coach holds, so that comparison is a model value only; a real party of 80 needs two coaches, and the £1000 budget is spent per coach.

4 Company A: $180 + 12n \le 1000$, so $12n \le 820$ and $n \le 68.3\ldots$; the greatest whole number is 68 students, costing $£996$ (69 would cost $£1008$). Company B: $420 + 8n \le 1000$, so $8n \le 580$ and $n \le 72.5$; the greatest whole number is 72 students, costing $£996$. For B the budget allows 72 and the coach seats 72, so the two restrictions meet exactly; for A the budget binds well before the capacity, cutting 4 seats.

5 The mathematics decides the cost once n is known: A is cheaper for $n \le 59$, the quotes are equal at $n = 60$, and B is cheaper for every $n \ge 61$; and only B can carry more than 68 students within £1000. So for an expected turnout of 61 to 72, hire B; for 59 or fewer, hire A. The committee must decide things the equations cannot: the likely value of n, how much of B's £420 booking fee is forfeited if numbers fall, whether the quotes include the driver's time, tolls and entry fees, and whether a third coach is needed. B carries more students for less at full capacity, but it is the riskier choice if the trip barely runs.

Prerequisite check

1 $3x = 15$, so $x = 5$. Check: $3(5) + 7 = 22$.

2 $2(x - 5) = 2x - 10$. Also $3(x + 2) - (x - 4) = 3x + 6 - x + 4 = 2x + 10$.

3 Multiply by 4: $x = 12$.

4 Plot $(0, -1)$ on the y-axis, one unit below the origin, and $(3, 5)$ three units right and five units up. The line through them rises 6 for a run of 3, so its gradient is 2 and its equation is $y = 2x - 1$.

5 $-2x = -4$, so $x = 2$.

6 $4(-2) - 3 = -8 - 3 = -11$.

Exercise 3A

1 **(a)** $5x = 30$, so $x = 6$. **(b)** $-2x = -8$, so $x = 4$.
(c) $2x = 8$, so $x = 4$. **(d)** $4x = 24$, so $x = 6$.

2 **(a)** $x + 4 = 9$, so $x = 5$. **(b)** $2x - 1 = 7$, so $x = 4$.
(c) $2x - 6 = x + 7$, so $x = 13$. **(d)** $8x + 4 = 3x + 24$, so $5x = 20$ and $x = 4$.

3 **(a)** Multiply by 12: $4x + 3x = 84$, so $7x = 84$ and $x = 12$.
(b) $x + 2 = 15$, so $x = 13$. **(c)** $2x - 1 = 3x - 6$, so $x = 5$.
(d) Multiply by 4: $2x - (x - 1) = 8$, so $x + 1 = 8$ and $x = 7$.

4 Multiply every term by 6: $2(2x + 1) - 3(x - 4) = 6$. Then $4x + 2 - 3x + 12 = 6$, so $x + 14 = 6$ and $x = -8$. Check: $\frac{-15}{3} - \frac{-12}{2} = -5 + 6 = 1$.
Common error: multiplying only the left-hand terms by 6 and forgetting the right-hand side.

5 **(a)** $3x = y + 7$, so $x = \frac{y + 7}{3}$. **(b)** $\frac{x}{4} = y - 2$, so $x = 4y - 8$.
(c) $3x = 12 - 2y$, so $x = \frac{12 - 2y}{3}$. **(d)** $5y = 2x + 1$, so $x = \frac{5y - 1}{2}$.

6 If $a = b$, then $a + c = b + c$ for any expression c. The move is reversible: subtracting c from both sides recovers $a = b$. Hence every solution of the original equation satisfies the new one and vice versa, so the solution set is unchanged.

7 The right-hand side was not multiplied by 2; the student divided only the left side. Correctly, $x + 3 = 10$, so $x = 7$. Check: $\frac{7 + 3}{2} = 5$.

8 One example is $3(x - 1) = 9$. Expanding gives $3x - 3 = 9$, then $3x = 12$ and $x = 4$. Check: $3(4 - 1) = 9$. Other valid equations are accepted.

9 **(a)** Both sides equal $3x + 6$, so the equation is an identity: every real number is a solution.
(b) Subtracting $2x$ gives $5 = -1$, which is false: no solution.
(c) Left side $4x - 4$; right side $4x - 6 + 2 = 4x - 4$. Identity: every real number is a solution.

10 Let the width be w cm; the length is $w + 4$ cm. Then $2(w + w + 4) = 44$, so $4w + 8 = 44$, $w = 9$. The dimensions are 9 cm by 13 cm. Check: $2(9 + 13) = 44$.

11 Let the integers be n, $n + 2$, $n + 4$. Then $3n + 6 = 81$, so $3n = 75$ and $n = 25$. The integers are 25, 27 and 29; their sum is 81.

12 Subtract u: $v - u = at$. Divide by $t \ne 0$: $a = \frac{v - u}{t}$. With $v = 28$, $u = 4$, $t = 6$: $a = \frac{24}{6} = 4\ \text{m/s}^2$.

13 For $x \neq 2$, multiply by $x-2$: $3 = 4(x-2)$, so $4x = 11$ and $x = \frac{11}{4} = 2.75$. The equation is undefined at $x = 2$, where the denominator is zero, so 2 cannot be a solution. Check: $\frac{3}{0.75} = 4$.

Exercise 3B

1 **(a)** $x > -3$, $(-3, \infty)$. **(b)** $x \leq 6$, $(-\infty, 6]$.
(c) $x \geq 5$, $[5, \infty)$. **(d)** $-x < -6$, so $x > 6$, $(6, \infty)$.

2 **(a)** Divide by -2, reversing: $x < -5$, $(-\infty, -5)$.
(b) Divide by -5, reversing: $x \geq -4$, $[-4, \infty)$.
(c) $-3x < -12$, so $x > 4$, $(4, \infty)$.
(d) $-2x \geq -6$, so $x \leq 3$, $(-\infty, 3]$.
Every division by a negative number reverses the symbol; test one value to confirm the direction.

3 **(a)** $x + 1 \leq 6$, so $x \leq 5$, $(-\infty, 5]$.
(b) $2x - 8 > x + 1$, so $x > 9$, $(9, \infty)$.
(c) $10x - 5 < 3x + 18$, so $7x < 23$ and $x < \frac{23}{7}$, $\left(-\infty, \frac{23}{7}\right)$.
(d) $4x - 8 \geq 6x - 3$, so $-2x \geq 5$ and $x \leq -\frac{5}{2}$, $\left(-\infty, -\frac{5}{2}\right]$.

4 **(a)** Subtract 1: $-3 \leq x < 3$. Integers: $-3, -2, -1, 0, 1, 2$.
(b) Add 1: $4 < 2x \leq 10$, so $2 < x \leq 5$. Integers: $3, 4, 5$.
(c) Subtract 1: $-7 \leq -3x \leq 3$. Divide by -3, reversing both: $\frac{7}{3} \geq x \geq -1$, that is $-1 \leq x \leq \frac{7}{3}$. Integers: $-1, 0, 1, 2$.

5 Take the true statement $2 < 5$. Multiplying by -1 gives -2 and -5, and $-2 > -5$. On the number line, multiplication by -1 reflects every point about 0, so left and right are exchanged and the order reverses.

6 Dividing by -4 reverses the symbol: $x > -3$. The student kept $x < -3$. Check $x = 0$: $-4(0) = 0 < 12$, true; but $x = -4$ gives $16 < 12$, false, so the student's set is wrong.

7 $3x - 5 \geq 7$, so $3x \geq 12$ and $x \geq 4$. On a number line, use a filled circle at 4 and shade to the right; the interval is $[4, \infty)$.

8 Substitute: $2(-4) + 9 = -8 + 9 = 1$. Since $1 \leq 1$ is true (equality is permitted), $x = -4$ is a solution. No solving is needed.

9 Let c be the number of crates: $240 + 55c \leq 900$, so $55c \leq 660$ and $c \leq 12$. The greatest number is 12 crates. Check: 13 crates would give $240 + 715 = 955$ kg, over the limit.

10 $42 - 3h \geq 12$, so $-3h \geq -30$ and $h \leq 10$. At most 10 whole hours; after 10 hours the charge is exactly 12%.

11 From $ax + 3 \leq 7$, $ax \leq 4$. The solution $x \geq -2$ points in the opposite direction to $x \leq \frac{4}{a}$, so $a < 0$; dividing by a reverses the symbol and gives $x \geq \frac{4}{a}$. Hence $\frac{4}{a} = -2$, so $a = -2$.
If $a = 0$, the inequality becomes $3 \leq 7$, true for every x, so the solution is all real numbers, not $x \geq -2$.

Exercise 3C

1 **(a)** $2x + (x + 3) = 12$, so $3x = 9$, $x = 3$, $y = 6$: $(3, 6)$.
(b) $3x - 1 = x + 5$, so $2x = 6$, $x = 3$, $y = 8$: $(3, 8)$.
(c) $2(y - 2) + 3y = 9$, so $5y = 13$, $y = \frac{13}{5}$, $x = \frac{3}{5}$: $\left(\frac{3}{5}, \frac{13}{5}\right)$.
(d) $4x - (2x - 3) = 6$, so $2x = 3$, $x = \frac{3}{2}$, $y = 0$: $\left(\frac{3}{2}, 0\right)$.

2 **(a)** $3(y + 4) + 2y = 17$, so $5y = 5$, $y = 1$, $x = 5$: $(5, 1)$.
(b) $(2y - 1) + 3y = 9$, so $5y = 10$, $y = 2$, $x = 3$: $(3, 2)$.
(c) $5x - 3(1 - 2x) = 8$, so $11x = 11$, $x = 1$, $y = -1$: $(1, -1)$.

3 $y = x - 1$ has gradient 1 and intercepts $(0, -1)$ and $(1, 0)$; $y = -2x + 8$ has gradient -2 and intercepts $(0, 8)$ and $(4, 0)$. From the graph the lines meet at approximately $(3, 2)$. Algebraically $x - 1 = -2x + 8$, so $3x = 9$, $x = 3$, and $y = 3 - 1 = 2$. Check: $2 = 3 - 1$ and $2 = -2(3) + 8$. The exact intersection is $(3, 2)$.

4 Replacing y by an expression in x eliminates y, leaving a single equation in the one unknown x. The second equation supplies the condition not already contained in that expression; substituting into the same equation would only reproduce an identity.

5 The working $2x + x + 3 = 12$ is correct, but it gives $3x = 9$, so $x = 3$, not $x = 5$. Then $y = x + 3 = 6$, and the solution is $(3, 6)$. Check: $2(3) + 6 = 12$. The pair $(5, 8)$ satisfies only $y = x + 3$, giving $18 \neq 12$ in the second equation.

6 Many answers are possible. For example $x + y = 1$ and $2x + y = 3$. Check $(2, -1)$: $2 + (-1) = 1$ and $2(2) + (-1) = 3$.

7 No solution. Distinct parallel lines never meet, so no point lies on both. Their equations have equal gradients but different intercepts, for example $y = 2x + 1$ and $y = 2x + 4$; substitution gives $1 = 4$, a contradiction, and elimination gives $0 = 3$.

8 Let the smaller number be x and the larger y. Then $x + y = 27$ and $y = 2x + 3$. Substituting: $3x + 3 = 27$, so $x = 8$ and $y = 19$. The numbers are 8 and 19.

9 Let the width be w and the length l. Then $2(l + w) = 30$, so $l + w = 15$, and $l = w + 3$. Hence $2w + 3 = 15$, $w = 6$ and $l = 9$. The rectangle is 6 cm by 9 cm.

10 Substitute $(2, 6)$ into $y = 2x + k$: $6 = 4 + k$, so $k = 2$. Substitute into $y = mx + 3$: $6 = 2m + 3$, so $m = 1.5$. The first equation determined k through its intercept; the second determined m through its

Exercise 3D

1 **(a)** Add: $2x = 12$, $x = 6$, $y = 3$: $(6, 3)$.

(b) Add: $5x = 20$, $x = 4$, $y = 3$: $(4, 3)$.

(c) Add: $6x = 24$, $x = 4$; $16 + 3y = 18$, $y = \frac{2}{3}$: $\left(4, \frac{2}{3}\right)$.

(d) Add: $8x = 24$, $x = 3$; $15 + 2y = 16$, $y = \frac{1}{2}$: $\left(3, \frac{1}{2}\right)$.

2 **(a)** $6x + 9y = 36$ and $6x + 4y = 26$; subtract: $5y = 10$, $y = 2$, $x = 3$: $(3, 2)$.

(b) $8x + 10y = 52$ and $15x - 10y = 5$; add: $23x = 57$, $x = \frac{57}{23}$, then $y = \frac{74}{23}$: approximately $(2.48, 3.22)$.

(c) $10x - 4y = 14$ and $3x + 4y = 25$; add: $13x = 39$, $x = 3$; $15 - 2y = 7$, $y = 4$: $(3, 4)$.

3 **(a)** The second equation is twice the first, so the lines coincide: infinitely many solutions.

(b) Doubling the first gives $6x + 2y = 14$, not 10, so the lines are parallel and distinct: no solution.

(c) The second equation is twice the first, so the lines coincide: infinitely many solutions.

4 Multiplying both sides by 2 gives an equivalent equation: any pair satisfying the original also satisfies the doubled equation, and dividing by 2 reverses the step. The set of pairs satisfying the system is therefore unchanged; only the appearance of the equation changes.

5 Subtracting as written gives one equation in two unknowns, $-x - y = -5$, which has infinitely many solutions and does not determine either variable. Eliminate instead: multiply by 3 and by 2 to get $6x + 9y = 39$ and $6x + 8y = 36$. Subtract: $y = 3$. Then $2x + 9 = 13$, so $x = 2$. The solution is $(2, 3)$.

6 If the system reduces to $0 = 5$, the statement is false for every pair, so there is no solution; the graphs are parallel distinct lines. If it reduces to $0 = 0$, the statement is true for every pair on the line, so there are infinitely many solutions; the graphs coincide.

7 Let a be adult tickets and c child tickets. Then $a + c = 100$ and $40a + 25c = 3100$. Substituting $a = 100 - c$: $4000 - 15c = 3100$, so $c = 60$ and $a = 40$. There were 40 adult and 60 child tickets.

8 Let x litres be the 20% solution and y litres the 50% solution. Then $x + y = 30$ and $0.2x + 0.5y = 0.3(30) = 9$. Multiplying by 10: $2x + 5y = 90$. Substituting $x = 30 - y$: $60 + 3y = 90$, so $y = 10$ and $x = 20$. Use 20 L of the 20% solution and 10 L of the 50% solution.

9 Write the lines as $y = -\frac{k}{2}x + 3$ and $y = -\frac{3}{4}x + \frac{9}{4}$. For no solution the gradients must be equal but the intercepts different: $-\frac{k}{2} = -\frac{3}{4}$, so $k = \frac{3}{2} = 1.5$. At $k = 1.5$, doubling $1.5x + 2y = 6$ gives $3x + 4y = 12$, which contradicts $3x + 4y = 9$; the lines are parallel and distinct, so there is no solution. For every other k the gradients differ and there is exactly one solution.

Exercise 3E

1 Let a adult and c child tickets. $a + c = 250$ and $30a + 18c = 5700$. Substituting $a = 250 - c$: $7500 - 12c = 5700$, so $c = 150$ and $a = 100$. 100 adult and 150 child tickets were sold.

2 Break-even: $240 + 6n = 15n$, so $240 = 9n$ and $n = \frac{80}{3} \approx 26.7$. At least 27 people are needed to avoid a loss. At 50 people, revenue $= 750$ and cost $= 240 + 300 = 540$, so the profit is £210.

3 Let x litres of 20% and y litres of 50% be used. $x + y = 30$ and $0.2x + 0.5y = 9$, or $2x + 5y = 90$. Substituting $x = 30 - y$ gives $60 + 3y = 90$, so $y = 10$ and $x = 20$. Use 20 L of the 20% solution and 10 L of the 50% solution.

4 Solve $18 + 0.04m = 30 + 0.01m$. Then $0.03m = 12$, so $m = 400$ minutes. At that point each plan costs $18 + 16 = £34$.

5 Let the width be w m and the length $2w + 10$ m. Then $2(w + 2w + 10) = 120$, so $6w + 20 = 120$, $w = \frac{50}{3} \approx 16.7$ m, and the length is $\frac{130}{3} \approx 43.3$ m. Check: $2\left(\frac{50}{3} + \frac{130}{3}\right) = 120$.

6 Items are discrete: 26.7 cases cannot be produced or sold, so the quantity must be a whole number. At 26 items the profit is negative; at 27 it is positive. The smallest whole number that avoids a loss is 27.

7 With price £35, revenue is $R = 35n$. Break-even: $1800 + 15n = 35n$, so $1800 = 20n$ and $n = 90$. The break-even rises from 72 to 90 because the contribution per unit falls from £25 to £20; 18 more units are needed to cover the same fixed cost.

8 The new profit is $15n + 600 - (240 + 6n) = 9n + 360$. Setting this equal to zero gives $n = -40$, outside the context: the sponsorship covers the fixed cost and more, so the venture is profitable even at $n = 0$, with profit £360. Sponsorship lowers the risk but does not change the profit per person, which remains £9, because it adds fixed revenue rather than altering the rate.

Investigation · Break-even and beyond

1 Using $C = 240 + 6n$, $R = 15n$, $P = 9n - 240$:

$n = 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50$

$C = 240, 270, 300, 330, 360, 390, 420, 450, 480, 510, 540$

$R = 0, 75, 150, 225, 300, 375, 450, 525, 600, 675, 750$

$P = -240, -195, -150, -105, -60, -15, 30, 75, 120, 165, 210$

2 Plot n horizontally and pounds vertically. The cost line starts at 240 and rises £6 per case; the revenue line starts at 0 and rises £15 per case. They cross between $n = 25$ and $n = 30$, at about $n = 27$, where both are close to £400. The estimate is $n \approx 27$.

3 $15n = 240 + 6n$, so $9n = 240$ and $n = \frac{80}{3} = 26\frac{2}{3} \approx 26.7$. The exact answer is not whole because the fixed cost is not a multiple of the £9 contribution. At 26 cases the profit is $234 - 240 = -£6$; at 27 it is $243 - 240 = £3$. The business must make and sell 27 cases to avoid a loss.

4 At break-even, $pn = F + vn$. So $pn - vn = F$, $n(p - v) = F$, and $n = \frac{F}{p - v}$, provided $p \neq v$.

5 For $p = 12$: $n = \frac{240}{12 - 6} = 40$. For $p = 18$: $n = \frac{240}{18 - 6} = 20$. For $p = 9$: $n = \frac{240}{9 - 6} = 80$. The last case has the smallest contribution per case (£3), so it needs the largest number of sales; at $p = 6$ the denominator would be zero and no quantity could cover the fixed cost.

6 Profit £450: $9n - 240 = 450$, so $9n = 690$ and $n = 76\frac{2}{3}$. Since cases are whole, 77 cases are needed. For target profit T: $9n - 240 = T$, so $n = \frac{240 + T}{9}$, rounded up to the next whole number.

→ **Generalise.** From $pn = F + vn$, subtract vn: $n(p - v) = F$, hence $n = \frac{F}{p - v}$. The denominator must be positive, $p > v$, so that each sale contributes a positive amount towards the fixed cost and the break-even quantity is positive. If $p < v$, every sale increases the loss and no quantity breaks even; if $p = v$, the contribution is zero and the fixed cost can never be recovered.

The quantity must be rounded *up* because production is whole-numbered and any quantity below the exact break-even still makes a loss. For target profit T, $pn = F + vn + T$, so $n = \frac{F + T}{p - v}$, again rounded up.

On the graph, raising F shifts the cost line up and raises the break-even; raising p steepens the revenue line and lowers it; raising v steepens the cost line and raises it. Changing p or v also changes the slope, while changing F changes only the intercept.

Modelling task · The festival budget

1 Let n be the number of visitors. Cost $C = 2600 + 4n$; income $I = 12n + 1000$; profit $P = I - C = 8n - 1600$, all in pounds.

2 Break-even: $8n - 1600 = 0$, so $n = 200$. At 200 visitors income equals cost. Below 200 the festival makes a loss, up to £1600 at zero visitors; above 200 it makes a profit.

3 Require $8n - 1600 \geq 1000$, so $8n \geq 2600$ and $n \geq 325$. Since $325 \leq 350$, the target is achievable within capacity. At 350 visitors the profit is $2800 - 1600 = £1200$.

4 With 300 visitors, cost is $2600 + 1200 = 3800$. For profit £1000, income must be £4800, so $300p + 1000 = 4800$, $300p = 3800$, $p = \frac{38}{3} \approx £12.67$. To ensure at least £1000, round up to £12.70 (or £13); £12.66 would give only £998.

5 Assumptions: every visitor pays the full £12; the £4 variable cost is constant per visitor; fixed costs and the £1000 sponsorship are guaranteed; capacity is 350. A rainy day lowers n, reducing profit and possibly pushing attendance below the break-even 200. A cap on the sponsor lowers income and raises the break-even: with sponsorship S, $P = 8n - 1600 - S$, so $n = \frac{1600 + S}{8}$. The recommendation should then require higher attendance, a higher price or lower costs.

Mixed practice 3

1 $7x = 28$, so $x = 4$.

2 $3x - 6 = 5x + 4$, so $-2x = 10$ and $x = -5$.

3 Multiply by 12: $4x + 3x = 84$, so $7x = 84$ and $x = 12$.

4 $-2x < 6$, so $x > -3$; the interval is $(-3, \infty)$.

5 Subtract 1: $-4 \leq 2x \leq 8$, so $-2 \leq x \leq 4$. Integers: $-2, -1, 0, 1, 2, 3, 4$.

6 $3x - 1 = x + 5$, so $2x = 6$, $x = 3$ and $y = 8$: $(3, 8)$.

7 From $x - y = 1$, $x = y + 1$. Then $2(y + 1) + 5y = 16$, so $7y = 14$, $y = 2$, $x = 3$: $(3, 2)$.

8 Add: $6x = 18$, so $x = 3$. Then $12 - 3y = 6$, so $y = 2$: $(3, 2)$.

9 Doubling the first equation gives $4x + 6y = 14$, but the second says $4x + 6y = 20$. The lines have the same gradient $-\frac{2}{3}$ but different intercepts, so they are parallel and distinct: no solution.

10 Dividing by -2 reverses the inequality, so $x > -4$, not $x < -4$. Check $x = 0$: $-2(0) = 0 < 8$ is true and $0 > -4$.

11 $4y = 12 - 3x$, so $y = 3 - \frac{3x}{4}$, or $y = \frac{12 - 3x}{4}$.

12 Let the numbers be x and y, with $x = 3y - 5$. Then $3y - 5 + y = 45$, so $4y = 50$, $y = 12.5$ and $x = 32.5$. The numbers are 32.5 and 12.5.

13 $140 + 28b \leq 500$, so $28b \leq 360$ and $b \leq 12.86$. The greatest whole number is 12 boxes; 13 boxes would give $140 + 364 = 504$ kg.

14 Distinct parallel lines have equal gradients m and different intercepts $c_1 \neq c_2$. Setting $mx + c_1 = mx + c_2$ gives $c_1 = c_2$, a contradiction, so no pair satisfies both. Graphically, the lines never meet, so there is no common point.

15 Revenue $= 180(25) = £4500$. Cost $= 900 + 7(180) = 900 + 1260 = £2160$. Profit $= £2340$.

16 $20 + 0.05m = 32 + 0.02m$, so $0.03m = 12$ and $m = 400$ minutes.

17 The given line has gradient -2, so the perpendicular gradient is $\frac{1}{2}$. Through $(3, -2)$: $y + 2 = \frac{1}{2}(x - 3)$, so $y = \frac{1}{2}x - \frac{7}{2}$, or $x - 2y = 7$. Check: $3 - 2(-2) = 7$, so $(3, -2)$ lies on it.

18 Profit $= R - C = pn - (F + vn) = (p - v)n - F$. Setting profit to zero: $(p - v)n = F$, so $n = \frac{F}{p - v}$. The business needs $p > v$ so that each sale contributes $p - v > 0$; otherwise profit does not increase with sales and the fixed cost cannot be recovered.

Criterion A · Knowing and understanding

1 $6x + 15 = 4x + 32 + 3 = 4x + 35$, so $2x = 20$ and $x = 10$. Check: $3(25) = 75$ and $4(18) + 3 = 75$. [3]

2 Multiply by 6: $3(x + 1) - 2(x - 3) = 12$. Then $3x + 3 - 2x + 6 = 12$, so $x + 9 = 12$ and $x = 3$. Check: $2 - 0 = 2$. [3]

3 $-4x > 16$, so $x < -4$. On the number line, an open circle at -4 with shading to the left; interval $(-\infty, -4)$. [3]

4 Subtract 1: $-6 < 3x \leq 9$, so $-2 < x \leq 3$. Integers: $-1, 0, 1, 2, 3$. [3]

5 From $x - y = 4$, $x = y + 4$. Then $3(y + 4) + 2y = 17$, so $5y = 5$, $y = 1$ and $x = 5$: $(5, 1)$. Check: $15 + 2 = 17$ and $5 - 1 = 4$. [4]

6 Multiply the second equation by 2: $4x - 10y = -8$. Subtract from the first: $13y = 26$, so $y = 2$. Then $4x + 6 = 18$, $x = 3$. Solution $(3, 2)$. Check: $12 + 6 = 18$ and $6 - 10 = -4$. [4]

7 Let a adult and c child tickets. $a + c = 250$, $30a + 18c = 5700$. Substituting $a = 250 - c$: $7500 - 12c = 5700$, so $c = 150$, $a = 100$. 100 adult and 150 child tickets. Check: $3000 + 2700 = 5700$. [5]

8 Doubling the first equation gives $4x + 6y = 14$, while the second states $4x + 6y = 20$. Subtracting gives $0 = 6$, impossible. Both lines have gradient $-\frac{2}{3}$ but different intercepts, so they are parallel and never meet. [2]

9 Dividing by -2 reverses the inequality: $x > -4$. Check $x = 0$: $-2(0) = 0 < 8$, true; the student's $x < -4$ would fail at $x = -5$, since $10 < 8$ is false. [3]

Criterion B · Investigating patterns

1 Costs in pounds for $m = 0, 100, 200, 300, 400, 500$:

Pay-as-you-go $= 0, 12, 24, 36, 48, 60$.

Plan A $= 18, 22, 26, 30, 34, 38$.

Plan B $= 30, 31, 32, 33, 34, 35$. [4]

2 $P(m) = 0.12m$; $A(m) = 18 + 0.04m$; $B(m) = 30 + 0.01m$, all in pounds. [3]

3 All three are straight lines. P passes through the origin with gradient 0.12; A has intercept 18 and gradient 0.04; B has intercept 30 and gradient 0.01. Label the intersections at $m = 225$ (P and A), $m \approx 273$ (P and B) and $m = 400$ (A and B). [5]

4 P and A: $0.12m = 18 + 0.04m$, so $0.08m = 18$, $m = 225$, cost £27.

P and B: $0.12m = 30 + 0.01m$, so $0.11m = 30$, $m = \frac{3000}{11} \approx 272.7$, cost $\approx$ £32.73.

A and B: $18 + 0.04m = 30 + 0.01m$, so $0.03m = 12$, $m = 400$, cost £34. [6]

5 For $0 \leq m < 225$, pay-as-you-go is cheapest (at $m = 0$ it costs £0). For $225 < m < 400$, Plan A is cheapest. For $m > 400$, Plan B is cheapest. At $m = 225$, P and A tie at £27, below B; at $m = 400$, A and B tie at £34, below P. The graph shows the lowest line changing at each intersection. [4]

6 Assumption: every minute is charged at the stated rate, with no rounding, data cap or extra fees. If minutes are rounded up in blocks, or a cap changes the rate, the intersection points shift and the cheapest plan for a given usage may change. [2]

Marking guidance. Table 4 marks (one per correct column set); expressions 3 (one per plan); graph 5 (axes and scales, three correct lines, labelled intersections); equations 6 (two per pair, with solution); intervals 4 (three intervals plus boundary values); assumption 2 (one stated, one consequence). Accept equivalent decimals and rounded intersection values with correct working.

Criterion C · Communicating

→ **Content required.** (i) Solving an inequality means finding every value that makes the statement true; the answer is a set, shown on a number line or in interval notation. (ii) Multiplying or dividing by a negative reverses the symbol because it reflects the number line about zero: $2 < 5$ but $-2 > -5$. (iii) Numerical example: $-3x \geq 12 \Rightarrow x \leq -4$; check $x = -5$ gives $15 \geq 12$, while $x = -3$ gives $9 \geq 12$, false. Algebraic example: $5 - 2x < 11 \Rightarrow -2x < 6 \Rightarrow x > -3$; check $x = 0$ gives $5 < 11$. (iv) Number line: open circle at -3, shade right for $x > -3$; filled circle if the endpoint is included. (v) Common error: keeping the symbol when dividing by a negative; avoid it by testing a value or by moving the variable term to the positive side.

→ **Model paragraph.** "An inequality is a question with many answers. Solving it means finding the whole set of values that keep the statement true, not a single number. The balance rules are the same as for equations, with one exception: multiplying or dividing by a negative number turns the number line around, so the symbol must turn too. For example, $2 < 5$, but multiplying by -1 gives $-2 > -5$. Always test one value from your answer set in the original inequality; if it fails, the direction is wrong."

Marking guidance. 20 marks: content (i)–(v) 8; accuracy of the two worked examples and checks 6; number line and notation 3; audience, structure and the "remember" box 3. A guide that states the reversal rule but gives no checked example cannot reach the top band. Accept any correct pair of examples; the numerical example must show the reflection, and the algebraic example must include a check.

Criterion D · Applying mathematics in real-life contexts

1 Operator A: $C_A = 240 + 6n$. Operator B: $C_B = 420 + 3n$, in pounds for n passengers. [3]

2 $240 + 6n = 420 + 3n$, so $3n = 180$ and $n = 60$. The cost is $240 + 360 = £600$ from either operator. [4]

3 At 40 passengers: A costs $240 + 240 = £480$, B costs $420 + 120 = £540$; A is £60 cheaper. At 80: A costs $240 + 480 = £720$, B costs $420 + 240 = £660$; B is £60 cheaper. The change occurs at the intersection $n = 60$. [3]

4 Budget: A requires $240 + 6n \leq 900$, so $n \leq 110$. B requires $420 + 3n \leq 900$, so $n \leq 160$, but capacity limits B to 120. Greatest numbers: A 110, B 120. [4]

5 At full capacity $n = 120$, revenue $= 8(120) = £960$. A: cost $240 + 720 = £960$, profit £0; break-even $240 + 6n = 8n$ gives $n = 120$. B: cost $420 + 360 = £780$, profit £180; break-even $420 + 3n = 8n$ gives $n = 84$. [5]

6 Recommend B at full capacity: it returns £180 profit and stays £120 inside the budget, whereas A only breaks even and cannot carry 120 within the £900 budget (its maximum is 110). Assumptions: all 120 seats are sold at £8; operating costs follow the quoted formulas; capacity and demand are independent. Fairness could change the choice: A's lower fixed charge favours a small group or a low-attendance day, and a subsidised fare for passengers unable to pay would make the profit criterion secondary. [5]

Marking guidance. Expressions 3; intersection 4 (equation, solution, cost, interpretation); comparisons 3 (both numbers and the direction of the difference); budget 4 (two inequalities, two maxima, capacity check); profit and break-even 5; recommendation 5 (choice, justification, two assumptions, one fairness point). For full marks the recommendation must use the computed values and distinguish a mathematical result from a value judgement.

CHAPTER 4 · FUNCTIONS, COORDINATES AND LINEAR MODELS

Opening problem · the fair fare

1 For a journey of d km, $A(d) = 4 + 2d$ and $B(d) = 8 + 1.5d$, the fares in pounds. The natural domain is $d \geq 0$, and in practice a journey has $d > 0$.

2 Both graphs are straight lines: A from $(0, 4)$ with gradient 2, B from $(0, 8)$ with gradient 1.5, meeting at $(8, 20)$. Each intercept is the flag-fall — the charge the moment the taxi starts, before any kilometre is driven. Neither line passes through the origin, so the fare is linear but not proportional to the distance.

3 Solve $4 + 2d = 8 + 1.5d$: $0.5d = 4$, so $d = 8$ km, and each company then charges $4 + 2(8) = £20$ (check: $8 + 1.5(8) = £20$). The intersection is the point where the two tariffs cost the same for the same distance, so it is where the cheaper choice switches: A below 8 km, B above 8 km.

4 5 km: $A(5) = 4 + 10 = £14$ against $B(5) = 8 + 7.5 = £15.50$, so A is cheaper. 12 km: $A(12) = 4 + 24 = £28$ against $B(12) = 8 + 18 = £26$, so B is cheaper. In terms of gradients, A costs £2 per kilometre and B only £1.50, so B closes the £4 flag-fall gap at $2 - 1.5 = 0.5$ pounds per kilometre and needs $4 \div 0.5 = 8$ km to close it. While $d < 8$ the steeper line has not yet caught up; once $d > 8$ the smaller gradient wins, however long the journey is.

5 Company A: $4 + 2d \leq 30$, so $2d \leq 26$ and $d \leq 13$: the greatest whole number is 13 km, and it costs exactly $£30$. Company B: $8 + 1.5d \leq 30$, so $1.5d \leq 22$ and $d \leq 14.6\ldots$: the greatest whole number is 14 km, costing $£29$ (15 km would be $£30.50$, over budget). The same £30 buys one kilometre further with B, because B's money is spent on distance rather than on the flag-fall.

6 Charging the same fare for the same distance is a fair rule in the sense that the price depends only on the service used, and it lets the two tariffs be compared directly at their crossing point $d = 8$. But the flag-fall is a fixed charge spread over the whole journey, so the average cost per kilometre is $\frac{4}{d} + 2$ for A and $\frac{8}{d} + 1.5$ for B: it falls as d grows. On 1 km A charges £6 and B £9.50, while on 10 km they charge £24 and £23 — a rule that is neutral over long trips punishes short ones, and the higher the flag-fall the worse that penalty. A per-kilometre-only fare would be fairer to short journeys, but it would leave the distance-independent costs (dispatch, the vehicle and driver tied up) unrecovered, so some flag-fall is defensible; fairness depends on which costs the charge is meant to reflect.

Prerequisite check

1 Plot $(-2, 3)$ two units left and three up; plot $(4, -1)$ four units right and one down. The line through them falls 4 units for a run of 6, so its gradient is $-\frac{2}{3}$.

2 At $x = 5$: $3(5) - 4 = 11$. At $x = -2$: $3(-2) - 4 = -10$.

3 $2x = 12$, so $x = 6$.

4 $6^2 + 8^2 = 36 + 64 = 100$, and $\sqrt{100} = 10$.

5 The midpoint is $\frac{2 + 10}{2} = 6$.

6 The next term is 19. The rule is "add 4"; the nth term is $4n - 1$.

Exercise 4A

1 $f(2) = 3(2) + 1 = 7$; $f(0) = 1$; $f(-4) = -12 + 1 = -11$; $f\left(\frac{1}{3}\right) = 1 + 1 = 2$.

2 $g(3) = 9 - 4 = 5$; $g(-3) = 9 - 4 = 5$; $g(0) = -4$; $g(2a) = (2a)^2 - 4 = 4a^2 - 4$.

3 $h(5) = \frac{6}{2} = 3$; $h(-3) = \frac{-2}{2} = -1$; $h(0) = \frac{1}{2}$.

4 **(a)** $4x - 1 = 11$, so $4x = 12$ and $x = 3$.

(b) $2x + 6 = 0$, so $x = -3$. **(c)** $x^2 + 1 = 5$, so $x^2 = 4$; with $x \geq 0$, $x = 2$.

5 **(a)** $x \neq -4$. **(b)** $x - 5 \geq 0$, so $x \geq 5$. **(c)** $2x - 1 \neq 0$, so $x \neq \frac{1}{2}$.

6 For $f(x) = x^2$, each input x gives exactly one output x^2. The relation $x = y^2$ gives two outputs for positive x: for example $x = 4$ gives $y = 2$ and $y = -2$. A vertical line through $x = 4$ meets the graph twice, so y is not a function of x.

7 The notation $f(2) = 2x - 3$ leaves x in the expression, as if x and 2 were interchangeable; $f(2)$ is a single output. Substitute 2 for every x: $f(2) = 2(2) - 3 = 1$. The value 1 is correct, but the written working is not.

8 The graph exists only for inputs from 1 to 5 inclusive: it lies between the vertical lines $x = 1$ and $x = 5$, with both endpoints included, and there are no points to the left of 1 or to the right of 5. A possible graph is the segment from $(1, 1)$ to $(5, 5)$ for $f(x) = x$.

9 One example is $f(x) = x^2$, whose range is $y \geq 0$. A square of a real number is never negative, and every non-negative value occurs (take $x = \sqrt{y}$), so no negative output is possible.

10 $C(300) = 18 + 0.04(300) = 18 + 12 = £30$. Solve $18 + 0.04m = 26$: $0.04m = 8$, so $m = 200$ minutes.

11 $h(1) = 45 - 5(1)^2 = 40$ m; $h(2) = 45 - 20 = 25$ m. The ground is $h = 0$: $45 - 5t^2 = 0$, so $t^2 = 9$ and $t = 3$ s (reject $t = -3$).

12 $f(2) = 2a + 3 = 11$, so $2a = 8$ and $a = 4$. Then $f(x) = 4x + 3 = 0$ gives $x = -\frac{3}{4}$. This is the x-intercept: the point $\left(-\frac{3}{4}, 0\right)$ where the graph crosses the x-axis.

Exercise 4B

1 **(a)** $\sqrt{3^2 + 4^2} = 5$. **(b)** $\sqrt{5^2 + (-4)^2} = \sqrt{41} \approx 6.40$. **(c)** $\sqrt{5^2 + 12^2} = 13$. **(d)** $\sqrt{6^2 + 8^2} = 10$.

2 **(a)** $(5, 3)$. **(b)** $(1, 1)$. **(c)** $(3, -3)$. **(d)** $(-4, 3)$.

3 Let $B = (x, y)$. Then $\frac{1 + x}{2} = 4$, so $x = 7$; $\frac{-3 + y}{2} = 1$, so $y = 5$. Hence $B = (7, 5)$.

4 $\sqrt{(3 - 0)^2 + (-2 - 0)^2} = \sqrt{9 + 4} = \sqrt{13}$.

5 The midpoint is halfway, so its coordinate is the mean of the endpoints: $(2 + 10)/2 = 6$. Subtracting gives the distance or change, not the halfway position: $(10 - 2)/2 = 4$, which is not the midpoint.

6 The student added the horizontal and vertical changes as if distance were their sum. The changes 3 and 4 are the legs of a right triangle, so $d = \sqrt{3^2 + 4^2} = 5$, not $3 + 4 = 7$.

7 Let the other endpoint be (x, y). $\frac{3 + x}{2} = 0$, so $x = -3$; $\frac{-5 + y}{2} = 0$, so $y = 5$. The other endpoint is $(-3, 5)$. The endpoints are reflections of each other in the origin, and the origin is the midpoint of the segment.

8 Yes: the distance is zero only when the horizontal and vertical differences are both zero. The two points then have the same coordinates, so they coincide.

9 $AB = 4$. $AC = \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5}$ and $BC = \sqrt{(-2)^2 + 4^2} = 2\sqrt{5}$. Two sides are equal, so the triangle is isosceles. Since $AC^2 + BC^2 = 40 \neq 16 = AB^2$, it is not right-angled.

10 Differences: 8 and 6. Distance $= \sqrt{8^2 + 6^2} = 10$ km. Midpoint $= \left(\frac{-2 + 6}{2}, \frac{3 + 9}{2}\right) = (2, 6)$.

11 $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Then $AM^2 = \left(\frac{x_2 - x_1}{2}\right)^2 + \left(\frac{y_2 - y_1}{2}\right)^2 = \frac{(x_2 - x_1)^2 + (y_2 - y_1)^2}{4}$. The same calculation for MB gives the same value, so $AM^2 = MB^2$ and, since lengths are non-negative, $AM = MB$.

Exercise 4C

1 **(a)** $\frac{7 - 1}{3 - 0} = 2$. **(b)** $\frac{7 - (-1)}{6 - 2} = 2$. **(c)** $\frac{-4 - 4}{1 - (-3)} = -2$. **(d)** The run is $5 - 5 = 0$, so the gradient is undefined (vertical line).

2 **(a)** $y - 4 = 3(x - 1)$, so $y = 3x + 1$.
(b) $y - 5 = -2(x - 2)$, so $y = -2x + 9$.
(c) $y - 1 = \frac{1}{2}(x + 4)$, so $y = \frac{1}{2}x + 3$.
(d) $y + 1 = -\frac{2}{3}(x - 3)$, so $y = -\frac{2}{3}x + 1$.

3 **(a)** $m = \frac{9 - 3}{4 - 1} = 2$, so $y = 2x + 1$.
(b) $m = \frac{-5 - 3}{2 - (-2)} = -2$, so $y = -2x - 1$.
(c) $m = \frac{0 - 4}{5 - 0} = -\frac{4}{5}$, so $y = -\frac{4}{5}x + 4$.

4 **(a)** Gradient 4, y-intercept -7.
(b) $y = -\frac{2}{3}x + 2$: gradient $-\frac{2}{3}$, intercept 2.
(c) $y = \frac{5}{2}x - 5$: gradient $\frac{5}{2}$, intercept -5.

5 A gradient of -3 means that for every 1 unit moved horizontally to the right, the graph moves 3 units vertically down. The line falls from left to right and is steeper than a line of gradient -1.

6 The student divided the horizontal change by the vertical change. Gradient is vertical change over horizontal change: $\frac{10 - 4}{3 - 1} = \frac{6}{2} = 3$, not $\frac{1}{3}$.

7 Substitute $(3, 1)$: $1 = 2(3) + c$, so $c = -5$. The line is $y = 2x - 5$, and it crosses the y-axis at $(0, -5)$.

8 Substitute $x = 4$: $2(4) - 1 = 7$, which equals the y-coordinate. The point $(4, 7)$ lies on the line; geometrically, it is one of the infinitely many points that make the equation true.

9 Gradient $= \frac{1.2}{8} = 0.15 = \frac{3}{20}$. The ramp rises 3 units for every 20 units horizontally, a ratio of $3 : 20$ (vertical to horizontal).

10 $C(m) = 18 + 0.04m$. The gradient is 0.04 pounds per minute, or 4 pence per minute: each extra minute adds £0.04. The y-intercept £18 is the fixed monthly charge when no minutes are used.

11 Substitute $(2, 7)$: $7 = 2m + 3$, so $m = 2$. The line is $y = 2x + 3$. Set $y = 0$: $2x + 3 = 0$, so $x = -\frac{3}{2}$; the x-intercept is $\left(-\frac{3}{2}, 0\right)$. This is exactly the solution of $mx + 3 = 0$: the x-intercept is where the output is zero.

Exercise 4D

1 **(a)** 5. **(b)** $-\frac{2}{3}$. **(c)** $y = -4x + 9$, so -4. **(d)** $y = \frac{2}{3}x - 2$, so $\frac{2}{3}$.

2 **(a)** $-\frac{1}{3}$. **(b)** 4. **(c)** The line has gradient $-\frac{2}{5}$, so the perpendicular gradient is $\frac{5}{2}$. **(d)** $y = 7$ is horizontal, so a perpendicular line is vertical: gradient undefined.

3 **(a)** $m = 2$: $y - 1 = 2(x - 2)$, so $y = 2x - 3$, which is parallel to $y = 2x - 5$ and distinct from it.
(b) $m = -1$: $y - 4 = -(x + 1)$, so $y = -x + 3$.
(c) $2x + y = 5$ has gradient -2: $y + 2 = -2(x - 3)$, so $y = -2x + 4$.

4 **(a)** Perpendicular gradient $-\frac{1}{2}$: $y-1=-\frac{1}{2}(x-4)$, so $y=-\frac{1}{2}x+3$.

(b) Perpendicular gradient -3: $y-5=-3x$, so $y=-3x+5$.

(c) $x+2y=4$ has gradient $-\frac{1}{2}$, so the perpendicular gradient is 2: $y-2=2(x-2)$, so $y=2x-2$.

5 A line of gradient m has direction vector $(1, m)$. Rotating it 90° gives $(-m, 1)$, whose gradient is $\frac{1}{-m}=-\frac{1}{m}$. The product is $m\left(-\frac{1}{m}\right)=-1$. A sketch shows the rise and run swapping roles, with one direction reversed.

6 The perpendicular gradient is the negative reciprocal, not the negative. For $m=4$, the correct gradient is $-\frac{1}{4}$, because $4\left(-\frac{1}{4}\right)=-1$; -4 would give a product of -16.

7 Rearrange $6x-2y=7$: $y=3x-\frac{7}{2}$. Both lines have gradient 3 but different intercepts (2 and $-\frac{7}{2}$), so they are parallel, not perpendicular.

8 Undefined gradient: the vertical line $x=3$. Gradient zero: the horizontal line $y=-2$. They meet at $(3,-2)$ at a right angle, so each is perpendicular to the other.

9 AB is horizontal (gradient 0); BC is vertical (undefined); CD is horizontal (0); DA is vertical (undefined). Each pair of consecutive sides is horizontal and vertical, hence perpendicular. The sides are all 6 units, so the garden is a square.

10 The perpendicular gradient is -10, since $\frac{1}{10}(-10)=-1$. Mathematically this is a drop of 10 m for every 1 m horizontally, an angle of about 84° below horizontal. That is far too steep for a wheelchair ramp (accessible gradients are around $1:12$), so -10 is not physically sensible; the negative sign also means the ramp runs downhill in the chosen direction.

11 $3x+4y=12$ has gradient $-\frac{3}{4}$, so L has gradient $\frac{4}{3}$. Through $(1,2)$: $y-2=\frac{4}{3}(x-1)$, so $y=\frac{4}{3}x+\frac{2}{3}$, or $4x-3y=-2$.

Substitute into $3x+4y=12$: $3x+4\left(\frac{4}{3}x+\frac{2}{3}\right)=12$, so $9x+16x+8=36$, $25x=28$, $x=\frac{28}{25}$. Then $y=\frac{54}{25}$. They meet at $\left(\frac{28}{25},\frac{54}{25}\right)=(1.12, 2.16)$.

Exercise 4E

1 **(a)** $k=\frac{15}{5}=3$, so $y=3(9)=27$.

(b) $k=\frac{42}{7}=6$, so $30=6x$ and $x=5$.

(c) $k=\frac{2.4}{0.6}=4$, so $y=4(2)=8$.

2 $C(d)=4+2d$. The gradient is £2 per kilometre: each extra kilometre costs £2. For 9 km, $C(9)=4+18=£22$.

3 $C(p)=120+0.05p$. For 800 pages, $C(800)=120+40=£160$.

4 The y-values increase by 8 for every 2 units of x, so the gradient is 4. Using $y=4x+b$ and $(1,8)$: $8=4+b$, so $b=4$ and $y=4x+4$. At $x=7$, $y=32$.

5 A linear relationship has a constant rate but may start at a non-zero value: $y=mx+c$. A directly proportional relationship passes through the origin, $y=kx$, so the ratio y/x is constant and doubling x doubles y. Example linear but not proportional: a taxi fare $C=4+2d$. Example proportional: the cost of apples at £3 per kg, $C=3m$.

6 A charge of £50 at 0 km is the fixed part of the cost: booking, handling, loading or call-out. It is reasonable if the firm incurs those costs before travelling, but it is an extrapolation to a "delivery" of zero distance; the intercept represents costs independent of distance, not a real journey.

7 The intercept 12000 is the population in 2020 ($t=0$). The gradient 350 is the increase in population per year. At $t=100$ (2120), the model gives $12000+35000=47000$. This is unreliable: it extrapolates a century beyond the data, assumes the same net growth every year, and ignores limits on housing, employment and migration.

8 Solve $18+0.04m=30+0.01m$: $0.03m=12$, so $m=400$ minutes, where both cost £34. For large m, the second plan is cheaper because its gradient 0.01 is smaller than 0.04; the fixed fees matter less as m grows. For $m<400$, the first plan is cheaper because its intercept is lower.

Investigation · A fair fare

1 Both sets of points lie on straight lines: Company A rises £2 for every extra kilometre, and Company B rises £1.50. The points are collinear, so each fare structure is linear: a fixed flag-fall plus a constant rate per kilometre.

2 Company A: $m=\frac{20-6}{8-1}=\frac{14}{7}=2$ pounds per kilometre. Company B: $m=\frac{20-9.5}{8-1}=\frac{10.5}{7}=1.5$ pounds per kilometre. Each is the cost of one additional kilometre.

3 Company A: at $d=0$, $F=6-2(1)=4$, so the flag-fall is £4. Company B: $F=9.5-1.5(1)=8$, so the flag-fall is £8. These are the charges before any distance is travelled.

4 $A(d)=4+2d$ and $B(d)=8+1.5d$. Equal fares: $4+2d=8+1.5d$, so $0.5d=4$ and $d=8$ km. At 8 km both charge $4+16=£20$, matching the table.

5 For $0 < d < 8$, Company A is cheaper: it starts £4 lower and the lines have not yet crossed. At $d = 8$ the fares are equal. For $d > 8$, Company B is cheaper because its lower gradient makes its fare grow more slowly. On the graph, A's line begins below B's but is steeper, so it crosses at $d = 8$.

6 Company A is fairer for short journeys: its flag-fall is £4 rather than £8, so at 1 km it charges £6 against £9.50, a difference of £3.50. A high flag-fall makes a short trip relatively expensive because the fixed charge is spread over few kilometres. A non-mathematical consideration is access for low-income passengers or the driver's waiting time, which may justify a higher flag-fall.

→ **Generalise.** With $F(d) = f + rd$, equal fares require $f_1 + r_1 d = f_2 + r_2 d$. Rearranging: $d(r_1 - r_2) = f_2 - f_1$, so $d = \dfrac{f_2 - f_1}{r_1 - r_2}$. The formula requires $r_1 \neq r_2$; if the rates are equal the lines are parallel, so they either never meet (different flag-falls) or coincide (equal flag-falls).

A larger flag-fall makes short journeys relatively more expensive because the fixed charge dominates when d is small. The average cost per kilometre is $\dfrac{f}{d} + r$, which falls as d grows. Test: $\dfrac{8-4}{2-1.5} = \dfrac{4}{0.5} = 8$ km, as found.

Modelling task · The courier's route

1 $DP = \sqrt{6^2 + 2^2} = \sqrt{40} \approx 6.32$ km. $PQ = \sqrt{4^2 + 4^2} = \sqrt{32} \approx 5.66$ km. Total route $D \to P \to Q$ is 11.98 km (2 d.p.).

2 The midpoint of $D(0, 0)$ and $Q(10, 6)$ is $(5, 3)$. It is the point halfway along the direct route, 5.83 km from each end; it could serve as a meeting, refuelling or transfer point.

3 DP: gradient $\dfrac{2-0}{6-0} = \dfrac{1}{3}$, a shallow rise of 1 km north for every 3 km east. PQ: gradient $\dfrac{6-2}{10-6} = 1$, a 45° rise of 1 km north for every 1 km east.

4 Courier A: $C_A = 3 + 1.2d$; Courier B: $C_B = 5 + 0.9d$, in pounds for d km. At $d = 11.98$: $C_A = 3 + 14.38 = £17.38$; $C_B = 5 + 10.78 = £15.78$.

5 Both are within the £18 budget: A costs £17.38 and B costs £15.78. For £18, A can cover $3 + 1.2d \leq 18$, so $d \leq 12.5$ km; B can cover $5 + 0.9d \leq 18$, so $d \leq 14.44$ km.

6 Recommend B: it completes the 11.98 km route for £15.78, £1.60 cheaper than A, and can cover up to 14.44 km within budget. Assumptions: straight-line distances equal road distances; the rates stay constant; there are no waiting, toll or surcharge costs. A non-cost factor such as reliability, insurance cover or speed could change the choice.

Mixed practice 4

1 $f(4) = 12 - 2 = 10$; $f(-1) = -3 - 2 = -5$; $f(0) = -2$.

2 $3x - 2 = 10$, so $3x = 12$ and $x = 4$.

3 $x + 2 \neq 0$, so $x \neq -2$.

4 $\sqrt{(5-1)^2 + (5-2)^2} = \sqrt{16 + 9} = 5$.

5 $\left(\dfrac{-3+5}{2}, \dfrac{4-2}{2}\right) = (1, 1)$.

6 $\dfrac{7-(-1)}{6-2} = \dfrac{8}{4} = 2$.

7 $y = -2x + 3$.

8 $4x - 2y = 8$ gives $y = 2x - 4$, so the gradient is 2.

9 $2x + 5y = 10$ has gradient $-\dfrac{2}{5}$, so the perpendicular gradient is $\dfrac{5}{2}$.

10 The correct midpoint is $(4, 7)$: the averages are $(2 + 6)/2 = 4$ and $(5 + 9)/2 = 7$. A difference-based method uses $6 - 2 = 4$ and $9 - 5 = 4$, which describe the change between the points, not their position. It happens to give 4 and 7 here only because half the changes, $(2, 2)$, added to $(2, 5)$, lands on $(4, 7)$; that is an anchored displacement calculation, not a midpoint formula. Without anchoring or halving, it gives $(4, 4)$, and it fails for other pairs.

11 $k = \dfrac{18}{6} = 3$, so $y = 3(10) = 30$.

12 The gradient £1.50 per kilometre is the cost of each extra kilometre; the intercept £5 is the flag-fall. For 12 km, $C = 5 + 18 = £23$.

13 Parallel gradient 3: $y - 2 = 3(x - 1)$, so $y = 3x - 1$.

14 The given gradient is $-\dfrac{1}{2}$, so the perpendicular gradient is 2: $y - 1 = 2(x - 2)$, so $y = 2x - 3$.

15 Base $= 4$. The other sides are $\sqrt{2^2 + 3^2} = \sqrt{13}$ each, so the triangle is isosceles. Since $13 + 13 \neq 16$ and $16 + 13 \neq 13$, it is not right-angled.

16 Distance $= \sqrt{8^2 + 6^2} = 10$; midpoint $= \left(\dfrac{-2+6}{2}, \dfrac{3+9}{2}\right) = (2, 6)$.

17 $\dfrac{-1-3}{4-k} = 2$, so $-4 = 2(4 - k)$, $-4 = 8 - 2k$, $2k = 12$ and $k = 6$. Check: $\dfrac{-4}{-2} = 2$.

18 $M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$. Then $AM^2 = \left(\dfrac{x_2 - x_1}{2}\right)^2 + \left(\dfrac{y_2 - y_1}{2}\right)^2$. The same expression gives MB^2, so $AM^2 = MB^2$ and $AM = MB$. The midpoint is therefore equidistant from the endpoints.

Criterion A · Knowing and understanding

1 $f(2) = 4 - 6 = -2$; $f(-1) = 1 + 3 = 4$. [3]

2 The denominator cannot be zero, so $x \neq 5$. [2]

3 $\sqrt{(5-(-1))^2 + (10-2)^2} = \sqrt{36 + 64} = 10$. [3]

4 $\left(\dfrac{3+7}{2}, \dfrac{-4+2}{2}\right) = (5, -1)$. [3]

5 $m = \dfrac{-3-5}{6-2} = -2$. Then $y - 5 = -2(x - 2)$, so $y = -2x + 9$. [4]

6 Parallel gradient 2: $y + 1 = 2(x - 4)$, so $y = 2x - 9$. [4]

7 The given gradient is -3, so the perpendicular gradient is $\frac{1}{3}$. Then $y-3=\frac{1}{3}(x-1)$, so $y=\frac{1}{3}x+\frac{8}{3}$, or $x-3y=-8$. [4]

8 The gradient is £0.40 per minute: each extra minute costs 40 pence. The intercept £25 is the fixed charge when $m=0$. $C(120)=25+48=£73$. [4]

9 The student divided the horizontal change by the vertical change. Correctly, $m=\frac{9-3}{5-2}=\frac{6}{3}=2$. [3]

Criterion B · Investigating patterns

1 For $k=-2,-1,0,1,2,3$:
y-intercept $=k$: $-2,-1,0,1,2,3$.
x-intercept $=-\frac{k}{2}$: $1,0.5,0,-0.5,-1,-1.5$. [5]

2 The x-intercept is always half the y-intercept in size and opposite in sign: $x=-\frac{k}{2}$. As k increases by 1, the x-intercept decreases by 0.5; as k decreases, it increases. [3]

3 Set $y=0$: $0=2x+k$, so $x=-\frac{k}{2}$. Check $k=2$: $x=-1$, and $0=2(-1)+2$. Check $k=-2$: $x=1$, and $0=2(1)-2$. [4]

4 The intercepts are $A(0,k)$ and $B\left(-\frac{k}{2},0\right)$. The horizontal difference is $\left|-\frac{k}{2}\right|=\frac{|k|}{2}$; the vertical difference is $|k|$. By Pythagoras, $AB=\sqrt{\frac{k^2}{4}+k^2}=\sqrt{\frac{5k^2}{4}}=\frac{\sqrt{5}}{2}|k|$. The absolute value is needed because a distance is non-negative even when $k<0$. [6]

5 Area $=\frac{1}{2}\left|\frac{k}{2}\right||k|=\frac{k^2}{4}$. For $k>0$, the area increases as k increases, and it grows quadratically, not linearly. For $k<0$, increasing k towards 0 decreases the area; the areas for k and $-k$ are equal. At $k=0$ the line passes through the origin and the "triangle" has zero area. [6]

Marking guidance. Table 5 (both intercepts for all six values); pattern 3 (relationship and direction); formula 4 (derivation and two checks); distance 6 (coordinates, differences, Pythagoras, simplification, absolute value); area 6 (formula, correct variation for positive and negative k, zero case). The top band requires the general result, not just numerical examples.

Criterion C · Communicating

→ **Content required.** (i) Gradient is the rate of change: the vertical change per unit horizontal change, with units of output per input. Its sign gives direction and its size gives steepness. (ii) From a graph, choose two clear points and calculate rise over run; from coordinates, $m=\frac{y_2-y_1}{x_2-x_1}$, subtracting in the same order.
(iii) Positive gradients rise left to right; negative gradients fall; zero is horizontal; undefined is vertical. (iv) Worked example: through $(1,4)$ and $(3,10)$, $m=\frac{10-4}{3-1}=3$, so $y-4=3(x-1)$ and $y=3x+1$; check $x=3$ gives $y=10$. (v) Common error: inverting the differences, as in $\frac{3-1}{10-4}$; avoid it by labelling Δy on top and Δx on the bottom, then checking the sign against the graph.

→ **Model paragraph.** "Gradient measures how fast a graph changes, not where it is. On a straight line, every step of 1 unit to the right produces the same vertical change, and that constant change is the gradient. A positive gradient climbs, a negative one falls, a horizontal line has gradient 0, and a vertical line has no gradient because the run is zero. The formula $m=\frac{y_2-y_1}{x_2-x_1}$ is just 'rise over run' written with coordinates; keep the points in the same order and the sign will take care of itself."

Marking guidance. 20 marks: content (i)–(v) 8; worked example and check 5; diagram and the four cases 4; audience, structure and "remember" box 3. The example must show substitution into the original points, not only the final equation. A guide that confuses gradient with the y-intercept cannot reach the top band.

Criterion D · Applying mathematics in real-life contexts

1 $CM=\sqrt{(10-2)^2+(9-3)^2}=\sqrt{64+36}=10$ km. On the grid, $C(2,3)$ and $M(10,9)$ are joined by a straight road 10 km long. [4]

2 $B=\left(\frac{2+10}{2},\frac{3+9}{2}\right)=(6,6)$. Because B is the midpoint, $CB=BM=5$ km, so the stop is equally distant from the clinic and the market. [3]

3 Gradient $=\frac{9-3}{10-2}=\frac{6}{8}=\frac{3}{4}$. On the map, the road rises 3 km north for every 4 km east, an angle of about $36.9°$ to the horizontal. [3]

4 The feeder is perpendicular to CM, so its gradient is $-\frac{4}{3}$. Through $S(8,3)$: $y-3=-\frac{4}{3}(x-8)$, so $y=-\frac{4}{3}x+\frac{41}{3}$, or $4x+3y=41$. [5]

5 $CB = 5$ km. $BS = \sqrt{(8-6)^2 + (3-6)^2} = \sqrt{4+9} = \sqrt{13} \approx 3.61$ km. Route $C \to B \to S$ is 8.61 km (2 d.p.).

Firm A: $5 + 1.3(8.61) = £16.19$. Firm B: $7 + 0.9(8.61) = £14.75$. Only Firm B stays within the £15 budget; A exceeds it by £1.19. [5]

6 Recommend Firm B: it is the only firm within the £15 budget and is £1.44 cheaper for the route. Assumptions: the grid distances equal road distances; the quoted rates are constant; there are no waiting or extra charges; the bus travels directly. A non-cost factor such as reliability, safety, wheelchair accessibility or timetable could change the decision, even at a higher fare. [4]

Marking guidance. Distance 4 (coordinates, squares, root, units); midpoint 3 (coordinates plus equal-distance explanation); gradient 3 (value plus interpretation with units); perpendicular 5 (negative reciprocal, point substitution, equation); route and costs 5 (two lengths, total, two costs, budget decision); recommendation 4 (choice, justification, two assumptions, one non-cost factor). Accept the equation in either gradient–intercept or general form.

CHAPTER 5 · QUADRATIC EXPRESSIONS AND EQUATIONS

Opening problem · the school farm pen

1 All 26 m is used, so $2(l + w) = 26$, hence $l + w = 13$ and the length is $l = 13 - w$.

For $w = 1, 2, 3, 4, 5, 6$ the lengths are $12, 11, 10, 9, 8, 7$ m and $A = 12, 22, 30, 36, 40, 42$ m^2. The area increases, but by ever less: the steps are $+10, +8, +6, +4, +2$, falling by 2 each time. The rate of change of A changes at a constant rate, which is the signature of a quadratic; the rise is slowing, so the maximum is close, and by symmetry $w = 7$ gives 42 m^2 again.

2 As a product, $A(w) = w(13 - w)$ m^2. Expanded in standard form, $A = -w^2 + 13w$, with $a = -1$, $b = 13$, $c = 0$. The context gives $0 < w < 13$: both sides must be positive, and $A = 0$ at $w = 0$ and at $w = 13$.

3 $w(13 - w) = 40$, so $13w - w^2 = 40$, i.e. $w^2 - 13w + 40 = 0$, and $(w - 5)(w - 8) = 0$: $w = 5$ or $w = 8$. Check: $5 \times 8 = 40$ and $8 \times 5 = 40$. The two widths describe one pen: $l = 13 - w$ swaps the labels, so 5 m by 8 m and 8 m by 5 m are the same rectangle turned through 90°. Algebraically $A(w) = A(13 - w)$, so the parabola is symmetric about $w = 6.5$ and every area below the maximum comes in a pair.

4 The table gives 42 m^2 at $w = 6$ and at $w = 7$, so the peak lies between them, near 6.5 m. Completing the square: $A = -w^2 + 13w = -\left(w^2 - 13w\right) = -\left[(w - 6.5)^2 - 42.25\right] = 42.25 - (w - 6.5)^2$. Because $(w - 6.5)^2 \geq 0$ for every w, $A \leq 42.25$, with equality only when $w = 6.5$. The greatest possible area is 42.25 m^2, in a square pen 6.5 m by 6.5 m.

5 $w(13 - w) \geq 36$ gives $13w - w^2 - 36 \geq 0$. Multiply by -1, reversing the sign: $w^2 - 13w + 36 \leq 0$, so $(w - 4)(w - 9) \leq 0$, hence $4 \leq w \leq 9$. On the number line the solution is the closed interval from 4 to 9, both endpoints filled in: they are the widths that give exactly 36 m^2 (4×9 and 9×4), and between them the parabola sits on or above 36, reaching 42.25 m^2 at $w = 6.5$. Outside that interval the pen is too narrow or, equivalently, too long. Intersecting with the domain $0 < w < 13$ changes nothing, so admissible widths are $4 \leq w \leq 9$.

6 Recommend $w = 6.5$ m: the completed square shows it is the only width that reaches 42.25 m^2, it satisfies the committee's 36 m^2 condition with room to spare, and every alternative is smaller. If posts must be set at whole metres, $w = 6$ or $w = 7$ both give 42 m^2, a loss of only 0.25 m^2. The algebra cannot say whether a square suits the animals: a long narrow run may be wanted for exercise, a gate or an existing shed wall changes how much fencing is needed and where it is placed, and access for a wheelbarrow or a tractor may rule the square out. It also assumes a perfect rectangle, that all 26 m must be used, and that no fencing is wasted on gates or overlaps.

Prerequisite check

1 $(x + 3)(x - 4) = x^2 - x - 12$.

2 $(x - 2)(x - 3)$.

3 $3x = 18$, so $x = 6$.

4 $\sqrt{49} = 7$; $5^2 - 4(2)(3) = 25 - 24 = 1$.

5 $(x - 5)^2 = x^2 - 10x + 25$.

6 $x = \pm 4$.

A square root of 16 is 4, but the equation $x^2 = 16$ has both roots 4 and -4.

Exercise 5A

1 (a) $x^2 + 3x - 10$; (b) $2x^2 + 5x - 3$; (c) $6x^2 + 11x - 10$.

2 (a) $(x + 5)(x + 2)$; (b) $(x - 2)(x - 4)$; (c) $3x(x - 5)$.

3 (a) $2x^2 - 3x - 4 = 0$, $a = 2$, $b = -3$, $c = -4$.

(b) $x^2 + x - 7 = 0$, $a = 1$, $b = 1$, $c = -7$.

(c) Multiply by -1: $2x^2 + x - 5 = 0$, $a = 2$, $b = 1$, $c = -5$.

4 (a) $x = \pm 6$; (b) $x = \pm\sqrt{5}$; (c) $x^2 = 4$, so $x = \pm 2$.

5 $2^2 - 2 - 2 = 0$, so $x = 2$ is a root. $x^2 - x - 2 = (x - 2)(x + 1)$, so $x - 2 = 0$ or $x + 1 = 0$. The other root is $x = -1$.

6 Factorising rewrites the expression as the product $(x - 2)(x - 3)$; the expression has a value for every x and no solutions.

Solving sets the expression equal to zero and uses the zero product property. Only $x = 2$ and $x = 3$ make the equation true.

7 $(x + 4)^2 = (x + 4)(x + 4) = x^2 + 8x + 16$. The missing middle term is $2 \cdot x \cdot 4 = 8x$. Test $x = 1$: $25 \neq 17$.

8 One solution, $x = 0$, repeated. $x^2 = (x - 0)(x - 0)$, so both factors give $x = 0$; the graph touches the x-axis at the origin and turns there.

9 Side $= \sqrt{49} = 7$ cm. For 50 cm^2, side $= \sqrt{50} = 5\sqrt{2}$ cm ≈ 7.07 cm.

10 $x^2 + 7x + 12 = (x + 3)(x + 4)$, so possible sides are $x + 3$ and $x + 4$ (with $x > -3$ for positive lengths).

11 $15t - 5t^2 = 5t(3 - t) = 0$, so $t = 0$ or $t = 3$. $t = 0$ is the launch; $t = 3$ is the return to the ground.

12 $2x + 3 = 0$, so $x = -\frac{3}{2}$. Check:
$\left(-\frac{3}{2} - 1\right)\left(2\left(-\frac{3}{2}\right) + 3\right) = \left(-\frac{5}{2}\right)(0) = 0$.

In 3(c), multiplying by -1 first avoids reading $a = -2$; the coefficients must be taken from one consistent standard form.

Exercise 5B

1 (a) $(x - 2)(x - 3) = 0$, $x = 2$ or 3; (b) $(x + 5)(x - 2) = 0$, $x = -5$ or 2; (c) $(x - 4)(x + 4) = 0$, $x = \pm 4$; (d) $(x + 3)^2 = 0$, $x = -3$ repeated.

2 (a) $(2x - 1)(x - 3) = 0$, $x = \frac{1}{2}$ or 3; (b) $(3x - 2)(x + 4) = 0$, $x = \frac{2}{3}$ or -4; (c) $(2x - 3)(2x + 3) = 0$, $x = \pm\frac{3}{2}$.

3 (a) $x(x-5)=0$, $x=0$ or 5; (b) $2x(x+3)=0$, $x=0$ or -3; (c) $3x(x-9)=0$, $x=0$ or 9.

4 (a) $2(x-2)^2=0$, $x=2$ repeated; (b) $(x+6)^2=0$, $x=-6$; (c) $x^2=4$, $x=\pm 2$.

5 Dividing by x assumes $x\neq 0$, which discards the root $x=0$. Correctly, $x^2-7x=x(x-7)=0$, so $x=0$ or $x=7$.

6 $(x-5)(x+5)=0$ has two different factors, so either may be zero: $x=5$ or $x=-5$.

$(x-5)^2=(x-5)(x-5)=0$ requires the single factor $x-5=0$, giving the repeated root $x=5$.

7 $(x-4)(x+2)=0$, so $x^2-2x-8=0$. Not unique: any non-zero multiple $k(x^2-2x-8)=0$ has the same roots.

8 $(x-3)^2=0$, so $x^2-6x+9=0$ (or any non-zero multiple).

9 $x(x+3)=54$, so $x^2+3x-54=0$, $(x+9)(x-6)=0$. $x=6$ (reject -9); dimensions 6 m by 9 m.

10 $n^2+5n=24$, so $n^2+5n-24=0$, $(n+8)(n-3)=0$. The numbers are 3 or -8.

11 Expand: $2x^2+5x-3=3x$, so $2x^2+2x-3=0$. $x=\dfrac{-2\pm\sqrt{4+24}}{4}=\dfrac{-1\pm\sqrt{7}}{2}$.

12 Let $u=x^2$: $u^2-5u+4=(u-1)(u-4)=0$, so $u=1$ or 4. Then $x=\pm 1$ or $x=\pm 2$: four real roots.

In questions 3 and 5, never divide by the variable; factorise so that the root $x=0$ is not lost.

Exercise 5C

1 (a) $(x+4)^2-16$; (b) $(x-3)^2-9$; (c) $(x+5)^2-18$.

2 (a) $(x+4)^2=9$, $x=-1$ or -7; (b) $(x-3)^2=4$, $x=1$ or 5; (c) $(x+1)^2=6$, $x=-1\pm\sqrt{6}$.

3 (a) $(x-2)^2=6$, $x=2\pm\sqrt{6}$; (b) $(x-2)^2=\dfrac{5}{2}$, $x=2\pm\dfrac{\sqrt{10}}{2}$; (c) $(x+1)^2=\dfrac{4}{3}$, $x=-1\pm\dfrac{2\sqrt{3}}{3}$.

4 $y=(x-6)^2-36+30=(x-6)^2-6$. Minimum -6 at $x=6$.

5 $x^2+6x+4=(x+3)^2-9+4=(x+3)^2-5$, so $(x+3)^2=5$ and $x=-3\pm\sqrt{5}$. The student used 4 instead of 5.

6 Completing the square does not need integer factors: $x^2+6x+2=(x+3)^2-7$, so $x=-3\pm\sqrt{7}$. The discriminant is $36-8=28$, not a square, so no integer factorisation exists.

7 $x^2+2x+5=(x+1)^2+4=0$, so $(x+1)^2=-4$. A square is never negative, so there is no real solution.

8 Length $=10-w$, so $A=w(10-w)=10w-w^2=25-(w-5)^2$. Since $(w-5)^2\geq 0$, $A\leq 25$, with equality at $w=5$. Maximum area 25 m^2, pen 5 m by 5 m.

9 $(x+4)^2=x^2+8x+16$ and $(x-4)^2=x^2-8x+16$, so $k=8$ or $k=-8$. Equivalently $\Delta=k^2-64=0$.

In 5C, the constant outside the bracket is not $(b/2)^2$ added but subtracted; check by expanding.

Exercise 5D

1 (a) $\Delta=20$, $x=3\pm\sqrt{5}$; (b) $\Delta=8$, $x=-1\pm\sqrt{2}$; (c) $\Delta=17$, $x=\dfrac{3\pm\sqrt{17}}{4}$.

2 (a) $\Delta=1>0$, two distinct real roots; (b) $\Delta=0$, one repeated root; (c) $\Delta=-8<0$, no real roots.

3 (a) $\Delta=49$, $x=\dfrac{1}{3}$ or -2; (b) $\Delta=20$, $x=2\pm\sqrt{5}$; (c) $\Delta=32$, $x=\dfrac{1\pm\sqrt{2}}{2}$.

4 $\Delta=9+32=41$, so $x=\dfrac{-3\pm\sqrt{41}}{4}$. Thus $x\approx 0.85$ or $x\approx -2.35$.

5 One root requires $\Delta=k^2-64=0$, so $k=\pm 8$.

6 No real roots requires $\Delta=16-8k<0$, so $k>2$.

7 $\Delta=3^2-4(1)(-1)=9+4=13$, not 5. Correct roots $x=\dfrac{-3\pm\sqrt{13}}{2}$. The student used $c=1$.

8 $\Delta=49-48=1$, so $x=\dfrac{7\pm 1}{2}$, giving $x=4$ or $x=3$. Factorising gives $(x-3)(x-4)=0$, the same roots.

9 (a) Factorise: $(2x-1)(x+3)=0$, so $x=\dfrac{1}{2}$ or -3; factorising is quickest.

(b) No integer factors; use the formula or completing the square, $x=-2\pm\sqrt{7}$.

(c) $x^2-10x+25=(x-5)^2$, a perfect square, so the repeated root 5 is immediate.

10 $(x+1)(x-2)=5$, so $x^2-x-7=0$ and $x=\dfrac{1\pm\sqrt{29}}{2}$. Since $x-2>0$, reject $\dfrac{1-\sqrt{29}}{2}\approx -2.19$. Hence $x=\dfrac{1+\sqrt{29}}{2}\approx 3.19$.

11 $30t-5t^2=40$, so $5t^2-30t+40=0$, $t^2-6t+8=0$, $(t-2)(t-4)=0$.

$t=2$ s (rising) and $t=4$ s (falling). The maximum height is 45 m at $t=3$ s, so 40 m is reached twice.

12 $\Delta=b^2-36>0$, so $|b|>6$. The positive integers below 10 that satisfy this are $b=7,8,9$.

Negative values work too: every integer $b\leq -7$ also gives two distinct real roots, because Δ depends on b^2. The sign of b shifts the roots, not their existence.

Exercise 5E

1 $n(n+1)=56$, so $n^2+n-56=0$, $(n+8)(n-7)=0$. The pairs are $7,8$ and $-8,-7$; both are consecutive integers with product 56, and the wording does not require positive integers.

2 $n(n+2)=99$, so $n^2+2n-99=0$, $(n+11)(n-9)=0$. Positive gives $n=9$; the integers are 9 and 11.

3 $w(w+4)=96$, so $w^2+4w-96=0$, $(w+12)(w-8)=0$. $w=8$ (reject -12); dimensions 8 m by 12 m.

4 $x^2+(x+1)^2=(x+2)^2$, so $2x^2+2x+1=x^2+4x+4$, $x^2-2x-3=0$, $(x-3)(x+1)=0$. $x=3$ (reject -1); sides 3, 4 and 5.

5 $25t - 5t^2 = 20$, so $t^2 - 5t + 4 = 0$, $(t-1)(t-4) = 0$: $t = 1$ s and $t = 4$ s.

$h = 31.25 - 5(t - 2.5)^2$, so the maximum height is 31.25 m at $t = 2.5$ s.

6 $x + \frac{1}{x} = \frac{5}{2}$; multiply by $2x$: $2x^2 + 2 = 5x$, so $2x^2 - 5x + 2 = 0$, $(2x-1)(x-2) = 0$. $x = \frac{1}{2}$ or $x = 2$; both are non-zero and check.

7 Let the two equal sides be w; the third side is $40 - 2w$. Then $w(40 - 2w) = 150$, so $w^2 - 20w + 75 = 0$, $(w-5)(w-15) = 0$.

The plots are 5 m by 30 m or 15 m by 10 m, both with 40 m of fencing. Recommend 15 m by 10 m: the same area is easier to work and access than a 5 m wide, 30 m deep strip.

8 $n(2n-3) = 35$, so $2n^2 - 3n - 35 = 0$, $(2n+7)(n-5) = 0$. $n = 5$ or $n = -\frac{7}{2}$. Check: $5 \times 7 = 35$ and $\left(-\frac{7}{2}\right)(-10) = 35$.

Investigation · The discriminant detective

1 For $b = -10, -8, -6, -4, 0, 4, 6, 8, 10$, $\Delta = 64, 28, 0, -20, -36, -20, 0, 28, 64$; the numbers of real roots are $2, 2, 1, 0, 0, 0, 1, 2, 2$.

2 Roots occur only when $\Delta \geq 0$, that is $|b| \geq 6$: two roots when $|b| > 6$, one repeated root when $|b| = 6$, none when $|b| < 6$. The pattern is symmetric in the sign of b, because Δ depends on b^2.

3 $b = 7$: $\Delta = 13 > 0$, two roots $x = \frac{-7 \pm \sqrt{13}}{2}$. $b = -6$: $\Delta = 0$, one root $x = 3$. $b = 0$: $\Delta = -36 < 0$, no real roots. The formula or substitution confirms each prediction.

4 $\Delta = b^2 - 36 = (b-6)(b+6)$. Two distinct real roots for $b < -6$ or $b > 6$; one repeated root for $b = \pm 6$; no real roots for $-6 < b < 6$.

5 For $x^2 + 6x + k = 0$, $\Delta = 36 - 4k$. Two roots when $k < 9$; one when $k = 9$; none when $k > 9$.

6 When $\Delta = 0$, completing the square gives $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \frac{\Delta}{4} = \left(x + \frac{b}{2}\right)^2$, a perfect square with the single root $x = -\frac{b}{2}$.

Here $b = \pm 6$ gives $(x \pm 3)^2$, and $k = 9$ gives $(x+3)^2$.

Modelling task · The market garden

1 Only three sides are fenced, so the depth is $30 - 2w$ and $A(w) = w(30 - 2w) = 30w - 2w^2$ m^2. Both sides must be positive: $w > 0$ and $30 - 2w > 0$, so $0 < w < 15$.

2 $A = -2(w^2 - 15w) = -2\left[(w - 7.5)^2 - 56.25\right] = -2(w - 7.5)^2 + 112.5$. The vertex is the maximum because $-2 < 0$: width 7.5 m, depth $30 - 15 = 15$ m, area 112.5 m^2.

3 $30w - 2w^2 = 100$, so $w^2 - 15w + 50 = 0$ and $(w-5)(w-10) = 0$: $w = 5$ or $w = 10$.

The same 100 m^2 comes from a narrow deep plot (5 m by 20 m) or a wide shallow one (10 m by 10 m). Both use exactly 30 m of fencing.

4 Fencing $30 \times 12 = £360$. Topsoil $112.5 \times 18 = £2{,}025$. Total £2,385.

5 Yes: $£2{,}385 \leq £2{,}500$, with £115 to spare.

With the fencing fixed at 30 m, £2,500 leaves $2{,}500 - 360 = £2{,}140$ for topsoil, which would buy $2140 \div 18 = 118.9$ m^2. That is more than the 112.5 m^2 the fencing can enclose, so the budget is not the binding constraint — the 30 m of fencing is.

6 Assumptions: the plot is exactly rectangular; the wall is straight, at least 15 m long and needs no fencing; the 30 m is used exactly, with no gate or waste; topsoil covers the whole plot to a uniform depth.

The model ignores practical factors such as an access gate, which costs fencing without adding area, and shading or drainage along the wall, which affect what the plot can actually grow.

Mixed practice 5

1 $3x^2 + 13x - 10$.

2 $2(x^2 - 9) = 2(x-3)(x+3)$.

3 $(x-5)(x-6) = 0$, so $x = 5$ or 6.

4 $x^2 = 9$, so $x = \pm 3$.

5 $2(x^2 - 6x + 9) - 8 = 2x^2 - 12x + 10$.

6 $(x+8)(x-2) = 0$, so $x = -8$ or 2.

7 $\Delta = (-3)^2 - 4(1)(5) = 9 - 20 = -11$.

8 $(3x-1)(x-2) = 0$, so $x = \frac{1}{3}$ or 2.

9 For real x, $x^2 \geq 0$, so $x^2 = -4$ is impossible. Equivalently $\Delta = 0 - 4(1)(4) = -16 < 0$.

10 $x^2 - 6x = x(x-6) = 0$, so $x = 0$ or $x = 6$. The root $x = 0$ is lost by dividing by x.

11 $y = (x+1)(x-4) = x^2 - 3x - 4$. Expanding: $\Delta = 9 + 16 = 25 > 0$, confirming two distinct roots.

12 Exactly one root requires $\Delta = b^2 - 4(2)(8) = b^2 - 64 = 0$, so $b = \pm 8$.

13 $x^2 - 2x - 4 = 0$ does not factorise with integers, so complete the square: $(x-1)^2 = 5$, $x = 1 \pm \sqrt{5}$.

14 $x^2 + 8x + 20 = (x+4)^2 - 16 + 20 = (x+4)^2 + 4$. Then $(x+4)^2 = -4$, impossible for real x; equivalently $\Delta = 64 - 80 = -16 < 0$.

15 $w + l = 13$ and $wl = 40$, so $w(13 - w) = 40$, $w^2 - 13w + 40 = 0$, $(w-5)(w-8) = 0$. Dimensions 5 m by 8 m.

16 $R = (300 - 15x)(12 + x) = 3600 + 120x - 15x^2 = 3840 - 15(x-4)^2$. The maximum is £3,840 at $x = 4$, so the price is £16.

17 $(x-p)(x-q) = x^2 - (p+q)x + pq$. Comparing with $x^2 + bx + c$: $-(p+q) = b$, so $p + q = -b$; and $pq = c$.

18 $\Delta = 9 + 56 = 65$, so $x = \frac{3 \pm \sqrt{65}}{4}$. Sum $= \frac{3 + \sqrt{65} + 3 - \sqrt{65}}{4} = \frac{6}{4} = \frac{3}{2}$.

Criterion A · Knowing and understanding

1 $2x^2 + 3x - 10x - 15 = 2x^2 - 7x - 15$. [3]

2 $3x(x-4)$. [3]

3 $(x-4)(x-5)=0$, so $x=4$ or 5. [3]

4 $(2x-1)(x+4)=0$, so $x=\frac{1}{2}$ or -4. [4]

5 $x^2+4x=(x+2)^2-4$, so $(x+2)^2=5$ and $x=-2\pm\sqrt{5}$. [4]

6 $\Delta=25+24=49$, so $x=\frac{5\pm7}{6}$: $x=2$ or $x=-\frac{1}{3}$. [4]

7 $\Delta=9-32=-23<0$, so there are no real roots. [4]

8 $x^2-5x=x(x-5)=0$, so $x=0$ or 5. Dividing by x assumes $x\neq0$, losing the root 0. [3]

9 One root requires $\Delta=k^2-100=0$, so $k=\pm10$. [2]

Criterion B · Investigating patterns

1 For $b=-9,-6,-3,0,3,6,9$: $\Delta=65,20,-7,-16,-7,20,65$; number of real roots $2,2,0,0,0,2,2$. [5]

2 Two roots when $|b|>4$; none when $|b|<4$; one repeated root when $|b|=4$. The table shows two roots at $b=\pm6,\pm9$ and none at $b=-3,0,3$; the pattern is symmetric in the sign of b. [4]

3 Two-root cases: $b=-9$, sum 9, product 4; $b=-6$, sum 6, product 4; $b=6$, sum -6, product 4; $b=9$, sum -9, product 4. [6]

4 For $x^2+bx+c=0$, sum of roots $=-b$ and product $=c$.

Check $b=5,c=4$: roots $-1,-4$, sum $-5=-b$, product $4=c$. Check $b=-5,c=4$: roots $1,4$, sum $5=-b$, product $4=c$. [5]

5 $(x-p)(x-q)=x^2-(p+q)x+pq$, so comparing coefficients gives $b=-(p+q)$ and $c=pq$, i.e. $p+q=-b$, $pq=c$.

Also $b^2-4c=(p+q)^2-4pq=(p-q)^2\geq0$: whenever real roots p,q exist, the discriminant is non-negative. Conversely $\Delta\geq0$ makes the formula real-valued, so the roots exist. [4]

Marking guidance. Q1 (5): one mark per correct discriminant pair and one for the full root count. Q2 (4): 2 for the rule, 2 for the symmetry explanation. Q3 (6): values 4, method/labels 2. Q4 (5): formulas 3, two verifications 2. Q5 (4): expansion and comparison 2, discriminant argument 2. Accept interval or inequality notation.

Criterion C · Communicating

→ **Sample guide.** A root is a value of x for which $ax^2+bx+c=0$. The discriminant $\Delta=b^2-4ac$ decides the count: $\Delta>0$ gives two roots, $\Delta=0$ one repeated root, $\Delta<0$ none, because the parabola meets the x-axis twice, once or not at all.

Take $x^2-5x+6=0$. Factorising: $(x-2)(x-3)=0$, so $x=2,3$. Formula: $\Delta=25-24=1$, $x=\frac{5\pm1}{2}=3,2$. The answers agree.

Sketches: $y=x^2-4$ cuts at $x=\pm2$; $y=x^2$ touches at the origin; $y=x^2+1$ lies above the axis. Label roots, vertex and axis.

Choose factorising for integer roots, completing the square when the vertex is needed or b is even, and the formula when nothing else is quick. Common error: dividing $x^2=5x$ by x and losing $x=0$; factorise instead, $x(x-5)=0$.

Marking guidance. 20 marks: (i) 4 for root definition and discriminant cases; (ii) 5 for two complete, agreeing solutions; (iii) 4 for three correctly labelled sketches; (iv) 4 for a justified comparison of methods; (v) 3 for a named error, an example and a prevention strategy. Deduct where the audience is not addressed or a diagram is unlabelled.

Criterion D · Applying mathematics in real-life contexts

1 Price $=12+x$ pounds; tickets $=300-15x$. Then $R=(12+x)(300-15x)=3600+120x-15x^2$. [4]

2 $R=-15(x^2-8x)+3600=-15((x-4)^2-16)+3600=3840-15(x-4)^2$. Maximum £3,840 when $x=4$, that is a price of £16. [5]

3 $3600+120x-15x^2=3600$, so $15x(8-x)=0$: $x=0$ or $x=8$. At £12, 300 tickets are sold; at £20, $300-120=180$ tickets are sold. Both give £3,600. [4]

4 The revenue graph is symmetric about $x=4$, and $x=0$ and $x=8$ are equally distant from it, so the revenues match. Recommend £12: it fills 300 of the 400 seats, giving a larger audience and better atmosphere; £20 leaves 220 seats empty for the same income. [4]

5 $3600+120x-15x^2=3900$, so $x^2-8x+20=0$. $\Delta=64-80=-16<0$: no real solution, so £3,900 is impossible; the maximum is £3,840. [4]

6 Assumptions: every £1 rise loses exactly 15 sales (linear demand); all tickets sell at one price; the hall's 400-seat capacity is not exceeded; costs are ignored. The survey may be unreliable if it was hypothetical, based on a small or unrepresentative sample, or if demand changes with publicity. [3]

Marking guidance. Q1 4 (expressions 2, expansion 2); Q2 5 (completing the square 3, price and maximum 2); Q3 4 (roots 2, interpretation 2); Q4 4 (symmetry reason 2, justified non-financial recommendation 2); Q5 4 (equation 2, discriminant and conclusion 2); Q6 3 (two assumptions 2, reliability factor 1). A recommendation without a non-financial reason scores at most 1 in Q4.

CHAPTER 6 · QUADRATIC FUNCTIONS AND GRAPHICAL REASONING

Opening problem · the tunnel arch

1 At $(0,0)$: $0 = a(0-6)^2 + 6 = 36a + 6$, so $a = -\dfrac{6}{36} = -\dfrac{1}{6}$ and $h = -\dfrac{1}{6}(x-6)^2 + 6$. Check: $h(12) = 0$, so the arch meets the road again at the right edge, 12 m across.

2 Maximum height 6 m (the vertex y-value); axis of symmetry $x = 6$. Physically the axis is the vertical line through the centre of the road: the arch is symmetric, so its highest point sits directly above the midline, 6 m from each edge.

3 $h(2) = -\dfrac{1}{6}(2-6)^2 + 6 = -\dfrac{16}{6} + 6 = \dfrac{10}{3} \approx 3.33$ m; $h(4) = -\dfrac{1}{6}(4-6)^2 + 6 = -\dfrac{4}{6} + 6 = \dfrac{16}{3} \approx 5.33$ m.

4 Yes. On $4 \le x \le 8$ the arch is lowest at the endpoints, furthest from the axis: $h(4) = h(8) = \dfrac{16}{3} \approx 5.33 \text{ m} > 4.8$ m. There is room, with about $5.33 - 4.8 = 0.53$ m to spare.

5 The van occupies $1 \le x \le 3.2$ and its top is 3.2 m up. At its left edge $h(1) = -\dfrac{25}{6} + 6 = \dfrac{11}{6} \approx 1.83 \text{ m} < 3.2$ m, so the top-left corner $(1,\ 3.2)$ lies above the arch and strikes it. Solving $h(x) = 3.2$: $(x-6)^2 = 16.8$, $x = 6 \pm 4.10$, i.e. $x \approx 1.90$ or 10.10; the arch is at least 3.2 m high only on $1.90 \le x \le 10.10$, and $x = 1$ is outside.

6 For example, shift the traffic lane about 1 m right so the van occupies $2 \le x \le 4.2$: the lower corner height is then $h(2) = \dfrac{10}{3} \approx 3.33 \text{ m} > 3.2$ m, so the van passes with roughly 0.13 m to spare. The arch equation itself is unchanged — only the interval on which $h(x) \ge 3.2$ is required moves. (If instead the tunnel were raised or the road lowered, the width at road level and the vertex height would change, so a and k would need re-fitting from the new measurements.)

Prerequisite check

1 $x^2 - 8x + 11 = (x-4)^2 - 16 + 11 = (x-4)^2 - 5$.

2 $(x-2)(x-3) = 0$, so $x = 2$ or 3.

3 $(2, -7)$.

4 $-3 < x < 3$.

5 $(5-3)^2 - 4 = 4 - 4 = 0$.

6 $(x-3)(x+2)$.

Exercise 6A

1 (a) roots $2, 6$; axis $x = 4$; vertex $(4, -4)$; y-intercept 12.

(b) roots $-3, 1$; axis $x = -1$; vertex $(-1, -4)$; y-intercept -3.

(c) roots $0, 4$; axis $x = 2$; vertex $(2, -4)$; y-intercept 0.

2 (a) axis $x = 4$, vertex $(4, -4)$, minimum.

(b) axis $x = 3$, vertex $(3, 4)$, maximum.

(c) axis $x = 1$, vertex $(1, 5)$, minimum.

3 (a) opens upwards; y-intercept 1.

(b) opens downwards; y-intercept 0.

(c) opens downwards; y-intercept 7.

4 $x = -\dfrac{b}{2a}$.

5 Points on a parabola with the same y-value are equidistant from the axis. Both roots have $y = 0$, so the axis is the perpendicular bisector of the segment joining them, at the mean $x = \dfrac{r+s}{2}$.

6 Axis $x = \dfrac{-2+6}{2} = 2$; the vertex has x-coordinate 2 .

7 Two errors: the x-coordinate is $-\dfrac{b}{2a} = 5$, not 10; and the y-coordinate is not $c = 21$ but $y(5) = 25 - 50 + 21 = -4$. Correct vertex $(5, -4)$.

8 No. If $a < 0$ the parabola opens downwards and $y \to -\infty$, so it has a maximum at the vertex and no minimum on its whole domain. A restricted domain could give an endpoint minimum.

9 $h = 40t - 5t^2 = -5(t-4)^2 + 80$. Maximum height 80 m at $t = 4$ s.

10 At $x = 4$, $y = 8$ m. At ground level, $8 - 0.5(x-4)^2 = 0$, so $(x-4)^2 = 16$, $x = 0$ or 8; width 8 m.

11 $(0, 3)$ and $(4, 3)$ are symmetric, so the axis is $x = 2$ and the vertex is $(2, -1)$.

$y = a(x-2)^2 - 1$; at $(0, 3)$, $4a - 1 = 3$, so $a = 1$. Equation $y = (x-2)^2 - 1 = x^2 - 4x + 3$.

Exercise 6B

1 (a) vertex $(4, 2)$, axis $x = 4$; (b) vertex $(-1, -6)$, axis $x = -1$; (c) vertex $(2, 1)$, axis $x = 2$.

2 (a) $y = (x-5)^2 - 1$; (b) $y = (x+2)^2 - 7$; (c) $y = 2(x-3)^2 - 5$.

3 (a) shift right 5; (b) shift up 4; (c) reflection in the x -axis; (d) shift left 2, vertical stretch factor 3, shift down 1.

4 (a) minimum -9; (b) maximum 8; (c) minimum 0.

5 The graph is the set of points $(u+7, u^2)$: each point of $y = x^2$ moves 7 right. The bracket is zero when $x = 7$, so the vertex moves from 0 to 7, not to -7.

6 $x^2 - 6x + 1 = (x-3)^2 - 9 + 1 = (x-3)^2 - 8$. The student kept $+1$ instead of subtracting 9. Vertex $(3, -8)$, not $(3, 1)$.

7 Same vertex and axis; different width and direction. A larger $|a|$ gives a narrower parabola; $a > 0$ opens up (minimum), $a < 0$ opens down (maximum). Opposite signs are reflections in the line $y = -3$.

8 $y = a(x+4)^2 + 6$ with $a < 0$. Not unique: any negative a works, for example $y = -(x+4)^2 + 6$ or $y = -2(x+4)^2 + 6$.

9 $h = -5(t^2 - 2.4t) = -5((t-1.2)^2 - 1.44) = 7.2 - 5(t-1.2)^2$. Maximum 7.2 m at $t = 1.2$ s.

10 Vertex $(6, 9)$. At ground, $0.25(x-6)^2 = 9$, so $(x-6)^2 = 36$, $x = 0$ or 12; width 12 m.

11 Vertex $(2, 1)$ gives $h = 2$, $k = 1$, so $y = a(x-2)^2 + 1$. At $(0, 5)$, $4a + 1 = 5$, so $a = 1$.

$y = (x-2)^2 + 1$: axis $x = 2$, y-intercept 5, no x-intercepts because the minimum is 1. Reflect $(0, 5)$ to $(4, 5)$; sketch an upward parabola with vertex $(2, 1)$.

Exercise 6C

1 (a) $2 < x < 6$; (b) $-1 \le x \le 4$; (c) $x < -3$ or $x > -1$.

2 (a) $-3 \le x \le 3$; (b) $x < -5$ or $x > 5$; (c) $-4 < x < 4$.

3 (a) $x \le 2$ or $x \ge 5$; (b) $-4 < x < 2$; (c) $(x-3)^2 > 0$, so $x \neq 3$ (all real x except 3).

4 $(2x+1)(x-3) \le 0$, so $-\frac{1}{2} \le x \le 3$.

5 Taking a square root gives $|x| < 3$, that is $-3 < x < 3$. The single inequality $x < 3$ also includes values such as $x = -4$, whose square is 16, not less than 9.

6 The student multiplied by -1 without reversing the inequality. Correctly $x^2 - 6x + 5 < 0$, so $(x-1)(x-5) < 0$ and $1 < x < 5$.

7 $x^2 - 4x < 0$, so $x(x-4) < 0$: the parabola is below the axis for $0 < x < 4$.

8 $x^2 - 4 \le 0$ gives $-2 \le x \le 2$, including the endpoints; $x^2 - 4 < 0$ gives $-2 < x < 2$, excluding them. At the roots the expression equals zero, which satisfies $\le$ but not $<$.

9 $24t - 5t^2 > 20$, so $5t^2 - 24t + 20 < 0$. The roots are $\frac{24 \pm 4\sqrt{11}}{10} = \frac{12 \pm 2\sqrt{11}}{5}$.
Hence $\frac{12 - 2\sqrt{11}}{5} < t < \frac{12 + 2\sqrt{11}}{5}$, approximately $1.07 < t < 3.73$ s.

10 $-x^2 + 12x - 20 \ge 0$, so $x^2 - 12x + 20 \le 0$, $(x-2)(x-10) \le 0$, giving $2 \le x \le 10$.

11 $x^2 - 6x + k = (x-3)^2 + k - 9$. Its minimum is $k - 9$, which must be positive, so $k > 9$. Equivalently $\Delta = 36 - 4k < 0$.

Exercise 6D

1 $w + l = 18$, so $A = w(18 - w) = 81 - (w-9)^2$. Maximum 81 m^2 for a 9 m by 9 m square.

2 $h = -5(t-3)^2 + 45$, so the maximum height is 45 m at $t = 3$ s. It lands when $5t(6 - t) = 0$, so $t = 6$ s.

3 Let the two equal sides be w; the side along the wall is $60 - 2w$. $A = w(60 - 2w) = 450 - 2(w-15)^2$, so the maximum 450 m^2 occurs at $w = 15$: 15 m by 30 m.

4 Price $10 + x$, tickets $400 - 20x$. $R = (10 + x)(400 - 20x) = 4500 - 20(x-5)^2$. Maximum £4,500 at $x = 5$, so the price is £15 (300 tickets).

5 $w(w + 2) = 80$, so $w^2 + 2w - 80 = 0$, $(w + 10)(w - 8) = 0$. $w = 8$ (reject -10); dimensions 8 m by 10 m.

6 Maximum height 6 m, at $x = 0$, where the x^2 term is zero.

At ground level $h = 0$, so $0.24x^2 = 6$, $x^2 = 25$ and $x = \pm 5$. The arch is $5 - (-5) = 10$ m wide at ground level, which is exactly the stated domain.

7 $P = -2(x-6)^2 + 18$, so the maximum profit is £18,000 at $x = 6$ (600 units).

8 $P = -10(x-6)^2 + 360$, whose vertex is $x = 6$, outside $7 \le x \le 12$. On this interval P decreases, so the maximum is at $x = 7$: $P = 840 - 490 = 350$, that is £350.

Exercise 6E

1 (a) $x^2 = x + 2$, $(x-2)(x+1) = 0$: $(2, 4)$ and $(-1, 1)$.
(b) $x^2 = 3x - 2$, $(x-1)(x-2) = 0$: $(1, 1)$ and $(2, 4)$.
(c) $x^2 - 1 = 2x + 3$, $x^2 - 2x - 4 = 0$, $x = 1 \pm \sqrt{5}$: points $(1 + \sqrt{5}, 5 + 2\sqrt{5})$ and $(1 - \sqrt{5}, 5 - 2\sqrt{5})$.

2 (a) At $x = 0$, parabola 3, line 1: the parabola is above.
(b) At $x = 0$, parabola -2, line 0: the line is above.

3 $0.5x^2 = 10 + 2x$, so $x^2 - 4x - 20 = 0$ and $x = 2 \pm 2\sqrt{6}$, approximately 6.90 or -2.90.

4 Setting the equations equal gives a quadratic. Its discriminant is positive for two intersections, zero for one (a tangent) and negative for none, because those are exactly the cases for two, one or no real solutions.

5 $0.5t^2 + 100 = 500 + 40t$, so $t^2 - 80t - 800 = 0$. $t = 40 \pm 20\sqrt{6}$, approximately 89.0 or -9.0. The positive time is about 89 years; the negative value is rejected.

6 $x^2 = 4$ gives $x = \pm 2$, so the intersections are $(-2, 4)$ and $(2, 4)$. The negative solution $x = -2$ is missing.

7 A linear model has fewer parameters, a constant rate that is easy to explain, and is less likely to fit random noise with spurious curvature. If it fits adequately over the required range, simplicity favours it.

8 Twelve years is far beyond the data, so the prediction is an extrapolation. A quadratic eventually falls without limit, which may not match reality; market conditions, competition and costs may change, and the model may only be valid over the observed domain.

9 $8 + 2d = 0.5d^2$, so $d^2 - 4d - 16 = 0$, $d = 2 \pm 2\sqrt{5}$. The positive distance is $2 + 2\sqrt{5} \approx 6.47$ km. At 6 km: taxi $8 + 12 = £20$, ride-share $0.5(36) = £18$, so the ride-share is cheaper.

10 $20t - 5t^2 = 10 + 2t$, so $5t^2 - 18t + 10 = 0$, $t = \frac{9 \pm \sqrt{31}}{5}$, approximately 0.69 s and 2.91 s.

11 $x^2 + 1 = 2x + c$, so $x^2 - 2x + (1 - c) = 0$. $\Delta = 4 - 4(1 - c) = 4c$.
If $c > 0$, $\Delta > 0$: two intersections. If $c = 0$, $\Delta = 0$: one intersection (tangent at $(1, 2)$). If $c < 0$, $\Delta < 0$: no intersection.

Investigation · The fountain arc

1 The points are symmetric about $x = 5$: heights at 0 and 10, 1 and 9, 2 and 8, 4 and 6 agree. The vertex is therefore at $x = 5$, with height about 5 m, so $(5, 5)$.

2 $h = a(x - 5)^2 + 5$. Using $(0, 0)$: $0 = 25a + 5$, so $a = -0.2$. Model $h = 5 - 0.2(x - 5)^2$.

3 $x = 1$: $5 - 0.2(16) = 1.8$; $x = 4$: $5 - 0.2(1) = 4.8$; $x = 9$: 1.8. All agree exactly with the table, so the fit is excellent.

4 $x = 3$: $5 - 0.2(4) = 4.2$; $x = 7$: 4.2. Both lie inside the measured range 0 to 10, so they are interpolations.

5 $5 - 0.2(x - 5)^2 = 4$, so $(x - 5)^2 = 5$ and $x = 5 \pm \sqrt{5} \approx 2.76$ or 7.24. The jet is 4 m high at about 2.76 m and 7.24 m from the nozzle, once rising and once falling.

6 The new vertex is $(5, 6)$ with the same landing points 0 and 10, so $0 = 25a + 6$, $a = -0.24$. The model becomes $h = 6 - 0.24(x - 5)^2$: higher and narrower.

In general $h = a(x - p)^2 + q$ and $a = \dfrac{h_1 - q}{(x_1 - p)^2}$. A real landing needs $a < 0$ and $q > 0$, giving $x = p \pm \sqrt{-\dfrac{q}{a}}$.

Modelling task · The school mural

1 The arch meets the ground at $(4, 0)$, so $0 = 5 - 16k$ and $k = \dfrac{5}{16} = 0.3125$. The model is $y = 5 - 0.3125x^2$.

2 At $x = 1$: $y = 5 - 0.3125 = 4.6875 \approx 4.69$ m. At $x = 3$: $y = 5 - 2.8125 = 2.1875 \approx 2.19$ m. The arch drops fastest away from the centre, which is why it is only 2.19 m high 3 m from the centre.

3 $5 - 0.3125x^2 \geq 2.5$ gives $0.3125x^2 \leq 2.5$, so $x^2 \leq 8$ and $-\sqrt{8} \leq x \leq \sqrt{8}$, that is $-2.83 \leq x \leq 2.83$.

The greatest width with 2.5 m of clearance is $2\sqrt{8} = 4\sqrt{2} \approx 5.66$ m.

4 The top corners must lie on the arch: $3 = 5 - 0.3125x^2$, so $x^2 = 6.4$, $x = \pm 2.53$. Greatest width $= 5.06$ m (3 s.f.).

Area $= 3 \times 5.06 = 15.18$ m^2; cost $= 15.18 \times 28 =$ £425 (to the nearest pound, £425.01).

5 No. A 6 m wide mural reaches $x = \pm 3$, where the arch is only 2.19 m high, below the 2.5 m requirement — and below the 3 m height used in step 4. The widest mural that keeps 2.5 m of clearance is 5.66 m, so 6 m is not achievable under this arch.

6 Assumptions: the arch is exactly parabolic and symmetric about the centre; the ground is level and the 8 m width is measured at the ground; the 5 m height is to the underside of the arch.

A real arch may be a circular arc or a catenary rather than a parabola, may have mouldings or a frame that reduce the clear space, and may sit on uneven paving — any of which changes the usable width at a given height.

Mixed practice 6

1 $(5, -4)$.

2 $y = (x + 4)^2 - 6$.

3 $x = -\dfrac{12}{2(-2)} = 3$.

4 $(x - 8)(x + 3) = 0$, so $x = 8$ or -3.

5 $-5 < x < 5$.

6 $(x - 5)(x + 2) \geq 0$, so $x \leq -2$ or $x \geq 5$.

7 Maximum 8 at $x = 3$.

8 $x^2 - 4 = 3x$, so $(x - 4)(x + 1) = 0$: $(4, 12)$ and $(-1, -3)$.

9 $y \geq -6$.

10 The vertex is $(4, 1)$; the 3 stretches the graph vertically and does not move it horizontally, so 12 is not the x-coordinate.

11 Axis $x = \dfrac{-3 + 7}{2} = 2$; the vertex has x-coordinate 2 .

12 With $a > 0$, the parabola is above the axis outside its two roots, so a solution of $y > 0$ consists of two rays, for example $x < -1$ or $x > 4$.

13 $h = -5(t - 1.8)^2 + 16.2$, so the maximum is 16.2 m at $t = 1.8$ s. It lands when $5t(3.6 - t) = 0$, $t = 3.6$ s.

14 The minimum is $9 - \dfrac{k^2}{4} = 0$, so $k^2 = 36$ and $k = \pm 6$. Equivalently $\Delta = k^2 - 36 = 0$.

15 $w + l = 14$, so $A = w(14 - w) = 49 - (w - 7)^2 \leq 49$. Equality at $w = 7$, a 7 m square.

16 $R = (30 - x)(500 + 25x) = 15625 - 25(x - 5)^2$. Maximum revenue at $x = 5$, so the price is £25.

17 $x^2 - 3x + 5 = x + c$, so $x^2 - 4x + (5 - c) = 0$. Tangency needs $\Delta = 16 - 4(5 - c) = 4c - 4 = 0$, so $c = 1$. Then $x = 2$ and $y = 3$: point $(2, 3)$.

18 $y = a(x - 1)(x - 5)$; at $(0, 10)$, $5a = 10$, so $a = 2$. $y = 2x^2 - 12x + 10 = 2(x - 3)^2 - 8$; vertex $(3, -8)$.

Criterion A · Knowing and understanding

1 $(x - 6)(x + 2) = 0$: roots $-2, 6$; axis $x = 2$; vertex $(2, -16)$. [4]

2 $y = (x - 3)^2 + 2$. [3]

3 Reflect in the x-axis and stretch vertically by factor 2; shift 3 left; shift 4 up. [4]

4 Maximum 5 (at $x = -2$). [2]

5 $(x - 4)(x + 2) \leq 0$, so $-2 \leq x \leq 4$. [4]

6 $x^2 - 16 > 0$, $(x - 4)(x + 4) > 0$, so $x < -4$ or $x > 4$. [3]

7 $x^2 - 2x = x + 4$, so $x^2 - 3x - 4 = 0$, $(x - 4)(x + 1) = 0$: $(4, 8)$ and $(-1, 3)$. [4]

8 The minimum is 2, since $(x - 5)^2 \geq 0$. Thus $y \geq 2$, so the graph never reaches $y = 0$; $(x - 5)^2 = -2$ has no real solution. [3]

9 Taking square roots gives $|x| < 3$, so $-3 < x < 3$. The error is ignoring the negative interval. [3]

Criterion B · Investigating patterns

1 $b=-4$: roots $0, 4$, axis $x=2$, vertex $(2,-4)$. $b=-2$: roots $0, 2$, axis $x=1$, vertex $(1,-1)$. $b=0$: root 0 repeated, axis $x=0$, vertex $(0,0)$. $b=2$: roots $0, -2$, axis $x=-1$, vertex $(-1,-1)$. $b=4$: roots $0, -4$, axis $x=-2$, vertex $(-2,-4)$. [6]

2 All curves pass through $(0,0)$. As b increases, the axis $x=-\frac{b}{2}$ moves steadily left, from $x=2$ to $x=-2$. The vertex y-value is $-4, -1, 0, -1, -4$, so it rises to the origin at $b=0$ and then falls; the vertex traces a downward curve. [4]

3 $y=x(x+b)$, so the roots are 0 and $-b$. The axis is $x=-\frac{b}{2}$ and the vertex is $\left(-\frac{b}{2}, -\frac{b^2}{4}\right)$. [5]

4 For $b=6$: $y=x^2+6x=(x+3)^2-9$. Roots 0 and -6 (formula $0, -b$); axis $x=-3$ (formula $-\frac{b}{2}$); vertex $(-3,-9)$ (formula $-\frac{b^2}{4}=-9$). All agree. [4]

5 Write the vertex as $(x,y)=\left(-\frac{b}{2}, -\frac{b^2}{4}\right)$. Since $b=-2x$, $y=-\frac{4x^2}{4}=-x^2$. Every vertex lies on $y=-x^2$, and every point of $y=-x^2$ arises from $b=-2x$, so the locus is exactly this downward parabola. [5]

Marking guidance. Q1 (6): 5 marks for the five complete rows (roots, axis and vertex) and 1 for correct layout or an additional check. Q2 (4): 2 for the axis movement, 2 for the vertex pattern and origin. Q3 (5): roots 2, axis 1, vertex 2. Q4 (4): completing the square 2, comparison 2. Q5 (5): elimination 3, converse/exact locus 2. Accept $b=-2x$ substituted into $y=-\frac{b^2}{4}$.

Criterion C · Communicating

→ **Sample guide.** In $y=a(x-h)^2+k$, the vertex is (h,k): h shifts the curve horizontally (opposite to its sign), k shifts it vertically, and a controls width and direction. If $a>0$ the parabola opens up and the vertex is a minimum; if $a<0$ it opens down and the vertex is a maximum; larger $|a|$ is narrower.

To sketch, read (h,k), draw the axis $x=h$, set $x=0$ for the y-intercept, and set $y=0$ for the roots. For $y=(x-3)^2-4$: vertex $(3,-4)$, axis $x=3$, roots $1, 5$, y-intercept 5, reflected point $(6,5)$.

For $y=-(x+1)^2+4$: vertex $(-1,4)$, a maximum, axis $x=-1$, roots $x=1, -3$, y-intercept 3. Label all of these.

The discriminant $\Delta=b^2-4ac$ predicts the number of x-intercepts: positive two, zero one, negative none. A common error is reading the vertex as $(-h,k)$; check by substituting $x=h$, which must give $y=k$.

Marking guidance. 20 marks: (i) 5 for the roles of a, h, k; (ii) 5 for a correct method and worked example; (iii) 4 for two labelled, correctly shaped sketches; (iv) 3 for the discriminant cases; (v) 3 for a specific error and remedy. Award communication marks for audience, sequencing and correct terminology; penalise unlabelled axes or a sketch contradicting its equation.

Criterion D · Applying mathematics in real-life contexts

1 At $x=2$: $6-0.24(4)=5.04$ m. At $x=4$: $6-0.24(16)=2.16$ m. [4]

2 $6-0.24x^2=0$, so $x^2=25$, $x=\pm5$. The solutions are the two ground edges; the greenhouse is 10 m wide. [3]

3 $6-0.24x^2\geq 2.5$, so $0.24x^2\leq 3.5$, $x^2\leq\frac{175}{12}$, $|x|\leq\frac{5\sqrt{21}}{6}\approx 3.82$. Usable width $=\frac{5\sqrt{21}}{3}\approx 7.64$ m. [5]

4 $6-0.24x^2\geq 2.1$, so $0.24x^2\leq 3.9$, $x^2\leq 16.25$, $|x|\leq\frac{\sqrt{65}}{2}\approx 4.03$. Greatest distance about 4.03 m from the centre line. [4]

5 At $x=\pm3$, $y=6-0.24(9)=3.84$ m, which is less than 4 m, so the model fails. Require $6-9k\geq 4$, so $9k\leq 2$ and $k\leq\frac{2}{9}\approx 0.222$; taking $k=\frac{2}{9}$ meets the requirement exactly. [5]

6 Assumptions: the cross-section is exactly parabolic and symmetric; the ground is level and the 10 m width is measured at ground level; the frame has negligible thickness. A real greenhouse may have vertical sides, a rounded ridge, uneven ground or a frame that reduces the clear space. [3]

Marking guidance. Q1 4 (two substitutions and units); Q2 3 (roots 2, meaning 1); Q3 5 (inequality solved 4, width 1); Q4 4 (inequality 2, exact and decimal 2); Q5 5 (showing failure 2, correct k 3); Q6 3 (two assumptions 2, limitation 1). The exact form is preferred; accept 3 s.f. decimals. In Q5, $k=\frac{2}{9}$ or any smaller positive value is acceptable if justified.

CHAPTER 7 · PATTERNS, SEQUENCES AND GENERALISATION

Opening problem · the amphitheatre

1 24, 27, 30, 33, 36, 39 — each row adds 3 seats.

2 Row 20 is 19 steps of 3 beyond row 1: $24 + 19 \times 3 = 24 + 57 = 81$ seats.

3 $r_n = 24 + 3(n-1) = 3n + 21$. Check: $r_{20} = 3(20) + 21 = 81$, agreeing with step 2.

4 $3n + 21 = 60$ gives $3n = 39$, $n = 13$: row 13 has exactly 60 seats. $3n + 21 = 100$ gives $n = \frac{79}{3} = 26\frac{1}{3}$, not a whole number, so no row has exactly 100 seats — indeed $r_n = 3(n + 7)$ is always a multiple of 3, and 100 is not.

5 $S_1 = 24$, $S_2 = 51$, $S_3 = 81$, $S_4 = 114$, $S_5 = 150$. First differences: 27, 30, 33, 36 (these are the row sizes themselves); second differences: 3, 3, 3 — constant, so the sequence of totals is quadratic.

6 Row 20 is the longest row with 81 seats and every other row is shorter, so the total is less than $20 \times 81 = 1{,}620$. Exact total: $S_{20} = \frac{20}{2}(24 + 81) = 10 \times 105 = 1{,}050$. The claim of about 1,000 is therefore reasonable — it under-estimates by 50 seats, about 5%.

Prerequisite check

1 $3(7) + 5 = 21 + 5 = 26$.

2 The terms increase by 5, so the next two are $19 + 5 = 24$ and $24 + 5 = 29$.

3 $38 - 17 = 21$.

4 $2n + 1 = 31$, so $2n = 30$ and $n = 15$.

5 $(n + 1)(n + 2) = n^2 + 2n + n + 2 = n^2 + 3n + 2$.

6 $x^2 + 3x - 10 = (x + 5)(x - 2) = 0$, so $x = -5$ or $x = 2$.

Exercise 7A

1 (a) Add 4: 19, 23, 27.
(b) Subtract 4: 18, 14, 10.
(c) Multiply by 3: 54, 162, 486.
(d) Square numbers: 25, 36, 49.
(e) The differences are 2, 3, 4, so add 5, 6, 7: 15, 21, 28.

2 (a) 7, 7, 7; constant, so linear.
(b) −3, −3, −3; constant, so linear.
(c) 3, 4, 5; not constant, so not linear.
(d) 4, 5, 6; not constant, so not linear.

3 (a) $u_n = n^2$, since the terms are $1^2, 2^2, 3^2, 4^2$.
(b) $u_n = 5n$.
(c) $u_n = 2^n$.
(d) $u_n = \frac{n(n+1)}{2}$, the triangular numbers.

4 If $u_{n+1} - u_n = d$ for every n, then $u_n = u_1 + (n-1)d = dn + (u_1 - d)$. This is a linear function of n, so its graph is a straight line: equal steps in position give equal steps in value.

5 The common difference is $9 - 5 = 4$, so $u_n = 4n + c$. From $u_1 = 5$, $c = 5 - 4 = 1$, hence $u_n = 4n + 1$. The 12th term is $u_{12} = 4(12) + 1 = 49$.

6 The differences are 2, 4, 8, not constant, so 2 is not a common difference. The ratios are 2, 2, 2, so the sequence is geometric with ratio 2. The student confused a common ratio with a common difference.
Common error: confusing a common ratio with a common difference.

7 The terms are 1, 5, 9, 13, so the common difference is 4 and $u_n = 4n - 3$. Hence $u_{10} = 4(10) - 3 = 37$. Each stage adds one dot to each of the four arms, so 4 dots are added each time.

8 The row numbers are the odd numbers $u_n = 2n - 1$, so row 8 has $2(8) - 1 = 15$ blocks. Every term is odd because $2n$ is even for every whole number n, and subtracting 1 makes it odd.

9 The first five terms are 500, 1,000, 2,000, 4,000, 8,000. The differences 500, 1,000, 2,000, 4,000 are not constant, so the sequence is not arithmetic. Each term is twice the previous term, so it is geometric with ratio 2.

10 The next first differences are 5 and 6, so the next terms are $11 + 5 = 16$ and $16 + 6 = 22$. The second differences are 1, 1, 1, 1, 1, all equal. A constant non-zero second difference signals a quadratic rule; here $u_n = \frac{n^2 - n + 2}{2}$.

Exercise 7B

1 (a) $d = 3$, $u_n = 3n + 1$.
(b) $d = 4$, $u_n = 4n + 7$.
(c) $d = 7$, $u_n = 7n - 5$.
(d) $d = -5$, $u_n = 55 - 5n$.

2 (a) $6(15) - 1 = 89$.
(b) $3 - 2(15) = -27$.
(c) $0.5(15) + 4 = 11.5$.

3 Here $d = 4$ and $u_n = 4n + 3$. Set $4n + 3 = 71$: $4n = 68$, so $n = 17$. The 17th term is 71.

4 Set $4n + 3 = 100$: $4n = 97$, so $n = 24.25$, not a positive integer. Hence 100 is not a term; the neighbouring terms are $u_{24} = 99$ and $u_{25} = 103$.

5 The 3rd and 8th terms differ by $33 - 13 = 20$ over 5 steps, so $d = 4$. Then $u_n = 4n + c$ and $13 = 4(3) + c$, giving $c = 1$. Hence $u_n = 4n + 1$.

6 The cost is $C = 15 + 0.08m$. For 200 minutes, $C = 15 + 16 = £31$. For a £45 bill, $0.08m = 30$, so $m = 375$ minutes.

7 At $n = 1$, $3n + 8 = 11$, not 8, so the constant is wrong. The common difference is 3, and c is the value at $n = 0$: $c = 8 - 3 = 5$. The correct rule is $u_n = 3n + 5$.
Common error: using the first term as the constant instead of $u_1 - d$.

8 $u_n = u_1 + (n-1)d = dn + (u_1 - d)$. Increasing n by 1 adds exactly d, so each unit step in position produces the same change d. Hence the coefficient of n is the common difference.

9 Solve $6n - 4 > 100$: $6n > 104$, so $n > 17.3$. The first whole number is $n = 18$, and $u_{18} = 108 - 4 = 104$. Check: $u_{17} = 98$, which is below 100.

10 The height is $h_n = 1.8 + 0.25n$ metres. Solve $1.8 + 0.25n > 4$: $0.25n > 2.2$, so $n > 8.8$. The first whole year is $n = 9$, when $h_9 = 1.8 + 2.25 = 4.05$ m.

11 $5a + b = 18$ and $12a + b = 39$. Subtracting gives $7a = 21$, so $a = 3$; then $15 + b = 18$, so $b = 3$. Hence $u_n = 3n + 3$.

12 Set $2n + 5 = 3n - 1$: $n = 6$. The common term is $2(6) + 5 = 17$.

Exercise 7C

1 (a) First $3, 5, 7$; second $2, 2$: quadratic.

(b) First $5, 9, 13$; second $4, 4$: quadratic.

(c) First $-3, -1, 1$; second $2, 2$: quadratic.

(d) First $7, 11, 15$; second $4, 4$: quadratic.

2 (a) $u_n = n^2$.

(b) $u_n = 2n^2$.

(c) Second difference 4, so $a = 2$; subtracting $2n^2$ leaves $3, 4, 5, 6 = n + 2$. Hence $u_n = 2n^2 + n + 2$.

(d) Second difference 4, so $a = 2$; subtracting $2n^2$ leaves $1, 1, 1, 1$. Hence $u_n = 2n^2 + 1$.

3 First differences $6, 8, 10$; second difference 2, so $a = 1$. Subtracting n^2 leaves $3, 6, 9, 12 = 3n$, so $u_n = n^2 + 3n$. Hence $u_{20} = 400 + 60 = 460$.

4 A constant second difference $2a$ means the first differences change by the same amount at every step, so they form an arithmetic sequence. Summing a linear first difference produces a quadratic term, so $u_n = an^2 + bn + c$ with $a \neq 0$; the constant second difference is $2a$.

5 The first differences $3, 5, 7$ are not constant, so the sequence is not linear. The second difference is constant 2, so it is quadratic. Subtracting n^2 from each term leaves $1, 1, 1, 1$, so $u_n = n^2 + 1$.

Common error: treating any increasing differences as linear growth.

6 First differences $20, 22, 24$; second difference 2, so $a = 1$. Subtracting n^2 leaves $17, 34, 51, 68 = 17n$, so $S_n = n^2 + 17n$. For 15 rows, $S_{15} = 225 + 255 = 480$.

7 $u_3 = 3^2 + 3k + 3 = 12 + 3k = 21$, so $3k = 9$ and $k = 3$.

8 The rule is $u_n = 2n^2 - n + 1$. Set $2n^2 - n + 1 = 142$: $2n^2 - n - 141 = 0$. The discriminant is $1 + 1{,}128 = 1{,}129$, not a perfect square, so there is no integer n. The neighbouring terms are $u_8 = 121$ and $u_9 = 154$.

9 Design 10: $10^2 + 4(10) = 140$ tiles. For 96 tiles, $n^2 + 4n = 96$, so $n^2 + 4n - 96 = 0$ and $(n + 12)(n - 8) = 0$. Thus $n = 8$: design 8.

10 The second difference is 6, so $2a = 6$ and $a = 3$. Using $u_1 = 4$ and $u_2 = 13$: $3 + b + c = 4$ and $12 + 2b + c = 13$. Subtracting gives $9 + b = 9$, so $b = 0$, then $c = 1$. Hence $u_n = 3n^2 + 1$.

11 $u_{n+1} - u_n = a(2n + 1) + b = 2an + (a + b)$. The next first difference is $2a(n + 1) + (a + b)$. Subtracting gives $2a$, which is independent of n. Hence the second difference is always $2a$.

Exercise 7D

1 (a) $3, 9, 15, 21, 27$.

(b) $20, 16, 12, 8, 4$.

(c) $2, 6, 18, 54, 162$.

(d) $1, 2, 3, 5, 8$.

2 The sequence is arithmetic with $d = 6$ and $u_1 = 3$, so $u_n = 6n - 3$. Hence $u_{20} = 120 - 3 = 117$.

3 $u_1 = 5$ and $u_{n+1} = u_n + 4$.

4 Let B_n be the balance after n weeks. Then $B_0 = 500$ and $B_{n+1} = B_n + 45$. The explicit rule is $B_n = 500 + 45n$. After 10 weeks, $B_{10} = 500 + 450 = £950$. Solve $500 + 45n > 1{,}100$: $45n > 600$, so $n > 13.3$. Week 14 is the first, with £1,130.

5 B2 holds 7. Filling down adds 3 at each step, so B10 holds $7 + 8(3) = 31$. The fill-down copies the relative formula **=B2+3** into each cell, so every cell adds 3 to the cell above and the column lists $7, 10, 13, \ldots, 31$.

6 $F_1 = 1$, $F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$. The next five terms are $8, 13, 21, 34, 55$. Each term depends on the two before it, so there is no simple polynomial rule in n; even the closed form involves the golden ratio.

7 The sequence $2, 6, 18, 54$ multiplies by 3; its differences $4, 12, 36$ are not constant, so adding 3 is wrong. The correct recursion is $u_1 = 2$, $u_{n+1} = 3u_n$.

Common error: choosing addition when the pattern is multiplicative.

8 $P_0 = 200$ and $P_{n+1} = 2P_n$. After 5 hours, $P_5 = 200 \times 2^5 = 200 \times 32 = 6{,}400$.

9 An explicit rule gives any term directly, such as u_{100}, without listing earlier terms. A recursive rule matches processes where the next value depends on the current one, and it is easy to enter in a spreadsheet; it also needs no closed formula.

10 The first four terms are $2, 4, 16, 256$. The differences $2, 12, 240$ are not constant and the ratios $2, 4, 16$ are not constant, so the sequence is neither arithmetic nor geometric. It squares each term: $u_{n+1} = u_n^2$.

11 $u_1 = 5$, $u_2 = 9$, $u_3 = 17$, $u_4 = 33$, $u_5 = 65$. If $u_n = 2^{n+1} + 1$, then $u_{n+1} = 2(2^{n+1} + 1) - 1 = 2^{n+2} + 1$; since $u_1 = 2^2 + 1 = 5$, the formula holds for all n.

Exercise 7E

1 $T_{10} = \dfrac{10(11)}{2} = 55$ and $T_{20} = \dfrac{20(21)}{2} = 210$.

2 The total is $15 + 14 + \cdots + 1 = T_{15} = \dfrac{15(16)}{2} = 120$ cans.

3 First differences $6, 8, 10$; second difference 2, so $a = 1$. Subtracting n^2 leaves $11, 14, 17, 20 = 3n + 8$, so $h_n = n^2 + 3n + 8$ cm. Week 8: $64 + 24 + 8 = 96$ cm. This is extrapolation beyond four weeks; a plant cannot accelerate indefinitely, and growth slows at maturity because of limited light, water and genetics.

4 $C = 25 + 0.10g$. For 30 GB, $C = 25 + 3 = £28$. For a £40 bill, $0.10g = 15$, so $g = 150$ GB.

5 $D_{12} = \dfrac{12(9)}{2} = 54$. For 35 diagonals, $\dfrac{n(n-3)}{2} = 35$, so $n^2 - 3n - 70 = 0$ and $(n - 10)(n + 7) = 0$. Thus $n = 10$: a decagon.

6 A row is a discrete object, so n must be a positive whole number. The quadratic equation is solved over all real numbers and may give a non-integer root such as 16.78; that root is not admissible as a row number and must be rounded according to the context.

7 Every real system has a ceiling: customers, resources, shelf space or demand eventually limit growth. A quadratic eventually predicts implausibly large profits, and the data cover only a short period, so the extrapolation is uncertain.

8 Interpolation predicts inside the data range, for example using $S_n = \dfrac{3}{2}n^2 + \dfrac{45}{2}n$ to find S_{10} between known totals. Extrapolation predicts outside the data, such as week 10 sales from weeks 1–4; it assumes the pattern continues and is riskier.

9 $d_9 = 4(9) + 6 = 42$ docks. For 70 docks, $4n + 6 = 70$, so $n = 16$.

10 $n^2 + 3n = 180$, so $n^2 + 3n - 180 = 0$ and $(n + 15)(n - 12) = 0$. Thus design 12. Design 15 uses $225 + 45 = 270$ tiles.

11 $\dfrac{n(n-1)}{2} = 66$, so $n^2 - n - 132 = 0$ and $(n - 12)(n + 11) = 0$. Hence $n = 12$ people. The answer must be a whole number because people are counted individually, and -11 is rejected.

Investigation · Counting a growing rectangle

1 $D_1 = 2 \times 3 = 6$, $D_2 = 3 \times 4 = 12$, $D_3 = 4 \times 5 = 20$, $D_4 = 5 \times 6 = 30$. The table is n : $1, 2, 3, 4$ with D_n : $6, 12, 20, 30$.

2 First differences $6, 8, 10$; second differences $2, 2$. The constant second difference 2 means the sequence is quadratic, with $2a = 2$, so $a = 1$.

3 $D_n = (n + 1)(n + 2) = n^2 + 3n + 2$. The n^2 term produces second difference $2a = 2$; the $3n$ term contributes a constant first difference 3 but zero second difference; the constant 2 contributes nothing. Hence the table's second difference is 2.

4 $D_{10} = 11 \times 12 = 132$ dots. This is extrapolation, because Figure 10 lies beyond the cases $n = 1$ to 4 that were counted.

5 $(n + 1)(n + 2) = 210$ gives $n^2 + 3n - 208 = 0$. The discriminant is $9 + 832 = 841 = 29^2$, so $n = \dfrac{-3 + 29}{2} = 13$ or $n = -16$. Only $n = 13$ is a positive whole figure number, so Figure 13.

6 With $n + 3$ rows and $n + 2$ columns, $D_n = (n + 3)(n + 2) = n^2 + 5n + 6$. The values are $12, 20, 30, 42$, so the first differences are $8, 10, 12, \ldots$, and the second difference is still 2.

→ The rows and columns are $n + 1$ and $n + 2$, both linear in n. Their product is quadratic: the $n \times n$ square contributes n^2, the two strips contribute $3n$, and the corner contributes 2. The coefficient of n^2 is $1 \times 1 = 1$, so the second difference is $2a = 2$. Adding 1 to n adds one row and one column; the number of dots added grows linearly with n, so the first differences increase by a constant and the second differences are constant.

Mixed practice 7

1 Add 7: $25, 32, 39$.

2 $d = 5$ and $c = 9 - 5 = 4$, so $u_n = 5n + 4$.

3 $u_{20} = 5(20) - 2 = 98$.

4 Here $u_n = 5n + 2$. Set $5n + 2 = 92$: $5n = 90$, so $n = 18$, a positive integer. Yes, 92 is the 18th term.

5 First differences $5, 7, 9$; second difference 2, so $a = 1$. Subtracting n^2 leaves $2, 4, 6, 8 = 2n$, so $u_n = n^2 + 2n$.

6 $d = -3$ and $u_1 = 12$, so $u_n = 12 - 3(n - 1) = 15 - 3n$.

7 First differences $4, 6, 8, 10$; second differences $2, 2, 2$.

8 Second difference 2, so $a = 1$; subtracting n^2 leaves $1, 2, 3, 4 = n$, so $u_n = n^2 + n$.

9 $u_1 = 5$, $u_2 = 3(5) - 2 = 13$, $u_3 = 37$, $u_4 = 109$.

10 The terms are $2, 3, 5, 8, 13, 21$, so $u_6 = 21$.

11 $4n + 7 = 63$, so $4n = 56$ and $n = 14$.

12 The first differences are $3, 5, 7$, which are not constant. Arithmetic sequences require a constant first difference; this sequence has constant second difference 2 and is quadratic, $u_n = n^2$.

13 At $n = 1$, $4n + 6 = 10$, not 6, so the constant is wrong. Since $d = 4$, $c = 6 - 4 = 2$, giving $u_n = 4n + 2$.

Common error: adding the first term instead of subtracting the common difference.

14 Design 12: $144 + 24 = 168$ tiles. For 120 tiles, $n^2 + 2n = 120$, so $(n + 12)(n - 10) = 0$ and $n = 10$: design 10.

15 The row numbers are triangular, so row 15 has $T_{15} = \dfrac{15(16)}{2} = 120$ cans.

16 $250 + 40n > 1{,}000$, so $40n > 750$ and $n > 18.75$. Week 19 is the first, with $250 + 760 = £1{,}010$; week 18 gives £970.

17 The second difference is 8, so $2a = 8$ and $a = 4$. Using $u_1 = 3$ and $u_2 = 14$: $4 + b + c = 3$ and $16 + 2b + c = 14$. Subtracting gives $12 + b = 11$, so $b = -1$, then $c = 0$. Hence $u_n = 4n^2 - n$.

18 For $u_n = n^2 + 3n + 2$, the first differences are $2n + 4$, namely $6, 8, 10, \ldots$, and the second difference is 2. Solve $n^2 + 3n + 2 > 500$: $n = 20$ gives 462 and $n = 21$ gives 506, so the first term greater than 500 is 506.

Criterion A · Knowing and understanding

1 $d = 12 - 7 = 5$, and $c = 7 - 5 = 2$, so $u_n = 5n + 2$.

2 $u_{30} = 4(30) - 3 = 120 - 3 = 117$.

3 $d = 17 - 20 = -3$, and $c = 20 - (-3) = 23$, so $u_n = 23 - 3n$.

4 First differences $4, 8, 12$; second difference 4, so $a = 2$. Subtracting $2n^2$ leaves $-1, -3, -5, -7 = 1 - 2n$, so $u_n = 2n^2 - 2n + 1$.

5 First differences $6, 10, 14$; second differences $4, 4$. The constant second difference 4 means the sequence is quadratic, with $2a = 4$, so $a = 2$.

6 The 4th and 9th terms differ by $38 - 18 = 20$ over 5 steps, so $d = 4$. Then $18 = 4(4) + c$, so $c = 2$, and $u_n = 4n + 2$.

7 $u_2 = 2(4) + 1 = 9$, $u_3 = 19$, $u_4 = 39$, $u_5 = 79$.

8 An arithmetic sequence has constant first differences, so all its second differences are zero. A non-zero constant second difference means the first differences are changing, so the sequence cannot be arithmetic; it is quadratic.

9 Design 15: $15^2 + 3(15) = 225 + 45 = 270$ tiles. For 180 tiles, $n^2 + 3n = 180$, so $(n + 15)(n - 12) = 0$ and $n = 12$: design 12.

Marking guidance. Award method marks for a correct difference or pair of equations, and the accuracy mark only for a fully correct rule. Q1: 1 mark for d, 1 for c, 1 for the rule. Q4 and Q6: 2 marks for the difference or simultaneous equations, 2 for the completed rule. Q8 requires both parts: zero second differences for arithmetic, and a changing first difference for a non-zero constant second difference. Accept equivalent forms such as $u_n = 5(n + 1) - 3$.

Criterion B · Investigating patterns

1 $M_1 = 4$, $M_2 = 12$, $M_3 = 24$, $M_4 = 40$. A correct table records n against M_n and shows the four counts; a sketch of the 2×2 and 3×3 grids supports the counts.

2 First differences $8, 12, 16$; second differences $4, 4$. The first differences increase by 4 each time, so the sequence is quadratic with $2a = 4$, giving $a = 2$.

3 The grid has $n + 1$ horizontal lines, each of n matches, and the same number of vertical lines, so $M_n = 2n(n + 1) = 2n^2 + 2n$. For $n = 4$: $2(16) + 2(4) = 40$, matching the count.

4 $M_{12} = 2(12)(13) = 312$. From side $n - 1$ to side n, one new horizontal and one new vertical line each add n matches, and each of the n existing lines in both directions gains one match: $2n + 2n = 4n$. This is exactly the first difference of $2n^2 + 2n$, and the same structural argument applies at every step, so the formula continues to hold.

5 The perimeter has n matches on each of four sides, so $P_n = 4n$, which is linear. The total count includes every internal and boundary match: $n + 1$ horizontal lines of n matches plus $n + 1$ vertical lines of n matches, giving $2n(n + 1)$. Both the number of lines and the length of each line grow with n, so their product grows like n^2; the perimeter counts only one line around the outside, so it grows like n.

Marking guidance. 24 marks: Q1 6 (four correct counts, 1 each; systematic table and description, 2), Q2 4 (first differences 2, second differences and conclusion 2), Q3 5 (formula 3, verification at $n = 4$ 2), Q4 4 (value 2, structural reason 2), Q5 5 (perimeter linear 2, quadratic explanation 3). A model response should move from counted values to differences, then to $M_n = 2n(n + 1)$, then justify the $4n$ increment structurally. Full marks require the structural explanation, not merely a numerical pattern.

Criterion C · Communicating

→ (i) Arithmetic sequences have a constant first difference; geometric sequences have a constant ratio; quadratic sequences have a constant non-zero second difference. For example, $4, 9, 14, 19$ is arithmetic ($d = 5$), $3, 6, 12, 24$ is geometric ($r = 2$), and $3, 8, 15, 24$ is quadratic (second difference 2).

(ii) For $4, 9, 14, 19$, $d = 5$ and $c = 4 - 5 = -1$, so $u_n = 5n - 1$. For $3, 8, 15, 24$, the first differences are $5, 7, 9$ and the second difference is 2, so $a = 1$; subtracting n^2 leaves $2, 4, 6, 8 = 2n$, so $u_n = n^2 + 2n$.

→ (iii) A difference table should show the terms $3, 8, 15, 24$, first differences $5, 7, 9$, and second differences $2, 2$. A labelled sketch plots $(1, 3), (2, 8), (3, 15), (4, 24)$ and shows them curving upward, not lying on a straight line.

(iv) Checking $n = 1$ is essential because the constant in $u_n = dn + c$ is the value at $n = 0$, not the first term. A rule that fits later terms can still fail at the first term; substituting $n = 1$ catches this immediately.

(v) A common error is writing $u_n = 5n + 4$ for $4, 9, 14, \ldots$, using the first term as the constant. Avoid it by using $u_n = u_1 + (n - 1)d$, or by testing the proposed rule at $n = 1$.

Marking guidance. 20 marks: (i) 4, (ii) 5, (iii) 4, (iv) 3, (v) 2, communication 2. Award (i) for all three classifications with the correct diagnostic; (ii) for both rules with working; (iii) for a correct table and a sensible labelled sketch; (iv) for explaining the $n = 0$ constant and the check; (v) for a specific error and a workable avoidance strategy. The response should be continuous prose, use u_n notation correctly, and be pitched at a student who missed the lessons.

Criterion D · Applying mathematics in real-life contexts

1 $d_8 = 4(8) + 6 = 38$ docks.

2 $d_n = 10 + 4(n - 1) = 4n + 6$.

3 $4n + 6 = 70$, so $4n = 64$ and $n = 16$. Row 16 has 70 docks.

4 The totals are $10, 24, 42, 64, \ldots$, with first differences $14, 18, 22$ and second difference 4. Hence $a = 2$; subtracting $2n^2$ leaves $8, 16, 24, 32 = 8n$, so $S_n = 2n^2 + 8n$. For 12 rows, $S_{12} = 2(144) + 96 = 384$ docks.

5 Solve $2n^2 + 8n \leq 500$, or $n^2 + 4n - 250 \leq 0$. $S_{13} = 338 + 104 = 442$ and $S_{14} = 392 + 112 = 504$, so 13 complete rows fit. Row 14 alone needs 62 docks, but only $500 - 442 = 58$ remain, so a 14th row cannot be completed.

6 $S_{20} = 2(400) + 160 = 960$ docks, not 1,000; the claim overstates the total by 40. Assumptions include a constant increase of 4 docks per row, no gaps or access aisles, every row fitting the site, and every dock being usable.

Marking guidance. 24 marks: Q1 4, Q2 4, Q3 4, Q4 5, Q5 4, Q6 3. Award method marks for the linear rule and the quadratic total, and accuracy marks for the correct values. Q5 must show the integer constraint and compare S_{13} with S_{14}, not merely round a decimal. Q6 full marks require both the numerical check and two genuine assumptions. A model response should state $d_n = 4n + 6$, derive $S_n = 2n^2 + 8n$, then interpret the 500-dock limit in whole rows.

CHAPTER 8 · GROWTH AND FINANCIAL MATHEMATICS

Opening problem · the long game

1 After n years the fund is $2{,}500 \times 1.06^n$: year 0 £2,500; year 1 £2,650; year 2 £2,809; year 3 £2,977.54; year 4 £3,156.19; year 5 £3,345.56; year 6 £3,546.30.

2 The rate is charged on the balance at the start of each year: new amount = old + $0.06\times$ old = $1.06\times$ old. It is £150 more only in year 1; later the yearly increases are larger — £159 in year 2, £168.54 in year 3 — because 6% of a bigger balance is a bigger amount.

3 End of year 5: £3,345.56, which is below £3,500; end of year 6: £3,546.30, which is above. So year 6. Algebraically: $1.06^n > \dfrac{3{,}500}{2{,}500} = 1.4$ gives $n > \log 1.4 \div \log 1.06 = 5.77$, so the first whole year is $n = 6$.

4 Simple account: $2{,}500 + 6 \times 150 = £3{,}400$. Compound: £3,546.30 — compound is £146.30 more. The difference arises because compound interest pays 6% on the growing balance (interest earns interest), while the simple account pays 6% of the original £2,500 every year.

5 The multiplier is $1 - 0.12 = 0.88$: $V = 2{,}500 \times 0.88^3 = 2{,}500 \times 0.681472 = £1{,}703.68$ after 3 years. Below £1,500 needs $0.88^n < \dfrac{1{,}500}{2{,}500} = 0.6$, i.e. $n > \log 0.6 \div \log 0.88 = 3.996$: year 4, when $V = 2{,}500 \times 0.88^4 = £1{,}499.24$.

6 The multipliers are 1.06 (fund) and 0.88 (equipment), whose product is $1.06 \times 0.88 = 0.9328$: a 6% gain followed by a 12% loss on the same amount is a fall of 6.72%, not 6% and not zero. Over the two holdings after year 1: $£2{,}650 + £2{,}200 = £4{,}850$, a 3% fall from £5,000. Percentages are always percentages of their own base amount, and the bases change each year, so changes combine by multiplying multipliers, not by adding percentages — the claim fails.

Prerequisite check

1 15% of 80 is $0.15 \times 80 = 12$.

2 $240 \times 1.05 = 252$.

3 $60 \times 0.88 = 52.80$.

4 $1.08^2 = 1.1664$.

5 $2^5 = 32$, so $n = 5$.

6 $0.85 = 1 - 0.15$, so this is a decrease of 15%.

Exercise 8A

1 (a) 1.07.
(b) 0.82.
(c) 1.025.
(d) 0.995.

2 The increase gives $240 \times 1.15 = 276$. The decrease gives $240 \times 0.85 = 204$. Each change is 15% of the same £240, so one adds £36 and the other subtracts £36; the results differ by £72. The multipliers 1.15 and 0.85 are not reciprocals.

3 $85 \times 1.08 = 91.80$, then $91.80 \times 0.90 = 82.62$. The combined multiplier is $1.08 \times 0.90 = 0.972$, a decrease of 2.8%. The final price is £82.62.

4 $18{,}000 \times 1.018^2 = 18{,}000 \times 1.036324 = 18{,}653.83$, so the population is 18,654 to the nearest person.

5 $0.80 \times 1.25 = 1.00$, so the final price equals the original. The 25% rise is taken on the reduced price, which is 80% of the original; 25% of that 80% is 20% of the original, exactly restoring the loss.

6 A 30% rise then a 30% fall gives $1.30 \times 0.70 = 0.91$, a 9% fall. A 25% fall then a 25% rise gives $0.75 \times 1.25 = 0.9375$, a 6.25% fall. The second result is larger: 93.75% of the original against 91%.

7 $1.10 \times 0.90 = 0.99$, so the value is 1% below the start, not equal to it. The 10% fall is calculated on the increased amount: 10% of 110 is 11, not 10. Starting at 100 gives $100 \to 110 \to 99$.
Common error: adding +10% and −10% as if they acted on the same base.

8 The hourly multiplier is $1 - 0.04 = 0.96$. After 6 hours, $0.96^6 = 0.7828$, so about 78.3% of the charge remains.

9 The multipliers are 0.70 and 0.80, so the combined multiplier is 0.56, a 44% discount, not 50%. The second 20% is taken from the already reduced price, so it removes only 20% of 70%, not 20% of the original.

10 Two increases of r% give $\left(1 + \dfrac{r}{100}\right)^2 = 1 + \dfrac{2r}{100} + \dfrac{r^2}{10{,}000}$. The percentage increase is therefore $2r + \dfrac{r^2}{100}$. The extra $\dfrac{r^2}{100}$ comes from the second increase being calculated on the first increase as well as on the original amount.

Exercise 8B

1 The multiplier is 1.05 and there are 3 years: $2{,}000 \times 1.05^3 = 2{,}000 \times 1.157625 = £2{,}315.25$. Interest earned: £315.25.

2 The multiplier is 1.032 and there are 6 years: $5{,}000 \times 1.032^6 = 5{,}000 \times 1.208031 = £6{,}040.16$. Interest earned: £1,040.16.

3 The multiplier is 1.028 and there are 4 years: $1{,}200 \times 1.028^4 = 1{,}200 \times 1.116793 = £1{,}340.15$. Interest earned: $1{,}340.15 - 1{,}200 = £140.15$.

4 Simple: $850(1 + 3 \times 0.035) = 850 \times 1.105 = £939.25$. Compound: $850 \times 1.035^3 = £942.41$. Compound interest is greater by £3.16.

5 The multiplier is 1.042 and there are 4 years: $3{,}200 \times 1.042^4 = 3{,}200 \times 1.178883 = £3{,}772.43$. Interest earned: £572.43.

6 Year 1: £2,120.00; year 2: £2,247.20; year 3: £2,382.03; year 4: £2,524.95. The balance first exceeds £2,500 after 4 complete years.

7 The monthly multiplier is 1.012 and there are 12 months: $1{,}500 \times 1.012^{12} = 1{,}500 \times 1.153894 = £1{,}730.84$. Interest earned: £230.84.

8 $1.06^{12} = 2.0122$, which is just over 2. So the investment doubles in 12 years, and the rule of 72 gives an accurate estimate at 6%.

9 $3000 \times 1.05 \times 8$ treats the 5% as a single multiplication and then scales by 8; that is neither simple nor compound interest. The correct expression is 3000×1.05^8, which gives £4,432.37.

Common error: using the exponent as a multiplier instead of a power.

10 $1.015^{12} = 1.195618$, so the effective annual rate is 19.56%. It exceeds $12 \times 1.5\% = 18\%$ because each month's interest earns interest in later months; the compounding adds about 1.56 percentage points.

Exercise 8C

1 The multiplier is $1 - 0.15 = 0.85$ and there are 3 years: $16{,}000 \times 0.85^3 = 16{,}000 \times 0.614125 = £9{,}826.00$. The loss is £6,174.00.

2 The multiplier is $1 - 0.25 = 0.75$ and there are 2 years: $1{,}200 \times 0.75^2 = 1{,}200 \times 0.5625 = £675.00$. The loss is £525.00.

3 $P \times 0.9^2 = 8{,}100$, so $P = \dfrac{8{,}100}{0.81} = £10{,}000$.

4 A 12% discount leaves 88%, so the original price is $\dfrac{66}{0.88} = £75$.

5 The price including 20% tax has multiplier 1.20, so the price before tax is $\dfrac{756}{1.20} = £630$.

6 Five years earlier means reversing five annual decreases: $\dfrac{24{,}000}{0.98^5} = \dfrac{24{,}000}{0.903921} = 26{,}551$ people, to the nearest person.

7 The salary before the rise is $\dfrac{31{,}200}{1.04} = £30{,}000$.

8 The combined multiplier is $0.75 \times 0.88 = 0.66$, so the original price is $\dfrac{66}{0.66} = £100$.

9 The student's method uses the wrong base. Subtracting 20% of the final value gives 80% of the final value, not the original. If the final value is 120, the original is $120 \div 1.20 = 100$; subtracting 20% of 120 gives 96, which is wrong.

Common error: reversing an increase by subtracting the percentage instead of dividing by the multiplier.

10 With annual rate r as a decimal, $5{,}000(1 + r)^2 = 5{,}408$, so $(1 + r)^2 = 1.0816$, $1 + r = 1.04$ and $r = 0.04$. The annual rate is 4%. Check: $5{,}000 \times 1.04^2 = 5{,}408$.

Exercise 8D

1 Monthly: $1.004^{12} = 1.049070$, an effective rate of 4.91%, giving £1,258.88 on £1,200. Annual: 1.05, giving £1,260.00. The 5% annual offer is slightly better.

2 Buying: $560 + 9 \times 24 = 560 + 216 = £776$. Contract: $32 \times 24 = £768$. The contract is cheaper by £8 over 24 months.

3 Offer A: interest $9{,}000 \times 0.06 \times 3 = £1{,}620$, total £10,620, monthly $\dfrac{10{,}620}{36} = £295$. Offer B: fee $9{,}000 \times 0.04 = £360$, total £9,360, monthly £260. B is cheaper by £1,260 in total and £35 per month.

4 Offer A: $3{,}000 \times 1.042^5 = £3{,}685.19$. Offer B: $3{,}150 \times 1.039^5 = £3{,}814.07$. B is ahead by £128.88 after 5 years.

5 $1.003^{12} = 1.036600$, an effective annual rate of 3.66%. The 3.7% annual account is higher, because 3.7% is greater than 3.66%.

6 Total paid: $24 \times 45 = £1{,}080$. Cost of credit: $1{,}080 - 1{,}000 = £80$. As a percentage of the purchase price: $\dfrac{80}{1{,}000} \times 100\% = 8\%$.

7 A lower monthly payment may hide a longer term, so the total paid can be larger even though each payment is smaller. Two quantities to compare are the total cost of credit and the effective annual rate (APR); fees and the deposit should also be included.

8 Three needed pieces: the amount borrowed, the repayment term, and whether the rate is flat or compound (and how often it is applied). Fees, a deposit, and the repayment schedule are also needed for an exact total.

9 Year 1: A £4,160.00, B £4,206.16. Year 2: A £4,326.40, B £4,357.58. Year 3: A £4,499.46, B £4,514.45. Year 4: A £4,679.43, B £4,676.98. A overtakes B in year 4, by £2.45.

Exercise 8E

1 Views double every 4 days, so $V = 800 \times 2^{t/4}$. At day 24, $V = 800 \times 2^6 = 800 \times 64 = 51{,}200$.

2 The real multiplier is $\dfrac{1.04}{1.03} = 1.009709$, about $+0.97\%$ per year. After 12 years, $1.009709^{12} = 1.1229$, a real increase of about 12.3%.

3 Rule of 72: $72 \div 2.5 = 28.8$ years. Check powers: $1.025^{28} = 1.9965$ and $1.025^{29} = 2.0464$. The first whole year at least double is year 29.

4 $1.08^{25} = 6.8485$, so prices would grow by a factor of about 6.85, or 585%. The claim may fail because 8% cannot continue indefinitely: incomes, interest rates, supply and demand change, and house prices vary by region.

5 Any three: the rate stays constant; interest is compounded once per year at that rate; no money is added or withdrawn; no tax or fees are charged; the rate is an annual rate, not a monthly one.

6 A constant percentage of a growing number requires ever-larger absolute increases. Once nearly all potential customers own the product, only replacement purchases remain, so sales must level off; the market is saturated and the exponential model over-predicts.

7 If prices double in 10 years, $(1+r)^{10}=2$, so $r=2^{0.1}-1=0.0718$, about 7.18% per year. This is mathematically consistent, but a sustained 7.2% annual rise for decades is high; it requires incomes and demand to keep pace, so the claim is optimistic rather than impossible.

8 $1{,}500\times1.22^2=1{,}500\times1.4884=£2{,}232.60$. Interest added: $2{,}232.60-1{,}500=£732.60$.

9 Linear: $1{,}000+200n$. Exponential: $1{,}000\times1.12^n$. At year 9, linear gives £2,800 and exponential £2,773.08; at year 10, exponential gives £3,105.85 against £3,000. The exponential model is first larger in year 10.

Investigation · How good is the rule of 72?

1 $1.06^{12}=2.0122$, so the first whole year is 12. $1.08^9=1.9990$ and $1.08^{10}=2.1589$, so the first whole year is 10. $1.12^6=1.9738$ and $1.12^7=2.2107$, so the first whole year is 7.

2 The differences are $0, 0, +0.6, 0, +1.0, +0.8, +1.0$. They are small and non-negative: $72\div r$ is at most one year below the first whole year that actually doubles. The gap tends to grow as the rate rises, because the approximation is weaker at higher rates.

3 Exact doubling times to one decimal place: 23.4, 17.7, 9.0, 7.3 and 6.1 years for 3%, 4%, 8%, 10% and 12%. At 3%: $70/3=23.3$ and $72/3=24.0$, so 70 is closer. At 4%: 17.5 and 18.0, so 70 is closer. At 8%: 8.75 and 9.0, so 72 is closer. At 10%: 7.0 and 7.2, so 72 is closer. At 12%: 5.83 and 6.0, so 72 is closer. Thus 70 suits 3–4% and 72 suits 8–12%.

4 $0.9^6=0.5314$ and $0.9^7=0.4783$, so the asset is first worth less than half after 7 whole years. The rule gives $72\div10=7.2$ years. The exact continuous halving time is $\ln0.5\div\ln0.9\approx6.58$ years, so the rule slightly overestimates, though it still rounds to the same first whole year, 7.

5 For small r, the yearly multiplier is $1+r$, and $(1+r)^n\approx e^{rn}$. Doubling needs $e^{rn}=2$, so $rn\approx\ln2=0.693$ and $n\approx\dfrac{69.3}{r}$. The doubling time is therefore roughly inversely proportional to the rate.

6 At 50%, the rule gives $72/50=1.44$ years, but $1.5^1=1.5$ and $1.5^2=2.25$, so the first whole year is 2; the exact continuous time is 1.71 years. The rule fails because $\ln(1+r)\approx r$ is poor at large r. It is useful roughly for rates from 2% to 15%, and best near 5–10%.

→ A doubling time of n years at rate r satisfies $(1+r)^n=2$, so $n\ln(1+r)=\ln2$ and $n=\dfrac{\ln2}{\ln(1+r)}$. For small r, $\ln(1+r)\approx r$, giving $n\approx\dfrac{0.693}{r}=\dfrac{69.3}{r}$. The constant is near 70, not 100, because 100 would imply that 1% doubles in 100 years, but $1.01^{100}=2.70$, not 2; the correct constant is $100\ln2\approx69.3$.

Mixed practice 8

1 1.065.

2 $350\times1.12=£392$.

3 $480\times0.65=£312$.

4 $2{,}500\times1.04^3=2{,}500\times1.124864=£2{,}812.16$.

5 $7{,}500\times1.025^4=7{,}500\times1.103813=£8{,}278.60$.

6 $12{,}000\times0.82^2=12{,}000\times0.6724=£8{,}068.80$.

7 $\dfrac{96}{0.75}=£128$.

8 $\dfrac{66}{1.20}=£55$.

9 Simple: $5{,}000(1+5\times0.05)=£6{,}250$, interest £1,250. Compound: $5{,}000\times1.05^5=£6{,}381.41$, interest £1,381.41. Compound interest is greater by £131.41.

10 $3{,}000\times1.045^n>3{,}600$, so $1.045^n>1.2$. $1.045^4=1.1925$ and $1.045^5=1.2462$, so 5 complete years are needed.

11 $1.15\times0.85=0.9775$, a decrease of 2.25%.

12 Compound interest multiplies by 1.05 ten times: $1.05^{10}=1.6289$, an increase of 62.89%, not 50%. The 50% figure is simple interest. The correct multiplier is 1.6289.

Common error: adding the annual rate for each year instead of raising the multiplier to a power.

13 Compound at 4%: $2{,}000\times1.04^6=£2{,}530.64$. Simple at 4.4%: $2{,}000(1+6\times0.044)=£2{,}528.00$. Compound gives £2.64 more.

14 Real multiplier: $\dfrac{1.035}{1.028}=1.006809$, a real increase of about 0.68%.

15 $0.80\times0.85=0.68$, an overall loss of 32%.

16 Total paid: $36\times245=£8{,}820$. Cost of credit: $8{,}820-8{,}000=£820$.

17 $(1+r)^{12}=2$, so $r=2^{1/12}-1=0.059463$, about 5.95% per year.

18 $1.045^{15}=1.9353$ and $1.045^{16}=2.0224$, so the first whole year is 16. The exact continuous doubling time is $\dfrac{\ln2}{\ln1.045}=15.75$ years, slightly less than 16 because $72\div4.5$ overestimates: 72 is slightly above $100\ln2\approx69.3$, and $\ln1.045$ is slightly below 0.045.

Criterion A · Knowing and understanding

1 (a) 1.045.

(b) 0.88.

(c) 1.008.

(d) 0.70.

2 The multiplier is 1.035 and there are 6 years: $4{,}500 \times 1.035^6 = 4{,}500 \times 1.229255 = £5{,}531.65$. Interest earned: £1,031.65.

3 The multiplier is 0.85 and there are 3 years: $18{,}000 \times 0.85^3 = 18{,}000 \times 0.614125 = £11{,}054.25$. The loss is £6,945.75.

4 The multiplier is 1.18, so the original price is $\frac{29{,}500}{1.18} = £25{,}000$.

5 Simple: $2{,}000(1 + 4 \times 0.06) = £2{,}480$. Compound: $2{,}000 \times 1.06^4 = £2{,}524.95$. Compound is greater by £44.95.

6 Year 9: $3{,}000 \times 1.08^9 = £5{,}997.01$. Year 10: £6,476.78. The first complete year above £6,000 is year 10.

7 Two 20% increases give $1.20^2 = 1.44$, an increase of 44%, not 40%. The second 20% is taken on the increased amount, so it adds 20% of 120%, which is 24% of the original.

8 $0.9 \times 1.1 = 0.99$. The 10% decrease acts on the original amount; the 10% increase acts on the reduced amount, which is only 90% of the original. The increase is therefore 9% of the original, one percentage point less than the decrease.

> **Marking guidance.** 30 marks: Q1 4 (1 per multiplier), Q2 4, Q3 4, Q4 4, Q5 5, Q6 4, Q7 3, Q8 2. Award method marks for the multiplier and power, and accuracy marks for the final value. Q5 needs both values and the difference. Q6 requires a table or powers showing year 9 below and year 10 above £6,000. Q7 and Q8 require an explanation of the changing base, not just the numerical result.

Criterion B · Investigating patterns

1 Day 0: 1 m²; day 1: 2 m²; day 2: 4 m²; day 3: 8 m²; day 4: 16 m²; day 5: 32 m²; day 6: 64 m². Each day's area is twice the previous day's area, so the ratios are constant at 2.

2 $A_d = 1 \times 2^d = 2^d$. The patch starts at 1 m² and doubles once per day, so after d days it has been multiplied by 2 exactly d times.

3 The pond is full when $2^d = 1{,}024 = 2^{10}$, so day 10. It is half covered on day 9, with 512 m², because the area doubles from day 9 to day 10; the day before full must be half full.

4 Starting at 4 m² gives $A_d = 4 \times 2^d = 2^{d+2}$. The pond is full when $d + 2 = 10$, so day 8. The larger start is equivalent to two extra doublings, saving two days.

5 After the pond is full, the area cannot keep doubling because there is no uncovered water left; the model would predict more than 1,024 m², which is impossible. With space limiting the lily, the graph would start exponential but bend and level off at 1,024 m², forming an S-shaped (logistic) curve rather than continuing upward.

> **Marking guidance.** 24 marks: Q1 6 (seven correct values, 4; pattern description, 2), Q2 4 (rule 2, structural explanation 2), Q3 5 (full day 2, half day 2, justification 1), Q4 4 (rule 2, day 8 with reasoning 2), Q5 5 (ceiling 2, graph description 3). A model response should state $A_d = 2^d$, use $2^{10} = 1{,}024$, and justify day 9 by the doubling step. Full marks require the structural link between doubling and the exponent, not just the table.

Criterion C · Communicating

→ (i) Simple interest is paid only on the principal: £2,000 at 5% for 3 years earns $2{,}000 \times 0.05 \times 3 = £300$, giving £2,300. Compound interest is paid on the principal and on interest already added: $2{,}000 \times 1.05^3 = £2{,}315.25$, so it earns £315.25, which is £15.25 more.

(ii) A percentage change is a multiplier: $+5\%$ is 1.05 and -5% is 0.95. Successive changes multiply, so a 20% rise then a 10% rise gives $1.20 \times 1.10 = 1.32$, a 32% increase, not 30%. The second percentage acts on a larger base.

→ (iii) To reverse a 20% increase, divide by 1.20, not subtract 20%. A final value of £120 corresponds to an original of $120 \div 1.20 = £100$; subtracting 20% of 120 would give £96, which is too small.

(iv) Two assumptions: the rate stays constant for the whole term, and no money is added or withdrawn (and interest is reinvested at the same rate). The model also assumes no tax or fees and annual compounding.

(v) A common error is adding percentages or using $P(1 + nr)$ for compound interest. Avoid it by writing the multiplier and the number of periods first, then evaluating $P(1 + r)^n$.

Marking guidance. 20 marks: (i) 5, (ii) 5, (iii) 4, (iv) 3, (v) 2, communication 1. Award (i) for both calculations and the difference; (ii) for the multiplier method and a correct counterexample to adding; (iii) for dividing by 1.20 with a numerical check; (iv) for two genuine assumptions; (v) for a specific error and a workable avoidance strategy. The response should be continuous prose, define P, r and n, and keep currency units throughout.

Criterion D · Applying mathematics in real-life contexts

1. Purchase plus maintenance: $850 + 4 \times 60 = 850 + 240 = £1{,}090$. After resale: $1{,}090 - 300 = £790$.
2. Subscription: $25 \times 12 \times 4 = 25 \times 48 = £1{,}200$.
3. Buying costs £790 and subscribing costs £1,200, so buying is cheaper by £410 over four years. A non-financial factor is convenience: a subscription includes maintenance and repairs, while owning requires time, storage and repair arrangements; flexibility and the option to keep the bike after four years also matter.
4. The battery multiplier is $1 - 0.08 = 0.92$. After 4 years, $0.92^4 = 0.716393$, so about 71.6% of the original capacity remains.
5. Buying is the better financial choice: £790 against £1,200, a saving of £410, and the owner still has a bicycle worth about £300 at the end. A student keeping the bike for four years should buy, provided the battery's 71.6% capacity is enough for their journeys. Assumptions: maintenance stays at £60 per year, the £300 resale value is achieved, the subscription price stays at £25 per month, and no major repairs are needed.

Marking guidance. 24 marks: Q1 5 (purchase 1, maintenance 2, resale 2), Q2 4, Q3 5 (comparison 3, non-financial factor 2), Q4 4 (multiplier 2, value 2), Q5 6 (recommendation 2, use of both totals 2, two assumptions 2). Award full marks only when the comparison is like-for-like over four years and includes the resale value. Q5 should acknowledge the battery degradation as a possible limitation, not ignore it.

CHAPTER 9 · CONGRUENCE, SIMILARITY AND TRANSFORMATIONS

Opening problem · the same shape, a different size

1 $3.6\text{ m} = 360\text{ cm}$ and $5.4\text{ m} = 540\text{ cm}$. Then $360 \div 12 = 30$ and $540 \div 18 = 30$: both pairs of corresponding sides are multiplied by 30 and every angle is a right angle in both, so the mural is an enlargement of scale factor 30. The area is multiplied by $30^2 = 900$.

2 Area $= 3.6 \times 5.4 = 19.44\text{ m}^2$ (check: $12 \times 18 = 216\text{ cm}^2$ and $216 \times 900 = 194{,}400\text{ cm}^2 = 19.44\text{ m}^2$). Paint needed: $19.44 \div 4 = 4.86 \approx 4.9$ litres.

3 They are similar — all angles equal and corresponding sides in the same ratio — but not congruent, because congruent figures must have equal corresponding sides and here the sides differ by a factor of 30. One measurement that would confirm similarity: the ratio of two adjacent sides of the mural, $3.6 : 5.4 = 2 : 3$, which matches the photograph's $12 : 18 = 2 : 3$; for rectangles (all angles 90°) a matching side ratio is enough.

4 $\dfrac{h}{11.4} = \dfrac{2.0}{2.4}$, so $h = 11.4 \times \dfrac{2.0}{2.4} = 9.5\text{ m}$. Both triangles are right-angled (vertical object on level ground) and the sun's rays are parallel, so the angle of elevation at the tip of the shadow — between the shadow and the ray to the sun — is the same for stick and flagpole; the triangles are similar by AA, hence matching sides are proportional.

5 None of its measurements change: a 90° rotation is an isometry, so side lengths, angles and area are preserved; only the position and orientation change (the coordinates of the vertices and which way each edge faces). Translations, rotations and reflections — and any combination of them — all produce congruent copies of a tile; an enlargement (with factor not equal to 1) does not.

6 Doubling the longest side is an enlargement of scale factor 2, and every length is then multiplied by 2, so the printer's charge, which is proportional to a length, doubles. Area is multiplied by the square of the length factor, $2^2 = 4$, so the signwriter's charge, which is proportional to area, is multiplied by 4.

Prerequisite check

1 $x = 6 \times \dfrac{4}{3} = 8$. Check: $\dfrac{8}{6} = \dfrac{4}{3}$.

2 Divide both parts by 6: $12 : 18 = 2 : 3$.

3 $4 \times 7 = 28\text{ cm}^2$.

4 $3^2 = 9$, $2^3 = 8$, $\sqrt{49} = 7$.

5 $\sqrt[3]{64} = 4$, because $4^3 = 64$.

6 $y = 0$: every point on the x-axis has y-coordinate 0.

Exercise 9A · Isometries and congruence

1 Translation vector $(3, -4)$: 3 right and 4 down. Check: $(2 + 3,\ 3 - 4) = (5, -1)$.

2 Add $(-2, 3)$ to each vertex: $P'(-1, 4)$, $Q'(2, 4)$, $R'(-1, 6)$.

3 In the x-axis: $(3, 4) \mapsto (3, -4)$. Then in the y-axis: $(3, -4) \mapsto (-3, -4)$. Final image $(-3, -4)$.

4 Quarter-turn clockwise, $(x, y) \mapsto (y, -x)$: $(5, 2) \mapsto (2, -5)$.

5 Translation, rotation and reflection are isometries. Enlargement with factor 3 is not: every length is tripled, so the image is similar but not congruent.

6 Similar but not congruent. $\dfrac{6}{3} = \dfrac{8}{4} = \dfrac{10}{5} = 2$, so the angles match and the sides are in a constant ratio $k = 2$; the sides are not equal, so the triangles are not congruent.

7 The student has confused size with orientation. A reflection is an isometry: every length and angle is unchanged, and only the sense of the figure is reversed (clockwise becomes anticlockwise). Corrected: "A reflection produces a mirror image of the same size and shape."

8 A translation perpendicular to the two lines, through $2 \times 4 = 8$ cm: each point moves twice the gap, in the direction from the first line towards the second.

9 A rotation of 180° about the origin. $(x, y) \mapsto (x, -y) \mapsto (-x, -y)$, which is a half-turn; for example $(2, 3) \mapsto (2, -3) \mapsto (-2, -3)$.

10 A rotation about the point where the lines meet, through $2 \times 35^\circ = 70^\circ$. Measure a ray's direction ϕ from the first mirror. Reflection in the first line sends ϕ to $-\phi$. The second line is at 35°, so that reflection sends $-\phi$ to $2(35^\circ) - (-\phi) = 70^\circ + \phi$. Every direction turns by 70° and the intersection point is fixed, so the composition is a rotation of 70°.

Exercise 9B · Similar triangles and scale factor

1 Yes. Both triangles contain the angles 50°, 60° and 70°; two equal pairs are enough (AA), and the order of listing does not matter.

2 $k = \dfrac{4}{6} = \dfrac{2}{3}$, so the matching side is $9 \times \dfrac{2}{3} = 6$ cm.

3 $k = \dfrac{DE}{AB} = \dfrac{6}{8} = 0.75$, so $EF = 0.75 \times 12 = 9$.

4 Yes: $\dfrac{14}{10} = 1.4$ and $\dfrac{21}{15} = 1.4$. The scale factor is 1.4.

5 The angles of a triangle total 180°, so once two angles are known the third is fixed. If two pairs match, all three pairs match; triangles with the same angles have the same shape, so corresponding sides are in a constant ratio.

6 $AB = 3 + 5 = 8$ cm and $k = \dfrac{3}{8}$, so $BC = \dfrac{6}{3/8} = 16$ cm.

Common error: using $DB = 5$ in the ratio. DB is not a side of the small triangle; the matching side of the large triangle is AB.

7 The scale factor must compare corresponding sides: the longest side matches the longest, not the shortest. Matching 5 with 6 in the triples $3, 4, 5$ and $6, 8, 10$ gives 1.2 instead of the true factor 2. Writing $\triangle ABC \sim \triangle DEF$ names the pairs, so $AB \leftrightarrow DE$ and $BC \leftrightarrow EF$ without guessing.

8 $\frac{1.8}{2.4} = 0.75$, so the tree is $0.75 \times 9.6 = 7.2$ m high. Assumption: both shadows are measured at the same time, so the Sun's rays are parallel, and the ground is level.

9 The vertically opposite angles are equal. The parallel lines make a second pair equal (alternate angles). Two pairs of equal angles give AA, so the triangles are similar; the third pair then also matches.

10 $AD : DB = 2 : 3$, so $AD : AB = 2 : 5$ and $DE : BC = 2 : 5$. The area ratio is $k^2 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$, that is $4 : 25$.

Exercise 9C · Enlargements, centres and scale factors

1 Multiply each coordinate by 3: $(6, 3)$, $(12, 3)$, $(6, 9)$.

2 Multiply each coordinate by $\frac{1}{2}$: $(1, 0.5)$, $(2, 0.5)$, $(1, 1.5)$.

3 Multiply by -1: $(-2, -3)$, the half-turn image of $(2, 3)$ about the origin.

4 $k = \frac{12.5}{5} = 2.5$.

5 Add twice the vector from $(1, 1)$ to each vertex: $A(2, 1) \mapsto (3, 1)$, $B(4, 1) \mapsto (7, 1)$, $C(2, 3) \mapsto (3, 5)$.

6 $k^2 = \frac{108}{12} = 9$, so $k = 3$ (the length factor is 3).

7 On the line through each vertex and its image. Join A to A' and B to B', extending the lines if necessary; their intersection is the centre. For $k < 0$ the centre lies between a vertex and its image, but the construction is the same.

8 A factor of $\frac{1}{2}$ multiplies every length by $\frac{1}{2}$, so the image is half the size, with one quarter of the area. "Twice as large" would need $k = 2$. The correct image is smaller and lies between the centre and the original.

9 Every length is doubled, so the image is twice as large. It lies on the opposite side of the centre, with $OP' = 2\,OP$, and is turned through $180°$; sides stay parallel and the sense of the labelling is preserved.

10 $\vec{OA} = (-1, -2) \mapsto (2, 4)$, so $A'(4, 7)$; $\vec{OB} = (1, -2) \mapsto (-2, 4)$, so $B'(0, 7)$; $\vec{OC} = (-1, -1) \mapsto (2, 2)$, so $C'(4, 5)$. Then $AB = 2$ and $A'B' = 4$; $AC = 1$ and $A'C' = 2$; $BC = \sqrt{5}$ and $B'C' = \sqrt{20} = 2\sqrt{5}$.

Exercise 9D · Coordinate rules and composition

1 $(3 - 2,\ -1 + 5) = (1, 4)$.

2 Reflection in $y = x$ swaps coordinates: $(4, -3) \mapsto (-3, 4)$.

3 $(x, y) \mapsto (-x, -y)$: $(2, 5) \mapsto (-2, -5)$.

4 $(x, y) \mapsto (-y, -x)$: $(6, 2) \mapsto (-2, -6)$.

5 $(1.5 \times 3,\ 1.5 \times (-2)) = (4.5, -3)$.

6 Reflection in $y = -x$: $(x, y) \mapsto (-y, -x)$ sends $(2, 3)$ to $(-3, -2)$. The other rules fail: $90°$ anticlockwise gives $(-3, 2)$, $90°$ clockwise gives $(3, -2)$, $180°$ gives $(-2, -3)$.

7 A rotation of $180°$ about the origin: $(x, y) \mapsto (x, -y) \mapsto (-x, -y)$.

8 Clockwise then x-axis: $(1, 2) \mapsto (2, -1) \mapsto (2, 1)$. Reverse order: $(1, 2) \mapsto (1, -2) \mapsto (-2, -1)$. The images differ, so order matters. The two composites are reflections in $y = x$ and in $y = -x$ respectively.

9 It must lie on the mirror line, where the perpendicular distance to the line is zero. In $y = x$, fixed points include $(0, 0)$, $(2, 2)$ and $(-1, -1)$.

10 Subtract the centre: $(3, 1) - (1, 1) = (2, 0)$. Rotate clockwise to $(0, -2)$, then add the centre: $(1, -1)$. Method: translate the centre to the origin, apply the origin rule, translate back.

Exercise 9E · Area and volume scale factors

1 $k = \frac{7}{4}$, so the areas are in the ratio $k^2 = \frac{49}{16}$, that is $16 : 49$.

Common error: giving $4 : 7$, the length ratio, instead of the area ratio. An area ratio is always the square of the length ratio.

2 $k = \frac{6}{2} = 3$, so the volumes are in the ratio $3^3 = 27$, that is $1 : 27$.

3 Area factor $\left(\frac{1}{20}\right)^2 = \frac{1}{400}$, that is $1 : 400$; volume factor $\frac{1}{8000}$, that is $1 : 8000$.

4 $k^2 = 4^2 = 16$.

5 $k = \sqrt[3]{\frac{125}{8}} = \frac{5}{2} = 2.5$.

6 $V = lbh$. After doubling, $V' = (2l)(2b)(2h) = 8lbh = 8V$: each of the three length factors contributes a factor 2, and $2 \times 2 \times 2 = 8$.

7 Model area $\times 50^2 = 0.072 \times 2500 = 180$ m^2, and $180 \div 2500 = 0.072$. The area ratio is $1 : 2500$, as required.

8 Volume depends on the cube of a length: $(3s)^3 = 27s^3$, so the volume is 27 times as large, not 3 times. Tripling is the length factor; the volume factor is $3^3 = 27$.

9 $k = \sqrt{\frac{64}{25}} = \frac{8}{5}$, so the volumes are in the ratio $\left(\frac{8}{5}\right)^3 = \frac{512}{125}$, that is $125 : 512$.

10 $k = \sqrt[3]{27} = 3$. The height is multiplied by 3, like every length; the surface area is multiplied by $3^2 = 9$.

Investigation · How far can a shadow measure?

1 Every ratio is 0.833 (exactly $\frac{5}{6}$). Flagpole row: shadow $L = 11.4$ m, $\frac{h}{L} = 0.833$, height $H = \frac{5}{6} \times 11.4 = 9.5$ m.

2 The ratio $\frac{h}{L}$ is constant at 0.833, however tall the stick. Each object, its shadow and a sun ray form a right-angled triangle; the ray makes the same angle with the ground for every object, and the right angle is shared, so all the triangles are similar by AA and corresponding sides are in the same ratio.

3 $H = kL$ with $k = 0.833 = \frac{5}{6}$, so $H = \frac{5}{6}L$.

4 Tree: $H = \frac{5}{6} \times 9.6 = 8.0$ m. Flagpole: $H = \frac{5}{6} \times 11.4 = 9.5$ m. In the diagram draw each object vertical, its shadow horizontal and the ray as the hypotenuse; the angle at the ground is the Sun's elevation, equal in both triangles.

5 The Sun's elevation changes during the day, so the angle at the ground changes and the shadows lengthen or shorten; the ratio is only constant at one moment. Assumptions: the ground is level; the objects are vertical; the rays are parallel. Test the first with a spirit level along the shadow, the second with a plumb line, and the third by measuring two sticks at once and comparing their ratios.

6 An error of 0.1 m in the shadow gives $\Delta H = 0.833 \times 0.1 \approx 0.08$ m, so $H = 8.0 \pm 0.1$ m, about 1%. For a very tall object the percentage error stays similar if the shadow is measured to the same relative precision, but the absolute uncertainty grows and a long shadow crosses uneven ground, so in practice the method becomes less reliable.

→ Similar triangles give $\frac{H}{L} = \frac{h}{l}$, hence $\frac{H}{h} = \frac{L}{l}$ and $H = \frac{h}{l}L$. Parallel rays make the angle at the ground equal in both triangles; with equal right angles, AA applies. The graph of H against L is a straight line through the origin with gradient $\frac{h}{l} = 0.833$: height is proportional to shadow length.

Mixed practice 9

1 $(1 + 4,\ -2 + 5) = (5, 3)$.

2 In the y-axis: $(-3, 4) \mapsto (3, 4)$. Then in $y = x$: $(3, 4) \mapsto (4, 3)$.

3 $(x, y) \mapsto (-y, x)$: $(5, 2) \mapsto (-2, 5)$.

4 Not congruent: two sides match but the third sides differ (7 and 8). Not similar either, since $\frac{5}{5} = \frac{6}{6} = 1$ but $\frac{7}{8} \neq 1$, so the sides are not all in one ratio.

5 Similar. Each triangle contains the same three angles $50°$, $60°$ and $70°$, so two pairs match and AA applies; the order in which the angles are listed is irrelevant.

6 $k = \frac{12}{8} = 1.5$, so the matching side is $6 \times 1.5 = 9$ cm.

7 Area factor $3^2 = 9$; volume factor $3^3 = 27$.

8 $k = \sqrt[3]{\frac{343}{125}} = \frac{7}{5}$, so the surface areas are in the ratio $\frac{49}{25}$, that is $49 : 25$.

9 $\frac{1.6}{2.0} = 0.8$, so $H = 0.8 \times 25 = 20$ m.

10 4 cm $\to 4 \times 25\,000 = 100\,000$ cm $= 1$ km; 6 cm $\to 1.5$ km. Area $= 1 \times 1.5 = 1.5$ km^2.

11 Doubling both dimensions multiplies the area by $2^2 = 4$: a 2×3 rectangle has area 6, its double has area $4 \times 6 = 24$. The correct factor is 4, not 2.

12 $(x, y) \mapsto (x, -y) \mapsto (-y, -x)$, which is a reflection in $y = -x$. Test $(2, 3) \mapsto (2, -3) \mapsto (-3, -2)$; the direct reflection also gives $(-3, -2)$.

13 Assuming the centre is the origin, $P' = (-1.5 \times 2,\ -1.5 \times (-4)) = (-3, 6)$. It lies on the opposite side of the centre, 1.5 times as far away, with lengths multiplied by 1.5 and the figure turned through $180°$.

14 All three pairs share one ratio: $\frac{6}{4} = 1.5$ and $\frac{12}{8} = 1.5$, so $x = 1.5 \times 6 = 9$.

15 $80 \times 1.25 = 100$, then $100 \times 0.8 = 80$. Final price £80, an overall change of 0%, because $1.25 \times 0.8 = 1$.

16 $3x - 6 = 2x + 5$, so $3x - 2x = 5 + 6$ and $x = 11$.

17 Let the lines meet at O at an angle θ, with directions α and $\alpha + \theta$ from a fixed axis. A ray in direction ϕ reflects to $2\alpha - \phi$ in the first line, then to $2(\alpha + \theta) - (2\alpha - \phi) = \phi + 2\theta$ in the second. Every direction turns by 2θ, and O is fixed (it lies on both lines), so the composition is a rotation about O through 2θ.

18 $\angle AEB = \angle CED$ (vertically opposite); $\angle ABE = \angle CDE$ and $\angle BAE = \angle DCE$ (alternate angles, $AB \parallel DC$). Two equal pairs give AA, so $\triangle ABE \sim \triangle CDE$. With $k = \frac{DC}{AB} = \frac{8}{12} = \frac{2}{3}$, the areas are in the ratio $k^2 = \frac{4}{9}$, that is $9 : 4$ for $ABE : CDE$.

Criterion A · Knowing and understanding

1 Translation vector $(3, -4)$, since $5 - 2 = 3$ and $-1 - 3 = -4$. [2]

2 In the x-axis: $(3, -2) \mapsto (3, 2)$. In the y-axis: $(3, -2) \mapsto (-3, -2)$. [3]

3 $(x, y) \mapsto (y, -x)$, so $(4, 1) \mapsto (1, -4)$. [2]

4 (a) Similar but not congruent: all angles are $90°$ and the sides are in the ratio $2 : 1$, but the side lengths differ. (b) Congruent (SSS), and therefore also similar. (c) Congruent: equal squares of side 4 cm. [6]

5 $k = \frac{DE}{AB} = \frac{6}{9} = \frac{2}{3}$, so $EF = \frac{2}{3} \times 12 = 8$ cm. [4]

6 Area factor $= 2.5^2 = 6.25$, so the new area is $48 \times 6.25 = 300$ cm^2. Check: $20 \div 8 = 2.5$. [4]

7 $k = \sqrt[3]{\frac{64}{27}} = \frac{4}{3}$, so the heights are in the ratio $3 : 4$ and the surface areas in the ratio $\left(\frac{4}{3}\right)^2 = \frac{16}{9}$, that is $9 : 16$. [5]

8 The 2×3 rectangle has area $6\ \text{cm}^2$; doubling gives 4×6, area $24\ \text{cm}^2$. The area is multiplied by 4, the square of the length factor, not by 2. [4]

Criterion B · Investigating patterns

1 Stage 0: $3, 4, 5$, perimeter $12\ \text{cm}$, area $6\ \text{cm}^2$. Stage 1: $6, 8, 10$, perimeter $24\ \text{cm}$, area $24\ \text{cm}^2$. Stage 2: $12, 16, 20$, perimeter $48\ \text{cm}$, area $96\ \text{cm}^2$. Stage 3: $24, 32, 40$, perimeter $96\ \text{cm}$, area $384\ \text{cm}^2$. [6]

2 $P_n = 12 \times 2^n$. The perimeter is a length, and each stage doubles every side, so the perimeter doubles: a geometric sequence with first term 12 and ratio 2. [4]

3 $A_n = 6 \times 4^n$. Each stage is an enlargement with $k = 2$, and area scales by $k^2 = 4$, so the area is multiplied by 4 every stage, starting from $6\ \text{cm}^2$. [5]

4 Area $= \frac{1}{2} \times \text{base} \times \text{height}$. Both the base and the height are lengths, so both double; the area gains a factor $2 \times 2 = 4$. Equivalently, a unit square of the grid becomes four squares when every length is doubled. [5]

5 Solve $6 \times 4^n > 1000$, so $4^n > 166.7$. Now $4^3 = 64$ and $4^4 = 256$: $A_3 = 384\ \text{cm}^2$, which is below 1000, and $A_4 = 6 \times 256 = 1536\ \text{cm}^2$, which is above it. The first such stage is stage 4. [4]

→ **Model response.** The table doubles both legs each stage, so the perimeter doubles (12, 24, 48, 96) while the area quadruples (6, 24, 96, 384). Because $P_n = 12 \times 2^n$ and $A_n = 6 \times 4^n = 6 \times 2^{2n}$, the area is the square of the length scale, reaching $1536\ \text{cm}^2$ at stage 4, the first above 1000.

Marking guidance (24). Table: 6 (1 per stage, plus 2 for correct perimeter and area columns). Perimeter rule: 4 (2 for 12×2^n, 2 for linking it to a length doubling). Area rule: 5 (3 for 6×4^n, 2 for the area factor $k^2 = 4$). Explanation of the factor 4: 5 (3 for the base–height argument, 2 for the link to k^2). First stage above 1000: 4 (2 for correct trial values, 2 for the justified conclusion). Accept $A_n = 6 \times 2^{2n}$ or 6×4^n; a consistent table with one arithmetic slip can still earn full method marks.

Criterion C · Communicating

i Congruent: same shape and same size — all corresponding sides and angles equal, related by a translation, rotation or reflection. Similar: same shape but not necessarily the same size — corresponding angles equal and corresponding sides in a constant ratio k. Congruence is the case $k = 1$.

ii Congruent example: $A(1, 1)$, $B(4, 1)$, $C(1, 3)$ translated by $(5, 2)$ gives $A'(6, 3)$, $B'(9, 3)$, $C'(6, 5)$. Sides are $3, 2, \sqrt{13}$ in both. Similar example: enlarging the same triangle from O by $k = 2$ gives $A'(2, 2)$, $B'(8, 2)$, $C'(2, 6)$ with sides $6, 4, 2\sqrt{13}$ — twice as long, so similar but not congruent.

iii Take a 2×3 rectangle, area $6\ \text{cm}^2$. Apply $k = 2$: 4×6, area $24\ \text{cm}^2 = 4 \times 6$. In general $(ka)(kb) = k^2ab$: the k appears twice because area is a product of two lengths.

iv Common error: assuming area doubles when lengths double, or matching the longest side of one figure with the shortest of the other. Avoid it by writing the vertex order $ABC \sim DEF$, stating which side corresponds to which, and checking that every ratio equals the same k.

v A map or a scale model: the real building cannot be congruent to a drawing, but a $1 : 100$ plan is similar, so every length can be read off and every area converted with $k^2 = 10\,000$. Similarity also measures inaccessible heights from shadows, where congruence is impossible.

→ **Sample paragraph.** "Two shapes are congruent when one fits exactly on the other after sliding, turning or reflecting; they have equal sides and equal angles. They are similar when the angles still match but every length is multiplied by the same factor k. In the grid, triangle $A(1, 1)$, $B(4, 1)$, $C(1, 3)$ translated by $(5, 2)$ is congruent to its image, while the enlargement by 2 is only similar, with sides $6, 4, 2\sqrt{13}$. Area grows faster than length: doubling a 2×3 rectangle gives 4×6, four times the area, because both dimensions double. A common mistake is to double the area when the sides double; check by multiplying the two new sides. Maps and models rely on similarity, since a drawing can never be the same size as a building."

Marking guidance (20). Definitions with $k = 1$ case: 4. Coordinate example with correct images and side checks: 5 (congruent 2, similar 3). Rectangle derivation of k^2: 4. Error and avoidance: 3. Real context with reasoning: 2. Communication: 2 for precise language, notation and a conclusion suited to a student who missed the lessons. A response that quotes k^2 without the rectangle calculation loses the derivation marks.

Criterion D · Applying mathematics in real-life contexts

1 Divide each dimension by 50: length $60 \div 50 = 1.2$ m, width $3 \div 50 = 0.06$ m, thickness $0.3 \div 50 = 0.006$ m. In centimetres: 120 cm, 6 cm, 0.6 cm. [6]

2 Real deck: $60 \times 3 = 180$ m^2. Model deck: $1.2 \times 0.06 = 0.072$ m^2. Then $0.072 : 180 = 1 : 2500$, and $50^2 = 2500$, so the area ratio is k^2, as required. [5]

3 Volume factor $= \left(\frac{1}{50}\right)^3 = \frac{1}{125\,000}$. Real volume 54 m$^3 = 54\,000\,000$ cm^3, so the model contains $\frac{54\,000\,000}{125\,000} = 432$ cm^3. Check: $120 \times 6 \times 0.6 = 432$ cm^3. [5]

4 $432 \times 0.02 = 8.64$, so the material costs £8.64. [4]

5 Two limitations. (i) Strength does not scale with geometry: weight grows as k^3 but cross-sectional area only as k^2, so the model is relatively far stronger than the bridge and cannot reveal its failure load. (ii) Material and loading are simplified: concrete curing, reinforcement, joints, wind, traffic and thermal movement are absent, so a model that stands says little about safety. The model is reliable for shape, fit and appearance, not for structural strength. [4]

→ **Model response.** The model is 1.2 m $\times$ 0.06 m $\times$ 0.006 m, with deck area 0.072 m^2 against 180 m^2 (ratio $1 : 2500$) and volume 432 cm^3 costing £8.64. It checks proportions and can show whether the parts fit, but it cannot test strength, because doubling every length multiplies weight by 8 while strength depends on area, multiplied by only 4.

Marking guidance (24). Dimensions: 6 (1 per correct value in each unit system, with units). Areas and ratio: 5 (2 for each area, 1 for linking to 50^2). Volume in cm^3: 5 (1 for the factor, 2 for conversion, 2 for the value). Cost: 4 (3 for the multiplication, 1 for the unit). Evaluation: 4 (2 per limitation with its effect). A candidate who works in centimetres throughout and converts correctly should not be penalised; missing units cost at most 2 overall.

CHAPTER 10 · PYTHAGORAS, TRIGONOMETRY AND MEASUREMENT

Opening problem · three measurements, one triangle

1 The slope is the hypotenuse: $s = \sqrt{9.0^2 + 0.75^2} = \sqrt{81 + 0.5625} = \sqrt{81.5625} = 9.0312\text{ m} \approx 9.03\text{ m}$, i.e. 903 cm to the nearest centimetre.

2 Rise to run: $0.75 : 9 = 1 : 12$ (divide both parts by 0.75). As a percentage of the run: $\frac{0.75}{9} = 0.0833 = 8.33\%$, so the gradient is about 8.3%.

3 The rafter is the hypotenuse: $\sqrt{3.6^2 + 2.4^2} = \sqrt{12.96 + 5.76} = \sqrt{18.72} = 4.33\text{ m}$ (3 s.f.). For the pitch use $\tan = \frac{\text{opposite}}{\text{adjacent}}$ with opposite $= 2.4$ m (rise) and adjacent $= 3.6$ m (half-span): $\tan\theta = \frac{2.4}{3.6} = \frac{2}{3}$, so $\theta = \tan^{-1}\left(\frac{2}{3}\right) = 33.7^\circ$ (33.69°).

4 Angle of elevation 28°, adjacent $= 85$ m, so $\tan 28^\circ = \frac{\text{opposite}}{85}$: height above the instrument $= 85 \times \tan 28^\circ = 85 \times 0.53171 = 45.2$ m (3 s.f., from 45.195 m). Adding the instrument height: $45.195 + 1.5 = 46.7$ m above the ground (3 s.f.).

5 Questions 1–3 (the ramp length, its ratio, the rafter length and the pitch angle) need only the two given lengths, because the right angle between them is built into the situation — Pythagoras gives the third side and a ratio of the two known sides gives the angle. Question 4 needs the measured angle of elevation as well, since only one length is known. When an angle is available, the ratios sine, cosine and tangent connect the sides: the angle fixes the triangle's shape and the one known length fixes its scale, so one side plus one acute angle is as good as two sides for finding everything else.

6 The proposed gradient is exactly $1 : 12$, and the rule allows anything no steeper than $1 : 12$, so the ramp is acceptable — just at the limit. With run shortened to 6.0 m: $0.75 : 6 = 1 : 8$ (divide by 0.75), which is steeper than $1 : 12$, so it would not be acceptable.

Prerequisite check

1 $9^2 + 12^2 = 81 + 144 = 225$.

2 $\sqrt{169} = 13$, because $13^2 = 169$.

3 $x = 0.6 \times 9 = 5.4$. Check: $\frac{5.4}{9} = 0.6$.

4 $180^\circ - 90^\circ - 37^\circ = 53^\circ$.

5 0.0473: the leading zeros are not significant, so the first three significant figures are 4, 7 and 2; the next digit is 6, so the 2 rounds up to 3.

6 The hypotenuse; it is opposite the right angle and is the longest side.

Exercise 10A · Pythagoras' theorem

1 $c^2 = 6^2 + 8^2 = 100$, so $c = 10$ cm.

2 $c^2 = 25 + 144 = 169$, so $c = 13$ m.

3 $b^2 = 17^2 - 8^2 = 289 - 64 = 225$, so $b = 15$ cm.

4 $b^2 = 26^2 - 10^2 = 676 - 100 = 576$, so $b = 24$ m.

5 $c^2 = 49 + 576 = 625$, so $c = 25$ cm.

6 $c = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.41$ m.

7 $c = \sqrt{4 + 9} = \sqrt{13} \approx 3.61$ cm.

8 (a) $9^2 + 12^2 = 81 + 144 = 225 = 15^2$, so right-angled. (b) $7^2 + 8^2 = 49 + 64 = 113$, but $11^2 = 121$; not right-angled.

9 $d^2 = 18^2 + 24^2 = 324 + 576 = 900$, so $d = 30$ m.

10 $h^2 = 5^2 - 3^2 = 16$, so $h = 4$ m up the wall.

11 $d^2 = 48^2 + 64^2 = 2304 + 4096 = 6400$, so $d = 80$ cm. The diagonal is the hypotenuse of the right-angled triangle formed by two sides, so it is longer than either side.

12 The height bisects the base: $h^2 = 10^2 - 8^2 = 36$, so $h = 6$ cm. Area $= \frac{1}{2} \times 16 \times 6 = 48\text{ cm}^2$.

13 Differences 6 and 8: $d = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. Sketch a right triangle with horizontal leg 6 and vertical leg 8.

14 $c = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13} \approx 7.21$ cm.

Common error: writing $\sqrt{6^2 + 8^2} = 6 + 8$. A square root is not distributed over a sum; square first, add, then root.

15 $\sqrt{a^2 + b^2} \neq a + b$: the student found $6^2 + 8^2 = 100$ and then added the sides. The square root must be taken: $c = \sqrt{100} = 10$, not 14.

16 Let the sides be n, $n + 1$, $n + 2$. Then $n^2 + (n + 1)^2 = (n + 2)^2$, so $2n^2 + 2n + 1 = n^2 + 4n + 4$ and $n^2 - 2n - 3 = 0$. Hence $(n - 3)(n + 1) = 0$, and $n = 3$ since $n > 0$. The only triple is $3, 4, 5$.

17 $x^2 + (x + 7)^2 = (x + 8)^2$, so $2x^2 + 14x + 49 = x^2 + 16x + 64$ and $x^2 - 2x - 15 = 0$. Hence $(x - 5)(x + 3) = 0$, so $x = 5$ and the sides are 5, 12 and 13.

Exercise 10B · Three dimensions

1 $d^2 = 1 + 4 + 4 = 9$, so $d = 3$ cm.

2 $d^2 = 4 + 9 + 36 = 49$, so $d = 7$ cm.

3 $d^2 = 16 + 16 + 49 = 81$, so $d = 9$ m.

4 $d^2 = 25 + 25 + 100 = 150$, so $d = \sqrt{150} \approx 12.25$ cm.

5 $d^2 = 64 + 81 + 144 = 289$, so $d = 17$ cm.

6 $d^2 = 4 + 9 + 36 = 49$, so $d = 7$.

7 Differences $3, 4, 12$: $d^2 = 9 + 16 + 144 = 169$, so $d = 13$.

8 $d^2 = 36 + 12.25 + 5.76 = 54.01$, so $d \approx 7.35$ m.

9 $d^2 = 900 + 1600 + 14\,400 = 16\,900$, so $d = 130$ cm.

10 $d^2 = 3025 + 1600 + 400 = 5025$, so $d \approx 70.9$ cm. No: a rigid 75 cm umbrella is longer than the greatest possible straight line in the case.

11 $d = 5\sqrt{3} \approx 8.66$ cm.

12 The student added the height to the base diagonal instead of forming a second right-angled triangle. Correct: $d^2 = 4^2 + 3^2 + 2^2 = 29$, so $d = \sqrt{29} \approx 5.39$ cm.

13 Base diagonal squared $= 6^2 + 6^2 = 72$. Then $h^2 = 11^2 - 72 = 121 - 72 = 49$, so $h = 7$ cm.

14 Unfold two adjacent walls into one rectangle: the route crosses a horizontal distance $4 + 3 = 7$ m while rising 2.5 m. Shortest wall route $= \sqrt{49 + 6.25} = \sqrt{55.25} \approx 7.43$ m. The space diagonal is $\sqrt{16 + 9 + 6.25} = \sqrt{31.25} \approx 5.59$ m, so the wall route is about 1.8 m longer.

Exercise 10C · Trigonometric ratios

1 $\sin\theta = \frac{3}{5} = 0.6$, $\cos\theta = \frac{4}{5} = 0.8$, $\tan\theta = \frac{3}{4} = 0.75$.

2 $\sin\theta = \frac{15}{17} \approx 0.882$, $\cos\theta = \frac{8}{17} \approx 0.471$, $\tan\theta = \frac{15}{8} = 1.875$.

3 $\sin 23^\circ \approx 0.3907$, $\cos 23^\circ \approx 0.9205$, $\tan 23^\circ \approx 0.4245$.

4 $\text{opp} = 12\sin 41^\circ \approx 12 \times 0.6561 = 7.87$ cm.

5 $\text{adj} = 20\cos 36^\circ \approx 20 \times 0.8090 = 16.18$ m.

6 $\text{opp} = 7\tan 52^\circ \approx 7 \times 1.2799 = 8.96$ cm.

7 $\sin\theta = \frac{4}{9} \approx 0.4444$, so $\theta \approx 26.4^\circ$.

8 $\cos\theta = \frac{12}{15} = 0.8$, so $\theta \approx 36.9^\circ$.

9 $\tan\theta = \frac{9}{5} = 1.8$, so $\theta \approx 60.9^\circ$.

10 $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\cos 45^\circ = \frac{\sqrt{2}}{2}$, $\tan 45^\circ = 1$.

11 The opposite side is shorter than the hypotenuse, because the hypotenuse is the longest side. So $\sin\theta = \frac{\text{opp}}{\text{hyp}} < 1$ for an acute angle, and the ratio is positive.

12 The calculator is in radian mode: $\tan(45 \text{ rad}) \approx 1.6198$. Switch the display to DEG; then $\tan 45^\circ = 1$.

Common error: trusting a trigonometric answer without checking the mode. A value of $\tan 45^\circ$ other than 1 signals radians.

13 $\sin 30^\circ = \frac{1}{2}$ and $\cos 60^\circ = \frac{1}{2}$, so the product is $\frac{1}{4}$.

14 $\sin 2^\circ \approx 0.0349$ and $\tan 2^\circ \approx 0.0349$. For a small angle the opposite side is almost equal to the arc, and $\cos 2^\circ \approx 0.9994 \approx 1$, so $\tan\theta = \frac{\sin\theta}{\cos\theta} \approx \sin\theta$.

15 $\frac{\sin\theta}{\cos\theta} = \frac{\text{opp/hyp}}{\text{adj/hyp}} = \frac{\text{opp}}{\text{adj}} = \tan\theta$. The hypotenuse cancels, so the identity holds for every acute angle.

16 Take hypotenuse 1 and opposite 0.35. Then adjacent $= \sqrt{1 - 0.35^2} = \sqrt{0.8775} \approx 0.937$, so $\cos\theta \approx 0.937$ and $\tan\theta = \frac{0.35}{0.9368} \approx 0.374$.

Exercise 10D · Solving right-angled triangles

1 $\text{opp} = 13\sin 32^\circ \approx 6.89$ cm; $\text{adj} = 13\cos 32^\circ \approx 11.02$ cm.

2 $\text{opp} = 18\tan 41^\circ \approx 15.65$ m.

3 $\text{hyp} = \frac{9}{\sin 27^\circ} \approx \frac{9}{0.4540} = 19.82$ cm.

4 $\sin\theta = \frac{5}{9}$, so $\theta \approx 33.7^\circ$.

5 $\cos\theta = \frac{14}{18} = 0.7778$, so $\theta \approx 38.9^\circ$.

6 $\tan\theta = \frac{11}{16} = 0.6875$, so $\theta \approx 34.5^\circ$.

7 $h = 8\sin 70^\circ \approx 7.52$ m.

8 $h = 40\sin 55^\circ \approx 32.77$ m.

9 Height: $h^2 = 10^2 - 6^2 = 64$, so $h = 8$ cm. Half the apex angle has $\sin\alpha = \frac{6}{10} = 0.6$, so $\alpha \approx 36.9^\circ$ and the apex is 73.7°. Base angles $= \frac{180^\circ - 73.7^\circ}{2} \approx 53.1^\circ$.

10 The student found the ratio $\sin\theta = 0.6$ but read it as the angle. The angle is $\theta = \sin^{-1} 0.6 \approx 36.9^\circ$, not 0.6°.

11 Base diagonal $= \sqrt{3^2 + 4^2} = 5$ cm and space diagonal $= 13$ cm. The angle α between the space diagonal and the base satisfies $\tan\alpha = \frac{12}{5} = 2.4$, so $\alpha \approx 67.4^\circ$.

12 $\sin\theta = 0.5$, so $\theta = 30^\circ$.

13 Opposite $= 1 \times \sin 30^\circ = \frac{1}{2}$; adjacent $= 1 \times \cos 30^\circ = \frac{\sqrt{3}}{2}$.

14 Let the height be h and the first distance d. Then $\tan 30^\circ = \frac{h}{d}$, so $d = \frac{h}{\tan 30^\circ} = h\sqrt{3}$. At one-third of the distance, $\tan\theta = \frac{h}{d/3} = 3\tan 30^\circ = \sqrt{3}$, so $\theta = 60^\circ$. The angle does not triple because it is not proportional to the distance: $\theta = \tan^{-1}\left(\frac{h}{d}\right)$, and the tangent is non-linear.

Exercise 10E · Elevation, depression and bearings

1 $h = 25\tan 36^\circ \approx 18.2$ m.

2 $d = \frac{80}{\tan 28^\circ} \approx \frac{80}{0.5317} = 150$ m.

3 The turn is 90°, so $AC = \sqrt{150^2 + 120^2} = \sqrt{36\,900} \approx 192$ km. The angle from the first leg has $\tan\alpha = \frac{120}{150} = 0.8$, so $\alpha \approx 38.7^\circ$ and the bearing is $030^\circ + 38.7^\circ \approx 069^\circ$.

4 $215^\circ - 180^\circ = 035^\circ$.

5 East $= 14\sin 72^\circ \approx 13.31$ km; north $= 14\cos 72^\circ \approx 4.33$ km.

6 The bearing 045° makes a 45° angle with north, so the boat is as far east of the point as the point is south of the lighthouse: 500 m. Distance $= \sqrt{500^2 + 500^2} \approx 707$ m.

7 $d = \frac{55}{\tan 32^\circ} \approx \frac{55}{0.6249} = 88.0$ m.

8 Rise $= 12\sin 5^\circ \approx 1.05$ m; run $= 12\cos 5^\circ \approx 11.95$ m.

9 $h = 18\tan 43^\circ \approx 16.8$ m.

10 Measuring from the vertical gives the complement of the angle of depression, so the cliff height is now adjacent to the marked angle and the horizontal distance is opposite it. A student who keeps the elevation formula $\tan\theta = \dfrac{h}{d}$ then computes $d = \dfrac{h}{\tan(90^\circ - \theta)} = h\tan\theta$, which is too small whenever the true angle of depression is under 45° (and too large when it is over 45°); either convert the angle back, or use $\tan(\text{marked angle}) = \dfrac{d}{h}$.

11 Let the height be h and $AP = d$. Then $h = d\tan 45^\circ = d$ and $h = (d+20)\tan 30^\circ$. So $d = \dfrac{d+20}{\sqrt{3}}$, giving $d(\sqrt{3}-1) = 20$ and $d \approx 27.3$ m; the height is 27.3 m.

12 $225^\circ = 180^\circ + 45^\circ$: from south, turn 45° towards the west, which is south-west. The opposite direction is $225^\circ - 180^\circ = 045^\circ$, north-east. On a compass rose, mark 225° clockwise from N and 045° on the opposite ray.

Investigation · How tall is the flagpole?

1 Position 1: $10\tan 46^\circ = 10.36$, $h = 11.86$ m. Position 2: $15\tan 35^\circ = 10.50$, $h = 12.00$ m. Position 3: $20\tan 28^\circ = 10.63$, $h = 12.13$ m. Position 4: $25\tan 23^\circ = 10.61$, $h = 12.11$ m.

2 Mean $= \dfrac{11.86 + 12.00 + 12.13 + 12.11}{4} = 12.03$ m. Range $= 12.13 - 11.86 = 0.27$ m, about 2% of the mean, so the four estimates agree closely enough to trust the method to roughly ± 0.15 m.

3 The angle is read to the nearest degree, and a 1° error changes $d\tan\theta$ by about 0.36 m at 10 m; the distances are rounded; the ground may not be level, so the horizontal baseline and the eye height vary; the top of the pole is not a sharp point. Each error is independent, so the four results scatter around the true height.

4 At eye level, draw a horizontal of length d to the pole and a vertical of length $h - e$ up to the top. The line of sight is the hypotenuse and θ is at the eye, so $\tan\theta = \dfrac{h-e}{d}$. Hence $h - e = d\tan\theta$ and $h = e + d\tan\theta$.

5 At $d = 10$ m: $10\tan 45^\circ = 10.00$ and $10\tan 47^\circ = 10.72$, so h ranges from 11.50 to 12.22 m — about ± 0.36 m. At $d = 25$ m: $25\tan 22^\circ = 10.10$ and $25\tan 24^\circ = 11.13$, so h ranges from 11.60 to 12.63 m — about ± 0.52 m. The 10 m position is more reliable: the sensitivity $d\sec^2\theta$ is smaller there, so the shorter baseline outweighs the larger angle.

6 Assumptions: the ground is level between observer and pole; the pole is vertical. Test the first with a spirit level or by measuring eye height at each position, and the second with a plumb line or by sighting from two directions. Improvement: measure each angle several times to 0.1° and average, and measure the distance with a tape rather than pacing.

→ For fixed θ, $h = e + (\tan\theta)d$ has the form $y = c + mx$: a straight line with gradient $m = \tan\theta$ and intercept e. The gradient is the height gained per extra metre of horizontal distance, the slope of the line of sight. When $d = 0$ the observer stands at the foot of the pole and the line of sight is horizontal at eye level, so $h = e$: the intercept must be the eye height.

Mixed practice 10

1 $c^2 = 36 + 64 = 100$, so $c = 10$ cm.

2 $b^2 = 41^2 - 9^2 = 1681 - 81 = 1600$, so $b = 40$ cm.

3 $d^2 = 4 + 9 + 36 = 49$, so $d = 7$ cm.

4 $\sin\theta = \dfrac{15}{17} \approx 0.882$, $\cos\theta = \dfrac{8}{17} \approx 0.471$, $\tan\theta = \dfrac{15}{8} = 1.875$.

5 $x = 9\tan 34^\circ \approx 9 \times 0.6745 = 6.07$.

6 $\theta = \sin^{-1} 0.42 \approx 24.8^\circ$.

7 $h = 4.8\sin 72^\circ \approx 4.57$ m.

8 $248^\circ - 180^\circ = 068^\circ$.

9 $d = \dfrac{70}{\tan 18^\circ} \approx \dfrac{70}{0.3249} = 215$ m.

10 $5^2 + 12^2 = 169$ gives the square of the hypotenuse, not the hypotenuse. Take the root: $c = \sqrt{169} = 13$, not 169.

11 $d^2 = 25 + 25 + 100 = 150$, so $d \approx 12.25$ cm.

12 The $6, 8, 10$ triangle has hypotenuse 10; the similar triangle has hypotenuse 30, so $k = 3$. Areas are in the ratio $k^2 = 9$, that is $9 : 1$.

13 $\tan\theta = \dfrac{1}{8} = 0.125$, so $\theta \approx 7.1^\circ$. Ramp length $= \sqrt{1.1^2 + 8.8^2} = \sqrt{78.65} \approx 8.87$ m (or $L = \dfrac{1.1}{\sin 7.1^\circ}$).

14 $h = 14\tan 38^\circ \approx 14 \times 0.7813 = 10.9$ m.

15 Distance $= \sqrt{80^2 + 60^2} = \sqrt{10\,000} = 100$ km. The angle east of north has $\tan\alpha = \dfrac{60}{80} = 0.75$, so $\alpha \approx 36.9^\circ$ and the bearing is 037°.

16 $\sin\theta = 0.45$, so $\theta \approx 26.7^\circ$.

17 $\sin 45^\circ \cos 45^\circ = \dfrac{\sqrt{2}}{2} \times \dfrac{\sqrt{2}}{2} = \dfrac{2}{4} = \dfrac{1}{2}$. $\tan 60^\circ \tan 30^\circ = \sqrt{3} \times \dfrac{1}{\sqrt{3}} = 1$.

18 Legs $a + b = 60 - 25 = 35$ and $a^2 + b^2 = 625$. Since $(a+b)^2 = 625 + 2ab$, $2ab = 1225 - 625 = 600$ and $ab = 300$. Solving $t^2 - 35t + 300 = 0$ gives $(t - 15)(t - 20) = 0$, so the legs are 15 cm and 20 cm.

Criterion A · Knowing and understanding

1 $c^2 = 7^2 + 24^2 = 49 + 576 = 625$, so $c = 25$ cm. [2]

2 $b^2 = 41^2 - 9^2 = 1681 - 81 = 1600$, so $b = 40$ cm. [2]

3 Longest side 26: $10^2 + 24^2 = 100 + 576 = 676 = 26^2$, so the triangle is right-angled. [2]

4 $d^2 = 2^2 + 3^2 + 6^2 = 4 + 9 + 36 = 49$, so $d = 7$ cm. [3]

5 $\sin\theta = \dfrac{5}{13}$, $\cos\theta = \dfrac{12}{13}$, $\tan\theta = \dfrac{5}{12}$. [3]

6 $x = 14\sin 38^\circ \approx 14 \times 0.6157 = 8.62$. [3]

7 $\theta = \tan^{-1}\left(\frac{7}{11}\right) \approx \tan^{-1} 0.6364 \approx 32.5^\circ$. [3]

8 $h = 7\sin 64^\circ \approx 7 \times 0.8988 = 6.29$ m. [3]

9 $137^\circ + 180^\circ = 317^\circ$. [2]

10 $8^2 + 6^2 = 100$ is c^2, not c. The hypotenuse is $\sqrt{100} = 10$, not 100. [3]

11 Depression equals elevation at the boat, so $\tan 25^\circ = \frac{60}{d}$. $d = \frac{60}{\tan 25^\circ} \approx \frac{60}{0.4663} = 128.7$ m, about 129 m. [4]

Criterion B · Investigating patterns

1 $n = 9$: second leg $\frac{81-1}{2} = 40$, hypotenuse 41. $n = 11$: $\frac{121-1}{2} = 60$, hypotenuse 61. $n = 13$: $\frac{169-1}{2} = 84$, hypotenuse 85. [5]

2 Square the odd number, subtract 1 and halve the result to get the second leg; add 1 to that for the hypotenuse. For $n = 15$: $\frac{225-1}{2} = 112$ and 113, so the triple is $15, 112, 113$. [4]

3 For odd n: $\left(n, \frac{n^2-1}{2}, \frac{n^2+1}{2}\right)$. [5]

4 Let $b = \frac{n^2-1}{2}$ and $c = \frac{n^2+1}{2}$. Then $b^2 = \frac{n^4 - 2n^2 + 1}{4}$ and $n^2 = \frac{4n^2}{4}$, so $n^2 + b^2 = \frac{n^4 + 2n^2 + 1}{4} = \left(\frac{n^2+1}{2}\right)^2 = c^2$. The converse of Pythagoras' theorem then makes the triangle right-angled. [6]

5 If n is odd, n^2 is odd, so $n^2 - 1$ and $n^2 + 1$ are even and halving gives whole numbers; if n were even, $n^2 - 1$ would be odd and the second leg would end in a half. For hypotenuse 145: $\frac{n^2+1}{2} = 145$, so $n^2 = 289$, $n = 17$, and the triple is $17, 144, 145$. [4]

→ **Model response.** The table extends to $9, 40, 41$; $11, 60, 61$; $13, 84, 85$, and the rule $\left(n, \frac{n^2-1}{2}, \frac{n^2+1}{2}\right)$ gives $15, 112, 113$. The algebra shows $n^2 + \left(\frac{n^2-1}{2}\right)^2 = \left(\frac{n^2+1}{2}\right)^2$, so every odd n produces a right triangle; when the hypotenuse is 145, $n = 17$ gives $17, 144, 145$.

Marking guidance (24). Table: 5 (1 per correct second leg, 1 for hypotenuse column consistency, 1 for all hypotenuses). Description and prediction: 4 (2 for the word pattern, 2 for $15, 112, 113$). Rule: 5 (3 for structure, 2 for correct notation for odd n). Proof: 6 (2 for substituting, 3 for the algebraic identity, 1 for invoking the converse). Odd condition and 145: 4 (2 for the parity argument, 2 for $17, 144, 145$). Accept equivalent forms such as $2k + 1$ with $n = 2k + 1$, provided the proof is complete.

Criterion C · Communicating

i Stand on level ground a measured horizontal distance d from the object, hold the clinometer at eye level, sight the top, and read the angle θ from the horizontal. Record your eye height e. Repeat from several positions and average.

ii Diagram: a horizontal line at eye level of length d, a vertical of length $h - e$ up to the top, the line of sight as hypotenuse and θ between the horizontal and the line of sight; mark e below the eye and h as the total height.

iii The triangle is right-angled at the pole. The side opposite θ is $h - e$ and the adjacent side is d, so $\tan\theta = \frac{h-e}{d}$. Hence $h - e = d\tan\theta$ and $h = e + d\tan\theta$.

iv With $d = 20$ m, $\theta = 32^\circ$ and $e = 1.5$ m: $h = 1.5 + 20\tan 32^\circ \approx 1.5 + 20 \times 0.6249 = 1.5 + 12.50 = 14.0$ m. Check that 14.0 m is taller than the observer and plausible for the object.

v Errors: (1) the angle is read to the nearest degree, giving about 0.5 m per degree at 20 m — reduce by averaging repeats or using a digital clinometer; (2) the distance is paced rather than taped — reduce by using a tape and measuring horizontally, not along sloping ground.

vi The method fails if the object leans, if the ground slopes so that the baseline is not horizontal, or if the top is not visible — for example a tree whose crown hides the highest point, or a tower seen through obstructions. Then the triangle is not the one assumed and the formula no longer applies.

→ **Sample paragraph.** "Measure a horizontal distance d from the foot, stand at eye height e, and sight the top with a clinometer, reading θ from the horizontal. The line of sight, the horizontal and the pole form a right triangle with opposite side $h - e$, so $\tan\theta = \frac{h-e}{d}$ and $h = e + d\tan\theta$. From 20 m away at 32°, with eye height 1.5 m, the height is $1.5 + 20\tan 32^\circ \approx 14.0$ m. Reading the angle to 1° changes the result by about 0.5 m at 20 m, so average several readings. The method fails on sloping ground or against a leaning object, where the assumed right triangle does not exist."

Marking guidance (20). Procedure: 4. Labelled diagram: 3 (right angle, d, $h - e$, e, θ in the correct place). Derivation: 5 (3 for the ratio, 2 for rearranging). Worked example with units: 3. Two errors with a remedy each: 3. Failure situation with reason: 2. Total 20; clarity and audience awareness are judged within each element, and a memorised formula without derivation scores at most 2 of the 5 derivation marks.

Criterion D · Applying mathematics in real-life contexts

1 Ramp length $= \sqrt{1.2^2 + 12^2} = \sqrt{1.44 + 144} = \sqrt{145.44} \approx 12.06$ m. Gradient $= 1.2 : 12 = 1 : 10$. Angle $= \tan^{-1}\left(\frac{1.2}{12}\right) = \tan^{-1} 0.1 \approx 5.7°$. [6]

2 For $1 : 12$, run $= 12 \times 1.2 = 14.4$ m. Ramp length $= \sqrt{1.2^2 + 14.4^2} = \sqrt{208.8} \approx 14.45$ m. The proposed design has a run of 12 m, not 14.4 m, and a gradient $1 : 10$, steeper than the recommended $1 : 12$; it does not meet the guidance and needs 2.4 m more run. [6]

3 Rise $= 1.7 \sin 28° \approx 1.7 \times 0.4695 = 0.80$ m. Horizontal projection $= 1.7 \cos 28° \approx 1.7 \times 0.8829 = 1.50$ m. [4]

4 Panel area $= 1.7 \times 1.0 = 1.7$ m^2. Incident energy $= 1.7 \times 1050 = 1785$ kWh per year. Electrical output $= 1785 \times 0.19 \times 0.92 \approx 312$ kWh per year. [5]

5 (i) The irradiation figure is an annual average for a horizontal surface; a panel on a 28° roof receives more in winter or less in summer depending on its orientation, so the true input could differ by 10–20%. (ii) The 8% loss allowance is optimistic: shading, soiling, high cell temperature and inverter losses can total 15–20%, lowering the output. The geometric calculations are exact for the dimensions, but the energy estimate is an upper bound. [3]

→ **Model response.** A 12 m run gives a 12.06 m ramp at 5.7°, gradient $1 : 10$, which fails the $1 : 12$ guidance; 14.4 m of run is needed, with a 14.45 m ramp. The panel rises 0.80 m and projects 1.50 m horizontally, so 1.7 m$^2 \times 1050$ kWh/m$^2 \times 0.19 \times 0.92 \approx 312$ kWh per year — a figure that should be treated as an upper bound, since tilt, orientation and larger real losses would reduce it.

Marking guidance (24). Ramp: 6 (2 for the length, 2 for the gradient $1 : 10$, 2 for the angle). Guidance comparison: 6 (2 for the 14.4 m run, 2 for 14.45 m, 2 for a correct judgement). Panel components: 4 (2 each, with units). Energy: 5 (2 for the area and irradiation, 2 for applying both factors, 1 for the unit and rounding). Evaluation: 3 (1.5 per assumption with its effect). Penalise a gradient written as $1 : 0.1$ or an angle found with the wrong ratio only once.

CHAPTER 11 · SURFACE AREA AND VOLUME

Opening problem · three vessels, one budget

1 $r = 2.2 \div 2 = 1.1$ m, so $V = \pi r^2 h = \pi(1.1)^2(3.0) = 3.63\pi \approx 11.4$ m^3 (3 s.f.).

In litres, $3.63\pi \times 1000 = 11\,404$ L $\approx 11\,400$ L (3 s.f.).

2 Closed cylinder: $S = 2\pi r^2 + 2\pi rh = 2\pi(1.1)^2 + 2\pi(1.1)(3.0) = 2.42\pi + 6.6\pi = 9.02\pi \approx 28.3$ m^2.

Curved surface alone is $2\pi rh = 6.6\pi \approx 20.7$ m^2, so the lid and base add $2.42\pi \approx 7.60$ m^2: $9.02 \div 6.6 = 1.37$, about 37% more sheet.

If the tank stands on a slab and needs no base, the sheet is $6.6\pi + 1.21\pi = 7.81\pi \approx 24.5$ m^2.

3 $h = \sqrt{l^2 - r^2} = \sqrt{2.0^2 - 1.2^2} = \sqrt{2.56} = 1.6$ m.

$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(1.2)^2(1.6) = 0.768\pi \approx 2.41$ m^3.

Curved sheet $= \pi rl = \pi(1.2)(2.0) = 2.4\pi \approx 7.54$ m^2.

4 $S = 4\pi r^2 = 4\pi(1.5)^2 = 9\pi \approx 28.3$ m^2.

$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(1.5)^3 = 4.5\pi \approx 14.1$ m^3.

5 Tank: $3.63\pi \div 9.02\pi \approx 0.402$ m^3/m^2. Hopper: $0.768\pi \div 2.4\pi = 0.320$ m^3/m^2 (with a base, $0.768\pi \div 3.84\pi = 0.200$). Holder: $4.5\pi \div 9\pi = 0.500$ m^3/m^2.

The sphere stores the most per square metre of material: it needs almost the same sheet as the tank (9π against 9.02π m^2) yet holds $14.1 - 11.4 = 2.7$ m^3 more. The ranking is unchanged whichever ends are counted.

6 Prediction: the capacity doubles, but the material increases by less than double.

$V = \pi(1.1)^2(6.0) = 7.26\pi \approx 22.8$ m^3 $= 22\,800$ L, exactly twice the original, because with r fixed V is proportional to h.

$S = 2.42\pi + 2\pi(1.1)(6.0) = 15.62\pi \approx 49.1$ m^2, a factor of $15.62 \div 9.02 = 1.73$. The curved part doubles (6.6π to 13.2π) but the lid and base stay at 2.42π m^2, so the total grows by only 73%. The tank becomes more efficient: V/S rises from 0.402 to $7.26\pi \div 15.62\pi = 0.465$ m^3/m^2.

Prerequisite check

1 $A = \pi r^2 = \pi(5)^2 = 25\pi$ cm^2.

2 $A = \frac{1}{2}bh = \frac{1}{2}(8)(6) = 24$ cm^2.

3 $c = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$ cm.

4 2.5 m^3 $= 2.5 \times 1000 = 2500$ L.

5 $3^2 + 4^2 = 9 + 16 = 25$ (the square of the hypotenuse 5).

6 $A = 7^2 = 49$ cm^2; $P = 4(7) = 28$ cm.

Exercise 11A

1 $S = 2(8 \times 5 + 8 \times 4 + 5 \times 4) = 2(92) = 184$ cm^2.

$V = 8 \times 5 \times 4 = 160$ cm^3.

2 $S = 6(6)^2 = 216$ cm^2; $V = 6^3 = 216$ cm^3.

3 Base area $= \frac{1}{2}(5)(12) = 30$ cm^2; $V = 30(20) = 600$ cm^3.

$P = 5 + 12 + 13 = 30$ cm, so $S = 2(30) + 30(20) = 660$ cm^2.

4 $V = \pi(4)^2(9) = 144\pi$ cm^3.

$S = 2\pi(16) + 2\pi(4)(9) = 32\pi + 72\pi = 104\pi$ cm^2.

5 $r = 5$ cm: $V = 300\pi \approx 942$ cm^3.

$S = 2\pi(25) + 2\pi(5)(12) = 170\pi \approx 534$ cm^2.

6 Curved area $= 2\pi(2.5)(8) = 40\pi \approx 126$ cm^2.

$V = \pi(2.5)^2(8) = 50\pi \approx 157$ cm^3.

7 $S = 2(100 + 200 + 200) = 1000$ cm^2; $V = 10 \times 10 \times 20 = 2000$ cm^3.

8 $V = AL = 24(15) = 360$ cm^3.

$S = 2(24) + 22(15) = 48 + 330 = 378$ cm^2.

9 Walls: $2(2.5)(1.2) + 2(1.8)(1.2) = 6.0 + 4.32 = 10.32$ m^2; ceiling: $2.5(1.8) = 4.5$ m^2. Painted area $= 14.82 \approx 14.8$ m^2.

Two coats $= 29.64$ m^2, so paint $= 29.64 \div 8.0 = 3.71 \approx 3.7$ L (buy 4 L).

10 $V = \pi(0.9)^2(2.4) = 1.944\pi \approx 6.11$ m^3 $= 6110$ L (3 s.f.).

$S = 2\pi(0.9)^2 + 2\pi(0.9)(2.4) = 5.94\pi \approx 18.7$ m^2.

11 Label area $= 2\pi(3.5)(11) = 77\pi \approx 242$ cm^2.

12 Base $30 \times 20 = 600$; two pairs of sides $2(30)(15) + 2(20)(15) = 900 + 600$. Material $= 2100$ cm^2.

Capacity $= 30 \times 20 \times 15 = 9000$ cm^3 $= 9$ L.

13 100π is only the curved surface; the two circular ends are missing. They add $2\pi(5)^2 = 50\pi$.

Correct total $= 150\pi \approx 471$ cm^2.

Common error: $2\pi rh$ is a closed cylinder only when the ends are added.

14 $1000 = \pi(4)^2 h$, so $h = 1000 \div 16\pi = 19.89 \approx 19.9$ cm.

$S = 2\pi(16) + 2\pi(4)(19.89) \approx 100.5 + 500.0 = 600.5 \approx 601$ cm^2.

15 $V = \frac{1}{2}\pi(1.0)^2(5.0) = 2.5\pi \approx 7.85$ m^3.

Area $= \pi rL + \pi r^2 = 5\pi + \pi = 6\pi \approx 18.8$ m^2.

16 With $h = 2r$: $S = 2\pi r^2 + 2\pi r(2r) = 6\pi r^2$; $V = \pi r^2(2r) = 2\pi r^3$.

$2\pi r^3 = 1000$, so $r = \sqrt[3]{1000/(2\pi)} = 5.419 \approx 5.42$ cm.
$S = 6\pi(5.419)^2 \approx 554$ cm^2.

17 $x^2 + 4xh = 1200$, so $h = \dfrac{1200 - x^2}{4x}$ and $V = x^2 h = 300x - \dfrac{x^3}{4}$.

$x = 20$: 4000 cm^3; $x = 25$: 3593.75 cm^3; $x = 30$: 2250 cm^3. Greatest at $x = 20$, $V = 4000$ cm^3 (the turning point of V is at $x = 20$).

Exercise 11B

1 $l = \sqrt{3^2 + 4^2} = 5$ cm.

$S = 8^2 + 4\left(\frac{1}{2} \cdot 8 \cdot 5\right) = 64 + 80 = 144$ cm^2; $V = \frac{1}{3}(64)(3) = 64$ cm^3.

2 $l = \sqrt{12^2 + 5^2} = 13$ cm.

$S = 100 + 4\left(\frac{1}{2} \cdot 10 \cdot 13\right) = 360$ cm^2; $V = \frac{1}{3}(100)(12) = 400$ cm^3.

3 $l = 5$ cm; $S = 9\pi + 15\pi = 24\pi \approx 75.4$ cm^2; $V = \frac{1}{3}\pi(9)(4) = 12\pi \approx 37.7$ cm^3.

4 $l = 10$ cm; $S = 36\pi + 60\pi = 96\pi \approx 302$ cm^2; $V = \frac{1}{3}\pi(36)(8) = 96\pi \approx 302$ cm^3.

5 $r = 7$, $l = 25$: $S = 49\pi + 175\pi = 224\pi \approx 704$ cm^2.
$V = \frac{1}{3}\pi(49)(24) = 392\pi \approx 1230$ cm^3.

6 $h = \sqrt{5^2 - 3^2} = 4$ cm; $V = \frac{1}{3}(36)(4) = 48$ cm^3.

7 $h = \sqrt{13^2 - 5^2} = 12$ cm; $V = \frac{1}{3}\pi(25)(12) = 100\pi \approx 314$ cm^3.

8 $V = \dfrac{9\pi}{3}(6^2 + 6 \cdot 4 + 4^2) = 3\pi(76) = 228\pi \approx 716$ cm^3.

9 $l = \sqrt{2.5^2 + 2.0^2} = \sqrt{10.25} \approx 3.20$ m.
Four faces $= 4\left(\frac{1}{2} \cdot 4.0 \cdot 3.20\right) \approx 25.6$ m^2; with 10% extra, $25.6 \times 1.10 \approx 28.2$ m^2.

10 $V = \frac{1}{3}\pi(4.0)^2(3.0) = 16\pi \approx 50.3$ m^3.

11 Capacity $= \frac{1}{3}\pi(3)^2(12) = 36\pi \approx 113$ cm^3.
$l = \sqrt{3^2 + 12^2} = \sqrt{153}$, curved area $= \pi(3)\sqrt{153} \approx 117$ cm^2.

12 The height 3 cm was used instead of the slant height. $l = \sqrt{4^2 + 3^2} = 5$ cm.
Correct curved area $= \pi(4)(5) = 20\pi \approx 62.8$ cm^2.
Common error: πrl uses the slant height, not the perpendicular height.

13 $324 = \frac{1}{3}(81)h$, so $h = 12$ cm.
$l = \sqrt{12^2 + 4.5^2} = \sqrt{164.25} \approx 12.8$ cm.

14 $1000 = \frac{1}{3}\pi(25)h$, so $h = \dfrac{120}{\pi} = 38.2$ cm.
$l = \sqrt{5^2 + 38.20^2} \approx 38.5$ cm, so $S = 25\pi + \pi(5)(38.5) \approx 684$ cm^2.

15 The cut is halfway up, so the small cone has linear factor $\frac{1}{2}$. Its volume factor is $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$.
The frustum therefore has $1 - \frac{1}{8} = \frac{7}{8}$ of the volume.

16 $V_{\text{cone}} = \frac{1}{3}\pi(36)(8) = 96\pi$; $V_{\text{cyl}} = 288\pi$, so V ratio $= 1 : 3$.
$S_{\text{cone}} = 36\pi + 60\pi = 96\pi$; $S_{\text{cyl}} = 72\pi + 96\pi = 168\pi$, ratio $= 4 : 7$.

17 $r = 3.0$, $l = \sqrt{3^2 + 2^2} = \sqrt{13} \approx 3.61$ m.
Curved area $= \pi(3)\sqrt{13} \approx 34.0$ m^2; with 12% overlap, $34.0 \times 1.12 \approx 38.1$ m^2.

Exercise 11C

1 $S = 4\pi(3)^2 = 36\pi$ cm^2; $V = \frac{4}{3}\pi(3)^3 = 36\pi$ cm^3.

2 $r = 5$: $S = 100\pi \approx 314$ cm^2; $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3} \approx 524$ cm^3.

3 $S = 4\pi(2.5)^2 = 25\pi \approx 78.5$ cm^2; $V = \frac{4}{3}\pi(15.625) \approx 65.4$ cm^3.

4 $S = 3\pi(6)^2 = 108\pi \approx 339$ cm^2; $V = \frac{2}{3}\pi(216) = 144\pi \approx 452$ cm^3.

5 $S = 3\pi(4)^2 = 48\pi \approx 151$ cm^2; $V = \frac{2}{3}\pi(64) \approx 134$ cm^3.

6 $\frac{4}{3}\pi r^3 = 36\pi$, so $r^3 = 27$, $r = 3$ cm. $S = 36\pi$ cm^2.

7 $4\pi r^2 = 100\pi$, so $r = 5$ cm. $V = \frac{4}{3}\pi(125) = \frac{500\pi}{3} \approx 524$ cm^3.

8 $V = 90\pi + 18\pi = 108\pi \approx 339$ cm^3.
Exposed area $= 60\pi + 9\pi + 18\pi = 87\pi \approx 273$ cm^2 (the joined circle is hidden).

9 $r = 11$: $S = 484\pi \approx 1520$ cm^2; $V = \frac{4}{3}\pi(1331) \approx 5580$ cm^3.

10 Curved area $= 2\pi(4)^2 = 32\pi \approx 100.5$ m^2; two coats $= 201$ m^2.
Paint $= 201 \div 4.0 = 50.3$ L, so about 51 L.

11 $1000 = \frac{4}{3}\pi r^3$, so $r^3 = 238.7$ and $r = 6.20$ cm.
$S = 4\pi(6.20)^2 \approx 484$ cm^2.

12 $2\pi r^2$ is the curved surface of a hemisphere, not a whole sphere. A sphere has twice this: $4\pi r^2$.
Correct $S = 4\pi(10)^2 = 400\pi \approx 1260$ cm^2.

13 $S = 4\pi(6371)^2 \approx 5.10 \times 10^8$ km^2.
Ocean $= 0.71(5.10 \times 10^8) \approx 3.62 \times 10^8$ km^2.

14 Inside and outside are both curved hemispheres: $2 \times 2\pi(9)^2 = 324\pi \approx 1020$ cm^2.
The thin rim is not included unless the question asks for the whole surface.

15 $V = 144\pi + 96\pi = 240\pi \approx 754$ cm^3.
Exposed area $=$ hemisphere $2\pi(36) = 72\pi$ plus cone $\pi(6)(10) = 60\pi$, total $132\pi \approx 415$ cm^2. The flat circle is hidden under the cone.

16 $S_{\text{sphere}} = 4\pi r^2$, $S_{\text{cube}} = 24r^2$, ratio $= \frac{\pi}{6}$.
$V_{\text{sphere}} = \frac{4}{3}\pi r^3$, $V_{\text{cube}} = 8r^3$, ratio $= \frac{\pi}{6}$.
The sphere occupies $\frac{\pi}{6} \approx 52\%$ of the cube in both measures; the ratio is independent of r.

17 Surface areas $1 : 4$; volumes $1 : 8$.
Larger volume $= 10 \times 8 = 80$ cm^3.

Exercise 11D

1 $k = 2$: area factor 4, volume factor 8.

2 $k = 3$: area factor 9, volume factor 27.

3 $k = 12 \div 8 = 1.5$: surface areas $2.25 : 1 = 9 : 4$; volumes $3.375 : 1 = 27 : 8$.

4 $k = 10 \div 4 = 2.5$: surface areas $6.25 : 1 = 25 : 4$; volumes $15.625 : 1 = 125 : 8$.

5 $k^3 = \frac{64}{27}$, so $k = \frac{4}{3}$. Area ratio $= k^2 = \frac{16}{9}$, i.e. $16 : 9$.

6 $k^2 = \frac{100}{25} = 4$, so $k = 2$. Volume ratio $= k^3 = 8 : 1$.

7 Area factor $= \left(\frac{1}{50}\right)^2 = \frac{1}{2500}$; volume factor $= \frac{1}{125\,000}$.

8 Real volume $= 250 \times 20^3 = 250 \times 8000 = 2\,000\,000$ cm$^3 = 2.0$ m^3.

9 Edge $= \sqrt[3]{64} = 4$ cm, $S = 6(4)^2 = 96$ cm^2.
Edge doubled to 8 cm: $S = 6(8)^2 = 384$ cm^2 (4×96).

10 Area factor 4, volume factor 8.

11 Each of the three dimensions contributes, so the volume factor is $2^3 = 8$, not 2. The area factor is $2^2 = 4$.

12 $k^3 = 4000 \div 500 = 8$, so $k = 2$. Larger area $= 300 \times 4 = 1200$ cm^2.

13 New $r = 6$, $h = 10$: $S = 2\pi(36) + 2\pi(6)(10) = 192\pi \approx 603\ \text{cm}^2$.

$V = \pi(36)(10) = 360\pi \approx 1130\ \text{cm}^3$.

14 $k^2 = 1.44$, so $k = 1.2$. Volume factor $= 1.2^3 = 1.728$, an increase of 72.8%.

15 $k^3 = \frac{8}{27}$, so $k = \frac{2}{3}$: heights 2 : 3.

Surface areas $k^2 = \frac{4}{9}$, i.e. 4 : 9.

16 $r = 1$ mm: $S = 4\pi \approx 12.6\ \text{mm}^2$, $V = \frac{4}{3}\pi \approx 4.19\ \text{mm}^3$, $S/V = 3\ \text{mm}^{-1}$.

$r = 2$ mm: $S = 16\pi \approx 50.3\ \text{mm}^2$, $V = \frac{32\pi}{3} \approx 33.5\ \text{mm}^3$, $S/V = 1.5\ \text{mm}^{-1}$. The smaller drop has twice the surface per unit volume, so it evaporates faster.

17 $k^2 = \frac{81}{36} = 2.25$, so $k = 1.5$. Volume factor $= 3.375$.

Larger volume $= 24 \times 3.375 = 81\ \text{cm}^3$.

18 $1\ \text{L} = 1000\ \text{cm}^3$, edge 10 cm, $S = 600\ \text{cm}^2$.

Doubled volume: edge $= \sqrt[3]{2000} = 12.6$ cm, $S = 6(12.6)^2 \approx 952\ \text{cm}^2$.

Lengths grow by $k = \sqrt[3]{2}$, so area grows by $k^2 = 1.59$, not by 2.

Exercise 11E

1 $S = 2(96 + 72 + 48) = 432\ \text{cm}^2$; $V = 576\ \text{cm}^3$.

Cube of equal volume: edge $= \sqrt[3]{576} = 8.32$ cm, $S \approx 415\ \text{cm}^2$. The cube is more efficient (smaller surface for the same volume).

2 $h = 2r$: $2\pi r^3 = 1000$, $r = 5.42$, $S = 6\pi r^2 \approx 554\ \text{cm}^2$.

$h = 4r$: $4\pi r^3 = 1000$, $r = 4.30$, $S = 10\pi r^2 \approx 581\ \text{cm}^2$.

The $h = 2r$ can uses less material.

3 $S = 2(25 + 50 + 50) = 250\ \text{cm}^2$; $V = 250\ \text{cm}^3$.

Cube: edge $= 6.30$ cm, $S \approx 238\ \text{cm}^2$, which is smaller.

4 Capacity $= 500\ \text{mL} = 0.5$ L. Costs per litre: cube $£0.034 \div 0.5 = £0.068$; cylinder $£0.031 \div 0.5 = £0.062$; flat box $£0.042 \div 0.5 = £0.084$.

5 Walls $= 2(6.0)(2.5) + 2(4.0)(2.5) = 30 + 20 = 50\ \text{m}^2$; ceiling $= 24\ \text{m}^2$; total $74\ \text{m}^2$.

Two coats $= 148\ \text{m}^2$; paint $= 148 \div 7.0 = 21.1$ L (buy 22 L). Cost $= 21.1 \times £9.50 \approx £201$ (or $£209$ for 22 whole litres).

6 Curved area $= 77\pi \approx 242\ \text{cm}^2$; label $= 0.85(242) \approx 206\ \text{cm}^2$.

7 Comparing prices, not prices per litre, is the error. 1.5 L: $£1.20 \div 1.5 = £0.80$ per litre; 2 L: $£1.50 \div 2 = £0.75$ per litre.

The 2 L bottle is cheaper per litre, although its price is higher.

8 $330 = \pi(3.3)^2 h$, so $h = 9.65$ cm.

$S = 2\pi(3.3)^2 + 2\pi(3.3)(9.65) \approx 268\ \text{cm}^2$. Material per litre $= 268 \div 0.330 \approx 813\ \text{cm}^2/\text{L}$.

9 Open box: $S = 20^2 + 4(20)(15) = 1600\ \text{cm}^2$; $V = 20 \times 20 \times 15 = 6000\ \text{cm}^3 = 6$ L.

A lid adds $400\ \text{cm}^2$, a 25% increase in material.

10 For a fixed volume the sphere has the least surface area, and for a fixed surface it encloses the most volume: every point is as close to the centre as possible.

It is rarely used for drinks because it rolls, does not stack or fit shelves, is awkward to hold and print, and wastes space in packing; a can or carton is more practical.

11 $h = 2r$: $S \approx 554\ \text{cm}^2$ (see 11E2).

$h = 3r$: $3\pi r^3 = 1000$, $r = 4.73$, $S = 8\pi r^2 \approx 563\ \text{cm}^2$. The $h = 2r$ cylinder is slightly smaller.

12 Before: $6(10)^2 = 600\ \text{cm}^2$. After: $8 \times 6(5)^2 = 1200\ \text{cm}^2$.

The area doubles: each of the three cuts creates two new $10 \times 10 = 100\ \text{cm}^2$ faces, adding $6 \times 100 = 600\ \text{cm}^2$.

Investigation · How much can one sheet hold?

1 A square of side x is removed from each end of a side, so the base is $20 - 2x$. The height is x, so $x > 0$; the base must have positive side, so $20 - 2x > 0$, i.e. $x < 10$. Hence $0 < x < 10$.

2 $x = 1$: $18^2(1) = 324$; $x = 2$: $16^2(2) = 512$; $x = 3$: $14^2(3) = 588$; $x = 3.5$: $13^2(3.5) = 591.5$; $x = 4$: $12^2(4) = 576$; $x = 5$: $10^2(5) = 500$; $x = 6$: $8^2(6) = 384$ (all cm^3).

3 The points rise from 324 to a peak between $x = 3$ and $x = 3.5$, then fall. The greatest volume is at about $x = 3.3$ cm.

4 $V = x(20 - 2x)^2$. At $x = \frac{10}{3} = 3.33$, $V = 3.33(13.33)^2 \approx 593\ \text{cm}^3$.

The base side is 13.3 cm and the height 3.33 cm: the base is four times the height.

5 For side S, $V = x(S - 2x)^2$. Substituting $x = \frac{S}{6}$ gives base $\frac{2S}{3}$, so $V = \frac{S}{6}\left(\frac{2S}{3}\right)^2 = \frac{2S^3}{27}$.

With $S = 20$: $V = \frac{2(8000)}{27} \approx 593\ \text{cm}^3$, as predicted.

6 A closed cylinder with the same volume $592.6\ \text{cm}^3$ is smallest when $h = 2r$: $2\pi r^3 = 592.6$, $r = 4.55$ cm, $S = 6\pi r^2 \approx 391\ \text{cm}^2$.

The open box uses about $356\ \text{cm}^2$, but the comparison is unfair: the box has no lid while the cylinder is closed. An open cylinder would use $5\pi r^2 \approx 325\ \text{cm}^2$, and thickness, seams and waste also differ.

→ **Generalise and justify.** At $x = 0$ the box has no height; at $x = 10$ the base has zero side, so $V = 0$ at both ends. Between them $V > 0$, and the table rises then falls with a single peak, consistent with a smooth curve.

A manufacturer might choose a smaller x: the maximum box is shallow and wide, so it is weaker, harder to stack and does not fit standard shelf sizes; seams and waste also reduce the saving.

Mixed practice 11

1 $S = 2(24 + 18 + 12) = 108\ \text{cm}^2$; $V = 72\ \text{cm}^3$.

2 $V = 250\pi \approx 785\ \text{cm}^3$; $S = 50\pi + 100\pi = 150\pi \approx 471\ \text{cm}^2$.

3 $l = 5$; $S = 16\pi + 20\pi = 36\pi \approx 113\ \text{cm}^2$; $V = 16\pi \approx 50.3\ \text{cm}^3$.

4 $S = 100\pi \approx 314\ \text{cm}^2$; $V = \frac{500\pi}{3} \approx 524\ \text{cm}^3$.

5 $l = 5$; $S = 36 + 4(15) = 96\ \text{cm}^2$; $V = 48\ \text{cm}^3$.

6 $S = 27\pi \approx 84.8\ \text{cm}^2$; $V = 18\pi \approx 56.5\ \text{cm}^3$.

7 Walls $= 2(5)(3) + 2(4)(3) = 54\ \text{m}^2$; ceiling $= 20\ \text{m}^2$; total $74\ \text{m}^2$. Two coats $= 148\ \text{m}^2$; paint $= 148 \div 8 = 18.5$ L.

8 $V = \pi(1.0)^2(3.0) = 3\pi \approx 9.42\ \text{m}^3 = 9420$ L.
$S = 2\pi(1) + 2\pi(1)(3) = 8\pi \approx 25.1\ \text{m}^2$.

9 $l = 13$; $S = 25\pi + 65\pi = 90\pi\ \text{cm}^2$; $V = 100\pi\ \text{cm}^3$.

10 $V = \frac{8\pi}{3}(81 + 54 + 36) = \frac{8\pi}{3}(171) = 456\pi \approx 1430\ \text{cm}^3$.

11 $V = 96\pi + \frac{128\pi}{3} = \frac{416\pi}{3} \approx 436\ \text{cm}^3$.
$S = 48\pi + 16\pi + 32\pi = 96\pi \approx 302\ \text{cm}^2$.

12 60π uses the height 12 in place of the slant height. $l = 13$.
Correct curved area $= \pi(5)(13) = 65\pi \approx 204\ \text{cm}^2$.

13 Area factor $= \frac{1}{400}$; volume factor $= \frac{1}{8000}$.
$3.2\ \text{m}^3 = 3\,200\,000\ \text{cm}^3$, so model volume $= 400\ \text{cm}^3$.

14 $k^2 = \frac{196}{49} = 4$, $k = 2$; volume ratio $= 8 : 1$.

15 $k = 1.2$: surface area factor 1.44, an increase of 44%.
Volume factor 1.728, an increase of 72.8%.

16 $2\pi r^3 = 1000$, $r = 5.42$ cm; $S = 6\pi r^2 \approx 554\ \text{cm}^2$.

17 $V_{\text{cyl}} = \pi(36)(12) = 432\pi$; $V_{\text{sphere}} = 288\pi$; empty space $= 144\pi \approx 452\ \text{cm}^3$.
Sphere $S = 144\pi \approx 452\ \text{cm}^2$; cylinder $S = 2\pi(36) + 2\pi(6)(12) = 216\pi \approx 679\ \text{cm}^2$.

18 The small cone is similar with volume ratio $\frac{1}{2}$, so its linear factor is $k = \sqrt[3]{0.5} = 0.794$.
The cut is $1 - 0.794 = 0.206$ of the height above the base, i.e. about 20.6% of the way up (79.4% down from the apex).

Criterion A · Knowing and understanding

1 $S = 2(54 + 36 + 24) = 228\ \text{cm}^2$; $V = 9 \times 6 \times 4 = 216\ \text{cm}^3$.

2 $S = 2\pi(9) + 2\pi(3)(8) = 18\pi + 48\pi = 66\pi\ \text{cm}^2$.
$V = \pi(9)(8) = 72\pi\ \text{cm}^3$.

3 $l = 10$ cm; $S = 36\pi + 60\pi = 96\pi\ \text{cm}^2$; $V = \frac{1}{3}\pi(36)(8) = 96\pi\ \text{cm}^3$.

4 $S = 4\pi(49) = 196\pi \approx 616\ \text{cm}^2$.
$V = \frac{4}{3}\pi(343) \approx 1437 \approx 1.44 \times 10^3\ \text{cm}^3$.

5 $l = 5$; $S = 64 + 4(20) = 144\ \text{cm}^2$; $V = \frac{1}{3}(64)(3) = 64\ \text{cm}^3$.

6 $S = 3\pi(25) = 75\pi \approx 236\ \text{cm}^2$; $V = \frac{2}{3}\pi(125) \approx 262\ \text{cm}^3$.

7 $V = \frac{12\pi}{3}(100 + 60 + 36) = 4\pi(196) = 784\pi \approx 2460\ \text{cm}^3$.

8 $V = 24\pi + \frac{32\pi}{3} = \frac{104\pi}{3} \approx 109\ \text{cm}^3$.
$S = 24\pi + 4\pi(4) = 40\pi \approx 126\ \text{cm}^2$, since the two hemispheres form one sphere and the joining circles are hidden.

9 Surface area factor 4; volume factor 8.

10 Area factor $\frac{1}{625}$; volume factor $\frac{1}{15\,625}$.
$1.25\ \text{m}^3 = 1\,250\,000\ \text{cm}^3$, so model $= 80\ \text{cm}^3$.

11 $2\pi r^2$ is only the curved surface of a hemisphere. A sphere has $4\pi r^2$.
Correct $S = 4\pi(36) = 144\pi \approx 452\ \text{cm}^2$.

Criterion B · Investigating patterns

1 For $k = 2$, $h = 2r$, so $V = \pi r^2(2r) = 2\pi r^3 = 1000$.
$r^3 = 159.15$, $r = 5.419 \approx 5.42$ cm. $S = 2\pi r^2 + 2\pi r(2r) = 6\pi r^2 = 6\pi(29.36) \approx 554\ \text{cm}^2$. The row is correct.

2 As k rises from 1 to 5, S falls from 586 to a minimum of 554 at $k = 2$, then rises to 563, 581 and 601. The change is not steady: it is a shallow U-shape with one minimum.

3 The minimum is at $k \approx 2$ (perhaps 2.1). It is the lowest table value, and the surface areas on both sides are higher (586 at $k = 1$, 563 at $k = 3$); a smooth curve through the points has its lowest point near $k = 2$.

4 $V = \pi r^2 h$, so $h = \dfrac{V}{\pi r^2}$. Then $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \dfrac{V}{\pi r^2} = 2\pi r^2 + \dfrac{2V}{r}$.
With $V = 1000\ \text{cm}^3$: $S = 2\pi r^2 + \dfrac{2000}{r}$.

5 Trend: S rises by about $20\ \text{cm}^2$ from $k = 4$ to $k = 5$, so predict about 620–$625\ \text{cm}^2$.
For $k = 6$: $6\pi r^3 = 1000$, $r = 3.757$ cm, $S = 14\pi r^2 \approx 621\ \text{cm}^2$. The prediction is confirmed.

6 At the minimum, $\dfrac{dS}{dr} = 4\pi r - \dfrac{2000}{r^2} = 0$, so $r^3 = \dfrac{500}{\pi}$, $r = 5.42$ cm and $h = \dfrac{1000}{\pi r^2} = 10.84\ \text{cm} = 2r$. The optimum is therefore a can as tall as it is wide.
A real can may be taller and narrower to suit standard diameters, filling and drinking, stacking and transport, label space, and thicker ends; these practical costs are not in the model.

→ **Marking guidance.** Q1–Q3 (10 marks): correct substitution, radius and surface area, a described minimum and a justified estimate. Q4–Q5 (9 marks): clear rearrangement, correct expression, sensible prediction checked by calculation. Q6 (5 marks): a mathematical reason for $h = 2r$ plus at least two practical limitations.

Criterion C · Communicating

i Proposed cylinder: $V = 10\ \text{m}^3$. Choosing the efficient proportion $h = 2r$ gives $2\pi r^3 = 10$, so $r = 1.17$ m and $h = 2.34$ m.
Proposed cuboid: a cube of side $\sqrt[3]{10} = 2.15$ m, or $2.5 \times 2.0 \times 2.0$ m.

ii Cylinder: $S = 2\pi(1.17)^2 + 2\pi(1.17)(2.34) \approx 8.6 + 17.2 = 25.7\ \text{m}^2$ (with lid).
Cube: $S = 6(2.15)^2 \approx 27.9\ \text{m}^2$. The $2.5 \times 2.0 \times 2.0$ cuboid gives $S = 2(5.0 + 5.0 + 4.0) = 28.0\ \text{m}^2$.

iii The cylinder uses about $2.2\ \text{m}^2$ less material, about 8% less. Its footprint is $\pi(1.17)^2 \approx 4.3\ \text{m}^2$, slightly less than the cube's $4.6\ \text{m}^2$.

iv Assumptions: the tank is closed and stands on level ground; wall thickness and seams are ignored; material cost is proportional to surface area; the tank is not buried or supported by a frame.

v Recommend the cylinder: it needs less sheet material for the same 10 m^3, has a slightly smaller footprint, and has no corners where stress concentrates. A cuboid is easier to build from flat panels and fits a rectangular compound, so the saving must be weighed against construction.

→ **Marking guidance.** 20 marks, about 4 per requirement: correct dimensions with units; two surface-area calculations; a comparison using area and footprint; two specific assumptions; a recommendation that follows from the numbers. A labelled sketch is required for full marks.

Criterion D · Applying mathematics in real-life contexts

1 Roof area $= 12 \times 8 = 96$ m^2. Rainfall $= 620$ mm $= 0.620$ m.

Volume $= 96 \times 0.620 = 59.52$ m^3; collected $= 0.85(59.52) = 50.6$ m^3.

2 Tank: $r = 1.5$ m, $h = 2.5$ m, $V = \pi(1.5)^2(2.5) = 17.67 \approx 17.7$ m^3.

Fills $= 50.59 \div 17.67 = 2.86 \approx 2.9$ times per year.

3 Saving $= 50.59 \times £1.80 \approx £91.1$ per year.

Payback $= £2400 \div £91.1 = 26.4 \approx 26$ years.

4 Rainfall: 620 mm is a long-term average; a dry year collects much less, so the saving varies and the payback may be longer. Collection: 85% assumes little evaporation, splashing or first-flush loss, and a roof material that is safe for the water. Use: the tank holds only 17.7 m^3, so without steady use much of the 50.6 m^3 overflows; the saving assumes all stored water replaces bought water. The tank also has maintenance and pump costs.

5 Recommend a smaller tank (about 5 m^3) or a staged installation: the full tank's 26-year payback is close to or beyond its service life, so the financial case is weak. The project still has educational, stormwater and water-restriction benefits. Distinguish the certain capital cost from the uncertain annual saving, and monitor rainfall and use for a year before enlarging.

Marking guidance. Criterion D (24 marks): Q1 5 marks for converting mm to m, multiplying by area and applying 85%; Q2 5 marks for correct capacity and division; Q3 5 marks for the saving and payback with units; Q4 5 marks for three specific assumptions, including rainfall and use; Q5 4 marks for separating financial payback from other benefits and making a justified recommendation. Accept consistent rounding; the key check is that units are m^3, litres or pounds throughout.

CHAPTER 12 · STATISTICS: DISTRIBUTIONS AND DATA

Opening problem · two classes, one average

1 Both scores total 630, so both means are $\frac{630}{9} = 70$ marks.

A: median = 5th value = 70; mode = 70, which occurs three times.

B: median = 70; there is no mode, because every score occurs once.

2 A: $85 - 55 = 30$ marks. B: $100 - 40 = 60$ marks, twice the range.

3 A: one dot at 55, 60, 65, 75, 80 and 85 and three stacked at 70 — a single cluster, symmetric about 70 and reaching 15 marks either side.

B: one dot at each of 40, 50, 60, 65, 70, 75, 80, 90 and 100 — flat, spread along the whole axis and with gaps at 85 and 95. Same centre, very different shape.

4 On centre neither class is better: mean and median are both 70, so the average alone cannot separate them.

A is the more consistent class: range 30 against 60, IQR $= 77.5 - 62.5 = 15$ against $85 - 55 = 30$, and $\sigma = \sqrt{700 \div 9} = 8.82$ against $\sigma = \sqrt{2850 \div 9} = 17.8$. B holds the two highest scores, 100 and 90, where A's best is 85, but also the two lowest, 40 and 50.

Judged on typical and dependable performance, Class A performed better; Class B is the better class only if the school is looking for the very highest scorers.

5 Taking a pass as 60 or above: A has 8 of the 9, $\frac{8}{9} = 0.889 = 88.9\%$; B has 7 of the 9, $\frac{7}{9} = 0.778 = 77.8\%$.

If 60 itself did not count as a pass the figures would be $\frac{7}{9} = 77.8\%$ *and* $\frac{6}{9} = 66.7\%$, *in the same order.*

6 How the data were produced: whether both classes took the same paper under the same conditions and with the same teacher; the full class sizes and whether all nine students sat the test; prior attainment; and whether the marks are exact or rounded.

How the results should be reported: whether these nine are the whole class or a sample of a larger cohort; the shape and spread of each distribution, not just the mean; and whether a pass rate from nine students is reliable enough to publish.

Prerequisite check

1 Mean $= \frac{4 + 7 + 9 + 10}{4} = \frac{30}{4} = 7.5$.

2 Ordered: 3, 4, 5, 8, 9; median = 5.

3 Mode = 7 (it occurs three times).

4 Range $= 24 - 9 = 15$.

5 $\frac{3}{8} = 0.375 = 37.5\%$.

6 $-2.5,\ -0.4,\ 1.8,\ 2.1$.

Exercise 12A

1 (a) discrete; (b) continuous; (c) qualitative; (d) continuous; (e) discrete; (f) qualitative.

2 (a) primary; (b) secondary; (c) primary; (d) secondary.

3 Population: all students at the school. Sample: for example, 60 students chosen at random from the school roll, or 10 chosen at random from each year group.

4 A census measures every member of the population; a sample measures a subset. Census advantage: complete coverage, so no sampling error. Sample advantage: cheaper and faster, and more detail can be collected from each person.

5 Examples: (a) number of siblings; (b) time taken to travel to school; (c) favourite type of music.

6 Goals are counted in separate whole events: 0, 1, 2, and so on, with no value between 1 and 2. Time is measured and can take any value in an interval, such as 12.37 minutes, so it is continuous.

7 "Don't you agree..." is leading: it invites agreement and supplies a reason on one side. A neutral version is: "At what time should the school day start? Give your reason."

A neutral question offers balanced options, e.g. later, earlier or unchanged, without arguing for one.

8 Everyone surveyed has already chosen to fly and may be a visitor rather than a resident. The sample excludes people who live locally and use other transport, and passengers on different routes have different needs, so their views need not represent the city's travellers.

9 Sampling fraction $= 90 \div 900 = 0.1$. Year 7: 30; Year 8: 32; Year 9: 28. Check: $30 + 32 + 28 = 90$.

10 Twenty classmates are a convenience sample, probably from one class or year group. They may differ from the whole school in age, gender and activity, and 20 is small, so the mean 1.62 m describes that group rather than all students.

11 Size reduces random variation, not bias. If the 100 people are chosen from one group or in a way that excludes others, the sample is still unrepresentative; a smaller random sample can be better.

12 Who commissioned the survey and who was asked; how the sample was selected and how large it was; how "teenager" was defined and in which country; the exact wording and options; when it was done; whether "prefer" means actual experience or opinion.

13 Reliability is consistency: repeating the measurement gives similar results. Validity is whether the measurement captures what it claims. The faulty balance always reads 0.5 kg too high, so repeated readings agree (reliable) but every reading is wrong (not valid).

14 No. $24/40 = 60\%$ is the proportion in that sample. The 40 students may not have been randomly chosen, and another sample would give a different proportion, so 60% is an estimate for the school, not a confirmed result.

15 Population: all students at the school. Method: a stratified random sample by year group, chosen from the roll. Questions: "How many minutes did you spend on homework last night?" and "On how many of the last five school nights did you do homework?" Avoid a leading question, a convenience sample and loss of anonymity; keep the recall period short.

16 Two-stage stratification guarantees that both year group and gender appear in the sample in the right proportions, so differences between boys and girls within a year can be examined. A practical difficulty is that it needs a list with both pieces of information, and some subgroups become very small, which adds cost and complexity.

17 Customers who contacted support are disproportionately those with a complaint, so dissatisfied customers are over-represented and the satisfied majority is missing. The result will underestimate overall satisfaction. A better sampling frame is a random sample of all customers, or of all recent purchasers, including those who never contacted support.

Exercise 12B

1 Midpoints: 0.5, 1.5, 2.5, 3.5. $\sum fx = 4(0.5) + 9(1.5) + 11(2.5) + 6(3.5) = 2 + 13.5 + 27.5 + 21 = 64$. Mean $= 64 \div 30 = 2.13$ kg (3 s.f.).

2 Midpoints 15, 25, 35: $\sum fx = 75 + 300 + 280 = 655$, $n = 25$, mean $= 26.2$.

3 The class starts at 20 and ends at 30. It includes 20 but not 30, so a value of exactly 30 belongs to the next class and no value is counted twice: $20 \leq x < 30$.

4 With equal widths, height represents frequency, so height 8 is a frequency of 8. The area is $5 \times 8 = 40$ square units, which also represents that frequency; area is proportional to frequency.

5 Frequency density $= 15 \div 10 = 1.5$ per unit of width.

6 Frequency $= 0.8 \times 20 = 16$.

7 Classes 140–150, 150–160, 160–170, 170–180, 180–190. A possible set of frequencies is 4, 9, 16, 14, 7, which sums to 50. The bars touch because the classes are adjacent intervals of a continuous variable; there is no gap between 149.9 and 150.

8 A long right tail pulls the mean towards the large values, so the mean is greater than the median.

9 With unequal widths, height must be frequency density, $f \div$ width, so that area, not height, represents frequency. Using frequency makes wide classes look too important and distorts the shape.

10 20–50: $20 + 20 = 40$ visitors. 50 or over: $8 + 4 = 12$ visitors.

11 The table records only class totals. The median position falls inside a class, and the individual values in that class are unknown, so the median can only be estimated by assuming the values are evenly spread.

12 The 20–30 class has frequency 20, but it is only 10 years wide, so its density is $20 \div 10 = 2.0$. The 30–50 class also has 20 visitors but is 20 years wide, density 1.0. Frequency alone favours narrow classes; the concentration in the twenties is real but the comparison needs density, and the data cannot show where inside 20–30 the values lie.

13 For example, frequencies 10, 20, 20 in classes 0–10, 10–20, 20–40 give densities 1.0, 2.0, 1.0, with a peak at 10–20. Regrouping the same data as 0–20 (30) and 20–40 (20) gives density 1.5 and 1.0: a flatter shape with the peak at the lower class. The shape depends on the chosen widths, so density must be used.

14 The labels 0–4, 5–9, 10–14 leave gaps (4 to 5, 9 to 10), so the bars would not touch. Use continuous boundaries such as 0–5, 5–10, 10–15, or −0.5 to 4.5, 4.5 to 9.5, and so on.

15 Widths 10, 20, 20, so densities are 1.0, 1.25 and 0.75.

Midpoints 5, 20, 40: $\sum fx = 50 + 500 + 600 = 1150$, $n = 50$, mean $= 23$.

16 Midpoints 5, 15, 25, 35: $\sum fx = 40 + 210 + 450 + 350 = 1050$, mean $= 1050 \div 50 = 21$ minutes.

The estimate assumes each value sits at its class midpoint. If waiting times cluster near the start of each class the true mean is lower; if near the end, higher.

17 For example: 200–220, 220–240, 240–260, 260–280, 280–300, 300–320, 320–340, 340–360, 360–380, 380–400. Start at 200 by rounding the smallest observation down to a multiple of 20; choose the range to cover the largest. Equal widths let height represent frequency, so bars are directly comparable and area remains proportional.

Exercise 12C

1 Cumulative frequencies: 5, 13, 25, 30.

2 Median: mean of 10th and 11th, position 10.5. Lower quartile: mean of 5th and 6th, position 5.5. Upper quartile: mean of 15th and 16th, position 15.5.

3 Minimum 3; $Q_1 = 6$; median $= 11$; $Q_3 = 15$; maximum 20.

4 IQR $= 31 - 18 = 13$; range $= 45 - 12 = 33$.

Fences: $18 - 1.5(13) = -1.5$ and $31 + 1.5(13) = 50.5$. There are no outliers.

5 Cumulative frequencies 6, 16, 30, 37, 40. The median is at position 20, which lies between the cumulative totals 16 (at 20 min) and 30 (at 30 min), so the median is in the class 20–30.

6 IQR $= 44 - 22 = 22$ minutes.

7 A cumulative frequency is a running total of non-negative frequencies, so it can stay the same but cannot decrease as the variable increases.

8 The cumulative total counts every value at or below the upper boundary. Plotting at the midpoint would count only part of the class and shift the curve to the left.

9 The midpoint is only a representative of the class. Unless the values are evenly spread, the true median can lie elsewhere; the estimate is poor for skewed data or for classes of unequal width.

10 Either the data are right-skewed, with a few unusually large values; or the maximum is a genuine extreme value/outlier; or the upper quartile is close to the median while the top quarter is spread over a long range.

11 The middle halves are equally wide and centred at the same value, but the tails differ: one set has longer whiskers or outliers, so its minimum or maximum is further from the box.

12 Yes. $1, 2, 3, 4, 5, 6, 7$ and $1, 2, 4, 4, 4, 6, 7$ both have minimum 1, $Q_1 = 2$, median 4, $Q_3 = 6$ and maximum 7, but different middle values.

13 Check whether it is a recording or typing error; whether it is plausible in the context (a genuine extreme case); whether the measurement method was consistent; and how it affects the conclusion. Report the analysis with and without it rather than deleting it silently.

14 The whiskers reach the most extreme values that are not outliers, not fixed quartile positions. If outliers are present, a whisker covers less than a quarter of the data; even without outliers, the values within a whisker need not be evenly spread.

15 Ordered: 32, 35, 37, 41, 44, 46, 48, 50, 52, 52, 55, 57, 59, 61, 63, 64.

Five-number summary: min 32, $Q_1 = 42.5$, median $= 51$, $Q_3 = 58$, max 64. IQR $= 15.5$; fences 19.25 and 81.25, so no outliers. Whiskers reach 32 and 64.

16 Cumulative frequencies 6, 18, 32, 47, 50. Positions: Q_1 at 12.5, median at 25, Q_3 at 37.5.

$Q_1 = 5 + \frac{12.5-6}{12}(5) = 7.71$; median $= 10 + \frac{25-18}{14}(5) = 12.5$; $Q_3 = 15 + \frac{37.5-32}{15}(5) = 16.83$. IQR ≈ 9.1 min.

17 A: median 9; $Q_1 = 4.5$; $Q_3 = 13.5$; IQR 9; range 16.

B: median 9; $Q_1 = 8$; $Q_3 = 10$; IQR 2; range 16.

The same median and range hide very different middles: B is concentrated near 9, while A is spread across the whole range. Range alone is therefore a poor description of spread.

18 The 20 values are ordered; the median uses the 10th and 11th, Q_1 the 5th and 6th, and Q_3 the 15th and 16th. Replacing the largest value leaves all these positions unchanged, so the median and quartiles do not change.

The sum, and therefore the mean, increases; the range increases because the maximum increases.

Exercise 12D

1 Range $= 18 - 4 = 14$. $n = 6$, median $= 10.5$, $Q_1 = 7$, $Q_3 = 15$, so IQR $= 8$.

2 Mean $= 6$. Deviations $-4, -2, 0, 2, 4$, squares sum to 40. $\sigma^2 = 40 \div 5 = 8$, so $\sigma = 2.83$.

3 $\sigma = 0$. Every value equals the mean 5, so every deviation is zero and there is no spread.

4 B has the larger standard deviation: its values lie much further from the mean of 10.

5 IQR $= 40 - 20 = 20$. Fences: $20 - 1.5(20) = -10$ and $40 + 1.5(20) = 70$.

6 Most values are likely between 46 and 54, within one standard deviation of the mean. This is not guaranteed: the distribution may be skewed or contain outliers, and the rule depends on the shape.

7 $\sum(x - \bar{x}) = \sum x - n\bar{x} = \sum x - n\left(\frac{\sum x}{n}\right) = 0$. Positive and negative deviations balance.

8 Squaring makes every deviation positive, so they cannot cancel, and it gives more weight to values far from the mean. The square root then returns the measure to the original units.

9 Mean increases by 10; standard deviation, range and IQR are unchanged, because every value and the mean shift together.

10 Mean doubles; standard deviation doubles; range doubles; IQR doubles. All are measured in the original units.

11 36 is the variance, not the standard deviation. Take the square root: $\sigma = \sqrt{36} = 6$.

12 Median and IQR. Incomes are right-skewed, so the mean and standard deviation are pulled upward by a few very high incomes and describe no typical person.

13 Check for a recording or data-entry error; check whether the value is genuine in context; check whether the measuring instrument or method changed; consider reporting results with and without it.

14 The class with standard deviation 2 is more consistent: its scores cluster more tightly around the mean. You would also want the sample sizes, the shapes of the distributions and the pass rates before deciding which class performed better.

15 $n = 10$, $\sum fx = 22$, mean $= 2.2$.

Squared deviations: $2(1.44) + 5(0.04) + 2(0.64) + 1(3.24) = 7.6$. $\sigma^2 = 0.76$, $\sigma = 0.872$.

16 One set is 7, 9, 10, 11, 13: mean 10, squared deviations $9 + 1 + 0 + 1 + 9 = 20$, $\sigma^2 = 4$, $\sigma = 2$.

It is not unique: many sets have the same mean and the same total of squared deviations, for example by moving values in opposite directions.

17 Reject outside 496–504 g. Machine A: mean 498, $\sigma = 2$, so 496 is one standard deviation below the mean and almost nothing exceeds 504; roughly 16% are rejected. Machine B: 496 and 504 are each 0.67σ from 500, so about half the bags lie outside the limits.

Recommend A, but adjust its setting upward to 500 g; this assumes roughly symmetric filling and that the standard deviation stays constant.

18 Adding c: new mean $\bar{x} + c$; deviations $(x + c) - (\bar{x} + c) = x - \bar{x}$ are unchanged, so σ is unchanged.

Multiplying by k: new mean $k\bar{x}$; deviations become $k(x - \bar{x})$, squares gain k^2, so new $\sigma = \sqrt{k^2\sigma^2} = |k|\sigma$.

Exercise 12E

1 Year 10: median ≈ 3.5 h, $Q_1 \approx 2.5$, $Q_3 \approx 5.0$, IQR ≈ 2.5 h. Year 12: median ≈ 2.5 h, $Q_1 \approx 1.5$, $Q_3 \approx 3.5$, IQR ≈ 2.0 h.

2 Year 10 has the higher median (3.5 h vs 2.5 h) and the greater IQR (2.5 h vs 2.0 h).

3 The bars are measured from 90, so 94 appears 4 units tall and 96 appears 6 units tall: a 50% increase in bar height for a change of about 2.1%. The truncated axis exaggerates the difference.

4 Class B is more consistent: standard deviation 3 is smaller than 8. It also has the higher mean (65 vs 60).

5 Who collected the data and from what sample; whether the sample was random and how large it was; whether the axis starts at zero; whether a measure of spread is shown; what exactly was measured.

6 (a) association; (b) causal claim; (c) association; (d) causal claim.

7 Overlap means many individuals in the lower-median group are above many in the higher-median group. A higher median describes the typical value, not every value, and the difference must be judged against the spread.

8 Without spread you cannot tell whether a difference is large compared with the variation, or whether the groups overlap substantially. The same mean difference can be trivial or important.

9 Perspective makes front slices look larger; the depth/height adds area or volume that grows faster than the angle; labels may be hard to match to slices; and a truncated or inconsistent scale can mislead.

10 Sports-club members are more active than the general student population and chose to join, so 85% is a biased estimate for all students in the town.

11 No. Hot summer weather is a confounder: it increases both ice-cream sales and swimming, and therefore drownings. The association does not show that ice cream causes drowning.

12 One paper may quote the mean, the other the median; or they may use different periods, areas or samples. Both can be correct for the quantity they measured.

13 The visible axis covers only 48–52, so a 2% change occupies a large fraction of the height. The graph should state that the axis is truncated, label the actual values, or start the axis at zero.

14 The mean is a balance point, not a mode. Scores could be spread widely on both sides of 70, with few or none equal to 70; the shape of the distribution is needed.

15 School B has the higher median (74 vs 68) and the smaller IQR (9 vs 18), so its typical score is higher and its middle group is more consistent. School A's larger sample (200 vs 25) makes its statistics more stable.

A researcher should obtain the full distributions, confirm that the samples are random and comparable, and compare pass rates. Limitations: different tests or cohorts, and the small sample at B.

16 Year 10 reports a higher median screen time, about 3.5 hours against 2.5 hours, so its typical student reports one hour more. The middle half of Year 10 is slightly more spread out (IQR 2.5 h vs 2.0 h), and its range is longer (7.0 h vs 5.5 h), with a maximum of 8 hours.

Both distributions are right-skewed, and the boxes overlap from 2.5 to 3.5 hours, so the groups are not cleanly separated: many Year 12 students report more than many Year 10 students. The data are self-reported and the samples are not described, so the comparison describes these groups rather than proving a difference between all Year 10 and Year 12 students.

17 Start the vertical axis at 0 and run it to at least 80 or 100, so bar heights are proportional to the means. Label both axes with units, state the sample size for each class, and add a measure of spread (standard deviation or IQR) or use dot/box plots. Include the test and the date, and avoid 3D effects.

18 Variables: start time (the intervention) and sleep duration, with bedtime, screen use, age and travel time as covariates. Comparison group: classes or schools keeping the 08:15 start. Measure sleep with a wearable or a sleep diary for two weeks before and after. Aim for several hundred students across several schools.

Reduce confounding by random assignment of schools or classes to start times, and by measuring the same students before and after. Ethical issues: informed consent from students and parents, privacy of sleep data, and fairness in disrupting families' schedules.

Investigation · Does a screen in bed shorten sleep?

1 Population: all students at the school (or all Year 10 students). Sample: the 28 students in the table, 14 in each group. The study is observational: the students chose their own screen habits and no treatment was assigned.

2 Screen: min 5.5, $Q_1 = 6.5$, median 7.0, $Q_3 = 8.0$, max 9.0; IQR 1.5; fences 4.25 and 10.25, no outliers.

No screen: min 6.5, $Q_1 = 7.5$, median 8.5, $Q_3 = 9.5$, max 10.5; IQR 2.0; fences 4.5 and 12.5, no outliers. Parallel box plots share the axis 5–11 hours.

3 The no-screen group's median is 1.5 hours higher (8.5 vs 7.0). Its middle half is slightly more spread out (IQR 2.0 vs 1.5) and its range is slightly longer (4.0 vs 3.5).

The distributions overlap between 7.5 and 9.0 hours: some students with screens sleep longer than some without. Typical sleep is lower with a screen, but the groups are not completely separated.

4 Supported: the median difference is exactly 1.5 hours, and the no-screen median is higher. Not supported: "reduces" is causal, but the data are observational, the groups were not assigned, and confounders may explain the difference. "By 1.5 hours" is also a group median, not an effect on every student.

5 Age/year group: older students may have later bedtimes, more screens and less sleep. Household rules or bedtime routine: stricter routines may reduce both screen use and sleep loss. Homework or stress: more late work could increase both screen use and shorter sleep. Each could produce the observed difference without the screen causing it.

6 Randomly assign volunteers to a screen-free-bedroom condition or to their usual routine for two weeks, and measure sleep objectively with a wearable plus a diary. Aim for at least 100–200 students, balanced across year groups.

Ethics: informed consent from students and parents, privacy of the sleep data, and the right to withdraw; do not deprive anyone of help if sleep problems are found.

7 A causal claim needs: random assignment to screen and control conditions, so confounders are balanced; a sufficiently large and representative sample; objective, consistent measurement of both screen use and sleep; and replication, ideally with a plausible mechanism and a dose–response pattern.

→ **Generalise and justify.** A difference between groups shows only that screen use and sleep are associated. Students who use screens may already differ in age, bedtime rules, workload or stress, and these could reduce sleep. A randomized study would make the groups similar in these respects, so a later difference could be attributed to the screen rather than to pre-existing differences.

Mixed practice 12

1 (a) discrete; (b) continuous; (c) qualitative; (d) discrete; (e) continuous; (f) discrete.

2 A census measures every member; a sample measures a subset. Census advantage: complete, no sampling error. Sample advantage: cheaper, faster and more detailed.

3 Mean $= 54 \div 6 = 9$; median $= (7 + 9) \div 2 = 8$; mode 7; range $= 15 - 4 = 11$.

4 Midpoints 10, 30, 50: $\sum fx = 50 + 270 + 300 = 620$; $n = 20$; mean $= 31$.

5 Densities $12 \div 5 = 2.4$ and $15 \div 10 = 1.5$. The first bar is taller (2.4 vs 1.5).

6 IQR $= 30 - 16 = 14$; range $= 42 - 10 = 32$; fences $16 - 21 = -5$ and $30 + 21 = 51$. No outliers.

7 Cumulative frequencies 4, 11, 20, 25. The median is at position 12.5, so it lies in the class 20–30.

8 Mean 7; deviations $-4, -2, 0, 2, 4$; squares sum to 40; $\sigma^2 = 8$; $\sigma = 2.83$.

9 Mean increases by 10. Standard deviation, range and IQR are unchanged.

10 IQR $= 38 - 20 = 18$; fences $20 - 27 = -7$ and $38 + 27 = 65$. Since $70 > 65$, 70 is an outlier.

Before removing it, check for a recording error, whether it is a genuine value in context, and whether the measurement was consistent; report the analysis with and without it.

11 Class A is more consistent: standard deviation 3 is much smaller than 11. You would also want the sample sizes, the shapes of the distributions and the pass rates.

12 The axis starts at 70, so 71 is 1 unit above the baseline and 74 is 4 units: the visible height quadruples for a change of 3 marks. Start the axis at 0, or label the truncated axis clearly and show the actual values.

13 Students choose tutoring, so they may already be more motivated or better supported. Prior attainment, parental support or income are confounders that affect both tutoring and scores; the study is observational, so "boosts" is too strong.

14 Only students who visit the school website can respond, so the sample is self-selected and excludes others. Those with strong views are more likely to reply, so the results may not represent all students.

15 Population: Year 10 students. Method: stratified random sample by class, chosen from the roll. Questions: "How many hours did you spend on homework last week?" and "How many minutes did you spend on a typical weeknight?" Avoid leading questions, convenience sampling and vague recall periods; keep answers anonymous.

16 Densities: $10 \div 10 = 1.0$, $20 \div 10 = 2.0$, $20 \div 20 = 1.0$. The 10–20 class has the greatest concentration; its bar is twice as tall as the others because its frequency is packed into the same 10-year width.

17 A mean above the median suggests a right-skewed distribution: a few expensive houses pull the mean up. A buyer should use the median $£280{,}000$, because it describes a typical price and is not distorted by the expensive tail.

18 A: mean 5, range 10, $\sigma = \sqrt{10} = 3.16$. B: mean 5, range 10, $\sigma = \sqrt{20} = 4.47$.

Both have the same extremes but B has more values at the extremes and fewer in the middle, so its spread is larger. Range uses only two values and ignores the middle, so it cannot distinguish these sets.

Criterion A · Knowing and understanding

1 (a) discrete; (b) continuous; (c) qualitative; (d) discrete; (e) continuous; (f) qualitative.

2 A census measures every member; a sample measures a subset. Census advantage: complete coverage, no sampling error. Sample advantage: cheaper, faster, and more detail per person.

3 Bias 1: the library sample is not representative — those students may be more studious. Bias 2: the question is leading ("Don't you agree…" plus a one-sided reason). Improvement: take a random sample from the full year list and ask a neutral question, e.g. "How many minutes of homework did you do last night?"

4 Sum $= 155$, mean $= 19.375$; median $= (18 + 20) \div 2 = 19$; mode 15; range $= 28 - 12 = 16$.

5 Midpoints 5, 15, 25: $\sum fx = 20 + 135 + 175 = 330$; $n = 20$; mean $= 16.5$.

6 Densities: $12 \div 5 = 2.4$, $15 \div 10 = 1.5$, $10 \div 10 = 1.0$. The 0–5 bar is tallest: it packs 12 values into half the width of the 5–15 class, so its density is greatest.

7 Cumulative frequencies 8, 24, 36, 40. Positions 10, 20, 30.

$Q_1 = 10 + \frac{2}{16}(10) = 11.25$; median $= 10 + \frac{12}{16}(10) = 17.5$; $Q_3 = 20 + \frac{6}{12}(10) = 25$. IQR $= 13.75$.

8 Minimum 3; $Q_1 = 6$; median 10; $Q_3 = 15$; maximum 18.

9 Mean 8; squares of deviations $16 + 4 + 0 + 4 + 16 = 40$; $\sigma^2 = 8$; $\sigma = 2.83$.

10 IQR $= 45 - 25 = 20$; fences $25 - 30 = -5$ and $45 + 30 = 75$. Since $80 > 75$, 80 is an outlier.

11 Both classes have the same mean, but Class A is much more consistent (standard deviation 4 vs 10). Class B's scores are more spread out, with more very high and very low scores.

Criterion C · Communicating

1 The question is whether students who use a screen in bed sleep less than those who do not. The source is Table 12.7: self-reported hours of sleep for 28 students, 14 in each group.

2 Screen group: median 7.0 h, IQR 1.5 h, range 3.5 h (5.5 to 9.0). No-screen group: median 8.5 h, IQR 2.0 h, range 4.0 h (6.5 to 10.5). The medians differ by 1.5 h; the no-screen group is slightly more spread out.

3 The distributions overlap between 7.5 and 9.0 hours: some screen users sleep longer than some non-users, so the groups are not completely separated. The difference is in the typical value, not in every individual.

4 The data are self-reported and the groups were not randomly assigned: students chose their habits, and age, bedtime rules, homework and stress could affect both screen use and sleep. The sample is only 28 students from one school.

5 "Far less" is not supported. A median difference of 1.5 hours is substantial, but the groups overlap and the data are observational, so they show an association, not a causal effect. A fair statement is: "In this sample, students who used a screen in bed reported a median of 1.5 hours less sleep; the groups overlap and the study cannot show that the screen caused the difference."

→ **Marking guidance.** 20 marks: 4 for a clear question and source; 6 for accurate median, IQR and range with units; 4 for a specific comment on overlap; 3 for a limitation tied to these data; 3 for a recommendation that follows from the statistics. A report that quotes means only, or claims causation, should not reach the top band.

Criterion D · Applying mathematics in real-life contexts

1 Screen: min 5.5, $Q_1 = 6.5$, median 7.0, $Q_3 = 8.0$, max 9.0; IQR 1.5 h.

No screen: min 6.5, $Q_1 = 7.5$, median 8.5, $Q_3 = 9.5$, max 10.5; IQR 2.0 h.

Typical sleep is 1.5 h higher without a screen; the no-screen middle half is 0.5 h wider, and its range is 4.0 h vs 3.5 h.

2 The data show an association between screen use and 1.5 h lower median sleep in this sample, not an effect of start time. There are no data on the 08:15 start, and the groups overlap, so "for every student" is unsupported. Even if a later start helped, the gain would be a group average, not a guaranteed 30 minutes each.

3 Variables: start time (independent) and sleep duration (dependent), with bedtime, screen use, age and travel time recorded. Comparison: classes or schools keeping 08:15. Measure sleep with a wearable plus a diary for two weeks before and after. Sample: several hundred students across several schools, randomly assigned to start times. Ethics: consent from students and parents, privacy, and fair treatment of the comparison group.

4 Measurement: self-reported sleep is affected by recall and by wanting to give a "good" answer; a wearable reduces this but can misclassify time in bed as sleep. Students: one school's volunteers may differ in age, commute and screen habits from the wider population, so results may not generalize. A novelty effect or seasonal change could also alter sleep during the trial.

5 Recommend a pilot, not an immediate permanent change. The evidence shows only an association between screens and shorter sleep; it does not establish that a later start adds 30 minutes. A pilot with random assignment and measured sleep would test the claim. Weigh the educational and wellbeing benefits against transport timetables, family work schedules, canteen and supervision costs, and disruption to after-school activities; decide using both the evidence and these values.

Marking guidance. Criterion D (24 marks): Q1 6 marks for two five-number summaries and a centre/spread comparison; Q2 5 marks for separating the association from the causal claim and noting the overlap; Q3 6 marks for variables, comparison group, measurement, sample size and ethics; Q4 4 marks for two limitations, one measurement and one sample; Q5 3 marks for a recommendation that separates statistical evidence from transport, family and cost considerations.

CHAPTER 13 · PROBABILITY: COMBINED EVENTS

Opening problem · a game with two dice

1 Totals 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 occur in 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1 ways.

Check: $1+2+3+4+5+6+5+4+3+2+1=36$, the 36 equally likely ordered pairs.

2 $P(\text{total } 7)=\frac{6}{36}=\frac{1}{6}\approx 0.167$

$P(\text{total } 8)=\frac{5}{36}\approx 0.139$

Both lie between 0 and 1, as required.

3 You: $72\times\frac{1}{6}=12$ wins. Your friend: $72\times\frac{5}{36}=10$ wins.

The other 50 games decide neither player. These are expected frequencies for 72 plays, not a guaranteed result of one run of 72.

4 No, it is not fair. $\frac{1}{6}=\frac{6}{36}>\frac{5}{36}$, so you win more often, by $\frac{1}{36}$ of the games — two extra wins in every 72.

Fairness belongs to the rules and their probabilities, not to one outcome: a single game always has a winner and a loser, and a short run can go either way by chance. The two events are mutually exclusive, so no play is a tie.

5 Let your friend win on doubles instead of on a total of 8: $(1,1),(2,2),\ldots,(6,6)$ is 6 of the 36 outcomes, so $P=\frac{6}{36}=\frac{1}{6}=P(\text{total } 7)$, and a double can never total 7, so the two chances are equal and never both happen.

Also acceptable: change your own target from 7 to 6, so each player has $\frac{5}{36}$.

6 The 11 totals are not equally likely outcomes, so dividing by 11 is invalid. The equally likely outcomes are the 36 ordered pairs, and a total of 2 or 12 can be made in only 1 way while a total of 7 can be made in 6.

$\frac{1}{11}\approx 0.091$ badly understates the true $P(7)=\frac{1}{6}\approx 0.167$. The rule $P(A)=\frac{n(A)}{n(S)}$ works only when every outcome in S is equally likely.

Prerequisite check

1 $\frac{3}{8}=0.375=37.5\%$

2 $\frac{18}{24}=\frac{3}{4}$ (divide both parts by 6).

3 $1-\frac{2}{7}=\frac{5}{7}$

4 $0.6\times 0.6=0.36$

5 $35\%=\frac{35}{100}=\frac{7}{20}$

6 AA, AB, BA, BB — four arrangements. Order matters, so AB and BA are different.

7 Four: HH, HT, TH, TT.

8 13 hearts, one quarter of the pack.

Exercise 13A

1 $S=\{1,2,3,4,5,6\}$, $n(S)=6$.

(a) $\frac{1}{6}$ (b) $\frac{3}{6}=\frac{1}{2}$ (c) $\frac{2}{6}=\frac{1}{3}$ (d) 0, since no face is greater than 6.

2 (a) $\frac{1}{8}$ (b) $\frac{4}{8}=\frac{1}{2}$ (c) 3 and 6, so $\frac{2}{8}=\frac{1}{4}$ (d) $\frac{7}{8}$

3 $6+4+5=15$ counters.

(a) $\frac{6}{15}=\frac{2}{5}$ (b) $\frac{4}{15}$ (c) $\frac{10}{15}=\frac{2}{3}$ (d) $\frac{10}{15}=\frac{2}{3}$

(c) and (d) are the same event: not yellow means red or blue.

4 $S=\{HH,HT,TH,TT\}$, $n(S)=4$.

(a) $\frac{1}{4}$ (b) $\frac{3}{4}$ (c) $\frac{2}{4}=\frac{1}{2}$

5 (a) $\frac{13}{52}=\frac{1}{4}$ (b) $\frac{4}{52}=\frac{1}{13}$ (c) $\frac{26}{52}=\frac{1}{2}$ (d) $\frac{39}{52}=\frac{3}{4}$

6 $1-0.18=0.82$

7 The coins are taken in order, so HT and TH are different outcomes; each of the four has probability $\frac{1}{4}$. Listing only HH, HT, TT would make the outcomes unequally likely, since HT then carries $\frac{1}{2}$.

8 Two possible outcomes need not be equally likely. The complement rule gives $P(\text{no rain})=1-0.4=0.6$, not 0.4.

9 (a) $\frac{6}{36}=\frac{1}{6}$ (b) $\frac{5}{36}$ (c) 10, 11, 12 give $3+2+1=6$ ways, so $\frac{1}{6}$ (d) 18 even totals, so $\frac{1}{2}$

Common error: two dice have 36 equally likely pairs. The eleven totals are not equally likely, so dividing by 11 is wrong.

10 $P(\text{greater than } 4)=\frac{2}{6}=\frac{1}{3}$, so $60\times\frac{1}{3}=20$ rolls. This is an average over many sets of 60 rolls, not a guarantee for one set.

11 No. $P(A)=\frac{n(A)}{n(S)}$ with $n(A)\le n(S)$, so $0\le P(A)\le 1$. A value of 1.2 would need more favourable outcomes than outcomes.

12 $1-\frac{10}{25}=\frac{15}{25}=\frac{3}{5}$

13 $\frac{7}{n}=0.35$, so $n=\frac{7}{0.35}=20$ counters.

14 The remainder is $240-96-72=72$.

(a) $\frac{96}{240}=\frac{2}{5}$ (b) $\frac{168}{240}=\frac{7}{10}$ (c) $\frac{72}{240}=\frac{3}{10}$

15 (a) AA, AB, AC, BA, BB, BC, CA, CB, CC — 9 outcomes.

(b) $\frac{3}{9}=\frac{1}{3}$ (c) $1-P(\text{no A})=1-\frac{4}{9}=\frac{5}{9}$

(d) The nine pairs are equally likely only when each face has probability $\frac{1}{3}$. A biased spinner gives unequal branch probabilities, so counting outcomes would no longer give the probability.

16 The claim is false. $P(\text{no 6 in six rolls}) = \left(\frac{5}{6}\right)^6 \approx 0.335$, so $P(\text{at least one 6}) \approx 0.665$: roughly one set of six rolls in three contains no 6 at all. Expected number 1 does not mean guaranteed number 1.

Exercise 13B

1 Phone and tablet 20; phone only $55 - 20 = 35$; tablet only $30 - 20 = 10$; neither $80 - 55 - 30 + 20 = 15$.

$P(\text{phone}) = \frac{55}{80} = \frac{11}{16}$, $P(\text{tablet}) = \frac{30}{80} = \frac{3}{8}$, $P(\text{both}) = \frac{20}{80} = \frac{1}{4}$.

Common error: $55 + 30 = 85$ *exceeds 80 because the 20 who own both are counted twice.*

2 (a) $\frac{18 + 15 - 7}{40} = \frac{26}{40} = \frac{13}{20}$ (b) $\frac{14}{40} = \frac{7}{20}$ (c) $\frac{11}{40}$

3 $\frac{26}{52} + \frac{4}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$

The two red aces belong to both events, so adding $\frac{26}{52} + \frac{4}{52}$ counts them twice.

4 Football only 11, tennis only 6, both 6, so 23 play at least one and $30 - 23 = 7$ play neither.

5 Newspaper only $48 - 12 = 36$, so $P = \frac{36}{120} = \frac{3}{10}$.

Neither $= 120 - (48 + 36 - 12) = 48$, so $P = \frac{48}{120} = \frac{2}{5}$.

6 $P(A \cup B) = 0.3 + 0.45 = 0.75$; $P(\text{neither}) = 0.25$.

7 Each of the four cells is one region of the Venn diagram, and both partition the same people, so either can be rebuilt from the other. The table makes marginal totals and arithmetic checks easy; the Venn diagram makes the overlap and the "neither" region visible.

8 The events overlap, so adding counts the overlap twice and the total exceeds 1. Since $P(A \cup B) \le 1$, the smallest possible overlap is $0.6 + 0.5 - 1 = 0.1$.

9 At least one subject $= 60 - 12 = 48$. So $25 + 30 - b = 48$, giving $b = 7$ who study both.

10 Mutually exclusive: "roll a 1" and "roll a 2". Not mutually exclusive: "roll an even number" and "roll a number greater than 3", which both occur on a 4 or a 6.

11 $70 + 40 - 25 = 85\%$, so 85% own at least one and 15% own neither.

12 $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$; $P(A \text{ only}) = 0.3$; $P(\text{neither}) = 0.3$.

13 At least one $= 108 + 72 - 36 = 144$.

$P(\text{language or art}) = \frac{144}{180} = \frac{4}{5}$; $P(\text{neither}) = \frac{36}{180} = \frac{1}{5}$; $P(\text{language only}) = \frac{72}{180} = \frac{2}{5}$.

14 (a) Gym only 90, swim only 40, both 50, neither 70.

(b) $\frac{90 + 40}{250} = \frac{130}{250} = \frac{13}{25}$

(c) $P(\text{swim} \mid \text{gym}) = \frac{50}{140} = \frac{5}{14} \approx 0.357$, so the claim is false: about 36%, not more than half.

15 At least one $= 100 - 20 = 80$, so $45 + 38 - b = 80$ and $b = 3$.

Bicycle only $= 42$, so $P = \frac{42}{100} = \frac{21}{50}$.

16 Since $P(A \cap B) \ge 0$, subtracting it cannot increase the sum, so $P(A \cup B) \le P(A) + P(B)$. Equality holds exactly when $P(A \cap B) = 0$, that is, when A and B are mutually exclusive.

Exercise 13C

1 $P(G) = \frac{4}{7}$, then 3 green remain among 6: $\frac{4}{7} \times \frac{3}{6} = \frac{12}{42} = \frac{2}{7}$.

2 $P(GY) + P(YG) = \frac{4}{7} \cdot \frac{3}{6} + \frac{3}{7} \cdot \frac{4}{6} = \frac{24}{42} = \frac{4}{7}$.

$P(\text{at least one yellow}) = 1 - P(GG) = 1 - \frac{2}{7} = \frac{5}{7}$.

3 Eight paths. $P(HHH) = \frac{1}{8}$; exactly two heads HHT, HTH, THH $= \frac{3}{8}$; at least one tail $= 1 - \frac{1}{8} = \frac{7}{8}$.

4 $P(HH) = 0.7^2 = 0.49$; exactly one head $= 2(0.7)(0.3) = 0.42$; at least one tail $= 1 - 0.49 = 0.51$.

5 $P(RR) = \frac{5}{8} \cdot \frac{5}{8} = \frac{25}{64}$; $P(RB) = \frac{5}{8} \cdot \frac{3}{8} = \frac{15}{64}$.

Same colour $= \frac{25}{64} + \frac{9}{64} = \frac{34}{64} = \frac{17}{32}$.

6 $P(\text{red then blue}) = 0.25 \times 0.45 = 0.1125 \approx 0.113$; $P(\text{green both}) = 0.3^2 = 0.09$.

7 With replacement the counter is returned, so every draw uses the original counts and the stages are independent. Without replacement the counter is missing, so both the number of that colour and the total fall by one, and the second stage is conditional on the first.

8 After one red counter has gone the bag holds 7 counters, not 8, and only 4 of them are red, so the branch should be $\frac{4}{7}$. A branch of $\frac{5}{5} = 1$ would claim the next counter is certain to be red.

9 Yes: $0.12 + 0.18 + 0.28 + 0.42 = 1.00$, and each first-stage pair sums to 1 (0.30 and 0.70). The paths are complete and the tree is consistent.

10 $P(BB) = \frac{6}{8} \cdot \frac{5}{7} = \frac{30}{56} = \frac{15}{28}$.

$P(\text{at least one red}) = 1 - \frac{15}{28} = \frac{13}{28}$.

11 $P(\text{both pass}) = 0.8^2 = 0.64$; $P(\text{exactly one}) = 2(0.8)(0.2) = 0.32$.

12 The complete paths are mutually exclusive and exhaust the sample space, so their probabilities must total 1. The check detects a missing path, a stale denominator or an arithmetic slip.

13 (b) $\frac{6}{8} \cdot \frac{5}{7} = \frac{15}{28}$ (c) $1 - \frac{15}{28} = \frac{13}{28}$

14 Both days: $0.4 \times 0.7 = 0.28$.

Exactly one: $0.4 \times 0.3 + 0.6 \times 0.2 = 0.12 + 0.12 = 0.24$.

15 $P(RR) = \dfrac{r}{r+b} \cdot \dfrac{r-1}{r+b-1}$, $P(RB) = \dfrac{r}{r+b} \cdot \dfrac{b}{r+b-1}$, $P(\text{one of each}) = \dfrac{2rb}{(r+b)(r+b-1)}$.
With $r = 3$, $b = 2$: $\frac{3}{10}$, $\frac{3}{10}$ and $\frac{12}{20} = \frac{3}{5}$, matching Example 13.9.

16 All the same: $2 \times \dfrac{3}{6} \cdot \dfrac{2}{5} \cdot \dfrac{1}{4} = \dfrac{2}{20} = \dfrac{1}{10}$.
Exactly two red: three orders, each $\dfrac{3}{6} \cdot \dfrac{2}{5} \cdot \dfrac{3}{4} = \dfrac{3}{20}$, so $\dfrac{9}{20}$.
Check: $\frac{1}{10} + \frac{9}{20} + \frac{9}{20} = 1$.

Exercise 13D

1 $P(A \cap B) = 0.5 \times 0.4 = 0.2$; $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

2 Yes. $0.6 \times 0.3 = 0.18 = P(C \cap D)$, so the multiplication rule holds.

3 $P(\text{head and } 6) = \dfrac{1}{2} \cdot \dfrac{1}{6} = \dfrac{1}{12}$; $P(\text{tail and even}) = \dfrac{1}{2} \cdot \dfrac{1}{2} = \dfrac{1}{4}$.

4 $\dfrac{4}{52} \cdot \dfrac{3}{51} = \dfrac{12}{2652} = \dfrac{1}{221} \approx 0.00452$

5 $0.1 \times 0.1 = 0.01$

6 $P(E \cup F) = 0.7 + 0.2 = 0.9$. They cannot also be independent: independence would need $P(E \cap F) = 0.14$, but mutually exclusive events have $P(E \cap F) = 0$.

7 Independent: "roll an even number" and "roll more than 4". $P = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$, and both occur only on a 6.
Mutually exclusive: "roll a 1" and "roll a 6"; they cannot both happen.

8 The coin has no memory; tosses are independent, so $P(\text{tails}) = \dfrac{1}{2}$ on the second toss.

9 4 red remain among 9 counters, so $P(\text{second red} \mid \text{first blue}) = \dfrac{4}{9}$.

10 $0.6 \times 0.5 = 0.3 = P(\text{both})$, so the events are independent.

11 Removing a counter changes the composition, so the second draw is conditional on the first. Replacing it restores the original probabilities, and the stages become independent.

12 No. 95% describes the chance of a correct test result given a person's status. With a rare disease, false positives from the many healthy people can outnumber true positives, so far fewer than 95% of positive results are genuine.

13 $P(BB) = \dfrac{5}{10} \cdot \dfrac{4}{9} = \dfrac{2}{9}$.
$P(\text{different}) = 1 - \left(\dfrac{6}{90} + \dfrac{2}{90} + \dfrac{20}{90}\right) = \dfrac{62}{90} = \dfrac{31}{45} \approx 0.689$.

14 (a) $1 - (0.6 + 0.45 - 0.3) = 0.25$
(b) $0.6 \times 0.45 = 0.27 \neq 0.30$, so the events are dependent: walking and packed lunches occur together slightly more often than independence predicts.

15 $P(A \cap B) = 0.4 + 0.5 - 0.7 = 0.2$, which equals 0.4×0.5. The events are independent.

16 $P(\text{sum } 7) = \dfrac{1}{6}$, $P(\text{first die } 3) = \dfrac{1}{6}$, $P(\text{both}) = \dfrac{1}{36}$ (only 3, 4).
$\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$, so the events are independent.

Exercise 13E

1 $300 \times \dfrac{1}{6} = 50$

2 $400 \times 0.35 = 140$

3 $500 \times \dfrac{7}{10} = 350$

4 Expected winnings $= 10 \times 0.1 = 1$; net gain $= 1 - 2 = -£1$, a loss of £1 per play on average.

5 Expected prize $\dfrac{300}{500} = 0.60$; net $0.60 - 2 = -1.40$, a loss of £1.40 per ticket.

6 40 heads and 40 tails.

7 Each roll has $P(6) = \frac{1}{6}$, so $60 \times \frac{1}{6} = 10$. Counts vary at random between sets of 60 rolls, and 13 is only 3 above the mean, well inside ordinary variation.

8 Over 100 plays the player expects to lose $100 \times 0.50 = £50$; the organizer expects to gain £50.

9 2.5 is a long-run average, not a prize. One play pays one of the listed amounts, never 2.5 pounds.

10 No. With only 20 trials the relative frequency is unstable; 8 successes instead of the expected 5 is plausible. A much larger simulation is needed before doubting the model.

11 Each trial produces exactly one outcome, so the expected frequencies add to $n \sum P = n \times 1 = n$.

12 The expected winnings equal the cost of playing.

13 $E = 6 \cdot \dfrac{2}{6} + 2 \cdot \dfrac{1}{6} = \dfrac{7}{3} \approx 2.33$.
Net $= 2.33 - 3 = -0.67$. Not fair: the player loses about £0.67 per play.

14 (a) $7(0.2) - 2 = -0.60$, a loss of 60 pence per play.
(b) $0.2p = 2$, so $p = £10$.
(c) Profit per play is £0.60, so 500 plays yield about £300, which exceeds £200: the game is suitable.

15 $E = 0.4(5) + 0.6(-2) = 2 - 1.2 = £0.80$.
Fair red payout: $0.4p = 1.2$, so $p = £3$.

16 Two dice cost £1; win £6 if the total is 7. Six of the 36 outcomes are sevens, so $P = \frac{1}{6}$ and $E = 6 \cdot \frac{1}{6} = 1$; net $1 - 1 = 0$.
Any design with $P(\text{win}) \times \text{prize} = 1$ works. A prize of £12 on a total of 12 would not: $P = \frac{1}{36}$, so the expected winnings would be only £0.33.

Investigation · Less than seven, or seven and above

1 The 36 ordered pairs, counted by total: 2:1, 3:2, 4:3, 5:4, 6:5, so 15 outcomes below 7; 7:6, 8:5, 9:4, 10:3, 11:2, 12:1, so 21 outcomes of 7 or more. $15 + 21 = 36$.

2 $P(\text{player}) = \dfrac{15}{36} = \dfrac{5}{12}$; $P(\text{stallholder}) = \dfrac{21}{36} = \dfrac{7}{12}$.

3 Player: $120 \times \frac{5}{12} = 50$. Stallholder: $120 \times \frac{7}{12} = 70$.

4 Record wins and losses in a frequency table and divide wins by 40. A typical run gives about 17 wins, a relative frequency of 0.425 against the theoretical 0.417.

5 Forty games is a small sample. Random variation produces runs of wins and losses, so the relative frequency only settles near $\frac{5}{12}$ after many more games; the two values are unlikely to be identical.

6 Count a total of 7 as a draw, replayed. The player wins on 2–6 (15 ways), the stallholder on 8–12 (15 ways), and 6 ways are draws. Each then has probability $\frac{15}{36} = \frac{5}{12}$ of winning, so neither player has an advantage.

7 Ten games is a very small sample. With $p = \frac{5}{12}$, three wins in ten is an ordinary outcome, and relative frequencies from 10 trials vary widely. Fairness is a property of the rules and payoffs, not of one short run.

→ Three conditions: (i) the sample space is complete and its outcomes are equally likely; (ii) the rules and payoffs are fixed in advance and applied to every play; (iii) each player's expected net gain is zero, or the two expected gains are equal.

The expected value connects the two: $E = \sum xP(x)$ weights every payoff by its probability, so a game is fair exactly when the long-run average gain is zero. Individual games still produce winners and losers; fairness describes the average, not the next result.

Mixed practice 13

1 $\frac{5}{10} = \frac{1}{2}$; not green $\frac{8}{10} = \frac{4}{5}$; red or blue $\frac{8}{10} = \frac{4}{5}$.

2 $1 - 0.12 = 0.88$

3 $60 + 45 - 25 = 80$, so $P(\text{tea or coffee}) = \frac{80}{100} = \frac{4}{5}$; $P(\text{neither}) = 0.2$.

4 $P(RR) = \frac{9}{25}$; $P(RB) = \frac{3}{5} \cdot \frac{2}{5} = \frac{6}{25}$; same colour $= \frac{9}{25} + \frac{4}{25} = \frac{13}{25}$.

5 $P(A \cap B) = 0.6 \times 0.5 = 0.3$; $P(A \cup B) = 0.6 + 0.5 - 0.3 = 0.8$.

6 $350 \times 0.4 = 140$

7 Order matters: HT and TH differ, so the four outcomes are HH, HT, TH, TT. Each has probability $\frac{1}{4}$; with three listed outcomes, HT would have probability $\frac{1}{2}$ and the outcomes would not be equally likely.

8 The events overlap, so adding counts the overlap twice and gives more than 1. Since a probability cannot exceed 1, the overlap is at least $0.7 + 0.6 - 1 = 0.3$.

9 $\frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} = \frac{1}{221} \approx 0.00452$

10 "At least one head" is the complement of "no heads", and "no heads" in three tosses is TTT, with probability $\frac{1}{8}$. So the probability is $1 - \frac{1}{8} = \frac{7}{8}$.

11 $E = 10(0.25) = 2.50$; net $2.50 - 3 = -0.50$. Not fair.

12 No. In 100 trials, 28 successes against an expected 25 is ordinary variation; a difference of 3 is well within what chance produces, so the model is not contradicted.

13 $P(A \cup B) = 0.45 + 0.5 - 0.2 = 0.75$; $P(\text{neither}) = 0.25$; $P(A \text{ only}) = 0.25$.

14 $0.3 \times 0.6 = 0.18$

15 Three orders (RRB, RBR, BRR), each $\frac{4}{10} \cdot \frac{3}{9} \cdot \frac{6}{8} = \frac{1}{10}$, so $P = \frac{3}{10}$.

16 $E = 6 \cdot \frac{2}{6} + 3 \cdot \frac{1}{6} = 2.5$; net $2.5 - 3 = -0.50$. The fair cost is £2.50.

17 Expected prize $\frac{250}{400} = 0.625$; net $0.625 - 2 = -1.375 \approx -£1.38$.

School profit $= 400 \times 2 - 250 = £550$, matching 400×1.375.

18 $6x - 15 = 4x + 7$, so $2x = 22$ and $x = 11$. Check: $3(17) = 51$ and $4(11) + 7 = 51$.

19 $P(A \cap B) = 0.5 + 0.4 - 0.8 = 0.1$, but $0.5 \times 0.4 = 0.2$. Since $0.1 \neq 0.2$, the events are dependent.

Criterion A · Knowing and understanding

1 $S = \{HH, HT, TH, TT\}$; $P(\text{two heads}) = \frac{1}{4}$.

2 $P(\text{red}) = \frac{6}{10} = \frac{3}{5}$; $P(\text{not blue}) = \frac{6}{10} = \frac{3}{5}$.

3 $P(\text{sport or instrument}) = \frac{90 + 60 - 30}{150} = \frac{120}{150} = \frac{4}{5}$.
$P(\text{neither}) = \frac{30}{150} = \frac{1}{5}$.

4 Cat only 12, dog only 8, both 6, neither $40 - 26 = 14$.
$P(\text{cat only}) = \frac{12}{40} = \frac{3}{10}$; $P(\text{neither}) = \frac{14}{40} = \frac{7}{20}$.

5 $P(\text{both green}) = \frac{4}{7} \cdot \frac{3}{6} = \frac{2}{7}$.
$P(\text{one of each}) = \frac{12}{42} + \frac{12}{42} = \frac{4}{7}$.

6 $P(A) \times P(B) = 0.5 \times 0.4 = 0.2 = P(A \cap B)$, so the events are independent.

7 $\frac{13}{52} \cdot \frac{12}{51} = \frac{156}{2652} = \frac{1}{17} \approx 0.0588$

8 $250 \times 0.28 = 70$

9 Expected winnings $= 8 \cdot \frac{2}{6} = \frac{8}{3} \approx 2.67$; net $= 2.67 - 2 = +0.67$.

The game is not fair: it favours the player by about £0.67 per play.

10 $P(\text{not } E) = 1 - 0.37 = 0.63$. A probability is the ratio of favourable outcomes to outcomes, so it cannot exceed 1; 1.2 would need more favourable outcomes than there are outcomes.

11 The statement is invalid. Two outcomes need not be equally likely: in a die game that pays only on a 6, $P(\text{win}) = \frac{1}{6}$. Probability comes from the sample space and the rules, not from counting the named outcomes.

Criterion B · Investigating patterns

1 No six: $0.833,\ 0.694,\ 0.579,\ 0.482,\ 0.402,\ 0.335$ for $n = 1$ to 6.

At least one six:
$0.167,\ 0.306,\ 0.421,\ 0.518,\ 0.598,\ 0.665$.

2 $P(\text{at least one six}) = 1 - \left(\frac{5}{6}\right)^n$.

Each roll misses a six with probability $\frac{5}{6}$; independence makes the chance of missing on all n rolls $\left(\frac{5}{6}\right)^n$; "at least one" is the complement of "none".

3 Need $\left(\frac{5}{6}\right)^n < 0.1$. Trials: $n = 10$, $P = 0.838$; $n = 11$, 0.865; $n = 12$, 0.888; $n = 13$, 0.907. The smallest is $n = 13$.

4 $n = 13$: $1 - 0.0935 = 0.9065 > 0.9$.

$n = 12$: $1 - 0.1122 = 0.8878 < 0.9$. The threshold is crossed between 12 and 13.

5 Assumptions: the die is fair (each face has probability $\frac{1}{6}$) and the rolls are independent.

As n grows, $\left(\frac{5}{6}\right)^n \to 0$, so the probability approaches 1: a six becomes almost certain, though never certain for a finite number of rolls.

Criterion C · Communicating

1 36 ordered pairs. The six with total 7 are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), so $P(7) = \frac{6}{36} = \frac{1}{6}$.

2 Expected winnings $= 5 \times \frac{1}{6} = \frac{5}{6} \approx £0.83$.

Expected net gain $= \frac{5}{6} - 2 = -\frac{7}{6} \approx -£1.17$.

3 Fair means the expected net gain is zero: expected winnings equal the £2 cost. Individual players still win or lose, but over many plays neither side profits.

4 The claim confuses the prize with the expected gain. £5 is paid only once in six plays, so the average return is £0.83 against a £2 cost; the stallholder keeps about £1.17 per play, some 58% of the takings. A prize "more than double the cost" is still far below the fair prize.

5 Set $p \times \frac{1}{6} = 2$, so $p = £12$. A £12 prize makes the expected winnings equal the cost.

A prize between £5 and £12, say £8, keeps a margin of £0.67 per play while reducing the player's expected loss.

Marking guidance. 20 marks: sample space and $P(7)$ 4; expected winnings and net gain 5; meaning of fair 3; evaluation of the claim 4; recommendation with justification 4. Communication is judged with the Criterion C descriptors: headings, precise language, correct units and a conclusion that follows from the numbers. A report that quotes £5 and £2 but never divides by 6 cannot score more than 2 for the evaluation, however well written.

Criterion D · Applying mathematics in real-life contexts

1 Sample space: 36 ordered pairs. Winning event "total is 7": six outcomes, so $P = \frac{1}{6}$.

An even total would give $P = \frac{1}{2}$, too generous to raise money with a £4 prize.

2 Player: $4 \times \frac{1}{6} - 2 = -\frac{4}{3} \approx -£1.33$ per play.

Stall: $2 - \frac{4}{6} = £1.33$ per play.

3 300 plays: income £600; expected prizes $50 \times 4 = £200$; profit £400; less the £40 stall fee $= £360$, which exceeds £300.

4 Expected wins $100 \times \frac{1}{6} \approx 17$. A simulated 14 wins gives relative frequency 0.14 against 0.167; a gap of this size is ordinary variation for 100 plays.

5 Use the game. Fairness: an expected loss of £1.33 per play is large, so publish the 1-in-6 odds, display the prize and consider a £1 consolation or a lower price. Ethics: advertising "win £4" without the odds misleads younger players; state the chance and the purpose of the fundraiser.

Marking guidance. 24 marks: sample space and probability 6; expected values 5; profit including the stall cost 5; simulation and comparison 4; recommendation, fairness and ethics 4. Accept any complete model: a different winning event scores full marks if the probability, expected values and profit are consistent. The £300 target must be shown after the £40 fee, and the ethical point must be specific, not a general statement about honesty.

CAPSTONE · DESIGNING A SUSTAINABLE SPACE

Opening problem · the courtyard brief

1 At 1:50, one centimetre on the plan is 50 cm in reality, so 18 m = 1800 cm → 1800 ÷ 50 = 36 cm and 12 m = 1200 cm → 24 cm: the drawing is 36 cm by 24 cm.

Check: plan area $36 \times 24 = 864$ cm², and $864 \times 50^2 = 2\,160\,000$ cm² = 216 m², the real courtyard area.

2 Courtyard area $18 \times 12 = 216$ m², so 40% means at least $0.40 \times 216 = 86.4$ m².

Impermeable: deck $8 \times 6 = 48$ m², path $1.5 \times 38.5 = 57.75$ m², tank footprint $\pi(0.8)^2 = 2.01$ m²; total $48 + 57.75 + 2.01 = 107.76$ m².

Permeable: $216 - 107.76 = 108.24$ m², which is $\frac{108.24}{216} \times 100 = 50.1\%$. The requirement is met, with 21.84 m² to spare.

3 One sail: $\frac{1}{2} \times 5 \times 4.5 = 11.25$ m²; two sails: 22.5 m², which is > 20 m², so the shade requirement is met by 2.5 m².

Shadow: $\tan 62^\circ = \frac{3.2}{s}$, so $s = \frac{3.2}{1.8807} \approx 1.70$ m.

4 620 mm = 0.62 m, so the rain falling on the roof is $120 \times 0.62 = 74.4$ m³; the harvest is $74.4 \times 0.85 = 63.24$ m³ = 63 240 L a year.

A full tank supplies $2000 \div 87 = 22.99$, so about 23 days — longer than the observed 17-day dry spell, so the tank is adequate for a dry spell though not for a whole year.

5 Subtotal from Table C.3: $1824 + 3200 + 1402.50 + 1074 + 2280 + 850 + 2772 + 3600 = £17{,}002.50$.

Contingency $= 0.10 \times 17{,}002.50 = £1{,}700.25$, so total $= £18{,}702.75$.

Against the £18,000 budget: $18{,}702.75 - 18{,}000 = £702.75$ over, about 3.9% over.

6 Remove one 6-seat table: saves £380 and leaves 30 seats, still above the 24 required.

Shrink the deck from 48 m² to 45 m² (7.5 m by 6 m): saves $3 \times 38 = £114$ and adds 3 m² of permeable surface.

Shorten the planters from 16.5 m to 14.5 m: saves $2 \times 85 = £170$, and 22 fewer bags of soil saves a further $22 \times 6 = £132$.

Saving $= 380 + 114 + 170 + 132 = £796$, more than the £702.75 needed. New subtotal $17{,}002.50 - 796 = £16{,}206.50$; contingency £1,620.65; total £17,827.15, which is £172.85 under budget. Shade stays 22.5 m², seats stay 30, and permeable area is $108.24 + 3 - 2.4 = 108.84$ m² or 50.4%, so no constraint is broken.

Prerequisite check

1 $6x - 15 - 2x + 8 = 4x - 7$.

2 $2x + 7 = 3x - 3$, so $x = 10$.

3 $(x - 3)(x - 4)$, since -3 and -4 multiply to $+12$ and add to -7.

4 $\frac{12 - 4}{5 - 1} = \frac{8}{4} = 2$.

5 Area $8 \times 6 = 48$ m²; diagonal $\sqrt{8^2 + 6^2} = \sqrt{100} = 10$ m.

6 $240 \times 1.15 = 276$.

7 Mean $45 \div 5 = 9$; range $14 - 4 = 10$.

8 $1 - P(\text{no heads}) = 1 - \frac{1}{4} = \frac{3}{4}$.

The four outcomes are HH, HT, TH, TT; only TT has no head.

Exercise C.1

1 $9.4 \times 50 = 470$ cm and $6.2 \times 50 = 310$ cm, so 4.7 m by 3.1 m. Area $= 4.7 \times 3.1 = 14.57$ m².

2 $3.5 \times 200 = 700$ cm = 7 m.

3 18 m → 36 cm and 12 m → 24 cm. With a 2 cm margin each side the sheet must be at least 40 cm by 28 cm, so A3 (42.0 cm by 29.7 cm). A4, at 29.7 cm by 21.0 cm, is too small.

4 Deck $7.5 \times 6 = 45$ m²; impermeable $45 + 54 + 2.0 = 101$ m².

Permeable $216 - 101 = 115$ m², which is $\frac{115}{216} = 53.2\%$. The 40% requirement is 86.4 m², so the design exceeds it by 28.6 m².

5 (a) $6n \geq 30$, so $n \geq 5$ tables.

(b) $C \leq 18{,}000$ (c) $11.25s \geq 20$, so $s \geq 2$ sails (d) $w \geq 1.5$

6 An assumption is accepted without proof: rainfall is spread evenly through the year, or quoted prices hold until the build. A constraint is a condition the design must meet: $C \leq 18{,}000$, at least 24 seats, at least 20 m² of shade.

7 Lengths scale by 50, but area is two-dimensional, so areas scale by $50^2 = 2500$. The real area is 2500 times the plan area, not 50 times.

8 8 m → 32 cm and 6 m → 24 cm. The area ratio is $1 : 25^2 = 1 : 625$: model area 0.0768 m² against 48 m².

9 Diameter on the plan $160 \div 50 = 3.2$ cm, so radius 1.6 cm.

Plan area $= \pi(1.6)^2 = 8.04$ cm²; real area $= \pi(0.8)^2 = 2.01$ m².

10 Impermeable area may be at most $0.60 \times 216 = 129.6$ m², so $A \leq 129.6 - 57.75 - 2.01 = 69.84$ m².

A deck of about 70 m² is the limit — for example 6 m by 11.6 m, since the brief confines the deck width to between 2 m and 6 m. A larger deck breaks the 40% rule however it is shaped.

Exercise C.2

1 $\frac{63}{180} = 0.35 = 35\%$; the pie-chart angle is $0.35 \times 360^\circ = 126^\circ$.

2 $\frac{108}{180} = 60\%$. With 240 students in the same proportions, $240 \times 0.60 = 144$ would say essential.

3 $P(\text{Year 10}) = \frac{60}{180} = \frac{1}{3}$; $P(\text{group work}) = \frac{63}{180} = \frac{7}{20}$. $P(\text{both}) = \frac{24}{180} = \frac{2}{15} \approx 0.133$; $P(\text{Year 10} \mid \text{group work}) = \frac{24}{63} = \frac{8}{21} \approx 0.381$.

4 $P(\text{not rain}) = 1 - \frac{102}{365} = \frac{263}{365} \approx 0.721$.

Expected rain days $= 190 \times \frac{102}{365} \approx 53.1$, about 53 days. Dry week $= 0.721^5 \approx 0.194$.

5 Bimodal: 9 days and 11 days each occur three times, so both are modal values; range $11 - 6 = 5$ days; mean $\frac{102}{12} = 8.5$ days.

No month has 8.5 rain days and the counts run from 6 to 11, so the mean describes the year, not a typical month.

6 $150 \times 0.580 \times 0.80 = 69.6\ \text{m}^3 = 69{,}600$ L.

Tank life $= 2500 \div 95 = 26.3$, so 26 whole days.

7 Contingency $0.10 \times 9776 = £977.60$; total £10,753.60.

This is £7,246.40 under the £18,000 budget.

8 New subtotal $9776 \times 1.08 = £10{,}558.08$; contingency £1,055.81; total £11,613.89, still under £18,000.

The rise is £782.08 and the contingency is £977.60, so the contingency covers it and £195.52 remains.

9 Students in the library at lunchtime are more likely to favour quiet study and less likely to represent sport or social users, so the sample is biased towards one use. A better method is a random sample stratified by year group, taken in tutor groups at different times of day.

10 Demand $= 120 \times 17 = 2{,}040$ L, which exceeds 2,000 L, so the tank fails by 40 L; it lasts $2000 \div 120 = 16.7$ days.

With a 20% margin, $2{,}040 \times 1.20 = 2{,}448$ L, so add about 448 L of storage.

Exercise C.3

1 Length $l = 12 - x$, so $A = x(12 - x) = 12x - x^2$.

Maximum 36 m² at $x = 6$: a 6 m by 6 m deck.

2 $A_{\max} = \frac{36^2}{16} = 81\ \text{m}^2$, given by a 9 m by 9 m square.

3 $s = \frac{2.8}{\tan 58^\circ} \approx 1.75$ m.

4 $c = \sqrt{7^2 + 1.5^2} = \sqrt{51.25} \approx 7.16$ m.

Angle $= \tan^{-1}\frac{1.5}{7} \approx 12.1^\circ$.

5 Contingency $0.12 \times 14{,}800 = £1{,}776$; total £16,576.

This exceeds the £16,500 budget by £76.

6 Tables $2 \times 380 = £760$; deck $4 \times 38 = £152$; planters $3 \times 85 = £255$.

Soil $1.62\ \text{m}^3 = 1{,}620$ L needs 33 bags at £6 $= £198$. Total saving £1,365, or £1,361.40 if soil is priced pro rata.

7 New subtotal $16{,}206.50 \times 1.06 = £17{,}178.89$; with 10% contingency, £18,896.78.

Overrun against £18,000 is £896.78.

8 Design A: $1.2 + 0.75 + 0.8 + 0.75 + 0.4 = 3.90$.

Design B: $0.9 + 1.0 + 0.8 + 0.6 + 0.5 = 3.80$. Design A is preferred on these weights.

9 One feature cannot establish sustainability. The tank addresses water, but the claim also depends on the proportion of permeable surface (50.4% here), the durability and origin of the materials, the shade that reduces heat, and whether the tank can meet demand through a dry spell. A tank that empties before a 17-day spell ends, or decking shipped long distances, would weaken the claim.

10 $2000 \div 100 = 20$ days, which covers 19 days with 100 L to spare.

With a 15% margin: $1{,}900 \times 1.15 = 2{,}185$ L, so add about 185 L.

11 Two sails give 22.5 m², exceeding 20 m² by 2.5 m², which is 12.5%.

One sail gives 11.25 m², falling short by 8.75 m², which is 43.75%.

Capstone practice

1 $3x - 12 = 2x + 5$, so $x = 17$.

2 $2x^2 - 8x = 2x(x - 4)$; the solutions are $x = 0$ and $x = 4$.

3 Vertex $(3, -4)$; roots $x = 1$ and $x = 5$.

4 Contingency $0.075 \times 18{,}000 = £1{,}350$; total £19,350.

5 $4000(1.035)^3 = £4{,}434.87$.

6 Height $= 5\sin 68^\circ \approx 4.64$ m; foot $= 5\cos 68^\circ \approx 1.87$ m from the wall.

7 $V = \pi(0.8)^2(2.2) = 4.423\ \text{m}^3 \approx 4{,}420$ L.

8 Mean 18; median 16.5; range 15; quartiles 15 and 21, so IQR 6.

9 $0.72^2 = 0.5184$; at least one wet day $= 1 - 0.5184 = 0.4816$.

10 Length $= 57.75 \div 1.5 = 38.5$ m.

New area $= 1.8 \times 38.5 = 69.3\ \text{m}^2$; extra $11.55 \times 48 = £554.40$.

Criterion A · Knowing and understanding

1 $8.6 \times 50 = 430$ cm and $5.4 \times 50 = 270$ cm, so 4.3 m by 2.7 m; area 11.61 m².

2 Impermeable $45 + 57.75 + 2.01 = 104.76\ \text{m}^2$; permeable $216 - 104.76 = 111.24\ \text{m}^2$.

$\frac{111.24}{216} = 51.5\%$, above the required 40% (86.4 m²) by 24.84 m².

3 $2 \times \frac{1}{2} \times 5 \times 4.5 = 22.5\ \text{m}^2$; exceeds 20 m² by 2.5 m², which is 12.5%.

4 $100 \times 0.600 \times 0.80 = 48\ \text{m}^3 = 48{,}000$ L.

5 $2000 \div 95 = 21.05$, so 21 whole days.

6 Contingency £1,540; total £16,940, which is £1,060 under budget.

7 $2x - 6 = x + 7$, so $x = 13$.

8 $y = (x - 4)^2 - 4$, so the vertex is $(4, -4)$.

9 Mean $\frac{72}{6} = 12$; median $\frac{10 + 12}{2} = 11$.

10 $0.7^5 = 0.168$ for five dry days; at least one wet day $= 1 - 0.168 = 0.832$.

Criterion B · Investigating patterns

1 $x = 4$: length 11, area 44. $x = 5$: 10, 50. $x = 6$: 9, 54. $x = 7$: 8, 56. $x = 7.5$: 7.5, 56.25. $x = 8$: 7, 56. $x = 9$: 6, 54. $x = 10$: 5, 50.

The perimeter is 30 m throughout.

2 $l = 15 - x$, so $A = x(15 - x) = 15x - x^2$.

Domain $0 < x < 15$ for a genuine rectangle; in the courtyard the 12 m width and the 1.5 m path restrict x further.

3 $A = 56.25 - (x - 7.5)^2$, so the maximum is 56.25 m² at $x = 7.5$ m, a 7.5 m square.

4 For perimeter P, $l = \frac{P}{2} - x$ and $A = x\left(\frac{P}{2} - x\right) = \frac{P^2}{16} - \left(x - \frac{P}{4}\right)^2$.

A square is never negative, so $A \leq \frac{P^2}{16}$, with equality when $x = \frac{P}{4}$ and $l = \frac{P}{4}$ — a square.

5 $P = 24$: $\frac{576}{16} = 36$ m², a 6 m square.

$P = 36$: $\frac{1296}{16} = 81$ m², a 9 m square.

6 A 7.5 m square plus the 1.5 m path uses 9 m of the 12 m width and a long frontage, leaving little room for the route, the sails and planting. Area is maximal but circulation, shade and sightlines may be worse: 7 m by 8 m loses only 0.25 m².

Criterion C · Communicating

1 State the four constraints with units: at least 24 seats, at least 20 m² of shade, at least 40% permeable surface (86.4 m² of 216 m²), and a 1.5 m accessible route, all within £18,000.

2 At 1:50 the courtyard is 36 cm by 24 cm. The plan must label the deck (7.5 m by 6 m), the 1.5 m path, 14.5 m of planters, the tank and the seating, with a scale bar and north point.

3 Expected values: permeable 108.8 m², or 50.4%; shade 22.5 m²; harvest 63,240 L a year against 15,660 L demand; total cost £17,827.15 including contingency. Each needs its formula, substitution and unit.

4 Compare on cost, capacity and sustainability: weighted scores 3.70 for the first proposal and 3.90 for the revision, with the weights stated. Show at least one criterion where the other option wins.

5 The recommendation must name the trade-off (a table removed to meet the budget leaves 30 seats) and one limitation (rainfall is a ten-year average and rain days cluster, so tank performance in a dry spring is uncertain).

6 Give sample size, date and source for the survey (180 students, March), the weather record (ten-year regional average) and the quotations (supplier prices, June).

SAMPLE RESPONSE OUTLINE

The brief is to convert the 18 m by 12 m courtyard within £18,000, providing at least 24 seats, 20 m² of shade, 40% permeable surface and a 1.5 m accessible route. I recommend the revised design: a 7.5 m by 6 m deck, the 38.5 m path, 14.5 m of planters, two sails and a 2,000 L tank. Permeable surface is 108.8 m², or 50.4%, leaving 22.4 m² of slack; the sails give 22.5 m² of shade, 12.5% above target; the roof harvests 63,240 L a year against a 15,660 L demand, although storage limits a single dry spell to about 23 days. Including 10% contingency the cost is £17,827.15, £172.85 under budget. On the weighted matrix the revision scores 3.90 against 3.70, mainly on cost. The trade-off is capacity: 30 seats and very little financial slack. The main limitation is that rainfall is a ten-year average and rain days cluster, so the tank may underperform in an unusually dry spring. Sources: student survey (180 responses, March); regional rainfall record (ten years to 2025); supplier quotations (June).

Marking guidance. 20 marks: brief and constraints 3; labelled plan 4; calculations 5; comparison of options 3; recommendation with trade-off and limitation 3; source note 2. The plan must carry a scale and dimensions to score above 2. Calculations score for method and units, not only for the final value. A recommendation that ignores the trade-off, or a "limitation" that is only a restatement of a strength, cannot reach the top band.

Criterion D · Applying mathematics in real-life contexts

1 $63 \times 2 = 126$ group-work visits per week, so $126 \div 5 = 25.2$ per school day. Against 30 seats this is about 84% average occupancy: enough on average, but lunchtimes are likely to exceed 30.

2 Rain days $= 190 \times \frac{102}{365} \approx 53$; dry week $= 0.721^5 \approx 0.194$, about one week in five.

Shade is still needed in a wet climate, and a dry week is common, so the design must offer shade and rain shelter together.

3 New demand $87 \times 1.30 = 113.1$ L per day. Over 17 days this is 1,922.7 L $\leq$ 2,000 L, so the tank copes with about 77 L (0.7 day) to spare. The margin is thin.

4 New total £18,896.78; overrun £896.78.

Remove one table (seats fall to 24, the minimum), 4 m² of deck and 3 m of planters with the soil. The base saving is £985, so at risen prices it is $985 \times 1.06 = £1{,}044.10$, and with contingency the total falls by £1,148.51. New total £17,748.27, which is £251.73 under budget.

5 Accessibility: a 1.5 m route is the minimum for a wheelchair to pass, and chairs at peak times can block it; the 7.5 m deck plus path leaves only 3 m for planting. Widen the entrance and turning points to 1.8 m and keep a clear 1.5 m zone.

Fairness: the survey counts only students present, and 63 of 180 prefer group work while 54 prefer quiet study; reserve two tables for quiet use.

6 Approve with conditions. Evidence: total £17,748 against £18,000; shade 22.5 m² above the 20 m² target; permeable surface 50.4% against 40%.

Limitation: the tank covers a 17-day spell at 30% higher demand by only 77 L, so a hotter spring would fail; monitor storage and add capacity if demand rises.

SAMPLE RESPONSE OUTLINE

I would approve the revised design with conditions. The evidence supports it: the total of £17,748 after the 6% rise and the trims is £252 under the £18,000 budget, shade is 22.5 m² against a 20 m² target, and permeable surface is 50.4% against 40%. Capacity is adequate on average — 25.2 group-work visits a day against 30 seats — but not at peak lunchtimes, when demand may exceed the seating. The tank covers a 17-day dry spell at 30% higher demand by only 77 L, so a hotter spring or a larger planting scheme would fail; I would add about 200 L of storage or reduce the planting. The trade-off is that removing a table to pay for the rise leaves exactly 24 seats, with no slack for a damaged table. One limitation: the rainfall figures are a ten-year average and rain days cluster, so the dry-spell calculation is a baseline, not a forecast.

Marking guidance. 20 marks: capacity 4; rain and shade interpretation 4; tank test 3; cost revision 4; accessibility or fairness 3; decision with evidence and limitation 2. Award method marks for a correct substitution even where an earlier error carries through, provided the interpretation stays consistent. The decision must cite two numerical pieces of evidence and one genuine limitation; an unsupported opinion scores 0 for the final part.

REFERENCE

R Mathematical notation

Notation is a compressed language. This table gives each symbol used in the course, with the reading that makes it unambiguous. Where a symbol has more than one common meaning, the meaning used in this book is stated.

Sets and number

$\mathbb{N}$ natural numbers: $1, 2, 3, \ldots$ (sometimes including 0; this book does not).

$\mathbb{Z}$ integers: whole numbers, positive, negative and zero.

$\mathbb{Q}$ rational numbers: those that can be written $\frac{a}{b}$ with integers a, b and $b \neq 0$.

$\mathbb{R}$ real numbers: every point on the number line.

$\in$, $\notin$ is an element of; is not an element of.

$\subset$ is a subset of, so $\mathbb{N} \subset \mathbb{Z}$.

$\cup$, $\cap$ union (in either set); intersection (in both sets).

$\varnothing$ the empty set, containing no elements.

$\{x : \ldots\}$ the set of all x such that the condition holds.

$<$, $\leq$, $>$, $\geq$ less than; less than or equal to; greater than; greater than or equal to.

$|x|$ modulus: the distance of x from zero, so $|-5| = 5$.

$\approx$, $\neq$ is approximately equal to; is not equal to.

$a \times 10^n$ standard form, with $1 \leq a < 10$ and n an integer.

Algebra and functions

a^n n factors of a; a is the base, n the index or exponent.

$a^{1/n}$, $a^{m/n}$ the nth root of a; the nth root raised to the power m.

$\sqrt{a}$, $\sqrt[3]{a}$ square root; cube root. The square root sign gives the non-negative root.

$\pm$ plus or minus; both signs must be considered.

$\equiv$ is identically equal to; true for every value of the variable.

$\Rightarrow$, $\Leftarrow$ implies; is implied by.

$\Leftrightarrow$ is equivalent to; each statement implies the other.

$\therefore$ therefore. $\blacksquare$ end of proof.

$f(x)$ the value of function f at input x. $f(3)$ means substitute $x = 3$.

$x \mapsto f(x)$ the rule that sends x to $f(x)$.

$y = mx + c$ a straight line with gradient m and y-intercept c.

$y = a(x - h)^2 + k$ a parabola with vertex (h, k); $a \neq 0$.

Δ the discriminant $b^2 - 4ac$ of a quadratic. Also a change in a quantity.

u_n the nth term of a sequence; u_1 is the first term.

d, r common difference of an arithmetic sequence; common ratio of a geometric sequence.

$n!$ factorial: $n(n-1)\cdots 2 \cdot 1$, with $0! = 1$.

Geometry and measure

AB the length of segment AB; AB may also name the segment itself.

$\angle ABC$ the angle at vertex B, between BA and BC.

$\triangle ABC$ the triangle with vertices A, B, C.

$\cong$ is congruent to: same shape and size.

$\sim$ is similar to: same shape, corresponding lengths in a fixed ratio.

$\parallel$, $\perp$ is parallel to; is perpendicular to.

$^\circ$ degrees. A right angle is 90°; a straight angle is 180°.

θ a Greek letter used for an unknown angle.

$\sin\theta$, $\cos\theta$, $\tan\theta$ trigonometric ratios of angle θ in a right-angled triangle.

k scale factor; k^2 and k^3 scale area and volume.

A, S, V area, surface area, volume. A_{plan} is the area on a scale drawing.

π the ratio of circumference to diameter.

mm, cm, m, km units of length; $1\text{ m} = 100\text{ cm}$, $1\text{ km} = 1000\text{ m}$.

mL, L, m³ units of volume; $1\text{ cm}^3 = 1\text{ mL}$, $1000\text{ L} = 1\text{ m}^3$.

Statistics and probability

$\bar{x}$, f, cf, E the mean of a data set; frequency; cumulative frequency; an event.

$A \cap B$, $A \cup B$, A' both A and B; A or B or both; the complement of A. $P(A)$ is a number from 0 to 1.

Q_1, Q_2, Q_3 lower quartile, median, upper quartile; $\text{IQR} = Q_3 - Q_1$.

REFERENCE

R Formulae and relationships

These are the relationships used in the course. A formula is a relationship, not a procedure: state what each letter represents and the conditions under which the relationship holds.

Number and accuracy

Percentage error $= \dfrac{|\text{measured} - \text{true}|}{\text{true}} \times 100\%$. If the true value is unknown, use the measured value as the denominator and say so.

Bounds for a measurement x to the nearest unit u: $x - \frac{u}{2} \le x < x + \frac{u}{2}$.

Recurring decimal $0.\overline{d} = \dfrac{d}{9}$, $0.\overline{ab} = \dfrac{ab}{99}$, and similarly for longer blocks.

Surds $\sqrt{a}\sqrt{b} = \sqrt{ab}$; $\dfrac{1}{\sqrt{a}} = \dfrac{\sqrt{a}}{a}$; $\dfrac{1}{a + \sqrt{b}} = \dfrac{a - \sqrt{b}}{a^2 - b}$.

Algebra and quadratics

Quadratic formula for $ax^2 + bx + c = 0$, $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant $\Delta = b^2 - 4ac$: two distinct real roots if $\Delta > 0$; one repeated root if $\Delta = 0$; no real roots if $\Delta < 0$.

Completing the square $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$.

Vertex of $y = ax^2 + bx + c$: $x = -\dfrac{b}{2a}$, $y = c - \dfrac{b^2}{4a}$.

Difference of squares $a^2 - b^2 = (a - b)(a + b)$.

Algebraic fraction simplify by factorising first: $\dfrac{ax + ay}{bx + by} = \dfrac{a(x + y)}{b(x + y)} = \dfrac{a}{b}$, provided $x + y \neq 0$.

Linear system $a_1x + b_1y = c_1$, $a_2x + b_2y = c_2$. It has a unique solution when $\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$.

Sequences and finance

Arithmetic sequence $u_n = u_1 + (n - 1)d$; the sum of n terms is $S_n = \dfrac{n}{2}\left(2u_1 + (n - 1)d\right)$.

Geometric sequence $u_n = u_1 r^{n-1}$; the sum of n terms is $S_n = \dfrac{u_1(r^n - 1)}{r - 1}$, $r \neq 1$.

Quadratic sequence constant second difference $2a$; the nth term has the form $an^2 + bn + c$.

Compound interest $A = P(1 + r)^n$, where r is the decimal rate per period and n the number of periods.

Depreciation $A = P(1 - r)^n$. **Appreciation** uses $1 + r$.

Reverse percentage original $= \dfrac{\text{final}}{\text{multiplier}}$. A 20% increase has multiplier 1.20; a 20% decrease has multiplier 0.80.

Coordinates and graphs

Gradient $m = \dfrac{y_2 - y_1}{x_2 - x_1}$. **Distance** $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Midpoint $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$.

Line $y - y_1 = m(x - x_1)$; parallel lines have equal gradients; perpendicular gradients satisfy $m_1 m_2 = -1$.

Direct variation $y = kx$; the graph is a line through the origin.

REFERENCE

R Geometry, statistics and probability

Triangles and trigonometry

Pythagoras in a right-angled triangle with hypotenuse c: $a^2 + b^2 = c^2$. Conversely, if $a^2 + b^2 = c^2$, the angle opposite c is a right angle.

Trigonometric ratios $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$.

Angle of elevation is measured upward from the horizontal; the angle of depression is measured downward from the horizontal.

Bearing is measured clockwise from north and written with three figures, for example 047°.

Similarity corresponding lengths scale by k; areas by k^2; volumes by k^3.

Congruence tests SSS, SAS, ASA/AAS, RHS. Similar triangles: AA, SSS (all three ratios equal), SAS (two ratios and the included angle).

Area of a triangle $\frac{1}{2} \times \text{base} \times \text{perpendicular height}$.

Perimeter, area and volume

Circle $C = 2\pi r$; $A = \pi r^2$.

Prism or cylinder $V = Ah$, where A is the cross-sectional area; $S = 2A + Ph$ for a prism with perimeter P.

Cylinder curved area $= 2\pi rh$; total surface area $= 2\pi r^2 + 2\pi rh$.

Cone $V = \frac{1}{3}\pi r^2 h$; curved area $= \pi rl$, where $l = \sqrt{r^2 + h^2}$.

Sphere $V = \frac{4}{3}\pi r^3$; $S = 4\pi r^2$. A hemisphere has half these values plus the flat circular face.

Pyramid $V = \frac{1}{3}Ah$, where A is the base area and h the perpendicular height.

Frustum $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ for a cone frustum; subtract the missing solid from the whole.

Composite solid add volumes; for surface area, add exposed faces only and subtract hidden joins.

Scale drawing real length $=$ plan length $\times$ scale factor; real area $=$ plan area $\times$ (scale factor)^2.

Statistics

Mean $\bar{x} = \dfrac{\sum x}{n}$. For grouped data, use class midpoints and $\bar{x} = \dfrac{\sum fx}{\sum f}$.

Median the middle value of an ordered set; for an even number of values, the mean of the two middle values.

Quartiles Q_1, Q_2, Q_3 divide the ordered data into four groups. $\text{IQR} = Q_3 - Q_1$.

Outlier a value below $Q_1 - 1.5\,\text{IQR}$ or above $Q_3 + 1.5\,\text{IQR}$.

Frequency density $= \dfrac{\text{frequency}}{\text{class width}}$; the area of each histogram bar equals its frequency.

Range $=$ largest $-$ smallest. The range is not robust; the IQR is.

Probability

Complement $P(A') = 1 - P(A)$.

Addition rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. For mutually exclusive events, $P(A \cap B) = 0$.

Independent events $P(A \cap B) = P(A)P(B)$; equivalently $P(A \mid B) = P(A)$.

Conditional probability $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, $P(B) \neq 0$.

Expected frequency $= np$, where n is the number of trials and p the probability of the event.

Expected value $= \sum xP(x)$. A game is fair when the expected net gain is zero.

Tree diagrams multiply along a path; add the probabilities of paths that give the same outcome.

REFERENCE

G Glossary

Accuracy. How close a measured or calculated value is to the true value. A result can be precise but inaccurate if it is consistently displaced.

Algebraic fraction. A fraction with algebraic expressions in the numerator, denominator or both. Simplify by factorising and cancelling, stating excluded values.

Angle of depression. The angle measured downward from the horizontal to a line of sight.

Angle of elevation. The angle measured upward from the horizontal to a line of sight.

Appreciation. An increase in value, often calculated at a compound rate: $A = P(1 + r)^n$.

Arithmetic sequence. A sequence with a constant difference between consecutive terms: $u_n = u_1 + (n - 1)d$.

Assumption. A statement accepted without proof so that a model can be built. It must be stated and, where possible, tested.

Axis of symmetry. The vertical line through the vertex of a parabola; for $y = a(x - h)^2 + k$, it is $x = h$.

Bearing. A direction measured clockwise from north and written with three figures, such as $047°$.

Bias. A systematic tendency for a sample, measurement or display to misrepresent the population.

Box plot. A display of the five-number summary: minimum, Q_1, median, Q_3, maximum.

Break-even. The point at which revenue equals total cost, so profit is zero.

Capacity. The volume a container can hold, usually measured in litres or millilitres.

Causation. A genuine cause-and-effect link. Correlation between two variables does not by itself establish causation.

Census. A survey that measures every member of a population.

Class interval. A range of values used to group continuous data, for example $20 \le x < 30$.

Common difference. The constant amount added to each term of an arithmetic sequence.

Completing the square. Rewriting $ax^2 + bx + c$ in the form $a(x - h)^2 + k$, which reveals the vertex and the maximum or minimum.

Composite solid. A solid formed by joining simpler solids. Its volume is the sum; its surface area includes only exposed faces.

Compound interest. Interest calculated on the principal and on previously earned interest: $A = P(1 + r)^n$.

Conditional probability. The probability of an event given that another event has occurred: $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$.

Cone. A solid with a circular base and a single vertex. $V = \frac{1}{3}\pi r^2 h$; curved surface area $= \pi r l$.

Congruence. Identical shape and size. Corresponding sides and angles are equal; the symbol is $\cong$.

Conjugate. The pair $a + \sqrt{b}$ and $a - \sqrt{b}$. Their product $a^2 - b$ is rational, which removes a surd from a denominator.

Constraint. A condition a design or solution must satisfy, such as a budget limit or a minimum number of seats.

Contingency. An allowance added to a budget for unexpected costs. It is part of the total, not spare money.

Continuous data. Data that can take any value in an interval, such as height or time; it is usually grouped for display.

Correlation. The strength and direction of a linear association between two variables, from -1 to 1.

Cosine. In a right-angled triangle, $\cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}}$.

Cumulative frequency. A running total of frequencies up to a given value or class boundary.

Cylinder. A prism with a circular cross-section. $V = \pi r^2 h$; total surface area $= 2\pi r^2 + 2\pi r h$.

Decision matrix. A table that scores options against weighted criteria and totals the weighted scores.

Degree. A unit of angle, $\frac{1}{360}$ of a full turn; also the highest power in a polynomial.

Dependent events. Events whose probabilities change when the other event is known to have occurred.

Depreciation. A decrease in value at a compound rate: $A = P(1 - r)^n$.

Direct variation. A relationship of the form $y = kx$, whose graph is a straight line through the origin.

Discrete data. Data that can take only separate, countable values, such as the number of seats.

Discriminant. For $ax^2 + bx + c = 0$, the value $\Delta = b^2 - 4ac$, which determines the number of real roots.

Distance formula. The distance between (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Domain. The set of input values for which a function is defined.

Factorable quadratic. A quadratic that can be written as the product of two linear factors with rational coefficients.

Factorising. Writing an expression as a product of factors, the reverse of expanding.

Feasible. Satisfying every constraint. A feasible design or solution is one that can actually be built or used under the stated conditions.

Fibonacci sequence. A sequence in which each term is the sum of the two previous terms: $1, 1, 2, 3, 5, 8, \ldots$

Five-number summary. The minimum, lower quartile, median, upper quartile and maximum of a data set.

Frequency density. Frequency divided by class width; used on a histogram so that bar area represents frequency.

Frustum. The part of a cone or pyramid between two parallel cuts. Its volume is found by subtracting the removed solid.

Function. A rule that assigns exactly one output to each input in its domain.

Function notation. The notation $f(x)$, meaning the output of function f for input x.

Geometric sequence. A sequence with a constant ratio between consecutive terms: $u_n = u_1 r^{n-1}$.

Gradient. The steepness of a line, $m = \dfrac{y_2 - y_1}{x_2 - x_1}$; the change in y for each unit change in x.

Grouped data. Data summarised in class intervals rather than listed individually.

Hemisphere. Half a sphere, including the flat circular face when surface area is required.

Histogram. A chart for continuous data in which frequency is represented by bar area and the horizontal axis is a continuous scale.

Hypotenuse. The side opposite the right angle in a right-angled triangle; the longest side.

Identity. An equation that is true for every value of the variable, such as $2(x + 3) \equiv 2x + 6$.

Image. The result of applying a transformation to a point, line or shape.

Inconsistent system. A system of equations with no common solution; its lines are parallel.

Independent events. Events for which the occurrence of one does not change the probability of the other: $P(A \cap B) = P(A)P(B)$.

Inequality. A statement that two quantities are not equal, using $<$, $\leq$, $>$ or $\geq$. Its solution is a set of values.

Interpolation. Estimating a value inside the range of known data, for example from a line of best fit.

Interquartile range. The range of the middle half of the data: $\text{IQR} = Q_3 - Q_1$.

Intersection. The set of elements common to two sets, written $A \cap B$.

Isometry. A transformation that preserves distance and angle: translation, rotation or reflection.

Like terms. Terms with the same variable factors and powers, which can be combined by adding or subtracting coefficients.

Linear model. A model of the form $y = mx + c$, in which a constant rate of change is assumed.

Lower bound. The smallest value a measurement could have, given its stated accuracy.

Midpoint. The point halfway between two points: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Modelling cycle. The iterative process of formulating a real problem, solving the model, interpreting the result, evaluating it and communicating it, then revising.

Monic quadratic. A quadratic whose coefficient of x^2 is 1, such as $x^2 - 5x + 6$.

Mutually exclusive. Events that cannot occur together, so $P(A \cap B) = 0$.

Net. A flat pattern that folds to form a solid; used to find surface area.

nth term. A formula that gives the term in position n of a sequence, usually in terms of n.

Objective. The quantity a design or model seeks to maximise or minimise, such as area or cost.

Outcome. A single possible result of a probability experiment.

Outlier. A value far from the rest of the data, often defined as below $Q_1 - 1.5\,\text{IQR}$ or above $Q_3 + 1.5\,\text{IQR}$.

Parabola. The U-shaped graph of a quadratic function.

Parameter. A quantity in a model that is fixed for a particular case but can be changed to test alternatives, such as a price per square metre.

Parallel lines. Lines in the same plane that never meet; they have equal gradients.

Perfect square. An expression that is the square of another expression, such as $x^2 + 6x + 9 = (x + 3)^2$.

Permeable surface. A surface that allows water to soak through, such as soil or gravel, rather than running off like concrete.

Perpendicular lines. Lines that meet at a right angle; their gradients satisfy $m_1 m_2 = -1$.

Point–gradient form. The equation $y - y_1 = m(x - x_1)$ of a line through (x_1, y_1) with gradient m.

Population. The whole group about which data are wanted; a sample is drawn from it.

Precision. How fine or consistent a measurement is, often shown by the number of significant figures or decimal places.

Primary data. Data collected by the person or group using them, for their own question.

Principal. The original amount invested or borrowed, before interest is added.

Prism. A solid with a constant cross-section; $V = Ah$, where A is the cross-sectional area.

Probability experiment. A process with uncertain outcomes, such as rolling a die or spinning a spinner.

Pyramid. A solid with a polygonal base and triangular faces meeting at a vertex; $V = \frac{1}{3}Ah$.

Pythagoras' theorem. In a right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides.

Qualitative data. Data that describe qualities or categories rather than numerical amounts.

Quantitative data. Numerical data that record an amount or count.

Quartile. One of the values that divide ordered data into four equal groups: Q_1, Q_2 (median), Q_3.

Range. The difference between the largest and smallest values; a simple but non-robust measure of spread.

Rationalising the denominator. Rewriting a fraction so that its denominator contains no surd, usually by multiplying by a conjugate.

Relative frequency. The experimental probability of an event: the number of successes divided by the number of trials.

Reliability. The extent to which a measurement or data source gives consistent results when repeated.

Reverse percentage. Finding the original amount before a percentage change, by dividing by the multiplier.

Rotation. A transformation that turns a shape about a fixed centre through a given angle and direction.

Root. A solution of an equation; on a graph, an x-intercept.

Sample. A subset of a population, selected to represent it.

Sample space. The set of all possible outcomes of a probability experiment.

SAS. Side–angle–side: a congruence test using two sides and the included angle.

Scale drawing. A drawing in which every length is a fixed fraction of the real length.

Scale factor. The multiplier that maps a shape to a similar shape; areas scale by its square and volumes by its cube.

Second difference. The difference between consecutive first differences; constant for a quadratic sequence.

Secondary data. Data collected by someone else, often for a different purpose.

Sensitivity. How much a conclusion changes when an input or assumption changes. A sensitive model has little slack.

Sequence. An ordered list of numbers, often generated by a rule.

Sign diagram. A number-line diagram showing where an expression is positive, zero or negative.

Similarity. Same shape but not necessarily the same size; corresponding lengths are in a fixed ratio.

Standard form. The notation $a \times 10^n$ with $1 \le a < 10$ and n an integer.

Surd. An irrational root left in exact form, such as $\sqrt{5}$ or $3\sqrt{2}$.

Surface area. The total area of all exposed faces of a solid.

Theoretical probability. Probability calculated from equally likely outcomes, without experiment.

Transformation. A change of position, size or orientation: translation, reflection, rotation or enlargement.

Tree diagram. A branching diagram showing successive events and their probabilities.

Upper bound. The largest value a measurement could have, given its stated accuracy.

Variable. A quantity that can change or take different values; in a model, a quantity the designer can choose or that varies with conditions.

Variance. The mean of the squared deviations from the mean; its square root is the standard deviation.

Venn diagram. A diagram using overlapping regions to show sets and their intersections.

Vertex. A corner of a shape; for a parabola, the turning point.

Vertex form. The form $y = a(x - h)^2 + k$ of a quadratic, showing the vertex (h, k).

Volume. The amount of space a solid occupies, measured in cubic units.

Volume scale factor. For similar solids with length scale factor k, volumes scale by k^3.

Weighted score. A total in which each score is multiplied by a weight reflecting its importance, then the products are added.

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